ISLAMIC REPUBLIC OF PAKISTAN
NATIONAL DISASTER MANAGEMENT AGENCY
(NDMA)

TECHNICAL SUPPORT PROJECT TO REVIEW AND UPDATE NATIONAL DISASTER PREVENTION PLAN (NDMP)

FINAL REPORT

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JAPAN INTERNATIONAL COOPERATION AGENCY

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OYO CORPORATION

PROJECT LOCATION MAP

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ABBREVIATIONS AND ACRONYMS

ADB	Asian Development Bank
ADRC	Asian Disaster Reduction Center
AJK (AJ&K)	Azad Jammu and Kashmir
AWS	Automatic Weather Station
BP	Basic Principle
CCI	Council of Common Interests
CCP	Climate Change Policy
CDA	Capital Development Authority
CENC	China Earthquake Networks Center
CFMIP	Cloud Feedback Model Intercomparison Project
CIA	Central Intelligence Agency
C/P	Counterpart
DCP	Data Collection Platform
DDMA	District Disaster Management Authority
DDMP	District Disaster Management Plan
DMA	Disaster Management Authority
DRR	Disaster Risk Reduction
DRRM	Disaster Risk Reduction Management
EAD	Economic Affairs Division
EMME14	2014 Earthquake Model of the Middle East
EM-DAT	Emergency Events Database
FFC	Federal Flood Commission
FFD	Flood Forecasting Division
FPSP-III	Flood Protection Sector Project-III
F/S	Feasibility Study
GB	Gilgit Baltistan
GBDMA	Gilgit Baltistan Disaster Management Authority
GCISC	Global Change Impact Studies Centre
GDP	Gross Domestic Product
GEM	Global Earthquake Model
GIS	Geographic Information System
GLOF	Glacial Lake Outburst Flood
GMRC	Glacial Monitoring Research Centre
GoP	Government of Pakistan
GPS	Global Positioning System
GSHAP	Global Seismic Hazard Assessment Program
GSP	Geological Survey of Pakistan
HFA	Hyogo Framework for Action
HKH	Hindu Kush-Himalayan
IDA	International Development Agency
INDC	Intended Nationally Determined Contribution
IOC	Intergovernmental Oceanographic Commission
JCC	Joint Coordination Committee
JICA	Japan International Cooperation Agency
KP	Khyber Pakhtunkhwa（KP）
MHVRA	Multi Hazard Vulnerability and Risk Assessment
M/M	Minutes of Meetings
MoCC	Ministry of Climate Change
MoPS	Ministry of Planning Development & Special Initiatives
MoWR	Ministry of Water Resources
MP	Master Plan
NAP	National Action Plan
NatCat	Natural Catastrophe
NCCP	National Climate Change Policy
NDMA	National Disaster Management Authority
NDMAct	National Disaster Management Act

NDMC	National Disaster Management Council
NDMC	National Drought Monitoring Center
NDMO	National Disaster Management Ordinance
NDMP	National Disaster Management Plan
NDMP-II	National Disaster Management Plan II
NDRMF	National Disaster Risk Management Fund
NED	NED (Nadirshaw Edulji Dinshaw) University of Engineering and Technology
NEOC	National Emergency Operation Center
NESPAK	National Engineering Services Pakistan
NFPP-IV	National Flood Protection Plan - IV
NIDM	National Institute of Disaster Management
NIED	National Research Institute for Earth Science and Disaster Resilience
NOAA	National Oceanic and Atmospheric Administration
NUST	National University of Science and Technology
OBS	Ocean Bottom Seismometer
OCHA	United Nations Office for the Coordination of Humanitarian Affairs
OJT	On-the-Job Training
PAGASA	Philippine Atmospheric, Geophysical and Astronomical Services Administration
Pak-EPA	Pakistan Environmental Protection Agency
PC	Planning Commission
PCRWR	Pakistan Council of Research in Water Resources
PDM	Project Design Matrix
PDMA	Provincial Disaster Management Authority
PDMP	Provincial Disaster Management Plan
PEC	Pakistan Engineering Council
PGA	Peak Ground Acceleration
PID(s)	Provincial Irrigation Department(s)
PKR	Pakistani Rupee
PMD	Pakistan Meteorological Department
PO	Plan of Operation
PSHA	Probabilistic Seismic Hazard Assessment
PTHA	Probabilistic Tsunami Hazard Assessment
R/D	Record of Discussion
RCP	Representative Concentration Pathways
RDA	Rawalpindi Development Authority
R-SMOG	Remote sensing for space-time mapping of smog in Punjab and identification of the underlying causes using geographic information system
SA	Spectrum Acceleration
SEDAC	Socioeconomic Data and Applications Center
SOP	Standard Operation Plan
SPARCO	Space & Upper Atmosphere Research Commission
SSP	Shared Socio-economic Pathways
S/GB/PDMA	State/Gilgit Balochistan/Provincial Disaster Management Authority
UNDP	United Nations Development Programme
USGS	United States Geological Survey
WAPDA	Water and Power Development Authority
WASA	Water and Sanitation Agency
WB	The World Bank
WCD	World Cities Database
WFB	The World Factbook
WWF	World Wide Fund for Nature
ZSP	Zoological Survey Department of Pakistan

CHAPTER 1 OUTLINE OF THE PROJECT

1.1 Background of the Project

Islamic Republic of Pakistan (hereinafter referred to as "Pakistan") has been repeatedly affected by natural disasters such as floods, earthquakes, landslides, and cyclones. Major disasters such as the 2005 Kashmir Earthquake and the 2010 devastating floods have caused significant devastation. To address these challenges, the Government of Pakistan has significantly reformed its conventional disaster management approach, which was primarily reactive and fragmented. In this context, the National Disaster Management Commission was established in 2007, followed by the enactment of the National Disaster Management Act (2010) and the establishment of the National Disaster Management Authority (NDMA). In addition, under JICA's development research type technical cooperation named "National Disaster Management Plan Formulation Project", Pakistan's first National Disaster Management Plan (hereinafter referred to as NDMP) was approved by the National Disaster Management Commission in 2013, with a focus on strengthening the disaster management system with a focus on prevention / mitigation and integrated-multi-hazard response.

Based on the NDMP, the Government of Pakistan is promoting the formulation of disaster management plans at the provincial and district levels, but its contribution to risk reduction projects is insufficient, and it is necessary to revise the initial plan considering the current disaster risk. Furthermore, in the “Sendai Framework for Disaster Risk Reduction 2015-2030” adopted at the United Nations World Conference on Disaster Risk Reduction (2015), highlights the urgent need to address new priorities such as “Promotion of Advance Investment in Disaster Risk Reduction”, “Strengthening Preparedness For Effective Emergency Response”, “Build Back Better” approaches. The priorities must be urgently incorporated in the current NDMP. In particular, flood risk is expected to rise due to ongoing climate change and rapid urbanization, making it critical to flood control measures to reduce to reduce flood risk and ensure sustainable development.

As described above, given the need to update the the NDMP, JICA has decided to implement this project with the aim of strengthening Pakistan's capacity to formulate a national disaster prevention plan, and a basic plan formulation study was conducted from August to October 2021, and both parties signed the R/D on 31st January 2022. The Project team for the “Technical Support Project to Review and Update National Disaster Management Plan” was dispatched in May 2022 to conduct a detailed planning survey. As a result, M/M was prepared to confirm the Project Design Matrix (hereinafter referred to as the PDM) and Plan of Operation (hereinafter referred to as the PO) and was duly signed by both parties on 29th June 2022.

1.2 Summary of the Project

1.2.1 Objective

The objective of the project is to strengthen the disaster management planning process and thereby contribute to the promotion of advance disaster prevention investment by preparing Substantial Contents for analyzing disaster risk, updating national disaster management plans, reviewing the draft pre-disaster prevention investment plan in the flood field, and coordinating with the related organizations.

1.2.2 Overall Goal

The overall goal is "Based on the updated National Disaster Management Plan (NDMP-II), priority projects with an emphasis on proactive disaster prevention investments will be defined and implementation of projects will begin within the country." The evaluation indicator of the overall goal is " An increase in the budget allocated for advance disaster prevention investment project.”." with the target value to be determined following the detailed planning survey.

1.2.3 Project Goal

The project goal is "to strengthen the disaster prevention planning process by working on revising the National Disaster Management Plan, which is implemented under the leadership, monitoring and coordination of NDMA." The following four indicators are used to evaluate the project goal:

- Technical review materials for NDMP-II, including measures to improve the current NDMP, are prepared.
- The principles and basic matters of advance disaster prevention investment plans are organized.
- An advance disaster prevention investment plan for flood control is formulated.
- Priority projects for NDMP-II are formulated.

1.3 Project Area

The project will be implemented across the entire country of Pakistan. The project office will be established in Islamabad, where NDMA is located.

1.4 C/P Agency, Cooperation Agency

A CP agency is the National Disaster Management Authority (NDMA).

Cooperating agencies include the Federal Flood Commission (FFC), Water and Power Development Authority (WAPDA), Pakistan Meteorological Department (PMD), and the Provincial Irrigation Departments (PIDs).), Ministry of Climate Change (MoCC), and Global Change Impact Studies Center (GCISC).

1.5 Task Description

1.5.1 Expected Results

Table 1.5.1 shows the four outcomes to be achieved through the Project.

Table 1.5.1 Project Outcomes

Outcome 1	NDMA's capabilities of information collection and analysis of natural hazards, situation of exposures and vulnerabilities are strengthened.
Outcome 2	Progress and issues of the current NDMP will be grasped, and the ability to formulate a more effective NDMP will be improved.
Outcome 3	A long list (draft) of priority projects for each type of disaster is prepared and NDMP-II technical review materials are prepared.
Outcome 4	Capacity as a central disaster prevention organization through disaster prevention centers is improved.

1.5.2 Outline of Activities

Table 1.5.2 summarizes the activities involved in each of the four outcomes in Table 1.5.1.

Table 1.5.2 Project Outcomes

Activities for Outcome 1	
1-1	Confirmation of hazard intensity and probability of occurrence, which are the basis for disaster risk information in the current NDMP.
1-2	Confirmation of vulnerability and exposure data underlying disaster risk information in the current NDMP.
1-3	Confirmation of the necessity to update and review disaster risk information in the current NDMP.
1-4	Perform a risk assessment of disaster risk information for updating and review which is confirmed in Activities 1-3.
Activities for Outcome 2	
2-1	Based on the results of the risk assessment in Activity 1-4, the basic strategy for Disaster Risk Reduction (DRR) aimed at by the current NDMP and its effectiveness will be reviewed.
2-2	The status of utilization and achievement of priority projects and human resource development plans of the current NDMP will be reviewed by relevant ministries and agencies, and issues will be identified.
2-3	After collecting and analyzing information on the issues of the current NDMP, improvement measures in NDMP-II will be examined.
2-4	The overall structure of NDMP-II including the above improvement measures will be examined.
Activities for Outcome 3	
3-1	Collect information on Projects for the investment for disaster risk reduction currently planned by ministries and agencies related to disaster prevention
3-2	The results of the risk assessment of Activity 1-4 and the results of the information collection of Activity 3-1 will be analyzed to identify additional projects that should be considered, and a draft project including budget size will be considered.
3-3	Based on Activity 3-1 and 3-2, A long list (draft) is prepared for each disaster type and the priority is considered.
3-4	The implementation promotion method of priority projects is examined, and the technical examination material of NDMP-II is prepared.
Activities for Activity 4	
4-1	Knowledge sharing on flood and earthquake related to capacity strengthening of the National Emergency Operation Center (NEOC) is conducted.
4-2	Workshops and training will be conducted in Pakistan and Japan.

1.6 Work Schedule and Amendment of Some Activities

The project period spans 38 months, from March 2022 to May 2025. Although the current R/D sets the total duration of each activity at 39 months, the actual duration of each activity (outcome 1 to outcome 4) is about 35 months, as the detailed planning survey needs to be carried out at the beginning of Project.

However, after the detailed planning study conducted during the project period, the current chairman did not approve the R/D and P/O signed with the previous chairman. As a result, after about a year of intermittent discussions, it was decided to update the R/D with an amendment (M/M) acceptable to both the chairman and JICA, while aligning the policies of both parties, in order to establish an appropriate disaster prevention system in Pakistan from January 2023 onwards. As a result, it was decided to resume support activities from the second half of 2024.

Initially, NDMA focused on "creating an annual plan that is easy to update and understand through NDMA's own efforts every year" and "prioritizing the strengthening of the National Emergency Operation Center (NEOC) through automation of decision-making systems using AI, etc." However, after the change in the R/D, based on the premise that long-term planning is essential for the development of disaster prevention systems in Pakistan, the project team supported the formulation of the annual NDMP (National Disaster Management Plan) while communicating sufficiently with the counterpart (C/P) to avoid misunderstandings at NDMA. In addition, regarding NDMA's aim to strengthen NEOC by automating decision-making using AI, the project team supported them to understand that decision-making is not necessarily based only on clear criteria, but is an extremely complex process that requires comprehensive consideration of the situation of victims and related parties, as well as objective data, in the midst of uncertainty.

After the above R/D change, the support activities were implemented through workshops and training in Japan, with the content of activities in Output 4 changed.

1.7 Changes of Project Activities

1.7.1 Final Response to the Changes in Policy by the Current NDMA Chairman

The current NDMA Chairman requested a change in the activities of this project, presenting the "formulation of an annual plan" and "strengthening the capacity of NEOC." Japan then implemented support activities based on the following three policies. For the formulation of an annual plan, the project team supported the creation of the mid- to long-term plan, keeping in mind the need for it.

- NDMA should have a mid- to long-term vision for improving disaster prevention capabilities throughout the country, clearly indicate the roles, goals, and necessary activities of each disaster prevention-related organization, and then oversee and manage activities aimed at achieving those goals (management of disaster prevention plans and implementation of measures). Therefore, annual disaster prevention plans should be formulated with a mid- to long-term vision.
- The construction of a decision-making system for disaster response, etc., to be constructed by NEOC should not necessarily be based on clear criteria alone, but should be understood to be an extremely complex process in which decisions are made in an objective and comprehensive manner, taking into account the situation of victims and related parties, as well as information shared by various related parties and organizations, in an uncertain environment. (Deepening understanding of the nature of decision-making systems)
- As the central disaster prevention organization, NDMA should grasp disaster risk areas based on information and results compiled by other ministries and agencies in order to respond quickly after a disaster occurs in Pakistan. It is also necessary to grasp the issues for risk assessment according to the type of disaster and cooperate and collaborate with related organizations to resolve the issues. (Promotion of efficient disaster risk management and disaster response)

1.7.2 Specific Changes made through Revised R/D

Following the scope cut of the project contents based on the optimization of the project contents in light of the current situation (the M/M on the scope cut of the R/D contents was handed over between NDMA and the JICA mission team in March 2024), discussions were held with NDMA's C/P on future activities in June 2024, and the following points were agreed upon.

- Pre-FS for flood control would not be conducted.
- Understanding of appropriate disaster prevention systems would be improved through support for strengthening NEOC.
- Training in Japan would be held only once.
- Explain the change in P/O to the person in charge in response to the above.

For the purpose of "support strengthening NEOC," seminars were held in Pakistan in October 2024 and February 2025. The training in Japan scheduled for May 2025 was canceled due to the outbreak of an international conflict.

CHAPTER 2 ACTIVITIES RELATED TO OUTCOME 1

Outcome 1	NDMA's capabilities of information collection and analysis of natural hazards, situation of exposures and vulnerabilities are strengthened.

2.1 Activity 1-1

Confirmation of hazard intensity and probability of occurrence, which are the basis for disaster risk information in the current NDMP.

2.1.1 Flood

2.1.1.1 Hazard Intensity in the Current NDMP

In the current NDMP, the Project Team reviewed the contents of basic data of the Flood Hazard & Risk Map (see Figure 2.1.1). Classification of hazards were based on the number of inundations caused by floods before 2011, the presence or absence of inundation and the number of people affected by floods during 2010 and 2011 as index data, and are rated on a scale of 5 for each district.

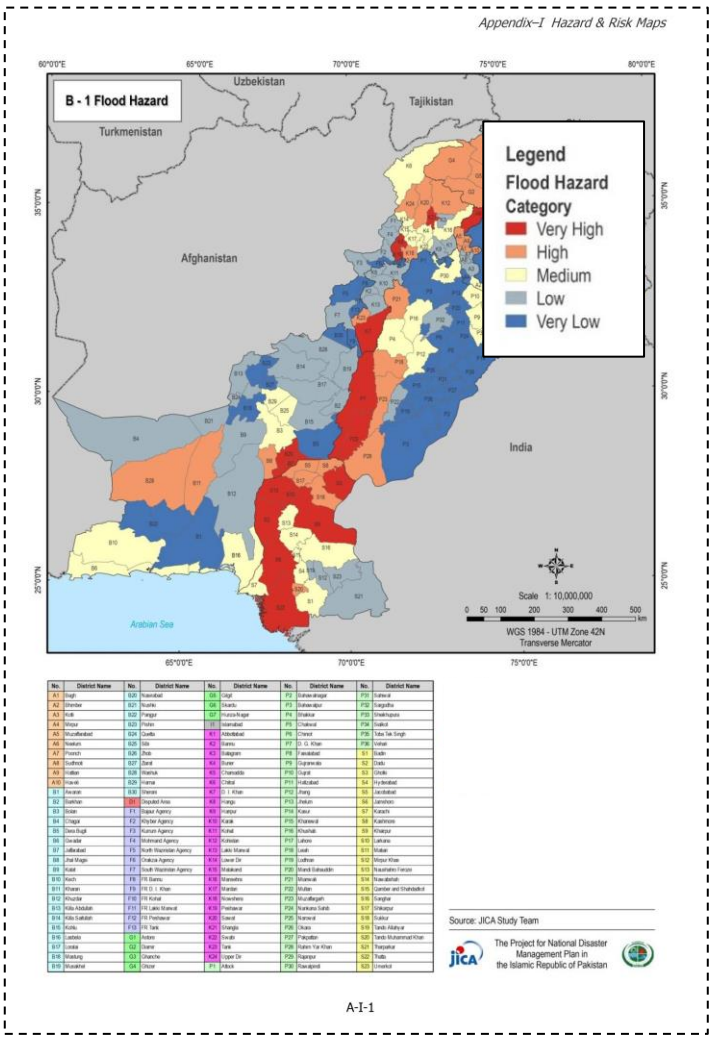
Figure 2.1.1 shows that numerous districts with high hazard intensity lie along the lower to middle reaches of the Indus River, , with additional hotspots present in its upper reaches..

2.1.1.2 History of Damage (Year of Occurrence, Scale of Damage, Etc.)

Table 2.1.1 presents the history if major floods, compiled from the current reports of NDMP, NDMA and the FFC. According to these reports, the number of deaths due to floods has continued to exceed 100 every year since 2010.

In recent years, the 2010 and 2022 floods inflicted significantly greater damage than other flood event.

For the 2010 floods, damage was primarily driven by the Indus River system's overflow, while the 2022 floods were mainly caused by intense regional rainfall.



Source: NDMP, Main Volume, A-1-1

Figure 2.1.1 Hazard & Risk Map (Example of Flood Hazard) indicated in the Current NDMP

Table 2.1.1 History of Past Flood Damage in Pakistan

Year	Value of Property Damaged (In Million Rs) Unadjusted	Adjusted	Lives Lost (persons)	Villages Affected (number)	Rivers Mainly Affected
2022	33 million people		1,730	Not Reported	
2021	Not Reported		198	Not Reported	
2020	1,193,353 people		423	Not Reported	
2019	Not Reported		235	Not Reported	
2018	Not Reported		135	Not Reported	Kabul, Chenab, Indus, Ravi
2017	Not Reported		271	Not Reported	Kabul, Chenab, Indus
2016	570		153	45	Chenab
2015	16,150		238	4,111	Indus
2014	41,800		367	3,798	Chenab, Jhelum
2013	190,000		333	8,297	Chenab, Sutlej, Indus
2012	250,800		571	14,159	**(Current NDMP was issued)**
2011	324,533		520	38,700	(Local Rainfall in Sindh/Balochistan)
2010	854,771		1,985	17,553	Kabul, Indus, Jhelum
2009	Not Reported		99	89	
2008	Not Reported		157	800	
2007	7,208.23		586	6,498	
2006	Not Reported		541	2,477	
2005	Not Reported		59	1,931	Indus
2004	15.00	15.00	85	47	
2003	5,175.00	5,175.00	484	4,376	
2001	450.00	450.00	219	50	Lai Nullah
1995	6,125.00	8,698.00	591	6,852	
1992	34,751.00	69,580.00	1,008	13,208	Indus, Jhelum, and Chenab
1988	6,879.00	25,630.00	508	1,000	Indus/Jhelum, & Chenab/Ravi/Sutlej
1978	4,478.00	51,489.00	393	9,199	
1976	5,880.00	80,504.00	425	18,390	Indus, Jhelum, Chenab and Ravi (Tarbela Dam completed in 1974)
1973	5,137.00	118,684.00	474	9,719	Indus, Jhelum, Chenab, Ravi and Sutlej (Chashma Reservoir completed in 1971) (Mangla Dam completed in 1967) (Warsak Dam completed in 1960)
1957	152.50	6,958.00	83	4,498	Chenab and Ravi
1956	155.50	7,356.00	160	11,609	Indus, Jhelum
1950	199.80	11,282.00	2,190	10,000	Chenab, Ravi and Sutlej

Source: Federal Flood Commission (FFC) for 1950 – 2010.
Preliminary DNA Floods 2010-2011, NDMA
"Value of Property Damaged" from 2012 - 2016 are based on Annual Report of FFC
"Lives Lost" and "Villages Affected" from 2012 – 2016 are based on Annual Report of NDMA
Source: NDMP, Main Volume, 2012 Project team added

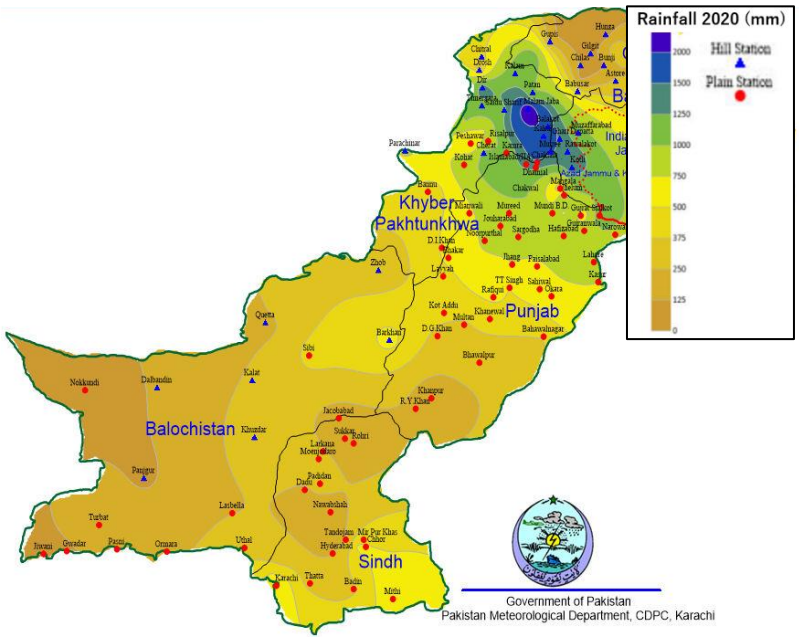
2.1.1.3 Flood Factors (Heavy Rain, Flash Flood, Overflow, GLOF, Etc.)

(1) Overflow from Rivers

1) Natural Breach

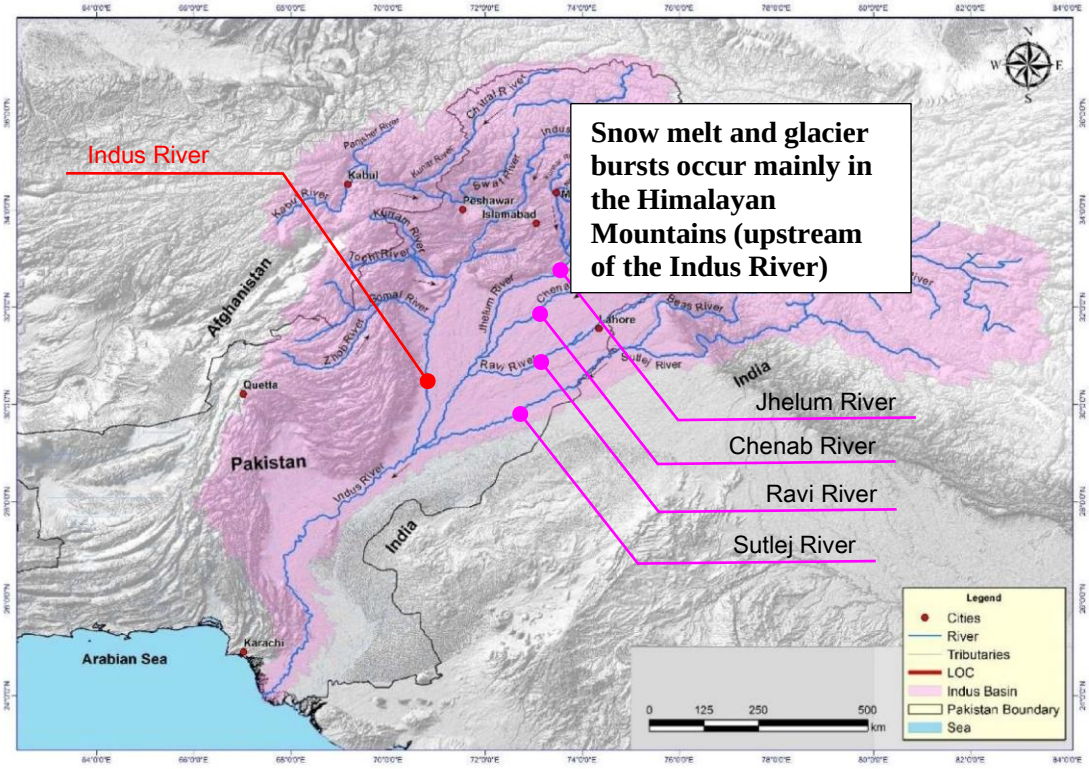
River line floods are caused by heavy rainfall concentrated in the upper reaches of rivers during the monsoon season (July to September). Low pressure caused by the monsoon current in the Bay of Bengal brings heavy rainfall to the mountains of western Himalaya and thus causes flooding of the Indus River and its main tributaries.

Figure 2.1.2 shows the annual rainfall distribution produced by the Pakistan Meteorological Department (PMD) in 2020, and Figure 2.1.3 shows the Indus River Basin map.



Source: State of Pakistan's Climate in 2020, PMD

Figure 2.1.2 Annual Precipitation Distribution in 20202.1



Source: Project Team added on Development of National Flood Protection Plan-IV (NFPP-IV) and Related Studies to Enhance Capacity Building of Federal Flood Commission-FFC, Final Report, 2019

Figure 2.1.3 Indus River Basin

Floods in the Indus River Basin results from mountains snowmelt, rainfall occurring alongside snowmelt, runoff from the Hill Torrent Basin, and glacier meltwater releases. While glaciers function as natural reservoirs, they can cause sudden increases in the level of the Indus River and its tributaries due to heavy rainfall and elevated temperatures. Since the impact of snowmelt on the flood peak is greater in the Chenab River than in other tributaries (Jhelum, Ravi and Sutlej), flood

damage occurs frequently along with heavy rainfall. Chenab River has no reservoirs such as dams in Pakistan compared to other rivers (Incidentally, there is a plan to construct the Chiniot Multipurpose Dam at Chiniot Tehsil located in the middle and lower Chenab.).

In addition, in the plain area extending over Punjab and Sindh provinces where the slope is gentle, large areas will be inundated if overflow occurs during flood. A large number of Hill Torrents have flowed into the western part of the low lying areas on the right bank of the Indus River (west of the boundary between Punjab and Sindh provinces and KPK and Balochistan provinces), causing serious problems in the low lying areas after heavy rainfall in the Hill Torrent basin. Although the affected area is usually limited, extensive inundation occurs in the low lying areas during a large flood.

"Development of National Flood Protection Plan-IV (NFPP-IV) and Related Studies to Enhance Capacity Building of Federal Flood Commission-FFC" briefly describes the locations and causes of natural breach in the 2010 flood. The main cause of natural breach was overflow, and it can be assumed that there were some areas where bund height was insufficient at that time. Breach caused by the destruction of scouring and seepage also occurred to a limited extent.

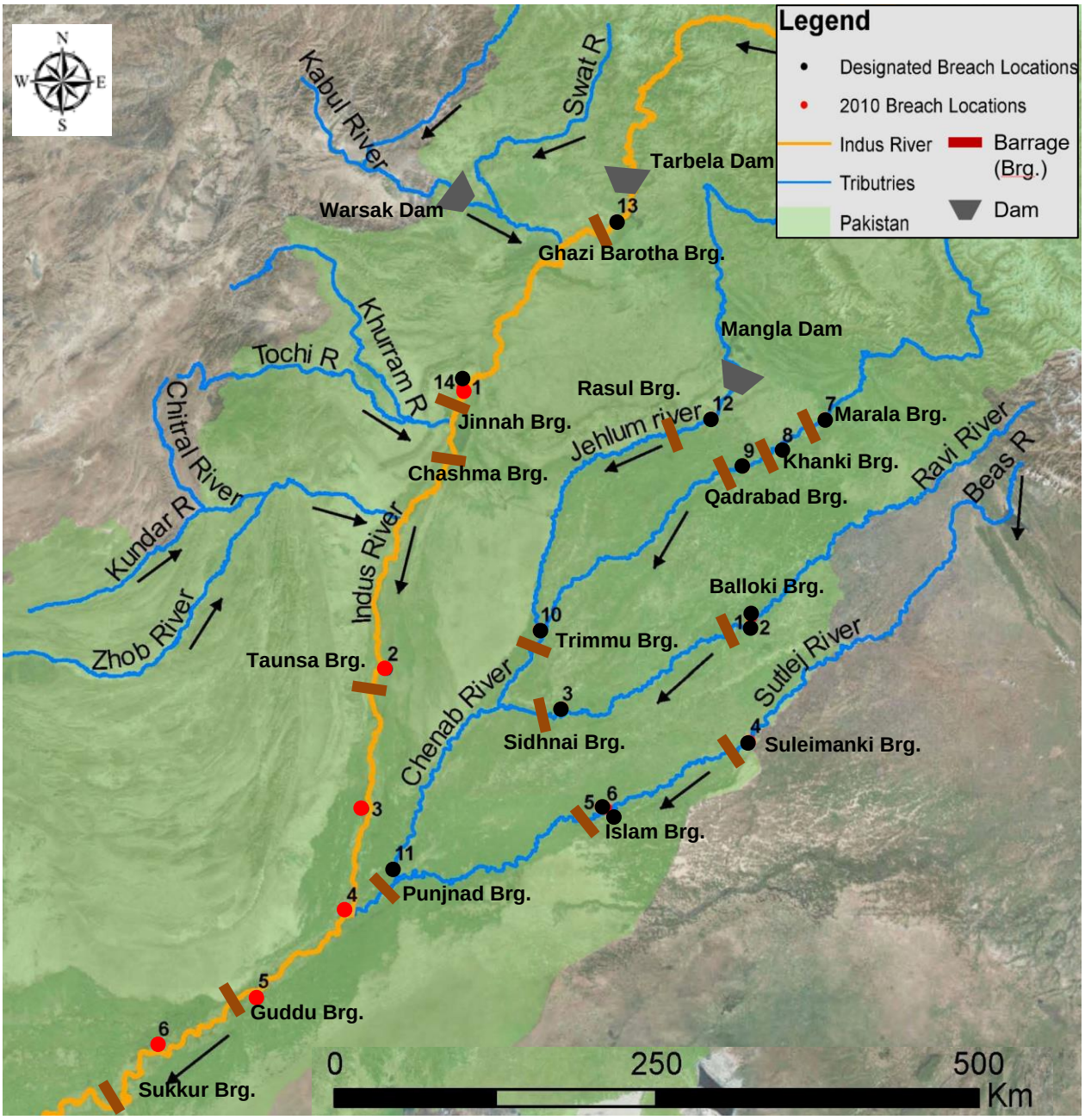
2) Intentional Breach (Man-made Breach)

In the main river and tributaries of the Indus River, a design discharge (water level) has been set at the major barrages. In the event of a flood exceeding this level, bund upstream of the barrage is intentionally cut off (called a Breach) to discharge the part of the flood flow out of the main river, protecting the barrage and simultaneously controlling the excessive increase of river discharge into the downstream channel. According to NFPP-IV, there are 14 designated breach locations, which are designated upstream of barrages. Table 2.1.2 lists the barrages with designated breach locations upstream of them, and Figure 2.1.4 shows the 14 designated breach locations.

Table 2.1.2 Barrages with Designated Breach Locations in the Upstream of them

Serial No.	River Name	Barrage Name	Design Discharge	
			[cu.ft/sec (cfs)]	[m^3/s]
1	Ravi	Balloki	225,000	6,368
2	Ravi	Balloki	225,000	6,368
3	Ravi	Sidhnai	175,000	4,953
4	Sutlej	Suleimanki	325,000	9,198
5	Sutlej	Islam	300,000	8,490
6	Sutlej	Islam	300,000	8,490
7	Chenab	Marala	1,100,000	31,130
8	Chenab	Khanki	800,000	22,640
9	Chenab	Qadirabad	900,000	25,470
10	Chenab	Trimmu	645,000	18,254
11	Chenab	Panjnad	700,000	19,810
12	Jhelum	Rasul	850,000	24,055
13	Indus	Ghazi Barotha	660,000	18,678
14	Indus	Jinnah	950,000	26,885

Source: Development of National Flood Protection Plan-IV (NFPP-IV) and Related Studies to Enhance Capacity Building of Federal Flood Commission-FFC, Final Report, 2019 Project Team



Source: Development of National Flood Protection Plan-IV (NFPP-IV) and Related Studies to Enhance Capacity Building of Federal Flood Commission-FFC, Final Report, 2019 Project Team

Figure 2.1.4 Location of Designated Breach Locations in the Upstream of them and Actual Breach Locations During the 2010 Flood

Due to the intentional bund breach, flood occurs in the area behind the breach point. In the 2010 flood, intentional bund breach was carried out at 4 sites in accordance with the Standard Operation Procedure (SOP) in the upstream side of Jinnah Barrage. In 2022, a bund breach was carried out at lake shore bund around Manchar Lake to prevent inundation of the surrounding urban area, but this was not planned in advance by SOP. Thus, it can be considered that intentional bund breach is classified into planned type and emergency type which are carried out at the discretion at the site as an emergency response.

(2) Flash Flood in Hill Torrent

Flash flood in Hill Torrent mainly occurs in mountainous regions and their adjacent foothills. They are characterized by a rapid surge in river discharge over a very short period of time, making them inherently brief but intense events. A flash flood phenomena is a short duration event that typically occurs within six hours of a triggering event such as heavy rain, dam failure, bund failure, rapid snowmelt, etc (as defined by the National Weather Service, US). In most cases, it occurs within two hours of the start of an event such as heavy rainfall. Flash flood occurs when the water level of a river rises rapidly, causing the flood to exceed bund and riverbanks. A Flash flood also generates a flow containing a large amount of debris at high velocity.

(3) Storm Surge

Sea level rise, influenced by weather conditions and tidal pattern , can lead to flooding in coastal areas. This storm surge damage is mainly due to the effects of severe cyclones or cyclones in the Arabian Sea, and winds from tropical storms and cyclones, or offshore cyclones, can cause flooding due to storm surges by creating sea water blows. There are records of storm surges caused by tropical cyclones from the Arabian Sea causing flooding on the Makran coast of Balochistan province and southeast Sindh province (see Table 2.1.3).

Table 2.1.3 Past Storm Surge Flood Damage in Pakistan

Storm Year Location	Storm Year	Location
Phet Storm	2010	Gwadar Area
BOB 03 Storm	2009	Karachi Area
BOB 06 Storm	2007	Karachi Area
Yemyin Storm	2007	Karachi Area
Onil Storm	2004	Thatta Area
Cyclone Storm	1999	Keti Bandar

Source: NFPP-IV and Related Studies to Enhance Capacity Building of Federal Flood Commission-FFC, Final Report, 2019

Pakistan has about 1,046 km (650 miles) of coastline along the Arabian Sea and the Gulf of Oman, located in the country's southwestern region. Cyclones in the Arabian Sea are relatively rare, and typically track towards Gujrat province of India; however, some do impact Pakistan coastal regions, particularly the province of Balochistan and Sindh.

Every year, the North Arabian Sea is prone to cyclones before the onset of the monsoon from mid-April to mid-July, and after the monsoon fades from mid-September to mid-December. Table 2.1.4 lists the record of major cyclones that have affected Pakistan in the past.

Table 2.1.4 Past Major Cyclones in Pakistan

Precipitation			Storm	Measurement Station
Rank	(mm)	(inch)		
1	370	14.57	Cyclone Phet 2010	Gwadar
2	285	11.22	Cyclone 2A 1999	Shah Bandar
3	245	9.64	Depression 2009	Karachi
4	191	7.51	Deep depression 2007	Karachi
5	145	5.71	Cyclone Onil 2004	Thatta
6	110	3.97	Cyclone Yemyin 2007	Karachi
7	43	1.69	Deep Depression 2007	Karachi
8	18	0.70	Depression 2009	Karachi

Source: Detailed History of Cyclones in Pakistan, Pakistan Weather Portal

(4) Inland Flood in Urban Areas

In urban areas, flooding is caused by riverine flood, flash flood, and storm surges. Inland flood due to poor drainage is an issue in urban areas.

If urban drainage systems do not have sufficient capacity to drain rainwater from the ground surface due to rainfall, heavy rain may cause inundation in urban areas. During urban flooding, roads can turn

into fast-flowing streams carrying drainage water, and basements may become inundated with sewage, posing serious risks to occupants. . Such urban floods has occurred in urban areas of drainage channel, especially during the monsoon season. It is an issue in areas with high population density, such as Karachi, Lahore, Faisalabad, Multan, and Hyderabad, where drainage systems are not functioning properly due to inadequate drainage planning and insufficient cross-sectional area of the existing channels.

(5) Glacial Lake Outburst Flood (GLOF)

Glacial Lake Outburst Flood (GLOF) is generally defined as the sudden release of glaciers or lake water dammed by glaciers. Such sudden floods have been reported in high-altitude areas globally and are believed to occur in Pakistan due to the following factors:

- Water from an ice-dammed lake suddenly drains under or through an ice dam
- Water overflows from a glacial lake, creating river-like flows containing ice, bedrock, and sediment
- Expansion and sudden collapse of glacial lakes

Spontaneous flooding from melting glaciers in mountainous areas has caused a devastating impact on downstream areas due to the large and rapid flow of the discharge, and such damage is more common in the Gilgit-Baltistan region. A list of past GLOF events is given in Table 2.1.5

Table 2.1.5 Past GLOF in Pakistan

Sr. No.	Year	Glacier	River
1	1533	Human	Upper Shyok
2	1750	Biafo Gl.	Braldu River
3	1780	Khumdan	Upper Shyok
4	1826	Khumdan	Upper Shyok
5	1833		Upper Shyok
6	1833	YashkukYaz	Chapursan [Hunza]
7	1835	Sultan Chussky	Shyok
8	1839	Khumdan	Upper Shyok
9	1842		Upper Shyok
10	1844		Ishkoman/ Ghizer/Gilgit
11	1848	Aktash or Khumdan	Upper Shyok
12	1851	Chungphar	Tarshing/ Astore
13	1855	Khumdan	Upper Shyok
14	1861	Karambar	Ishkoman
15	1865		Ishkoman
16	1871		Gilgit Hunza
17	1882	Chong or KichikKhumdan	Upper Shyok
18	1884		Shimshall
19	1892		Ishkoman, Shyok
20	1893	Karambar, Chhateboi	Ishkoman/ Ghizer
21	1893		Shimshall
22	1898		Upper Shyok
23	1901	Khumdan	Upper Shyok
24	1903	KichikKhumdan	Upper Shyok
25	1905	KichikKhumdan	Upper Shyok
26	1905	Khurdopin	Shimshall/ Hunza
27	1905	Karumbar	Iskoman/Gilgit
28	1906	Khurdopin	Shimshall/ Hunza
29	1907	Khurdopin	Shimshall/ Hunza
30	1924	Chong Khumdan	Upper Shyok
31	1926	Chong Khumdan	Upper Shyok
32	1927	KazYaz	Chapursan/ Hunza
33	1927	Khurdopin	Shimshall/ Hunza
34	1928	Kilik Valley	Hunza
35	1929	Chong Khumdan	Upper Shyok
36	1932	Chong Khumdan	Upper Shyok

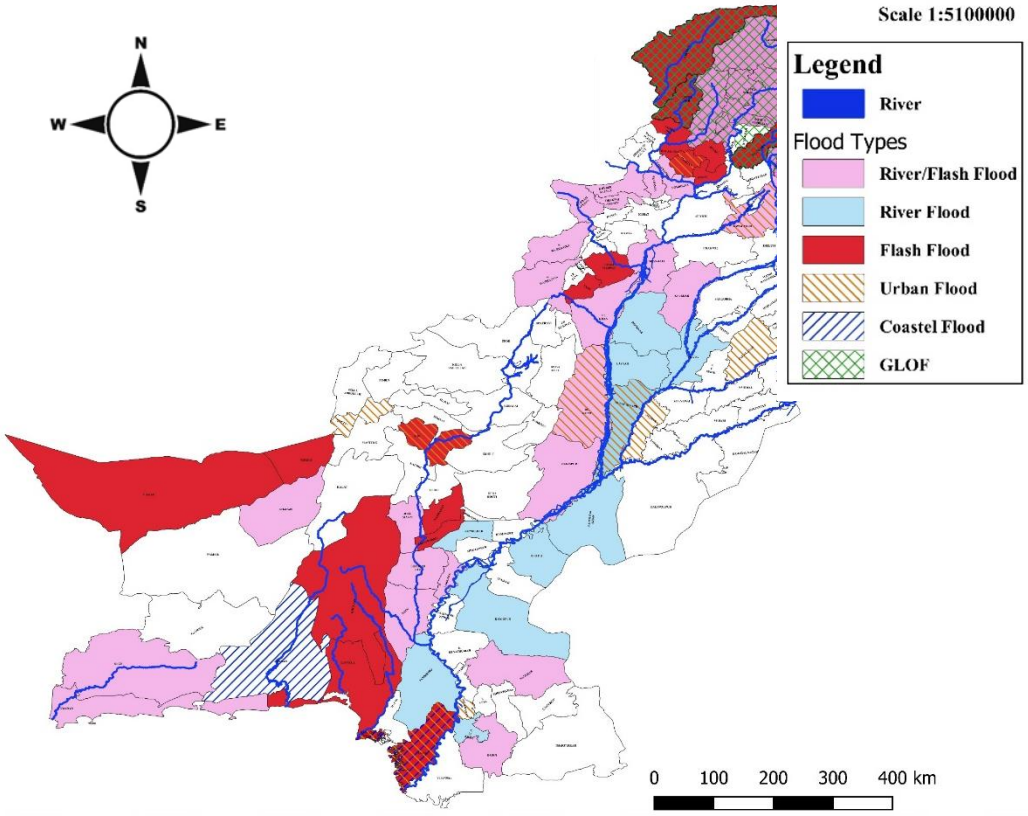
Sr. No.	Year	Glacier	River
37	1933	Chong Khumdan	Upper Shyok
38	1942	Khordopin	Shimshall/ Hunza
39	1959/60,1964	Khordopin	Shimshall/ Hunza
40	1967		Hunza
41	1972	Batura Gl.	Hunza
42	1973	Batura Gl.	Hunza
43	1974	Balt Bare	Baltar/Hunza
44	1979	Kvagar	Shagsgam
45	1980	Un-named or Khalti	Ghizer/Gilgit
46	1984	Gulagab	Iskoman
47	1990	Lokpar/Alling Tributary	Shyok-Hushe
48	1995	Sosat Glacier, Sosat Valley	Ghizer
49	1997	Kande	Kande/Hushe/Shyok
50	1999	Charti	Ghizer/Gilgit
51	2000	Khordopin or YukshinGardan Glaciers	Shimshall/ Hunza
52	2000	Kande	Kande/Hushe/Shyok
53	2005	Sosot/Gupis	Gilgit
54	2007	Ghulkin	Hunza
55	2008	Passu	Hunza
56	2008	Ghulkin	Hunza
57	2008	Ghulkin	Hunza
58	2008	Ghulkin	Hunza
59	2008	Ghulkin	Hunza
60	2009	Ghulkin	Hunza
61	2012	Sosot/Gupis	Hunza
62	2017	khordopin	Shimshall

Source: NFPP-IV and Related Studies to Enhance Capacity Building of Federal Flood Commission-FFC (Final Report, 2019); Diagnostic Study of Heavy Downpour in 2015 Flash Floods over Chitral Area, Northern Pakistan (Ahmad B, et al.)

2.1.1.4 Hazard Area

(1) Areas Where Damage from Various Types of Floods is a Problem

The areas where damage from various types of floods are an issues are shown in Figure 2.1.5 and Table 2.1.6. Hazard areas are those shown in Figure 2.1.5 and Table 2.1.6



Source: Created by the Project Team based on Table 2.1.6

Figure 2.1.5 Areas where Flood Damage is an Issue

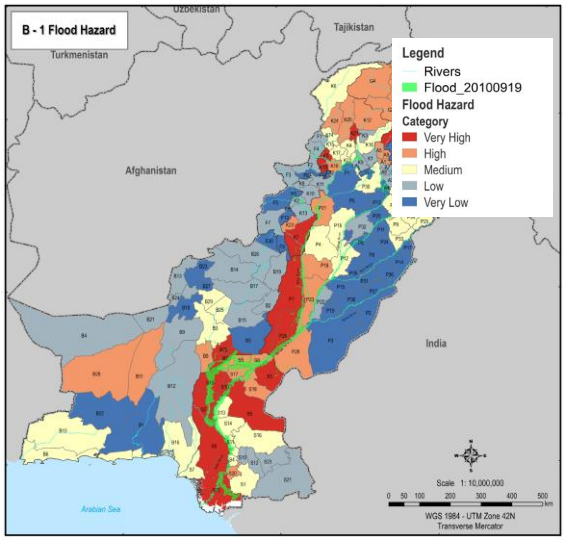
Table 2.1.6 Areas where Flood Damage is an Issue

Province	Type of Flood					
	River Flood/ Flash Flood[1)]	River Flood[1)]	Flash Flood[1)]	Urban Flood[2)]	Coastal Flood (Storm Surge) [1)]	GLOF[1)]
Khyber Pakhtunkhwa including Farmer FATA	Kohistan, Shangla, Swat, Charsadda, Peshawar, Nowshera, Dera Ismail Khan, Waziristan, South Waziristan, Khyber, Kurram and Orakzai Agencies		Mansehra, Buner, Swabi, Chitral, Lower & Upper Dir, Malakand Division, Mardan, Tank, Lakki Marwat and Kurram districts	Peshawar, Mardan, and Dera Ismail Khan		Chitral, Batagram, Kohistan, Mansehra, Swat, Shangla, and Upper Dir
AJK	Neelum, Muzaffarabad and Bagh		Poonch and Bhimber			Muzaffarabad and Neelum
Gilgit Baltistan	Diamer, Gilgit, Ghizer, Chilas, Hunza Nagar, Skardu and Ganche		Astore			Ghanche, Hunza-Nagar, Skardu, Astore, Diamir, Ghizer and Gilgit
Punjab	Rawalpindi, Mianwali, Dera Ghazi Khan, Rajanpur, Khushab, Gujrat, Sialkot, Gujranwala, Narowal and Sheikhupura	Bakkar, Layyah, Muzaffargarh, Rahim Yar Khan, and Jhang		Lahore, Rawalpindi, Multan, Faisalabad, Sialkot, Dera Ghazi Khan, Muzaffargarh and Rawalpindi		
Balochistan	Bolan, Jhal Magsi, Kharan, Kech and Gwadar		Sibi, Nasirabad, Jaffarabad, Khuzdar, Lasbela, Chagai (Dalbadin town) and Nushki	Quetta, Sibi, Dera Allah Yar Khan	Gawadar, Ketch, Lasbella, and Awaran	
Sindh	Larkana, Kamber-Shahdadkot, Dadu, Sanghar and Badin	Kashmore, Shikarpur, Jacobabad, Ghotki, Sukker, Jamshoro, Khairpur and Tando Muhammad Khan	Thatta and Karachi	Karachi, Hyderabad, Sukkur, Thatta, Jacobabad, Kashmor and Shikarpur	Badin, Karachi, Hyderabad, Thatta and Kashmore	

Source:
1) NDPM-I, Main Volume
2) Summarized from Final Report of Development of NFPP-IV and Related Studies to Enhance Capacity Building of Federal Flood Commission-FFC

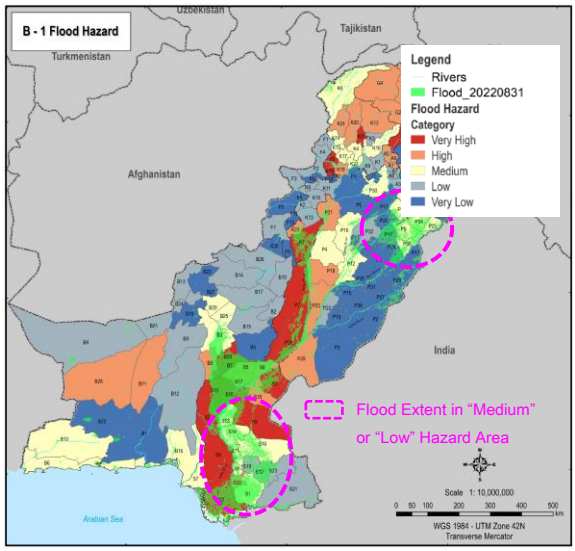
(2) Possibility of Hazard Area Transition

Figure 2.1.6 and Figure 2.1.7 illustrate the flood-affected areas due to 2010 and 2022 floods, superimposed on the corresponding hazard prone areas[1] by riverine floods. According to these figures, the flood prone areas from 2010 floods are generally located in the areas classified as "Very High" or" High" in the hazard prone areas, Final Report. . On the other hand, the flood prone areas from the 2022 floods are located in the areas classified as "Medium " or" Low" This is especially true in the areas along the left bank of the Indus River in Sindh Province and along the Ravi and Chenab Rivers in Punjab Province. Since hazard areas can vary based on the primary causes of flooding, it is essential to periodically reassess the hazard levels in each district to ensure accurate risk management and preparedness.



Source: United Nations Satellite Centre (UNOSAT)

Figure 2.1.6 Superposition of 2010 Flood Inundation Area and Flood (River Line Flood) Hazard Area



Source: United Nations Satellite Centre (UNOSAT)

Figure 2.1.7 Superposition of 2022 Flood Inundation Area and Flood (River Line Flood) Hazard Area

2.1.1.5 Probability Analysis of River Discharge and Rainfall

(1) River Discharge and Rainfall Data

The relevant government departments in Pakistan have not carried out probabilistic calculations of river discharge and rainfall, nor have they used the results of these calculations in flood management. The Flood Forecasting Division (FFD) in the PMD currently lacks data representing flood hazard levels and flood probability. In the hydrological analysis, flood arrival time of all major rivers is calculated based on physical properties such as elevation and slope (probably using the formula of flood arrival time), and this information is attached to the river system map to understand the flood arrival time during monsoon season (e.g. between Barrages). The following hydrological and meteorological raw data were collected from the PMD as it is necessary to perform probability analysis for river discharge and rainfall to inform effective flood control strategies. Daily rainfall data level can be used to understand the overall hydrological and meteorological conditions of Pakistan.

1. Climate Normal (64 stations)
    a. Monthly and annual average rainfall (mm)
    b. Monthly average minimum temperature (degree celsius)

1 Created with NDMP-I and still used by NDMA.

c. Monthly average maximum temperature (degree celsius)
d. Monthly and annual average wind speed (knots) and direction (degree)
e. Monthly and annual average relative humidity (%)

2. Daily Rainfall (64 stations)
   a. Daily rainfall (mm) during Monsoon Season (i.e., June 1 to September 30) from 2007 to 2022
3. Extreme Records (62 stations; same with # 2 above except for two stations in Islamabad Old Airport and Sh. Manda Quetta)
   a. All the data include the dates/years when they were recorded
   b. Monthly and annual highest 24-hr rainfall (mm) for 62 stations
   c. Monthly highest recorded rainfall (mm) for 62 stations
   d. Monthly and annual lowest and highest minimum temperature (°C) for 62 stations
   e. Monthly and annual lowest and highest maximum temperature (°C) for 62 stations
4. Monthly Rainfall (93 stations; 29 stations in addition to # 2 above)
   a. Monthly rainfall (mm) from 2010 to 2022
5. Water Level and Discharge (25 stations)
   a. Highest recorded flow (cusecs) from 1926 to 2022
   b. 6-hourly flow data (cusecs) during Monsoon Season (i.e., June 15 to September 30) from 1986 to 2022
6. River Shapefiles
   a. Chenab, Indus, Jhelum, Kabul, and Sutlej Basins (upstream)
   b. River alignments

In addition, the following geolocation information related to meteorological and hydrological stations was obtained from WAPDA, which includes peak discharge data for major dams. This data from WAPDA were used to compliment the information provided by PMD and for further analysis.

1. Hydrological Observatory
2. Flood Warning Station
3. Data Collection Platform (DCP) Station under GMRC (Glacial Monitoring Research Centre)
4. Dam Operation/Flood Control Procedures for Tarbella, Mangla and Chashma Dams

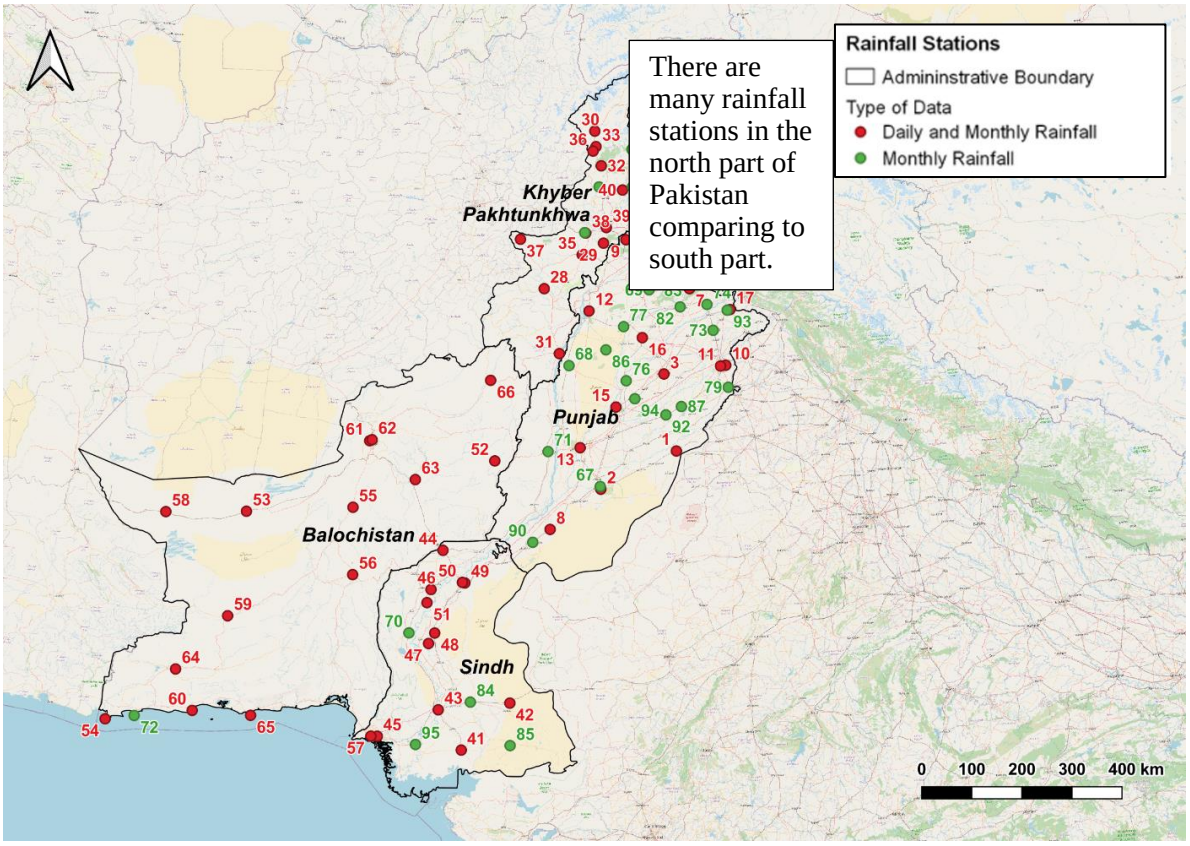
Table 2.1.7 provides a summary of the location details for the 95 stations with rainfall instruments included in the above meteorological dataset. Figure 2.1.8 illustrates a location map of these PMD rainfall stations.

Table 2.1.7 List of PMD Daily Rainfall Stations

Station Name	Province	Lat.	Long.	Data Received	Station Name	Province	Lat.	Long.	Data Received
1. Bahawalnagar	Punjab	30.00	73.30	1, 2, 3	49. Rohri	Sindh	27.68	68.90	1, 2, 3
2. Bahawalpur City	Punjab	29.35	71.71	1, 2, 3	50. Sukkur	Sindh	27.69	68.85	1, 2, 3
3. Faisalabad	Punjab	31.40	73.10	1, 2, 3	51. M.Jo.Daro	Sindh	27.32	68.13	1, 2, 3
4. Chaklala Airbase	Punjab	33.60	73.10	2, 3	52. Barkhan	Balochistan	29.89	69.52	1, 2, 3
5. Islamabad Old A/P	Punjab	33.55	72.83	1, 3	53. Dalbandin	Balochistan	28.90	64.40	1, 2, 3
6. Islamabad Zero point	Punjab	33.69	73.06	1, 2, 3	54. Jiwani	Balochistan	25.04	61.75	1, 2, 3
7. Jhelum	Punjab	32.92	73.72	1, 2, 3	55. Kalat	Balochistan	29.03	66.59	1, 2, 3
8. Khanpur	Punjab	28.64	70.65	1, 2, 3	56. Khuzdar	Balochistan	27.81	66.61	1, 2, 3
9. Kamra Airbase	Punjab	33.85	72.39	1, 2, 3	57. Lasbela	Balochistan	24.89	67.03	1, 2, 3
10. Lahore, Airport	Punjab	31.51	74.41	1, 2, 3	58. Nokkundi	Balochistan	28.83	62.75	1, 2, 3
11. Lahore, City	Punjab	31.50	74.30	1, 2, 3	59. Panjgur	Balochistan	27.00	64.10	1, 2, 3
12. Mianwali Airbase	Punjab	32.58	71.55	1, 2, 3	60. Pasni	Balochistan	25.27	63.47	1, 2, 3

Station Name	Province	Lat.	Long.	Data Received	Station Name	Province	Lat.	Long.	Data Received
13. Multan	Punjab	30.11	71.30	1, 2, 3	61. Quetta (Samungli)	Balochistan	30.24	66.91	1, 2, 3
14. Murree	Punjab	33.90	73.40	1, 2, 3	62. Quetta (Sh. Manda)	Balochistan	30.26	66.96	1, 3
15. Shorkot Airbase	Punjab	30.83	72.07	1, 2, 3	63. Sibbi	Balochistan	29.55	67.87	1, 2, 3
16. Sargodha Airbase	Punjab	32.07	72.67	1, 2, 3	64. Turbat	Balochistan	26.00	63.10	1, 2, 3
17. Sialkot Cantt	Punjab	32.51	74.56	1, 2, 3	65. Ormara	Balochistan	25.22	64.63	1, 2, 3
18. Astore	GB/AJK	35.33	74.85	1, 2, 3	66. Zhob	Balochistan	31.35	69.44	1, 2, 3
19. Bunji	GB/AJK	35.70	74.60	1, 2, 3	67. Bahawalpur A/P	Punjab	29.40	71.70	3
20. Chilas	GB/AJK	35.41	74.10	1, 2, 3	68. Bhakkar	Punjab	31.60	71.10	3
21. Garhi Dopatta	GB/AJK	34.23	73.61	1, 2, 3	69. Chakwal	Punjab	32.93	72.85	3
22. Gilgit	GB/AJK	35.90	74.30	1, 2, 3	70. Dadu	Sindh	26.77	67.77	3
23. Gupis	GB/AJK	36.23	73.44	1, 2, 3	71. DG Khan	Punjab	30.05	70.63	3
24. Kotli	GB/AJK	33.52	73.90	1, 2, 3	72. Gawadar	Balochistan	25.13	62.32	3
25. Muzaffarabad A/P	GB/AJK	34.36	73.47	1, 2, 3	73. Gujranwala	Punjab	32.15	74.18	3
26. Skardu	GB/AJK	35.30	75.60	1, 2, 3	74. Gujrat	Punjab	32.62	74.07	3
27. Balakot	KPK	34.55	73.35	1, 2, 3	75. Hunza	GB/AJK	36.31	74.65	3
28. Bannu	KPK	33.00	70.60	1, 2, 3	76. Jhang	Punjab	31.30	72.30	3
29. Cherat	KPK	33.80	71.90	1, 2, 3	77. Joharabad	Punjab	32.28	72.28	3
30. Chitral	KPK	35.83	71.78	1, 2, 3	78. Kalam	KPK	35.48	72.59	3
31. D.I. Khan	KPK	31.82	70.90	1, 2, 3	79. Kasur	Punjab	31.11	74.44	3
32. Dir	KPK	35.20	71.90	1, 2, 3	80. Lower Dir	KPK	34.82	71.84	3
33. Drosh	KPK	35.55	71.80	1, 2, 3	81. Malam Jabba	KPK	34.80	72.57	3
34. Kakul	KPK	34.18	73.25	1, 2, 3	82. Mandi Bahauddin	Punjab	32.60	73.50	3
35. Kohat Airbase	KPK	33.60	71.43	1, 2, 3	83. Mangla	Punjab	33.10	73.70	3
36. Mirkhani	KPK	35.47	71.73	2, 3	84. Mirpur Khas	Sindh	25.52	69.01	3
37. Parachinar	KPK	33.90	70.10	1, 2, 3	85. Mithi	Sindh	24.73	69.80	3
38. Peshawar Airbase	KPK	33.99	71.52	1, 2, 3	86. Noorpur Thal	Punjab	31.87	71.89	3
39. Risalpur	KPK	34.08	71.97	1, 2, 3	87. Okara	Punjab	30.80	73.44	3
40. Saidu Sharif	KPK	34.75	72.35	1, 2, 3	88. Pattan	KPK	35.10	73.00	3
41. Badin	Sindh	24.65	68.83	1, 2, 3	89. Peshawar City	KPK	34.00	71.50	3
42. Chhor	Sindh	25.50	69.80	1, 2, 3	90. Rahim Yar Khan	Punjab	28.41	70.29	3
43. Hyderabad	Sindh	25.38	68.37	1, 2, 3	91. Rawalakot	AJK	33.85	73.75	3
44. Jacobabad	Sindh	28.27	68.45	1, 2, 3	92. Sahiwal	Punjab	30.66	73.11	3
45. Karachi A/P	Sindh	24.89	67.16	1, 2, 3	93. Sialkot Airport	Punjab	32.50	74.50	3
46. Larkana	Sindh	27.56	68.21	1, 2, 3	94. T.T. Singh	Punjab	30.97	72.47	3
47. Sh.B.Abad	Sindh	26.58	68.17	1, 2, 3	95. Thatta	Sindh	24.75	67.92	3
48. Padidan	Sindh	26.77	68.29	1, 2, 3					

**Not included in Climate Normal and Extreme Records*
***Not included in Daily Rainfall and Extreme Records*
Source: Received from PMD



Source: Created by the project team based on the table above

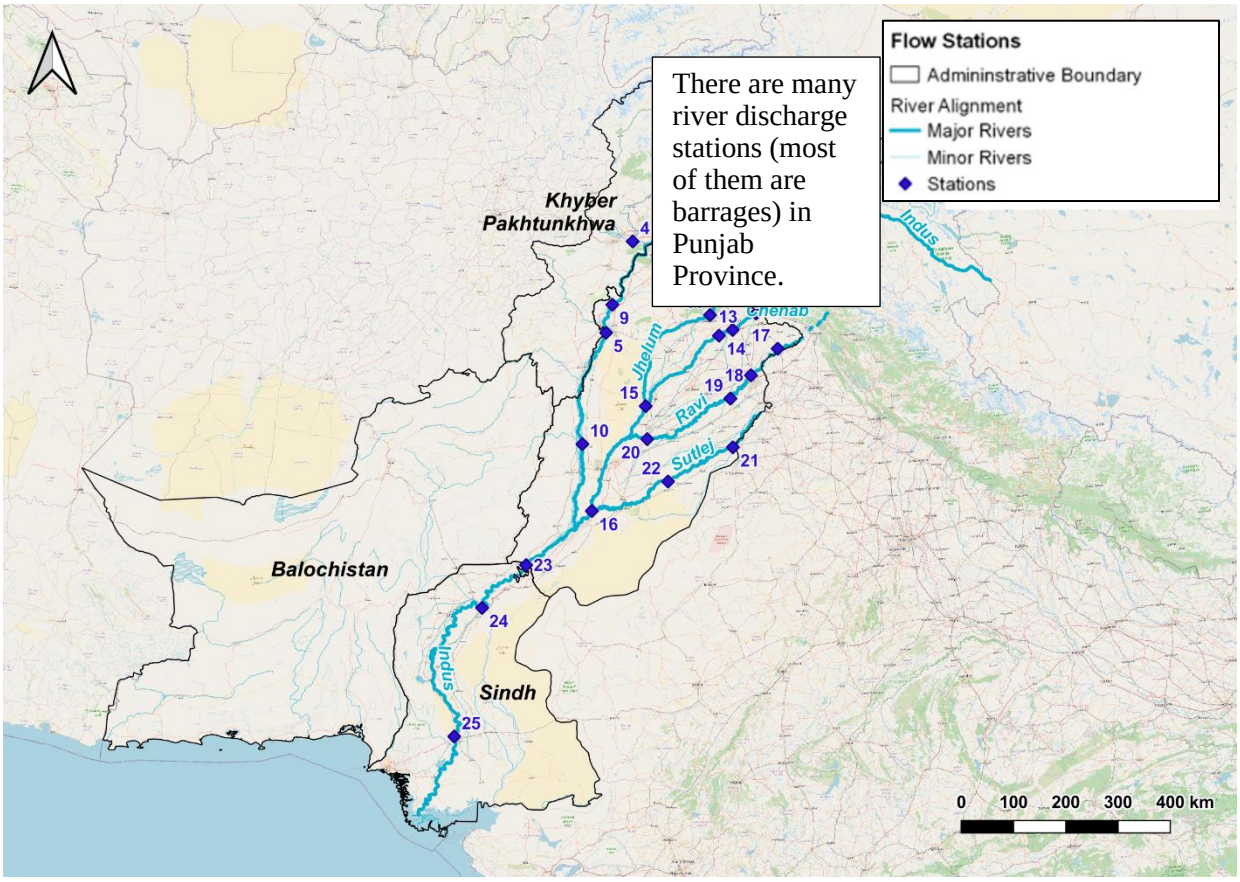
Figure 2.1.8 Location Map of PMD Rainfall Stations

For river discharge data, PMD receives data from the following WAPDA stations, which are mainly used for collecting and disseminating flood related information. Most of WAPDA's 25 stations are located along the major rivers of Indus, Jhelum, Chenab, Ravi, and Sutlej. Table 2.1.8 summarizes the location of these stations. Figure 2.1.9 illustrates the location map of WAPDA discharge observatories.

Table 2.1.8 List of WAPDA Stations in Discharge

Station Name	River Basin	Latitude	Longitude	Station Name	River Basin	Latitude	Longitude
1. Tarbela	Indus	34.11	72.73	14. Q.Abad	Chenab	32.32	73.69
2. Skardu	Indus	35.32	75.55	15. Trimmu	Chenab	31.14	72.15
3. Besham	Indus	34.91	72.87	16. Panjnad	Chenab	29.35	71.02
4. Kabul	Kabul	34.01	71.98	17. Jassar	Ravi	32.05	74.87
5. Chashma	Indus	32.44	71.38	18. Shahdara	Ravi	31.61	74.30
6. Mangla	Jhelum	33.15	73.65	19. Balloki	Ravi	31.22	73.86
7. C.Kallas	Jhelum	34.20	73.47	20. Sidhnai	Ravi	30.57	72.16
8. Kotli	Jhelum	33.37	73.86	21. Sulemanki	Sutlej	30.38	73.87
9. Kalabagh	Indus	32.92	71.52	22. Islam	Sutlej	29.83	72.55
10. Taunsa	Indus	30.51	70.86	23. Guddu	Indus	28.42	69.71
11. Rasul	Jhelum	32.68	73.52	24. Sukkur	Indus	27.68	68.85
12. Marala	Chenab	32.67	74.46	25. Kotri	Indus	25.44	68.32
13. Khanki	Chenab	32.41	73.97				

Source: Received from PMD



Source: Created by the project team based on the above table

Figure 2.1.9 Location Map of WAPDA Discharge Observatory

(2) Rainfall and Discharge Probability Analysis

For effective planning purposes, the data used for probability analysis needs to ideally cover a longer period of time (e.g., 30 years). The daily rainfall data obtained from PMD spans 31 years, while the monthly rainfall covers 16 years. However, data from certain rainfall stations, which contains numerous gaps, are not suitable for analysis.

The annual extreme data used in the probability calculation are arranged as 1-day, 2-day, 3-day, 5-day, 7-day and monthly rainfall, considering the vastness of the Indus River and its tributaries. Table 2.1.9 shows the annual maximum rainfall recorded during the major flood events of 2010 and 2022.

Table 2.1.9 Annual Maximum Rainfall in 2010 and 2022

No	Station Name	Maximum Rainfall (mm) in 2010						Maximum Rainfall (mm) in 2022					
		Monthly	7-day	5-day	3-day	2-day	1-day	Monthly	7-day	5-day	3-day	2-day	1-day
1	Bahawalnagar	68.0	40.0	30.0	28.0	28.0	28.0	297.0	146.0	138.0	128.0	89.0	80.0
2	Bahawalpur City	161.6	93.0	86.4	85.0	85.0	49.0	212.3	135.3	126.4	113.2	85.7	51.7
3	Faisalabad	243.1	168.4	117.8	117.8	111.8	91.7	105.1	86.6	86.6	55.4	55.4	54.4
4	Chaklala Airbase	383.3	209.0	209.0	206.0	186.6	156.2	481.9	192.3	165.5	143.5	119.0	84.5
5	Islamabad Zero point	344.8	209.0	209.0	206.0	186.6	156.2	449.0	170.8	145.5	119.8	82.8	80.8
6	Jhelum	259.0	162.0	150.6	131.1	105.7	91.6	383.7	179.7	136.7	121.0	117.0	111.0
7	Khanpur	276.1	251.5	246.9	217.3	170.2	121.0	300.4	122.8	113.8	72.2	72.2	61.0
8	Kamra Airbase	517.0	337.0	337.0	333.0	308.0	240.0	349.0	152.8	151.0	133.2	93.2	76.7
9	Lahore, Airport	307.7	242.0	240.0	238.0	236.0	138.0	317.4	225.9	225.9	207.5	207.5	158.0
10	Lahore, City	288.0	235.1	234.7	220.4	220.0	122.0	240.9	114.8	105.7	105.7	105.7	71.0

No	Station Name	Maximum Rainfall (mm) in 2010						Maximum Rainfall (mm) in 2022					
		Monthly	**7-day**	**5-day**	**3-day**	**2-day**	**1-day**	**Monthly**	**7-day**	**5-day**	**3-day**	**2-day**	**1-day**
11	Mianwali Airbase	530.6	271.0	271.0	271.0	221.0	190.0	199.1	129.0	99.0	97.0	95.0	95.0
12	Multan	158.1	151.8	140.7	120.0	120.0	120.0	183.1	110.2	110.2	77.4	59.0	36.6
13	Murree	579.9	388.6	373.6	367.6	270.6	231.0	402.5	202.0	188.0	109.0	106.0	64.0
14	Shorkot Airbase	231.5	169.0	159.0	88.0	88.0	87.0	121.9	66.6	66.6	66.6	52.6	52.6
15	Sargodha Airbase	217.0	169.6	169.6	169.6	138.0	87.0	189.1	93.0	90.0	75.0	45.0	42.0
16	Sialkot Cantt	415.4	255.2	253.2	192.4	132.4	127.4	423.1	223.9	221.4	206.2	191.5	156.2
17	Astore	177.6	49.9	44.8	40.7	36.7	31.1	58.9	47.8	46.8	34.2	32.1	29.7
18	Bunji	91.6	48.2	47.1	42.0	35.8	30.8	51.0	26.0	26.0	20.0	20.0	13.0
19	Chilas	76.9	40.4	40.4	38.4	34.4	26.4	26.1	15.8	15.8	12.0	10.0	9.6
20	Garhi Dopatta	570.5	379.8	346.4	331.4	215.4	189.4	278.3	128.0	112.4	106.4	86.4	65.4
21	Gilgit	60.7	50.1	50.0	38.0	33.0	20.0	60.8	42.1	40.7	31.9	26.1	14.6
22	Gupis	69.0	54.0	54.0	54.0	52.0	32.0	98.8	60.0	52.0	49.0	36.0	19.0
23	Kotli	416.4	259.2	210.2	157.6	108.6	102.6	360.0	177.0	165.0	95.0	90.0	75.0
24	Muzaffarabad A/P	554.6	269.4	267.4	229.0	138.0	103.0	289.3	251.3	236.3	185.0	185.0	170.0
25	Skardu	124.3	21.3	21.3	21.3	20.9	17.7	53.7	13.1	11.9	10.5	7.7	6.2
26	Balakot	357.4	260.9	255.9	179.9	134.7	90.1	352.4	202.0	137.0	95.0	93.0	68.0
27	Bannu	312.6	158.2	158.2	158.2	152.0	101.0	113.5	64.0	54.0	53.0	44.0	40.0
28	Cherat	393.0	387.0	372.0	371.0	338.0	257.0	182.0	81.0	80.0	80.0	76.0	54.0
29	Chitral	151.8	61.6	60.4	60.4	54.6	41.4	110.0	94.7	85.6	74.9	53.3	31.6
30	DI Khan	376.0	250.3	219.0	205.5	188.5	150.0	223.0	129.2	129.2	118.2	118.2	67.6
31	Dir	395.0	309.0	309.0	284.0	227.0	149.0	361.0	209.0	196.0	168.0	140.0	90.0
32	Drosh	131.9	99.4	99.4	99.4	84.4	61.2	142.3	103.9	94.5	84.8	64.8	42.8
33	Kakul	389.2	189.0	187.1	185.6	154.5	123.6	300.0	187.0	164.0	139.0	126.0	92.0
34	Kohat Airbase	0.0	315.0	305.0	276.0	262.0	233.0	130.0	65.0	57.0	48.0	48.0	48.0
35	Mirkhani	0.0	47.0	42.0	42.0	42.0	27.0	213.0	146.0	118.0	104.0	92.0	55.0
36	Parachinar	345.0	194.2	140.0	134.2	109.2	79.0	166.1	109.0	92.0	88.0	54.0	35.0
37	Peshawar Airbase	125.0	356.0	341.0	341.0	333.0	274.0	107.5	64.7	53.2	49.7	49.7	49.7
38	Risalpur	0.0	428.0	415.0	406.0	401.0	280.0	156.0	125.0	125.0	123.0	96.0	51.0
39	Saidu Sharif	472.0	338.0	338.0	334.0	290.0	187.0	178.0	130.0	111.0	110.0	75.0	48.0
40	Badin	163.0	135.8	111.0	109.0	109.0	102.0	335.8	262.0	260.0	249.0	245.0	219.0
41	Chhor	148.2	134.0	77.0	75.2	74.0	55.0	503.9	327.9	272.9	201.9	174.9	130.9
42	Hyderabad	85.9	76.5	76.5	76.5	76.5	70.3	247.0	137.0	127.0	99.0	83.0	59.0
43	Jacobabad	132.1	119.0	71.0	71.0	62.0	57.0	493.0	355.0	261.0	193.0	145.0	110.0
44	Karachi A/P	120.4	105.6	105.6	105.6	105.6	77.1	348.4	176.6	155.5	121.5	120.9	81.7
45	Larkana	207.2	141.0	140.0	135.0	134.0	131.0	738.3	565.3	488.3	288.0	210.0	157.3
46	Sh.B.Abad	138.6	72.0	72.0	72.0	72.0	70.0	488.2	383.0	234.0	214.0	184.0	123.0
47	Padidan	92.6	71.0	70.0	70.0	70.0	65.0	1228.5	829.4	784.9	648.6	595.0	355.0
48	Rohri	67.0	62.0	46.0	30.0	27.0	18.0	420.0	303.0	248.0	177.0	153.0	90.0
49	Sukkur	44.9	40.3	33.5	27.0	19.0	10.4	377.5	292.0	267.0	178.0	129.0	82.0
50	M.Jo.Daro	42.1	27.5	27.5	22.0	22.0	22.0	779.5	573.0	537.0	335.0	228.0	125.0
51	Barkhan	142.0	101.0	85.0	56.0	56.0	56.0	312.0	188.0	175.0	132.0	132.0	110.0
52	Dalbandin	9.0	0.0	0.0	0.0	0.0	0.0	33.0	33.0	29.0	29.0	29.0	15.0
53	Jiwani	194.0	194.0	194.0	194.0	194.0	165.0	63.0	46.0	46.0	43.0	36.0	34.0

No	Station Name	Maximum Rainfall (mm) in 2010						Maximum Rainfall (mm) in 2022					
		Monthly	7-day	5-day	3-day	2-day	1-day	Monthly	7-day	5-day	3-day	2-day	1-day
54	Kalat	0.0	2.0	2.0	2.0	2.0	2.0	361.0	284.0	204.0	204.0	179.0	132.0
55	Khuzdar	59.3	78.9	78.9	76.7	57.4	57.0	237.1	184.7	144.4	118.8	116.8	90.4
56	Lasbela	0.0	36.7	36.7	33.2	31.1	27.9	404.7	270.3	253.5	220.0	192.0	118.0
57	Nokkundi	7.5	0.0	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	0.0	0.0
58	Panjgur	43.5	0.0	0.0	0.0	0.0	0.0	186.0	105.0	78.0	63.0	63.0	40.0
59	Pasni	139.0	139.0	139.0	139.0	139.0	79.0	185.0	109.0	90.0	82.0	48.0	43.0
60	Quetta (Samungli)	0.0	5.0	5.0	5.0	5.0	5.0	208.0	184.0	149.0	149.0	129.0	86.0
61	Sibbi	107.6	65.5	65.1	65.1	65.1	58.0	243.0	145.0	125.0	125.0	109.0	81.0
62	Turbat	0.0	43.3	43.3	43.3	43.3	43.3	119.0	55.0	55.0	55.0	51.0	41.3
63	Ormara	84.0	80.0	80.0	80.0	80.0	60.0	221.0	104.0	104.0	104.0	88.0	62.0
64	Zhob	118.4	89.0	82.0	75.0	73.0	73.0	161.0	106.0	73.0	57.0	51.0	51.0
65	Bahawalpur A/P	111.8	-	-	-	-	-	192.9	98.5	76.1	69.1	68.1	58.3
66	Bhakkar	192.2	-	-	-	-	-	284.2	187.0	187.0	173.0	173.0	166.0
67	Chakwal	178.3	-	-	-	-	-	383.6	162.4	154.2	116.3	113.3	105.4
68	Dadu	0.0	-	-	-	-	-	337.0	217.0	164.0	126.0	110.0	80.0
69	DG Khan	175.0	-	-	-	-	-	184.8	119.0	113.8	88.8	74.8	41.0
70	Gawadar	386.0	-	-	-	-	-	151.2	83.2	75.5	70.5	60.5	58.0
71	Gujranwala	0.0	-	-	-	-	-	494.0	189.8	189.8	160.7	160.7	160.7
72	Gujrat	0.0	-	-	-	-	-	424.0	204.0	142.0	106.0	104.0	72.0
73	Hunza	68.1	-	-	-	-	-	105.1	75.1	74.5	42.8	42.8	40.0
74	Jhang	157.0	-	-	-	-	-	230.1	190.3	190.3	187.2	149.6	149.6
75	Joharabad	219.5	-	-	-	-	-	145.9	74.9	74.9	74.9	73.1	48.1
76	Kalam	474.6	-	-	-	-	-	268.1	214.0	194.1	155.5	114.0	71.5
77	Kasur	0.0	-	-	-	-	-	344.5	208.5	163.5	143.0	134.0	80.0
78	Lower Dir	424.0	-	-	-	-	-	278.0	201.0	152.0	150.0	122.0	80.0
79	Malam Jabba	0.0	-	-	-	-	-	256.0	146.0	132.0	128.0	93.0	54.0
80	Mandi Bahauddin	262.1	-	-	-	-	-	327.6	166.3	138.3	96.0	95.0	87.3
81	Mangla	288.9	-	-	-	-	-	391.2	182.7	178.4	168.1	167.5	166.4
82	Mirpur Khas	112.8	-	-	-	-	-	304.0	197.0	146.0	137.0	109.0	84.0
83	Mithi	0.0	-	-	-	-	-	273.0	157.0	138.0	127.0	113.0	67.0
84	Noorpur Thal	314.2	-	-	-	-	-	116.3	73.0	67.0	43.0	43.0	42.0
85	Okara	92.0	-	-	-	-	-	160.1	105.0	102.0	102.0	102.0	102.0
86	Pattan	483.0	-	-	-	-	-	136.0	80.0	57.0	57.0	49.0	25.0
87	Peshawar City	0.0	-	-	-	-	-	110.5	63.0	44.0	38.0	38.0	38.0
88	Rahim Yar Khan	177.4	-	-	-	-	-	224.4	135.5	97.0	80.0	79.5	58.0
89	Rawalakot	526.1	-	-	-	-	-	207.5	77.0	63.8	44.0	37.1	29.0
90	Sahiwal	129.6	-	-	-	-	-	132.8	57.8	57.8	57.8	54.0	54.0
91	Sialkot Airport	354.5	-	-	-	-	-	352.4	174.8	162.4	155.8	139.8	114.5
92	T.T. Singh	222.0	-	-	-	-	-	110.4	64.8	64.8	57.3	57.3	55.3
93	Thatta	181.1	-	-	-	-	-	297.3	147.7	145.3	142.2	137.0	131.0

Source: Received from PMD

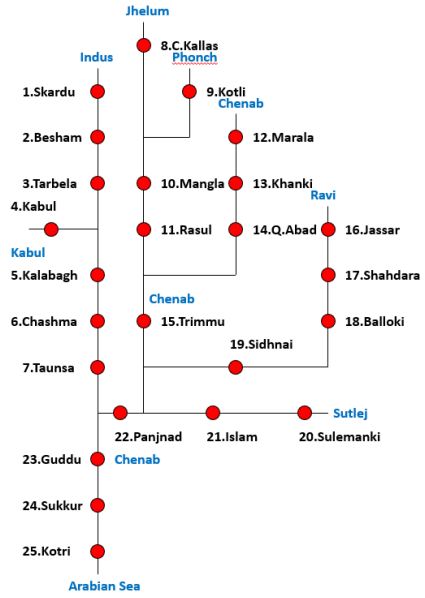
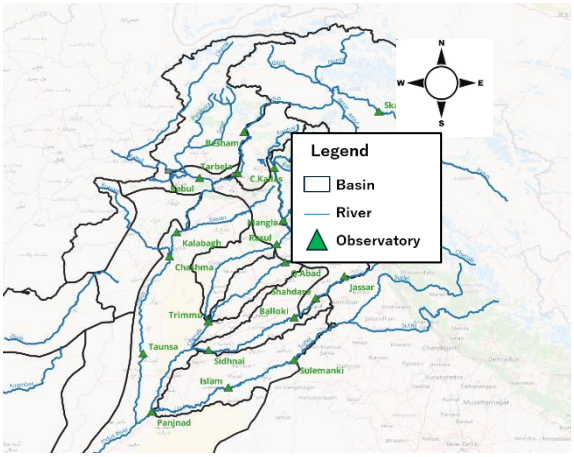
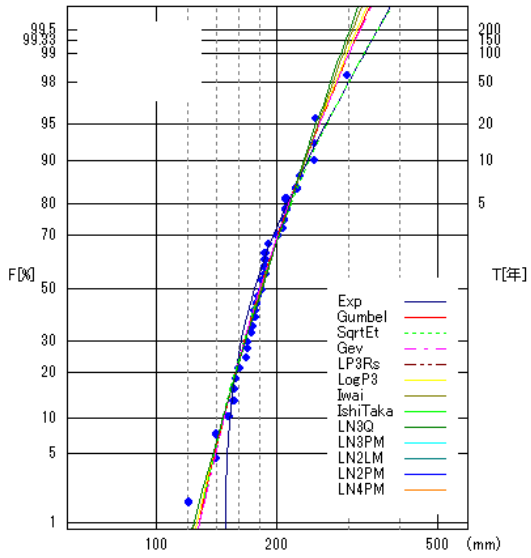
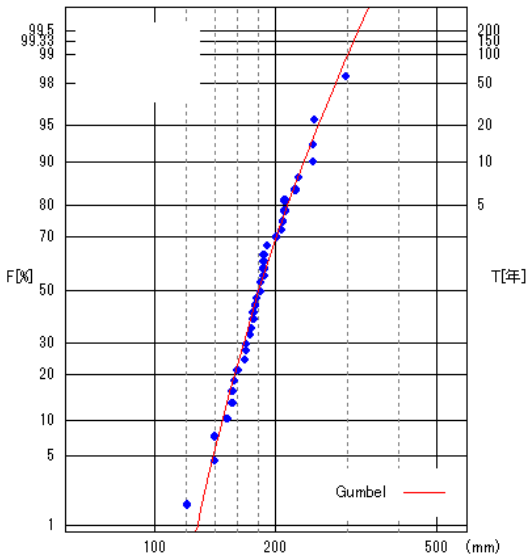
On the other hand, discharge data obtained through PMD spans 96 years. The table below shows the annual maximum discharge and historical peak discharge values for the major flood events of 2010 and 2022.

Table 2.1.10 Annual Maximum Discharge and Historical Maximum Discharge for 2010 and 2022

Number	Station Name	River Basin	Historical Peak Flow		Year of Peak Flow	Peak Flow in 2010		Peak Flow in 2022	
			(cfs)	(m^3/s)		(cfs)	(m^3/s)	(cfs)	(m^3/s)
1	Tarbela	Indus	875,000	24,763	1929	832,000	23,546	407,000	11,518
2	Skardu	Indus	296,700	8,397	2010	296,700	8,397	211,500	5,985
3	Besham	Indus	713,500	20,192	2010	713,500	20,192	403,400	11,416
4	Kabul	Kabul	391,300	11,074	1990	249,000	7,047	336,600	9,526
5	Chashma	Indus	1,038,873	29,400	2010	1,038,873	29,400	542,765	15,360
6	Mangla	Jhelum	1,060,000	29,998	1929	305,579	8,648	123,000	3,481
7	C.Kallas	Jhelum	332,390	9,407	2010	332,390	9,407	162,747	4,606
8	Kotli	Jhelum	403,872	11,430	1997	76,230	2,157	25,103	710
9	Kalabagh	Indus	950,005	26,885	1942	937,453	26,530	512,643	14,508
101	Taunsa	Indus	959,991	27,168	2010	959,991	27,168	624,159	17,664
11	Rasul	Jhelum	952,170	26,946	1992	263,795	7,465	37,045	1,048
12	Marala	Chenab	1,100,000	31,130	1957	315,245	8,921	225,837	6,391
13	Khanki	Chenab	1,086,460	30,747	1957	334,437	9,465	216,908	6,138
14	Q.Abad	Chenab	948,530	26,843	1992	328,983	9,310	221,150	6,259
15	Trimmu	Chenab	944,000	26,715	1928	328,926	9,309	144,147	4,079
16	Panjnad	Chenab	802,516	22,711	1973	310,117	8,776	121,764	3,446
17	Jassar	Ravi	680,000	19,244	1955	21,100	597	63,740	1,804
18	Shahdara	Ravi	576,000	16,301	1988	41,900	1,186	31,415	889
19	Balloki	Ravi	389,845	11,033	1988	69,935	1,979	55,245	1,563
20	Sidhnai	Ravi	330,210	9,345	1988	35,265	998	26,552	751
21	Sulemanki	Sutlej	598,872	16,948	1955	58,301	1,650	24,785	701
22	Islam	Sutlej	492,581	13,940	1955	31,454	890	13,326	377
23	Guddu	Indus	1,199,672	33,951	1976	1,148,738	32,509	576,076	16,303
24	Sukkur	Indus	1,166,574	33,014	1986	1,130,995	32,007	579,754	16,407
25	Kotri	Indus	980,329	27,743	1956	964,897	27,307	626,193	17,721

Note) 1 cfs = 0.0283m^3/s
Source: Received from PMD

For each rainfall and discharge station, the optimum probability distribution was selected, and the rainfall and discharge of each probability year were obtained by referring to "Guidelines for Small and Medium Scale River Planning (Ministry of Land, Infrastructure, Transport and Tourism)". Figure 2.1.10 shows probability distribution for the Skardu discharge observatory serving as an example of the probability analysis conducted for river discharge data.

Discharge Station Name: 01_Skardu (River: Indus)	Latitude: 35.325° , Longitude: 75.551°
	
Optimal Probability Distribution: Gumbel	Available Years of Data: 35
SLSC Value: 0.027	Jackknife Estimation Error: 538m^3/s
Probability Distribution Selection: The probability distribution with the smallest SLSC Value and Jackknife among Gumbel, Gev and SqrtEt was selected.	
All Probability Distribution	Optimal Probability Distribution
	

Source: Project team

Figure 2.1.10 Example of Probability Analysis (Discharge Observatory: 01_Skardu)

Table 2.1.11 shows the probability scale of each discharge observatory determined by the above procedure. Figure 2.1.11 shows 100-year probability discharge of each station. However, it is important to note that the observation data collected at the downstream of the dam site such as Tarbela Dam and Mangla Dam, are influenced by the dam operations.

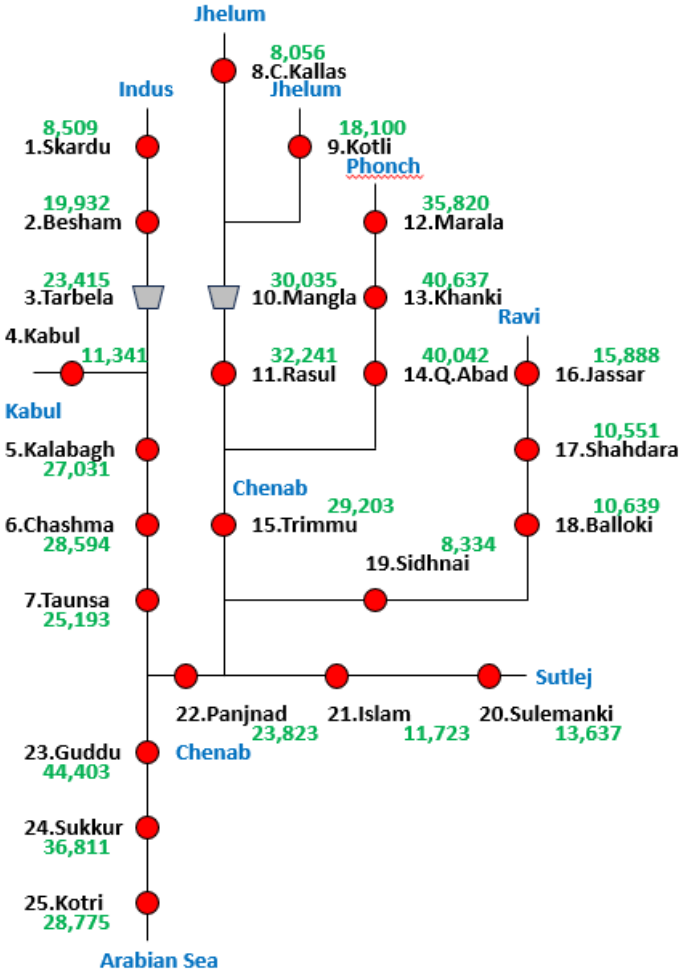
Table 2.1.11 Probability Scale of Each Discharge Observatory

No	River	Observatory	Analysis				X-years Discharge											
			Available Year of Data	Probability Distribution	SLSC Value	Jackknife Estimation Error	2	3	5	10	20	30	50	80	100	150	200	400
1	Indus	Skardu	35	Gumbel	0.027	538	5,125	5,553	6,029	6,629	7,204	7,535	7,948	8,328	8,509	8,832	9,064	9,619
2	Indus	Besham	35	SqrtEt	0.062	2,076	10,364	11,414	12,640	14,263	15,905	16,888	18,157	19,354	19,932	21,005	21,781	23,704
3	Indus	Tarbela	49	Gev	0.040	3,557	10,168	11,126	12,371	14,255	16,463	17,947	20,054	22,262	23,415	25,692	27,467	32,360
4	Kabul	Kabul	37	Gev	0.041	2,260	3,285	3,908	4,703	5,881	7,235	8,130	9,381	10,675	11,341	12,649	13,657	16,398
5	Indus	Kalabagh	95	Gumbel	0.026	1,283	13,561	15,268	17,168	19,555	21,846	23,163	24,811	26,317	27,031	28,328	29,245	31,454
6	Indus	Chashma	52	SqrtEt	0.030	1,835	12,791	14,473	16,455	19,111	21,829	23,466	25,592	27,611	28,594	30,415	31,740	35,030
7	Indus	Taunsa	63	Gumbel	0.030	1,441	12,037	13,702	15,557	17,890	20,127	21,413	23,021	24,494	25,193	26,456	27,354	29,511
8	Jhelum	C.Kallas	10	LN3Q	0.062	2,871	1,696	2,098	2,665	3,574	4,664	5,400	6,436	7,507	8,056	9,129	9,953	12,156
9	Phonch	Kotli	10	Gev	0.079	6,683	903	1,337	2,050	3,491	5,794	7,742	11,120	15,478	18,100	24,038	29,387	47,657
10	Jhelum	Mangla	96	Gev	0.024	3,477	4,771	6,258	8,291	11,562	15,656	18,547	22,826	27,515	30,035	35,166	39,289	51,176
11	Jhelum	Rasul	97	Gev	0.033	3,387	3,681	5,196	7,323	10,840	15,387	18,675	23,644	29,206	32,241	38,502	43,610	58,669
12	Chenab	Marala	97	Gev	0.037	2,631	7,696	9,662	12,244	16,183	20,858	24,018	28,543	33,320	35,820	40,793	44,686	55,503
13	Chenab	Khanki	97	Gev	0.040	3,123	7,603	9,744	12,609	17,083	22,517	26,264	31,703	37,545	40,637	46,850	51,771	65,672
14	Chenab	Q.Abad	53	LN3Q	0.032	4,423	7,790	10,313	13,646	18,612	24,199	27,793	32,689	37,582	40,042	44,746	48,277	57,468
15	Chenab	Trimmu	93	Gev	0.028	2,492	6,516	8,418	10,800	14,240	18,080	20,563	23,978	27,442	29,203	32,615	35,212	42,135
16	Ravi	Jassar	74	Gev	0.029	2,206	2,362	3,307	4,550	6,448	8,696	10,217	12,391	14,688	15,888	18,275	20,147	25,338
17	Ravi	Shahdara	95	Gev	0.026	1,447	1,696	2,180	2,857	3,964	5,383	6,400	7,929	9,628	10,551	12,445	13,986	18,491
18	Ravi	Balloki	97	Gev	0.029	1,022	2,141	2,667	3,381	4,508	5,895	6,864	8,283	9,820	10,639	12,295	13,615	17,381
19	Ravi	Sidhnai	97	Gev	0.021	966	1,322	1,713	2,251	3,135	4,262	5,066	6,272	7,611	8,334	9,823	11,026	14,546
20	Sutlej	Sulemanki	97	Gumbel	0.039	886	3,429	4,723	6,162	7,971	9,707	10,706	11,955	13,096	13,637	14,620	15,316	16,990
21	Sutlej	Islam	97	LogP3	0.035	2,042	2,209	3,534	5,125	7,082	8,772	9,639	10,607	11,389	11,723	12,281	12,635	13,368
22	Chenab	Panjnad	97	Gev	0.021	1,659	7,184	9,319	11,689	14,654	17,480	19,097	21,113	22,953	23,823	25,394	26,504	29,169
23	Indus	Guddu	53	Gev	0.046	2,546	15,356	19,012	23,092	28,229	33,170	36,019	39,584	42,851	44,403	47,218	49,214	54,028
24	Indus	Sukkur	92	Gev	0.036	2,044	15,990	19,210	22,574	26,490	29,948	31,814	34,025	35,939	36,811	38,338	39,377	41,747
25	Indus	Kotri	91	Gev	0.020	1,758	10,947	13,646	16,489	19,824	22,795	24,409	26,334	28,011	28,775	30,123	31,043	33,153

: Available year of data is short (less than 20 years) or the accuracy of analysis is slightly questionable (SLSC value greater than 0.040).

: 100 year probability discharge

Source: Project Team



Source: Project Team

Figure 2.1.11 100-Year Probability Discharge of Each Discharge Station

Similarly, the probability scales for each discharge observatory were calculated for various time intervals, including 1-day, 2-day, 3-day, 5-day, 7-day and monthly durations. As an example, the probability scale for 7-day is shown in Table 2.1.12 and 100-year probability rainfall of each rainfall station is shown in Figure 2.1.12.

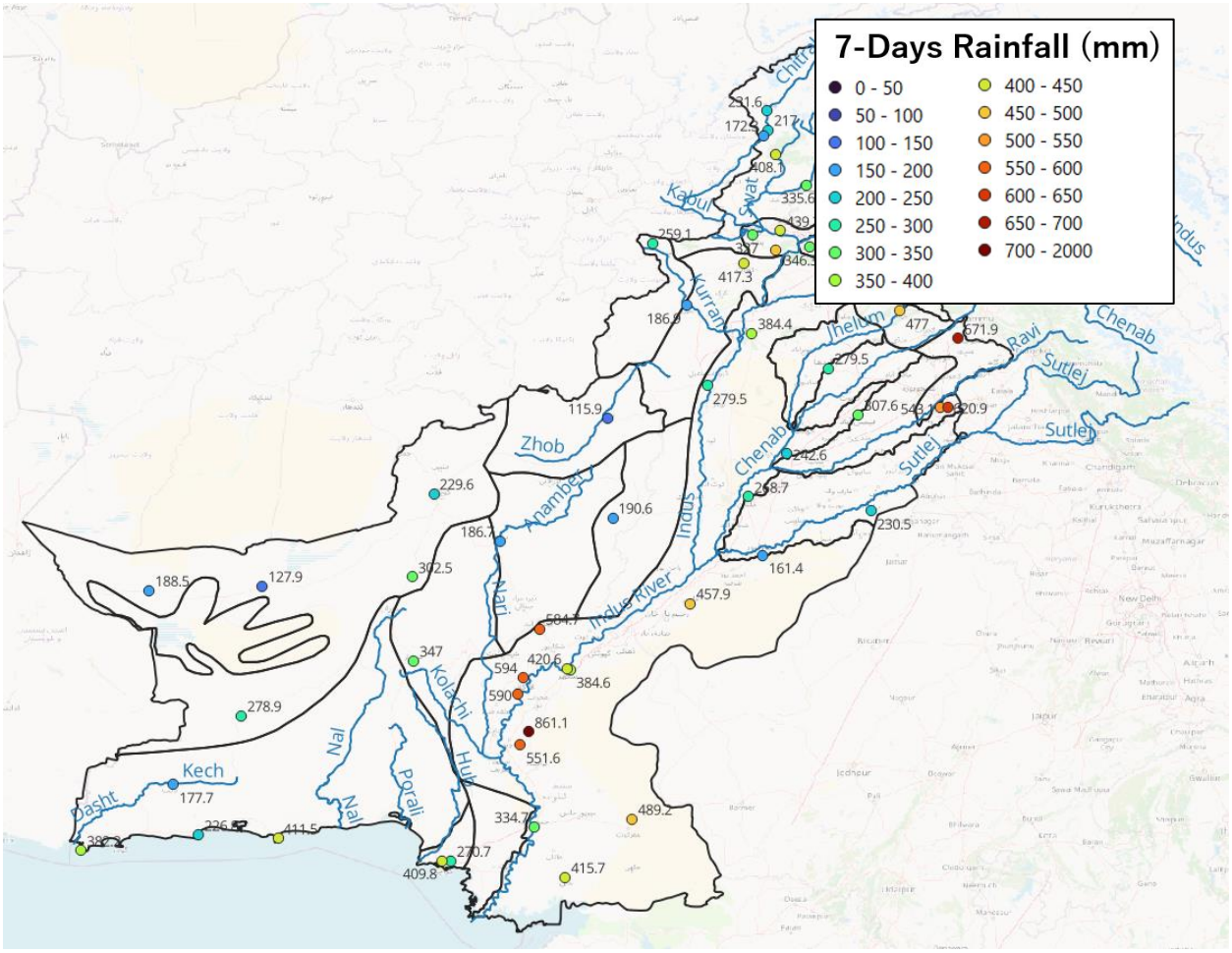
Table 2.1.12 Probability Scale of Each Rainfall Observatory (7-Day Rainfall)

Basin	No	Observatory	Analysis				X-years Rainfall											
			Available Years of Data	Probability Distribution	SLSC Value	Jackknife Estibation Error	2	3	5	10	20	30	50	80	100	150	200	400
1.GilGit-Baltistan Hill	1	Skardu	31	Gev	0.031	29.9	33.2	45.5	61.0	83.8	109.6	126.5	149.9	173.9	186.2	210.2	228.5	277.9
	3	Gupis	31	LogP3	0.041	124.3	40.4	63.8	96.8	147.7	205.9	243.2	293.8	343.7	368.5	415.5	450.3	539.0
	4	Gilgit	31	LogP3	0.038	17.1	26.0	33.8	43.9	58.8	75.6	86.4	101.3	116.3	123.9	138.5	149.6	178.9
	5	Bunji	31	SqrtEt	0.040	13.9	30.2	39.0	50.0	65.4	81.8	92.0	105.4	118.5	124.9	137.0	145.8	168.1
	6	Astore	31	Gumbel	0.023	18.1	56.7	72.8	90.7	113.2	134.7	147.1	162.6	176.8	183.6	195.8	204.4	225.2
	7	Chilas	31	LogP3	0.038	44.9	29.6	45.7	64.9	88.8	110.2	121.5	134.5	145.3	150.1	158.1	163.4	174.9
2.Kabul River Hill	8	Chitral	31	Gumbel	0.043	27.0	55.0	77.3	102.3	133.6	163.6	180.9	202.4	222.2	231.6	248.6	260.6	289.6
	9	Drosh	31	Gumbel	0.028	19.8	60.8	80.6	102.6	130.3	156.9	172.1	191.2	208.7	217.0	232.0	242.6	268.3
	10	Mirkhani	16	LN3Q	0.039	76.2	22.7	30.0	41.2	60.6	85.7	103.5	129.6	157.5	172.3	201.6	224.7	288.4
	12	Dir	31	Gumbel	0.023	43.6	136.3	170.8	209.1	257.3	303.5	330.1	363.3	393.7	408.1	434.3	452.8	497.4
	14	Saidu_Sharif	31	SqrtEt	0.030	35.1	117.2	139.3	165.8	201.9	239.6	262.6	292.7	321.5	335.6	361.9	381.0	429.0
3.Null	17	Kakul	31	Gev	0.038	75.2	162.9	183.8	209.9	247.4	289.0	315.8	352.6	389.8	408.7	445.2	472.9	546.5
	18	Balakot	31	SqrtEt	0.029	57.9	195.0	228.4	268.2	322.3	378.4	412.6	457.1	499.7	520.5	559.2	587.4	657.9
4.Azad Jammu & Kashmir Hill	19	Muzaffarabad	31	LogP3	0.044	34.6	192.3	229.3	269.4	318.2	363.1	388.3	419.1	446.7	459.7	482.9	499.2	537.9
	20	G.Dopatta	31	Gev	0.027	58.7	162.7	198.8	241.7	299.8	360.4	397.5	446.4	493.7	517.1	560.9	593.2	675.4
	22	Kotli	31	SqrtEt	0.024	55.8	167.6	202.4	244.4	302.2	362.7	399.7	448.3	495.0	517.8	560.5	591.6	669.8
5.Pothohar Hill	23	Kamra	31	Gev	0.029	35.7	182.3	210.9	239.5	271.3	297.8	311.6	327.3	340.4	346.3	356.2	362.8	377.4
	24	Murree	31	Gumbel	0.021	44.7	203.8	240.6	281.7	333.2	382.7	411.1	446.7	479.3	494.7	522.6	542.5	590.2
	25	Islamabad	31	Gev	0.031	146.8	212.1	251.1	300.3	371.9	452.5	505.0	577.6	651.6	689.4	762.9	819.1	969.8
	28	Jhelum	31	Gev	0.028	81.0	162.8	192.0	227.6	277.3	330.8	364.5	409.8	454.6	477.0	519.8	551.9	635.2
6.Rachna & Chaj Doab Hill	31	Sialkot	31	Gev	0.029	85.3	184.6	226.7	279.0	353.8	436.5	489.5	562.0	635.0	671.9	743.2	797.2	940.1
8.Chenab	35	Sargodha	31	Gev	0.019	38.7	98.4	117.2	139.4	169.4	200.3	219.2	244.0	267.8	279.5	301.5	317.6	358.4
9.Ravi	37	Lahore	31	Gev	0.030	88.1	153.2	188.9	232.6	293.9	360.3	402.2	458.8	514.9	543.1	596.9	637.3	742.7
	38	Lahore_Airport	31	SqrtEt	0.026	74.8	162.4	205.9	259.5	334.3	413.7	462.8	527.6	590.2	620.9	678.6	720.8	827.2
	39	Faisalabad	31	Gev	0.028	43.6	94.8	112.0	133.8	165.6	201.5	225.0	257.4	290.6	307.6	340.6	365.9	433.8
10.Sutlej	44	Bahawalnagar	31	Gumbel	0.032	19.6	81.4	100.3	121.3	147.7	173.1	187.7	205.9	222.6	230.5	244.8	255.0	279.4
	45	Multan	31	Gev	0.040	30.8	60.0	76.6	97.6	128.4	163.6	186.6	218.7	251.7	268.7	301.7	327.2	395.8
	46	Bahawalpur	31	Gumbel	0.025	16.2	54.1	67.7	82.9	101.9	120.1	130.6	143.7	155.7	161.4	171.7	179.0	196.6
11.Indus River	48	Peshawar	31	Gev	0.039	85.3	79.5	100.1	126.0	163.4	205.3	232.4	269.7	307.7	327.0	364.5	393.0	469.2
	50	Risalpur	31	Gev	0.030	93.5	118.8	143.9	176.0	223.2	277.2	312.6	362.1	413.0	439.2	490.5	529.9	636.5
	51	Cherat	31	Gev	0.033	92.8	94.9	118.0	149.6	200.4	264.0	308.9	375.4	448.2	487.3	567.0	631.0	815.6
	52	Chaklala_Airbase	16	LogP3	0.049	79.6	209.2	247.2	284.4	324.1	355.9	371.9	389.9	404.7	411.1	422.1	429.3	445.1
	53	Mianwali	31	Gev	0.027	61.5	118.5	140.8	168.7	209.0	254.0	283.1	323.1	363.7	384.4	424.4	454.9	536.1
	54	Di_Khan	31	SqrtEt	0.033	34.1	69.7	89.4	113.8	148.0	184.3	206.8	236.6	265.4	279.5	306.1	325.5	374.6
	57	Shorkot_Airbase	16	Gev	0.039	26.7	84.7	109.3	134.9	164.7	190.9	205.0	221.7	236.1	242.6	254.0	261.8	279.5
	59	Khanpur	31	LogP3	0.027	72.8	47.5	75.3	114.7	176.8	249.0	296.2	360.8	425.5	457.9	519.9	566.2	685.7
	61	Rohri	31	Gev	0.028	73.2	41.4	62.0	90.1	134.9	190.8	229.9	287.7	350.8	384.6	453.2	508.1	665.9
	62	Sukkur	23	Gev	0.040	89.0	33.4	52.8	80.4	126.8	187.8	232.5	300.8	378.1	420.6	509.0	581.8	798.6
	63	Jacobabad	31	Gev	0.031	127.4	39.9	63.6	98.3	159.1	242.5	305.6	404.6	520.0	584.7	721.7	836.8	1189.9
	64	Larkana	31	LogP3	0.025	196.8	38.2	68.0	114.4	193.3	291.2	357.5	450.5	545.6	594.0	687.2	757.6	941.3
	65	Mjo_Daro	30	Gev	0.032	158.5	34.8	56.0	88.2	146.3	229.1	293.3	396.6	519.8	590.0	741.2	870.3	1276.4
	67	Padidan	31	Gev	0.037	213.7	42.9	71.6	115.8	197.6	316.9	411.4	565.7	753.1	861.1	1096.3	1299.4	1950.0
	68	Nawabshah	31	Gev	0.038	81.8	56.6	87.6	129.3	195.2	276.2	332.5	414.9	504.1	551.6	647.4	723.8	941.1
	69	Chhor	31	Gev	0.025	65.8	105.9	142.4	186.6	247.8	313.2	354.0	408.7	462.4	489.2	540.1	578.0	676.2
	72	Badin	31	Gev	0.026	59.7	100.3	135.1	175.5	228.8	282.7	315.1	356.8	396.5	415.7	451.4	477.4	542.2
	73	Hyderabad	16	LN3Q	0.042	182.8	92.9	123.9	158.3	201.2	242.1	265.6	294.9	321.9	334.7	358.0	374.6	414.9
12.Fata Hill	74	Parachinar	31	Gev	0.038	43.7	102.5	120.5	141.2	168.0	194.8	210.6	230.9	249.9	259.1	276.1	288.3	318.4
13.D.I. Khan Hill	75	Kohat	31	Gev	0.038	79.4	75.9	95.4	122.2	165.7	220.7	259.8	318.2	382.5	417.3	488.3	545.7	712.3
	76	Bannu	23	Gumbel	0.028	19.4	74.2	88.5	104.4	124.4	143.5	154.6	168.3	181.0	186.9	197.8	205.5	223.9
14.Zhob-Loralai Hill	77	Zhob	31	Gev	0.019	13.0	49.9	60.5	71.5	84.0	94.9	100.7	107.5	113.3	115.9	120.5	123.6	130.5
15.D.G. Khan Hill	78	Barkhan	31	Gumbel	0.029	19.9	75.4	90.0	106.3	126.7	146.3	157.5	171.6	184.5	190.6	201.7	209.5	228.4
16.Kacchi Basin Hill	79	Sibbi	31	Gev	0.029	26.3	52.5	68.1	85.9	109.0	131.9	145.4	162.7	178.9	186.7	201.0	211.3	236.8
17.Khithar Hill	80	Khuzdar	31	Gev	0.025	83.4	53.2	71.0	95.2	133.8	181.6	215.2	264.5	318.3	347.0	405.4	452.0	585.7
18.Kharan Closed Desert Hill	81	Quetta_Samungli	28	Gumbel	0.038	25.4	49.3	72.1	97.5	129.5	160.1	177.8	199.8	220.0	229.6	246.9	259.2	288.8
	83	Kalat	31	Gev	0.028	77.1	47.9	67.5	92.6	129.8	172.6	200.9	240.5	281.4	302.5	344.0	375.9	462.8
	84	Dalbandin	31	Gumbel	0.043	17.2	25.5	38.5	52.9	71.1	88.5	98.5	111.0	122.4	127.9	137.7	144.7	161.5
	85	Nokkundi	19	LN3Q	0.036	50.5	8.4	15.5	27.1	48.5	78.0	99.7	132.6	169.0	188.5	228.3	260.0	350.1
	86	Panjgur	16	LN3Q	0.042	97.7	16.1	26.3	43.1	74.2	117.2	148.9	196.9	250.2	278.9	337.2	383.8	516.2
19.Makran Coastal Hill	87	Turbat	8	Gev	0.042	80.2	36.3	51.2	68.8	92.3	116.4	131.1	150.3	168.7	177.7	194.6	206.9	238.1
	88	Jiwani	31	LN2LM	0.037	79.8	34.6	54.0	82.5	129.9	189.1	229.8	288.4	350.1	382.2	445.5	494.5	627.9
	90	Pasni	8	Gev	0.052	41.2	62.7	87.3	113.3	143.9	171.2	186.2	204.0	219.6	226.8	239.4	248.0	267.9
	91	Ormara	28	Gev	0.032	97.2	36.5	57.7	86.9	134.5	194.8	237.8	302.1	373.1	411.5	490.2	553.8	738.9
21.Karachi Area Hill	93	Karachi	31	Gev	0.040	26.1	85.1	113.4	143.1	177.9	208.8	225.5	245.5	262.8	270.7	284.6	294.1	315.7
	94	Lasbela	16	Gev	0.040	113.5	40.1	55.7	78.7	119.3	175.6	218.4	286.0	365.2	409.8	504.6	584.5	831.4

: Available year of data is short (less than 20 years) or the accuracy of analysis is slightly questionable (SLSC value greater than 0.040).

: 100 year probability Rainfall

Source: Project Team



Source: Project Team

Figure 2.1.12 100-Year Probability Rainfall of Each Rainfall Station (7-Day Rainfall)

2.1.1.6 Issues In Disaster Risk Analysis

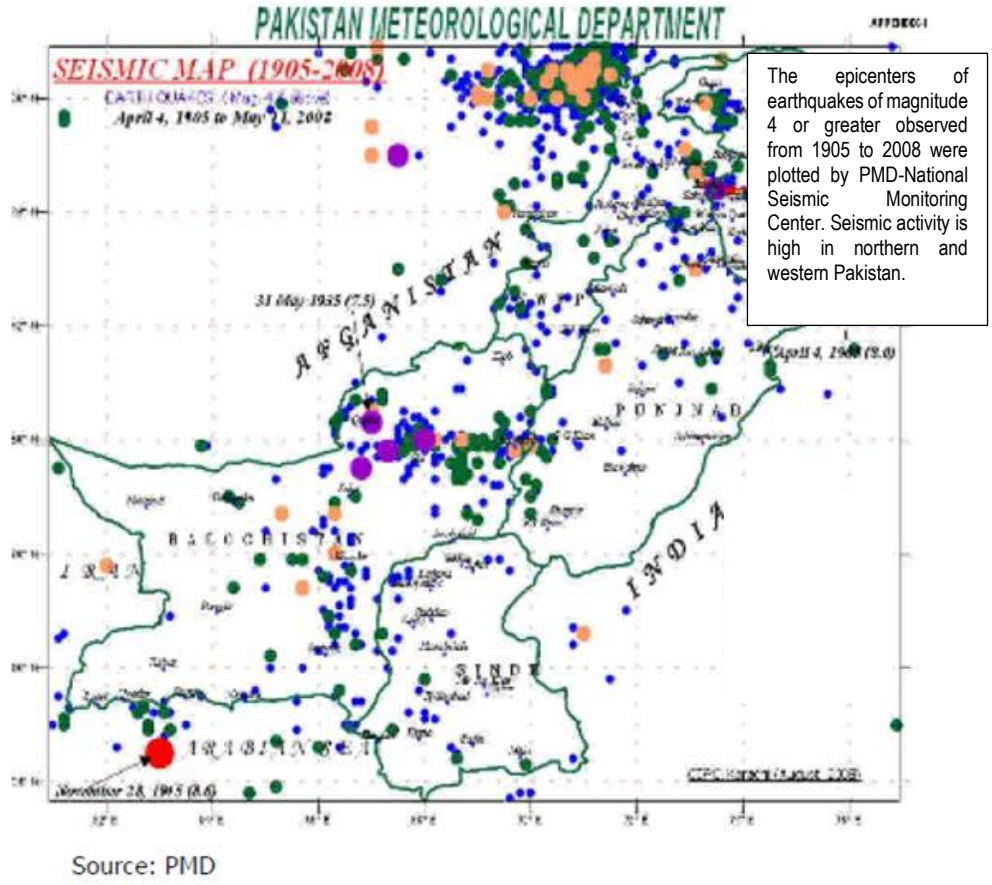
In the current NDMP analysis, hazards have been evaluated based on the number of disasters at district level. At the prefectural level, there are many areas that are not actually damaged (see Figure 2.3.2 below), as a result, it is not possible to reliably disseminate hazard information to DRR personnel at the prefectural level or below. In principle, it is necessary to create a list of projects with more realistic hazard areas. Specifically, it is desirable to create a hazard map of NFPP-IV (only within river channels), a flood map based on satellite image analysis by the Pakistan Space and Upper Atmosphere Research Council (Space & Upper Atmosphere Research Commission: SUPARCO), and a hazard map including floodplain based on the probability of occurrence.

However, there is insufficient information on the damage caused by flood events in each district since 2011. As shown in Table 2.1.1, no statistics on the property damage or the number of affected villages have been collected since 2016. Therefore, it is difficult to re-evaluate the hazard levels based on the data since 2011 due to the lack of comprehensive damage statistics and other relevant information. In addition, SPARCO is often reluctant to share the original data on flood maps derived from satellite image analysis and hazard maps, which are required for assessing the probability of occurrence with NDMA.

2.1.2 Earthquakes

2.1.2.1 Seismic Activity, Seismic Observation, Damage History (Year of Occurrence, Scale of Damage, etc.)

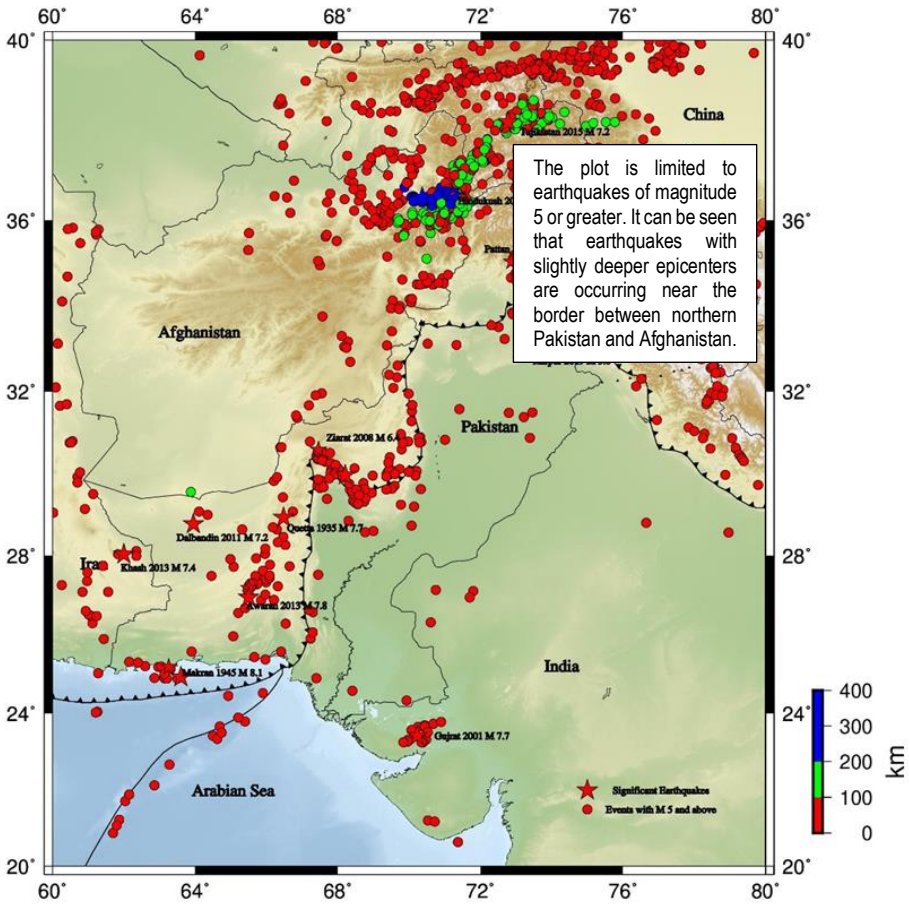
Seismic activity data, forming the basis for hazard assessment, have been collected for earthquakes of magnitude 4 and above from 1905 to 2008. The data are compiled by PMD and show high seismic activity in northern and western Pakistan. Figure 2.1.13 shows a map of seismic activity covering the period from 1905-2008, based on data obtained from the PMD.



Source: Current NDMP (2012)

Figure 2.1.13 Seismic Activity Map (1905 – 2008)

Two interviews were conducted with National Seismic Monitoring Center at PMD on September 15, 2022, and September 27, 2022, during which the latest earthquake catalogue data for Pakistan (1905 – 2022) was obtained to verify recent seismic activity. Figure 2.1.14 shows a plot of earthquake catalog data with a magnitude 5 or greater. Several earthquakes with slightly deeper epicenter are observed near the border between northern Pakistan and Afghanistan.

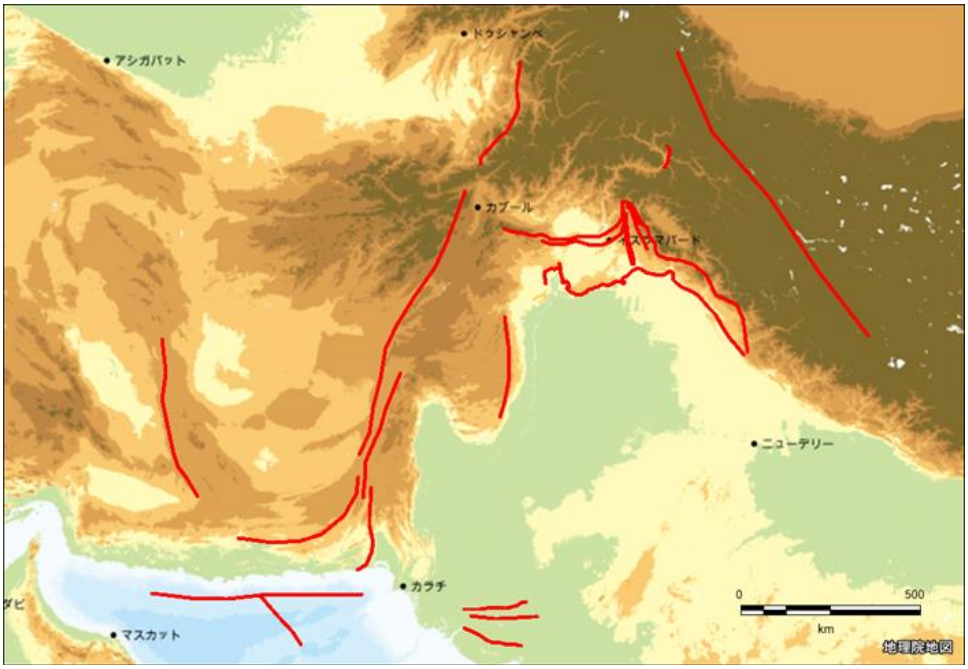


Source: Courtesy of PMD

Figure 2.1.14 Seismic Activity in Pakistan Including the Latest Data Up to 2022. Shows Earthquakes of Magnitude 5 Or Greater from 1905 to 2022.

The project team consulted PMD to verify whether a peak ground acceleration (PGA) hazard map, similar to the seismic hazard map (Spectral acceleration) included in the Pakistan Building Codes 2021, was available. As a result, it was found that PMD did not conduct the analysis and that staff and students of NED University of Engineering & Technology in Karachi owned the updated version (2021) of Seismic Hazard and Zonation 2007.

The project team also consulted PMD about active fault maps in Pakistan. The project team obtained shapefiles of major faults in Pakistan, and the data was illustrated in GIS as shown in Figure 2.1.15. But the project team did not get information on the probability of occurrence or the time of occurrence of each fault. The project team were informed that more detailed data on active faults could be obtained by contacting Geological Survey of Pakistan (GSP).



Source: PMD

Figure 2.1.15 Distribution of Major Faults (PMD)

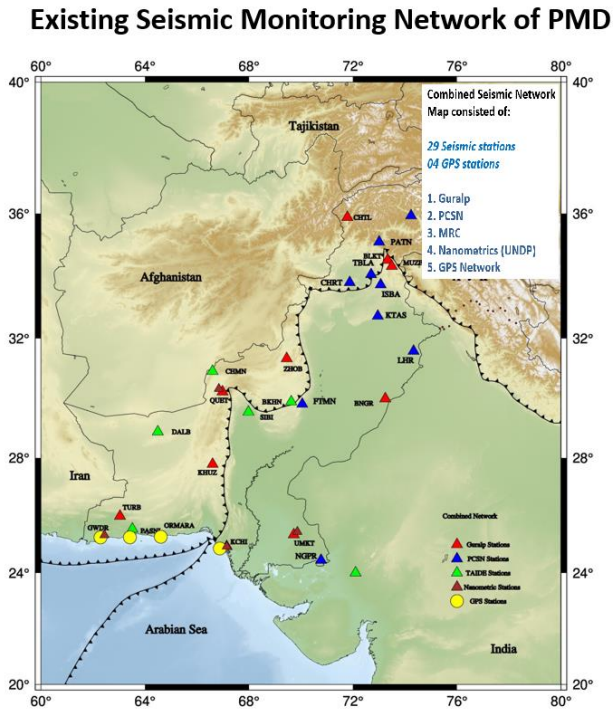
Table 2.1.13 shows damage from the 2005 Pakistan earthquake, which caused severe damage mainly in northern Pakistan, in the current NDMP.

Table 2.1.13 Damage from the 2005 Pakistan Earthquake

		a	b	c	d	e	f			
Province	District	Population in 1998	Houses in 1998	Houses full damage	Houses partial damage	Deaths	Injuries	MMI	Full damage ratio (%) =c/b	Death ratio (%) =e/a
KP	Abbottabad	880,666	153,819	6,961	27,051	515	1,730	8	4.5	0.1
	Batagram	307,278	44,585	28,712	8,656	3,232	3,279	10	64.4	1.1
	Kohistan	472,570	74,087	4,350	18,395	661	639		5.9	0.1
	Mansehra	1,152,839	203,109	31,323	43,282	24,511	30,585	10	15.4	2.1
	Shangla	434,563	67,003	15,661	10,281	423	957	5	23.4	0.1
AJK	Bagh	393,000	59,623	47,619	18,226	8,157	6,644	9	79.9	2.1
	Murpur	334,000		0	0	6	11	7		
	Muzaffarabad	746,000	123,679	108,157	17,120	33,724	21,374	10	87.4	4.5
	Neelum			3,692	7,215	447	1,013	9		
	Rawalakot (Poonch)			15,086	25,405	1,025	1,909	9		
	Sudhnoti	224,000		429	1,719	4	16	9		
	Total			261,990	177,350	72,705	68,157			

MMI: Modified Mercalli Intensity
Source: ADB and WB, 2005, Census data, 1998.

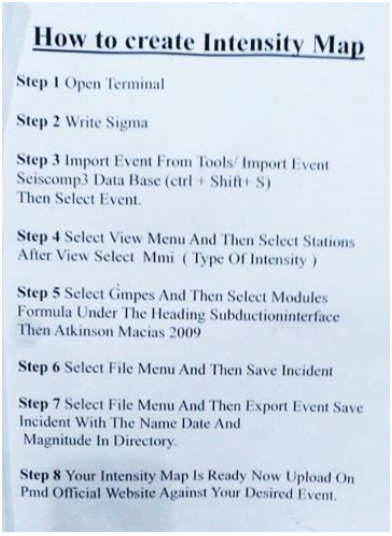
PMD maintains an earthquake observation network to determine earthquake hypocenter and simple seismic intensity assessment around the hypocenter (See Figure 2.1.16). Power outages occur frequently, however, a system is in place to back up the data. On the other hand, there is a problem that the necessary strong-motion seismometers have not been developed for the observation of strong earthquakes. The seismic observation network consists of 22~25 stations using broad-band seismometers. It does not include strong-motion seismometers, and the above-mentioned broad-band seismometers for hypocenter determination have been pointed out to be saturated (a situation in which they cannot record accurately because the handling limit of the instruments is exceeded) in the case of strong earthquakes. Therefore, it was confirmed that the development of a strong-motion seismometer network is an issue in the current NDMP in order to promote risk assessment by issuing seismic intensity bulletins in each region and analyzing the relationship between damage to buildings and strong-motion records. According to the staff of PMD, they plan to increase the number of seismic stations in the near future, but specific information was not obtained.



Source: Supplied by PMD

Figure 2.1.16 Seismic Observation Network in PMD

PMD uses a processing method to create seismic intensity maps based on the identification of hypocenters from seismic observations (Figure 2.1.17). This processing method attenuates the distance concentrically from the epicenter location, and does not consider the ground amplification factor. Since the amplification factor of the ground differs greatly between the alluvial lowlands and mountains, it is necessary to examine the amplification factors of the ground across Pakistan as part of the current NDMP.



How to create Intensity Map

Step 1 Open Terminal

Step 2 Write Sigma

Step 3 Import Event From Tools/ Import Event Seiscomp3 Data Base (ctrl + Shift+ S) Then Select Event.

Step 4 Select View Menu And Then Select Stations After View Select Mmi (Type Of Intensity)

Step 5 Select Gmpes And Then Select Modules Formula Under The Heading Subductioninterface Then Atkinson Macias 2009

Step 6 Select File Menu And Then Save Incident

Step 7 Select File Menu And Then Export Event Save Incident With The Name Date And Magnitude In Directory.

Step 8 Your Intensity Map Is Ready Now Upload On Pmd Official Website Against Your Desired Event.

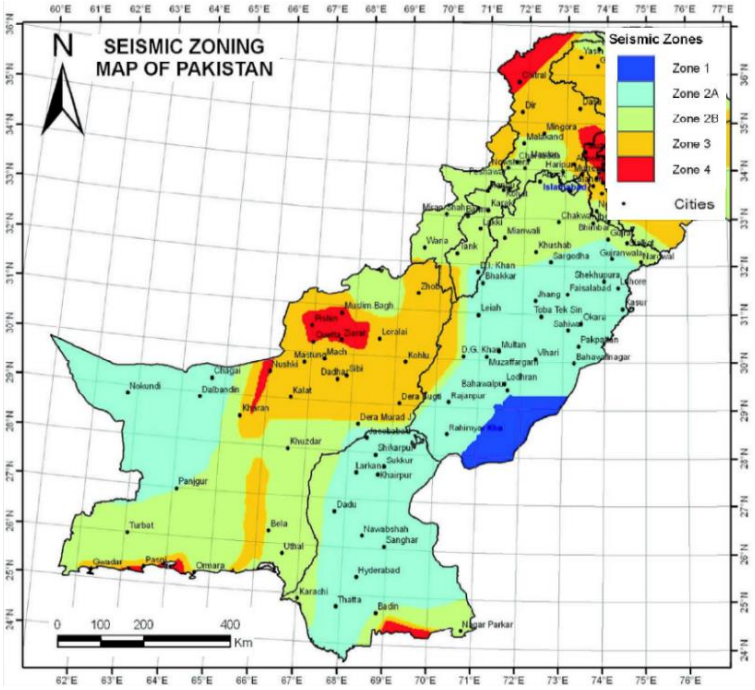
Source: Courtesy of PMD

Figure 2.1.17 Procedure for Preparing Seismic Intensity Maps In the PMD

2.1.2.2 Hazard Area and Intensity

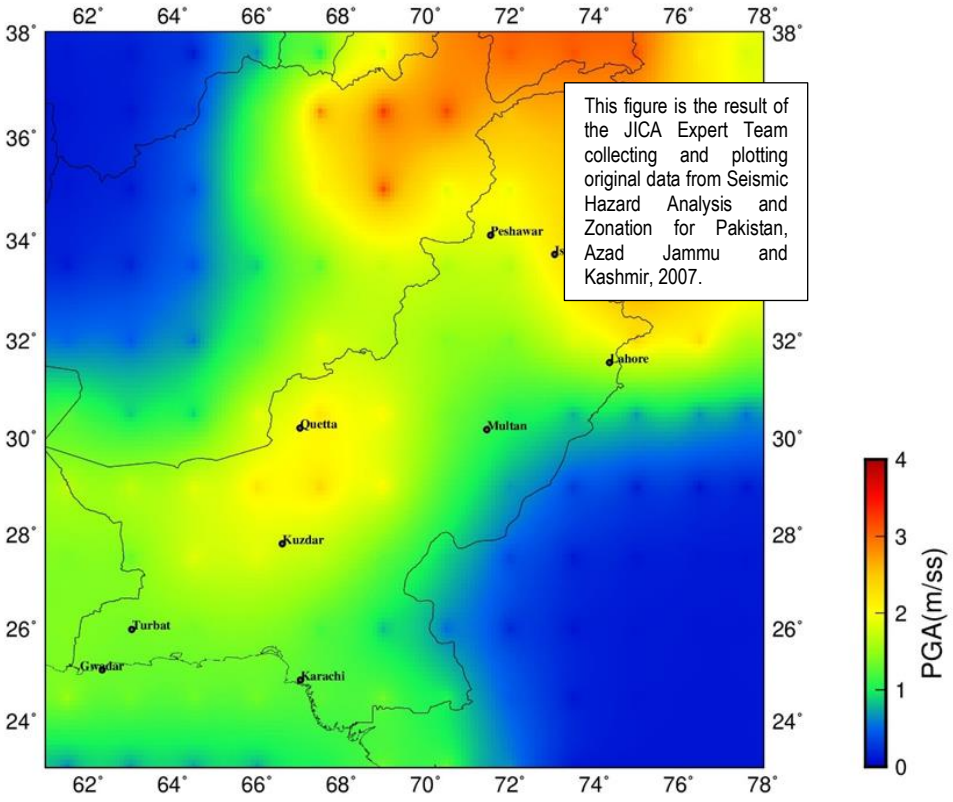
(1) Seismic Hazard Maps in the Current NDMP

Seismic zoning maps covering the entire country were developed by the Ministry of Housing and Works during the preparation of the 2007 Building Codes. Figure 2.1.18 shows the seismic zoning map of Pakistan. The entire country is divided into 5 seismic zones— Zone 1, Zone 2A, Zone 2B, Zone 3 and Zone 4 —each zone assigned a specific range of PGA values. The NDMA has prepared seismic hazard assessment maps for Muzaffarabad and Mansehra in northern KPK, and the Karachi Building Control Authority has collected seismic risk information for Karachi. Likewise, the PMD has developed a seismic hazard map using their earthquake observation database. (see Figure 2.1.19)



Source: Building Code of Pakistan

Figure 2.1.18 Seismic Zoning Map of Pakistan



Source: PMD Seismic Hazard Assessment (PGA) Data: Plotted from the original data of Seismic Hazard Analysis and Zonation for Pakistan, Azad Jammu and Kashmir, 2007

Figure 2.1.19 Seismic Hazard (PGA) Distribution of Pakistan

(2) Latest Seismic Hazard Maps

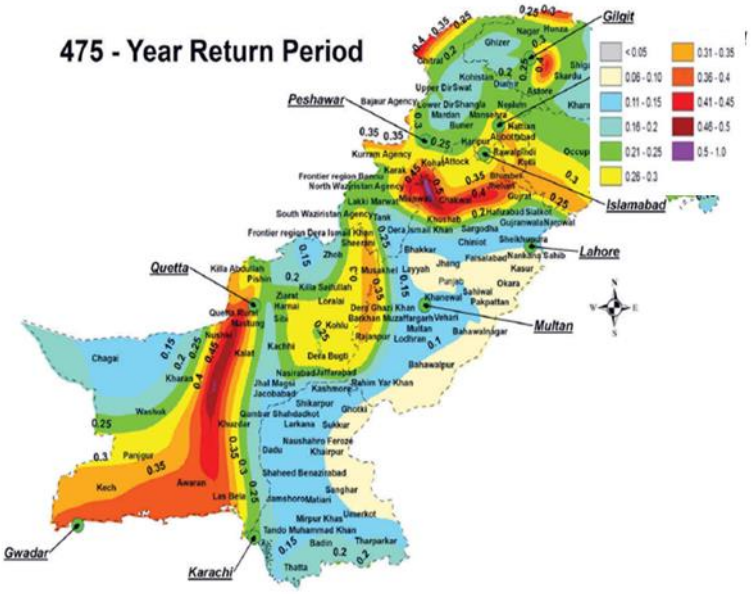
This paper investigates the latest results of seismic hazard maps in Pakistan, which serve as base information for seismic risk assessment.

1) Research Results at University of Karachi (NED University of Engineering and Technology)

The following documents containing the latest seismic hazard maps were obtained from NED University of Engineering and Technology.

- Muhammad Ahmed, Sarosh Hashmat Lodi & Muhammad Masood Rafi, Probabilistic Seismic Hazard Analysis Based Zoning Map of Pakistan, Journal of Earthquake Engineering, 26:1, 271-306, DOI: 10.1080/13632469.2019.1684401, 2022.

In this study, the latest seismic activity data and inland fault data are used, applying a Poisson process as the probabilistic model. Pakistan was divided into 24 zones (assuming constant seismic activity in one zone). It is considered to be one of the most reliable seismic hazard maps in Pakistan as of 2022 Figure 2.1.20 shows the PGA distribution corresponding to a 475-year return period.



Source: Muhammad Ahmed, Sarosh Hashmat Lodi & Muhammad Masood Rafi, Probabilistic Seismic Hazard Analysis Based Zoning Map of Pakistan, Journal of Earthquake Engineering, 26:1, 271-306, DOI: 10.1080/13632469.2019.1684401, 2022

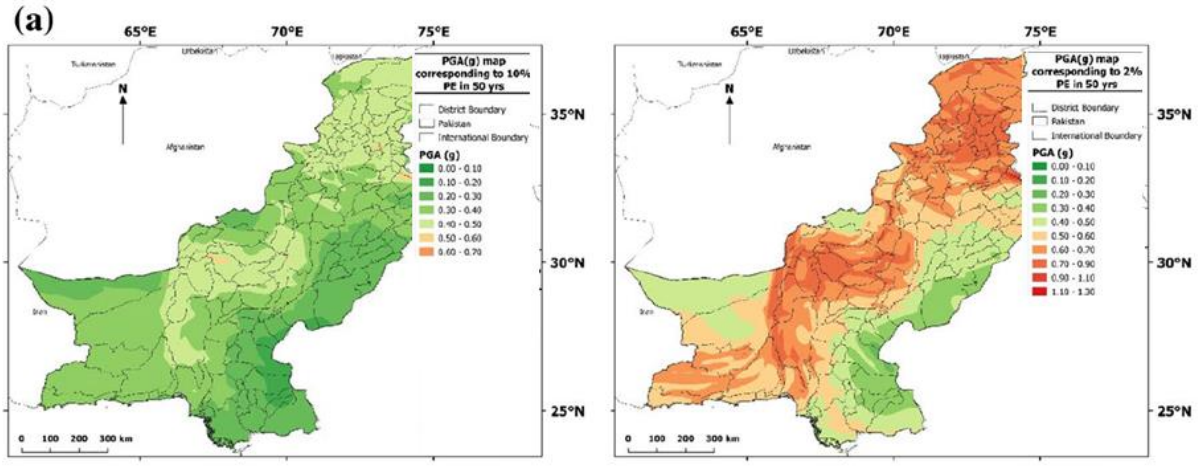
Figure 2.1.20 Distribution of PGA (Return Period 475 Years)

2) Research Results at National University of Science and Technology (NUST National University of Sciences and Technology)

The project team obtained the following literature on the latest seismic hazard maps conducted at NUST.

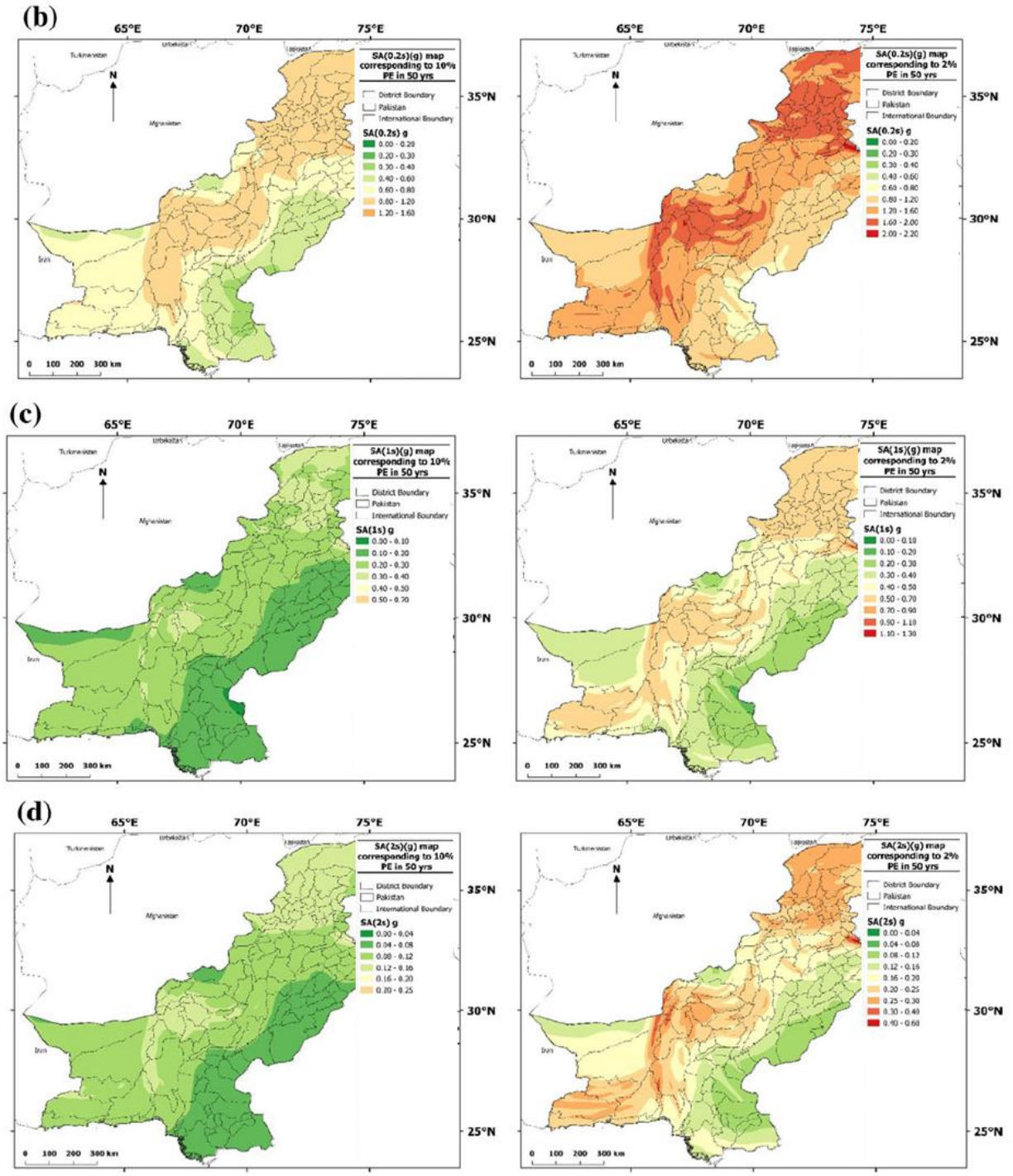
- Asad Ur Rahman, Fawad Ahmed Najam, Saeed Zaman, Atif Rasheed and Irfan Ahmad Rana, An updated probabilistic seismic hazard assessment (PSHA) for Pakistan, Bulletin of Earthquake Engineering, 19, pages 1625–1662, 2021.

In this paper, the latest Probabilistic Seismic Hazard Assessment (Probabilistic Seismic Hazard Assessment: PSHA) for the whole of Pakistan is developed using methods developed for the US seismic hazard map and the Middle East seismic model (EMME14). The base earthquake catalog is also updated, and background seismicity is modeled using both conventional zoning and spatially smoothed grids. The fault model in the crust is based on the Global Earthquake Model (GEM) data (slip velocity), and the earthquake occurring in the Makran Subduction Zone (The Sunda/Java Trench and the subduction zone off the coast of Pakistan and Iran that generated the Indian Ocean tsunami in 2004) is also modeled. The peak ground acceleration (PGA) and acceleration response spectrum (SA) in natural period at 0.2 s, 1 s, and 2 s at engineering bedrock are estimated using the attenuation equation with Next Generation Attenuation (NGA) models. This is considered to be one of the latest advancements in foundational seismic hazard assessment, and can be utilized for structural design of new buildings and performance evaluation of existing buildings, as well as for policy formulation and planning for effective regional disaster risk reduction. Figure 2.1.21 – Figure 2.1.24 shows a series of seismic hazard maps described above.



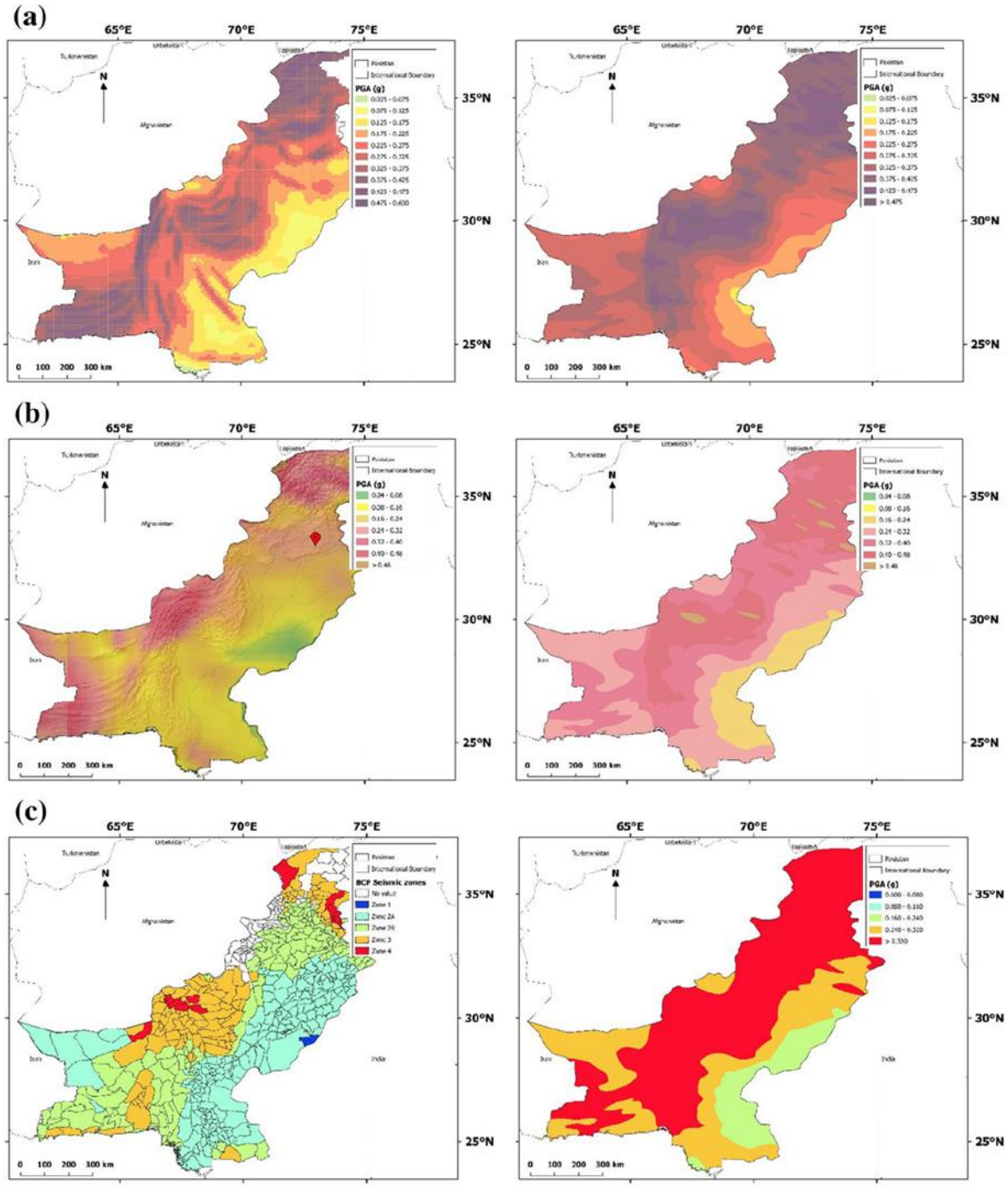
Source: Asad Ur Rahman, Fawad Ahmed Najam, Saeed Zaman, Atif Rasheed and Irfan Ahmad Rana, An updated probabilistic seismic hazard assessment (PSHA) for Pakistan, Bulletin of Earthquake Engineering, 19, pages1625 ‑1662, 2021.

Figure 2.1.21 Seismic Hazard Map Prepared in Rahman et al., (2021).
((a) PGA) Corresponds to a 50 Year Exceedance Probability of 10% (Left) and 2% (Right).



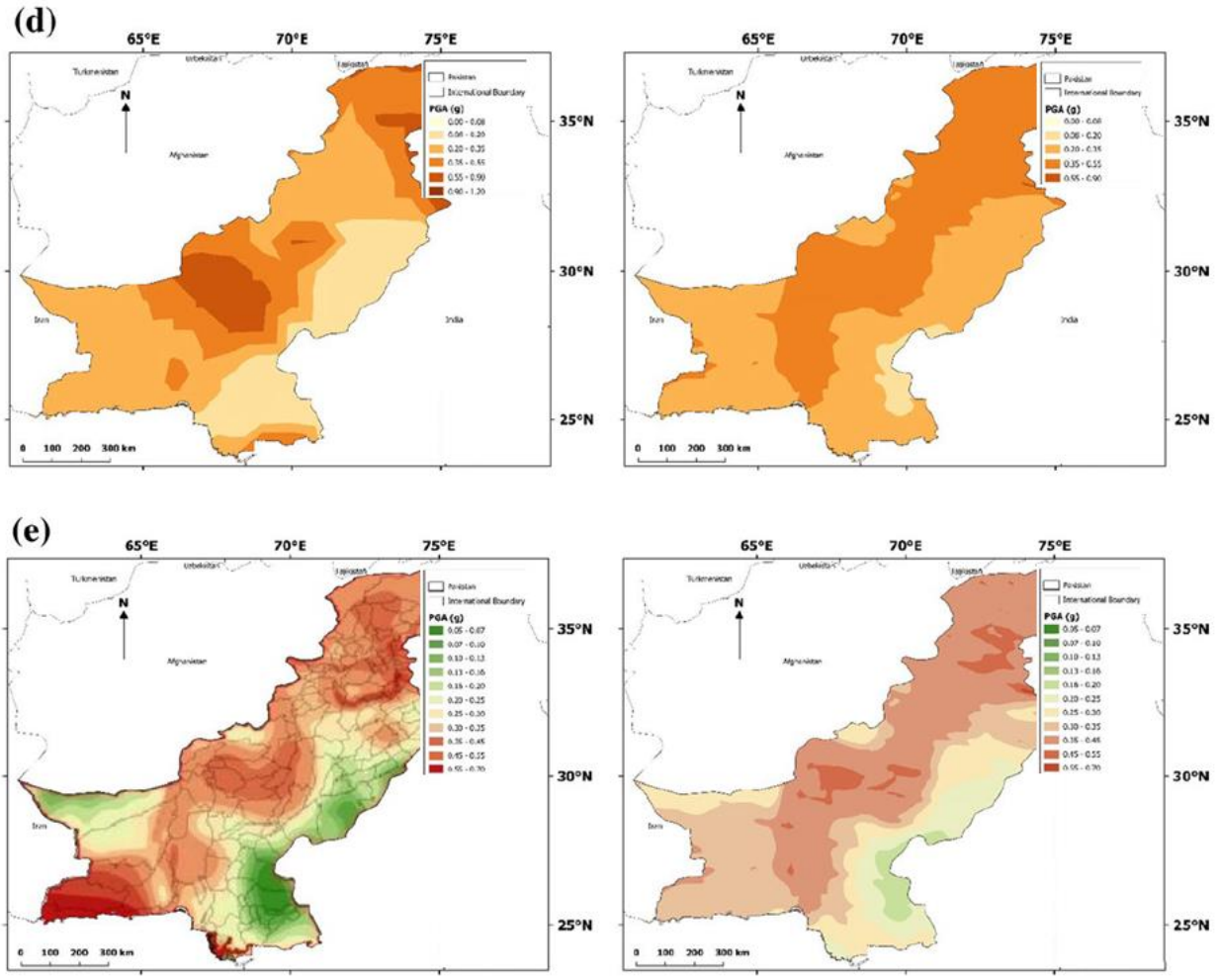
Source: Asad Ur Rahman, Fawad Ahmed Najam, Saeed Zaman, Atif Rasheed and Irfan Ahmad Rana, An updated probabilistic seismic hazard assessment (PSHA) for Pakistan, Bulletin of Earthquake Engineering, 19, pages1625–1662, 2021.

Figure 2.1.22 Seismic Hazard Map Prepared in Rahman et al., (2021). (Corresponds to a 50 Year Exceedance Probability of 10% (Left) and 2% (Right)):
(b) SA (natural period 0.2 s), (c) SA (natural period 1 s), (d) SA (natural period 2 s)



Source: Asad Ur Rahman, Fawad Ahmed Najam, Saeed Zaman, Atif Rasheed and Irfan Ahmad Rana, An updated probabilistic seismic hazard assessment (PSHA) for Pakistan, Bulletin of Earthquake Engineering, 19, pages1625 –1662, 2021.

Figure 2.1.23 Comparison of Seismic Hazard Maps for Pakistan (50 Year Exceedance Probability of 10%, Equivalent to 475 Year Return Period) (a): EMME (2014) (b):GSHAP (c): NESPAK (Building Code of Pakistan 2007



Source: Asad Ur Rahman, Fawad Ahmed Najam, Saeed Zaman, Atif Rasheed and Irfan Ahmad Rana, An updated probabilistic seismic hazard assessment (PSHA) for Pakistan, Bulletin of Earthquake Engineering, 19, pages1625 ‑1662, 2021.

Figure 2.1.24 Comparison of Earthquake Hazard Maps for Pakistan (50 Year Exceedance Probability of 10%, Equivalent to 475 Year Return Period) (d): PMD and NORSAR (e): Zaman (2012) (left) and Rahman et al., (2021)

Since the data used in this research paper represents some of the latest earthquake hazard data available in Pakistan, the project team obtained the earthquake catalog, model, and hazard map data directly from the author at NUST University of Technology.

Based on the seismic hazard data outlined in 1) or 2), the project team will proceed with the seismic risk assessment for the current NDMP update.

2.1.2.3 Probability of Occurrence

Seismic zoning is determined by considering geological and seismological and seismological characteristics. It is assumed that the seismic activity is homogeneous in one zone and independent of time, and the annual occurrence frequency for each magnitude is expressed by a regression equation. Based on this regression equation, the probability of occurrence of earthquakes is given.

2.1.3 Tsunami

2.1.3.1 Damage History (Year of occurrence, scale of damage, etc.)

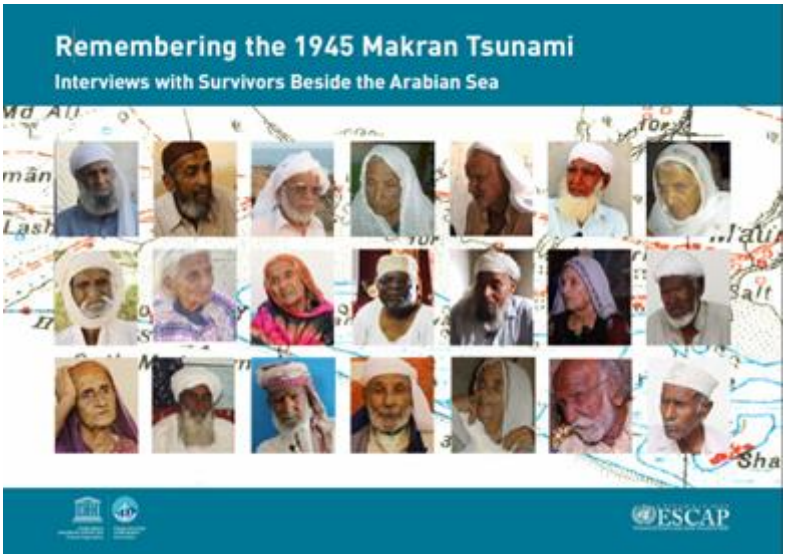
The current NDMP summarizes tsunami run-up heights from historical tsunamis in the Arabian Sea in Table 2.1.14.

Table 2.1.14 Tsunami Run-Up Height Due to Historical Tsunamis in the Arabian Sea

Date	Time	M	Run up (m)	Location
1883/08/27	02:59	(Volcano)	0.50	Karachi
1945/11/27	21:56:40	8.3	15.24	Pasni, Ormara
1945/11/27	21:56:40	8.3	1.37	Karachi

Source: National Geophysical Data Centre

The 1945 tsunami in the Makran subduction zone is estimated to have reached the coast within 20 minutes. Tsunami simulations conducted by PMD have estimated the extent of inundation in each coastal city. Assuming an earthquake of Mw 8.5 at a distance of 20 km from the coast, the tsunami reaches the coastal city of Gwadar in about 22 minutes and the run-up height is estimated to be 3.7m. The documents (including historical documents) regarding the 1945 Makran tsunami, which the project team obtained, provide valuable testimony about the damage caused by the event. (See Figure 2.1.25)



Source: ITIC (International Tsunami Information Center)

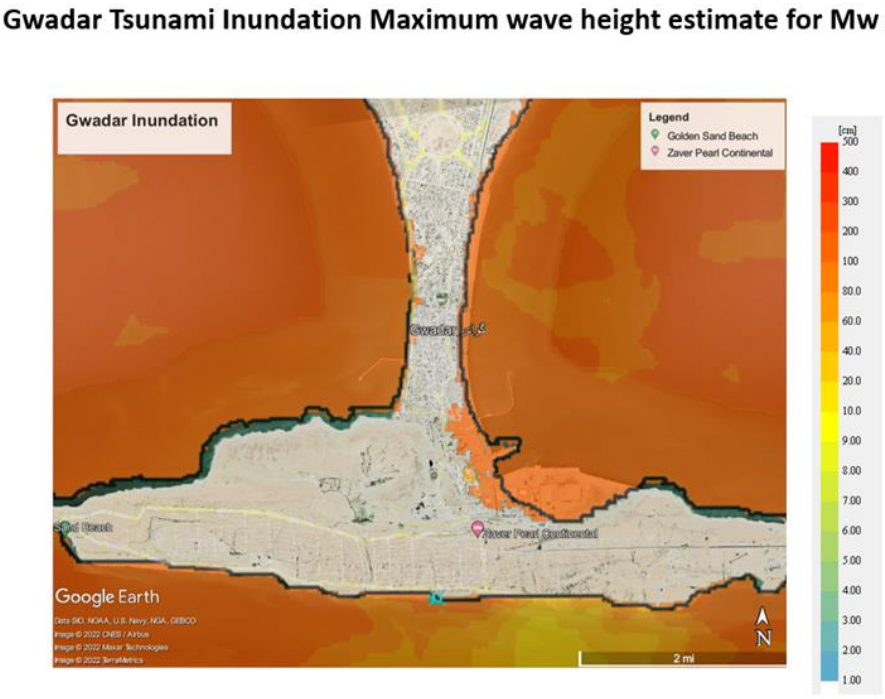
Figure 2.1.25 Documentation of the 1945 Makran Tsunami

2.1.3.2 Hazard Area and Intensity

Based on the simulation results of Rafi et al. (2010), the current NDMP evaluates tsunami run-up height, run-up distance, tsunami arrival time, and tsunami duration for only one coastal city, i.e. Gwadar. Figure 2.1.26 shows an example of an inundation simulation. The project team interviewed the PMD, which conducted the simulation on probabilistic tsunami hazard.

- A fault model published by NOAA (National Oceanic and Atmospheric Administration) was established along the Makran Trench located south of Pakistan, and inundation simulations were conducted for 4 major coastal cities (Gwadar, Karachi, Ormara and Pasni). These tsunami simulations are based on deterministic methods for tsunami hazard assessments, rather than probabilistic tsunami hazard assessments.

Since probabilistic tsunami hazard information is not currently included in the NDMP, it is important to establish a program to introduce and implement this framework.



Source: Courtesy of PMD

Figure 2.1.26 Results of Inundation Simulation Conducted by PMD (Gwadar)

2.1.3.3 Probability of Occurrence

Probabilistic tsunami hazard assessments for major cities along the coast of Pakistan have been carried out in research papers, focusing on earthquakes occurring in the Makran subduction zone (Momeni et al., 2020, Hoechner et al., 2015, Salah et al., 2020) from other countries (Japan, Germany, Iran, etc.), and the corresponding earthquake occurrence probabilities have been given. However, since the current NDMP is based on scenario-based (deterministic) simulation by PMD, the earthquake occurrence probabilities are not given at present.

2.1.3.4 Issues in Disaster Risk Analysis

PMD has previously carried out tsunami hazard assessments using deterministic methods; however, probabilistic tsunami hazard assessments have not been performed. Since probabilistic tsunami hazard information is not included in the current NDMP, the project team introduced the probabilistic tsunami hazard assessment conducted in Japan to PMD. As the assessment target area, it is necessary to evaluate the hazard of the entire coast of Pakistan, not just one coastal city listed in the current NDMA. Since there are no examples of probabilistic tsunami hazard assessments by research institutes or ministries in Pakistan, a framework and programming for probabilistic tsunami hazard assessments are necessary. In order to analyze probabilistic tsunami hazard, it is necessary to set the probability of occurrence of fault models corresponding to trench-type earthquakes.

When the project team checked with the PMD staff, it was confirmed that there are no supporting documents in Pakistan such as the "long-term assessment" published by the Japan Earthquake Research Commission. In addition, it is pointed out that high-resolution topographic data are necessary for tsunami simulation.

2.1.4 Other Disasters

Following the discussions with NDMA, the candidate the candidate disasters to be included in NDMP 2024 have been identified as: 1. Flood, 2. Earthquake, 3. Tsunami, 4. Smog, 5. Heat wave, 6. Cyclone, 7. Snow avalanche, 8. Locust infestation, 9. Plague, 10. Industrial and technological hazards, 12. Fire, etc. For the

three risk factors for other disasters, the data preparation of the individual factors on the Pakistan side has not progressed, so these are summarized in Activiti 1-3 and Activity 1-4.

2.2 Activity 1-2

Confirmation of vulnerability and exposure data underlying disaster risk information in the current NDMP.

2.2.1 Verification of Exposure Data

2.2.1.1 Verification of Exposure Data in Flood Risk Assessment

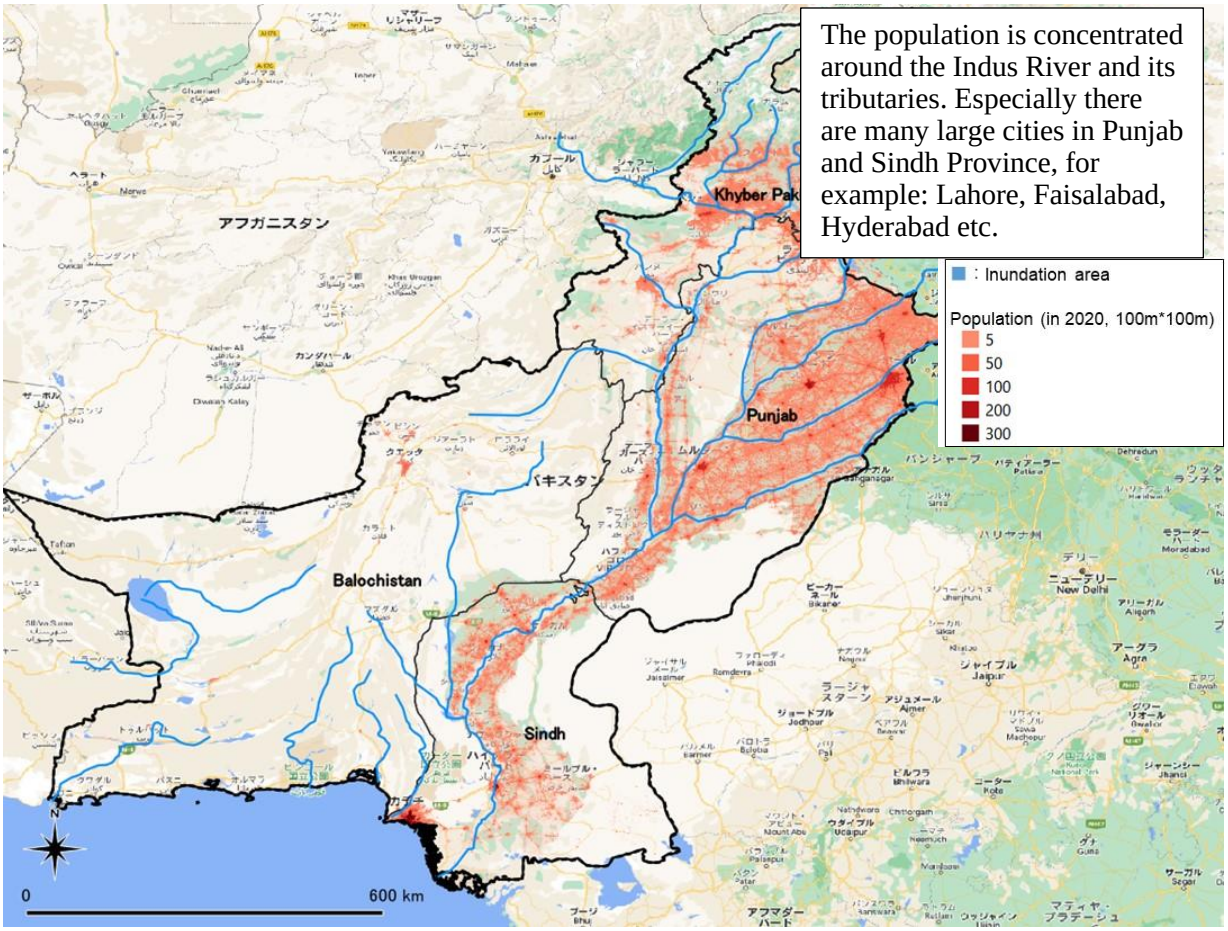
The contents of foundation data in the Hazard & Risk Map of the current NDMP were reviewed. About the exposure data, the population density and crop yield (Wheat, rice, and cotton) of each district were evaluated in five stages in each district unit as foundation data. According to foundation data, exposure data were scored for each province in the categories listed in Table 2.2.1.

Table 2.2.1 Exposure Data Used in Current NDMP

Score	Population Density (person/$m^{2)}$	Wheat Yield (kg/ha)	Rice Yield (kg/ha)	Cotton Yield (Bale*/ha)
1	< 26	< 2250	< 500	1
2	26 - 138	2250 - 2750	500 - 1000	2
3	138 - 261	2750 - 3000	1000 - 1250	3
4	261 - 479	3000 - 3250	1250 - 1500	4
5	479 <	3250 <	1500 <	5

**1 Bale in Pakistan = 170 kg*
Source: Summarized by the Project Team from foundation data of the current NDMP

Population distribution, which serves is foundational information for estimating exposure data, is available on the existing website as shown in Figure 2.2.1.

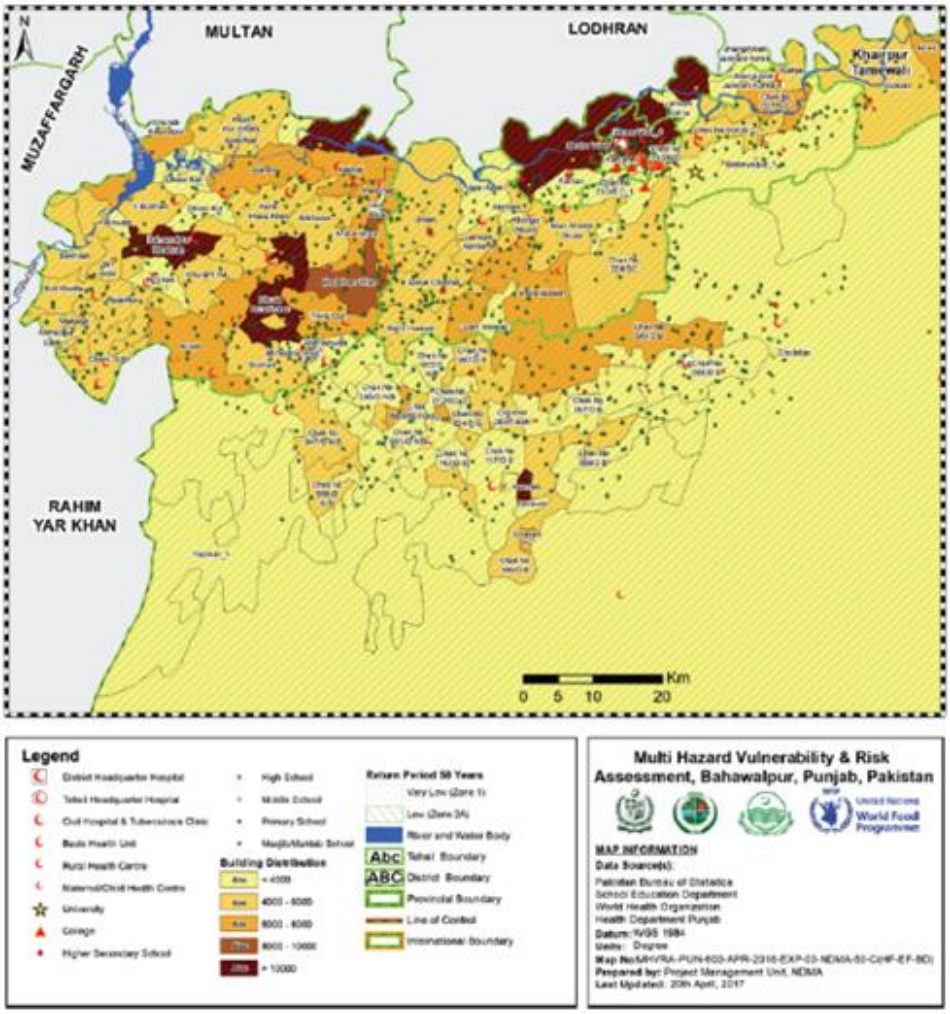


Source: WorldPop (in 2020)

Figure 2.2.1 Population Distribution (2020)

2.2.1.2 Review of Exposure Data in Earthquake Risk Assessment

The Multi Hazard & Vulnerability Risk Assessment (MHVRA) report, which is implemented as part of the activities proposed in the current NDMP, confirms that population statistics from the census and residential building types included in the survey are mapped using GIS. An example is shown in Figure 2.2.2. The legend of the figure confirms that the statistics are presented as Tehsil units. In the current NDMP update, the project team purchased and utilized 2017 census data released by the Bureau of Statistics.



Source: Source: MHVRA report

Figure 2.2.2 Example Building Distribution Map (Color-Coded by Building Quantity, Hatches are Earthquake Hazard Zones)

2.2.2 Vulnerability Data (Flood Field)

2.2.2.1 Vulnerability Data Types

The current NDMP does not incorporate individual vulnerability data. However, vulnerability related to flooding and the availability of relevant data can be summarized as follows.

- **Inundation Duration and Depth in Past Flood Events**
  >Due to the lack of existing data, it is difficult to compile and incorporate them into the analysis.
- **Strength and Durability of Existing Bund**
  >Since the quality of bunds at each location is unclear due to the lack of existing data, it is not advisable to conduct analysis based on the estimation.
- **Sufficiency of Damage Mitigation Measures Such as Evacuation Facilities and Systems in Case of Floods**
  >It is an option to consider whether or not there are evacuation centers available in case of floods securing routes to evacuation centers, and whether or not evacuation drills are implemented in case of flood damage. However, the degree of sufficiency of damage mitigation measures for each region is unknown and it is difficult to appropriately consider.
- **Sufficiency of Measures for Early Recovery and Reconstruction in Case of Floods**

>It is an option to consider whether or not there are facilities for natural or forced drainage, and whether or not there are transportation routes that can transport materials. However, although there are few relevant data, it is assumed that there are no regional differences.

Therefore, as mentioned above, it is difficult to accurately reflect the vulnerability of any of the items due to the limited availability of data and minimal regional variation .

2.2.2.2 How to Use Vulnerability Data

In case of flood, vulnerability shall be considered in risk analysis because information should be collected and aggregated when considering countermeasures in each small- and medium-scale basin which have issues. In the nationwide risk analysis, vulnerability shall be classified only as landside of the bunds (kutcha houses) or riverside of the bunds (bricks and concrete structures). In the future, it is necessary to consider including vulnerability in the risk analysis if the level of information that can grasp and introduce the vulnerability of each house or crop can be obtained. In addition, the quality of bund, damage mitigation measures in the event of a flood, and the degree of adequacy of measures for early recovery and reconstruction are unclear, so they will not be used in this NDMP update.

2.2.3 Vulnerability Data (Earthquake Field)

2.2.3.1 Verification of Vulnerability Data in Earthquake Risk Assessment

Regarding the vulnerability of buildings, it has been confirmed that the quantity of building type is organized by earthquake hazard zone. (See Figure 2.2.3) However, it is only the stage of organizing the building inventory information which is the basis of vulnerability assessment. Vulnerability is generally evaluated using the damage function such as Vulnerability Function, and Fragility Curve, etc.; however, the data did not include results of such evaluations.

Zone 1	Zone 2A	Zone 2B	Zone 3	Zone 4	Zone 1	Zone 2A	Zone 2B	Zone 3	Zone 4	Zone 1	Zone 2A	Zone 2B	Zone 3	Zone 4	Zone 1	Zone 2A	Zone 2B	Zone 3	Zone 4
0	0	4,949	0	0	0	0	1,253	0	0	0	0	2,844	0	0	0	0	851	0	0
0	0	6,943	0	0	0	0	3,343	0	0	0	0	2,825	0	0	0	0	775	0	0
0	0	5,068	0	0	0	0	3,254	0	0	0	0	1,485	0	0	0	0	329	0	0
0	0	15,688	0	0	0	0	9,642	0	0	0	0	3,142	0	0	0	0	2,904	0	0
0	0	4,712	0	0	0	0	1,944	0	0	0	0	2,396	0	0	0	0	371	0	0
0	0	6,859	0	0	0	0	2,428	0	0	0	0	3,559	0	0	0	0	872	0	0
0	0	5,018	0	0	0	0	1,915	0	0	0	0	2,786	0	0	0	0	318	0	0
0	0	4,935	0	0	0	0	1,079	0	0	0	0	2,314	0	0	0	0	1,542	0	0
0	0	5,672	0	0	0	0	2,404	0	0	0	0	2,958	0	0	0	0	310	0	0
0	0	4,832	0	0	0	0	2,391	0	0	0	0	2,197	0	0	0	0	244	0	0
0	0	5,864	0	0	0	0	1,184	0	0	0	0	3,654	0	0	0	0	1,025	0	0
0	0	4,167	0	0	0	0	1,771	0	0	0	0	2,048	0	0	0	0	348	0	0
0	0	7,158	0	0	0	0	3,050	0	0	0	0	2,938	0	0	0	0	1,171	0	0
0	0	4,197	0	0	0	0	1,624	0	0	0	0	2,116	0	0	0	0	456	0	0

Source: Source: MHVRA Report

Figure 2.2.3 Example of Organizing Statistical Data by Building Type in Tehsil Units by Earthquake Hazard Zone

2.2.4 Other Disasters

As described in Activity 1-1, this topic is summarized in Activity 1-3 and Activity 1-4.

2.3 Activity 1-3

Confirmation of the necessity to update and review disaster risk information in the current NDMP.

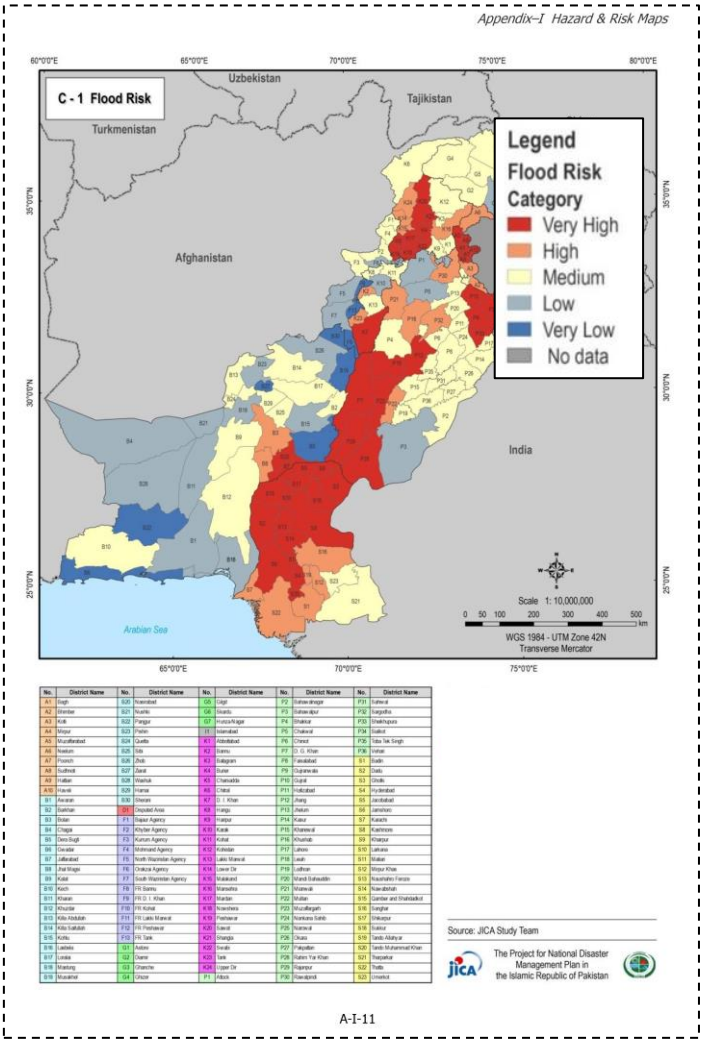
2.3.1 Flood

2.3.1.1 Risk Calculation Method

(1) Risk Calculation Method and Issues in the Current NDMP

Figure 2.3.1 shows the flood risk assessment in the current NDMP. The flood risk in the current NDMP is calculated using "Hazard x Exposure Data." The hazard and exposure data considered here are described above.

Figure 2.3.1 shows that districts with high flood risk are distributed from the downstream to the middle of the Indus River, showing the same tendency as the hazard distribution shown in Figure 2.1.1. On the other hand, the upstream region of the Indus River —identified as having relatively high hazard intensity in Figure 2.1.1 —is assessed as having a moderate level of flood risk.



Source: NDMP, Main Volume, A-1-11

Figure 2.3.1 Hazard & Risk Map Described in the current NDMP(Example of Flood Risk)

The issues in the current NDMP are as follows.

- Consider the feasibility of conducting risk assessment based on the integtration of hazard, exposure and vulnerability data.
- With regard to hazards, the flood events since 2011 have been documented, and the hazard assessment has been updated accordingly.
- With regard to exposure data, the population density at the district level is used. However, the area at the district level is wide, and it is difficult to fully consider the population distribution along rivers directly damaged by floods and close to population and industrial clusters (see Figure 2.3.2).

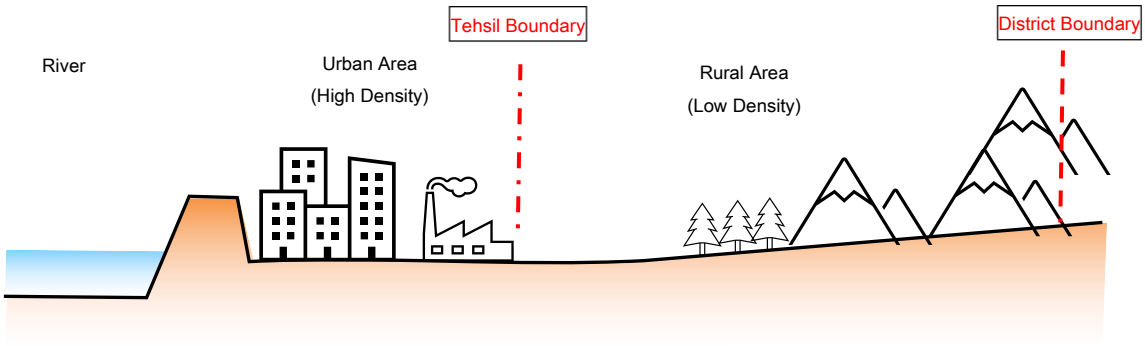
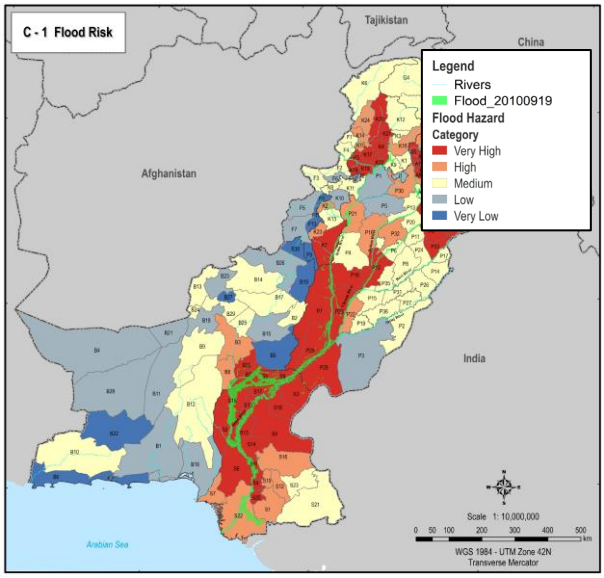


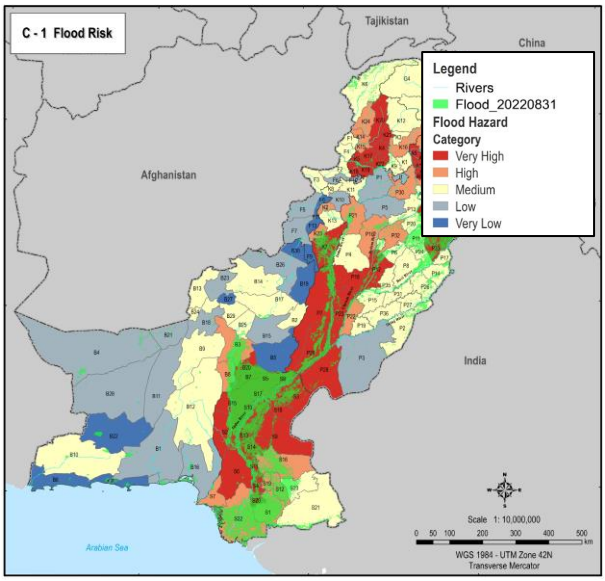
Figure 2.3.2 Example of Insufficient Accuracy with District Level Population Density

Figure 2.3.3 and Figure 2.3.4 show the flood areas due to 2010 and 2022 floods superimposed on the risk areas[2] by river line flood. According to these figures, the flood areas due to 2010 floods are generally located in the areas classified as "Very High" or" High" among the risk areas by river line flood. Although the flood areas due to 2022 floods show similar trends, it can be seen that the flood areas also extend to the areas classified as "Medium " or" Low" in Hafizabad District and Kasur District in Punjab Province. Although these districts are located along the Chenab and Ravi Rivers, the risk levels are not high because the original hazard levels are "Medium " or" Low." Therefore, the risk levels may be raised if the hazard levels can be re-evaluated by considering the flood records after 2011 and future flood events and the risk maps that are expected to be developed by SUPARCO, Punjab-PID, etc.



Source: United Nations Satellite Centre (UNOSAT)

Figure 2.3.3 Superposition of 2010 Flood Areas and Flood (Riverine Flood) Risk Areas



Source: United Nations Satellite Centre (UNOSAT)

Figure 2.3.4 Superposition of 2022 Flood Areas and Flood (Riverine Flood) Risk Areas

(2) Proposed Risk Calculation Method in the Current NDMP

In assessing risk, it is desirable to calculate vulnerability in addition to hazard and exposure data. However, vulnerability is not considered in the national assessment because, as mentioned above, it is unclear and there is generally no difference between regions. Therefore, in the current NDMP, vulnerability related to organizations, institutions, and assets (buildings, farmland, etc.) is considered

2 Created with NDMP-I and still used by NDMA.

as equivalent across the country, and risk is expressed as the distribution of hazard and exposure data. As in the current NDMP, vulnerability is calculated using "hazard x exposure " without considering individual foundation data in this project.

2.3.1.2 Policy for Update and Review

Based on the following points, it is considered that the analysis using the results at the formulation of the current NDMP is sufficient to show the national project.

- With regard to hazard data, there is no damage data available from 2011 onwards. Although the 2022 floods affected areas previously not classified as high hazard zones, it is unlikely that meteorological and hydrological conditions have changed significantly over the past decade .
- With regard to exposure data, it is expected to improve the accuracy of risk analysis by using population density at the Tehsil level and the corresponding hazard information, rather than using population density at the district level. However, as mentioned above, it is difficult to update and improve the accuracy of risk analysis beyond the current NDMP because sufficient data for updating hazards have not been accumulated in NDMA, etc. Therefore, the analysis at the district level is the basis as in the current NDMP.
- As with the current NDMP, individual foundation data (vulnerability of houses and crops) will be recognized as a uniform level across the country. In this NDMP update work, regional vulnerabilities cannot be utilized for risk analysis.

2.3.2 Earthquake

2.3.2.1 Risk Calculation Method

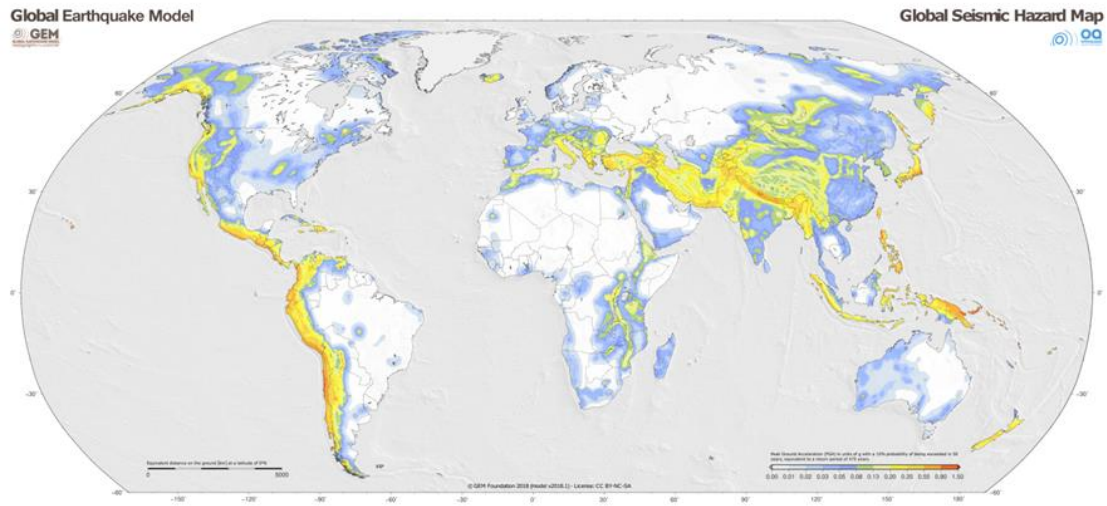
(1) Earthquake Risk Map of Pakistan Using Global Data

A list of earthquake risk and high-risk cities in Pakistan has been prepared in "Information Collection and Verification Survey on Promotion of Investment in Disaster Prevention for Urban Resilience Against Global Earthquakes (2022)" of your organization. In this project, earthquake risk is evaluated using the geomagnetic amplification factor in addition to population data and PGA values, which can be applied globally from the viewpoint of a unified evaluation of JICA-supported countries. The risk index calculated from the population, PGA value, and geomagnetic amplification factor of each city is used to evaluate earthquake risk, and 20 cities with high earthquake risk are selected in each country. At the same time, a geographical distribution map illustrating the risk index for each country is being prepared. It should be noted that, in this case, earthquake risk is considered as "hazard x exposure" and vulnerability is not considered as the same reason as for floods. The data used for the risk evaluation are as follows.

- Population data: World Cities Database (WCD) [3]
  Only the population of local governments in urban areas is counted and tabulated, and this data is used to calculate the index by city.
- Population density data: Socio-economic Data and Applications Center (SEDAC) [4]
  It is used for the distribution map of the INDEX.
- PGA data: Global Earthquake Model (GEM) [5]
  It was used to calculate INDEX for both city and distribution maps.
  The GEM Global Seismic Hazard Map shows the geographical distribution of Peak Ground Acceleration with a 50-year exceedance probability of 10%, calculated for a ground with an S-wave velocity of 760-800 m/s. The recurrence period corresponds to 475 years.

3 https://simplemaps.com/data/world-cities
4 https://sedac.ciesin.columbia.edu/data/sets/browse
5 https://www.globalquakemodel.org/gem

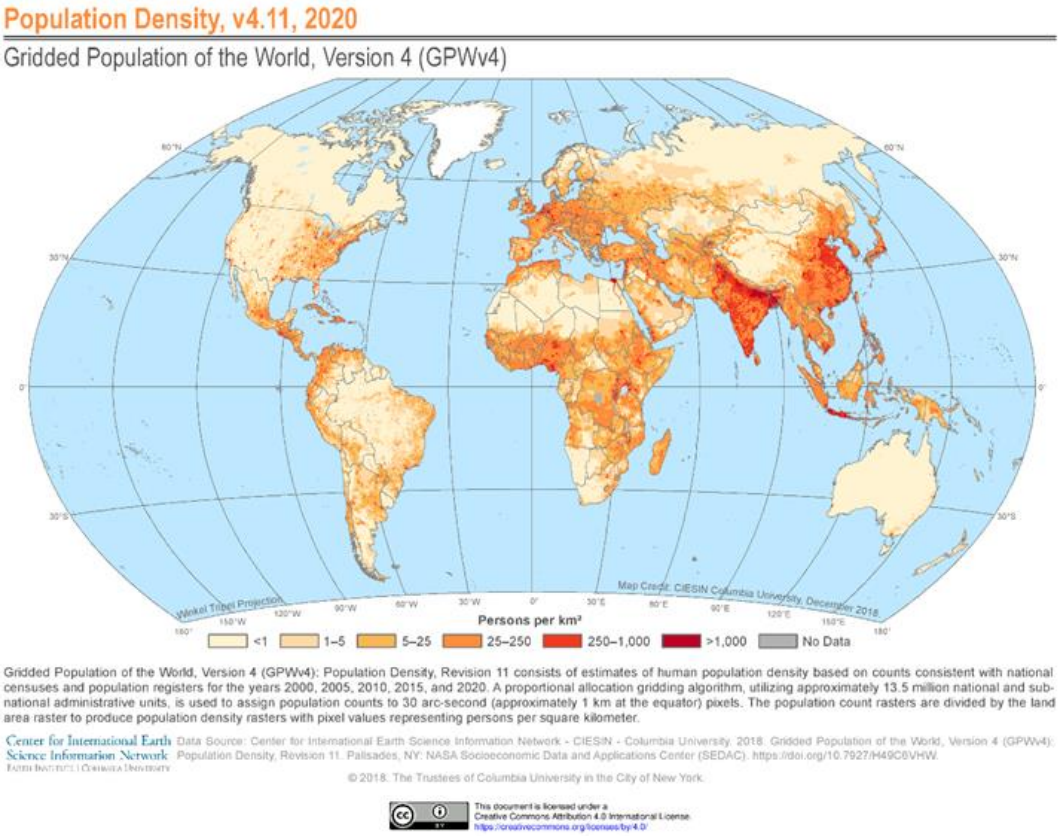
- VS30 data: Average S-wave velocity model for surface 30 m (USGS (the time-averaged shear-wave velocity to 30 m depth)) [6]
  To consider the ground amplification factor in the earthquake risk assessment, the USGS Average S-wave velocity model for surface 30 m was used to calculate the ground amplification factor.
  The population density data (SEDAC) is in the form of a 30 second grid, and the PGA data is in the form of discrete data. To calculate these data at the same point, linear interpolation is applied to the PGA data to create the same 30 second grid data as the population density data, and INDEX is calculated. Figure 2.3.5 and Figure 2.3.6 show the global PGA and population density distributions.



Source: Global Earthquake Model (GEM) Global Seismic Hazard Map (version 2018.1)

Figure 2.3.5 Global Earthquake Hazard Map of GEM
(Shown is the distribution of seismic acceleration (g) corresponding to a simulation period of 475 years.)

6 https://earthquake.usgs.gov/data/vs30



Source: Socioeconomic Data and Applications Center (SEDAC)

Figure 2.3.6 Population Density Map (v4.11,2020)

1) Calculation Method of Ground Amplification Factor

To account for the ground amplification factor in seismic risk assessment, the USGS 30 m surface mean S-wave velocity model (the time-averaged shear-wave velocity to a depth of 30 m) was used to calculate ground amplification factor. The details of the data are shown in Figure 2.3.5 and Figure 2.3.6.

USGS VS30 Data
• Grid resolution: 30 arc-seconds (0.0083333333 degrees) • Latitude span: -56.0 to 84.0 degrees • Longitude span: -180 to 180 degrees • Vs30 units: meters per second • Vs30 range: 98 to 2197 m/s • Vs30 in water-covered areas: 600 m/s

Source: USGS, Vs30 Models and Data, https://earthquake.usgs.gov/data/vs30/

Figure 2.3.7 Details of USGS Surface 30 m Average S-Wave Velocity Model Data

Ground amplification factor (Amp) was calculated using USGS VS30 data and Borcherdt et al. *(*Borcherdt, R. D., Gibbs, J. F., and Fumal, T. E.: Progress on ground motion predictions for the San Francisco Bay region, California, U.S. Geol. Survey Cir. 807, 1979, pp. 13*) equation (1).

$$Amp = -11.4 \log (S\text{-vel, m/s}) + 33.6 \quad (1)$$

Amp: Ground amplification factor
S-vel: S-wave velocity

When Amp is less than 1, Amp=1 is used.

2) Earthquake Risk Distribution Chart

The project team used population density (people/km^2) data in a 30 second grid for analysis (Data from NASA's Socioeconomic Data and Applications Center: SEDAC) and PGA distribution of ground amplification factor and discrete data. The project team created grid data by applying linear interpolation to the PGA distribution. The grid after interpolation is the same 30 second grid as the population density data. Combining these data with capital data, the project team used INDEX (the multiplication of population, PGA, and ground amplification factor (Amp)) shown in Equation (2) as a hazard index of earthquake risk to create a hazard distribution.

$$\text{INDEX} = \text{Population density (SEDAC)} * \text{PGA (GEM)} * \text{Amp} \quad (2)$$

3) Extraction of Cities with High Earthquake Risk (INDEX)

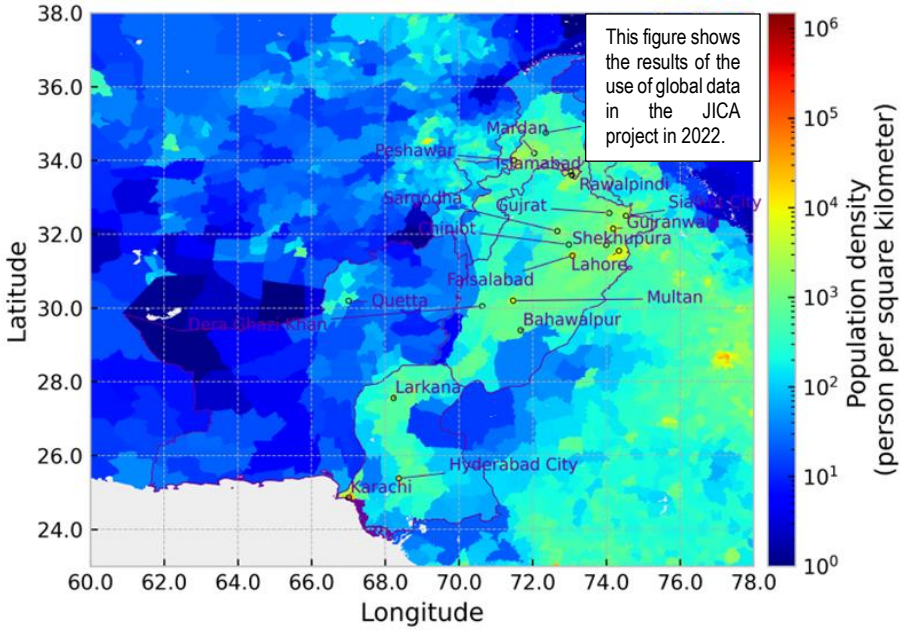
To evaluate earthquake risk in major cities, the project team used population data by city (https://simplemaps.com/data/world-cities), PGA data (GEM), and ground amplification factor (Amp), and calculated the hazard index (by city) using Equation (3).

$$\text{INDEX (by city)} = \text{Population (WCD)} * \text{PGA (GEM)} * \text{Amp} \quad (3)$$

The PGA value is extracted from the coordinates of each city and used to calculate INDEX (by city). 20 cities with large INDEX (by city) were extracted and ranked. The INDEX used for the distribution map uses population density data (SEDAC data), while the INDEX used for the city list uses population data (World Cities Database (WCD)). The following is a list of cities with high earthquake risk in Pakistan, based on global data (Table 2.3.1) and risk maps (Figure 2.3.8 to Figure 2.3.11.)

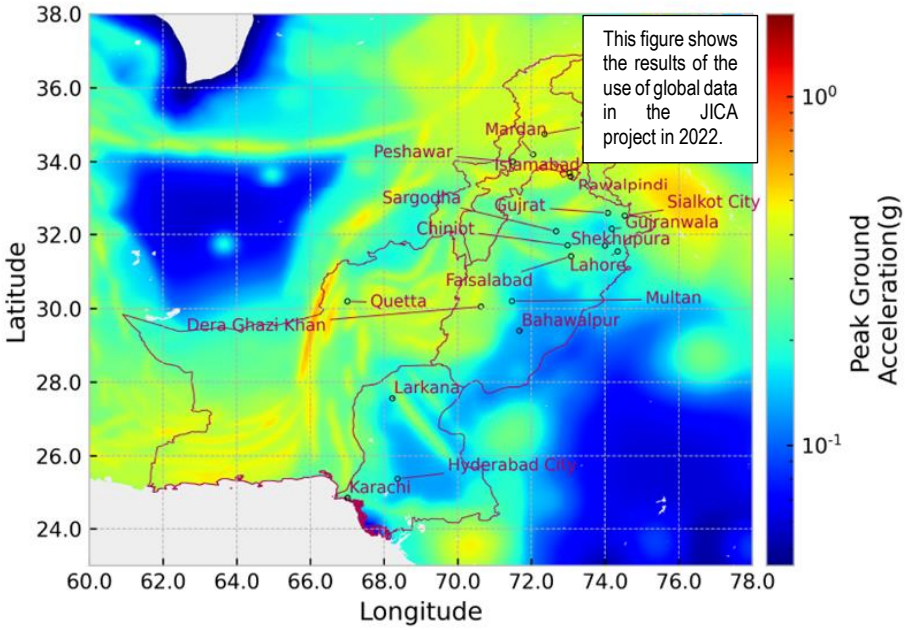
Table 2.3.1 Cities with High Earthquake Risk in Pakistan

Cityname	Population (WCD)	Lon	Lat	PGA (nearest)	Ground amplification	INDEX
Karachi	14,835,000	67.01	24.86	0.2248	5.9810	19,945,965
Lahore	11,021,000	74.34	31.55	0.1965	6.9331	15,014,608
Faisalabad	3,203,846	73.08	31.42	0.2581	6.5568	5,421,861
Rawalpindi	2,098,231	73.07	33.60	0.4109	5.7535	4,960,442
Peshawar	1,970,042	71.50	34.00	0.3376	5.6145	3,734,094
Islamabad	1,014,825	73.04	33.70	0.4620	5.4676	2,563,491
Gujranwala	2,027,001	74.18	32.15	0.1791	6.3813	2,316,639
Multan	1,871,843	71.47	30.20	0.1718	5.9804	1,923,195
Hyderabad City	1,732,693	68.37	25.38	0.1479	6.8100	1,745,160
Quetta	1,001,205	67.01	30.19	0.2847	4.6097	1,313,961
Sialkot City	655,852	74.53	32.50	0.2621	6.0442	1,038,984
Saidu Sharif	1,860,310	72.36	34.75	0.4310	1.1990	961,362
Dera Ghazi Khan	399,064	70.63	30.05	0.3314	6.7221	888,995
Sargodha	659,862	72.67	32.08	0.1997	6.7221	885,798
Bahawalpur	762,111	71.67	29.40	0.1329	6.7976	688,491
Mardan	300,424	72.04	34.20	0.3403	6.4716	661,616
Gujrat	390,533	74.08	32.57	0.2336	6.8918	628,728
Larkana	490,508	68.23	27.56	0.1871	6.8276	626,598
Chiniot	477,781	72.98	31.72	0.2017	6.3264	609,662
Shekhupura	473,129	74.00	31.71	0.1955	6.4576	597,305



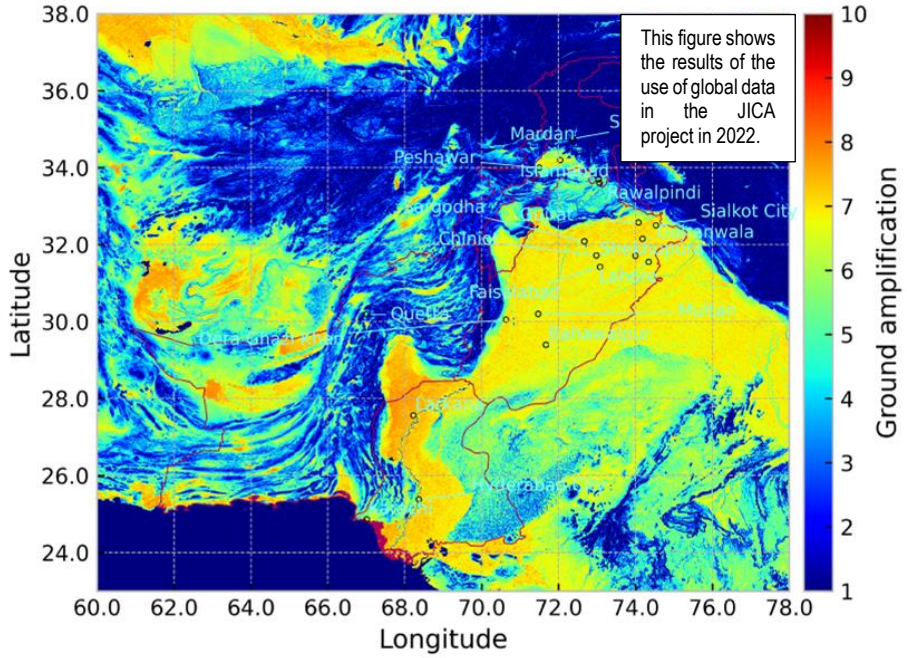
Source: "Information Collection and Verification Survey on Promotion of Investment in Disaster Prevention for Urban Resilience Against Global Earthquakes (2022)"

Figure 2.3.8 Population Density Distribution of Pakistan



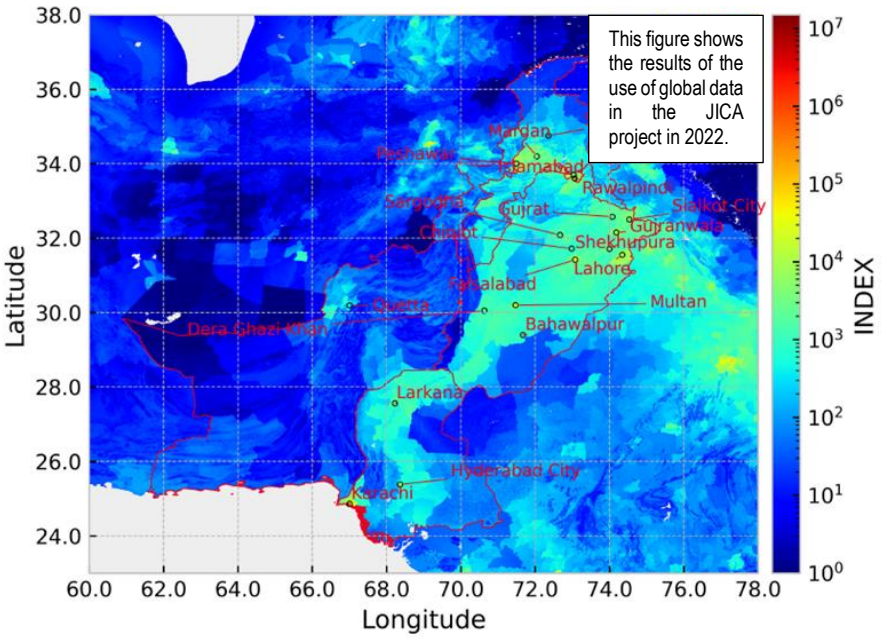
Source: Information gathering and verification study on promoting investment in disaster prevention for urban resilience against global earthquake disaster (2022)

Figure 2.3.9 PGA Distribution in Pakistan



Source: "Information Collection and Verification Survey on Promotion of Investment in Disaster Prevention for Urban Resilience Against Global Earthquakes (2022)"

Figure 2.3.10 Ground Amplification Rate Distribution in Pakistan



Source: "Information Collection and Verification Survey on Promotion of Investment in Disaster Prevention for Urban Resilience Against Global Earthquakes (2022)"

Figure 2.3.11 Earthquake Risk Distribution in Pakistan
(Earthquake Risk (INDEX) =PGA x Population Density x Ground Amplification Rate)

2.3.2.2 Update and Review Policy

(1) Collecting Information on Earthquake Risk Assessment

1) Building Vulnerability Assessment Methodology

According to the Earthquake Hazard Assessment Methodology in the MHVRA Policy Guideline, the assessment should be based on the scores shown in Table 2.3.2.

Table 2.3.2 Vulnerability Scores by Building Type

Table 23: Building Vulnerability Scoring for Floods and Earthquakes Based on Building Type

Sr.	Building Types	Vulnerability Scoring	
		Flood	Earthquake
aa.	Katcha	6.5	7
bb.	Semi-Pacca	5.0	6
cc.	Pacca	2.5	3

Source: MHVRA Policy Guideline

This score is a relative evaluation, and although a certain level of evaluation is given in a multi-hazard comprehensive evaluation, quantitative values such as Economic Loss cannot be calculated, making it difficult to estimate the actual damage. Kacha house is close to mud house (including unburned brick: Mud house), and Pacca is close to brick (concrete or burned brick).

2) Important Facilities in Cities

In the MHVRA report, it was confirmed that statistical information on general buildings and location data of critical facilities (Schools, hospitals, factories, commercial areas, etc.) were compiled as GIS data. Since these sources include organizations affiliated with the district

governments (For example, Health Department Punjab), efforts were made to obtain the data. However, as described in Chapter 1, coordination with NDMA proved challenging. Given the slow progress of district level MHVRA, and to maintain consistency with the national-level analysis, it was determined that direct use of this information was not necessary..

3) Other Information Collection from Existing Materials

1. The project team purchased population and building data (in excel format, etc.) from the Bureau of Statistics, and compiled it into GIS data at the Tehsil level.
2. The probabilistic seismic motion map published in Building Code 2021 is based on the results of the following research, which the project team obtained from researchers at NED University of Engineering and Technology.

Title: Probabilistic Seismic Hazard Analysis Based Zoning Map of Pakistan Pages 271-306 | Received 16 Jan 2019, Accepted 21 Oct 2019, Published online: 11 Nov 2019 Authors: Muhammad Ahmed, Sarosh Hashmat Lodi & Muhammad Masood Rafi, Department of Earthquake Engineering, NED University of Engineering and Technology, Karachi, Pakistan (E-mail: rafi-m@neduet.edu.pk)

(2) Confirmation of Necessity of Update and Review of Seismic Risk Assessment

As a result of the above verification, it has been concluded that updating and revising the seismic risk assessment for the future version of the NDMP. The specific recommendations are as follows.

1. **Update of Seismic Hazard Data (National level response has been carried out):** Building Code 2021 contains seismic hazard maps that have very different trends from those used in the current NDMP. In updating the current NDMP, it is necessary to acquire the latest probabilistic earthquake ground motion map data and carry out the seismic risk assessment accordingly.
2. **Review of the seismic hazard assessment method (measures have been implemented at the national level):** The current NDMP shows a policy to consider the ground amplification factor, but there is no result. Therefore, for the NDMP update, it is necessary to collect and model various datasets to evaluate the ground amplification characteristics such as geological maps and ground maps, to prepare a ground amplification factor map for the whole of Pakistan. And the project team can prepare an accurate seismic hazard map on the ground surface by multiplying the amplification factor with the seismic hazard (peak acceleration at the engineering bedrock, etc.).
3. **Review of the building vulnerability assessment method (recommendations for the future):** It is recommended to introduce a damage function to calculate the damage ratio of buildings based on ground motion strength (e.g. maximum acceleration, etc.) at the ground surface. As a result, it becomes possible to calculate the economic loses and quantitatively asses the effectiveness of disaster prevention and mitigation measures For the future NDMP update, it is necessary to systematically to accumulate and integrate the vulnerability data to support more accurate risk assessments and planning.
4. **Update of the earthquake risk assessment (recommendations for the future):** As mentioned above, in addition to reviewing the earthquake hazard and vulnerability assessments, the update of the earthquake risk assessment will be carried out by incorporating the latest census information. This result will enable Government of Pakistan to realize how much it can reduce its financial burden in the event of a disaster by making advance investment for disaster prevention in order to evaluate the economic loss.. This will be a first step towards popularizing the consistent approach from earthquake risk assessment to advance investment for disaster prevention.

(3) Policy to Implement Activity 1-4

For implementing Activity 1-4, the only existing data available on a national scale was population census data from the Census. While this data includes information on the number of households and population by housing construction type, it is not sufficient for the number of buildings, as it does not include non-residential buildings. In addition, damage functions for assessing building vulnerability have been identified, but developing an inventory of buildings corresponding to these functions remains to be solved.

Therefore, as a nationwide earthquake risk assessment, the policy is to show the seismic intensity exposure population at Tehsil level. The standard seismic intensity is MMI seismic intensity grade IX (9) or higher (PGA ≥ 300 cm/s2) at maximum ground acceleration (PGA) with a 50-year exceedance probability of 10% in probabilistic ground motions (according to NED University of Engineering and Technology) adopted in Building Code 2021.

2.3.3 Tsunami

2.3.3.1 Risk Calculation Method

Tsunami risk is not currently assessed in the current NDMP, and foundation data, such as exposure data, are not sufficiently available. Therefore, it is necessary to establish a framework for conducting a comprehensive tsunami risk assessment in the future update of the NDMP.

Based on tsunami inundation distribution data obtained by tsunami simulation conducted by PMD and State Disaster Management Agency (Provincial Disaster Management Authority: PDMA), inundation distribution map of the target city and region is prepared. The following items should be implemented for the tsunami risk assessment.

(1) Calculation of Damage Due to Inundation

Calculation of assets in the inundation mesh

1. Foundation Quantity: Population, number of households, number of employees in business establishments, number of agricultural and fishery households, total floor area, and field area
2. Asset Value: Residential properties, household items, depreciation and inventory assets of business establishments, depreciation and inventory assets of agricultural and fishery households, and crop assets
   =>Calculate the total asset value by multiplying the asset unit price
   =>Calculate damage based on the calculated asset value
   foundation Quantity data should be obtained as much as possible from the Bureau of Statistics, etc.

(2) Calculate Damage

Calculate the following damage based on the calculated asset value.

1. Direct damage (Damage to General Assets, Crop Damage, Agricultural Land Damage, Damage to Public Civil Engineering Facilities, etc.)
2. Indirect damage (Loss due to business suspension and emergency response costs for both home and business)

As described above, there are a wide range of damage types and their targets related to tsunami risk assessment, and it is estimated that there are complex interrelationships among different types of damage. However, these damages can be classified based on the following points. The risk analysis focusing on human damage and building damage is effective and appropriate as a starting point for the renewal.

- Direct damage (Direct damage such as human or building damage)
- Indirect damage (economic damage that cannot be seen such as business suspension)

- Primary damage (direct damage by tsunami)
- Secondary damage (damage caused by drifting debris)

- Assets that cannot be relocated when a tsunami strikes (e.g. houses, offices, infrastructure facilities, etc.)
- Assets that can be protected through evacuation actions (e.g. people and Transportation such as cars, and trains)

In calculating amount of damage, it is necessary to classify the damage ratio according to the depth of inundation. For example, the damage ratio may be set to 0.5 in areas with an inundation depth of less than 2 m and 1.0 in areas with an inundation depth of more than 2 m.

2.3.3.2 Policies For Updating and Reviewing

Since tsunami risk assessment has not been included in the current NDMP, it is necessary to carry out tsunami risk assessment in the future NDMP using basic exposure data such as population data and building data based on inundation distribution data. This will enable us to grasp the general picture of tsunami risk along the coast of Pakistan.

As a more practical method of utilizing tsunami hazard data, evacuation simulation using Tsunami Early Warning System and others implemented in PMD can be mentioned. It is possible to select model cities such as Gwadar and Karachi and carry out evacuation simulation based on arrival time and inundation distribution of the 1945 Makran tsunami, and consider the location of evacuation places. However, there are no known cases of evacuation simulations conducted in Pakistan to date. This should be considered as one of the key area for development in the future update of the NDMP.

2.3.3.3 Recommendations For Tsunami Risk Reassessment and Policy to Implement 1-4

(1) Recommendations for Tsunami Risk Reassessment

For tsunami risk assessment, it is necessary to collect foundation quantity data (exposure data) included in the national census such as population data and building data. Based on these basic quantitative data and inundation distribution data obtained through tsunami simulation by organizations such as PMD or PDMA Balochistan, the overall trend of tsunami risk distribution can be identified. In order to calculate the tsunami damage in detail, data on land use such as paddy field area and field area are required in addition to the data on the number of employees in business establishments and the number of agricultural and fishery households. If these data are available, it is possible to calculate the tsunami damage by multiplying them by the unit price of assets. However, it is considered that the calculation of Tsunami damage is more effective in advance disaster mitigation investment if the scope is limited at the Tehsil or municipal level. Therefore, it is important to assess Tsunami risk along coast of Pakistan using exposure data, such as population and building data, as these localized administrative units.

(2) Policy to Implement Activity 1-4

For implementing Activity 1-4, the only nationally available datasets was the population data obtained from the national census. Although it includes information on the number of households and population by housing type, it is not sufficient as the number of buildings other than houses is not included. Therefore, the policy is to indicate the exposed population at the Tehsil level for future risk assessment.

2.3.4 Other Disasters

For these disasters, which were not included in the current NDMP, information collection, research, and analysis are less developed compared to those for floods and earthquakes. Therefore, NDMP 2024 does not refer to mitigation and reconstruction stages (4RF). Instead, activities and project proposals are mainly focus on preparedness and response.

2.4 Activity 1-4

Perform a risk assessment of disaster risk information for updating and review which is confirmed in Activities 1-3.

2.4.1 Flood

As described above, the risk assessment at the formulation of the current NDMP is used. However, how to update the risk analysis in the future would be suggested to NDMA by the technical explanation of the risk analysis at the formulation of the current NDMP.

2.4.2 Earthquake

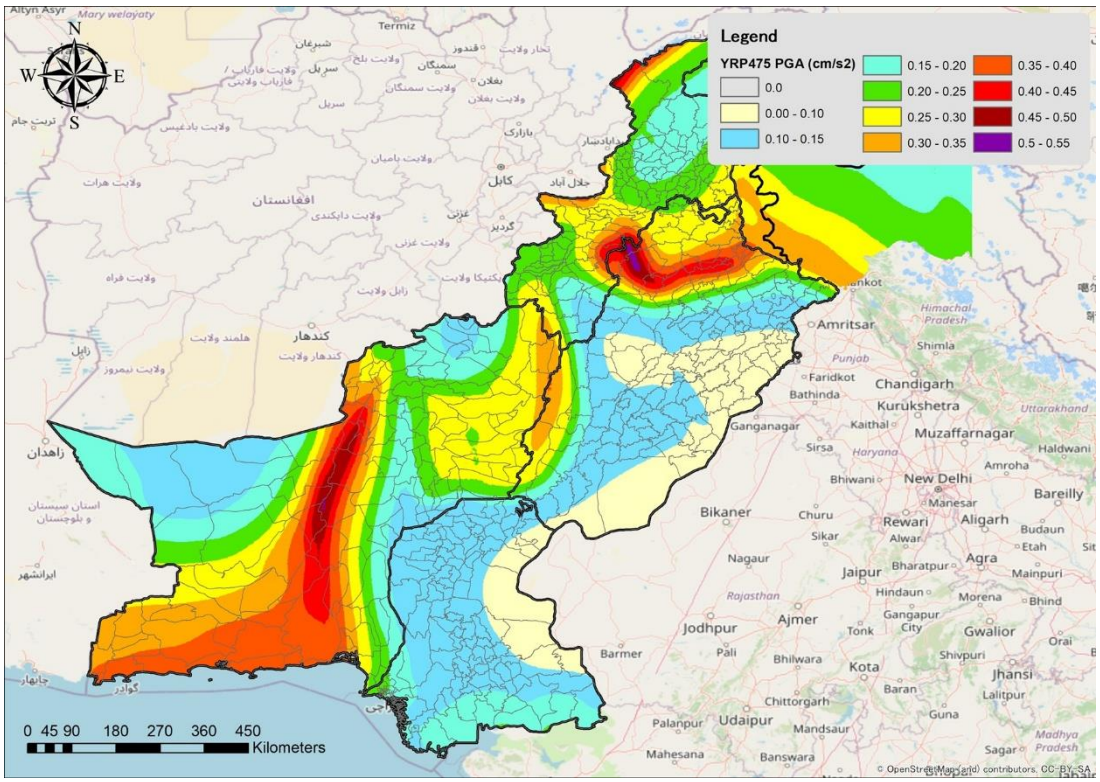
As a probabilistic hazard map of earthquakes, the project team investigated the earthquake hazard map published in the latest Building Code document. As a result, the project team obtained GIS data of the earthquake hazard map from NED University of Engineering and Technology. Among them, the project team adopted the maximum acceleration map, which is the most common index. (10% exceedance probability in 50 (475-year replication period), corresponding to an engineering basis of Vs30=760 (m/sec)). However, since the amplification factor of the surface ground was not considered, the project team estimated the amplification factor of the ground using Vs30 data (average S-wave velocity up to 30 m depth) of the U.S. Geological Survey (USGS) by applying an existing empirical method (Midorikawa et al., 1994)[7]. The population data of Census 2017 purchased in this project was converted into a GIS database at the Tehsil level. Therefore, the maximum acceleration and Vs30 data were also converted into average data at Tehsil level. Figure 2.4.1 shows the original maximum acceleration map (engineering basis), and through Figure 2.4.2 to Figure 2.4.4 show the maximum acceleration at Tehsil level, Vs30 maps related to ground amplification, and the maximum acceleration at the ground surface (MMI seismic scale is shown in the legend), which is the maximum acceleration multiplied by the ground amplification factor. Figure 2.4.5 shows a map (seismic intensity population exposure) in which Tehsil with MMI seismic intensity of 9 or higher is extracted from Figure 2.4.4 and colored by population. This map can be used for disaster prevention investment in the future for disaster mitigation measures, because the actual number of people affected by an earthquake is regarded as an earthquake risk and the volume can be grasped. For reference, Figure 2.4.6 shows a map of an index (using the common logarithm) based on population density multiplied by the maximum acceleration at the ground surface, which is an index in the current NDMP. Only urban areas with high population density tend to be given the highest rank, which is not a convenient index.

The geographical names at the Tehsil level shown in Figure 2.4.5, including provinces and districts, can be extracted from the GIS database organized in this project. Table 2.4.1 shows a summary of Output 1 in the earthquake field, and Table 2.4.2 shows a list of the exposed population at the district level.

Table 2.4.1 Summary of Activities Related to Output 1 in the earthquake field

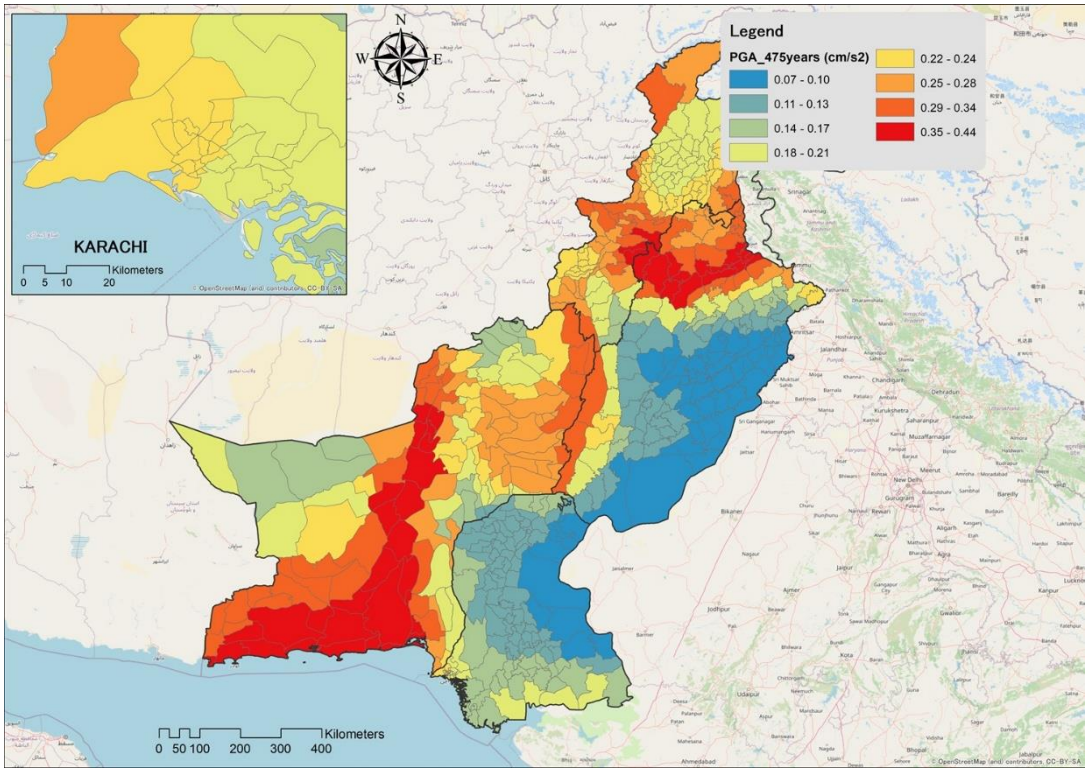
[Earthquake field] Activities related to Output 1	
1-1 Confirmation of hazard intensity and probability of occurrence	Original source data of a series of earthquake hazard maps published in Building Standards Act 2021 were obtained from NED University of Engineering and Technology.
1-2 Verification of vulnerability and exposure data	Confirmation of population in Census 2017 and population by housing construction method and database. Building data could not be confirmed at the national level.
1-3 Confirmation of the necessity of updating and reviewing disaster risk information	Since earthquake hazard maps are values based on engineering foundations, it is necessary to consider ground amplification. Not a relative risk assessment, but an exposure population that can be utilized for disaster mitigation is presented.
1-4 Risk assessment of disaster risk information that needs to be reviewed	A map of seismic intensity exposure population is prepared as a risk assessment as a national version (* In order to conduct the case study in the pilot area instead of the nationwide version, it is desirable to confirm the ground information by three-dimensional processing based on existing ground information.)

[7] Midorikawa, S., Matsuoka, M. and Sakugawa, K. (1994): Site Effectson on Strong Motion Records Observed during the 1987 Chiba-ken-toho-oki, Japan Earthquake, Proc. Ninth Japan Earthquake Engineering Symposium, 3, 85-90.



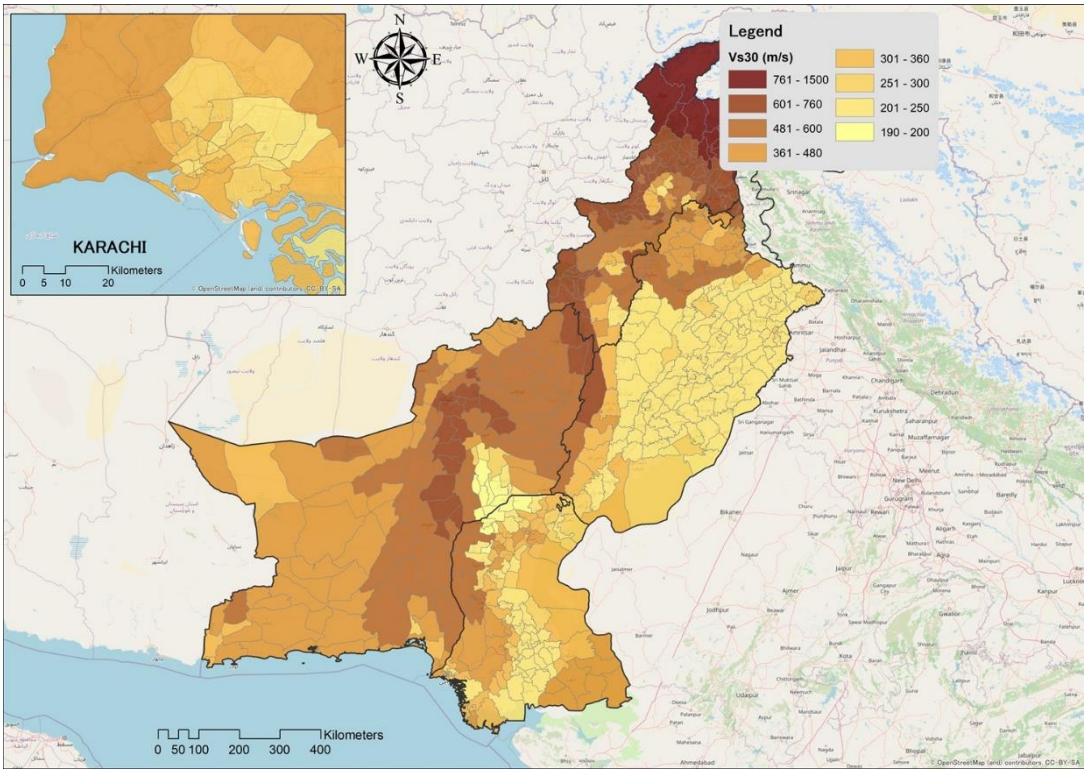
Source: NED University of Engineering and Technology

Figure 2.4.1 Peak Acceleration Map on Engineering Bedrock



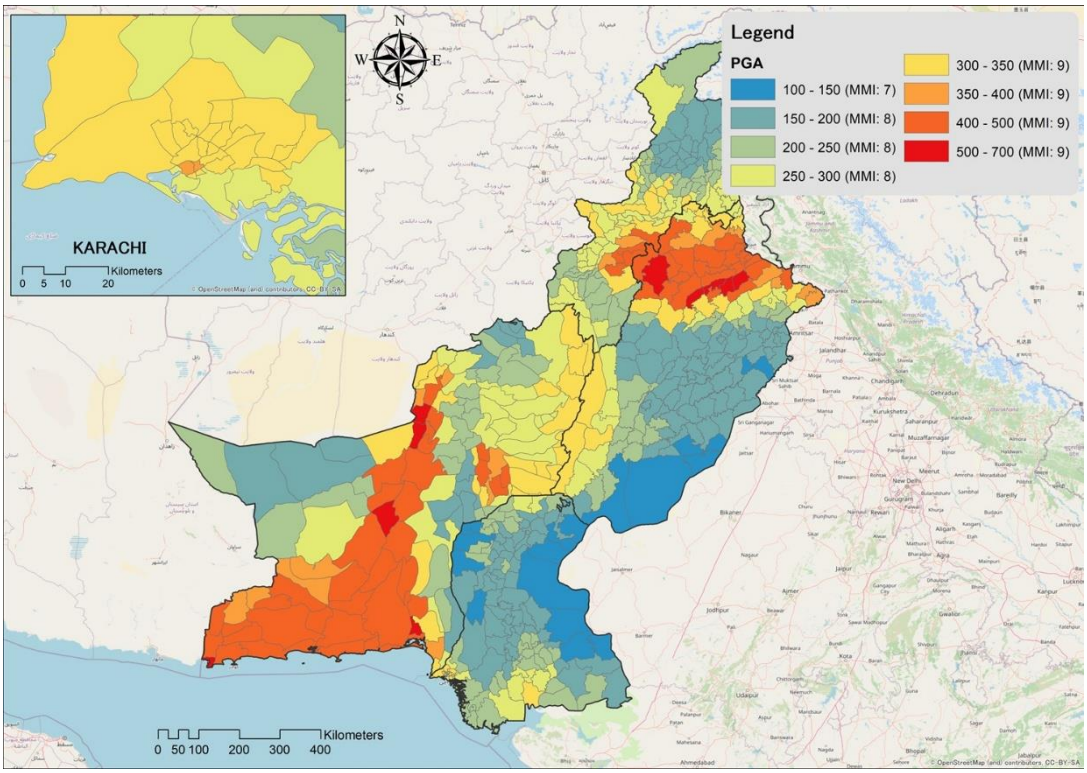
Source: Project Team

Figure 2.4.2 Peak Acceleration Map at Tehsil Level



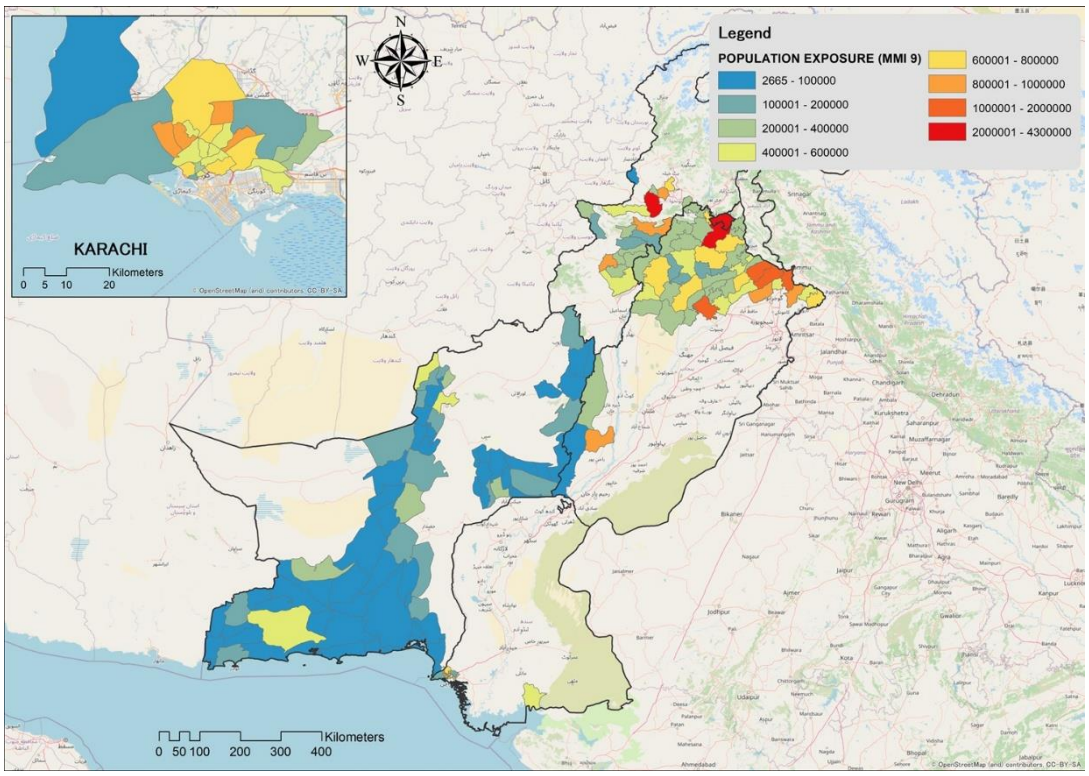
Source: USGS, Project Team Processing

Figure 2.4.3 Vs30 Map for Ground Amplification at Tehsil level



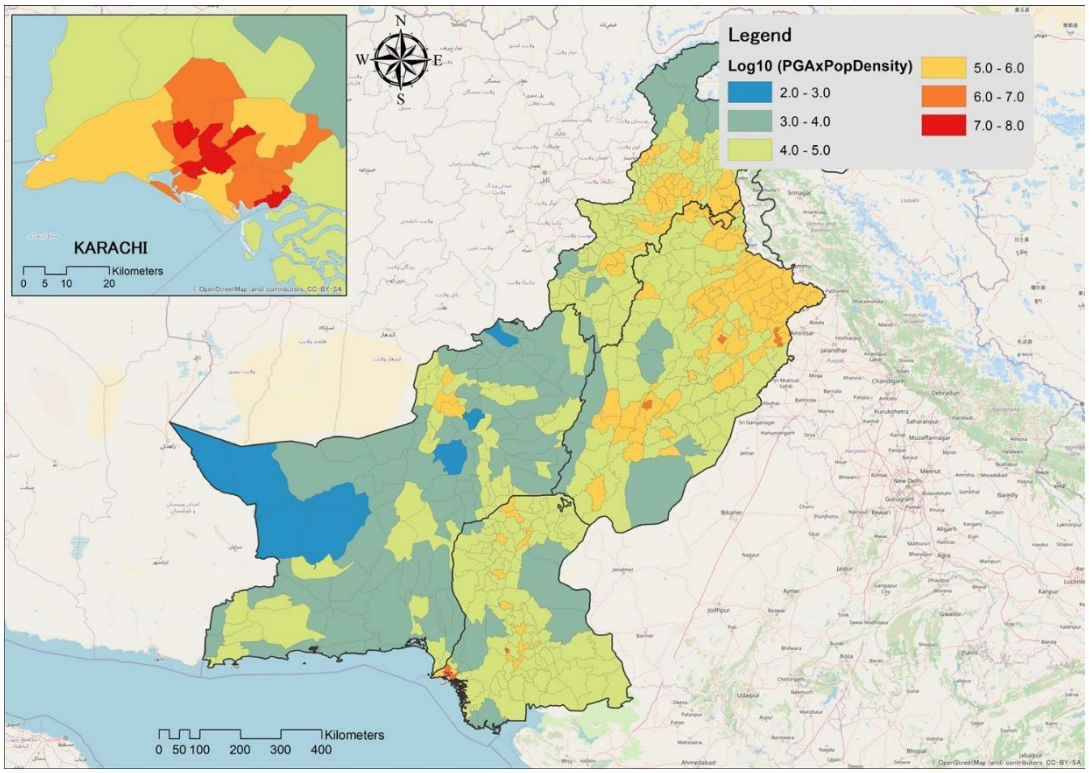
Source: Project Team

Figure 2.4.4 Peak Ground Acceleration (PGA) Map at Tehsil Level (MMI Intensity Scale Shown in Legend)



Source: Project Team

Figure 2.4.5 Population Exposure to Seismic Intensity Map at Tehsil Level (MMI Intensity ≥ 9)



Source: Project Team

Figure 2.4.6 Population Density ×PGA Index Map at Tehsil Level

Table 2.4.2 List of Districts with High Earthquake Risk
(The intensity exposure population is calculated from the Tehsil level in Figure 2.4.5.)

Province/District	**Seismic Intensity Exposure Population**	**Province/District**	**Seismic Intensity Exposure Population**
BALOCHISTAN	**6585528**	**KPK**	**10395950**
Awaran	121821	Bannu	1167071
Barkhan	171025	Charsadda	1184795
Dera Bugti	272481	Hangu	270399
Gwadar	262253	Karak	705362
Jhal Magsi	9441	Kohat	992427
Kachhi	131013	Lakki Marwat	875744
Kalat	395629	Mardan	627172
Kech	907182	Peshawar	4267198
Kharan	130096	Swabi	305782
Khuzdar	268600	**PUNJAB**	**26176087**
Killa Abdullah	696613	Attock	1451175
Lasbela	312854	Bhakkar	346591
Loralai	40867	Chakwal	1495463
Mastung	200911	Dera ghazi khan	212652
Musakhel	167243	Gujranwala	830272
Nasirabad	487847	Gujrat	2756289
Nushki	178947	Jhelum	1222403
Panjgur	315353	Khushab	1280372
Pishin	496014	Mandi Bahauddin	1594039
Quetta	546727	Mianwali	1542601
Sherani	152952	Narowal	1111036
Sibi	44490	Rajanpur	883580
Sohbatpur	200426	Rawalpindi	5050068
Washuk	74743	Sargodha	3351725
FATA（Now included in KPK）	**880968**	Sialkot	3047821
Khyber	443449	**SINDH**	**13412432**
Kurram	386365	Badin	454576
Mohmand	51154	Central Karachi	2971382
ISLAMABAD	**2003368**	East Karachi	2875315
Islamabad	2003368	Korangi	1451415
		Malir	526469
		South Karachi	1627325
		West Karachi	3505950

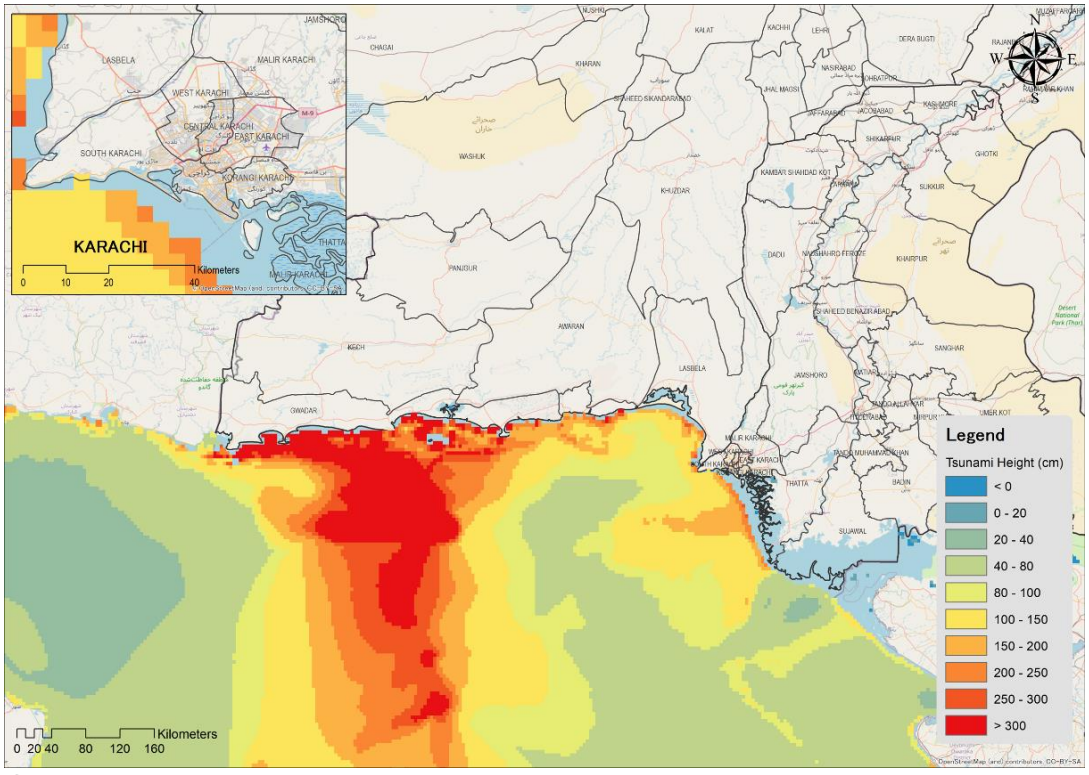
2.4.3 Tsunami

The PMD headquarters in Islamabad has experience conducting tsunami simulations using ComMIT, a scenario-based tsunami modeling tool. Regarding probabilistic evaluation, it is hoped that the PMD will exchange technical opinions with the PMD and eventually contribute to technical cooperation with the NDMA, which is the counterpart. Figure 2.4.7 shows the results of PMD simulation using the source model assuming the 1945 tsunami disaster. Figure 2.4.8 shows the Tehsil and exposed population affected by the simulation results.

Table 2.4.3 shows the summary of Output 1 in the tsunami field, and Table 2.4.4 lists the exposed population affected by tsunami risk for each district.

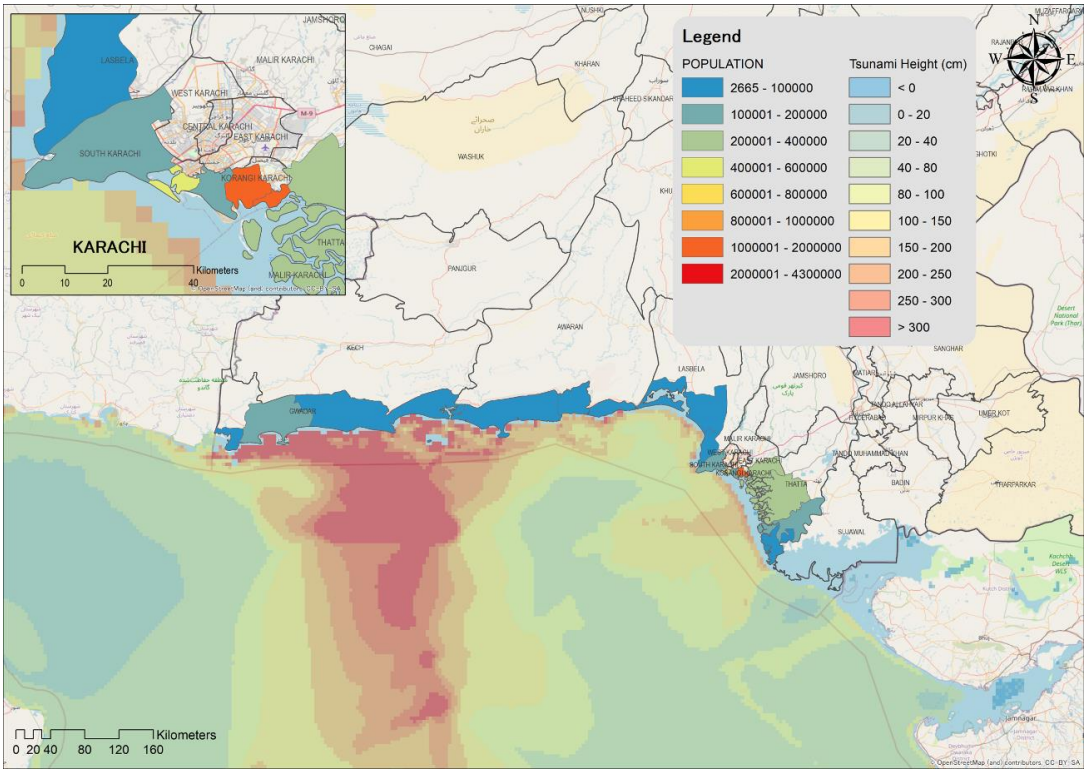
Table 2.4.3 Summary of Output 1 Activities in the Tsunami Field

【 Tsunami Field 】 Outcome 1 Activities	
1-1 Confirmation of hazard intensity and probability of occurrence	Currently, probabilistic simulations have not been performed. Therefore, tsunami hazard assessment map data (GeoTIF format, available in GIS) based on scenario theory were obtained from PMD.
1-2 Confirmation of vulnerability and exposure data	Confirmation of population of census 2017 and population by housing construction was made into a GIS database and utilized as the exposed population. Building data could not be confirmed at the national level.
1-3 Confirmation of the necessity of updating and reviewing disaster risk information	First, it is necessary to carry out probabilistic tsunami inundation simulation. Therefore, it is necessary to establish a committee of domestic experts to examine the geological evaluation of earthquake faults that cause tsunamis as information from foundation.
1-4 Risk assessment of disaster risk information that needs to be reviewed	A map of the population exposed to tsunami inundation at Tehsil level was prepared as a risk assessment. (* It is necessary to extract inundated areas in more detail than Tehsil in coastal areas, and to grasp the groups of buildings and population affected by inundation in detail.)



Source: PMD

Figure 2.4.7 Tsunami Hazard Map Based on Scenario Theory



Source: Project team

Figure 2.4.8 Tsunami Exposure Population Map at Tehsil Level

Table 2.4.4 List of Districts with High Tsunami Risk (The exposed population is calculated by aggregating the Tehsil level values in Figure 2.4.8.)

Province/District	Name of Tehsil	Exposed population
BALOCHISTAN		**351709**
Gwadar	Gwadar, Jiwani, Ormara, Pasni	251367
Lasbela	Gaddani, Liari, Sonmiani	100342
SINDH		**3741038**
Korangi	Korangi	1126141
Malir	Bin Qasim, Ibrahim Hydri	1126141
South Karachi	Saddar	1303878
Thatta	Ghorabari, Keti Bunder, Mirpur Sakro	248098
West Karachi	Harbour, Mauripur	1055780

2.4.4 Other Disasters

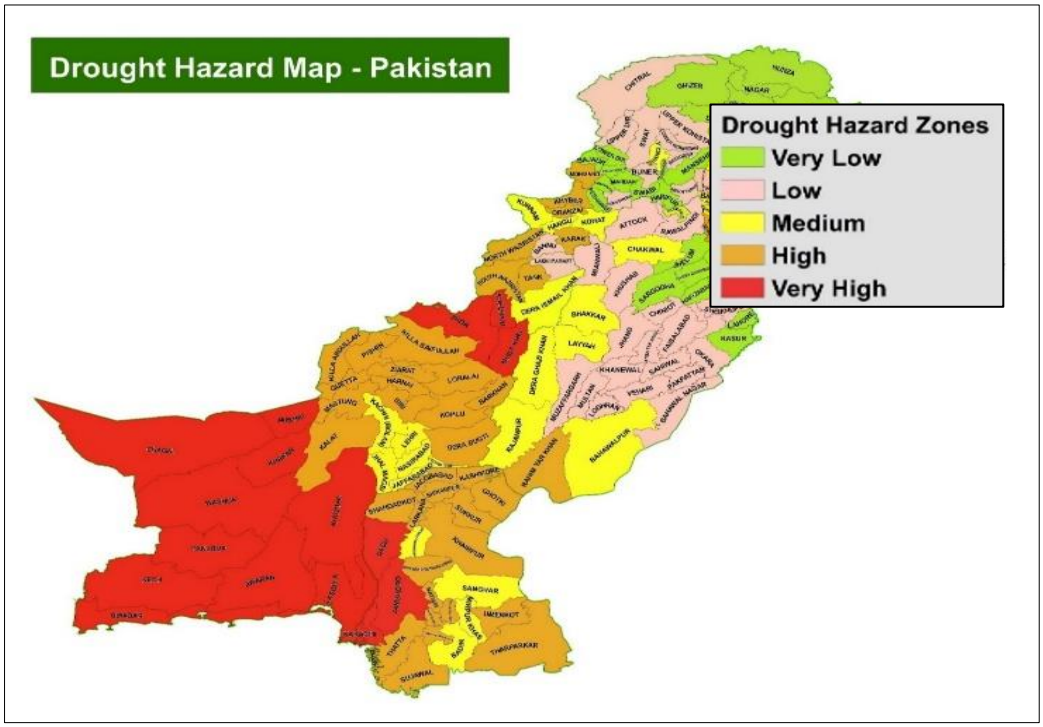
Regarding other disasters, and in light of the reasons mentioned in Activity 1-3, the project team proposed projects and actions mainly focused on Response and 4RF framework. While detailed risk analysis are not performed, for these hazards, available damage information is summarized below for reference.

2.4.4.1 Drought

In recent years, severe drought damage has been reported in Balochistan, Sindh, and Punjab (south part) provinces where the average annual rainfall is only 200 ~ 250 mm. Severe drought damage occurred in

Punjab in 1899, 1920 and 1935, KPK in 1902 and 1951, and Sindh in 1871, 1881, 1899, 1931, 1947 and 1999. Nationally, the most severe drought occurred from 1998 to 2001.

Figure 2.4.9 shows the potential of drought hazard by district. As shown in this figure, the potential of drought hazard is higher in the southern regions with climate categories close to semi-arid areas. According to the Indus River System Authority (IRSA), the distribution of irrigation water from the Indus River (water rights during normal and drought conditions) is stipulated in the Act. However, the distribution of water discharged from the Tarbella Dam is not properly controlled in some cases, and Sindh Province (located in the downstream) tends to be drained after water is withdrawn from Punjab Province. In particular, it is necessary to implement measures to appropriately grasp the amount of water discharged from the Tarbella Dam and water rights during drought conditions.



Source: NDMA2024

Figure 2.4.9 Potential of Drought Damage by District

2.4.4.2 Heat Waves

Karachi experienced severe heat waves in 2014 and 2015. In June 2015, Karachi city recorded more than 1,200 deaths, 65,000 people affected, and 1,200 cases of fever. During this period, temperatures of 40 ~ 49^0C were recorded. It is possible that heat waves are becoming more common due to global warming. Karachi being the most populous city, may be at higher risk due to its industrial activities. Some studies indicate that factors such as quality and type of housing, population size, low-income level, and lack of awareness could have increased the vulnerability particularly during the June 2015 heat wave. Therefore, the residents of Karachi need to provide many shady and shared spaces through greening and optimal urban design. They also need timely access to water and fans for the most vulnerable people such as the elderly, the sick, and children, and education and training on proper measures against heat stroke.

2.4.4.3 Smog

Smog is mainly caused by vehicle and industrial emissions, excessive waste production and garbage incineration, dust from the surrounding desert, excessive fuel consumption due to population growth and cold winters, fires on agricultural land, and smoke from brick kilns around Lahore. The problem has been found to worsen in recent years between December and January. The air becomes muddy and difficult to breathe, causing severe disruption to human activities, and can cause highway and traffic accidents, airplane

disruptions, and health problems, especially in the mornings when people are forced to stay home. In addition, excessive pollution makes it difficult to breathe even indoors, putting housewives and children at great risk for complications of the eyes, skin, lungs, ears, nose and throat.

Lahore and Peshawar in particular were ranked among the top 10 most polluted cities in the world in 2023. Pollution in these cities stems from a combination of factors, including burning of crop residues on agricultural land, industrial emissions, increased traffic, adverse weather conditions that trap pollutants, dust from construction activities, and traditional brick kilns using inefficient techniques. These factors release large amounts of particulate matter and pollutants into the atmosphere, causing a persistent problem of smog in cities.

Therefore, it is important to assess the potential hazards of smog, identify the vulnerable victims, and establish specific actions to mitigate the negative effects and damage of smog, especially in the districts listed in Table 2.4.5.

Table 2.4.5 Areas at High Smog Risk

Province	Districts
Punjab	Lahore, Faisalabad, Sheikhupura, Nankana Sahib, Kasur, Gujrat, Narowal, Sialkot are vulnerable to smog hazards during winter season.

Source: Project Team

2.4.4.4 Locust Infestation

In Pakistan, over the past 100 years, the infestation of the desert locust occurred in 1926, 1952, 1962 and 1992. These swarms invaded all crops and pasture vegetation, and it combined with 2 to 3 years of drought and a long-term decrease in water, caused relatively heavy damage. In recent years, from 2019 to 2020, severe attacks caused heavy damage in 54 districts of the country (31 districts were affected in Balochistan province, 8 in KPK province, 10 in Punjab province and 5 in Sindh province). In Pakistan, the breeding areas of the desert locust account for 38% of the country (60% of Balochistan, 25% of Sindh and 15% of Punjab). If the breeding areas of the desert locust are not controlled, the whole country is threatened by the infestation caused by the locust.

Table 2.4.6 lists the areas with high-probability of occurrence of locust infestation. Even if the areas where the risk occurs are whole farmland, it is necessary to analyze and organize the areas (hazards) that need breeding control in the future. However, it is necessary to consider and study the balance between the level of control of the locust and the ecosystem, and to take careful countermeasures.

Table 2.4.6 Locust Potential Areas

Province	Districts
Balochistan	Chaghai, Kharan, Panjgur, Khuzadar, Awaran, Pishin, Barkan, Harani, Kohlu, Washuk, Gwadar, Ketch and Lasbela
Sindh	Badin, Sukkur, Khairpur, Shaheed Benazir Abad, Sanghar, Tharparker, and Ghotki
Punjab	Rahim Yar Khan, Bawhawalpur, Bahwalnagar, Bhakkar, Khushab, Rajinpur, and Muzaffargarh
Khyber Pakhtunkhwa	Dera Ismail Khan, Bannur, Lakki, Orakzai, Kurram and Tank

Source: Project Team

2.4.4.5 Fires

In the past 20 years, there has been a massive migration from rural to urban areas, which has led to the expansion of slum areas in Pakistan's urban areas. Furthermore, many buildings have been constructed without complying with building codes. In addition, there has been a rise in unlicensed LPG gas distributors and the sale of petroleum products in residential areas is becoming widespread in cities. These practices pose a major fire risk in urban areas. Fire risk exists in all houses, but cities in Pakistan, with high concentration of industrial facilities, CNG stations, LPG gas distribution shops, and warehouses tend to have a relatively high fire risk. In addition, Pakistan has various types of forests, ranging from mangroves on the coast to alpine vegetation in the north. Among these, subtropical broad-leaved evergreen shrublands and subtropical pine forests (chill pines) pose the highest fire risk.

Government of Pakistan needs to accumulate disaster prevention information on natural and urban disasters, conduct risk analysis, and propose and implement specific countermeasures from the perspective of both economic development and nature conservation.

Government of Pakistan needs to accumulate disaster prevention information on natural and urban disasters, conduct risk analysis, and propose and implement specific countermeasures from the branches of Economic Development and Nature Conservation.

2.4.4.6 Infectious Diseases and Epidemics (Especially Epidemics Like Dengue)

The first dengue outbreak was recorded in 1994, and since then, the disease has periodically emerged, especially at the beginning and end of summer, affecting various parts of the country, such as Lahore and Rawalpindi. In 2010, there was a dengue outbreak in Pakistan, with 16,580 confirmed cases and 257 deaths reported in Lahore, and about 5,000 cases and 60 deaths reported in other areas. The 3 provinces (KPK, Punjab and Sindh) were suffering from dengue, and more than 19,000 cases and 30 deaths were reported in 2019. According to National Institute of Health in Pakistan, as of 2019, 147,200 cases and more than 800 deaths were reported from 1995 to 2019.

Regarding communicable diseases such as dengue and COVID, it is necessary to carry out a risk analysis based on information from the National Institute of Health and other relevant institutions, and to clearly designate the institutions in charge, and to analyze and build preparedness and response methods at the national and provincial levels.

2.4.4.7 Industrial and Technical Disasters

These include oil leaks, chemical substance spills, biological disasters, radiation leaks, nuclear power destruction, water quality accidents, and other man-made disasters and accidents caused by industrial expansion. In particular, industrial cities such as Karachi, Hyderabad, Lahore, Faisalabad, Gujrat, Gujranwala, and Sialkot are vulnerable to industrial hazards. The risks of human disasters caused by the chemical industry and marine accidents at ports are considered particularly dangerous. In Karachi in July 2003, the wreck of the Tsunami Spirit, a former Greek ship, caused catastrophic environmental and health damage to businesses, port workers, and surrounding communities. About 28,000 tons of oil spilled throughout the port area, which adversely affected marine life and the surrounding ecosystem. In the weeks following the incident, local residents reported headaches, nausea, and respiratory problems, and it took several months for the authorities to clean up the oil-contaminated area.

In Pakistan, it is necessary to clearly define the organization in charge of risk analysis and countermeasure planning for man-made disasters associated with industrial development, and to prepare and respond accordingly based on the results of the risk analysis.

2.4.4.8 Traffic Accidents

According to the JICA report in 2006, the following information is available. No new information was available for this project.

(1) Road Traffic

Pakistan experience a high rate of accidents with the resulting economic losses estimated to amount to approximately 2% of the country gross domestic product (GDP). For this reason, national road accident countermeasures are necessary, but there is a lack of reliable accident statistics, making it difficult to take strategic action. During 1986 to 2006, total number of vehicle owned has nearly doubled. According to statistical data, the annual death toll from road accidents is estimated to be about 5,000, while the actual number is estimated to be 7,000 ~ 10,000. In particular, many accidents are caused by buses and overloaded trucks.

Although improvements in road facilities and road safety measures have been implemented, further measures are needed. Specifically, it is required to train drivers and establish a licensing system, set up specifications and safety structures for commercial vehicles, strengthen the vehicle inspection system, make seat belts and helmets mandatory, and operate highly trained traffic police and conduct road safety campaigns. Practical application of traffic laws is expected to improve the behavior of road users in a relatively short period of time. Furthermore, various organizations and their training are needed to ensure road safety. In other words, effective cooperation is required among the federal government, and Provincial government, the National Highway Authority (NHA), police, industry, transportation companies, and drivers.

(2) Rail Transportation

Traffic accidents are on the rise due to the increasing traffic volume. Overloading not only damages roads but also contributes to accidents. During 2001 to 2006, railways also experienced 3 major accidents, and improvement of the signaling system is necessary. There were 17 railway projects proposed by JICA, and the cost estimate was 146 billion rupees, but most of them were not implemented by 2006. In particular, although there were major accidents in 1990 and 1992, improvement of the signaling system was not addressed at all. It is necessary to reorganize information management and invest in preparedness measures to enhance disaster resilience and response capabilities.

(3) Aviation

Regarding air traffic control to ensure aviation safety, the JICA report states that in Pakistan, airway-monitoring radars are installed at 6 locations to monitor flying airplane at various altitudes throughout the country. However, because northern Pakistan is a mountainous region, radar signals are blocked, and airplane at low altitudes such as propeller planes can hardly be seen. Furthermore, communication is not possible. In order to ensure that airplane in these mountainous regions can fly safely, it is recommended that ADS and data links be established.

In December 2016, an airplane crashed into a mountain in KPK province, resulting in death of about 50 people. Similarly in June 2020, another airplane crashed into a residential area near Karachi, killing about 100 people. In the future, it is necessary to prevent such accidents by examining investigation reports of international organizations, etc. and investing in advance countermeasures.

CHAPTER 3 ACTIVITIES RELATED TO OUTCOME 2

Outcome 2	Progress and issues of the current NDMP will be grasped, and the ability to formulate a more effective NDMP will be improved.

3.1 Activity 2-1

Based on the results of the risk assessment in Activity 1-4, the basic strategy for Disaster Risk Reduction (DRR) aimed at by the current NDMP and its effectiveness will be reviewed.

3.1.1 The Results of the Risk Assessment and the Basic Strategy for DRR According to the Disaster Type

The results of the risk assessment, summarized in Activity 1, form the basis for the following disaster specific strategies.

(1) Flood

The flood is the disaster type which requires the largest amount of advance investment. So, risk analysis is very important to understand the priority of the investment. However, information that serves as an indicator of the accuracy of risk analysis is insufficient, especially the accumulation and organization of disaster-related history. Also, the national-level flood area analysis focuses only on the river channel and does not include hazard analysis in the floodplain. Furthermore, there are challenges in implementing flood control plans and integrated flood management for each project in the NFPP-IV. For example, NFPP-IV is not adequately summarized from the viewpoint of integrated flood management.

In the implementation of flood control measures, it is necessary to recognize the importance of advance investment through the Flood Protection Sector Project-III (FPSP-III). At the same time, appropriate risk analyses must be carried out and the impacts of climate change must be taken into consideration when planning and designing. In addition, various activities should be carried out to formulate NFPP-V and local medium- to long-term plans so that integrated flood management can be realized through national and local cooperation and coordination. Furthermore, existing plans such as NFPP-IV should be reviewed and implemented to the greatest extent possible, and guidelines, legal systems, and organizational strengthening should be carried out to contribute to their implementation.

(2) Earthquakes and Tsunamis

Observations, research, and surveys have been carried out by universities and PMDs, and the relevant organizations have been prepared to use the latest risk analysis methods. Therefore, it is necessary to improve the accuracy of risk analysis by updating the observation and analysis methods, and at the same time to propose activities to reduce the damage by building EWS, improving the legal system, and developing various guidelines and technical guidelines based on the results of the risk analysis. As for tsunami, in the very long term, measures for structures in coastal areas are also possible. However, NDMP 2024 does not recommend large-scale structural measures against tsunamis, taking into consideration of the financial situation and the environmental conditions. These include nature-based solutions, such as planting mangroves in shallow coastal areas like Karachi, combined with the development and integration of effective tsunami warning systems to reduce potential damage.

(3) Other Disasters

The current NDMP lacks an introduction of the types of disasters that should be recognized as Multi-Hazards, so the progress of activities of related organizations against other disasters is still low. Therefore, it is necessary to incorporate activities in the direction of investigation, research, and human resource development for each element of risk as first in NDMP2024. In addition, since all the organizations visited in the project attach importance (or universalize) to human resource development and employment, it is not necessary to recommend human resource development as in

the current NDMP. The development of EWS for various types of disasters is actively being promoted, and the NDMA attaches importance to the construction of Multi-EWS that contributes to response activities. Therefore, EWS will be systematically organized and proposed for all types of major disasters.

3.2 Activity 2-2

The status of utilization and achievement of priority projects and human resource development plans of the current NDMP will be reviewed by relevant ministries and agencies, and issues will be identified.

3.2.1 Achievement of Comprehensive Issues

3.2.1.1 Status of Utilization and Achievement

The current NDMP was formulated in 2012, which was 2 years after the NDMAct was formulated by the Government of Pakistan in 2010, and NDMA was established in 2007. The current NDMP sets the roles of almost all relevant organizations for the disasters described here, and it sets various general activities to cover these organizations. However, no specific projects or activities have been proposed for all the disasters identified in this update. On the other hand, the 18th Constitutional Amendment, enacted in the same year of the current NDMP.

The progress of the current NDMP remains limited. It is difficult to quantitatively evaluate the progress of the current NDMP due to the extensive use of wording such as Multi-Hazard. However, even the PMD, whose role is directly related to disaster prevention activities, has achieved about 60% of its priority actions in Intervention 4 Establish a multi-hazard early warning system," which is the place of its main role. In addition, the NFPP-IV, mainly managed by the FFC, is expected to be promoted by the FPSP-III which was approved by the MoPS-PC in June 2023. However, as of the end of 2023, investment in the implementation of the project has just started. In other words, on the whole, the achievement rate of the current NDMA is quite low.

3.2.1.2 Factors and Challenges for Low Progress

The main reasons for the slow progress of the current NDMP include (1) insufficient project monitoring and coordination caused by the impacts of COVID-19 and other factors, and (2) weakening of the coordination capacity of related organizations such as local organizations due to the promotion of decentralization, according to the letter written by the current NDMA Chair to the EAD at the end of 2023 to change the policy of this project and according to the persons in charge of this project NDMA. For the factor (2), The NDMAct declares the exercise of influence on DMA related organizations that mainly implement preparedness and response, but it does not specifically mention other major disaster prevention related organizations, which is considered to be one of the factors for the slow progress of Mitigation and Rehabilitation. Regarding Mitigation and Rehabilitation, focusing on flood countermeasures, which have the largest inputs, budgetary measures were not taken for more than 5 years after the formulation of the 10 years plan, NFPP-IV (2018). However, after the flood damage in 2022, budgetary measures by domestic and foreign capital are finally being promoted. This delay may be attributed to the lack of awareness of advance investment in disaster risk reduction by the budget allocation organizations. According to the FFC, the budget allocation organizations have not maintained the recognition that "disaster risk reduction is a development issue" as stated in the Sendai Framework for Disaster Risk Reduction.

The current NDMP activities, formulated in 2012, follow the policies of the Hyogo Framework for Action (2005-2015). The Sendai Framework for Disaster Risk Reduction 2015-2030, formulated at the United Nations World Conference on Disaster Risk Reduction (2015), has new priorities. Especially, it is necessary to address "promoting pre-investment in disaster risk reduction," "strengthening preparations for effective emergency response," and "building back better."

3.2.1.3 Implementation System and Issues for Disaster Risk Reduction Projects

After the formulation of the current NDMP, Government of Pakistan has promoted the formulation of disaster management plans at the province and district levels based on the NDMAct. Originally, it was assumed that the NDMP had a policy for implementing disaster management activities and projects, and that local disaster risk reduction plans were formulated aligned with the policy. However, the PDMA seems to be independently undertaking policy formulation and investment for disaster preparedness and response (preparedness and response) including initiatives not currently addressed by the existing NDMP.

On the other hand, the Disaster Management Authorities (DMAs) does not have the ability to directly implement plan formulation and design, so disaster risk reduction related organizations such as FFC, PID, PMD, WAPDA, WASA and C&W are responsible for disaster risk reduction. As a result, disaster risk reduction is primarily managed by Disaster Management Authorities (DMAs) for preparedness and emergency response, while mitigation efforts are handled by other relevant agencies, with responsibilities distributed based on the scale and nature of the disaster risk.

From the above, although it is clear that the activities of the disaster management cycle need cooperation with other related organizations, the coordination and monitoring system by NDMA was not sufficient. It is necessary to construct a legal system and organizational system to compensate for the system deficiencies for the implementation of the disaster prevention cycle activities.

3.2.2 Flood (Utilization Status, Achievement Status, Review Results, and Issues)

In the flood management sector, FFC, PMD and WAPDA are the main responsible organizations. Interviews and consultations were conducted with these key organizations.

3.2.2.1 Status of Utilization

(1) Status of Utilization In FFC

The Fourth National Flood Protection Plan (NFPP-IV) was approved on May 2, 2017 by the Council of Common Interests (CCI), a highest inter-state coordinating body under the Prime Minister. The NFPP-IV was prepared after the issuance of the current NDMP, and although the text of the NFPP-IV does not specifically describe the relationship with the current NDMP, the 5 schemes shown in Table 3.2.1 are proposed as non-structural measures in relation to the NDMA. It is also mentioned that the 2 schemes shown in Table 3.2.1 are based on Intervention 4 and 10 of the NDMP.

Table 3.2.1 Schemes Related to NDMA Included in PC-1 of NFPP-IV

Sr. No.	Name of Scheme/Problem Area	Total Investment (Million Rupees)	Remarks
1	Enhance the Capacity of Government Organizations Responsible for Disaster Management.	10,100.00	Priority includes NIDM, National Disaster Response Force, Capacity Building of Disaster Management Agencies
2	Develop Capacity of Stakeholders in Post Disaster Recovery. Micro Level Multi-Hazard Vulnerability and Risk Assessment of Flood Prone Districts in the Country.	2,620.00	Align with Intervention 10 of NDMP around 125 Districts across the country
3	Strengthen Early Warning Dissemination System and Establishment of Linkages between NDMA, PMD, PDMAs and DDMAs.	1,900.00	Aligned with Intervention 4 of NDMP
4	Enhance Disaster Risk Management Capacity in Urban Areas and Establish a National Emergency Response System.	1,100.00	Capacity Building and Enhancement of USAR Teams & Enhancement of Emergency Response System
5	Prepare Disaster Management Plans at Various Levels.	2,600.00	

Source: Project Team excerpt from ANNEX-4 of Final Report, Development of National Flood Protection Plan-IV (NFPP-IV) and Related Studies to Enhance Capacity Building of Federal Flood Commission-FFC

3.2.2.2 Status of Achievement

(1) Status of Achievement In FFC

1) Status of the Current NDMP

The Project Team reviewed the implementation of the status of current NDMP with the FFC, and the key findings are summarized below.

- Regarding Intervention 3, the Flood Risk Mapping along major rivers (in the area surrounded by the second line bunds) was prepared in the study for the formulation of the NFPP-IV. The analysis of vulnerable areas for flash flood and GLOF and the preparation of hazard maps were included in the program to be implemented in line with the NFPP-IV, but they have not been implemented at the time of the interview survey.
- Regarding Intervention 4, as in Intervention 3, the preparation of hazard maps and capacity enhancement in vulnerable areas for flash flood have not been implemented. The development of landslide hazard maps is planned as part of projects under the FPSP-III.
- Regarding Intervention 7, the NFPP-IV was formed and approved by the MoPS-CCI as a comprehensive flood management plan, but it has not been implemented because the budget was not allocated. On the other hand, the rehabilitation project of the existing barrages by PID and the construction of new dams by WAPDA are underway, and the FFC providing review, guidance, and approving these projects.
- Seminars and workshops aimed at strengthening the capacity of the FFC have been held by various donors and other organizations. In the Technical Advisor on Flood Management, which was implemented until September 2023, and The Project for Capacity Development of Effective River Dikes Management Response to 2022 Flood, which is currently being implemented, seminars and workshops have been held for FFC staff and have contributed to strengthening the capacity of FFCs.

Source: Pakistan Flood Management Advisor Program

Photo 3.2.1 Workshop Conducted by Technical Advisor on Flood Management (2022.11)

2) Progress of NFPP-IV

One of the activities specified in Intervention 7 of the current NDMP is the formulation of NFPP-IV and implementation of various projects based on it.

The NFPP-IV was approved by the MoPC-CCI, the highest federal interstate coordinating body, on May 2, 2017 (an agreement was reached to allocate a total of Rs. 332.2 billion in a 5: 5 ratio between the federal and state budgets). However, as of December 2022, no government budget has been allocated and no concrete projects have been started.

In 2019, the FFC submitted a budget application to the MoPS-Planning Commission (PC) for PKR 332bil. for the entire NFPP-IV project, but it was not accepted. According to FFC officials, the reason for this was not only that many public investment projects in other sectors had already been allocated, but also because such projects are perceived as not delivering direct or immediate economic benefits.."

Therefore, the FFC selected the priority projects from the NFPP-IV projects and reorganized them as FPSP-III (PKR95bil), and applied for implementation funds through EAD to several donors including Asian Development Bank and JICA. However, as of December 2022, the FFC had not secured a budget. A consultancy team dispatched by ADB from November to March 2022 formed the FPSP-III business. The final result was approximately 194.6 bil. PKR, which was approved by MoPS-PC in June 2023.

(2) Progress in PMD

The progress of the current NDMP was reviewed during a meeting with PMD-FFD, which oversees the PMD Planning Division and flood-related components. The results were as follows. Only flood-related information is shown here and other meteorological disasters are summarized in 3.2.3 .

- The establishment of the Specialized Medium Range Weather Forecasting Center has been completed, the establishment of the Regional Flood Warning Centre and the establishment of the Urban/Flash Flood Guidance System in major cities are planned.
- The analysis of flood-vulnerable areas and the development of hazard maps have not yet been undertaken.
- Although the GLOF team is conducting investigation and research funded by UNDP in Islamabad, no specific projects related to EWS have been implemented. there is a lack of collaboration and information sharing with WAPDA, which is also involved in addressing GLOFs.

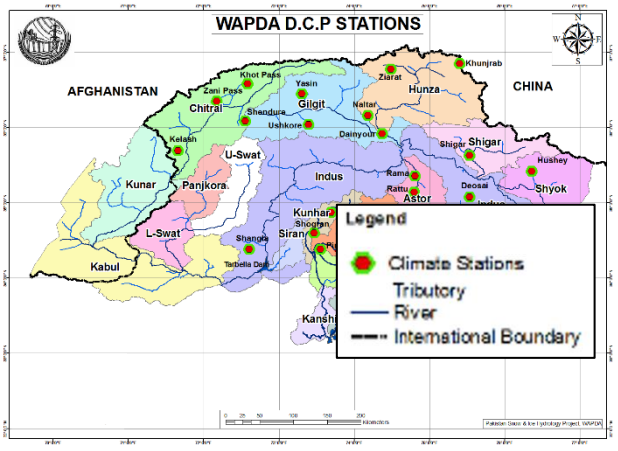
(3) Progress in WAPDA

The progress of the current NDMP was reviewed during a meeting with WAPDA. Based on the information obtained from PMD and FFC, the results were as follows:

- Regarding Intervention 3, 40 most vulnerable glaciers and 36 glacial lakes are identified and monitored under the Tarbela 4th Extension HPP Project (T4). Mass transfer monitoring is being conducted for 5 huge glaciers. In addition, WAPDA is monitoring glaciers so support disaster preparedness among stakeholders, and an expansion of this monitoring system is currently under consideration. These studies will be carried over by the Tarbela 5th Extension HPP Project (T5), which will receive 4 to 5 million USD in funding from the World Bank.
- In addition to the above, the Golen Gol Project in District Chitral has been implemented. There was considerable damage to the power plant due to GLOF event triggered by an upstream glacier. The early warning system was installed and is now fully operational in order to prevent further incidents.
- Regarding Intervention 4, the Glacier Monitoring Research Centre was established, which was implemented before the current NDMP (see Figure 3.2.1 for the location). In addition, GLOF countermeasures (mainly monitoring and pre-warning) are being implemented based on WAPDA's own plans and policies. The main objective of WAPDA's GLOF response is to protect WAPDA facilities from GLOF damage.
- Regarding Intervention 7, revision of the Tarbela Dam Standard Operating Procedures (SOPs) is currently underway, along with the ongoing construction of new dams under the Mohmad

Dam Project[1] and the Diamer Basha Dam Project[2]..

- In addition to the above, there are plans to strengthen and automate a number of stations in accordance with the Flood Telemetry Master Plan (WAPDA-ADB) (see Table 3.2.2). Improvement of the existing telemetry network by WAPDA is one of the projects described in NFPP-IV, and implementation of the program described in NFPP-IV is required by Intervention-7 of the current NDMP.



Note D.C.P: Data Collection Platform.
Source: WAPDA

Figure 3.2.1 Location map of glacier stations by WAPDA

Table 3.2.2 Stations covered by the Flood Telemetry Master Plan (WAPDA-ADB)

PROVINCE/FLA	PROPOSED MASTER PLAN FOR FLOOD TELEMETRY NETWORK TOTAL HYDROLOGICAL STATIONS : 679	
	UP-GRADATION of EXISTING Manual TOTAL: 237	NEWLY PROPOSED TOTAL: 442
BALOCHISTAN	19	188
PUNJAB	75	95
KPK	77	79
GB	29	55
SINDH	19	10
AJK	18	15

Source: Presentation On Flood 2022, to JICA Advisory Team, WAPDA

(4) Progress in PID

1) Progress of Current NDMP

The progress of the current NDMP was confirmed by the meeting with PIDs of Sindh, Punjab and Balochistan. The results were as follows:

- About Strategy 1 “Formulate/update disaster management plans at national, provincial, district and community or TMA levels” in Intervention 2.
  - In Sindh, an Annual Flood Fighting Plan is developed with the involvement of the Sindh-PID and FFC. The plan is updated annually by the respective Executive Engineers in each area between March and April.
  - In Punjab, the Annual Flood Fighting Plan in which FFC, Punjab-PID and Hydraulic Structure Safety Evaluation Unit (HSSEU) are involved is also updated every year. The plan includes the vulnerable points list, flood control materials list, and the necessary materials and personnel for each flood level.
  - Balochistan-PID prepares and updates the Monsoon Plan covering the period from June to August, and the Winter Plan covering December 15 to March 15 every year.
- About Strategy 2 "Preparation of contingency plans specific to specific disasters" (Strategies2 “Develop hazard specific contingency plans” in Intervention 2.
  - The PIDs in Sindh, Punjab and Balochistan share a common understanding that they prepare a Flood Contingency Plan following a major flood event. The corresponding Implementation Plan is then incorporated into the Annual Flood Fighting Plan.

1 http://www.wapda.gov.pk/index.php/projects/water-sector/under-construction/mohmand-dam-hydropower-project
2 http://www.wapda.gov.pk/index.php/projects/water-sector/under-construction/diamer-basha-dam

- According to the local staff in the project team, the Executive Engineer prepares a Damage Report after every flood disaster, and the corresponding Implementation Plan then incorporated into the Annual Flood Fighting Plan.

・ About Strategy 3 "Strengthen early warning dissemination system" in Intervention 4.

- Sindh-PID is considering building its own system apart from the system provided by PMD. The name of the system planned by Sindh-PID is "Flood Forecast and Early Warning System", which aims to provide information on the Hill Torrent basins in particular. However, the plan has not been implemented due to financial constraints. Similarly, Sukkur Barrage Office under the Sindh-PID has expressed the need for real time information on the Hill Torrent originating from Balochistan and Punjab Province.
- Punjab-PID and PMD-FFD, based in Lahore, operate the Early Warning Dissemination System in coordination with other relevant organizations (SUPARCO, Pak Army, PMD, WAPDA, NDMA and PDMA Punjab). PMD-FFD issues flood warnings to all stakeholders upon receiving the updated flood related information.
- Balochistan-PID uses flood forecasting and warning system from PMD and also uses mobile phone, social media and radio broadcast networks in the three districts of Kharan, Turbat and Gwadar.

・ The following table summarizes Priority Actions/Program 4-1 "Establishment of comprehensive and integrated flood management/protection plan in Pakistan" in Intervention 7,

Table 3.2.3 Summary of Intervention 7 Program 4-1 Interviews (PIDs)

Priority Actions/Programs	Province	Results
A review of breaching mechanism of the river embarkments (flood bunds) and a review of designated breaching points	Sindh	No artificial breach site has been established. ・It is difficult to get consensus from residents for political reasons.
	Punjab	Completed. ・A committee of experts from Punjab-PID, Highways Department, Pak Army and Pakistan Railway has been established. ・This committee takes into account all flood conditions and makes the judgement of artificial breach.
	Balochistan	Completed. ・There are about 500 vulnerable point locations across Balochistan Province and Balochistan-PID manages several Planned/Forced Breaching Points.
Revision and updating of NFPP-IV	Sindh	Almost complete. ・The NFPP-IV was developed between 2017 and 2018 and is currently being revised following the 2022 flood. ・Sindh-PID has already submitted a revised version of the project list to the FFC.
	Punjab	Near to completion. ・The NFPP-IV was developed between 2017 and 2018 and is currently being revised following the 2022 flood. ・Punjab-PID has already submitted a revised version of the project list to the FFC.
	Balochistan	Near to completion. ・The NFPP-IV was developed between 2017 and 2018 and is currently being revised following the flood in 2022. ・Balochistan-PID has already submitted a revised version of the project list to the FFC.
Flood plain zoning and management	All	It is the responsibility of the FFC and the FFC has started this program. ・Both Sindh-PID, Punjab-PID and Balochistan-PID agree on the above recognition.
Capacity building of the stakeholders in flood mitigation	Sindh	It is in progress. ・Newcomer training, inside training and stakeholder training (several days) are being conducted.
	Punjab	It is in progress.

Priority Actions/Programs	Province	Results
		・Several inside training sessions have been held before the flood season begins. ・Simulation exercises for flood emergency response have been conducted with Civil Administration, Pak Army, and other relevant parties.
	Balochistan	It is in progress. ・Several inside training sessions have been held before the flood season begins.

・ The following table summarizes Priority Actions/Program 4-2 "Construction and rehabilitation of flood control/mitigation structures" in Intervention 7,

Table 3.2.4 Summary of Intervention 7 Program 4-2 Interviews (PIDs)

Priority Actions/Programs	Province	Results
Implementation of NFPP-IV	Sindh	It is in progress. ・In Sindh-PID, the revision and update of NFPP-IV is in progress (to be completed in 2025), and the revision and update of FPSP-III has been completed.
	Punjab	It is in progress. ・The Annual Development Program (ADP) is a program under the umbrella of NFPP-IV and FPSP-III, and ADP is also under revision after the 2022 flood.
	Balochistan	It is in progress. ・The NFPP-IV was developed between 2017 and 2018, but the revision of the NFPP-IV is in progress because many conditions have changed due to the 2022 flood. After approval by the Competent Authority, several projects will be implemented accordingly.
Retrofitting of existing dams, barrages and flood protection works to increase floodwater retarding capacity and ability to accommodate design discharges.	Sindh	It is in progress. ・Sindh-PID is implementing "Sindh Barrage Improvement Project (SBIP)", which has been completed in Guddu Barrage and is in progress in Sukkur Barrage. ・Kotri Barrage Office in Sindh-PID is proposing "Important Nature of Works Required at Kotri Barrage Head Works Jamshoro". ・Under "Sindh Emergency and Resilience Project (SERP)" funded by the World Bank, Sukkur Barrage Office in Sindh-PID has already started several flood control projects including major renovations such as replacement of the Barrage Gate.
	Punjab	It is in progress. ・Rehabilitation of 13 barrages and irrigation and flood control facilities has been carried out. ・Many projects still need to be carried out, but few have been carried out due to lack of funds.
	Balochistan	Few have been carried out. ・Since there are about 1,040 Check/Small Dams for Grand Water Recharge Purpose and 8~10 Medium/Large Multipurpose Dams in Balochistan Province, many projects need to be carried out, but few have been carried out.
Constructing of new flood protection works, dams and barrages to increase flood mitigation/protection capacity.	Sindh	There are no plans for the construction of new flood control facilities.
	Punjab	Few projects have been implemented. ・Many projects need to be implemented, but few have been implemented due to lack of funds.
	Balochistan	Some are in progress. ・Some projects are needed for flood protection works, but some important projects have not been started due to lack of funds. ・In 2019, construction of about 15 Medium/Large Dams started under the Federal PSDP through the Ministry of Water Resources. In this project, 2% of the project cost is reserved for maintenance of the structures.

2) Analysis of Interviews

Although PID in Sindh, Punjab and Balochistan provinces were not aware of the priority actions related to the current NDMP and each Intervention in it, projects and activities related to each priority action were being implemented. In the field of DRR, PIDs are in charge of flood control, so they cooperate with the FFC, a federal government organizations, in planning, designing and budget allocation of projects. In addition, flood structural mitigation measures are being implemented across all provinces, targeting the main stem and key tributaries of the Indus River, Hill Torrents, Irrigation Channels, and Drainage Channel. Some structural mitigation measures for each province are listed in NFPP-IV and FPSP-III. In addition, I as the FFC recognized as the lead agency for non-structural measures such as zonation of flood areas, PID does not seem to carry out any regulations based on the study and the results of the study. (Actually, the organization of Punjab-PID which is in charge of analysis is working on the hazard map.)

In Punjab and Balochistan, a committee consisting of experts from various organizations, such as PID, Highways Department, Pak Army, Pakistan Railway, etc., has been established. According to responses received, this committee is responsible for making decisions regarding artificial breaches, taking into account all prevailing flood conditions. In regard to artificial breach, the planned site of artificial breach is decided after a preliminary study (the planned site of artificial breach is set in NFPP-IV). On the other hand, Sindh Province has a different approach from other provinces. During the flood in 2022, breaches occurred at several locations in the key embankments, including the FP Bund designed to protect against Hill Torrents, the Manchar Containing Bund (MCB) intend to prevent overflow from Manchar Lake, and the L.S. Bund in bund near Sewan along the main river of Indus were cut off at some of the locations. In this way, artificial breaches are often taken along bunds to protect barrages and populated areas, but in Sindh Province, the answer was that "the site of designated breach has not been determined." It is possible that the government intended not to have a designated to avoid confusion among local residents if it was announced as a designated site.

Regarding flood warning, Punjab and Balochistan provinces responded that they were operating an Early Warning Dissemination System in cooperation with PMD-FFD Lahore and other related organizations (SUPARCO, Pak Army, PMD, WAPDA, NDMA, PDNA). On the other hand, in Sindh Province, although Sindh-PID is not currently in charge of flood warning, Sindh-PID was planning to issue its own flood warning. In particular, Sindh-PID was aiming to improve flood information in the Hill Torrent, and Sindh-PID recognized that there is room for improvement on the flood warning issued by PMD. In addition, although they communicate with PMD at the time of floods, they do not communicate with PMD much during other periods.

3.2.3 Meteorological Disasters (Utilization Status, Achievement Status, Review Results, and Issues)

3.2.3.1 Utilization and Achievement Status

(1) Progress in PMD

The progress of the current NDMP was confirmed by the meeting with PMD. The results were as follows:

- About Intervention 3, only the earthquake hazard map has been prepared by PMD. The development of the Vulnerability Atlas has not yet been undertaken by PMD.
- About Intervention 4, enhancement of flood forecasting capability, installation of weather radar, additional AWS and broadcasting stations have been implemented. The deployment of weather radar systems is either in the proposal stage or pending due to funding constraints.
  The deployment of tide observation facilities and systems, and the deployment of real-time rainfall and discharge observation network were not implemented. It was answered that the preparation of hazard maps for flash flood and landslide hazard zones is not the responsibility of PMD.
- About weather radars, Islamabad and Karachi weather radars have been completed and delivered to PMD with grant aid from Japan. The weather radars of Multan and Sukkur are also planned

to be implemented with grant aid from Japan, and although preparatory studies have already been completed for both projects, adjustments are currently being made due to the effects of recent global inflation and exchange rate fluctuations.

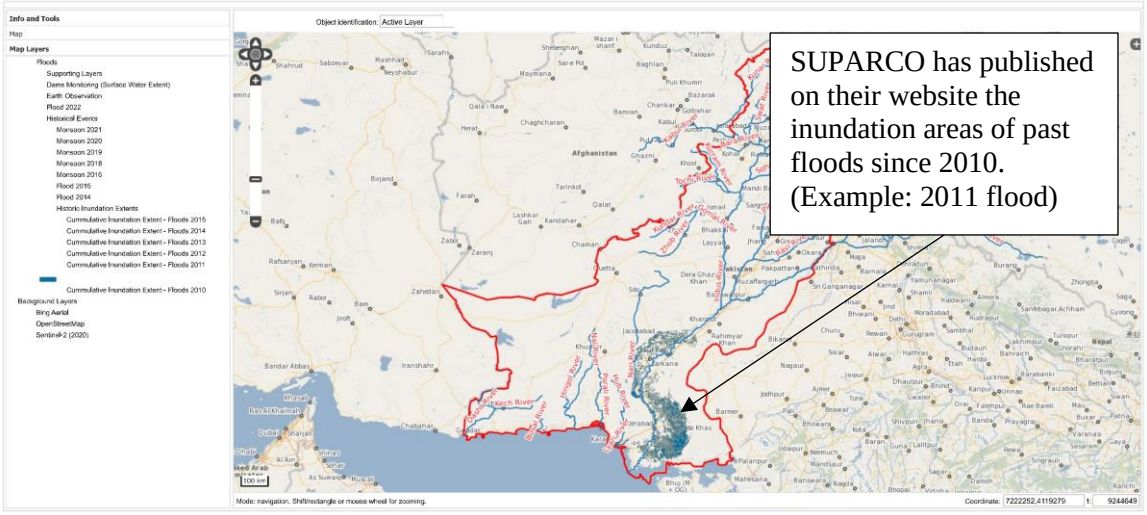
- About the expansion of automated weather observation networks, projects for installation at 45 locations in Sindh and Punjab provinces have already been completed by the Economic and Social Development Program Project (JICS Project) of Japan.

(2) Progress in MoCC

Based on the information obtained from PMD and the published sources on the website, the Ministry of Climate Change (MoCC) with support from United Nations Development Programme (UNDP), is implementing the project "Scaling-up of Glacial Lake Outburst Flood (GLOF) Risk Reduction in Northern Pakistan (GLOF II)[3]" under the intervention 3. This project focuses on the deployment of an early warning system for GLOF events, with partial involvement from the FFC and PMD. Progress in SUPARCO

According to publicly available information on SUPARCO's website, SUPARCO has implemented a project related to the enhancement of flood forecasts for the Indus River using satellite images under Intervention 4. Additionally, SUPARCO has launched a dedicated webpage titled “Disaster Watch”[4] to share relevant data and updates. On the website, a map showing the inundation area of the past floods since 2010 is available.

According to the interviews with SUPARCO officials and the website, a project to build a NatCat (Natural Catastrophe) Model was on-going with funding from the NDRMF (National Disaster Risk Management Fund). Currently, the project targets four types of disasters: earthquakes, river floods, droughts, and tropical cyclones. The NatCat Model provides quantitative estimates of the expected losses from hazard events of various types, intensities, and durations, and will eventually be made available as open source[5]. Figure 3.2.3 shows the Vulnerability Map published on the project's website. Similar to the Flood Vulnerability shown in Figure 3.2.3, hazard maps for earthquakes, cyclones, and droughts are also available. Most of these hazard maps align with the priorities outlined in the current NDMP.



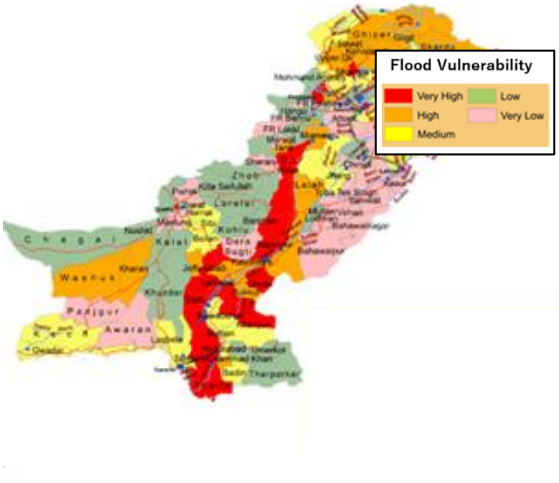
Source: Disaster Watch-SACRED-SUPARCO (sgs-superco.gov.pk)

Figure 3.2.2 Information published on Disaster Watch (Example of Flood Extent in 2011)

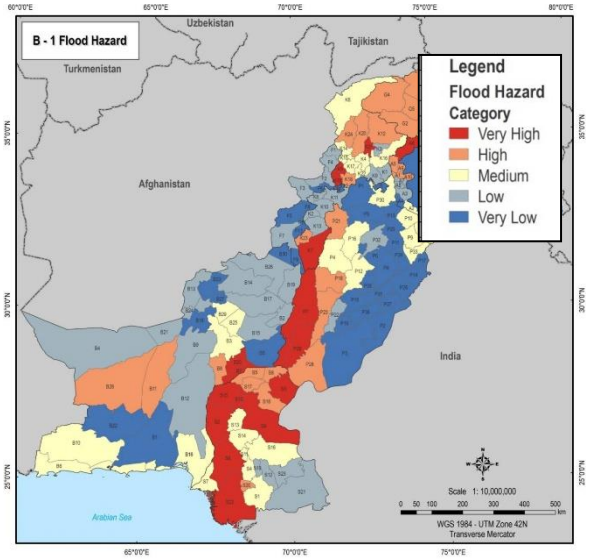
3 Scaling-up of Glacial Lake Outburst Flood (GLOF) risk reduction in Northern Pakistan | United Nations Development Programme (undp.org)

4 DisasterWatch - SACRED-SUPARCO (sgs-suparco.gov.pk)

5 NatCat Model • NDRMF – National Disaster & Risk Management Fund(https://www.ndrmf.pk/natcat-model/)



Source: NatCat Model ・NDRMF ‐National Disaster & Risk Management Fund (https://www.ndrmf.pk/natcat-model/)



Source: Current NDMP

Figure 3.2.3 Vulnerability Map Published on NatCat Model Web Site and Hazard Map under Current NDMP

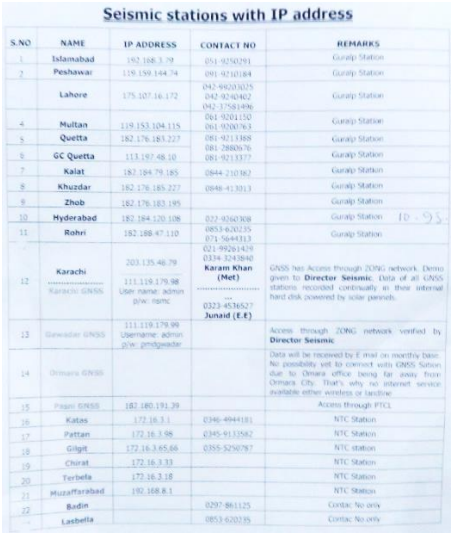
3.2.4 Earthquakes (Utilization Status, Achievement Status, Review Results, and Issues)

3.2.4.1 Status of Utilization and Achievement

Preparation of earthquake hazard and risk data is important, because it serves as the foundation for DRR activities across all interventions. In PMD, the project team confirmed the development of earthquake observation network and seismic intensity indication which contribute to early warning, and also confirmed the earthquake hazard evaluation. The status and issues are described below.

(1) Situation in PMD

- In Intervention-4, the development of a seismic network was proposed. The seismic network has been developed for the purpose of seismic source determination and simple seismic intensity assessment around the epicenter. Although power outages often occur, a system is in place to back up the data. According to the list of seismic observation shown in Figure 3.2.4, there are 23 observation points. Figure 3.2.5 shows the network of seismic observations in PMD.
- On the other hand, the development of strong seismometers, which are necessary for detecting major earthquakes, has not yet progressed. Since the above-mentioned seismometers (broadband type, etc.) for seismic source determination become saturated in the event of a strong earthquake, it was confirmed that the development and strengthening of the strong seismometer network is an issue in the current NDMP update for the purpose of providing seismic intensity information in each region, and for the purpose of analyzing the relationship between damage to buildings and strong earthquake records for risk assessment. It has been confirmed that the development of this robust earthquake observation network is currently in progress.



Seismic stations with IP address

S.NO	NAME	IP ADDRESS	CONTACT NO	REMARKS
1	Islamabad	192.168.3.79	051-9250291	Guralp Station
2	Peshawar	119.159.144.74	091-9210184	Guralp Station
3	Lahore	175.107.16.172	042-99203025 042-9240402 042-37581496	Guralp Station
4	Multan	119.153.104.115	061-9201150 061-9200763	Guralp Station
5	Quetta	182.176.183.227	081-9211368 081-2880676	Guralp Station
6	GC Quetta	113.197.48.10	081-9213377	Guralp Station
7	Kalat	182.184.79.185	0844-210362	Guralp Station
8	Khuzdar	182.176.185.227	0848-413013	Guralp Station
9	Zhob	182.176.183.195		Guralp Station
10	Hyderabad	182.184.120.108	022-9260308	Guralp Station
11	Rohri	182.188.47.110	0853-620235 071-5644313	Guralp Station
12	Karachi Karachi GNSS	203.135.48.79 111.119.179.98 User name: admin pw: nsmc	021-99261439 0334-3243840 **Karam Khan (Met)** ... 0323-4536527 **Junaid (E.E)**	GNSS has Access through ZONG network. Demo given to **Director Seismic**. Data of all GNSS stations recorded continually in their internal hard disk powered by solar panels.
13	Gawadar GNSS	111.119.179.99 Username: admin pw: pmdgwadar		Access through ZONG network verified by **Director Seismic**
14	Ormara GNSS			Data will be received by E mail on monthly base. No possibility yet to connect with GNSS Station due to Ormara office being far away from Ormara City. That's why no internet service available either wireless or landline
15	Pasni GNSS	182.180.191.39		Access through PTCL
16	Katas	172.16.3.1	0346-4944181	NTC Station
17	Pattan	172.16.3.98	0345-9133582	NTC Station
18	Gilgit	172.16.3.65,66	0355-5250787	NTC station
19	Chorat	172.16.3.33		NTC Station
20	Terbella	172.16.3.18		NTC Station
21	Muzaffarabad	192.168.8.1		NTC Station
22	Badin		0297-861125	Contac No only
	Lasbella		0853-620235	Contac No only

Figure 3.2.4 List of Seismic Observation Networks in PMD

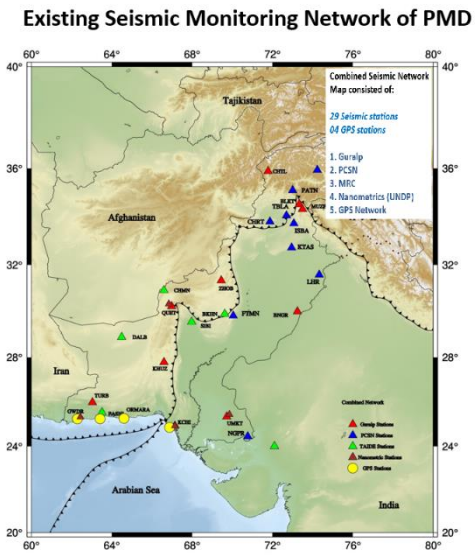


Figure 3.2.5 Map of Seismic Observation Networks in PMD

- Regarding the identification of hypocenter based on seismic observation, a processing method is included. However, in this processing, it is a method of attenuating distance concentrically from the location of hypocenter, and ground amplification factor is not considered. Since ground amplification factor varies significantly between alluvial lowlands and mountainous areas, it is necessary to include an assessment of ground amplification factor across Pakistan in the update of the current NDMP.

(2) Situation at PEC

It was confirmed that the earthquake hazard map from the MHVRA Report, included in the current NDMP, is incorporated into the Building Code of 2007. The probability of occurrence is 10% over 50 years, and the zone classification is based on the maximum surface acceleration (PGA). (See Figure 3.2.6)

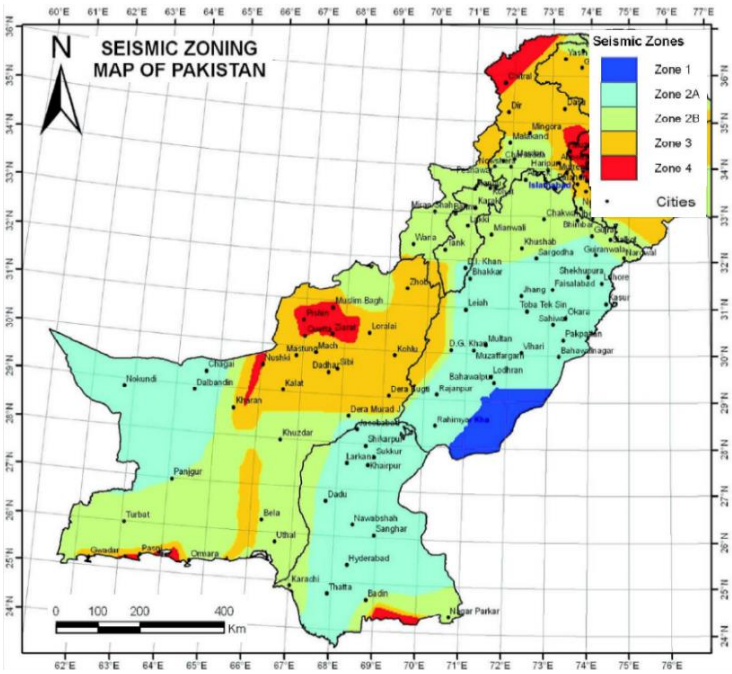
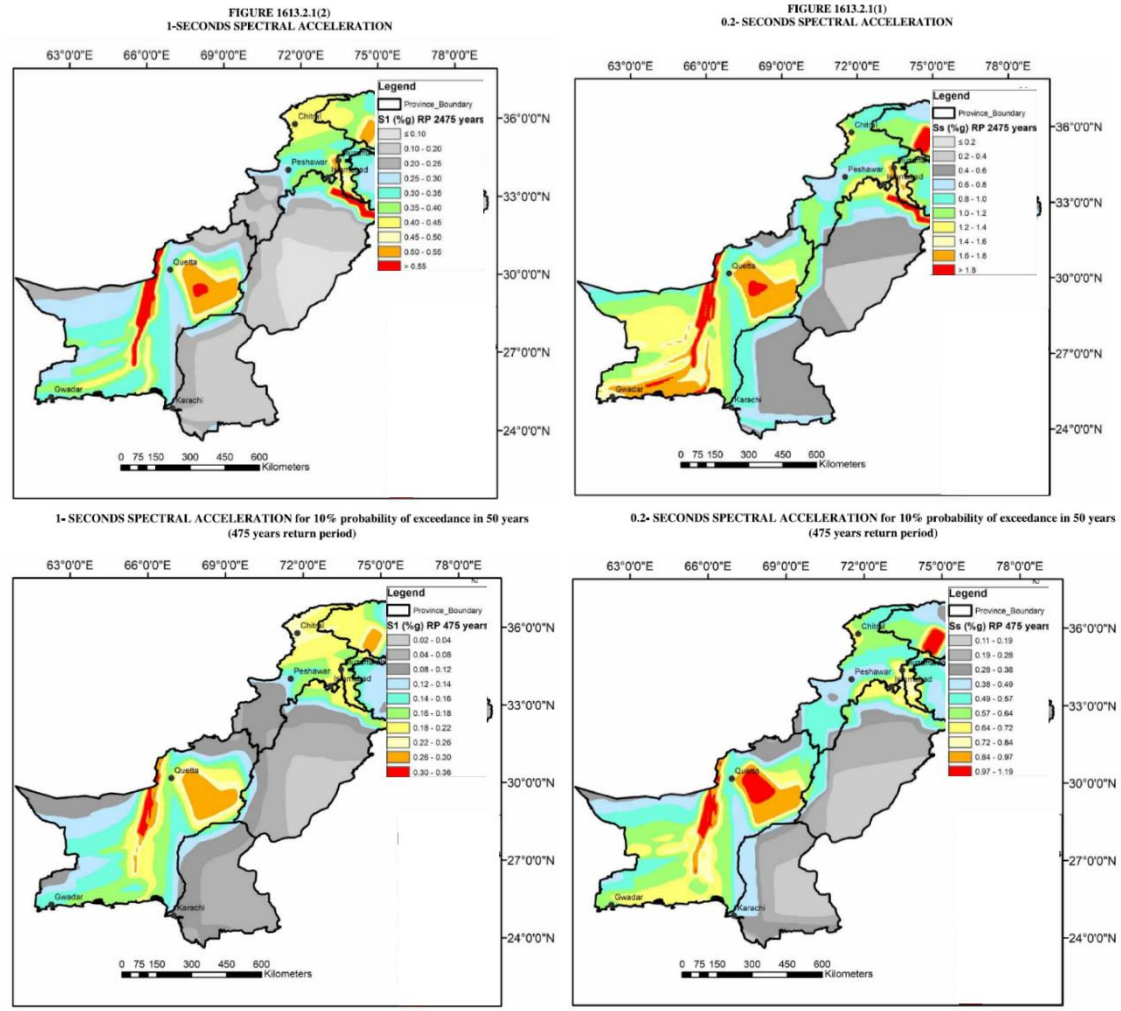


Figure 3.2.6 Example of National Earthquake Hazard Map Published in Building Code 2007

Further, upon reviewing the source, it was confirmed that the Pakistan Engineering Council (PEC) published the latest edition of Building Code in 2021, during which the earthquake hazard map was also updated.

Figure 3.2.7 shows the earthquake hazard map of Pakistan in Building Code as an example of earthquake hazard map to be considered in updating the current NDMP. The expression method was classified by zone in 2007, but in 2021, it was classified by spectral period (natural period: 0.2 seconds for short period and 1.0 seconds for long period), and the policy was adopted for the construction of general low-rise buildings and high-rise buildings. The probability of occurrence has also been changed, and the probability of exceeding 2% in 50 is adopted in Building Code 2021 text. The source of the earthquake hazard map adopted by PEC in the Building Code was found to be as follows.

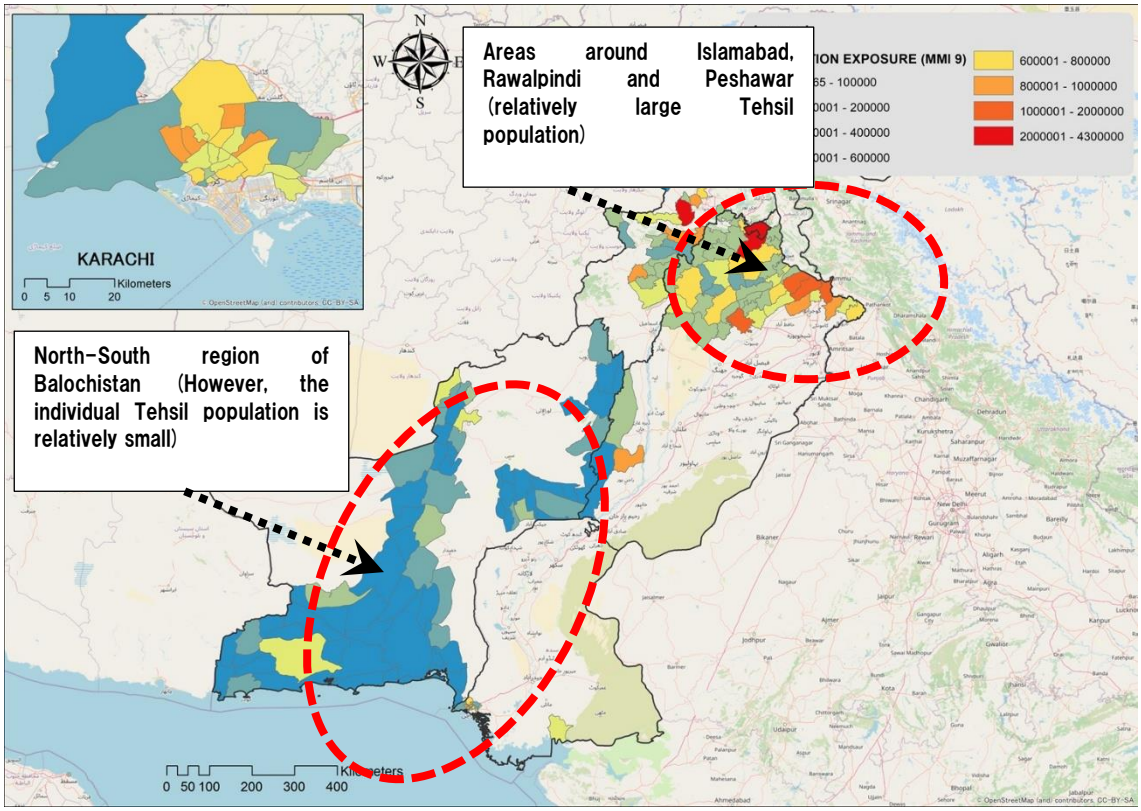
Name of paper	Probabilistic Seismic Hazard Analysis Based Zoning Map of Pakistan, Pages 271-306 | Received 16 Jan 2019, Accepted 21 Oct 2019, Published online: 11 Nov 2019
Author	Muhammad Ahmed,Sarosh Hashmat Lodi &Muhammad Masood Rafi
Affiliation	Department of Earthquake Engineering, NED University of Engineering and Technology, Karachi, Pakistan(E-mail: rafi-m@neduet.edu.pk)



(2% probability of exceedance in 50, (upper left): long period, (upper right): short period)
(10% probability of exceedance in 50, (bottom left): long period, (bottom right): short period)

Figure 3.2.7 Example of the National Earthquake Hazard Map Published in the Building Code 2021

From the comparative examination of these drawings, it was confirmed that information on earthquake hazard, which is one of the basis data for disaster risk assessment, has been updated. The seismic intensity exposure population map was published in Output 1-4 for the areas with particularly high risk. Figure 3.2.8 shows this figure again.



Source: Project team

Figure 3.2.8 Population Exposure to Seismic Intensity Map at Tehsil Level (MMI Intensity 9 and above)

(3) NDMA situation

According to the MHVRA report—an activity proposed under the current NDMP—the seismic hazard assessment is based on the information described in Building Standards Act of 2007. Vulnerability mapping was conducted using GIS, incorporating population data from the census and statistical information on residential building types collected through surveys.. It is confirmed that the statistical information is in Tehsil units. The Bureau of Statistics confirmed that the new census data was completed in 2017, and it is necessary to prepare the data based on the latest information when updating the current NDMP.

Regarding the vulnerability of buildings, it was confirmed that the quantity of building type is organized by earthquake hazard zone. However, it is only the stage of organizing the building inventory which is the basis of the vulnerability assessment. The vulnerability assessment is generally evaluated by damage function (Vulnerability Function, Fragility Curve, etc.), but the vulnerability assessment result could not be confirmed in the document.

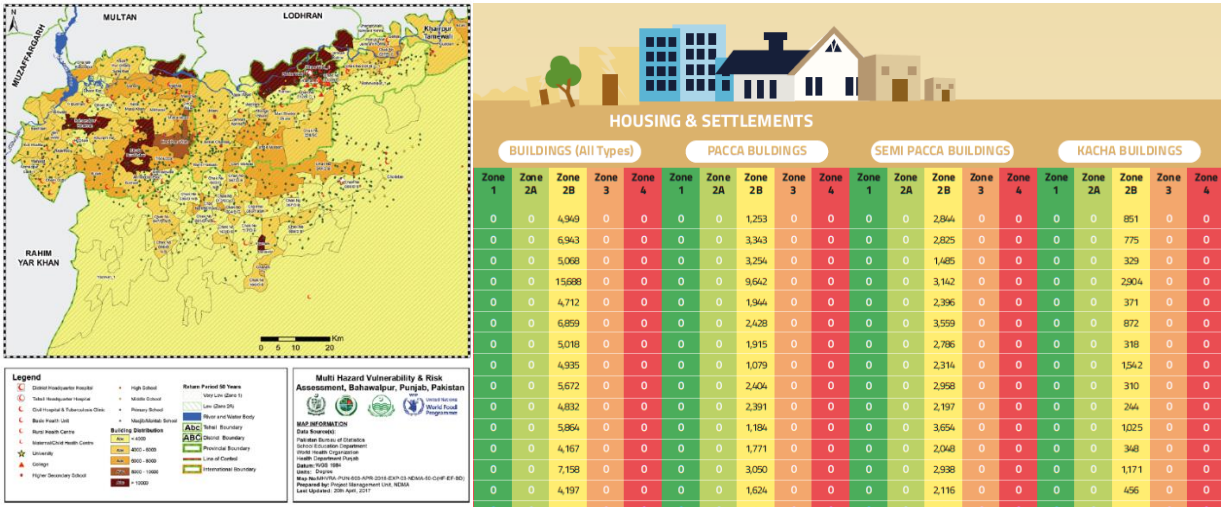


Figure 3.2.9 Expression of Building Vulnerability in MHVRA Report

(Left: Building distribution map (Color-coded by building quantity, hatches are earthquake hazard zones) Right: Statistical data by building type in Tehsil units organized by earthquake hazard zone)

In addition, in the MHVRA Policy Guideline, Earthquake Hazard Assessment Methodology clearly states the following policy on "evaluation of amplification characteristics based on S-wave velocity structure of ground". However, the MHVRA report does not present the corresponding results. Nevertheless, since a geological map is included, it is expected to support future evaluations of ground amplification characteristics.

(8) Geotechnical investigation data (borehole, SPT) should be obtained for the target region to perform site-soil profiling on the basis of shear wave velocity and specific weight of soil.

(9) Acceleration records should be retrieved from the locally available or online global strong ground motions database[2], specifying the search criterion on the basis of fault focal mechanism, earthquake magnitude and source-to-site distance.

(10) The acceleration records should be analyzed and processed for the target region hazard compatibility (i.e. spectral matching of the acceleration records with the code specified spectrum).

(11) The compatible accelerograms should be employed for site soil response analysis, using dynamic time history analysis of 1D soil column, to obtain the site-specific ground motion on site soil surface. The ratio of the Site Amplified Ground Motion to the bedrock/outcrop motion gives estimate of site amplification factor.

3.2.4.2 Status of Achievement

From the perspective of Intervention-3: Establish a National Hazard and Vulnerability Assessment, a probabilistic earthquake ground motion hazard assessment has been carried out at the national level. However, the following technical issues has been identified.

- Since the ground surface as a standard for seismic hazard assessment is the hard rock layer (called engineering bedrock) as a standard in the attenuation equation, it is necessary to multiply the ground motion value at the hard rock layer by the degree of ground amplification.
- Building vulnerability information is currently being organized based on census data.. In the future, it is necessary to prepare a comprehensive building inventory aligned with the damage to the damage function of each structural type, tailored to specific regional context.

From the viewpoint of Intervention-4: Establish a Multi-hazard Early Warning System, certain results based on the seismic observation of PMD are recognized. However, the following technical issues have been identified.

- It is necessary to strengthen the development of the seismic observation network to improve the accuracy in determining the epicenter.
- It is necessary to consider the degree of ground amplification when estimating the areal ground motion.
- It is desirable to develop the strong motion observation network in order to observe strong ground motion and immediately display the seismic intensity distribution. (Telecommunication facilities are in progress.)

In addition, confirmation of the status of efforts based on the interview survey is as follows.

Name of Province/Region	Type of disaster	Efforts for earthquake and tsunami disaster prevention	Status of achievement and risk recognition
Punjab Province	Earthquake	・ Sediment disaster is a concern as a secondary disaster after the earthquake, and the Public Works Department has already carried out road hazard mapping. ・ Building materials have been tested in the Materials Testing Laboratory of the Public Works Department.	・ Achievement of earthquake countermeasures such as earthquake resistance improvement is limited. ・ It is necessary to identify the area where the risk should be examined in detail. ・ Awareness of earthquake risk in Lahore is rather low.
Sindh Province	Tsunami Earthquake	・ Hazardous buildings were listed by the Technical Committee in 2016. ・ A one-stop service is provided for both private and public buildings outside the Cantonment area and military lands.	・ Efforts by universities are active, and an earthquake hazard assessment map has been adopted in the Building Code, and the status of achievement is moderate. ・ Awareness of earthquake and tsunami is high.
Balochistan Province	Earthquake	・ In the monitoring of building construction, inspection certificates are issued and supervised at each stage, such as compliance with building standards, confirmation of labor works, and final inspection.	・ The status of achievement is limited. ・ Awareness of risk to earthquake is high.
KPK Province	Earthquake	・ After the 2005 Northern Earthquake, seismic reinforcement of buildings has been implemented with the participation of the University of Peshawar. ・ Earthquake Engineering Center of the University of Peshawar has been set up. ・ Technical training for Public Works Department staff is being conducted.	・ Seismic reinforcement with the participation of the University has been conducted, and the achievement of countermeasures is judged to be at least moderate. ・ There are times when shaking is felt, and the risk perception of earthquake is very high.

Name of Province/Region	Type of disaster	Efforts for earthquake and tsunami disaster prevention	Status of achievement and risk recognition
AJK Region	Earthquake	・ Seismic construction was carried out as a model in the One UN project after the Northern Earthquake in 2005. ・ 5-10 potential danger points of slope disaster of mountain road after earthquake were listed. ・ Inventory preparation of important structures for earthquake resistance was carried out.	・ Earthquake resistant construction is carried out, although it is a model, and the achievement status is moderate. ・ Awareness of earthquake risk is high.

3.2.5 Tsunami (Utilization Status, Achievement Status, Review Results, and Issues)

3.2.5.1 Utilization and Achievement Status

In the context of Intervention-4, a GPS observation network by PMD has been established for tsunami observation along the coast. The early warning system based on this observation network has achieved certain results. In addition, it has been reported that tsunami evacuation towers have been constructed and operated in some coastal areas.

In the context of Intervention-3, a deterministic tsunami inundation simulation has been conducted in coastal areas under the UNDP-funded project "Strengthening Tsunami and Earthquake Preparedness in Pakistan" (2021). The counterpart is NED University of Engineering and Technology.

60 scenarios for Tsunami inundation simulation were set for Gwadar Province and 38 scenarios for Karachi, including the model that actually generated the tsunami in 1945. Table 3.2.5 lists the simulation results compiled in this project.

Table 3.2.5 List of Tsunami Inundation Results Based on Various Deterministic Scenarios

(a) Inundation Depth due to Scenarios for Gwadar

Scenario Ref	Mw	Inundation depth (m)
Byrne	8.27	0–3 m along the shore line. Up to 1.5 m inundation depth, significant inundation.
Worst case scenario		
RA_gem1a	9.22	0–>3 (mostly >3 along entire coast, neck not completely inundated, 0–1m)
RA_gem1b	9.07	Almost similar to above with a little less inundation depth in specific areas
RA_gem1c	9.07	0–>3 inundation depths greater than previous 2 scenarios, specifically western bay. Most the inundated areas along the neck are inundated with depths b/w 0.5–1.5m
RA_gem2a	9.02	0–>3 (mostly >3 along entire coast, very less inundation along the neck mostly no inundation or b/w 0–0.5 m)
RA_gem2b	8.87	0–>3 (mostly >3 along entire coast, very less inundation along the neck mostly no inundation or b/w 0–1.5 m)
RA_gem2c	8.87	Almost similar to previous scenario
RA_gem3a	8.85	0–>3 along the eastern bay and 1.5– 2m mostly along the western bay. Most inundation near the Dor Kaur area. Neck is mostly un-inundated.
RA_gem3b	8.69	0–>3 along the shores, most of the neck and Dor Kaur area is inundated by a depth of 0–3m.
RA_gem3c	8.69	0–>3 along the shores, most of the neck and Dor Kaur area is inundated by a depth of 0–3m.
For early warning thresholds		
RA0_c	6.75	0–1.5 m along shoreline
RA0_d	6.75	0–1.5 m along shoreline
RA0_e	6.75	0–1.5 m along shoreline
RA1_c	7	0–1.5 m along shoreline
RA1_c_G2	7	0–1.5 m along shoreline
RA1_d	7	0–2 m along shoreline
RA1_d_G1	7	0–1.5 m along shoreline
RA1_d_G2	7	0–1.5 m along shoreline
RA1_e	7	0–2 m along shoreline
RA1_e_G1	7	0–1.5 m along shoreline
RA5_e_G1	8	0–3 m along shoreline some inundation.
RA6_c	8.25	0–>3 m along shoreline, extreme inundation (0–>3 m).
RA6_d	8.25	0–3 m along shoreline significant inundation (0–2 m).
RA6_d_G1	8.25	0–>3 m along shoreline, severe inundation (0–>3 m).
RA6_d_G2	8.25	0–>3 m along shoreline significant inundation (0–1.5 m).
RA6_e_G1	8.25	0–>3 m along shoreline significant inundation (0–1.5 m). >3m at few locations.
RA7_a	8.5	0–3 m along shoreline significant inundation (0–3 m).
RA7_b	8.5	0–>3 (mostly >3 along entire coast, neck completely inundated, 0.5–3 m mostly with eastern side being more affected).
RA7_b_G1	8.5	0–>3 m along shoreline significant inundation (0–2.5 m).
RA7_b_G2	8.5	0–>3 m along shoreline significant inundation (0–1.5 m). >3m at few locations.
RA7_c	8.5	0–3 m along shoreline significant inundation (0–3 m).
RA7_c_G1	8.5	0–3 m along shoreline significant inundation (0–1.5 m). >3m at few locations.

Scenario Ref	Mw	Inundation depth (m)
RA2_c	7.25	0–3 m along shoreline very little inundation
RA2_c_G2	7.25	0–1.5 m along shoreline
RA2_d	7.25	0–3 m along shoreline very little inundation
RA2_d_G1	7.25	0–3 m along the shores, no significant inundation, 2 m and greater mostly along the western bay.
RA2_d_G2	7.25	0–1.5 m along shoreline
RA2_e	7.25	0–3 m along shoreline very little inundation.
RA2_e_G1	7.25	0–2 m along the shores, no significant inundation, 2 m mostly along the western bay.
RA3_c	7.5	0–3 m along shoreline some inundation.
RA3_c_G2	7.5	0–1.5 m along shoreline
RA3_d	7.5	0–3 m along shoreline some inundation.
RA3_d_G1	7.5	0–3 m along the shores, no significant inundation, greater than 2m mostly along the western bay.
RA3_d_G2	7.5	0–2 (0–1.5 mostly, 2 at few places)
RA3_e	7.5	0–3 m along shoreline some inundation.
RA3_e_G1	7.5	0–3 m along the shores, no significant inundation, 2m and greater than 2m mostly along the western bay. Very few places with 3m
RA4_c	7.75	0–3 m along shoreline some inundation (0–1 m).
RA4_c_G2	7.75	0–3 m along shoreline little inundation, 3m only along very few places extreme south west.
RA4_d	7.75	0–>3 m along shoreline significant inundation (0–1.5 m).
RA4_d_G1	7.75	0–>3 m along shoreline significant inundation (0–1.5 m) with >3m in south west
RA4_d_G2	7.75	0–3 m along shoreline some inundation. 3m mainly along the eastern jetty.
RA4_e	7.75	0–3 m along shoreline some inundation (0–0.5 m).
RA4_e_G1	7.75	0–3 m along the shores, no significant inundation, 2m and greater than 2m mostly along the western bay. Very few places with 3m
RA5_c	8	0–>3 m along shoreline significant inundation (0–3 m).
RA5_c_G2	8	0–3 m along the shores, some inundation, 3m mostly along the eastern bay and south west.
RA5_d	8	0–>3 m along shoreline, inundation (0–>3 m).
RA5_d_G1	8	0–>3 m along shoreline significant inundation (0–>3 m). Most inundation is along the western side and Dor Kaur on the east.
RA5_d_G2	8	0–3 m along the shores, some inundation, 3m mostly along the eastern bay.
RA5_e	8	0–3 m along shoreline some inundation (0–0.5 m).

(b) Inundation Depth due to Scenarios for

Scenario Id	Magnitude	Inundation depth (min–max) (m)
Byrne 1945	8.27	0–1
Worst case scenarios		
RA_gem1a	9.22	0–3 (worst along Defence Phase VIII, Manora, sandspit and KPT)
RA_gem1b	9.07	0–3 (b/w 0-1.5 mostly)
RA_gem1c	9.07	0–>3 (many places such as KPT, Manora, Defence Phase VIII and Sandspit are inundated by 2.5 m or more).
RA_gem2a	9.02	0–>3 (many places such as KPT, Manora, Defence Phase VIII and Sandspit are inundated by 2.5 m or more.
RA_gem2b	8.87	0–>3 (mostly b/w 0-1.5) only a very small area to the extreme west where inundation is greater than 3 m.
RA_gem2c	8.87	0–>3 (many places such as KPT, Manora, Defence Phase VIII and Sandspit are inundated by 2.5m or more.
RA_gem3a	8.85	0–2
RA_gem3b	8.69	0–1.5
RA_gem3c	8.69	0–2 (mostly between 0.5–1.5)
For early warning thresholds		
RA1_a	7	No inundation
RA1_a_K1	7	No inundation
RA1_b	7	No inundation
RA1_b_K1	7	No inundation
RA2_a	7.25	No inundation
RA2_a_K1	7.25	No inundation
RA2_b	7.25	No inundation
RA2_b_K1	7.25	No inundation
RA3_a	7.5	Insignificant inundation.
RA3_a_K1	7.5	No inundation
RA3_b	7.5	Insignificant inundation.
RA3_b_K1	7.5	No inundation
RA4_a	7.75	Insignificant inundation.
RA4_a_K1	7.75	Insignificant inundation b/w 0–0.5
RA4_b	7.75	Insignificant inundation.
RA4_b_K1	7.75	Insignificant inundation b/w 0–0.5
RA5_a	8	0–2 (mostly 0–1.5), (only 1 place to extreme west, near Mubarak Village has 2)
RA5_a_K1	8	0–0.5
RA5_b	8	0–1.5 (mostly 0–0.5 but Abdul Rahman Goth, Goth Jumma and Mubarak Village are inundated with 1.5 m water depth)
RA5_b_K1	8	Insignificant inundation b/w 0–0.5
RA6_a	8.25	0–3 Significant inundation (0–0.5 mostly, 2 m at Manora and Defence Phase VIII, 3 m at Mubarak Village and nearby areas)
RA6_a_K1	8.25	0–1.5 (mostly 0–0.5)
RA6_b	8.25	0–1.5 Significant inundation
RA6_b_K1	8.25	0–1 m (mostly 0–0.5, 1 m at very few locations)
RA7_a	8.5	0–3 Significant inundation (0–1.5 mostly, 2 m at Manora and Defence Phase VIII, 2.5 m at Hawke's Bay, Paradise Point and Goth Jumma, 3 m at only near Mubarak Village')
RA7_a_K1	8.5	0–1.5 (mostly 0–1, 1.5 m depth at Defence Phase VIII, KPT, Manora, Hawke's Bay, French beach, Paradise Point, near Goth Jumma and Mubarak Village Defence Phase VIII)
RA7_b	8.5	0–1.5 m (mostly 0–1)
RA7_b_K1	8.5	0–1.5 m (mostly 0–0.5)

The inundation prediction map is also included in this report. The inundation depth along the Karachi coast is 0~1.5 m and reaches 2 m only in the westernmost area. The reasons why inundation is not severe are as follows:

- Most of the tsunami energy propagates to the south or north of the fault rupture area and only a small amount propagates to Karachi.
- In the coastal area of Karachi, there is a shallow water (0~170 m) called Shelf that extends 50 km in width, and the tsunami energy is rapidly reduced by friction on the seabed of this shallow water.

Figure 3.2.10 shows the inundation area along the Karachi coast based on multiple simulation results.

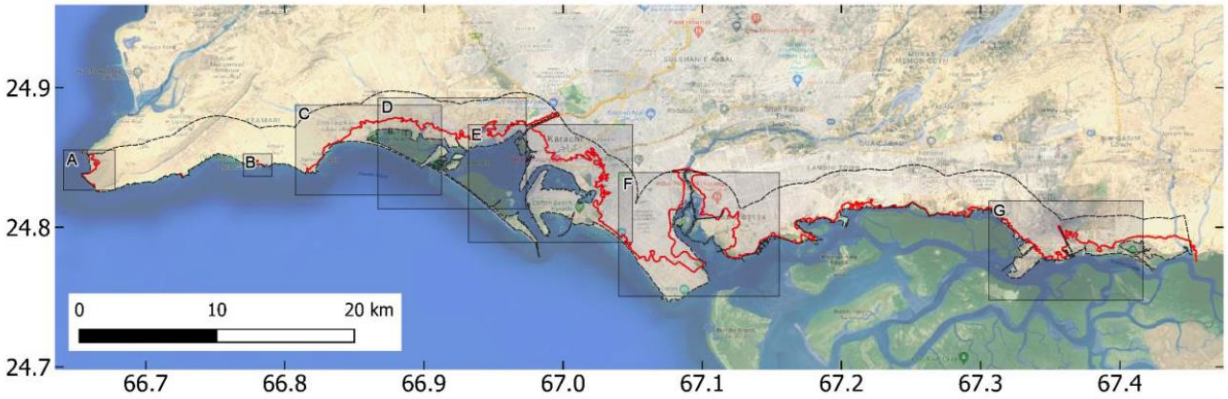


Figure 3.2.10 Tsunami Inundation Area Along the Karachi Coast Shown in the UNDP Report

On the other hand, the inundation depth along the coast of Gwadar ranges from 0~3 m in most of the cases, due to the steeply sloping seabed topography in the area. However, the complex seabed topography makes it challenging to accurately interpret the results of tsunami propagation simulations. The maximum inundation depth does not occur at the maximum magnitude (Mw9.22), and the maximum inundation depth is estimated at the scenario of Mw8.25, and the inundation depth is estimated to exceed 3 m near the Tombolo Peninsula, which projects south from Gwadar.

Other issues identified during the on-site interviews in Sindh Province are as follows:

- Inadequate tsunami countermeasures for the nuclear power plant located approximately 40 km from the center of Karachi City. .
- Insufficient liquefaction countermeasures for the coastal landfill areas.
- There is a possibility that the functions of ports and logistics will be stopped by tsunami.
- There is a possibility that the Central District of Karachi City will become an isolated island on land if all bridges fall.

3.2.5.2 Status of Achievement

From the perspective of Intervention-3, the UNDP-supported project "Strengthening Tsunami and Earthquake Preparedness in Pakistan" (2021), conducted deterministic tsunami inundation simulations in coastal areas. As a future priority, it is necessary to carry out probabilistic simulations to consider countermeasures.

Furthermore, based on the probabilistic simulation results, it is expected to consider concrete countermeasures such as tsunami countermeasures using both green infrastructure and gray infrastructure utilizing shallow water along the Karachi coast.

Based on Intervention-4, a GPS observation network by the PMD has been established for tsunami monitoring along the coastline, and it has demonstrated some effectiveness in Early Warning. As a future priority, according to the PMD staff, it is expected to develop a system to rapidly estimate and disseminate

the maximum inundation depth and affected areas using real-time observation data, supported by a pre-compiled database of multiple inundation simulation scenarios.

3.2.6 Human Resource Development (Utilization Status, Achievement Status, Review Results, and Issues)

The objective of the current NDMP Human Resource Development Plan is "A system of human resource development required in the field of disaster management will be established and human resources that promote “self-help help”, “mutual-help”, and “public-help” will be developed through improved coordination” and it is considered that it aimed to develop the overall capacity of human resource in the field of disaster management. Therefore, the organizations expected to implement the activities were not those in charge of specialized fields such as disaster types and meteorology, but the National Institute of Disaster Management (NIDM) under the NDMA, which are in charge of overall coordination of disaster management; Rescue1122 and Civil Defense, which handle rescue and emergency response; and Education Departments, oversee disaster education, all play critical role in human resource development but require better integration and coordination under the current plan. For the organizations expected to implement the Human Resource Development Plan of the current NDMP, questionnaires focusing on human resource development activities were prepared, and a survey was planned to be conducted on the implementation status of the current NDMP and future plans. However, since the letter to send the questionnaires was not issued by the NDMA, informal responses to the questionnaires were received from several organizations, but no responses were received from others. Therefore, the assessment of the utilization and achievement status of the Human Resource Development Plan was conducted through the results of general interviews with each organization regarding the current NDMP and information (annual reports, etc.) available on the websites of each organization.

3.2.6.1 Utilization Status and Achievement Status

(1) NDMA-NIDM

The NIDM's human resource development activities were verified from the annual reports published by NDMA. There were very few activities implemented in the stages of the current NDMP, but the implemented initiatives gradually increased starting around 2015. Since the current NDMP's Human Resource Development Plan was based on the assumption that NIDM would be established and functioning soon, NDMA has little role in human resource development activities. Although NIDM officially resumed activities in 2013, there were no independent facilities, and the activities seemed to be largely limited to project based training. In fact, NDMA has carried out various human resource development activities during the 10 years of the current NDMP.

However, very limited activities were funded directly by the NDMA's own budget, with most of its human resource development efforts being part of projects implemented jointly with international organizations and NGOs. As a result, there is no strategy or consistency in the activities, and the targets and contents are scattered. The targets audience includes officers from the federal government down to district and tehsil levels, with training contents covering from general DRM to simulation exercises and first aid training. Organizational capacity of NIDM has also been strengthened, but this is also dependent on supports from donors and UNDP, SDC (Swiss Agency for Development and Corporation) and JICA have supported the capacity building of NIDM. The only activities consistently carried out by NDMA using its own budget are awareness activities on the National Resilience Day, held every October.

The following activities have been implemented by NIDM:

- Activities implemented a lot
  - DRM related training for government officers at Federal Provincial (PDMA), and at district department level. (Some implemented through NIDM)
  - General awareness activities such as activities related to National Resilience Day
- Activities implemented sometimes

- Promotion of education on disaster risk reduction at basic educational institutions (mainly promotion of Pakistan School Safety Framework)
- USAR (Urban Search and Rescue) team training, etc.

・ Activities implemented infrequently

- Strengthening the organizational capacity of NIDM
- Promotion of DRM-related research
- Enhancement of DRM-related subjects in the curricula of government training institutions
- DRM-related training, rescue training, etc. at community level
- Promotion of education on disaster risk reduction in higher education institutions

(2) S/GB/PDMA, Rescue 1122, Civil Defense

Information on S/GB/PDMA was obtained from KPK, Balochistan, and GB provinces. Requests for similar data were made to NIDM and other provincial authorities, but they did not provide the information. One reason for this is that human resource development activities is not systematically organized by any of the organizations. PDMAs, with which information was shared, have been conducting training on DRM over the past 10 years, but like NDMA and NIDM, most of the training seems to be conducted as part of the activities of projects supported by international organizations and NGOs. On the other hand, PDMA in Balochistan Province, with its own budget, conducts flood response training every year, and it also conducts DRM training, and earthquake and tsunami response training. In addition, according to the websites of each organization, like NDMA, awareness activities are conducted on National Resilience Day.

As for Rescue1122 and Civil Defense, information was collected only from KPK Province. Rescue1122 conducts a lot of community based disaster management, DRM, rescue, fire training, first aid training, and awareness activities for schools and local communities. Civil Defense also conducts rescue, fire training, first aid training, and awareness activities. In addition to KPK Province, Rescue1122 in Punjab Province operates its own Training Academy, providing training related to rescue not only to rescue personnel but also for local communities.

(3) Progress of the Human Resources Development Plan (Intervention 5 of the current NDMP)

Progress of the priority activities of the current NDMP Human Resources Development Plan (Intervention 5) is summarized in Table 3.2.6.

Table 3.2.6 Summary of Intervention 5 Achievements

Strategies	Priority Actions/ Programs	Responsible Organizations	Achievements
1. Develop NIDM (National Institute of Disaster Management) to promote human resource development in the field of disaster management	1.1. Setting up of organization of NIDM	NDMA	• NIDM was suspended when the current NDMP was formulated, but officially resumed in 2013. • UNDP, SDC and JICA supported the strengthening of the organization. • Most of the staff were supported by donors.
	1.2. Construction of NIDM	NDMA	• No construction of the NIDM has been confirmed.
	1.3. Enhancement of coordination and partnership with stakeholders	NIDM	• Although there is a relationship between NIDM and DRM related organizations, there is no regular coordination.
	1.4. Streamlining of DRM capacity building	Relevant organizations	• NDMA/NIDM has no regular DRM training. • Balochistan PDMA conducts flood training every year.
	1.5. Enhancing the quality of DRM training	NIDM	• No activities to improve the quality of training by NIDM were confirmed.

Strategies	Priority Actions/ Programs	Responsible Organizations	Achievements
	1.6. Establishment of library	NIDM	• The existence of the library was not confirmed.
	1.7. Promotion of research	NIDM	• There is a little research support by NDMA.
2. Enhance the capacity of government agencies in charge of disaster management	2.1. DRM courses for NDMA, GB/S/PDMAs, and DDMAs staff	NIDM	• NDMA and NIDM have been conducting various trainings on the project basis of international organizations.
	2.2. Capacity enhancement of urban search and rescue teams	City District Government	• Training has been conducted by NDMA and other organizations once every two years.
	2.3. Implementation of regular refresher training for district fire brigades	Provincial Fire Brigade, Provincial Civil Defense	• Rescue 1122 and Civil Defense in each province train their own staff.
	2.4. DRM workshops for TMA staff	NIDM, DDMA	• Training for TMA staff was not confirmed.
3. Promote mainstreaming DRR through capacity enhancement of governmental officers	3.1. DRM workshops for relevant ministries	NIDM	• The NDMA mainly provides DRM training for federal government officers and politicians almost every year with the support of donors.
	3.2. DRM workshops for relevant departments of provincial governments	PDMAs	• It seems that the NDMA and PDMA provide DRM training with the support of donors, but the number is small.
	3.3. DRM workshops for district governments	DDMAs	• It seems that the NDMA and PDMA provide DRM training with the support of donors and others. • DRM training by DDMA was not confirmed.
	3.4. DRM subjects into curriculum of government training institutes	NIDM	• Minimal progress was made.
4. Develop the capacity of communities to cope with disasters	4.1. DRM workshops for community leaders	DDMAs, NGO	• As part of projects, NDMA has been conducting some community training. • No training was confirmed by DDMAs. • NGOs seem to be conducting these training, but it was not confirmed.
	4.2. Search and rescue training for members of community emergency response teams	District Fire Brigade, District Civil Defense, NGOs	• Rescue 1122 and Civil Defense in each province (the activities of each office at the district level were not confirmed) are conducting community rescue training.
5. Raise people's awareness of disaster management	5.1. Awareness campaigns	NIDM, GB/S/PDMAs, DDMAs, TMAs	• On National Resilience Day in October, NDMA and PDMAs have held ceremonies and awareness campaigns.
	5.2. Promotion of disaster education at schools	NIDM, Provincial Education Department	• NDMA has widely implemented the Pakistan School Safety Framework. • In the latter years, NDMA has promoted the inclusion of education on disaster risk reduction in school education.
	5.3. Promotion of disaster education in higher education	NIDM, Provincial Education Department, Universities	• NDMA staff provided some lectures. • There are not many universities offering courses on DRM, but there are several. (University of Peshawar, University of Balochistan, etc.)

3.2.6.2 Review Results

Several priority activities of the current NDMP have been implemented, which can be considered as achievements. It is evident that the volume of human resource development activities has increased compared to when the current NDMP was first formulated. However, since many human resource

development activities related to disaster management are conducted as part of the projects led by the international organizations and others, they are implemented according to the objectives of those projects, making it difficult to say they fully align with the goals of the current NDMP. For example, the NDMA at the federal level is carrying out training for the community level. On the other hand, any examples of activities that did not depend on projects and were carried out on their own budget were not confirmed, except for the awareness activities of National Resilience Day and the flood drills in Balochistan. In the annual reports of NDMA, it was stated that PC-1 was prepared and submitted to MoPS-PC several times to implement the Human Resource Development Plan of the current NDMP. However, it is assumed that PC-1 was not approved as a result.

Compared with human resource development activities related to general disaster management, regarding emergency response capacity improvement such as rescue training and firefighting training, organizations in charge of emergency response have been established in each province after the current NMDP was formulated, and training of their own personnel and training of the community have been carried out by them.

Regarding the organizations implementing human resource development activities, activities at the federal and provincial levels were confirmed by these organizations, however, activities at the district level (mainly by District Disaster Management Authority (DDMA) were not identified. In many provinces, DDMAs are not permanent organizations. Even if they are permanent organizations, they have only a few staff. As a result, it is difficult for DDMAs to take the initiative for human resource development activities.

In addition, since NIDM still has limited functions, there is no organization to confirm and coordinate human resource development activities implemented by each organization. Therefore, each organization implements the activities on its own, and NIDM does not provide advice on improving the contents of the activities.

Another activity that is rarely implemented is the addition of disaster management-related subjects to training at government training institutions. For the government to recognize the importance of implementing disaster management activities and to implement them, it is necessary that government officers deepen their understanding of disaster management. This activity is recommended to support that goal. JICA's "Project for Capacity Development of Disaster Management" also made similar recommendations.

As for the promotion of education on disaster risk reduction by educational institutions, for primary and secondary education, because information could not be obtained from the education departments of each province, whether the activities were implemented was not confirmed. Although the activities of each provincial government for higher education were not confirmed, it was confirmed that several universities and research institutes conducted research and education on disaster risk reduction.

3.2.7 Rainwater Drainage (Utilization Status, Achievement Status, Review Results, and Issues)

3.2.7.1 Positioning of Rainwater Drainage (Urban Flooding) In the Current NDMP

The achievement status of the current NDMP is reviewed by first summarizing the contents of the current NDMP regarding rainwater drainage (urban flooding).
The objective ii) “To mitigate damages from recurring disasters such as floods, urban flooding, earthquakes, tsunamis, droughts, landslides, avalanches, GLOFs, and cyclones with storm surges, etc. (Main report p. ES-2)" is one of the 4 key goals listed in the current NDMP, with a specific focus on reducing disaster related damage, including that caused by urban flooding.

It has been pointed out that the frequency of heavy rainfall and the intensity of rainfall are expected to increase due to the effects of climate change, and the disaster of urban flooding is likely to increase accordingly. In response to the increase in disaster risk due to climate change, the policy for reducing and minimizing damage in the National Climate Change Policy (2012) is cited. One of the 22 policy recommendations includes the review and upgrade of rainwater drainage

Standards for large cities such as Karachi and Lahore "Redesign and upgrade the storm drainage capacity of major cities, especially Karachi and Lahore, keeping in view the likely increase in climate change-related

short duration intense rainfall events. (Main report p. 22-23)."

The current NDMP has priority actions based on the basic policy classified into 10 interventions. Among them, the items related to urban flooding are extracted as follows.

Table 3.2.7 Policies and Priority Actions Related to Urban Flooding in the Current NDMP

Intervention	Strategies	Priority Actions /Programs	Responsible Organizations
2	2. Develop hazard specific contingency plans	2.1. Develop hazard specific contingency plans at national and provincial levels	NDMA, GB/S/PDMAs, Federal Ministries, Department and Authorities
3	2. Conduct detailed multi–hazard vulnerability and risk analysis/assessments at local levels	2.1. Preparation of vulnerability analysis and creation of hazard maps for selected districts and cities	NDMA in association with GB/S/PDMAs, DDMA, FFC, PMD, GSP and ERRA
4	1. Strengthen forecasting and early warning systems	1.4. Establishment of a flash flood* forecasting and warning system including local flash flood forecasting centres (LFFFC)	PMD in consultation with NDMA/GB/S/PDMAs
7	4. Implement appropriate structural measures in flood prone areas taking into account comprehensive and integrated flood management plans	4.2. Construction and rehabilitation of flood control/ mitigation structures	FFC/PID/WAPDA/PMDs /Planning Commission

**The early warning of Intervention-4 was investigated by replacing flash flood with urban flood (inland water) because there was no keyword of urban flood in the priority action on the hazard map.*

3.2.7.2 Review of Progress of the Current NDMP

In this project, the review was carried out by internet, literature survey and interview survey.

(1) Disaster Type Emergency Response Plan (Intervention2-2)

The NDMA is responsible for developing emergency response plans at the national level, while the PDMAs are responsible for developing plans at the provincial level. Although there seems to be no emergency response plan specific to urban flooding, emergency response plans for floods have been developed at the national and province levels, and some of them mentioned about the urban flooding.

Since 2008, the NDMA has annually developed the National Monsoon Contingency Plan at the National level, with the exception of 2010 and 2011. The plan, published in July 2023, provides a summary of the previous year's flood damage and disaster response initiatives, flood impact projections based on PMD's 2023 monsoon rainfall forecast, and response guidelines for each stage of the disaster management cycle (Mitigation, Preparedness, Response, Recovery). The plan is updated annually based on the information and lessons learned from the previous year's flood disaster response. The appendix includes records of the previous year's distribution of relief supplies, various disaster hazard maps at the district level, flood arrival times of major rivers, disaster status reporting formats for NDMA to collect disaster information, and telephone numbers of major contacts in the event of a disaster. Response guidelines for urban floods include the following:

Disaster Mitigation
- Identification of low-lying areas prone to flooding in urban areas

Preparedness
- Promoting understanding of flood management, floodplain development regulations, and effective urban planning through human resource development by local governments
- Widening rainwater drainage channels, dredging, and removal of obstacles to reduce the risk of urban flooding
- Establishing inspection, management, and maintenance plans for drainage stations and emergency response plans
- Implementation of training and re-education programs to improve the skills and knowledge of technical personnel involved in flood management and drainage
- Supply of reliable backup power to prevent the operation of sewage treatment plants, drainage station pumps, and other facilities from shutting down in the event of a power failure to enable efficient drainage and sewage management in the

event of a flood
- Establishment of an expert committee, consisting of stakeholders, experts, and volunteers, at the municipal level to plan and implement emergency response plans, especially in large cities, to promote coordinated and proactive efforts to respond to floods in urban areas

Response
- Establishment of coordination mechanisms for rapid and adequate response to urban flooding
- Mobilization of search and rescue teams equipped with flood rescue equipment and urban flood search and rescue tools
- Prioritize the rescue and evacuation of flooded buildings, vehicles and others exposed to hazards
- Search and rescue operations in coordination with local authorities, emergency services, urban defense departments and urban flood search and rescue teams
- Request air support through NDMA when aerial search and rescue operations are needed in urban flood affected areas
- Establish communication systems to provide real-time updates and instructions to affected areas
- Conduct post-disaster assessments to identify areas for improvement in urban flood search and rescue strategies

Disaster Mitigation (Relief)
- Establish temporary shelters and evacuation centers with necessary equipment for urban displaced people
- Ensure immediate medical assistance by establishing medical response teams and deploying mobile medical units to affected areas
- Provide emergency relief supplies such as food, clean water and basic necessities to affected areas in partnership with relevant departments and organizations
- Conduct rapid damage surveys to identify critical infrastructure and prioritize recovery efforts
- Clean, unclog and repair drainage systems to mitigate urban flooding
- Coordinate with relevant departments to ensure the restoration of essential services such as electricity, water supply and communications networks
- Conduct public awareness campaigns to educate communities about urban flood risk, safe sanitation practices and disease prevention
- Provide support for the rehabilitation and reconstruction of damaged homes, utilities and infrastructure in urban areas

Early Recovery
- Conduct emergency assessments to identify early recovery needs in urban areas, focusing on emergency evacuation, temporary shelter and basic necessities
- Support the restoration of basic services such as water supply, sanitation and electricity to affected urban residents
- Support the restoration of damaged housing and infrastructure, prioritizing the most vulnerable communities
- Promoting community-led initiatives to clean up urban wastewater systems and manage waste to reduce future flood risk
- Supporting local businesses and economic recovery through financial assistance and revitalization programs
- Strengthening collaboration and cooperation among government agencies, NGOs, local community organizations and other stakeholders for effective early recovery response

Similarly, province-level contingency plans have been developed by the PDMAs, such as Punjab's Flood Contingency Plan (2021), and Sindh Provincial Monsoon/Floods Contingency Plan (2012).

(2) Multi-Hazard, Vulnerability and Risk Assessment (Intervention 3-2)

PMD and the relevant stormwater drainage department are responsible for conducting hazards, vulnerability and risk assessments for urban flooding caused by inland water.

According to a consultation with PMD Islamabad on September 15, 2023, PMD is also responsible for urban flood hazard assessment, however, progress has been limited due to budget constraints.

In many cities except Karachi, WASA is the department in charge of stormwater drainage, and information on vulnerability such as areas prone to inland water damage and inundation depth during past floods has been organized in the stormwater drainage MP. On the other hand, there is no information leading to economic evaluation such as damage amount, so it is considered that risk assessment has not progressed. In addition, inland water inundation simulation modeling rainfall, topography and stormwater drainage network have not been carried out, and inland water hazard maps have not been developed yet.

(3) Strengthening Early Warning System (Intervention 4-1)

For flood forecasting, the PMD is the responsible ministry, and only they are authorized to issue weather and flood forecasts. The NDMA also urges PDMAs (Provincial), GDMA (Gilgit-Baltistan) and SDMA (State) to refrain from issuing warnings based on open-source weather information.

During a consultation with PMD Islamabad on September 15, 2023, it was confirmed that PMD-FFD

also undertakes urban flood forecasting; however, a unified system for the entire country is currently lacking. Currently, there is only one Urban Flooding Center for Islamabad, which was established with JICA support. PMD would like to establish such facilities in other major cities, but due to lack of budget, planning and implementation are not in sight. Urban floods occur almost every year in major cities such as Karachi, Lahore, Quetta, Hyderabad and Faisalabad, but there is no pre-warning for them.

(4) Implementation of Appropriate Structural Countermeasures Considering Comprehensive and Integrated Flood Management Plan (Intervention 7-4)

Regarding structural countermeasures for urban inland water inundation, the respective stormwater drainage department in each city have developed Master Plans and are implementing structural countermeasures accordingly. Specifically, the following structural countermeasures are implemented.

- Renovation of existing drainage channel (Cleaning, widening, networking, etc.)
- Renovation of existing pumping station (Repairing, extension, etc.)
- Construction of new drainage channels
- Construction of new pumping stations
- Construction of rainwater storage facilities

3.2.8 EWS (Utilization Status, Achievement Status, Review Results, and Issues)

The Early Warning System (EWS), which encompasses multiple types of disasters, is summarized here as an independent item. PMD and NDMA are the main organizations responsible for Early Warning Systems, With FFC, WAPDA, and PID are also playing partial roles. A summary of interview findings with these organizations is provided below.

3.2.8.1 Overview of EWS

(1) Types and Overview of EWS

Table 3.2.8 shows the types and overview of EWS existing in Pakistan. The contents of each EWS are summarized in the following page.

Table 3.2.8 Types and Overview of EWS

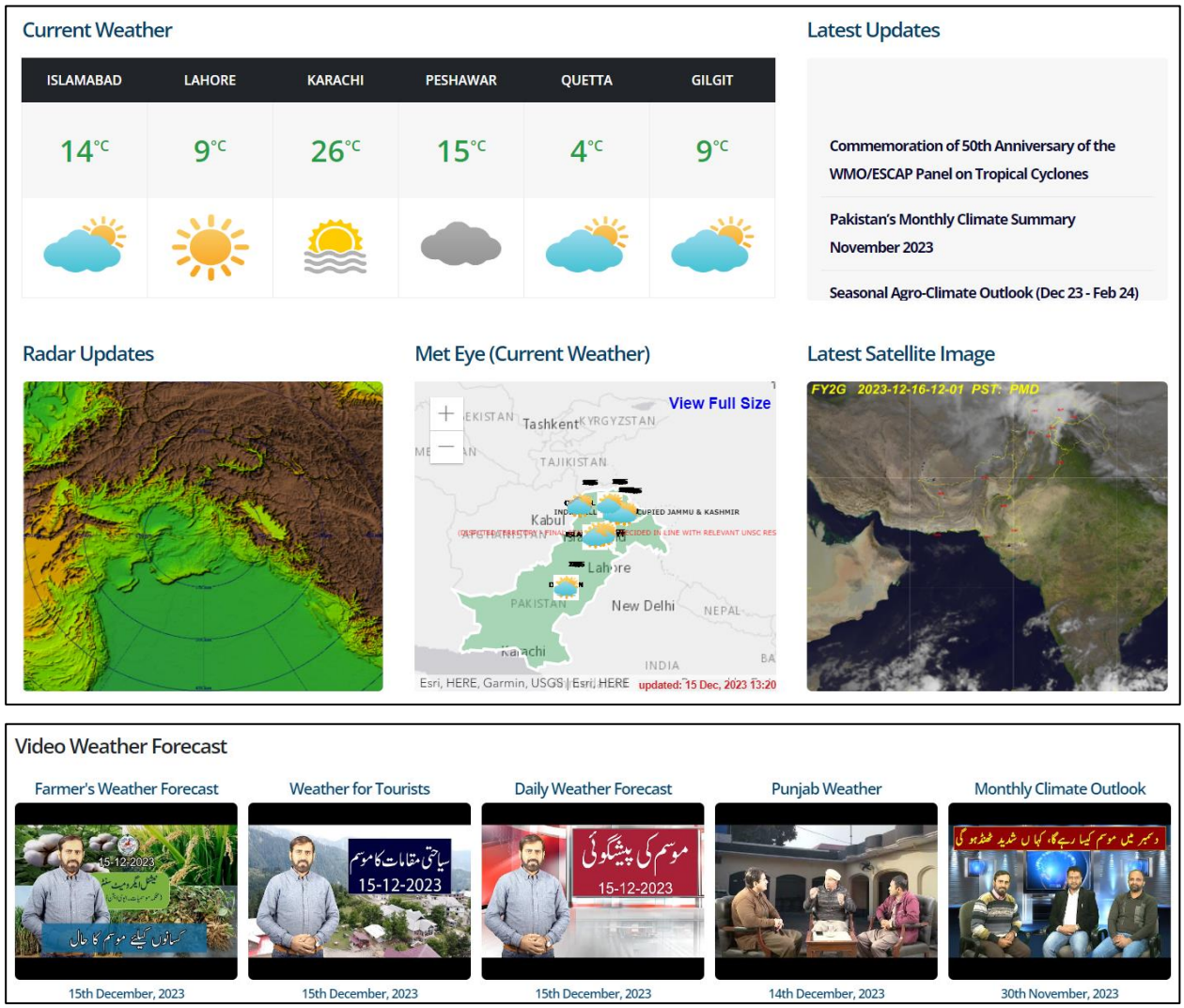
Types of EWS	Overview
Numerical Weather Forecast System	**Numerical Weather Prediction System** Weather prediction is carried out using numerical weather prediction model with the weather observation network owned by PMD as input condition. This system also includes the following two systems. ・Heavy Rain Warning System ・Heat Wave Early Warning System
Flood Early Warning System	**Flood Early Warning System** The forecast of the river discharge is carried out using the runoff model, river channel model, empirical formula, etc. as input condition of the rainfall of the weather forecast.
Earthquake & Tsunami Early Warning System	**Earthquake and Tsunami Early Warning System** Earthquake monitoring is carried out. Tsunami prediction (simulation) is also carried out in case of earthquake.
Cyclone Early Warning System	**Cyclone Early Warning System** Tropical Cyclone Warning Center (TCWC), Karachi in PMD issues warnings on cyclone through PMD's website.
GLOF Early Warning System	**Glacial Lake Outburst Flood (GLOF) Early Warning System** Based on information on possible glacial lakes outburst and temperature information, alert for areas where GLOF may occur.
Drought Monitoring & Early Warning System	**Drought Monitoring and Early Warning System** National Drought Monitoring Center, Islamabad in PMD monitors drought conditions and issues bulletins and recommendations.

Source: Prepared by the project team based on PMD introductory presentation

(2) Numerical Weather Forecast System & Flood Early Warning System

Numerical weather prediction systems and Flood Early Warning System are summarized below. Weather and flood information predicted by PMD-FFD are published on the PMD website.

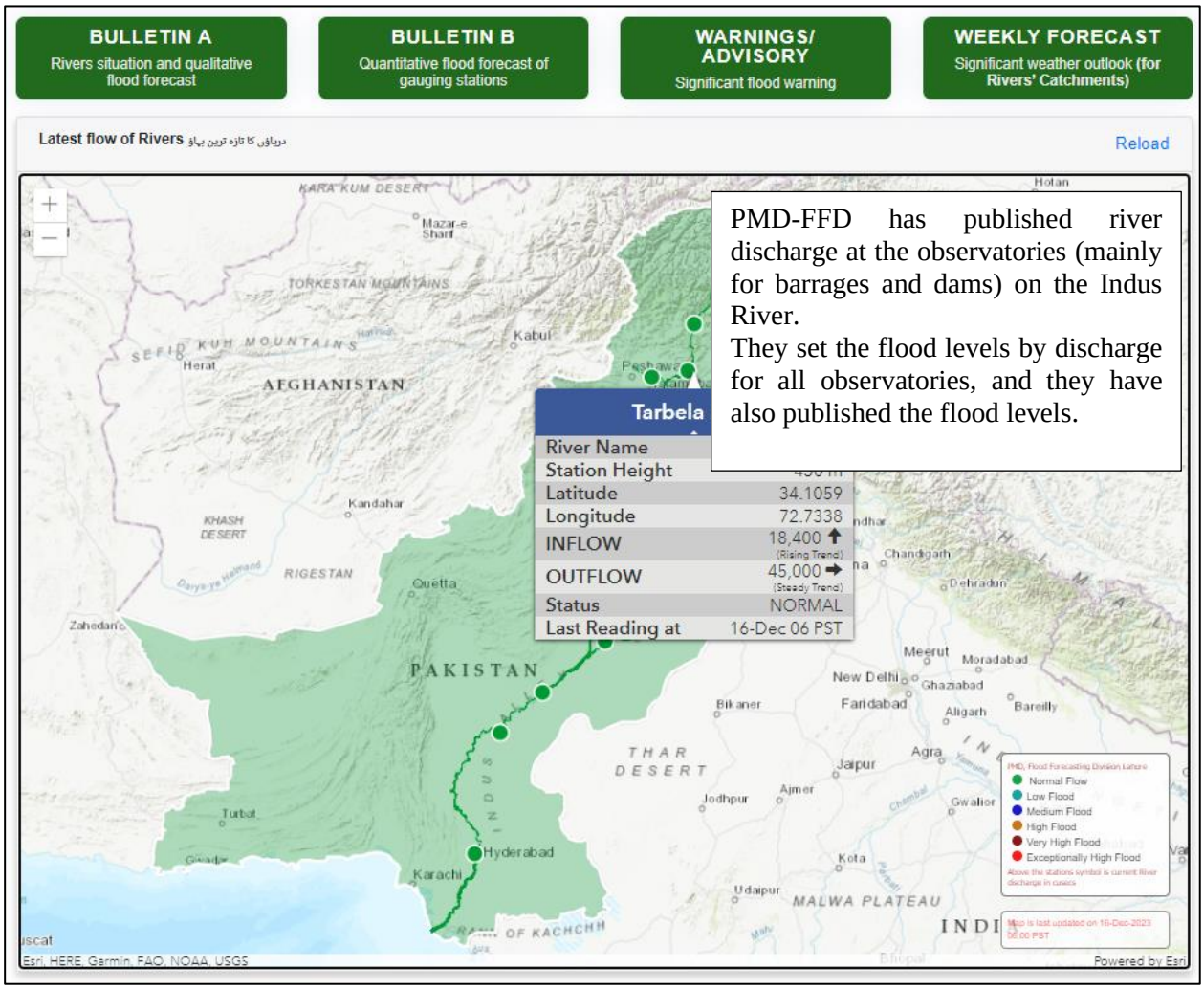
Figure 3.2.11 shows the PMD website. Information such as current weather, update information, radar information, satellite images, and weather forecast videos are published here.



Source: PMD Home Screen (https://www.pmd.gov.pk/en/)

Figure 3.2.11 PMD Website

Figure 3.2.12 shows the PMD-FFD website. The map shows river discharge in Barrages and Dams across six levels, ranging from Normal Flow to Exceptionally High Flood., Additional information is available by clicking on the Barrage or Dam points.. Qualitative information on river floods can be shown in “Bulletin A”, quantitative information on river floods in “Bulletin B”, warnings and advisories in “Warnings/Advisory”, and weekly forecast information in “Weekly Forecast”.



Source: PMD FFD Lahore Home Screen (https://ffd.pmd.gov.pk/home)

Figure 3.2.12 (1) PMD-FFD Website

Bulletin A: Rivers situation and qualitative flood forecast

Flood Season-2023 has been concluded. This is the last Flood Bulletin of Flood Season-2023, consequently no Flood Bulletin will be issued till 14th June-2024.

GOVERNMENT OF PAKISTAN
PAKISTAN METEOROLOGICAL DEPARTMENT
FLOOD FORECASTING DIVISION
46-JAIL ROAD LAHORE-54000

Dated: - 15th October, 2023
Time: - 10:40 hours (PST)

BULLETIN No. A-123/23

1. President Secretariat, Islamabad.
2. PM House, Islamabad.
3. Federal Minister for Water Resources, Islamabad.
4. Governor (All Provinces).
5. Chairman NDMA, Islamabad.
6. Chief Minister (All Provinces).
7. PM Secretariat, Islamabad.
8. Secretary, Aviation Division, Islamabad.
9. Secretary, Ministry for Water Resources, Islamabad.
10. Secretary, Ministry of Information, Islamabad.
11. Chairman NHA, Islamabad.
12. Chief Secretary (All Provinces).
13. PDMA'S (D.G'S, All Provinces & A.JK).
14. Chairman, Federal Flood Commission, Islamabad.
15. Chairman, Indus River System Authority, Islamabad.
16. Chairman, PCRWR, Islamabad.
17. The Secretary General, Red Crescent, Islamabad.
18. Secretary, Irrigation Department, (All Provinces).
19. D.G. Engineers, Eng. Directorate GHQ, Rawalpindi.
20. Pakistan Commissioner for Indus Waters (PCIW), Islamabad.
21. Chief Engineer, Mangla/Tarbela/Chashma.
22. Chief Engineer, Irrigation (All Provinces).
23. Chief Engineer, (All Barrages).
24. Chief Engineer, C.D.O. (Muzaffarabad).
25. Commissioner for Indus Waters, Lahore.
26. All other concerned.

SUBJECT: -RIVERS SITUATION AND QUALITATIVE FLOOD FORECAST:

I: HYDROLOGICAL SITUATION at 0600 PST:

All the Major Rivers are flowing in the state of their normal flow. Flows are likely to increase in all the Major Rivers, while Flash flooding is expected in Rivers Swat & Panjkora along with their associated tributaries during 15th to 17th October, 2023.

II: METEOROLOGICAL FEATURES:

a. Yesterday's strong trough of westerly wave over Northern parts of the country persists.
b. Seasonal low lies over Northeastern Balochistan.

III: WEATHER FORECAST FOR NEXT 24-HOURS:

Scattered wind thunderstorm/rain of moderate intensity with isolated heavy falls is expected over the upper catchments of all the Major Rivers along with Islamabad, Khyber Pakhtunkhwa & Punjab. Scattered wind thunderstorm/rain of moderate intensity is also expected over upper Sindh and Northeastern Balochistan

IV: WEATHER OUTLOOK FOR NEXT 48 HOURS:

Wet spell is likely to continue.

V: RIVERS CATCHMENTS RAINFALL RECORDED (MM) DURING LAST 24 HRS (UPTO 0800 PST):

Indus:	Malam Jabba=17, Bannu=13, Kohat (Airbase)=09, Saidu Sharif & Besham=07(each), Khawagoobo Bridge= 06, Bagrote & Parachinar=03(each), Murree & Dir=02(each), Bosak Bridge, Cherat & Mianwali (Airbase)=01(each), Islamabad (Zeropoint, Chaklala Airbase), Astore, Arandu Nullah, Chowni Bridge Chitral, Chilas, Peshawar (Airbase) & Bhakkar=Trace(each)
Jhelum:	Chakwal=52, Mandibahauddin=14, Mangla=12, Qilla Rohtas & Palandri=10(each), Kallar=08, Jhelum=06, Kotli & Talhatta=05(each), Muzaffarabad City & Rawalakot=02 (each)
Chenab:	Sialkot (Cantt=19, Airport=15), Gujrat & Hafizabad=15(each), Gujranwala=11, Marala=Trace
Ravi:	Lahore [Qurtba chowk=112, Gulshan e Ravi=91, Iqbal Town=86, Airport & Shahi Qilla=83(each), Lucksham=77, Samanabad=70, Township=60, Jail Road=59, Gulburg=43, Upper Mall=42, Johar Town=41, Mughal Pura=39, Taj Pura=22, Shahdara=20, Misri Shah=19], Narowal= 63, Kotnaina=05, Shakar Garh=02, Faisalabad=Trace
Sutlej:	Kasur=12
Others:	-Nil-

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FFD Lahore

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Chief Meteorologist: 04299200208 & Duty Officer 24/7 hrs: 04299200139, 04299205367-70 and Fax:04299200209 & 5367

Bulletin B: Quantitative flood forecast of gauging stations

Flood Season-2023 has been concluded. This is the last Flood Bulletin of Flood Season-2023, consequently no Flood Bulletin will be issued till 14th June-2024.

GOVERNMENT OF PAKISTAN
PAKISTAN METEOROLOGICAL DEPARTMENT
FLOOD FORECASTING DIVISION
46-JAIL ROAD LAHORE-54000

BULLETIN No. B-123/23

Dated: - 15th October, 2023
Time: - 10:40 hours (PST)

I: WEATHER/RAINFALL FORECAST FOR NEXT 24-HRS:

RIVERS	Upper Catchments (Above Rim Stations)	Lower Catchments (Below Rim Stations)
INDUS	Scattered wind thunderstorm/rain	Scattered wind thunderstorm/rain (Rawalpindi Division)
JHELUM	-do-	-do- (Sargodha Division)
CHENAB	-do-	-do- (Gujranwala Division)
RAVI	-do-	-do- (Lahore Division)
SUTLEJ	-do-	-do- (Bahawalpur Division)

RESERVOIRS LEVEL (FEET) at (0600 PST)

TARBELA
Current: 1528.98
Max. Conservation: 1550.00

MANGLA
Current: 1213.75
Max Conservation: 1242.00

II: QUANTITATIVE FLOOD FORECAST OF GAUGING STATIONS (IN THOUSANDS OF CUSECS)

RIVERS	Stations	Design Capacity	Actual Observations at 0600 PST Inflow	Actual Observations at 0600 PST Outflow	Quantitative Forecast for Next 24-hrs (Inflow)	Qualitative Forecasted Flood Level (Inflow)	Max Flood Peaks (In Thousands of Cusecs) Historical		Flood Season 2023
INDUS	Tarbela	1500	38.0	29.1	35-40	Below Low	832.0	(2010)	398.0
	Kalabagh	950	60.9	53.9	No Sig. change	-do-	950.0	(1942)	373.4
	Chashma	950	57.4	56.4	-do-	-do-	1038.9	(2010)	419.7
	Taunsa	1100	57.9	57.9	-do-	-do-	960.0	(2010)	439.9
	Guddu	1200	65.3	57.0	-do-	-do-	1199.7	(1976)	482.9
	Sukkur	1500	56.5	20.9	-do-	-do-	1166.6	(1986)	448.1
	Kotri	875	19.5	2.4	-do-	-do-	980.3	(1956)	258.2
KABUL	Nowshera	-	10.4	10.4	10-20	-do-	391.3	(1990)	100.6
JHELUM	Mangla	1060	10.0	32.7	10-30	-do-	1090.0	(1992)	140.0
	Rasul	850	37.3	15.8	No Sig. change	-do-	952.2	(1992)	37.0
CHENAB	Marala	1100	11.8	4.0	15-30	-do-	1100.0	(1957)	211.0
	Khanki	1100	8.7	1.9	No Sig. change	-do-	1086.5	(1959)	211.5
	Qadirabad	807	20.3	2.3	-do-	-do-	948.5	(1992)	203.7
	Trimmu	645	17.5	2.2	-do-	-do-	944.0	(1928)	125.2
	Punjnad	700	11.3	0.0	-do-	-do-	802.5	(1973)	159.5
RAVI	Jassar	275	1.3	1.3	-do-	-do-	680.0	(1955)	71.0
	Shahdara	250	1.8	1.8	-do-	-do-	576.0	(1988)	39.8
	Balloki	225	24.5	0.0	-do-	-do-	389.8	(1988)	74.2
	Sidhnai	150	8.5	0.0	-do-	-do-	330.2	(1988)	58.3
SUTLEJ	G.S. Wala	--	0.0	0.0	-do-	-do-	**	**	278.3
	Sulemanki	325	18.6	5.2	-do-	-do-	598.9	(1955)	191.1
	Islam	300	1.0	0.0	-do-	-do-	492.6	(1955)	151.9

Isolated= 20 to 30% of area Scattered= 30 to 50% of area | Widespread= 90 to 100% of area R= Rising, F= Falling
Note: Concerned Departments may contact FFD, Lahore for rectification up-dation (if any)

(Chaudhary Muhammad Aslam)
Chief Meteorologist
FFD Lahore

www.ffd.pmd.gov.pk | Email: ffdlhr@yahoo.com
Chief Meteorologist: 04299200208 & Duty Officer 24/7 hrs: 04299200139, 04299205367-70 and Fax:04299200209 & 5367

Warnings/Advisory

The sample could not be posted because no warnings/advisories were issued at the time of the survey.

Weekly Forecast: Singnificant weather outlook

- **LAST BULLETIN**

GOVERNMENT OF PAKISTAN
PAKISTAN METEOROLOGICAL DEPARTMENT
FLOOD FORECASTING DIVISION
46-JAIL ROAD LAHORE-54000

BULLETIN No. C-120/23

Dated: - 12th October, 2023
Time: - 10:45 hours (PST)

SUBJECT: - WEEKLY SIGNIFICANT WEATHER OUTLOOK FOR RIVERS CATCHMENTS (13th October to 15th October, 2023)

- **RAINFALL OUTLOOK:**

No high flood generating rainfall is expected over the upper catchments of all the Major Rivers.

- **FLOOD SITUATION IN MAJOR RIVERS:**

No high flood situation is expected in any of the Major Rivers.

Note: This forecast is updated on daily basis.

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FFD Lahore

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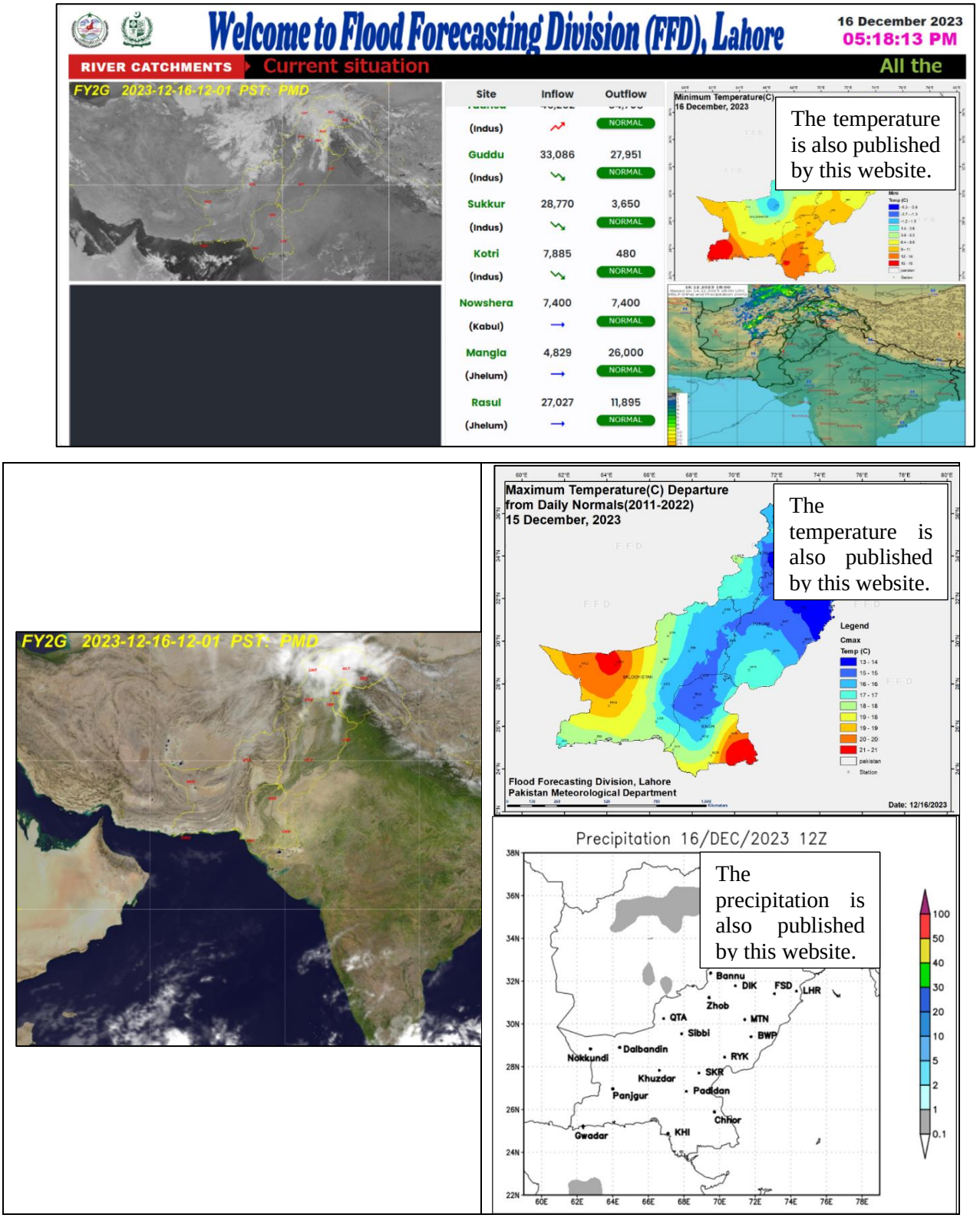
Chief Meteorologist: 04299200208 & Duty Officer 24/7 hrs: 04299200139, 04299205367-70 and Fax:04299200209 & 5367

Source: PMD FFD Lahore home screen (https://ffd.pmd.gov.pk/home)

Figure 3.2.12 (2) PMD-FFD Website

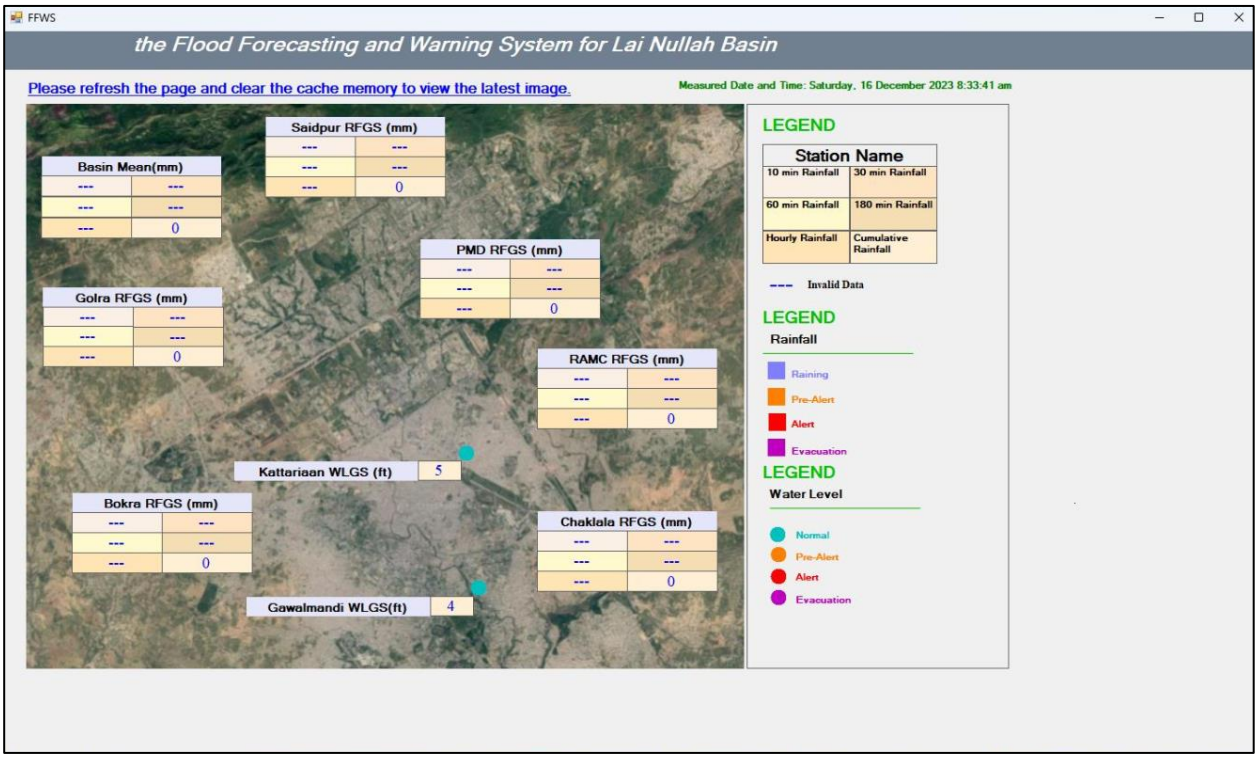
Figure 3.2.13 shows the dashboard screen of PMD-FFD website. This platform provides satellite images, inflow and outflow data at Barrages and Dams, temperature and rainfall distribution, along with other relevant information.



Source: PMD FFD Lahore Dashboard (https://ffd.pmd.gov.pk/dashboard)

Figure 3.2.13 Dashboard Screen of PMD-FFD Website

Figure 3.2.14 shows the website of the Lai Nullah Basin Flood Prediction Warning System, established with a grant from Japan. It publishes rainfall and discharge data for the Lai Nullah basin located in Islamabad and Rawalpindi.

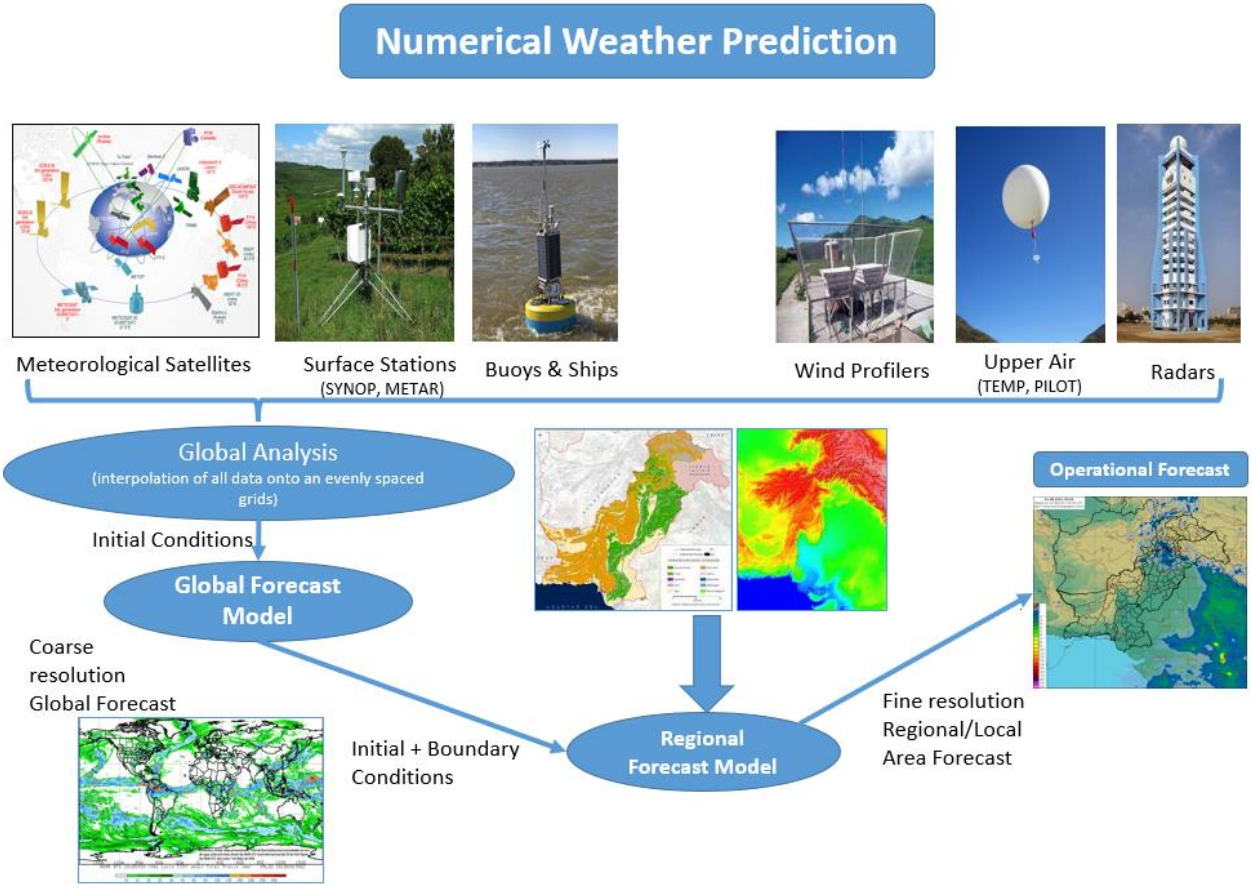


Source: PMD Project for Strengthening Flood Risk Management in Lai Nullah Basin (https://ffws.pmd.gov.pk/rtr.html)

Figure 3.2.14 Lai Nullah Basin Flood Prediction Warning System

Figure 3.2.15 shows an overview of numerical weather prediction. Weather forecasts are generated using a global model with the observational data such as ground observation stations, radar rainfall, satellite images, etc., and then downscaled and high-resolution weather prediction is performed using a regional forecast model.

Flood prediction is performed using IFAS Model, FUSE Model, PHAST Model, regression model, etc. with the above weather prediction as input conditions.



Source: PMD

Figure 3.2.15 Overview of Numerical Weather Prediction

(3) Earthquake & Tsunami Early Warning System

The earthquake monitoring system and the tsunami early warning system are also available on the PMD website.

Figure 3.2.16 shows the earthquake historical information page from the earthquake monitoring system. It provides information such as the date, region, magnitude, depth of epicenter, and latitude and longitude of the epicenter. The monthly earthquake occurrence chart is also available as shown in Figure 3.2.17.

Figure 3.2.18 shows the tsunami early warning system. In case of emergency, this warning is released to the coastal areas of the Arabian Sea. The tsunami simulation is performed as shown in Figure 3.2.19.

Events (10187)

Date	Time (utc)	Region	Magnitude	Depth (km)	Latitude	Longitude	Mode	Map
16/12/2023	06:45:58	Southern Sumatra, Indonesia	5.2	66	0.32 S	98.87 E	A	-
16/12/2023	06:10:54	South of Fiji Islands	5.4	535	24.28 S	179.26	A	-
16/12/2023	04:57:51	Tajikistan	3.6	89	37.72 N	73.05 E	M	-
16/12/2023	01:22:54	New Ireland Region, P.N.G	4.8	377	3.38 S	152.15 E	A	-
15/12/2023	19:55:49	Hindu Kush Region,Afghanistan	4.1	130	36.57 N	70.60 E	M	-
15/12/2023	16:35:35	East of Philippine Islands	4.8	3654	7.11 N	128.18	A	-
15/12/2023	16:05:29	Philippine Islands Region	5.3	10	8.93 N	127.60 E	A	-
15/12/2023	03:43:56	32 Km West of Mastung	3.9	18	29.80 N	66.51 E	M	-
15/12/2023	03:17:57	Hindu Kush Region,Afghanistan	3.7	170	36.57 N	70.25 E	M	-
15/12/2023	01:43:48	Tajikistan-Xinjiang Border Region	3.9	170	38.38 N	74.45 E	M	-
15/12/2023	00:02:09	Banda Sea	5.1	155	6.97 S	129.60 E	A	-
14/12/2023	22:44:08	Philippine Islands Region	4.8	63	8.63 N	127.51 E	A	-
14/12/2023	22:19:22	Eastern Kashmir, 60 Km NE of Jammu	3.2	45	33.11 N	75.30 E	M	-
14/12/2023	21:49:27	Afghanistan-Tajikistan Border Region	3.8	180	36.73 N	71.47 E	M	-
14/12/2023	20:45:09	Hindu Kush Region,Afghanistan	3.8	210	35.89 N	66.18 E	M	-

Event No. 271

Region: Eastern Kashmir, 60 Km NE of Jammu

Date	Origin Time	Magnitude	Latitude	Longitude	Depth
14 December, 2023	22:19:22	3.2	33.11 N	75.30 E	45 km

Station	Distance (°)	Azimuth (°)	Phase	Time (utc)
MUZF	218.0	310	P	22:19:52
MUZF	218.0	310	S	22:20:17
SKRD	244.0	7	P	22:19:57
SKRD	244.0	7	S	22:20:22

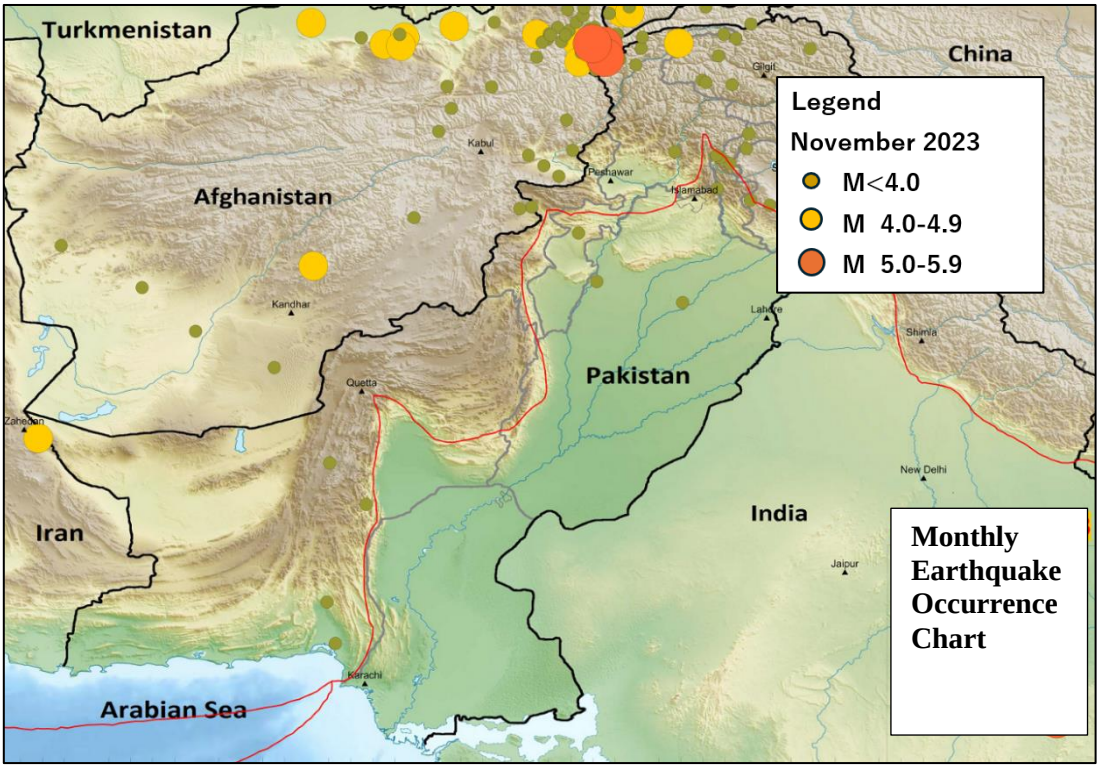
Event No. 272

Region: Banda Sea

Date	Origin Time	Magnitude	Latitude	Longitude	Depth
15 December, 2023	00:02:09	5.1	6.97 S	129.60 E	155 km

Source: PMD National Seismic Monitoring Centre, Islamabad (https://seismic.pmd.gov.pk/index.php)

Figure 3.2.16 Earthquake Monitoring System (Historical Earthquake Information)



Source: PMD National Seismic Monitoring Centre, Islamabad (https://seismic.pmd.gov.pk/index.php)

Figure 3.2.17 Earthquake Monitoring System (Monthly Earthquake Occurrence Chart)

(SEVERE TSUNAMI THREAT)
TSUNAMI BULLETIN-01
Issued at 11:05:00 PST

National Tsunami Warning Centre (NTWC), Karachi of Pakistan Meteorological Department has recorded an Earthquake with the following preliminary seismological parameters:

Origin time 08-09-2019 at 11:00:00 PST
Magnitude: 8.7
Depth: 10 Km
Latitude: 24.80° N
Longitude: 62.20° E
Location: Off Coast of Pakistan

EVALUATION:

. Based on the historical data, earthquakes of this size may cause wide-spread inland damages and Destructive Tsunami generation in the Arabian Sea and along coastline of Pakistan. However, it is not known that a tsunami has been generated. This bulletin is based only on the earthquake evaluation. Monitoring of Sea Level Gauges is under way to determine if a tsunami has been triggered. **People in coastal areas of Balochistan especially Jiwani, Gwadar, Pasni, Ormara and neighborhood are STRONGLY ADVISED TO EVACUATE IMMEDIATELY** to higher grounds or go far inland. **People in coastal areas of Sindh are advised to standby for evacuation.** Boats and Ships at sea are advised to stay in the deeper parts of the sea until the threat is over. If there is sufficient time, boats and ships in harbors and bays are advised to go to the deeper parts of the sea until the threat is over.

UPDATES

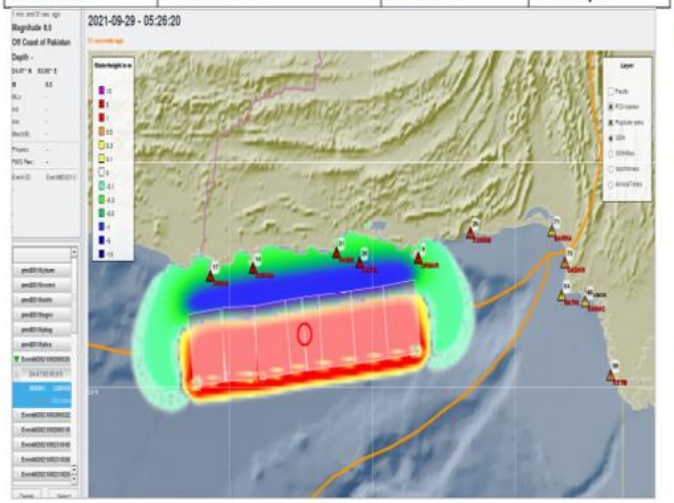
Additional bulletins will be issued by NTWC, Karachi for this event as more information becomes available.

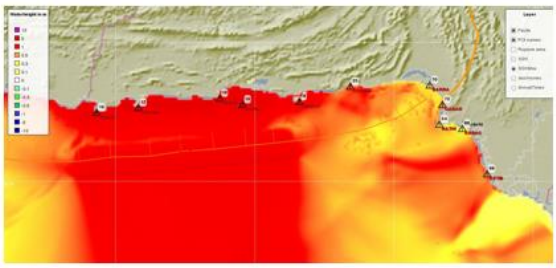
Sd/=
Duty Seismologist

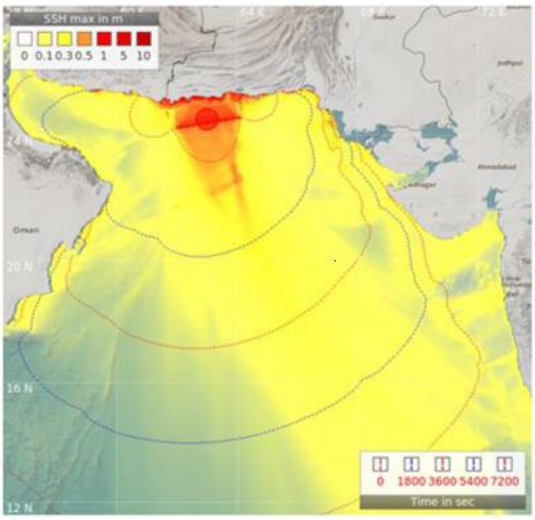
Source: PMD

Figure 3.2.18 Tsunami Early Warning System (Warning)

Coastal Point	Maximum Wave Height(m)	Arrival Time (min.)	Wave Speed (m/s)
Gwadar	4.3 meters	14 min	4-6 m/s
Pasni	3.0 meters	18 min	4-6 m/s
Jiwani	3.4 meters	17 min	4-6 m/s
Ormara	3.5 meters	9 min	4-6 m/s
Karachi	0.4 meters	87 min	4-6 m/s







Makran Subduction Zone Mw 8.5 Simulation (Arrival Times (minutes)

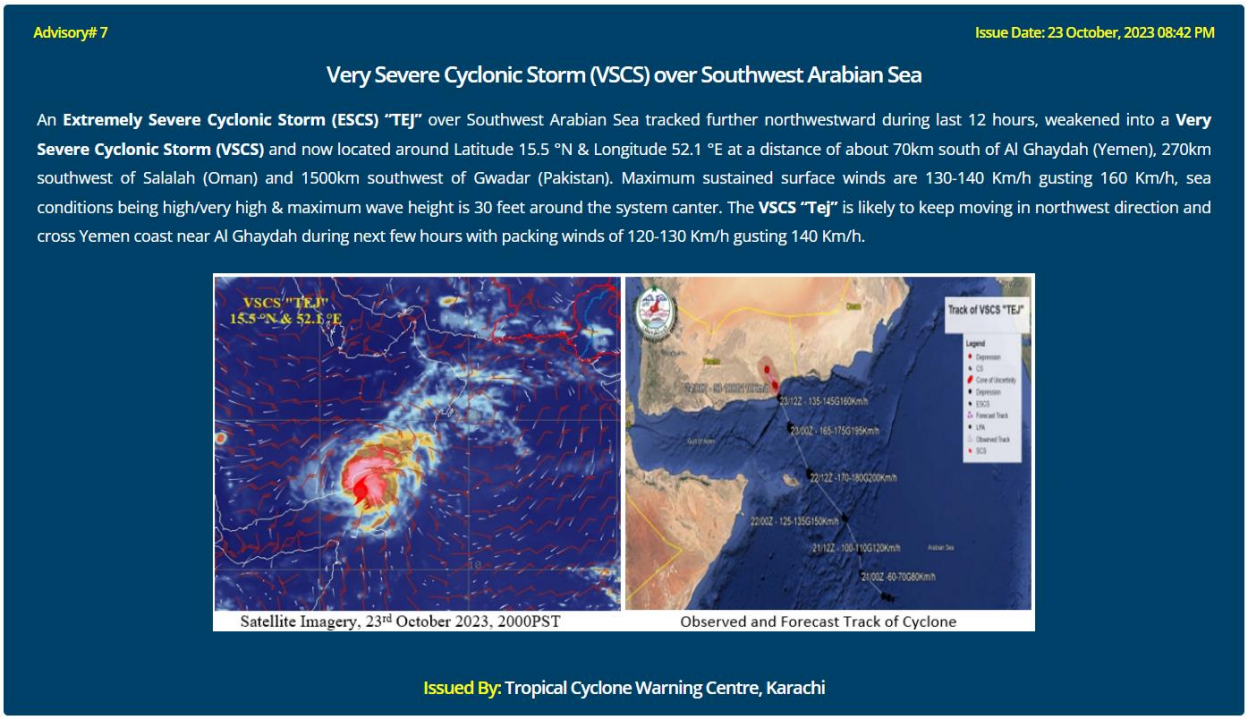
Source: PMD

Figure 3.2.19 Tsunami Simulation

(4) Cyclone Early Warning System

The Cyclone Early Warning System monitors cyclones originating from the northern Arabian Sea and issues warnings to all disaster management stakeholders. Recently, PMD has successfully tracked and forecasted Severe Cyclonic Storm "Shaheen" (September 30 to October 4, 2021).

Figure 3.2.20 shows the recommendations within the Tropical Cyclone Warning System, covering six stations (Tej, Shaheen, Gujab, Tauktae, Kyarr, Maha).



Source: PMD Tropical Cyclone Warning System
(https://www.pmd.gov.pk/en/tropical-cyclone-warning-centre/advisories.php?cyclone=TEJ)

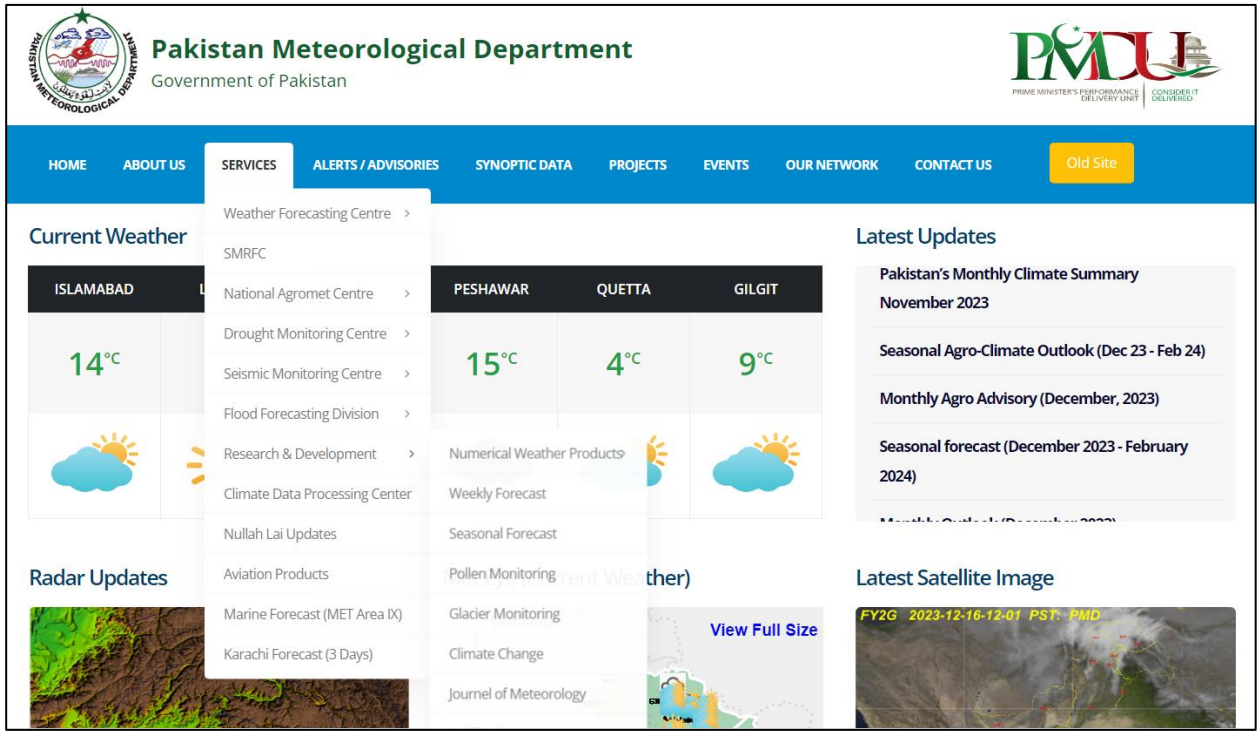
Figure 3.2.20 Recommendations in the Tropical Cyclone Warning System

(5) GLOF Early Warning System

The GLOF Early Warning System provides early warning of glacial lake outburst floods to stakeholders at the local, provincial, and federal levels. The GLOF Early Warning System has been established in three valleys, Bindogal and Golain in Chitral and Bagrot in Gilgit. The GLOF event at Shishper Glacier in Karimabad, Hunza, was dealt with the support of GBDMA and other relevant organizations.

WAPDA conducts more advanced research on glaciers, glacial lakes, and GLOF compared to PMD. It is therefore recommended that PMD and WAPDA collaborate in developing an effective GLOF early warning system in the future.

Figure 3.2.21 shows the PMD Home Screen. Although a "Glacier Monitoring" section is exist, it is not accessible as of December 16, 2023.



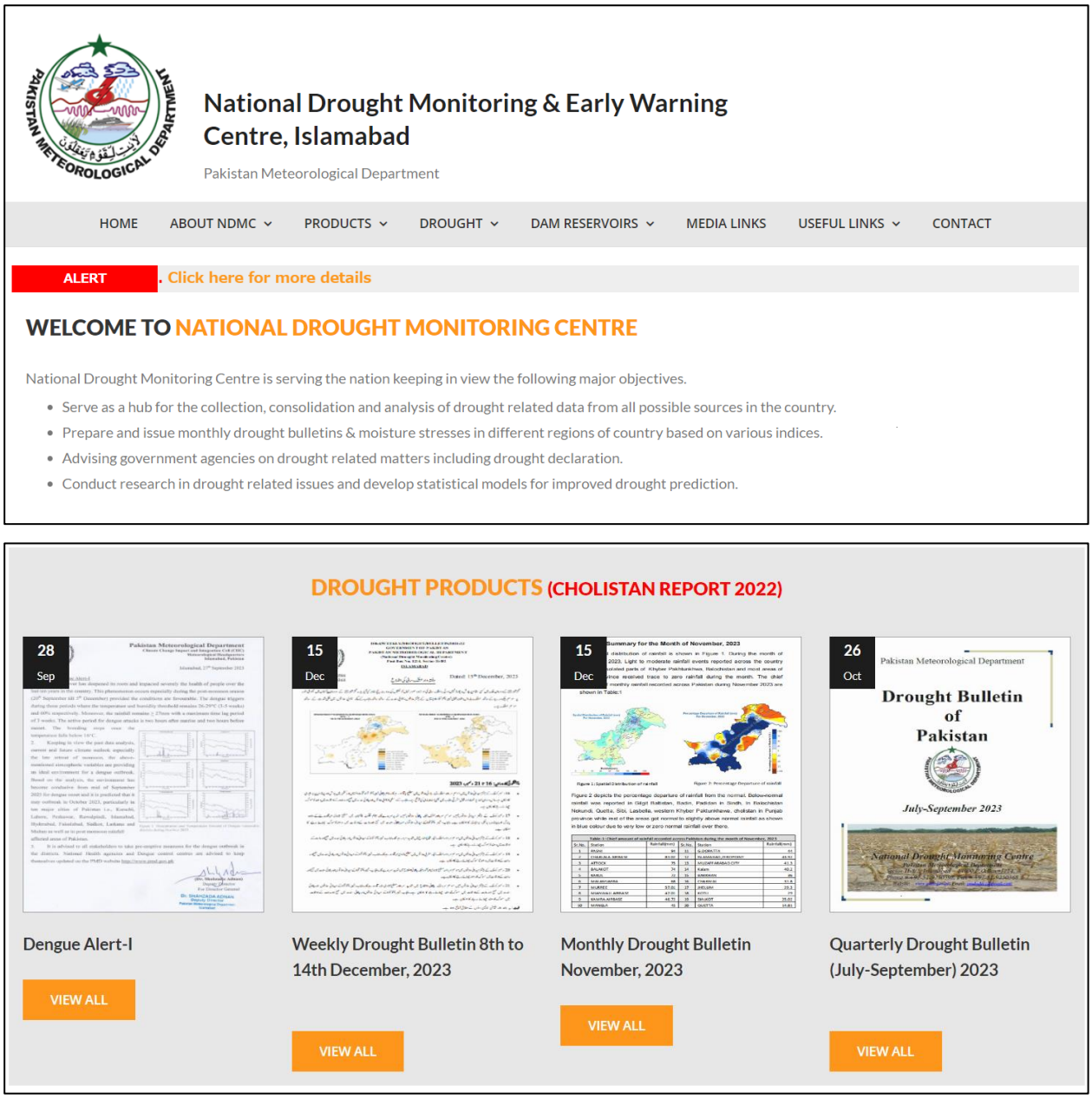
Source: PMD Home Screen (https://www.pmd.gov.pk/en/)

Figure 3.2.21 PMD Home Screen (Glacier Monitoring)

(6) Drought Monitoring & Early Warning System

National Drought Monitoring Center, Islamabad monitors drought conditions in Pakistan and issues drought bulletins and recommendations to relevant departments at the federal and provincial levels.

Figure 3.2.22 shows the website of National Drought Monitoring Center, Islamabad, Figure 3.2.23 shows the part of the drought monitoring report, and Figure 3.2.24 shows the drought monitoring map (comparison of rainfall distribution with previous years). Both are available on this website.



Source: National Drought Monitoring Centre, Islamabad (https://ndmc.pmd.gov.pk/new/advisories.php)

Figure 3.2.22 Website of the National Drought Monitoring Center, Islamabad

Government of Pakistan
Ministry of Aviation
Pakistan Meteorological Department
National Drought Monitoring & Early Warning Centre (NDMC)
Meteorological Headquarters
Islamabad, Pakistan

DR-5/DRT/ADV/ALERT/2021-22

Islamabad, 7th September, 2023

SUBJECT: DROUGHT WATCH (2023)

As predicted by the Pakistan Meteorological Department (PMD), the country overall received normal rainfall up till now. August 2023 was the 2nd driest month in Pakistan in the last 63 years as -65.9% below normal rainfall was recorded in the country. The highest deficit in rainfall was experienced in Sindh and Balochistan during the month. The rainfall departure (%) detail is as under:

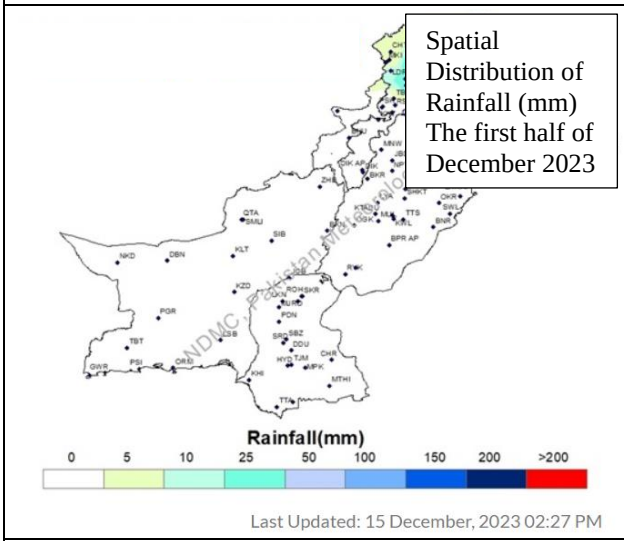
Table-1 Area weighted rainfall departure (%) from June till 7th September 2023

	June	July	August	(1-7) September	July to 7th September	June to 7th September
Sindh	385.1	142.6	-98.3	-99.6	20.4	48.6
Balochistan	199.1	111.4	-89.4	-70.7	19.7	45.0
Punjab	172.3	47.0	-58.0	-91.2	-8.6	13.5
Khyber-PK	88.3	19.2	-51.3	-71.3	-19.1	-4.2
Gilgit-B	8.9	233.4	28.2	75.9	116.1	89.2
Azad J&K	48.4	30.9	-54.9	-58.7	-11.8	-2.5
Pakistan	**160.3**	**69.7**	**-65.9**	**-80.2**	**0.5**	**20.8**
**Normal period (1961-2010)*						

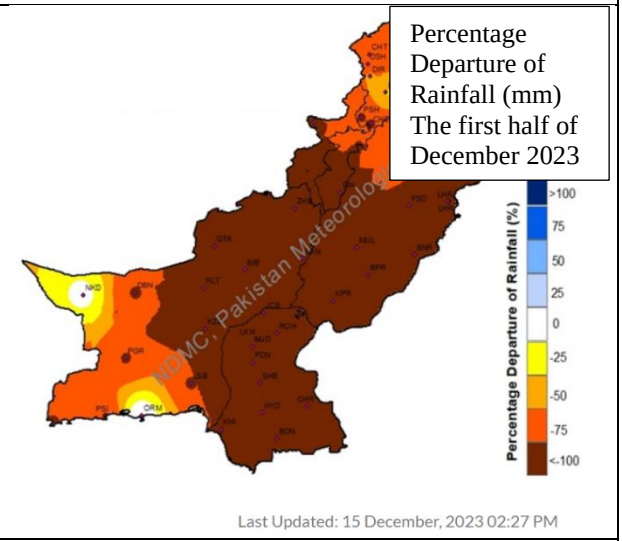
Source: National Drought Monitoring Centre, Islamabad (https://ndmc.pmd.gov.pk/new/advisories.php)

Figure 3.2.23 Drought Monitoring Report (Partial)

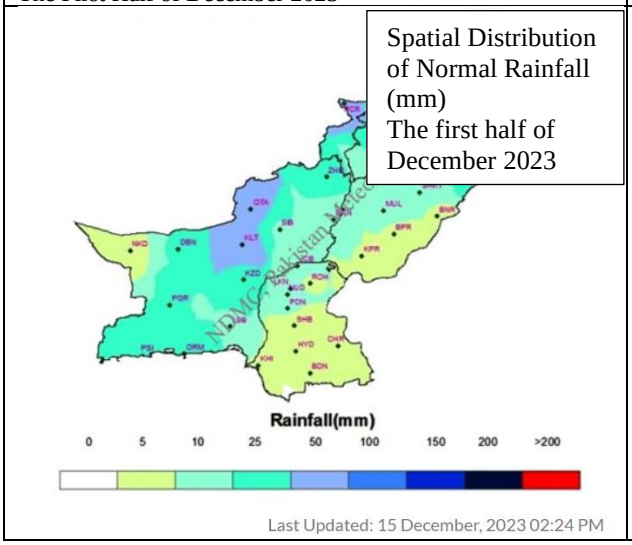
Spatial Distribution of Rainfall (mm)
The First Half of December 2023



Percentage Departure of Rainfall (mm)
The First Half of December 2023



Spatial Distribution of Normal Rainfall (mm)
The First Half of December 2023



Source: National Drought Monitoring Centre, Islamabad (https://ndmc.pmd.gov.pk/new/advisories.php)

Figure 3.2.24 Drought Monitoring Map (Comparison of Rainfall Distribution with the Previous Year)

3.2.8.2 Utilization and Achievement Status

Based on the results of consultations with PMD and WAPDA, the utilization and achievement status of the current NDMP has been summarized. The status of EWS utilization and achievements summarized here corresponds to the Intervention 4 "Multi-Hazard Early Warning System" in the current NDMP.

Table 3.2.9 Overview of EWS Interviews

Date of Interview	Interviewee Organization	Contents of the Interview
November 14, 2023	WAPDA	・Current NDMP utilization and achievement ・Research activities related to glaciers, glacial lakes and GLOF
November 15, 2023	PMD FFD Lahore	・Current NDMP utilization and achievement ・Mainly forecasting activities (weather forecast, river water level forecast, etc.)
November 24, 2023	PMD Islamabad	・Current NDMP utilization and achievement ・Contents related to EWS in general and future project
November 27, 2023		

Source: Project Team

Table 3.2.10 (1) Current NDMP Utilization and Achievement Status (EWS)

Strategies	Priority Actions/Programs	Interview Target organization	Status of utilization and achievement
1. Strengthen Forecasting and Early Warning Systems.	1.1. Establishment of a Specialized Medium Range Forecasting Center (SMRFC) with the Meteorological Radar Station at Islamabad, Upper Air Observations at Designated Locations.	PMD Islamabad	● SMRFC has already been established, but improvement is needed. ● Islamabad radar was installed in 2019. ● Upper Air Observations are conducted, but only wind direction can be observed. In addition, observation points and observation frequency are not high.
	1.2. Strengthening of Flood Forecasting by Satellite Information and Hazard Maps of Indus River.	PMD Islamabad	● Satellite Information is not digitized, although satellite images are utilized. (It is not utilized as an input condition for numerical weather prediction)
	1.3. Establishment of a River Flood Forecast and Warning System, Including Establishment of Regional Flood Forecasting and Warning Centers (RFFWC).	PMD FFD Lahore	● River Flood Forecast and Warning System is in operation and continues to be improved. ● PMD-FFD Lahore carries out weather prediction and flood prediction, and when there is a possibility of flood, all flood related parties are notified by mobile phone, internet, fax, WhatsApp, etc. ● Weather forecasting is performed by downscaling the global model to the Weather Research & Forecasting (WRF) model. River water levels and discharge forecasts are also performed by using the IFAS (Integrated Flood Analysis System) Model, FUSE (Framework for Understanding Structural Errors) Model, PHAST Model, and sometimes regression models. ● The outputs of the above models are reviewed to determine which model to use based on the experience of PMD-FFD Lahore staff. ● Meteorological data in India is mainly obtained from satellites and radar, but weather-related information is also collected from news, electronic media and social media.
		PMD Islamabad	● RFFWC has not been established.

■: Implemented or in operation.
■: Implemented or in operation, but needs improvement.
■: Planned.
■: Hardly implemented.
Source: Compiled by the project team based on interviews

Table 3.2.10 (2) Current NDMP utilization and achievement (EWS)

Strategies	Priority Actions/Programs	Interviews Target organization	Progress
1. Strengthen Forecasting and Early Warning Systems.	1.4. Establishment of a Flash Flood Forecast and Warning System, Including Establishment of Local Flash Flood Forecasting Centers (LFFFC).	PMD FFD Lahore	● Flash Flood Forecast and Warning System has been established only for the Lai Nullah, an urban river in Islamabad and Rawalpindi. ● Technically, it is possible to develop similar models for urban rivers and Hill Torrents by utilizing the river water level and discharge prediction technology used in the River Flood Forecast and Warning System.
		PMD Islamabad	● LFFFC has not been established.
	1.5. Establishment of Landslide Forecast and Warning System.	PMD Islamabad	● Landslide Forecast and Warning System has not been developed.
	1.6. Establishment of a New Tide Level Monitoring Network Including a Data Communication System.	PMD Islamabad	● Tide Level Monitoring Network has not been constructed in PMD. ● PNHD has established the Tide Level Monitoring Network.
	1.7. Establishment of a GLOF and Snow Melt Flash Flood Forecast and Warning System with an Avalanche Advisory Information System.	WAPDA	● WAPDA has identified and monitored 40 most vulnerable glaciers and 36 glacial lakes in the Tarbela 4th Extension HPP Project (T4) and Tarbela 5th Extension HPP Project (T5). These include 5 huge glaciers, for which mass movement monitoring has been conducted. ● Furthermore, in the Golen Gol Project, an upstream glacier caused GLOF, which caused damage to the power plant, and EWS has been established and is now fully functioning. The EWS does not cover all areas where GLOF can occur, but only covers the structures of WAPDA.
		PMD Islamabad	● The GLOF2 Project exists in the PMD, which observes 24 valleys, and alerts the area where GLOF can occur based on information about possible breaking glacial lakes and temperatures.
		PMD Islamabad	● The Avalanche Advisory Information System has not been established.

■: Implemented or in operation.
■: Implemented or in operation, but needs improvement.
■: Planned.
■: Hardly implemented.

Source: Compiled by the project team based on interviews

Table 3.2.10 (3) Current NDMP Utilization and Achievement (EWS)

Strategies	Priority Actions/Programs	Interviews Target organization	Progress
1. Strengthen Forecasting and Early Warning Systems.	1.8. Establishment of a Seismic Intensity Reporting System Including a Data Communication System.	PMD Islamabad	● Although the Seismic Intensity Reporting System has already been established, further improvements are required.
	1.9. Establishment of a Meteorological Radar Systems along Coastal Area.	PMD Islamabad	● Regarding the installation and replacement of radars, since there were many changes compared to the current NDMP, the plans and priorities at the time of the interview are summarized in Table 3.2.11 and Figure 3.2.25.
	1.10. Establishment of a Meteorological Radar System at Designated Locations.		
	1.11. Replacement of a Meteorological Radar Systems at D.I. Khan, R.Y. Khan and Karachi.		
	1.12. Expansion and Additional Installation of an Automatic Weather Observation System (AWS).	PMD Islamabad	● There are currently 85 AWS stations. However, the number of stations does not meet the observation density set by the World Meteorological Organization (WMO).
	1.13. Expansion of the Real-Time Rainfall & Water Level Observation Network.	PMD Islamabad	● PMD monitors rainfall and WAPDA monitors water levels. ● In order to build and operate the Urban Flood Model, a system that collects both rainfall and water level in real time is required.

■: Implemented or in operation.
■: Implemented or in operation, but needs improvement.
■: Planned.
■: Hardly implemented.

Source: Compiled by the project team based on interviews

Table 3.2.10 (4) Current NDMP Utilization and Achievement (EWS)

Strategies	Priority Actions/Programs	Interviews Target Organizations	Progress
3. Strengthen Early Warning Dissemination System.	3.1. Innovation of the Communication System between PMD and NDMA, and among DMAs (NDMA-PDMAs-DDMAs), including the Use of Mobile and Radio Broadcast Networks.	PMD Islamabad	● For weather-related disasters, the PMD communicates to related organizations by mobile phone, internet, fax, WhatsApp, etc.
	3.2. Development of the EWS National Plan, Guidelines and SOPs for Health Emergency Preparedness and Response (HEPR).	PMD Islamabad	● In the 2022 interview, it was reported that the development of the EWS National Plan, Guideline, and SOP for Health Emergency Preparedness and Response (HEPR) is under way.
	3.3. Weather Information Broadcasting System and Weather Information Broadcasting Program Production System and Installation of GTS in SMRFC Project.	PMD Islamabad	● For the Weather Information Broadcasting System and Weather Information Broadcasting Program Production System, PMD disseminates social media and website weather information from normal times. ● On the other hand, NDMA disseminates disaster information only in case of emergency.
			● For the Global Telecommunication System (GTS), PMD Climate Data Processing Center (CDPC), Karachi collects data every 3 hours. The facility also meets the WMO standard.
	3.4. Finalization of SOP of Cyclone EWS	PMD Islamabad	● In the 2022 Interview, it is confirmed that the SOP on Cyclone EWS has been finalized. ● PMD Tropical Cyclone Warning Center (TCWC), Karachi issues an alert on cyclone via the PMD website.

■: Implemented or in operation.
■: Implemented or in operation, but needs improvement.
■: Planned.
■: Hardly implemented.

Source: Compiled by the project team based on interviews

Table 3.2.10 (5) Current NDMP utilization and achievement (EWS)

Strategies	Priority Actions/Programs	Interviews Target organization	Progress
4. Develop Capacity of Early Warning and Evacuation Systems	4.1. Enhancement of Research Activities for Snow/Glacial/Glacial Lakes.	WAPDA PMD Islamabad	● WAPDA has been implementing projects such as Tarbela 4th Extension HPP Project (T4), Tarbela 5th Extension HPP Project (T5), and the Golen Gol Project, etc. These projects mainly focus on research related to glaciers, glacial lakes, and GLOF, etc. ● Similarly, under the GLOF-II project, PMD has issued observations and warnings at 24 valleys . ● WAPDA's Project carries out more advanced research, but at the time of the interview, WAPDA and PMD have not cooperated in research on glaciers, glacial lakes, and GLOF.
	4.2. Establishment of Weather Forecast Guidance System.	PMD Islamabad	● The Weather Forecast Guidance System is in operation, but needs to be improved due to the limited information available at present.
	4.3. Enhancement of Community Enlightenment of EWS with Execution of Training and Drills.	PMD Islamabad	● PMD acknowledge that educating local residents is not within the mandate and consider it the responsibility of NDMA, PDMAs and DDMAs.
	4.4. Education Program for Advanced Meteorology and Hydrology for PMD Staff.	PMD Islamabad	● In PMD, there are 31 PhDs by professional education courses.

■: Implemented or in operation.
■: Implemented or in operation, but needs improvement.
■: Planned.
■: Hardly implemented.

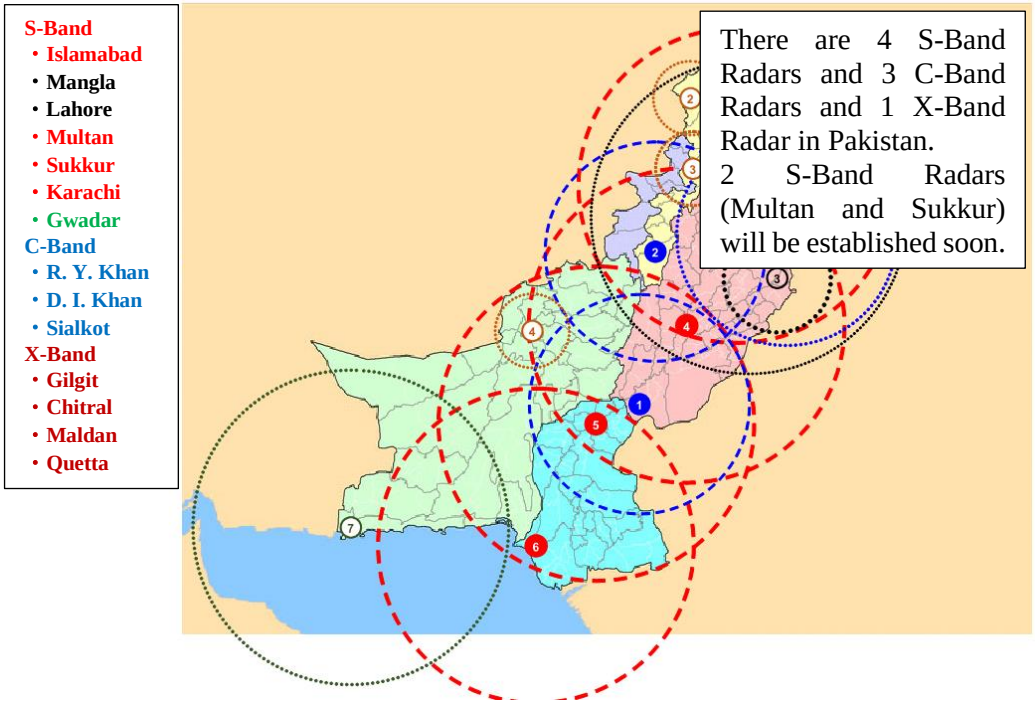
Source: Compiled by the project team based on interviews

Table 3.2.11 List of Existing and Planned Radars in Pakistan

City of radar installation	Radar type	Year of installation	Need for replacement	Priority
Islamabad	S-Band	2019	Not required	-
Mangla	S-Band	2004	Not required	-
Lahore	S-Band	1997	Required	2
Multan	S-Band	2026	-	1
Sukkur	S-Band	2027	-	1
Karachi	S-Band	2020	Not required	-
Gwadar	S-Band	TBD	-	3
R. Y. Khan	C-Band	1999	Required	2
D. I. Khan	C-Band	1999	Required	2
Sialkot	C-Band	2005	Under consideration	2
Gilgit	X-Band	TBD	-	3
Chitral	X-Band	TBD	-	3
Mardan	X-Band	2018	Not required	-
Quetta	X-Band	TBD	-	3
Kalat	-	-	-	4
Zhob	-	-	-	4
Dalbandin	-	-	-	4
Swat	-	-	-	4
Mansehra	-	-	-	4
Muzaffarabad	-	-	-	4

:Establish of New Radar
:Replacement of Existing Radar

Source: Table prepared by the International Meteorological Consultant (IMC) and added by the project team based on interviews



Source: International Meteorological Consultant (IMC)

Figure 3.2.25 Locations of Existing and Planned Radars in Pakistan

3.3 Activity 2-3

After collecting and analyzing information on the issues of the current NDMP, improvement measures in NDMP-II will be examined.

The following improvement measures will be summarized in the technical review document.

3.3.1 Floods (Issues and Improvement Measures)

3.3.1.1 Technical Issues

(1) Establishment and Implementation of a Plan to Enable Earlier Effect of Pre-Investment in Disaster Risk Reduction

1) Issues

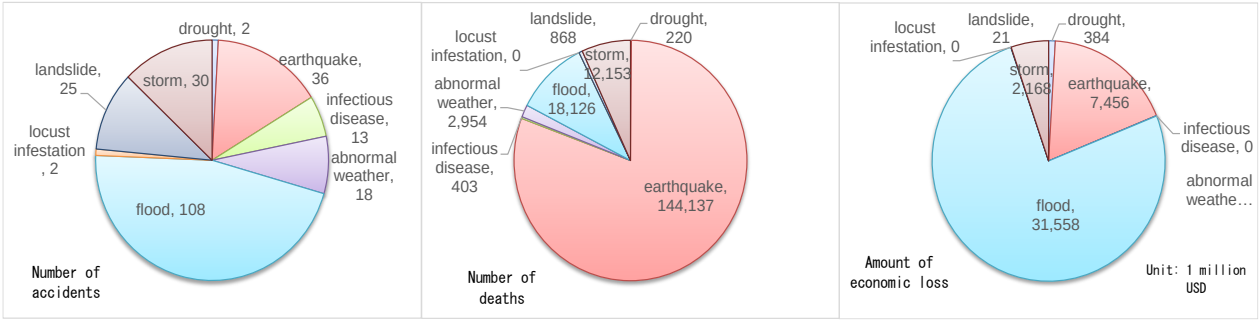
Flood damage occurs every year in Pakistan, causing severe losses. Compared with other types of disaster, the frequency of such losses is remarkable and the annual damage is also high. This fact is organized as statistical information in the current NDMP (see Table 3.3.1). Figure 3.3.1 shows that the number of disasters, the number of fatalities, and the number of economic losses in Pakistan are significantly higher than other types of disasters in terms of both the number of disasters and the number of economic losses caused by floods. The Sendai Framework for Disaster Risk Reduction, established in 2015, provides guidelines for reducing the economic losses caused by disasters. Based on the statistical data in Figure 3.3.1, it is clear that reducing the flood damage is the top priorities in Pakistan. However, the impression which the Project Team got thru the dialogue with C/P agencies and related organizations is that they do not necessarily have such a common understanding.

Table 3.3.1 Past disasters in Pakistan

Type	Event/ year	Deaths/ event	Affected/ event	Loss*/ event
Droughts	0.03	143	2,200,000	247,000
Earthquakes	0.66	3,900	339,198	274,553
Epidemics	0.31	27	1,712	
Extreme temp	0.41	101	48	
Floods	1.72	136	565,236	33,908
Mass mov. dry	0.03	50		
Mass mov. wet	0.55	33	227	
Storms	0.59	85	128,641	95,937

* US$ 1,000
Source: EM-DAT, 1980-2008

Source: Current NDMP



Source: Compiled by the Project Team from EM-DAT data from 1900 to November 2021

Figure 3.3.1 Numbers of Disasters, Numbers of Deaths, and Economic Losses in Pakistan

2) Measures for Improvement

The Sendai Framework for Disaster Risk Reduction shows the importance of advanced investment in disaster risk reduction. Such active advanced investment in disaster risk reduction is necessary to promote smooth economic growth in developing countries. The advance investment in flood prevention in Pakistan is very effective from the viewpoint of concentrating the limited budget and achieving the effect (reducing damage) efficiently and early. In order to formulate National Disaster Management Plan with the aim of realizing the early effect of advanced investment in disaster risk reduction, it is necessary to make a long list after setting the improvement target after 10 years on Integrated Flood Management (IFM), and present the effective implementation procedure.

(2) Clarification of Policy on Implementation of Integrated Flood Management

1) Issues

In Pakistan, bunds and dams & barrages (whose main purpose is water utilization) have been developed in the Indus River and its main tributaries. At present, many flood control projects have been proposed in NFPP-IV and FPSP-III.

However, it can be said that facilities are not being developed in accordance with river plans which have concrete priority for the implementation, because in case of bunds, for example, the design water level is simply set to be higher than the flood level experienced in the past, and the projects proposed in NFPP-IV are deemed only a compilation of the project lists proposed by PIDs, etc. In addition, in the event of a large-scale flood, it has been traditionally done to protect downstream facilities and urban areas from flood damage by intentionally breach the existing bund. However, some of these actions are not stipulated in SOP due to the difficulty of explaining it to the local residents.

For these reasons, there is no consistent policy or development plan for flood control measures in the river system at present, and there is no policy on how large-scale design flood will be dealt with by hard measures such as bunds. On the other hand, Pakistan is also aware of such issues, and although the importance of integrated flood management is mentioned in the "Development of NFPP-IV and Related Studies to Enhance Capacity Building of Federal Flood Commission-FFC," it has not yet been implemented.

2) Measures for Improvement

The FFC, not the NDMA, is responsible for formulating specific policies and plans for river management and river improvement. However, it is recommended that National Disaster Management Plan also describes a policy on the implementation of integrated flood management in line with the FFC, and clarify the level of responsibility of the relevant organizations.

(3) Responding to Climate Change

1) Issues

The current NDMP recommends the implementation of reduction of the impact and mitigation measures indicated in the National Climate Change Policy as a policy to address climate change.

On the other hand, in Pakistan, there are currently no basic policies or long-term implementation targets under the river plan. Therefore, when considering the river plan, future increases in rainfall and sea level rise due to climate change have not been considered in the national planning stage. In Pakistan, the GCISC under the PMD and MoCC conducts research and studies on climate change, but there are no officially authorized values or models for conditions for increases in rainfall and sea level rise.

Average temperatures rising due to climate change will also affect increases in rainfall and sea level rise. Currently, some rivers under the direct control of the Japanese national government have started to review their basic policies. In the case of the Philippines, specific values for sea level rise considering climate change have already been described in the design standards. Regarding rainfall, the Philippine Atmospheric, Geophysical and Astronomical Services Administration(PAGASA) has predicted future weather events based on the RCP 4.5 scenario (Representative Concentration Pathways, IPCC Fifth Report), and it is expected to increase rainfall based on this scenario in river planning.

2) Measures for Improvement

In the future, it is desirable that the scale of external forces (not the value of external forces but the probability scale) to be considered for each location is not different, and that the increase in rainfall and sea level rise due to climate change shall be studied and discussed across ministries and agencies to establish an authorized response policy.

Although the specific examination will be the task of the PMD, GCISC under the MoCC, and FFC, it is necessary in National Disaster Management Plan to clearly describe the level of responsibility of each relevant organization and monitor the progress of the activities through consultation with the NDMA and related organizations.

3.3.1.2 Organizational and Institutional Issues

(1) Clarification of Government Agencies and Organizations in Charge

1) Issues

In the current NDMP, the main responsibility for certain activities is assigned to several related organizations, making it unclear which organization should take the lead to carry out the activity. Therefore, some activities are not progressing effectively.

The main organizations involved in flood management are FFC, PID, PMD, and WAPDA. However, the progress on activities outlined in the current NDMP varies, influenced by differences in each agency's financial resources and organizational capacity, such as the number of engineers.

2) Measures for Improvement

It is necessary to clarify the division of responsibilities of each organization along with allocation of a budgets that align with their respective activities is essential.. In addition, it is necessary for the NDMA to take the lead in monitoring and following up on the rough progress of the activities, including the organizations related to budget allocation. Considering the progress of the current NDMA and the stance of the NDMA to date, it may be necessary to revise the current NDMAct or establish a new organization to establish such a system.

(2) Close Coordination among Responsible Government Offices and Relevant Organizations

1) Issues

As mentioned above, in the current NDMP, the primary responsibility for one activity is assigned to multiple related organizations, and it is not clear which organization will take the initiative and advance the activity. This situation, in part, makes it difficult to say that coordination among related organizations in charge is sufficient, which further delays the progress of the activities.

In addition, as for the lack of coordination among related organizations, it seemed to be recognized as an issue by some offices in Pakistan during the interview survey with related organizations. For example, a GCISC official commented, "The current NDMP is deemed not progressing well due to the overlapping of responsibilities and insufficient coordination among relevant agencies."

2) Measures for Improvement

It is necessary to establish a mechanism that allows similar activities to be shared and information to be shared while maintaining close coordination among related organizations. For this, it is important to coordinate the guidance and advice provided by the NDMA to the relevant organizations.

(3) Improvement of Awareness Regarding the Importance of Advanced Investment in Disaster Risk Reduction

1) Issues

The ratio of disaster risk reduction related budget to Pakistan's GDP is lower than that of neighboring countries. It can be considered that there are multiple factors for this. The following factors can be cited as examples.

- The recognition of flood countermeasures as a development priority by the federal government is low
- Even if a budget application for a project is submitted to the federal government, many other projects have already submitted budget applications, and the allocation of budget is delayed

- According to the arrangement of NFPP-IV (listing of projects by each province), it was not considered to be an investment for the transboundary rivers, it was not recognized the project with which the provincial governments should be related, and it was understood that it should be carried out by each provincial government budget. As a result, no budget was allocated by the federal government

From the examples of the above factors, it can be considered that the recognition of investment in disaster risk reduction including flood countermeasures as important for the economic growth of a country is still low, especially in the ministries and agencies in charge of budget allocation.

2) Measures for Improvement

As mentioned above, in Pakistan, the economic loss caused by floods in particular accounts for about 3/4 of the economic loss caused by past disasters (see Figure 3.3.1). Due to economic development and climate change in the future, there is a high possibility that floods will cause serious economic damage in the future.

Advanced investment in flood countermeasures will effectively reduce the overall economic damage caused by future disasters and contribute to smoother economic growth in Pakistan. It is necessary to consistently communicate these benefits to the C/P and flood project related organizations, as well as the ministries and departments in charge of budget allocation, and to promote awareness reform.

3.3.2 Earthquakes (Issues and Measures for Improvement)

3.3.2.1 Technical Issues and Measures for Improvement in Hazard and Risk Assessment

It is an important approach to examine and implement disaster prevention and mitigation measures for earthquakes by assessing the vulnerability to earthquake disasters in each region and conducting risk assessment. Since multi-hazard assessment is part of Intervention-3: Establish a National Hazard and Vulnerability Assessment, the following issues related to this item have been identified.

- Earthquake Hazard: The key issue is to update the probabilistic earthquake hazard assessment currently based on the outdated Building Standards Act. Additionally, scenario-based earthquake ground motion simulations are needed for a more accurate regional earthquake risk assessment.
- [Improvement] Use the national-level earthquake hazard information published in the new Building Standards Act, which is publicly evaluated. In addition, in regions with high risk of earthquakes, deterministic earthquake hazard assessment assuming a specific scenario is also very important, because it can be used as foundation data to carry out earthquake damage estimation considering regional characteristics in the region.
- Building Inventory: The Multi Hazard & Vulnerability Risk Assessment Report confirms that the earthquake hazard assessment incorporates the provision of Building Standards Act 2007, while vulnerability information is derived from demographic data based on the census and statistical data on residential buildings collected through the survey.
- [Improvement] It was confirmed that the 2017 Bureau of Statistics data was compiled using the latest census. For the current NDMP, it is necessary to prepare the building inventory at the Tehsil level, incorporating updated residential building information from the new census.
- Vulnerability of buildings: It is necessary to carry out vulnerability assessment for each building type. Vulnerability of structures in the earthquake damage assessment are generally evaluated by damage function (Vulnerability Function, and Fragility Curve, etc.).
- [Improvement] It is necessary to construct a damage function based on earthquake damage information in Pakistan or similar countries in cooperation with research institutes such as universities.

3.3.2.2 Issues and Improvement Measures in Institutional and Policy Aspects

(1) Interviews with RDA (Rawalpindi Development Authority)

Issues and improvement measures obtained from interviews with RDA (Rawalpindi Development Authority) are shown.

1. Seismic hazard maps, recognized as one of the outcome of the current NDMP, were updated and incorporated into the Building Code in 2021. Therefore, the project team checked whether the results were actually recognized by the local government at the site where the application for building approval of the local government was implemented. As a result, it was confirmed that the updated version of the earthquake hazard map was not sufficiently recognized.
   [Measures for improvement] It is necessary to display the earthquake hazard and risk assessment map especially in the area with high earthquake hazard assessment, and to disseminate the hazard map to local governments and local residents.
2. It was inferred that the relevant stakeholders had a basic understanding of the earthquake hazard map, but it was unlikely that they handled it on GIS, which is essential for confirming the hazard category of the construction area.
   [Measures for improvement] In the consultation with GIS staff, it is suggested that the importance of confirming the earthquake hazard category in high-rise building construction, etc., and sufficiently communicating the hazard and risk assessment conducted by the central government to local governments, should be discussed through the update of the current NDMP.
3. Regarding the earthquake hazard and risk assessment, there is a possibility that local governments are preparing data on buildings using GIS, because MHVRA is published in limited districts in addition to the map in Building Code. It was confirmed that there is a GIS engineer at Rawalpindi Development Authority (RDA) for the work of application for confirmation, etc., but it was not possible to sufficiently confirm the specific data handled.
   [Measures for improvement] A key issue identified in the current NDMP is the need to conduct additional surveys aimed at enabling local governments to develop GIS based data on buildings and population by region, and to address technical issues.
4. From the perspective of earthquake hazard and risk, the project team interviewed the criteria used for approving new buildings constructions through stakeholders interviews. The answer was that all buildings are judged based on 9.2 on the Richter scale, but 9.2 is a moment magnitude unique to one earthquake, and it seems to be inappropriate as a ground motion as a standard for buildings.
   It is necessary to set appropriate standards in Pakistan in the future.
5. When the project team heard about earthquake data, which is the basis for approval of new buildings, RDA did not own earthquake-related data, but used data owned by PEC (Pakistan Engineering Council) or information based on that data.
   [Measures for improvement] It is necessary to prepare digital data such as GIS so that earthquake hazard map shown in Building Code can be utilized smoothly in business.

(2) Interviews in Each State

In addition, issues based on interviews with each province/region regarding earthquakes (Tsunami partly) are shown below. Projects or activities to address these issues will be proposed during the update of current NDMP.

State	Type of disaster	Efforts for earthquake and tsunami disaster prevention (republished)	Issues for earthquake and tsunami disaster prevention
Punjab Province	Earthquake	・ Sediment disaster is feared as a secondary disaster after the earthquake, and the Public Works Department has carried out road hazard mapping. ・ Building materials have been tested in the Materials Testing Laboratory of the Public Works Department.	・ Seismic resistance of private buildings is generally weaker than that of public buildings. ・ It is necessary to improve the enforceable administrative system (Application for confirmation, on-site examination, etc.) for safe construction. ・ It is necessary to carry out ground investigation for seismic strengthening of foundation structures.
Sindh Province	Tsunami Earthquake	・ Hazardous buildings were listed by the Technical Committee in 2016. ・ A one-stop service is provided to apply for the certification of both private and public buildings outside the Cantonment area and the military land.	・ Tsunami countermeasures are necessary for the nuclear power plant located about 40 km from the center of Karachi city. ・ There is concern about liquefaction of the coastal landfill. ・ There is concern that logistics will stop due to port malfunction in case of tsunami. ・ There is concern that the Central District of Karachi will become an island on land in case of bridge fall.
Balochistan Province	Earthquake	・ In the monitoring of building construction, the inspection certificate is issued for each stage, such as conformity with building standards, confirmation of labor works, and final inspection.	・ Risk assessment by microzoning in Quetta metropolitan area is necessary. ・ Seismic design and earthquake-resistant technology (including site) are generally required in both private and public sectors. ・ The monitoring system at the time of construction is different in each development agency, and it is necessary to standardize it because engineers are confused.
KPK Province	Earthquake	・ Seismic reinforcement of the building after the 2005 Kashmir Earthquake has been carried out with the participation of the University of Peshawar. ・ Seismic engineering center at the University of Engineering and Technology at Peshawar has been set up. ・ Technical training for Public Works Department staff has been carried out.	・ Ad hoc development has been carried out after the 2005 Kashmir Earthquake. ・ Concerns about slope disaster mitigation as a secondary disaster ・ Seismic design technology is insufficient. Site inspection has been carried out based on the checklist, but it is insufficient. ・ Almost 80% of private buildings are non-engineered and have no on-site inspection. Building codes are often not observed. ・ Drilling technology is inadequate. ・ There is a need to develop guidelines for supervision techniques (How to wind reinforcing bar, concrete mix, etc.) and to improve the ability of contractors and workers. ・ Monitoring and repair is not implemented in 100%. ・ Priority roads are designated in the event of a disaster, but alternative roads are designated only as secondary roads.

State	Type of disaster	Efforts for earthquake and tsunami disaster prevention (republished)	Issues for earthquake and tsunami disaster prevention
AJK region	Earthquake	・ Construction of Seismic resistant public buildings was carried out as a model in the One UN DRM project after the 2005 Kashmir Earthquake ・ A list of 5-10 potential hazard sites for landslide on mountain roads after the earthquake was carried out. ・ Inventory preparation of important structures for earthquake resistance was carried out.	・ On-site construction monitoring is almost unimplemented. ・ Examination of private buildings is almost unimplemented. ・ There are concerns about landslides on mountain roads after an earthquake.

3.3.3 Tsunami (Issues and Measures for Improvement)

3.3.3.1 Technical Issues and Measures for Improvement in Hazard and Risk Assessment

[Issues] A tsunami inundation simulation has been conducted along the coast of Pakistan; however, it is based on a deterministic scenario anlaysis.

[Measures for Improvement] A probabilistic tsunami inundation simulation needs to be conducted in the future to provide a scientific basis for implementing proactive disaster prevention measures. It is also important to systematically collect and compile relevant data to support this effort. .

3.3.3.2 Institutional and Policy Issues and Measures for Improvement

The issues from the above-mentioned on-site consultation in Sindh Province are as follows.

- Tsunami countermeasures for the nuclear power plant located about 40 km from the central Karachi are currently insufficiently.
- Liquefaction countermeasures for the coastal landfill are not sufficiently taken.
- There is a risk that the functions of ports and logistics will stop due to tsunami.
- There is a risk that the Central District of Karachi City will could become isolated if all connecting bridges were to collapse.

[Measures for improvement]

Since the residential areas and important facilities are located along the Karachi coastal area, it is essential to implement tsunami protection measures in the future. However, only the results of deterministic tsunami simulation were carried out, and it is necessary to take measures based on the results of probabilistic tsunami inundation simulation, which is used as a scientific basis. For example, in cases where the tsunami damage areas are confined to specific sections of the city's coastline, it is expected that measures will be adopted with consideration for the living environment, such as raising the ground level and integrated green infrastructure together.

3.3.4 Human Resource Development (Issues and Improvement Measures)

(1) Implementation of Human Resource Development Activities with Own Budget

1) Issues

Although NDMA and PDMAs have undertaken many human resource development activities, most have been part of projects funded by international organizations and other partners, with only a limited number of financed through their own budgets. Human resource development in disaster management has a wide variety of targets (all citizens can be targets), and it is necessary to implement it continuously. Therefore, it is necessary to establish a sustainable system for implementing human resource development activities, regardless of the external project support.

2) Measures for Improvement

It is necessary for NDMA and PDMAs must make efforts to to secure funding within their regular

budget to ensure the continuous implementation of human resource development activities, while also making effective use of projects in implementing necessary human resource development activities.

(2) Promotion of Efficient Implementation of Human Resource Development Activities

1) Issues

Apart from

Apart from project-based initiatives, the resources available for human resource development in disaster management in Pakistan—whether human, physical, or financial—are notably limited. . On the other hand, as mentioned above, it is not an exaggeration to say that the targets of human resource development activities are all citizens of the country. Therefore, it is necessary to carry out human resource development activities efficiently and strategically across the country.

2) Measures for Improvement

In order to carry out human resource development activities efficiently in the field of disaster management, it is necessary to have an organization that can grasp the content of human resource development activities carried out by each organization and coordinate the whole. NDMA or NIDM is suitable for this role. NDMA-NIDM should ensure organizations involved in human resource development activities regularly share their reports or participate in online coordination meetings to facilitate collaboration and oversight. By grasping the content and coordinating the activities in this way, it is possible to clarify the roles (content and targets) of each organization carrying out human resource development activities, avoid duplication from the planning stage, and identify areas lacking. Moreover, when projects by international organizations are carrying out human resource development activities, Pakistan government will be able to accurately propose the necessary activities to them, and the projects will be able to effectively utilize the proposals for the implementation of human resource development activities.

Another efficient approach to implementing human resource development activities is to incorporate them into the existing programs. In this regard, it is important to promote the inclusion of of disaster risk reduction education within the regular education system. This applies not only to primary, secondary, and higher education, but also to educational institutions for government officers. In particular, in order to allocate the government budget for activities in the field of disaster management, it is necessary for many government officers to recognize that budget allocation to disaster management is important. Therefore, it is necessary to take actions to incorporate disaster management subjects into the curriculum of educational institutions that train government officers.

The above activities do not require a special budget and can be implemented immediately.

3.3.5 Urban Rainwater Drainage (Issues and Improvement Measures)

The following issues were identified in rainwater drainage systems in Pakistan. Below is an overview of these issues along with proposed improvement measures..

- Low maintenance levels of rainwater drainage facilities
- Insufficient measures against illegal dumping in drainage channels
- Lack of coordination among relevant organizations
- Promotion of the development of non-structural soft measures
- Addressing climate change impacts

3.3.5.1 Low Maintenance Rate of Rainwater Drainage Facility

(1) Issues

In large cities such as Karachi and Lahore, the development of rainwater drainage facilities are not progressing, and urban flooding caused by inland water occurs frequently

The delay in implementing rainwater drainage measures is believed to stem from two main reasons: limited damage before rapid urbanization due to relatively low rainfall, and slower development of drainage infrastructure compared to essential services like water supply. In addition, due to unplanned residential land development due to rapid urbanization, it is very difficult to widen existing drainage channels and expropriate land for the construction of new drainage channels, which is a major hindrance to the improvement of the development rate.

(2) Improvement Measures

Since the development of rainwater drainage facilities requires a substantial budget, it is important to promote efficient development by making full sue of existing facilities. According to the review of current rainwater drainage M/P in Lahore, Rawalpindi and Multan, the cleaning and renovation of existing rainwater drainage facility (drainage channel, pump station), the development of a new drainage channel pumping station, and the construction of an underground rainwater storage facility for the purpose of early flood mitigation are planned, and efficient facility development is being attempted. On the other hand, the policy is to shift the existing combined sewerage system to a diverted sewerage system. However, although the conversion to a diverted sewerage system has merits in terms of water quality improvement, it requires a significant budget, and at present, most funding relies on overseas aid. In Japan, there are cases where water quality improvement measures have been implemented with the combined sewerage system, and in the past, there are cases where the plan to switch to a diverted sewerage system was reviewed and the combined sewerage system was adopted. Therefore, Japan's approach to improving combined sewerage systems is considered a valuable reference for Pakistan. In order to improve efficiency and early realization of rainwater drainage facility, it is recommended to promote investment in hard countermeasures including improvement measures of combined sewerage systems.

In order to promote investment in rainwater drainage countermeasures, it is desirable to evaluate the project effect economically. In Pakistan, information such as inundation location and inundation depth of past urban floods are recorded, but the amount of damage is often not organized. In addition, inundation hazard maps have not been prepared, and it is not possible to show the effect of rainwater drainage countermeasures based on inundation simulation. In addition to the development of inundation hazard maps, it is desirable to develop a mechanism to evaluate the project effect of rainwater drainage countermeasures economically.

3.3.5.2 Measures Against Illegal Dumping in Drainage Channel

(1) Issues

In urban areas of drainage channels, garbage is often illegally dumped. In many cases, the rainwater drainage channel and sewerage system are clogged and blocked due to the garbage, and the ability of the water to flow is reduced due to the obstruction of the flow. As a result, the drainage function is lowered, and inundation of inland water is aggravated.

(2) Improvement Measures

In Pakistan, waste treatment plants often operate below capacity, and in many cases, the waste collection system by the government is not functioning sufficiently. Under such circumstances, it is difficult to reduce illegal dumping in drainage channel. Therefore, establishing an effective waste collection system must be prioritized,. At the same time, awareness activities should be carried out to reduce illegal dumping in drainage channels.

3.3.5.3 Lack of Coordination with Related Organizations

(1) Issues

Road ditches have been established along major roads to manage rainwater drainage from both roads and private land. However, there is insufficient coordination between the management of Road ditch and WASA, which is in charge of rainwater drainage. According to an interview with WASA Lahore (conducted on October 5, 2023), some roads ditches do not meet the required rainwater drainage

standards required by the rainwater drainage standards, and there may be inconsistency between the drainage standards of road managers and those of rainwater drainage managers.

(2) Improvement Measures

Rainwater drainage measures need to be carried out in cooperation with many related organizations such as rainwater drainage managers, sewage managers, river managers, road managers, etc. Therefore, coordination with related organizations is indispensable. First, they should investigate whether there are good examples of coordination in Pakistan, and if there are good examples, they should share each other. If there is no such system, it is possible to establish a committee for rainwater drainage measures in some drainage area as a model case, and request participants from all related organizations to coordinate.

3.3.5.4 Promotion of the Development of Soft Measures

(1) Issues

In Pakistan, there are currently no inland water hazard maps, pre-warning systems for inland flooding, and city-level emergency response plans for urban flooding. It is desirable to promote the soft countermeasures in order to reduce the damage caused by inland flooding, implement efficient countermeasures, and promote self-help of residents.

(2) Improvement Measures

In the National Water Policy formulated by the Ministry of Water Resources (MoWR) in 2018, as one of the climate change adaptation measures for urban flooding, MoWR has proposed the establishment of a pre-warning system specifically for urban flooding to accurately predict extreme events caused by climate change. According to a discussion with the PMD (held on September 15, 2023), the PMD-FFD is responsible for issuing pre-warning for urban flooding; however, at present this service is limited to the city pf Islamabad. The FFD intends to install a pre-warning system in other major cities where urban flooding is a problem, but the development has not progressed due to lack of budget. It is considered that horizontal deployment of the existing pre-warning system in Islamabad or introduction of a simpler pre-warning system (As an example, “Flood Kikikuru” of the Japan Meteorological Agency) should be considered.

For inland water hazard maps, it is desirable to prepare them based on simulation using a model that can reproduce inland water inundation. However, even if there is no budget or necessary data for inundation simulation, there is a simple method to estimate inland water inundation using topographic information and inundation records (Guide for preparing inland water hazard map (draft), Ministry of Land, Infrastructure, Transport and Tourism, 2009). An inland water hazard map can lead to reduction of inland water damage by promoting self-help (improvement of evacuation behavior, advance preparation of water shutoff plates and sandbags, etc.), and as mentioned above, it can also be used to evaluate the effects of rainwater drainage measures and to consider alternatives.

An emergency response plan refers to a plan that specifies measures to be taken in the event of a disaster with the aim of minimizing damage in the event of a disaster. National and provincial level plans have been developed for urban flooding. However, urban flooding has been developed as part of the emergency response plans for river flooding, and it seems that city level plans have not yet been developed. Unlike river flooding, which effects large areas across multiple administrative regions, urban flooding is confined within a city limits and involve fewer related organizations for disaster response coordination, and therefore may be less of priority than emergency response plans for river flooding. On the other hand, it is desirable to develop an emergency response plan for urban flooding, because urban flooding occurs frequently and causes severe damage.

3.3.5.5 Climate Change Response

(1) Issues

In Pakistan, climate change in expected to increase both the frequency and intensity of heavy rainfall events. In the National Climate Change Policy (2028), one of the policies is to review and update

rainwater drainage planning standards for large cities such as Karachi and Lahore. However, as mentioned above, the maintenance rate of rainwater drainage facility is low at present, and it has not been considered raising the planning target (planned rainfall) considering the impact of climate change.

(2) Improvement Measures

Rainwater drainage facility requires a long extension of the facility and requires planning, design, construction, and maintenance in cooperation with sewerage and other related facilities. Therefore, it takes a great deal of effort, time, and cost to raise the planning target rainfall after the construction. Therefore, if future urban planning and climate change are likely to increase the design rainfall levels, it is recommended to consider additional rainwater drainage measures that are necessary when the planning target rainfall is raised. As an example, the increased discharge if the planned target rainfall were raised should be estimated, and it will be compared with the flow capacity of the original planned target rainfall. It is necessary to estimate the cost and feasibility of dealing the raised target discharge with new rainwater drainage facilities, as well as the damage that would occur if there were no measurements. Regarding the increase in planned rainfall due to the effects of climate change, the evaluation results of MoCC and the case of Japan (Proposal for the promotion of urban inundation countermeasures by sewerage based on climate change (2020)), can serve as vaulable reference.

3.3.6 EWS (Issues and Measures for Improvement)

3.3.6.1 Technical Issues

(1) Expansion of Observation Network

1) Issues

AWS observatory data serves as the fundamental information for EWS. By increasing the number of surface rainfall observation stations, the accuracy of radar rainfall can be improved by accurately calibrating the radar system. In addition, the accuracy of the flood forecasting system using this rainfall information as input conditions will be also improved.

However, at the time of the interview, there were only 85 AWS observation stations across Pakistan's vast area of about 796,000km^2 from the viewpoint of flood monitoring. Increasing the number of AWS stations is expected to enhance the accuracy of the accuracy of the EWS.

As for the radar system, 8 radars are installed at present, and judging from the observation range of the radars, they cover almost all of Pakistan. However, it is necessary to strengthen the observation network by installing new radars, considering that the radar cannot observe the amount of rainfall beyond the shielding such as mountains, and that in general, the accuracy is higher at the location closer to the radar installation site. In addition, some radar sites have been installed for about 20 years, and it is necessary to be replaced.

Regarding the aerial observations, currently only the wind direction can be monitored, and both the observation network and frequency are inadequate. I Therefore, improvement is necessary considering the influence of monsoon from abroad (upper Indus River Basin).

2) Improvement Measures

As for the AWS observatory, 300 additional sites will be added under the Modernization of Hydro-met Services Project (MHSP) planned by PMD. After the implementation of this project, it is desirable to continue to add AWS observatories, giving priority to locations with large rainfall characteristics that are significantly different from those in the surrounding area, or near important cities.

Regarding the radar system, it is necessary to conduct the project systematically according to the priority order of the radar sites to be newly built and replaced.

Regarding the aerial observation, it is necessary to strengthen the aerial observation by radiosondes, etc. as soon as the budget is secured.

(2) Expansion and Improvement of Systems and Models

1) Issues

According to the interviews with the PMD, most of the EWS and related systems mentioned in the current NDMP have already been established; however, many require expansion and enhancement. Specifically, the following EWS and related systems need to be upgraded and imrpoved.

- Numerical Weather Forecast System
  - It is also necessary to improve weather forecasting systems and digitize satellite images.
- Flood Early Warning System
- Earthquake and Tsunami Early Warning System
  - Improvement of seismic intensity warning system is also necessary.
- Cyclone Early Warning System
- GLOF Early Warning System
  - Research on glaciers, glacial lakes, and GLOF is also necessary.
- Drought Monitoring and Early Warning System

In addition, although PMD recognized their necessity, the following EWS and related systems had not been established at the time of the interview, and development is also required.

- Landslide Forecast and Warning System
- Avalanche Advisory Information System

2) Improvement Measures

It is necessary to prioritize which system should be expanded and improved, considering the damage by the disaster and the damage reduction by system improvement.

(3) Establishment of Flood Forecasting Centers

1) Issues

Flood forecasting centers, such as Regional Flood Forecasting Centers (RFFWC) and Local Flash Flood Forecasting Centers (LFFFC), which are limited to specific regions, have not been established, and their response to floods outside the Indus River and its main tributaries (Jhelum, River, Chenab River, Ravi and, Sutlej River) is insufficient.

These flood forecasting centers can establish systems to address floods from the local Hill Torrent and urban areas, as well as issue regional forecast warnings.

2) Improvement Measures

In the EWS program planned by PMD, there is a project that establish Urban/Flash Flood Early Warning Centers in Major Cities in Pakistan, which corresponds to the establishment of flood forecasting centers mentioned above. In the long term, it is important to expand the number of such flood forecasting centers and to contribute to the reduction of human damage caused by floods.

3.3.6.2 Organizational and Institutional Issues

(1) Strengthening Inter-Organizational Cooperation

1) Issues

There is insufficient information sharing and technical cooperation among EWS organizations (PMD, WAPDA, FFC, PIDs, and DMAs). For example, PMD and WAPDA independently conduct monitoring and research activities on glaciers, glacial lakes, and GLOFs, but they do not share data or research findings with each other. In addition, PIDs have started to construct their own flood

inundation models in the HEC (hydrological simulation model), but there is no cooperation with PMD in the process of development.

2) Improvement Measures

As an exception to the above, during the four months monsoon season each year, staff from flood related organizations (WAPDA, PIDs, PDMAs) gather at PMD-FFD Lahore to share information. By using these opportunities to share the research and investigation status of each organization, it is possible to maximize the potential of each organization. To ensure ongoing cooperation and effective use of results, coordination by NDMA or FFC is essential.

(2) Professional Education of Staff of Each Organization

1) Issues

In the PMD, 31 staff members have completed doctoral level courses through specialized education courses supported by international education programs. However, the current NDMP does not set targets for specialized education in organizations outside PMD, and it appears that such training is not provided to staff in those other organizations..

2) Improvement Measures

In the future, it is necessary to continue specialized education courses in the PMD and expand the target staff. In addition, other EWS organizations (PMD, WAPDA, FFC, PIDs, DMAs) should have specialized education courses. These activities can contribute to the improvement of disaster prevention capacity in Pakistan.

(3) Awareness-Raising Activities for Local Residents

1) Issues

According to the interviews conducted by the PMD, awareness raising activities for local residents are not the work of the PMD, but of the NDMA, PDMA and DDMA. Awareness-raising activities for local residents should be carried out mainly by the NDMA, PDMA and DDMA.

2) Improvement Measures

In the presentation "Early Warning & DRM System of Pakistan, Learning from Good Practices" by Mr. Idrees member DRR at NDMA at the South Asia Hydro-met Forum, NDMA-NIDM, Civil Defense, and the Provincial Education Department are allocated the budget to be responsible for the promotion of training, education, and awareness related to disaster risk reduction. It is advisable for these organizations to carry out planned awareness activities targeted at local residents.

3.4 Activities 2-4

The overall structure of NDMP-II including the above improvement measures will be examined.

3.4.1 Overall Structure of NDMP-II

Following the suggestion of the Chairman NDMA, it was decided to formulate an annual plan (NDMP 2024) consisting of about 70 pages for easier updates, instead of a long-term plan like NDMP-II. Then, the overall structure of NDMP2024 was examined by overviewing the improvement measures described in Section 3-3, and it was established in consultation with NDMA. The official name of NDMP was changed to National Disaster Mitigation Plan as proposed by the Chairman.

3.4.1.1 Organization of Projects and Activities by Disaster Type

Since the current NDMP was in the early stage of disaster management activities, projects were organized and formulated according to the 10 basic policies of the Hyogo Framework for Action. However, as explained in Section 3-3, differences have emerged in the progress of projects, the state of information development, and the level of technology introduced for each type of disaster. Under such circumstances, it is difficult to categorize all types of disasters under the same plan at the same time, as in the current NDMP, using foundation strategies. In particular, it is difficult to grasp the contents of projects and activities at various levels of response if they are organized in the same table. Therefore, in NDMP 2024, activities are organized according to the disaster type.

3.4.1.2 NDMA2024 Chapters

Chapter 1 describes the disaster risk of each type of disaster as "Explanation of Disaster Mitigation Needs". Chapter 2 describes the "Disaster Management System in Pakistan" and explains the strategies under the categories of "Mitigation and Preparedness," "Preparedness and Response," and "Recovery and Rehabilitation, etc." Chapter 3 describes the "Strategy and Implementation Policy for Disaster Management" and describes the measures (interventions) based on each strategy and the activities or projects (actions) to achieve them.

3.4.1.3 Clarification of Responsibilities

In the implementation of each Action, NDMP 2024 specifies the main organization and supporting organizations. The main organization is responsible for implementing the priority actions, and the supporting organizations are responsible for providing information, confirmation, inspection and guidance to the main organization during the implementing the Action. This division of roles is made after confirming the current status of each disaster management organization.

3.4.1.4 Posting of Annual Plan

In NDMP 2024, only the annual plan was presented. However, for floods, earthquakes and tsunami, where concrete activities have already begun and improvements can be easily foreseen, a long-term project list was prepared. Each action includes a specific start date at the quarterly level.

3.4.1.5 Disaster Prevention Cycle and Responsible Agency

At present, in Pakistan, the DMA organizations mainly carry out "Preparedness and Response" and other key agencies carry out "Mitigation and Preparedness."

CHAPTER 4 ACTIVITIES RELATED TO OUTCOME 3

Outcome 3	A long list (draft) of priority projects for each type of disaster is prepared and NDMP-II technical review materials are prepared.

Projects identified through these activities are classified and listed as projects in NDMP 2024. In other words, the specific names of projects classified here are not presented in NDMA2024, but activities that consolidate similar projects are listed to ensure flexibility. A long list (draft) is prepared, but only activities that will be implemented in 2024 will be listed in NDMP 2024 according to the Chairman's suggestion.

4.1 Activity 3-1

Collect information on Projects for the investment for disaster risk reduction currently planned by ministries and agencies related to disaster prevention

4.1.1 Flood

4.1.1.1 Collect and Organize Information from Related Organizations and Existing Documents

(1) Rawalpindi Development Authority (RDA) and Water and Sanitation Agency (WASA-Rawalpindi)

Project relating the disaster risk reduction implemented or currently proposed by Rawalpindi Development Authority (RDA) and Water and Sanitation Authority (WASA) under RDA were confirmed. The outline is as follows.

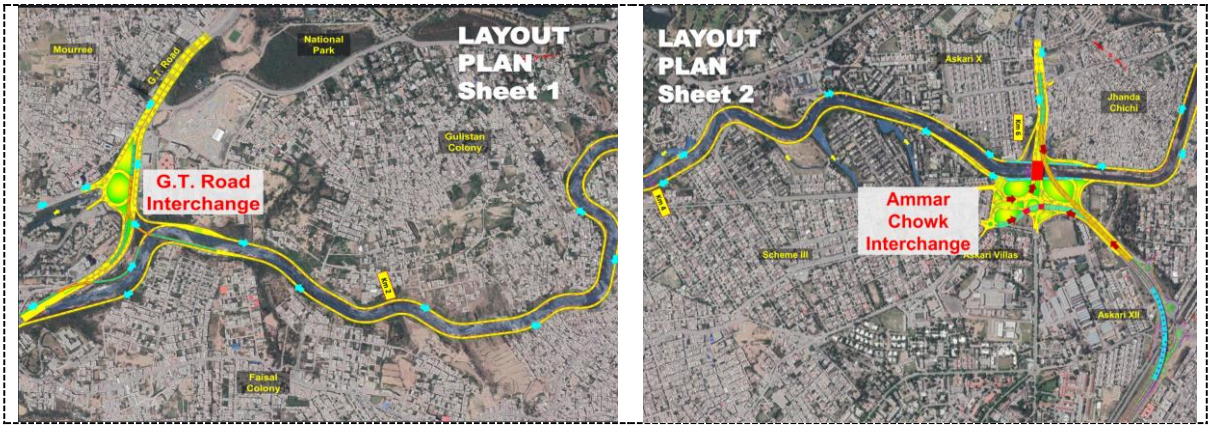
- The Comprehensive Flood Mitigation and Environment Improvement Plan of the Lai Nullah Basin proposed both hard and soft components. However, to-date, only the warning system under the soft component has been developed, while the remaining components have not yet been implemented..
- Although there is a plan (Lai Expressway and Flood Channel Project) that integrates the improvement of the Rai Nullah River, with the development of adjacent road infrastructure, it has not yet been implemented.

Project Component

Road Component:
- **11 Km with 2 Interchanges, 14 Flyovers and 3 Underpasses**
- **12.5 Meter Carriageway on both sides**

Flood Channel:
- **Width 40 to 50 meters with 7.5 meters flow depth and 0.5 m free board with RCC Concrete Bed Lining and Embankment with box type structure.**
- **To accommodate 100 years flood return period from 47000 - 72000 cusecs.**

Sewage Disposal:
- **2 trunk sewers with size 36 inches at Katarian to 72 inches at Moti Mehal.**
- **Conduit from 2.5m x 2m from Moti Mehal to Ammar Chowk**



Source: Excerpt from materials provided by WASA-Rawalpindi

Figure 4.1.1 Outline of the LAI EXPRESSWAY AND FLOOD CHANNEL PROJECT

- Since the existing drainage plan is based on simplified calculations; therefore, a more comprehensive drainage plan that considers the situation of the entire drainage target area is being developed. A comprehensive M/P survey was conducted in 2016) to support this efforts, and related are included in the NFPP-IV.

In addition, the following information was provided regarding projects that are considered necessary to be implemented in the future although no specific plan has been made.

- In order to cope with the water shortage in the dry season, they are studying structural measures to store and treat road drainage during the rainy season and use it during the dry season.

(2) FFC

As of December 2022, no government budget has been allocated to the NFPP-IV, which was formulated in 2015 with an estimated budget of PKR 362 billion. Consequently, the National Flood Control Project has not yet started. Therefore, the FFC selected priority projects from the NPFF-IV and reorganized them under the FPSP-III. The FFC then submitted funding requests for implementation through the EAD to various donors, including the ADB and JICA. However, as of March 2022, no budget had been secured. In response, a consultancy team dispatched by ADB from November 2022 to March 2023 formed the FPSP-III project, which was eventually about 194.6 bil. The PKR budget was approved by the MoSP-PC in June 2023.

Flood protection work accounted for 80% of the projects under the FPSP-III proposal so far (similar to the NFPP-IV). According to the Bund Manual, flood protection work refers to the maintenance of river bund and water systems, which suggests that restoration and construction of these projects are major projects for flood control measures.

Table 4.1.1 NFPP and FPSP

Cost in Million Rupees

No.	Plan/Project	Period	Proposed Schemes		Executed Schemes		Remarks
			No.	Estimated Cost	No.	Estimated Cost	
1	NFPP-I	1978-1988	840	9,500	311	1,730	100% by GoP
2	NFPP-II	1988-1998	735	8,500	180	1,419	100% by GoP
2A	FPSP-I	1989-1997	256	4,556	256	4,735	80% by ADB & 20% by GoP
3	NFPP-III	1998-2008	439	11,703	383	4,292	100% by GoP
3A	FPSP-II	1998-2007	391	13,877	101	4,165	80% by ADB & 20% by GoP
3B	Lai Nullah Flood Forecasting & Warning System		1	348	1	348	97% by Japanese Grant & 3% by GoP
NFPP-I, II, III			2,014	29,703	874	7,441	
ADB: Asian Development Bank, GoP: Government of Pakistan							

(3) WAPDA

As of the interview survey, WAPDA operates 679 manual monitoring stations and 59 AWS stations. There are 679 observation points, and 59 of them have been automated. In the future, many stations will be strengthened and automated according to the plan of Pakistan National Master Plan for Flood Telemetry Network (ADB, 2021.9).

WAPDA did not provide a specific list of future projects to be implemented. On the other hand, NFPP-IV includes projects shown in Table 4.1.2, such as strengthening the flood observation network, strengthening glacier observation, and dam maintenance. The overall budget for these projects is also smaller compared to those under the PIDs.

Table 4.1.2 Projects included in NFPP-IV by Measure (WAPDA)

No.	Scheme Name/Project Name	Budget (Million Rupees)
1	Improvements in Surface Water Hydrology Project Network.	550.00
2	Improvements/Expansion of Flood Telemetric Network including Operational Expenditures.	710.00
3	Improvements in Snow Gauging Stations Network and O/M Expenditures.	550.00
4	Development of Glacier Monitoring Research Center (GMRC)	200.00
5	Improvements of HF Radios (18 Nos.) under WRMD and O/M Expenditures	150.00
6	Feasibility Study - Chiniot Dam on Chenab River	137.00
	Total	2297.00

Source: Project Team summarized from NFPP-IV

(4) Provincial Irrigation Departments (PIDs)

1) Sindh-PID

Regarding the disaster risk reduction investment projects currently proposed by Sindh-PID, there are improvement projects for three barrages in Sindh Province and improvement projects for the along in Indus River. The list of disaster risk reduction investment projects obtained is as shown in Table 4.1.3 The components included in each project described in Table 4.1.3 are shown in Table 4.1.4 to Table 4.1.9

According to Table 4.1.3, significant investment has been directed towards projects related to repair and reinforcement of existing bund in Sindh Province. In addition, projects related to improvement of existing barrage (Improvement, reinforcement and upgrading of systems) are also increasing.

During the interview survey to Sindh-PID, each of the three existing barrage in Sindh Province has its own issues, and they seemed to have a strong desire to focus on projects to solve them. This can be said to be due to the idea that if barrage were to suffer damage, it would greatly affect the water use for agriculture in ordinary times, and would have a serious impact on the local industry and economy.

Moreover, projects such as improvement of existing barrages are not included in the list of projects under NFPP-IV and FPSP-III. The reason for this is that the improvement of barrage requires a large amount of money, and the province did not include it in the list of projects of NFPP-IV and FPSP-III, even though it considers it a high priority.

Table 4.1.3 Projects for the investment for disaster risk reduction in Sindh-PID

Project for the Investment for Disaster Risk Reduction and Name of List	Description of the Project for the investment for Disaster risk reduction	Remarks
Sindh Barrage Improvement Project (SBIP)	Renovation of existing barrage, including replacement of gate. Renovation work is already on-going at Guddu Barrage.	

Project for the Investment for Disaster Risk Reduction and Name of List	Description of the Project for the investment for Disaster risk reduction	Remarks
Barrage Management Unit Requirement	・Improvement project at Guddu, Sukkur and Kotri Barrages ・Establishment of Hydrology & Research Wing and Advisor Office Karachi	・List of Projects for the investment for Disaster risk reduction prepared by Barrage Management Unit ・Cost: 9,400 mil. PKR
Flood Protection Schemes being Implemented through Normal Emergent Flood Programme 2022-23 under PSDP	・Under PSDP, 2 projects related to bund improvement, including the installation of stone pitching and the enhancement of earth work, are on-going.	・List of projects announced at the Pre Flood Coordination Conference on May 18, 2023 ・Cost: 629 mil. PKR
Flood Emergent Works Included in ADP 2022-23	・Under ADP, 10 projects related to bund improvement, including the Sindh Flood Emergency Reconstruction Project for Bands, are on-going.	・List of projects announced at the Pre Flood Coordination Conference on May 18, 2023 ・Cost: 16,514 mil. PKR
Important Nature of Works Required at Kotri Barrage Head Works	・6 projects related to facilities around Kotri Barrage (Stone Apron extensions, Regulator, etc.) are being proposed.	・List of projects proposed by Sindh-PID (Kotri Barrage Office) ・Cost: 6,420 mil. PKR
IRC Proposal: Construction of Stone Stud & Apron Along SM Bund Mile 12/0 to 12/7, Raising & Strengthening Along SM Bund Mile 12/0 to 14/0; J-Spur Along SM Bund Mile 11/6 (BAKHRI)	・2 projects related to bund improvement (Installation of Stone Stud & Apron, J-Spur) to vulnerable point downstream of Sukkur Barrage are being proposed.	・List of projects submitted by Sindh-PID (Sukkur Barrage Office) to IRC ・Cost: 4,781 mil. PKR
Maintenance Works on Critical Point at Sukkur Barrage	・1 project is being carried out for the maintenance of Sukkur Barrage (installation of Diaphragm Wall).	・Project proposed by Sindh-PID (Sukkur Barrage Office) at Flood Contingent 2023 ・Cost: 63 mil. PRK

Source: Project Team

Table 4.1.4 Barrage Management Unit Requirement

No	Infrastructure	No	Project Name	Budget (mil. PKR)
1	Kotri Barrage	1	Kotri Barrage Scada	400
		2	Gate Control System of 44 Gates along Canal Head Regulator	700
		3	Activators Motors for Motorized Gate System	200
		4	Piezometers Replacement (100 Numbers)	500
		5	GPR and Geophysical Survey for Barrage Submerged Component	300
		6	Barrage Sediment Monitoring System	200
		7	Sediment Monitoring Unit	500
		8	Barrage Emergency Response Equipment's	450
		9	Bulk Head Gates and Pontoon Repair	300
		10	Amphibious dragger procurement	450
		11	Barge Boat	450
2	Sukkur Barrage	1	Sukkur Barrage Scada	700
3	Guddu Barrage	1	Guddu Barrage Scada	600
4	Hydrology & Research Wing	1	Office & Research Facility upgradation (Hyderabad & Karachi)	500
		2	River Flow Measurement ADCP Equipment's	600
		3	River Ranger Boats	300
		4	Establishment of Gauging Unit Kotri	150
		5	Establishment of Gauging Unit Sukkur	200
		6	Establishment of Gauging Unit Guddu	150
		7	Establishment of Gauging Unit Chachran (Punjab)	400
		8	Establishment of Gauging Unit Taunsa (Punjab)	350
5	Advisor Office Karachi	1	Office upgradation and Equipment's	250
		2	Data Center	600
—	**Total**	—	**Total**	**9,400**

Source: Barrage Management Unit

Table 4.1.5 Flood Protection Schemes being Implemented through Normal Emergent Flood Programme 2022-23 under PSDP

No	Project Name	Budget (mil. PKR)	Remarks
1	Raising & Strengthening of S.M Bund from Mile 80/0 to 123/0 (different reaches) in Dadu Division Shaheed Benazirabad	375	Work in Progress
2	Construction of Stone Pitching & Strengthening Earth Work along S.M Bund from Mile 38/2 to 40/0 to 42/5, Recouping of Stone Pitching from Mile 40/0 to 42/0 and Stone Apron 42/0 To 42/5 (Dad Wah).	254	Under tendering process
—	**Total**	**629**	

Source: Sindh-PID

Table 4.1.6 Flood Emergent Works Included in ADP 2022-23

No	Project Name	Budget (mil. PKR)	Remarks
1	Restoration of Flood Damages to Gaj Diversion Bund at RD-2+700, 5+000, 12+000 & 13+000 in Southern Dadu Division (SDG # 6)	344	Work in Progress
2	Construction of T-Head Spurs at LS Bund Mile 40/4+500 Moria Loop Bund Mile 0/0+400 & 0/1+600 in Northern Dadu Division - Revised on 18.10.19 (SDG # 6)	919	Work in Progress
3	Providing Stone Apron, Stone Pitching, Earth Work along L.S Bund Mile 18/0 to 20/0 in Northern Dadu Division Larkana. Total Cost Rs.241.586 Million (GOP Share 65%) & (GOS Share 35%) - NDRMF	90	Work in Progress
4	Recoupment of Damaged T-Head Spur along Agani Akil Loop Bund 2/6+250, Stone Apron at Mole of 0/4 and 0/7 Mole Spurs and 09 Nos. Stone Studs in Larkana Sub Division. Total Cost Rs.416.306 Million (GOP Share 65%) & (GOS Share 35%) - NDRMF	117	Work in Progress
5	Retaining wall along Gaj Diversion Bund in Southern Dadu Division.	2,281	Work in Progress
6	Strengthening/Earth work from RD-20.0 to 75.0 & Stone Pitching from RD-20.0 to 75.0 along F.P. Bund in Southern Dadu Division.	702	Work in Progress
7	Providing 04 Nos. J-Head Spurs with Apron and Stone Pitching for Centralizing the River Flow Up-stream of Guddu Barrage Region.	1,608	Work in Progress
8	Re-sectioning and Sand Coring of Gharaghat Bund mile 0/0 to 2/4 and Haibat Bund mile 0/0 to 5/0 in Begari Division Kashmore.	880	Work in Progress
9	Construction of J-Spurs U/S Left Guide Bank to Centralize the Flow of Indus River Water under Thatta Sujawal Road Bridge (Doolah Bridge) in Kalri Baghar Division Thatta.	1,948	Work in Progress
10	Sindh Flood Emergency Reconstruction Project for Bunds & Canals (Flood 2010-11) ADB Loan Scheme (Tando Muhammad Khan, Shaheed Benzair Abad, Jamshoro, Hyderabad, Ghotki, Khairpur, Nowsehro. Feroze, Matiari, Larkana, Kamber, Dadu, Sujawal, Thatta, Kashmore, Sukkur, Shikarpur and Rahim Yar Khan (Punjab) - (Revised Total Cost Rs 26,905 M, ADB Share Rs.19,279.493 M & GoS share Rs.7,625.507 M) (Revised on 03.07.14) (SDG # 6)	7,626	Work in Progress
—	**Total**	**16,514**	

Source: Sindh-PID

Table 4.1.7 Important Nature of Works Required at Kotri Barrage Head Works

No	Project Name	Budget (mil. PKR)
1	Rehabilitation of Piezometer at Main Barrage & Head Regulators of Kotri Barrage Jamshoro.	800
2	Rehabilitation of Solar System Installed at Kotri Barrage Head Works Jamshoro.	120
3	Construction of Weir Sub-Division Office Jamshoro and Renovation of Old Whessoe Store at Kotri Barrage	250
4	Extension of Stone Apron in Upstream of Kotri Barrage.	2,450
5	Reconstruction/Rehabilitation of Old Fuleli Regulator	2,000
6	Stone dumping along Left Divide Wall (Pit)	800
—	Total	6,420

Source: Sindh-PID (Kotri Barrage Office)

Table 4.1.8 IRC Proposal (Sukkur Barrage Office)

No	Project Name	Budget (mil. PKR)
1	Construction of 5 No Stone Studs & Apron along SM Bund Mile 12/0 to 12/7, Raising and Strengthening along SM Bund Mile 12/0 to 14/0	1,791
2	Construction of J Spur along SM Bund Mile 11/6 (Bakhri)	2,004
3	Other Cost: Contractor Premium, Design & Supervision, Work Charge & Contingencies, Misc (Safeguards etc.)	986
—	Total	4,781

Source: Sindh-PID (Sukkur Barrage Office)

Table 4.1.9 Maintenance Works on Critical Point at Sukkur Barrage

No	Project Name	Budget (mil. PKR)	Remarks
1	Construction of Diaphragm Wall Along R.N Bund for Strengthening of Bund, Stone Dumping & Earth Work Along Satin-jo-astan Loop Bund No. I & II and Recoupment/stone Pitching Along Left Guide Bank	63	Proposal approved by the Full Indus River Commission in the 180th meeting & PC-I submitted to Govt. for approval & financing.

Source: Sindh-PID (Sukkur Barrage Office)

In addition, Table 4.1.10 shows the schemes listed under NFPP-IV projects, classified into two main types: bund installation and erosion protection. From the arrangement in Table 4.1.10, it can be seen

that schemes related to improvement/upgrade of bunds in Sindh Province have a high priority, with a large number and budget.

Table 4.1.10 Projects included in NFPP-IV by Types of Measures (Sindh Province)

Classification	Number of schemes			Budget (Million Rupees)		
	Phase I	Phase II	Total	Phase I	Phase II	Total
Bund Installation	2	-	2	3,120	-	3,120
Bund Reinforcement/Improvement	35	-	35	12,372	-	12,372
bund Rehabilitation	6	-	6	1,805	-	1,805
Erosion Protection	13	-	13	4,435	-	4,435
Bund Installation + Erosion Protection	0	-	0	0	-	0
Flood Gate	3	-	3	644	-	644
Other	1	-	1	500	-	500
Total	**60**	**-**	**60**	**22,875**	**-**	**22,875**

Source: Project Team summarized from NFPP-IV

From the above, it can be said that improvement of the existing barrage and strengthening/improvement of the existing bund are high priority tasks in Sindh-PID.

2) Punjab-PID

Regarding the currently proposed projects for the investment for disaster risk reduction by Punjab-PID, A list of projects mainly for construction of dams in Hill Torrent Area and construction/renovation of dispersion structure was obtained. Table 4.1.11 gives an overview of the project list, and Table 4.1.12 lists the projects included in the list.

From Table 4.1.11 and Table 4.1.12, it can be seen that Punjab-PID focuses on the project for Hill Torrent. According to Punjab-PID, they would like to utilize the Hill Torrent countermeasures in terms of water use in ordinary time, and they would be more receptive to investment that is effective in ordinary time, not just for simple disaster risk reduction purposes.

Table 4.1.11 Projects for the investment for disaster risk reduction in Punjab-PID

Project for the Investment for Disaster Risk Reduction and Name of List	Description of the Project for the investment for Disaster risk reduction	Remarks
List of Works/Schemes Along Hill Torrents	・Proposing 12 projects for the Hill Torrent Area in Punjab Province, Mainly the construction of Dam as well as the construction and renovation of Dispersion Structure.	・List of projects related to Hill Torrent proposed in NFPP-IV ・Cost: 80,000 mil. PRK

Source: Project Team

Table 4.1.12 List of Works/Schemes Along Hill Torrents

No	Project Name	Cost (mil. PKR)
1	Rehabilitation of Flood Carrying Channel in area of CRBC Canal (Taunsa Sharif)	2,000
2	Construction/Rehabilitation of Dispersion Structures at Kaura Hill Torrent	4,000
3	Construction/Rehabilitation of Dispersion Structures at Sanghar Hill Torrent	4,000
4	Construction of Vidore Dam	25,000
5	Construction of Dam at Mithawan Hill Torrent	25,000
6	Construction/Rehabilitation of Dispersion Structures in Kaha Hill Torrent	4,500
7	Construction/Rehabilitation of Dispersion Structures at Sori Shumali Hill Torrent	2,500
8	Construction/Rehabilitation of Dispersion Structures at Sori Janubi Hill Torrent	1,500
9	Construction/Rehabilitation of Dispersion Structures at Pitok & Zangi Hill Torrent	1,500
10	Construction of Drainage System for Sori Shumali, Sori Janubi, Pitok and Zangi Hill Torrents	3,000
11	Remodeling/Enhancing Capacity of Qutab Drain	4,000
12	Bhokker Head Bund installation RD 175-264	3,000
—	**Total**	**80,000**

Source: Punjab-PID

Table 4.1.11 shows the schemes included in NFPP-IV projects, organized into categories of bund installation and erosion protection. As shown in Table 4.1.11, both the number and budget of schemes related to bund installation and erosion protection are larger in Punjab. The concentration of erosion protection in Phase-I indicates that erosion protection is a high priority.

Table 4.1.13 NFPP-IV Projects by Measures (Punjab State)

Classification	Number of Schemes			Budget (Million Rupees)		
	Phase I	Phase II	Total	Phase I	Phase II	Total
Bund Installation	10	10	20	5,965	14,210	20,175
Bund Reinforcement/Improvement	10	1	11	3,135	11,750	14,885
Bund Rehabilitation	1	0	1	95	0	95
Erosion Protection	27	0	27	11,588	0	11,588
Bund Installation + Erosion Protection	2	0	2	1,400	0	1,400
Others	3	0	3	1,167	0	1,167
Total/Total	**53**	**11**	**64**	**23,350**	**25,960**	**49,310**

Source: Project Team summarized from NFPP-IV

From the above, it can be said that in Punjab-PID, response to Hill Torrent, riverside erosion protection, construction of bund and improvement/upgrade of existing bunds are high priority issues.

3) Balochistan-PID

The list of projects approved by PC-I for the currently proposed projects for the investment for disaster risk reduction by Balochistan-PID was obtained. Table 4.1.14 gives an overview of the project list and Table 4.1.15 lists the projects included in the list.

Since 20 projects are proposed mainly for the installation of a dam (Small Dam/Check Dam) in Balochistan, the allocated budget is larger than other provinces. The main purpose of the dam construction in the region is to capture surface water from floods caused by Hill Torrents and use it for regular water supply needs. The other projects are described as the Flood Irrigation Scheme and do not include projects specialized in specific countermeasures such as bund reinforcement or installation of water control.

Table 4.1.14 Projects for the investment for disaster risk reduction in Balochistan-PID

Project for the Investment for Disaster Risk Reduction and Name of List	Description of the Project for the investment for Disaster risk reduction	Remarks
PC-I's Ready for Approval	・Proposing 20 projects, mainly for the construction of Dam in Balochistan Province.	・PC-I's Ready for Approval Project ・Cost: 248,451 mil. PRK

Source: Project Team

Table 4.1.15 PC-I's Ready for Approval

No	Project Name	Cost (mil. PKR)
1	Construction of Aspi Kahan Flood Irrigation Scheme, District Kech, Balochistan	8,926
2	Construction of Koshk Dam, District Khuzdar, Balochistan	8,663
3	Construction of Pellar Dam, District Awaran. Balochistan	33,247
4	Construction of Khazran Ghar/Nar Dam, District Khuzdar, Balochistan	7,293
5	Construction of Soolar Dam, District Gwadar, Balochistan	7,589
6	Construction of Sukkha Dam, District Gwadar, Balochistan	6,782
7	Construction of Kud Dam, District Lasbela, Balochistan	12,162
8	Construction of Kolachi Dam, District Khuzdar, Balochistan	6,384
9	Construction of Sorgaz Dam, District Khuzdar, Balochistan	12,468
10	Construction of Sukleji Dam, District Kachhi, Balochistan	16,935
11	Construction of Badinzai Dam, District Zhob, Balochistan	22,049
12	Construction of Ashkan Dam, District Mastung, Balochistan	643
13	Construction of Dadday Dam, District Kech, Balochistan	1,128
14	Dasht River Flood Irrigation Scheme, District Gwadar, Balochistan	2,230
15	Construction of Gadibar Dam, District Loralai, Balochistan	47,775
16	Construction of Duki Dam District Duki, Balochistan	17,694

No	Project Name	Cost (mil. PKR)
17	Construction of Tali Tangi Dam, District Sibi, Balochistan	10,735
18	Construction of Barkhan Dam, District Barkhan, Balochistan	5,647
19	Construction of Ghazlai Dam, District Killa Saifullah, Balochistan	6,027
20	Construction of Dhoreri Dam, District Jhal Magsi, Balochistan	14,074
—	Total	248,451

Source: Balochistan-PID

Table 4.1.16 shows the schemes listed under the NFPP-IV projects, classified into bund installation and erosion protection. In Balochistan, the number and budget of schemes related to bund installation and flood protection are larger than those in Table 4.1.16. In addition, bund installation is concentrated in Phase-I, which indicates that bund installation is a high priority. Regarding flood protection, it is not specified in the NFPP-IV whether is the measures are intended for overflow control or erosion protection; however, it is assumed that the projects address both aspects..

Table 4.1.16 Projects included in NFPP-IV by Measure (Balochistan)

Classification	Number of schemes			Budget (Million Rupees)		
	Phase I	Phase II	Total	Phase I	Phase II	Total
Bund Installation	125	0	125	10,716	0	10,716
Bund Rehabilitation	1	0	1	20	0	20
Flood Protection	102	134	236	3,364	5,766	9,130
Erosion Protection	8	0	8	550	0	550
Drainage	2	0	2	60	0	60
Coastal Conservation	3	0	3	500	0	500
Others	2	0	2	2,475	0	2,475
Total/Total	**243**	**134**	**377**	**17,685**	**5,766**	**23,451**

Source: Project Team summarized from NFPP-IV

Based on the above, it can be concluded that flood protection in Balochistan under the PID is a high priority, including dam construction to manage Hill Torrents, bund construction, and measures addressing both overflow and erosion protection. .

4) KP-PID

In KP Province, a list of projects that need to be implemented in the future is not available.

Here, Table 4.1.17 shows the schemes included in the list of projects included in the NFPP-IV, classified into bund and erosion protection. Compared with the list in Table 4.1.17, the number and budget of schemes related to riverbank heightening and protection and flood protection are larger in KP Province. In addition, since riverbank heightening and protection and flood protection are concentrated in Phase-I, it can be seen that they are high priority. Regarding flood protection, it is not specified in the NFPP-IV whether it is for overflow or erosion protection, but it is assumed that it is related to both.

Table 4.1.17 List of Projects included in the NFPP-IV by Measures (KP Province)

Classification	Number of schemes			Budget (Million Rupees)		
	Phase I	Phase II	Total	Phase I	Phase II	Total
Riverbank Heightening	2	0	2	310	0	310
Bund Installation	3	0	3	2,460	0	2,460
Riverbank Heightening + Protection	32	0	32	3,800	0	3,800
Erosion Protection	7	0	7	2,316	0	2,316
Flood Protection	27	25	52	10,949	67,916	78,865
Flood Control	1	0	1	165	0	165
Total	**72**	**25**	**97**	**20,000**	**67,916**	**87,916**

Source: Project Team summarized from NFPP-IV

Based on the above, it can be concluded that raising and protecting riverbanks, along with overall flood protection, are high priority issues in the KP-PID.

4.1.2 Earthquake

Table 4.1.18 provides an overview of hazard risk assessment, disaster management plan, and Emergency Operation Center (EOC) installation status based on interviews conducted by each province.

Table 4.1.18 Earthquake and Tsunami Risk Assessment, Updated Disaster Management Plan, and EOC Installation Status

State region	Risk assessment	Updated disaster prevention plan	EOC	Special notes
Punjab Province	・Multi Hazard Vulnerability Risk Assessment: MHVRA2021 was conducted (not disclosed) by ADB support. ・20 district-specific maps were also created. ・Although it is multi-hazard analysis, flooding is focused. ・Regarding earthquakes, public facilities and road networks are shown based on the Seismic Zoning Map.	・None disaster management plan based on a latest risk assessment exists. ・Draft PDMP2023-2028, 20 DDMPs draft has not yet been approved. ・Monsoon Contingency Plan2022 and Winter Contingency Plan2023 are available.	・Provincial EOC is the highest quality in the state ・36 District EOCs have networks, 20 sites are supported by WB, 16 sites are funded by the Punjab Government. ・No earthquake information was displayed during monsoon season. ・Rescue1122 EOC is also considered of high quality.	・Rescue 1122 and the emergency services academy are headed by Dr. Rizwan. ・Rescue1122 conducts safety inspections of high-rise buildings, sports, and health facilities by state law.
Sindh Province	・MHVRA2022 (publically available) was Supported by SPARCO, WB. ・Seismic Zoning Map shows risk using population data. ・Tsunami risk uses NED University inundation prediction map.	・MHVRA was implemented in 20 districts supported by SPARCO. ・D.M.Plans 2023-2032 and Mitigation Plan includes in it.	・PEOC is installed by WB support (about 1 large TV and poor) ・20 sites of DEOC are upgraded by WB support.	・2 regional offices have been established at Hyderabad and Sukkur to divide and govern the district, each equipped with the large depots to support operations ・2 more regional offices to be established.
Balochistan Province	・Baluchistan Disaster Management Project 2017-2021 was supported by WB. ・District Disaster Management Plans were developed.	・There is no recent disaster prevention plan (last plan was published in 2008.) ・Risk Assessment & DDMP for Gwadar, Lasbella was implemented. ・There are Hazard specific response plans for cyclones and heat waves but none for earthquakes and tsunamis.	・PEOC is installed by WB support.	・Disaster Prevention Department has been established at the University of Quetta (up to Masters) ・PMD formulates earthquake and tsunami SOP. ・Regional office locates in Turbat.
KP Province	・MHVRA has not been implemented. ・D.I.Khan, Chitral District only implemented by Supported by UN Habitat. ・The University of Peshawar, Departments of Geology, Geography, and Disaster Preparedness and Management carry out various types of research, including earthquakes, floods, sediment and GLOF disasters. ・Joint research with disaster prevention organizations and planning of disaster prevention departments was implemented.	・PDMP & Policy are in progress. ・Road Map towards Safer KPK 2022 was developed (target is 5 years). ・DDMPs 2019 -2022 was developed for Swat, Nowshera, Charsadda, Shangla District.	・PEOC is installed and renovated 3 years ago by state funds.	・University of Peshawar CDPM: Center for Disaster Preparedness and Management ・Earthquake Engineering Center in Univ of Eng & Technology ・School of Disaster Management, NUST University Risalpur Campus ・The above three universities have courses of undergraduate, master's and doctorate.
Gilgit-Baltistan Province	・MHVRA implementation was approved. ・GLOF Risk Assessment UNDP GLOF 2 projects, 5 years was implemented.	・GB Disaster Management Plan 2023 completed by ICIMOD support.	・PEOC is in operation.	・An enthusiastic Director leads organizations from the beginning

State region	Risk assessment	Updated disaster prevention plan	EOC	Special notes
AJK region	・DCRIP: Disaster and Climate Resilience Improvement Project was implemented by WB support. ・Land use regulations based on landslide risk assessment were implemented.	・DRM Plan2017 (strategy level) was implemented by Islamic Relief Support. ・SDMPs 2022 was implemented in 10DDMPs (all districts).	・PEOC was installed in 10DOCs by WB Support.	・Rescue1122 is not installed in the 3 most vulnerable districts. ・Disaster Prevention Bureau does not have its own HP.
Overall Characteristics Summary	・No detailed risk maps at city level exists.	・Sindh and Gilgit - Baluchistan have a Mitigation Plan.	・Punjab has a network of EOCs.	・KP has universities contributing to disaster management.

Note:Only Balochistan and Sindh have coastlines

- With support from the World Food Programme (WFP), the World Bank, the Asian Development Bank and other partners, district level hazard and risk maps have been developed for Sindh and Punjab. Flood disaster is the main focus, howeverl earthquake hazard maps have also been developed, which are classified by different colors of earthquake probability (50, 100, 250, 475, 1000 by probability, etc.) and overlaid on population density, land use, important facilities and road networks. From the outputs it is possible to compare which district has a high degree of risk at the national level. Since information on ground susceptibility and buildings has not been input, studies on seismic intensity and damage to buildings have not been conducted. According to the role distribution list, individuals designated as seismology experts are, in fact, not actual experts in the field of seismology. . Furthermore, Seismologists were also not involved in the development of hazard map for Sindh and even have not examined it (PMD seismologists also employ physics majors to study for master's courses and participate in training at overseas universities and research institutes.).

- In this NDMP-II, the ground surface acceleration (PGA) (which is the source data of seismic susceptibility revised as new information in the building code) is used to revise it, and the earthquake risk and tsunami risk are examined by overlaying the exposed population. Therefore, more vulnerable districts can be extracted based on the revised data (Exposed population is taken into account, so megacities with high vulnerability are extracted).

- As a future step, megacities identified as highly vulnerable, where population and economic activities are concentrated, can be examined at the city level. By incorporating data on ground seismic susceptibility to the ground and predicting the potential damage to buildings, more specific and scientifically based investment plans can be formulated at the city level.

4.1.3 Tsunami

A provincial disaster management plan (2023-2032) based on the risk maps has been developed in Sindh State in particular. Tsunami-related investment plans include:

- Improvement of coastal seawalls (also contributing to cyclones)
- Establishment of tsunami observation and preliminary warning centers
- Implementation of F/S at the Preliminary Warning Center
- List of buildings that could serve as multi-purpose tsunami shelters
- Installation of tide level observation equipment in coastal areas

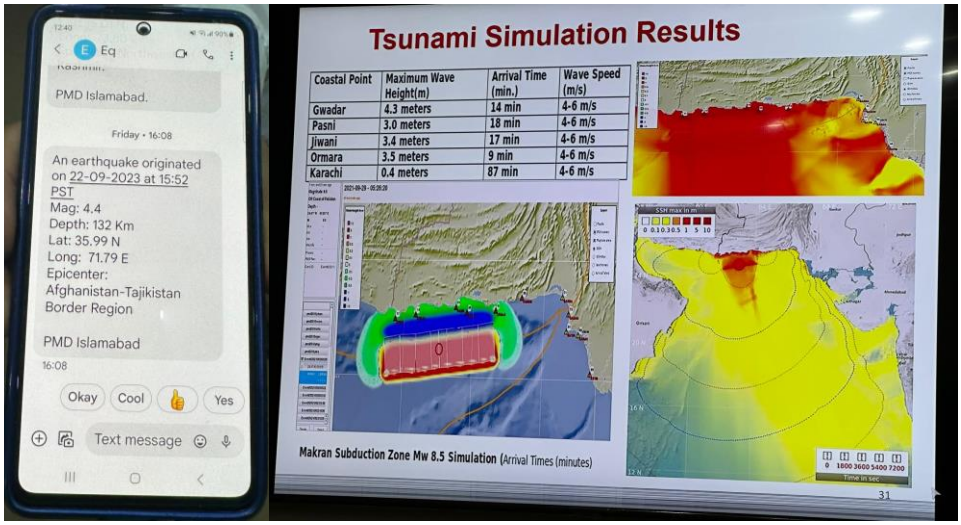
In addition, the following information was obtained through interviews with the PMD Karachi branch.

- There is a tsunami monitoring center in the PMD Karachi office, where tsunami observation is carried out. A tsunami and earthquake SOPs (an emergency response manual for the PMD) has been established, and evacuation drills using tsunami sirens (4 units) and dissemination of mobile phone messaging are also conducted. However, the estimated tsunami arrival time is in the coastal area of Balochistan province very short —ranging from 4 to 17 minutes..

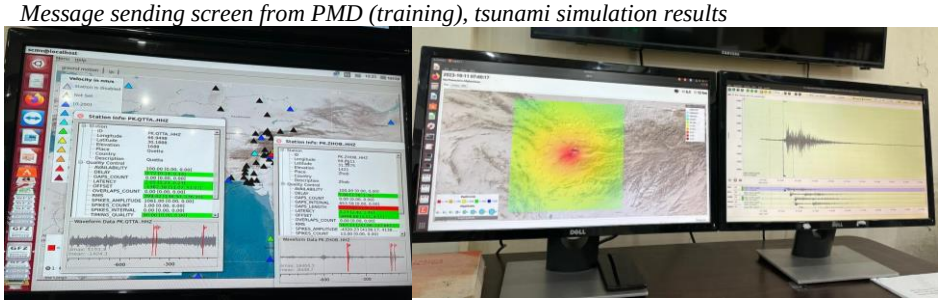
- PMD's future investment plans include the following.

- GNSS is installed at the above 10 strong-motion seismometer stations to monitor crustal movements.
- Equipment (geophones, etc.) for understanding ground conditions is deployed to indicate seismic intensity.
- Tsunami simulation software and human resource development for software utilization.

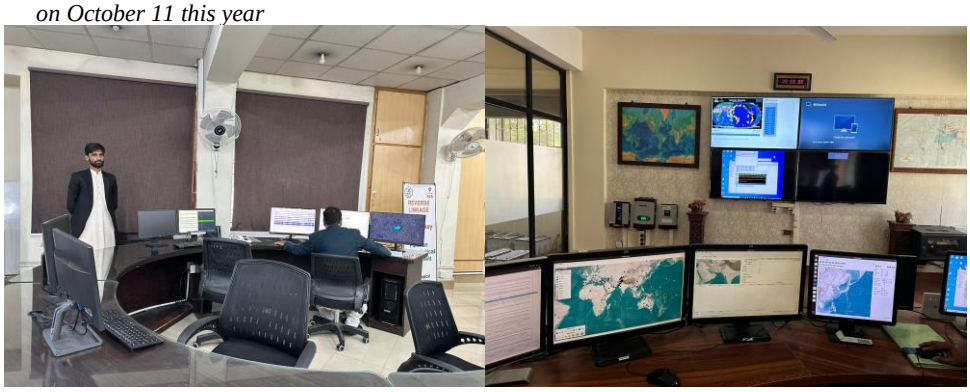
For reference, the following photos were taken during the interviews.



Message sending screen from PMD (training), tsunami simulation results



Seismic observation data, waveform and epicenter of the earthquake that occurred in northwestern Afghanistan on October 11 this year



Left: Renewed Earthquake Observation Center (Islamabad) Right: Tsunami Observation Center (Karachi)

4.1.4 Human Resource Development

The availability of existing plans was confirmed mainly through the websites of the State/Gilgit Balochistan/Provincial Disaster Management Authorities (S/GB/PDMAs), Rescue 1122 (emergency rescue services Department), and Civil Defence in each province, which serve as the main implementing organizations of Intervention 5 of the current NDMP. The plans for human resource development activities in the field of disaster management in the other main implementing agencies (NDMA, NIDM, Provincial Education Departments, etc.) were not confirmed.

Table 4.1.19 Existence of Human Resource Development Plan on Disaster Management for Relevant Agencies in Each Province

Province	Organization/Authority	Provincial government affiliation	Whether or not there is a plan and how to handle the human resources development area	Secured budget (Reviewed Annual Development Programme 2023-2024 and its equivalent)	Remarks
Punjab	PDMA	Relief Department	No Provincial Disaster Management Plan (PDMP) on Website	No budget for PDMA was found in the ADP	-
	Rescue 1122	Emergency Service Department (Rescue 1122 Department)	No plans found on website	Budget for Capacity Building activities (Rescue Professional and Community Rescue Team) found under the ADP	It owns Emergency Services Academy. provides Medical First Responder, Collapsed Structure Search and Rescue, and Training for Instructors in addition to 6-month Basic Course.
	Civil Defence	Home Department	No plans found on website	No Home Department budget found on ADP	-
Sindh	PDMA	Rehabilitation Department	There is an MHVRA Informed Disaster Management Plan (2023-2032), but there is no overall human resource development plan because the plan is developed by disaster type.	PDMA has a budget under the ADP, but there is no budget that seems to be related to human resource development.	The World Bank is implementing the Sindh Resilience Project, which includes capacity building activities.
	SERS (Sindh Emergency Rescue Service)	Rehabilitation Department	No plans found on the website.	ADP does not have a budget for SERS.	The World Bank project has recently strengthened SERS.
	Civil Defence	Home Department	No plans found on the website	No budget found on ADP for Civil Defence	Budget listed on the website, but only Recurrent. It is mentioned on the website that technical training (Training of firefighting, First Aid/Casualty, Emergency preparedness, Earthquake awareness) can be provided, but it is not budgeted. So, it is assumed that it will be provided upon request.
KPK	PDMA	Relief, Rehabilitation and Settlement Department	Provincial DRR Policy and Provincial Disaster Management Plan are in process (based on the interview survey)	ADP has a budget for PDMA, but budget that seems to be related to human resource development is not on ADP.	-
	Rescue 1122	Relief, Rehabilitation and Settlement	No plan found on the website	There is budget for infrastructure development, but no	Academy will be built in the Accelerated Implementation

		Department		budget that seems to be related to human resource development.	Programme (AIP).
	Civil Defence	Relief, Rehabilitation and Settlement Department	No plans found on the website	No budget for Civil Defence found on ADP	-
Balochistan	PDMA	Board of Revenue	Provincial Disaster Management Policy is in progress (based on the interview survey)	There is a budget for PDMA in the Public Sector Development Programme (PSDP), but there is no budget related to human resource development.	-
	Rescue 1122	Part of PDMA	No own website	PSDP has budget only for infrastructure	Rescue 1122 works only in Quetta. Balochistan is broadly covered by MERC 1122 which mainly works along the highway.
	MERC (Medical Emergency Response Centre) 1122	Health Department	No plans found on website	No budget found on PSDP for MERC 1122	-
	Civil Defence	Not confirmed	Website not found	No Civil Defence budget found on PSDP	(Balochistan government site are not accessible.)
GB	GBDMA	Home and Prisons Department	Disaster Risk Management Plan 2023 (2023-2027) with plans for human resource development	"Strengthening of DRM System" and "Establishment of Comprehensive DM System" are found on ADP, but details are unclear	UNDP GLOF II is conducting training. In 7 districts with many disasters, Assistant Directors are assigned to DDMAs.
	Rescue 1122	Home and Prisons Department	No plans found on the website.	ADP has budget for Rescue 1122 (infrastructure development), but no budget for human resource development.	-
	Civil Defence	No confirmation.	No website found.	No Civil Defence budget found on ADP	-
AJ&K	SDMA	Relief, Disaster Management and Civil Defence Department	State Disaster Risk Management Plan 2017 (2016 – 2020) with plans for human resource development	In a section on Civil Defence & Disaster Management on ADP, "Targeting training of 150 volunteers for water rescue" is included.	-
	Rescue 1122	Part of SDMA	No own Website	No budget for Rescue 1122 found on ADP	-
	Civil Defence	Not confirmed	No Website found	Budget for Civil Defence (infrastructure development) found on ADP, but no budget for human resource development	-

As shown in the table above, plans for human resource development in the field of disaster management could not be confirmed for most organizations. Only GBDMA in Gilgit-Baluchistan has a solid plan. According to the Disaster Risk Management Plan 2023 of Gilgit-Baltistan, the human resource development activities planned from 2023 to 2027 are shown in the following table.

Table 4.1.20 Human Resource Development Activities of the Disaster Risk Management Plan 2023 of Gilgit-Baltistan

Strategies	Priority Actions/ Programmes	Responsible Organizations	Supporting Organizations
Implement drills and training of disaster management activities with the organisations/line departments to improve their capacity	Each department conducts drills and training as per their role. GBDMA should oversee the drill exercises.	GBDMA, Line Departments, DDMAs	I/NNGOs, Rescue-1122, civil defence, etc.
Strengthen early warning dissemination system	Conduct simulation/drill exercises on safe evacuation	GBDMA	NDMA. PMD/PID/ NGOs/PTA/PEMRA/DDMAs/ DMCs
Develop capacity for early warning and evacuation systems	Improve community engagement in EWS though training and drills	GBDMA, DDMAs	DMCs, I/NNGOs, Irrigation, Agriculture, LG&RD
Raise public awareness of disaster management	• Awareness campaigns including walks, seminars, competitions, posters, DRM exhibitions, global event days, special coverage in print and electronic media, social media messages, etc. for the general public • Appreciation and incentive schemes for best practices in disaster management by various actors including I/NGOs, civil society, govt. institutions, academia, media, community representatives, etc. • Improve knowledge and understanding of available technologies for disaster reduction and MHVRA among parliamentarians, policy makers, and professionals through seminars, workshops, etc.	GBDMA, GBEPA	P&D, DDMAs, DMCs, Education Dept, media
	Include courses on disaster management in higher education	GBDMA	Education Department, HEC, Universities/Media/ UN/ NGOs
Training and research programme through Disaster Management Institutions	• Develop a pool of trainers through regular ToT programmes • Develop and standardise training materials on DRM related subjects, i.e. CBDRM, SBDRM, climate-induced disaster management, etc. • Trainings for government employees on DRM related subjects • Trainings for line departments on DRM related subjects like controlling soil erosion, climate change adaptation, bioengineering techniques for flood, and landslides management • Trainings for electronic and print media persons on disaster reporting • Trainings and capacity development for post-disaster recovery, rehabilitation and reconstruction	GBDMA	KIU, UoB, PEC, Works Dept. Rescue 1122, GB-Scouts

Source: Disaster Risk Management Plan 2023 Gilgit-Baltistan

Rescue 1122 in Punjab, Rescue 1122 in KPK (limited to Academy construction), and SDMA in AJ & K are the organizations that either allocate budgets for human resource development activities in their Annual Development Programmes or have existing, though outdated, plans in place. But it is difficult to confirm the details of the budget or the plan. Although not funded by the government, it seems that human resource development activities are being carried out by PDMA Sindh as part of the World Bank project; however, the specific details of these activities have not been confirmed.

4.1.5 Urban Disasters and Land Use

4.1.5.1 Urban Disasters in General

In the field of urban disasters and land use planning, the following two types of surveys have been conducted to collect information on projects for the investment in disaster risk reduction under the current NDMP.

Table 4.1.21 Survey to Collect Information to Examine the Draft of Projects for the investment for disaster risk reduction

No.	Type	Investigation and work
1	Literature investigation	Collection, confirmation, and analysis of data on existing legislation in Pakistan regarding land use considering urban disasters, and related plans developed by government agencies and donors in Pakistan
2	Interviews	Confirmation of the implementation status of Projects for the investment for disaster risk reduction in local governments as described in the current NDMP.

Source: Project Team

(1) Literature Survey

Among the data collected by the project, information related to land use control in the context of disaster risks was identified within existing legal frameworks and planning documents. The following table shows the data that have been confirmed.

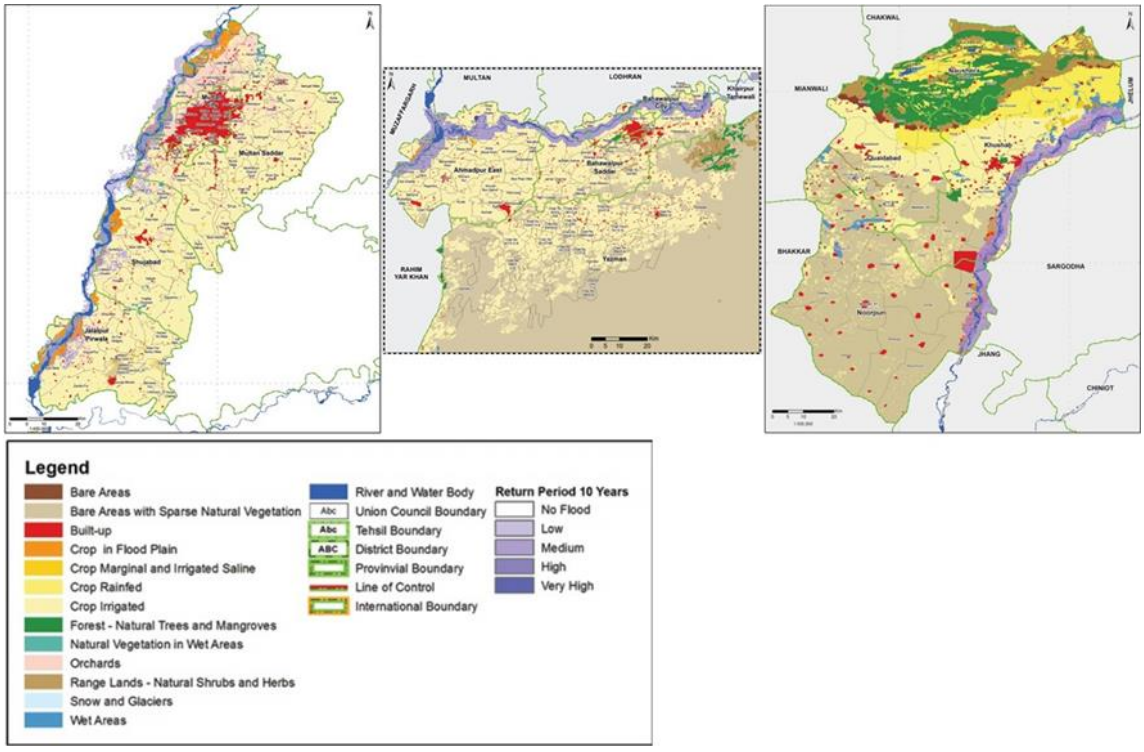
Table 4.1.22 Literatures on Land Use Control Considering Disasters

No.	Name	Source	Year
1	National Disaster Management Plan (NDMP)	NDMA	2012
2	National Disaster Response Plan (NDRP)	NDMA	2019
3	National Policy Guidelines on Vulnerable Groupe in Disasters	NDMA	
4	National Climate Change Policy	PMD	2021
5	Pakistan Environmental Protection act 1997	Government of Pakistan	1997
6	Review of Initial Environmental Examination and Environmental Impact Assessment Regulations	Government of Pakistan	2000
7	Pakistan Climate Change Act	Government of Pakistan	2017
8	National Environmental Policy	Government of Pakistan	2005
9	Land Acquisition Act 1894	Government of Pakistan	1894
10	National Resettlement Policy (Draft)	Government of Pakistan	2002
11	Training Manual for Disaster Risk Management	NDMA	2010
12	Building Code of Pakistan-2021	Pakistan Engineering Council	2021
13	Building Code of Pakistan - Seismic Provisions-2007	Pakistan Engineering Council	2007
14	Building Code of Pakistan- Energy Provisions-2011	Pakistan Engineering Council	2011
15	Pakistan Electric - Telecommunication Safety Code (PETSAC)-2014	Pakistan Engineering Council	2014
16	Building Code of Pakistan- Fire Safety Provisions-2016	Pakistan Engineering Council	2016
17	Seismic Hazard Assessment of Mansehra	NDMA	2009
18	Seismic Hazard Assessment of Muzaffarabad	NDMA	2009
19	Site Specific Probabilistic Seismic Hazard Analysis of Mansehra Urban Area	NDMA	2009
20	Multi Hzarad Vulnerability and Risk Assessment (MHVRA)	NDMA/WFP	2015
21	The Urgent Development Study on Rehabilitation and Reconstruction in Muzaffarabad City	JICA	2007
22	Annex of the Project Formation Investigation Report for Disaster Risk Reduction and Management	JICA	2008

Source: Project Team

The results of confirmation of the above documents are as follows.

- No. 1 "National Disaster Management Plan (NDMP)": Land use related action plans were described by each project entity.
- No. 11 "Training Manual for Disaster Risk Management": Participatory DRRM (Disaster Risk Reduction and Management) content on land use considering disasters and land use as non-structural measures was confirmed.
- No. 22 " Annex of the Project Formation Investigation Report for Disaster Risk Reduction and Management, JICA, 2008": The necessity of land use regulation considering disasters by local governments was pointed out.
- No. 20 "Multi Hazard Vulnerability & Risk Assessment (MHVRA)": Analysis by overlapping land use and hazard maps for each disaster type was confirmed (figure below). The disaster types covered in this document are flood damage, sediment disaster, cyclone, drought, and tsunami. By overlapping these hazard maps with geographic information on population distribution and infrastructure, the objective was to analyze and visualize where and what kinds of hazards exist. In this document, analysis by overlapping flood hazard maps (Annual probability of exceedance: 10, 30, 100 years) and built-up areas were performed. Figure 4.1.2 shows, from left to right, urban areas exposed to flood risk in Multan, Bahawalpur and Khushab cities (excess probability year: 100 years). This figure clearly indicates the presence of flood risk in and around urban areas.



Source: MHVRA (Current NDMP Multi Hazard & Vulnerability Risk Assessment Report)

Figure 4.1.2 Example of Analysis by Overlaying Hazard Maps for Each Land Use and Disaster Type

(2) Interview Survey

In order to confirm the implementation status of the projects for the investment for disaster risk reduction described in the current NDMP, interviews and information/data collection surveys were conducted with the local governments officials. The following table provides an overview of the survey.

Table 4.1.23 Overview of the Interview Survey

Item	Contents
Period	Phase 1: Mid-September to late September 2022 Phase 2: Late September to early October 2023
Purpose	Confirmation of the implementation status of Projects for the investment for disaster risk reduction described in the current NDMP In particular, collection of information on land use control in consideration of disasters
Subject and date of survey	1. September 15, 2022: Urban Planning Directorate, Capital Development Authority (CDA) 2. September 19, 2022: Design Directorate, Capital Development Authority (CDA) 3. September 20, 2022: Directorate of Land Use, Building, Rawalpindi Development Authority (RDA) 4. September 23, 2023: Karachi Development Authority (KDA), Sindh Master Plan Master Plan Authority (SMPA) 5. October 6, 2023: HUD & PHED: Housing Urban Development & Public Health Engineering Department, Government of The Punjab 6. October 6, 2023: Lahore Development Authority
Survey item	1. Use of disaster hazard and risk information in land use planning 2. Guidelines for land use in areas with disaster risk 3. Request for materials 4. Specific projects related to disaster prevention 5. Actual status of land use regulations in consideration of disasters

Source: Project Team

The following is a summary of the results of the interviews.

Table 4.1.24 Summary of the Results of the Interviews (Urban Planning Directorate, Capital Development Authority (CDA))

No.	Survey item	Survey result
1	Use of disaster hazard and risk information in land use planning	• CDA recognizes flood disasters and earthquakes (including heat waves in recent years) as major disasters. • Large-scale flood disasters occurred in the 1990s, and large-scale flood disasters have occurred in recent years. Based on this, a hydrological survey (E11 hydrological survey) was conducted under the direction of the federal government.
2	Guidelines for land use in areas with disaster risk	• Although flood disasters have occurred in settlements near urban rivers, Urban Planning Directorate of the Capital Development Authority (CDA) has not taken measures to regulate land use in consideration of disasters. However, in light of recent flood disasters, the authority is now considering measures to regulate land use in disaster-prone areas.
3	Request for materials	The following materials were provided by Urban Planning Directorate, Capital Development Authority (CDA). Zoning map is available from CDA by a letter from NDMA or JICA. 1. The Islamabad Laws: This is a document of the ordinance by CDA. Chapter 4 is land use related (land acquisition for public projects, CDA territory, and land authority). Specific land use regulations are described in "DEVELOPMENT OF PRIVATE HOUSING/FARM HOUSING SCHEMES IN ZONES 2, 4 & 5_2020," and “Building Control Regulations 2020”. 2. Building Control Regulations: Chapter 1 is definitions of terms, chapter 2.1 is building uses, chapter 2.2 is building inspections, and later inspection standards similar to those in Japan (volume, building coverage ratio, height, setback, etc.) 3. Revised Modalities & Procedures (2020) for development of Private Housing/Farm Housing Schemes in Zones 2, 4 & 5 of ICT Zoning Plan: There are technical and site standards similar to the development permit system in Japan. 4. ICT (Islamabad Capital Territory) Zoning Regulations 2005: Restrictions on change of use and restrictions on development over a certain area. Building inspections are based on Building Control Regulations.
4	Specific projects related to disaster prevention	• CDA conducted a hydrological survey based on recent flood disasters. Based on the survey results, projects related to disaster prevention are under consideration.
5	Actual status of land use regulations in consideration of disasters	• Urban Planning Directorate, Capital Development Authority (CDA) does not operate land use regulations and systems considering disasters.

Source: Project Team

Table 4.1.25 Summary of the Results of the Interviews (Design Directorate, Capital Development Authority (CDA))

No.	Survey item	Survey result
1	Use of disaster hazard and risk information in land use planning	• In July 2021, 2 deaths due to flood disaster occurred in E11 sector of Islamabad. E11 is a private housing society, and the administratively managed the river's rights-of-way (ROW) was narrowed due to unauthorized private development and did not comply with CDA's development permit standards. • "The Study on Comprehensive Flood Mitigation and Environmental Improvement Plan for the Lai Nullah Basin, JICA, 2003" pointed out the necessity of flood risk and structural measures in E11 area. • Based on the findings of "E11 Hydrological Survey" by the federal government, CDA advised residents of E11 area to avoid living in areas identified as having flood risk.
2	Guidelines for land use in areas with disaster risk	• The submission of traffic impact assessment and hydrological impact assessment is required to obtain building and development permits.
3	Request for materials	• Report of "E11 Hydrological Survey": Available with CDA and can be obtained by writing a letter from NDMA or JICA
4	Specific projects related to disaster prevention	• As flood risk countermeasures, "E11 Hydrological Survey" is being carried out by the Design and Consulting Bureau (DD&C) of the military. Specific countermeasures are still under consideration.
5	Actual status of land use regulations in consideration of disasters	• Land use regulation considering disasters is not particularly implemented.

Source: Project Team

Table 4.1.26 Summary of the Results of the Interviews (Directorate of Land Use, Building, Rawalpindi Development Authority (RDA))

No.	Survey item	Survey result
1	Use of disaster hazard and risk information in land use planning	• Types of Hazards of Concern in Rawalpindi: Floods and Water Pollution • Rawalpindi Development Authority (RDA) does not possess a flood hazard map
2	Guidelines for land use in areas with disaster risk	• Since disasters are not considered a major concern in Rawalpindi,, Rawalpindi Development Authority (RDA) has not prepared guidelines for disaster risk management.
3	Request for materials	The following materials were provided by the Urban Planning Directorate, Capital Development Authority (CDA). In addition, the project team requested Rawalpindi Development Authority (RDA) to share materials and data such as land use area maps, city planning documents, and land use regulation guidelines. • Building And Zoning Regulations, 2007: Building and Development Review Standards.
4	Specific projects related to disaster prevention	• Technical and budgetary support is provided to Rawalpindi Development Authority (RDA) by Punjab Province (not by the federal government).
5	Actual status of land use regulations in consideration of disasters	• Land use regulations in consideration of disasters are not specifically implemented.

Source: Project Team

Table 4.1.27 Summary of the Results of the Interviews (Karachi Development Authority (KDA), Sindh Master Plan Master Plan Authority (SMPA))

No.	Survey item	Survey result
1	Use of disaster hazard and risk information in land use planning	• SMPA does not have information on the results of natural disaster surveys such as hazard/risk maps. • The results of natural disaster risk assessments conducted by the Sindh Province Government are not shared between Sindh Province and SMPA because there is no coordination mechanism for national and local disaster management plans and related plans. • As there is no hazard/risk map, it is not possible to identify high-risk areas and there is no

		regulation of building and development activities in high-risk areas.
2	Guidelines for land use in areas with disaster risk	• The building and development permit system in the SMPA planning area is based on "Karachi Building and Town Planning Regulations-2002". • The ordinance does not include criteria for land use in areas of disaster risk, for example, building and development activities are permitted in areas of high flood risk.
3	Request for materials	SMPA/KDA provided the following materials to the project team: 0_ About SMPA – 2023(Word) 1_Questionnaire Sheet_Filled_SMPA (Word) 2_Karachi Strategic Master Plan 2020 (PDF) 3_Karachi Building and Town Planning Regulations-2002 (PDF) 4_Greater Karachi Plan – 1952 (PDF) 5_Karachi Development Plan (1974-85) (PDF) 6_Karachi Development Plan 2000 (PDF) 7_Project List (hard copy/Scanned PDF) 8_Land Use Zoning Maps (5 maps, hard copy)
4	Specific projects related to disaster prevention	• In the material provided by SMPA/KDA (7_Project List), 14 projects related to disaster management proposed by Karachi City were described. • The projects included the establishment of a disaster management organization, improvement of drainage channel, drainage channels, dams and canals, and water storage tanks, improvement of topographic maps and geographical information including elevation data, cleaning of waterways, and introduction of a monitoring system for illegal dumping into waterways.
5	Actual status of land use regulations in consideration of disasters	• The regulation of building and development activities in the area under the jurisdiction of SMPA/KDA is based on "Karachi Strategic Master Plan 2020" and is operated based on the "Karachi Building and Town Planning Regulations-2002". • The above does not specifically regulate building and development activities in areas with high disaster risk. Therefore, land use regulation considering disaster is not implemented.

Source: Project Team

Table 4.1.28 Summary of the Results of the Interviews (Housing Urban Development & Public Health Engineering Department, Government of The Punjab)

No.	Survey item	Survey result
1	Use of disaster hazard and risk information in land use planning	• The organizations responsible for hazard mapping are Provincial Disaster Management Authority (PDMA) and District Disaster Management Authorities (DDMAs). The organization responsible for the "flood" hazard map is Punjab Irrigation Department (PID). Housing Urban Development & Public Health Engineering Department (HUD & PHED) considers that PDMA, DDMA and PID should reflect hazard maps and risk information in plans related to disaster risk management such as land use and housing development.
2	Guidelines for land use in areas with disaster risk	• In Section 7 (Evaluation Criteria) of Chapter II (Application for Approval) of the "Notice concerning the exercise of powers under section 43 of the Punjab Urban Development Act", one of the criteria is the absence of flood hazard. The judgment of the absence of flood hazard is made by Punjab Irrigation Department (PID).
3	Request for materials	The following material was provided to the project team by HUD & PHED: 1. "Notification in exercise of the powers conferred under section 43 of the Punjab Development of Cities Act"
4	Specific projects related to disaster prevention	• HUD & PHED is implementing the Affordable Housing Scheme with the support of the World Bank (WB) to eliminate the shortage of 5 million housing units. This housing project considers disaster preparedness and mitigation. • To address recent flood disasters, river management facilities have been equipped with a devices installed above the anticipated inundation level. (Example: the equipment related to electronic and the drain pump facility is installed at a position higher than the assumed inundation depth in case of a power failure, and the electric power is supplied by sunlight.)
5	Actual status of land use regulations in consideration of disasters	• Efforts to enforce regulations against illegal construction are ongoing throughout Punjab Province.. However, there are challenges in regulating housing development in areas with high disaster risk.

Source: Project Team

Table 4.1.29 Summary of the Results of the Interviews (Lahore Development Authority)

No.	Survey item	Survey result
1	Use of disaster hazard and risk information in land use planning	• Described in "Master Plan of Lahore 2050" which is available from the Lahore Development Authority (LDA) through an official letter from the project team.
2	Guidelines for land use in areas with disaster risk	• The Lahore Development Authority (LDA) reviews building and development activities based on "Land Use Regulations 2020". However, the disaster risk management criteria within this regulation are limited in scope. Specifically, regarding the development of health and education facilities as well as industrial parks and facilities in agricultural areas, review of the development proposal is carried out only when the area to be developed is judged to have no flood risk.
3	Request for materials	Lahore Development Authority (LDA) provided the following materials to the project team. 1. Land Use Map (Existing Land Use & Proposed Land Use), Master Plan of Lahore 2050 2. Land Use Regulations 2020
4	Specific projects related to disaster prevention	• There is a case of buffer zone designation for development regulation. • There is a project to improve open and green spaces. This project is considered to contribute to disaster risk management. • The area along the Ravi River in Lahore is administered by the "Ravi Urban Development Authority (RUDA)".
5	Actual status of land use regulations in consideration of disasters	• The LDA basically does not permit development in areas where there is a risk of disaster. Specifically, if a building or development application submitted to the LDA is in a flood risk area such as around a river, the LDA will contact Punjab Irrigation Department (PID). If the PID judges the area as a flood risk area, the LDA will reject the building or development application. • According to the legal system, if illegal construction is confirmed in a flood risk area, the unauthorized building will be removed based on the land use regulations.

Source: Project Team

The findings obtained through the above interviews and data collection are as follows.

- In Pakistan, decentralization has been promulgated since the 18th constitutional amendment. Regarding the land use control such as city planning and building regulation with disaster considerations—the legal authority and budget are delegated to the state government and the development authority (DA) of each municipality. In Japan, prefectures and municipalities follow the legal system developed by the central ministries and agencies. As a result, they can expect support from the government in terms of budget and technology, and they do not rely on the government as the legal basis for urban planning and building administration.
- The subject of this survey has an operational building and development review system. While some unlicensed building and development activities still occur, the Development Authorities (DAs) of the respective municipality generally reviews and approves private building and development projects. Building and development regulations based on building standards, development standards, and designation of use areas also exist. According to the collected materials, regulations are specified for each use area, covering building and land use, floor area ratio, building coverage ratio, building height regulation, and technical and location standards such as building setback from the front road .
- As mentioned above, Pakistan has a legal system similar to Japan for application for building approval and development review. Building owners and developers of real estate, etc. apply for building and development activities to the Development Authority in advance and obtain permission to do so. These review systems are in operation and are functioning within local governments. Building and development activities are guided by technical and locational standards and are regulated based on zoning designations such as residential, commercial, industrial areas. However, building and development activities are not regulated in consideration of disasters like in Japan, as shown below.
- Under the Building Standards Act, local governments have the authority to designated disaster

risk areas. In these designated zones, residential construction is prohibited, and construction of structures necessary for disaster prevention is mandated.

- Based on the Act on Regional Development for Tsunami Disaster Management, local governments can designate special warning zones for tsunami disasters. In designated areas, both the use and structural specification of buildings are subject to regulation.
- Based on the Landslide Disaster Prevention Law, prefectural governors can designate special warning zones for landslide disasters. In designated areas, development activities are subject to specific restrictions, and buildings containing living spaces must comply with structural regulations..

In general, there is room for improvement in land use regulations with respect of disaster risk considerations. In particular, it is necessary to utilize hazard and risk information (reflected in planning and projects), strengthen regulations in areas with high disaster risk, and incorporate disaster risk assessment into building and development permit conditions. In addition, t was suggested that effective disaster countermeasures require strong coordination among federal and provincial governments, as well as among disaster risk management agencies within provincial governments, such as PDMAs, DDMAs, PID, WASA, DAs, etc..

4.1.5.2 Rainwater Drainage (Urban Flooding)

In rainwater drainage (urban flooding) sector, information was gathered through online sources, literature reviews, and interview surveys. Table 4.1.30 shows the outline of the information gathered through these surveys. Regarding the interview survey, the department responsible for rainwater drainage in Karachi and Lahore, the Pakistan's largest cities and frequent sites of urban flooding, was targeted.

Table 4.1.30 Outline of Information Gathering for "Rainwater Drainage (Urban Flooding)"

No.	Type	Survey and work
1	Internet and literature survey	Information gathering and analysis of websites (NDMA, FFC, PMD, WASA, KWSC, KMC, etc.) of administrative organizations related to rainwater drainage (urban flooding), laws and guidelines related to rainwater drainage plan, rainwater drainage plan, and rainwater drainage countermeasure project. Table 4.1.31 lists the collected data.
2	Interview survey	Interview surveys were conducted with the following organizations. ***Karachi:*** • Master Plan Department, Sindh Master Plan Authority (SMPA) • Karachi Water and Sewerage Corporation (KWSC) • Karachi Metropolitan Corporation (KMC) ***Lahore:*** • Water and Sanitation Authority (WASA) • Housing, Urban Development & Public Health Department, Government of Punjab • Lahore Development Authority (LDA)

Source: Project Team

Table 4.1.31 List of Research Documents

Source	Level	Reference Name	Year
NDMA	National	National Disaster Management Plan (NDMP)	2012
MoCC	National	National Climate Change Policy	2012
MoWR	National	National Water Policy	2018
NDMA	National	National Monsoon Contingency Plan 2023	2023
FFC	National	National Flood Protection Plan-IV	2017
FFC	National	Annual Report 2021	2022
Punjab Emergency Service	Punjab Province	Flood Contingency Plan -2021	2021

Source	Level	Reference Name	Year
Rehabilitation Dept. of Gov. of Sindh, PDMA	Sindh Province	Sindh Provincial Monsoon/Floods Contingency Plan 2012	2012
Planning & Development Dept., Gov. of Sindh	Sindh Province	Planning Manual	2023
PHED Punjab	Punjab Province	Design Criteria for Water and Sanitation Agencies & PHED Punjab	2017
WASA Lahore	Lahore	Final Master Plan Report -Drainage (Preparation of Master Plan for Water Supply, Sewerage and Drainage System for Lahore)	2019
WASA Rawalpindi	Rawalpindi	Sewerage and Drainage Master Plan (Comprehensive Master Planning for Water Supply, Sewerage, Drainage and Waste Water Treatment System in Rawalpindi)	2016
WASA Multan	Multan	Master Plan -Sewerage and Storm Water Drainage System (Master Planning of Water Supply, Sewerage and Drainage System of WASA Multan)	2017
World Bank	National	Pakistan – Getting More from Water	2019
World Bank	Karachi	Project Appraisal Document on a Proposed Loan in the Amount of US$40 Million to the Islamic Republic of Pakistan for a Karachi Water and Sewerage Services Improvement Project (KWSSIP)	2019
JICA	National	Final Report on Water Supply, Sewerage and Wastewater Sector Information Collection and Confirmation	2021
JICA	Karachi	Final Report on Karachi City Water Supply and Sewerage Improvement Plan	2008
JICA	Lahore	Preparatory Study Report on Lahore City Sewage and Drainage Equipment Emergency Recovery Plan	2010
JICA	Multan	Preparatory Study Report on Multan Sewage and Drainage Capacity Improvement Plan (preliminary release)	2022
JICA	Faisalabad	Final Report on Faisalabad Water Supply, Sewage and Drainage Master Plan Project	2019
JICA	National	Completion Report on Flood Management Advisor Project	2023

Source: Project Team

(1) Rainwater Drainage Project in Karachi

The sewerage system in Karachi is undergoing a transition from a confluence system to a split-flow system. The sewerage system is mainly handled by Karachi Water and Sewerage Corporation (KWSC) (formerly known as KWSB) and rainwater drainage by Karachi Metropolitan Corporation (KMC). According to interviews with KWSC and KMC conducted on November 20 and 21, 2023, Karachi is divided into 6 District Municipal Corporations (DMC), with responsibilities for rainwater drainage distributed among these municipal entities. In addition, the Defense Housing Authority (DHA) has jurisdiction over the sewage and rainwater drainage in the military area (Clifton Cantonment). Table 4.1.32 shows the rainwater drainage projects planned or implemented by KMC.

DHA has been constructing a 74 km drainage channel since September 2021, and the Karachi Water and Sewerage Services Improvement Project (KWSSIP), jointly financed by the World Bank and the Asian Infrastructure Investment Bank, is scheduled to be implemented from 2019 to 2031.

Table 4.1.32 Rainwater Drainage Projects Implemented and Planned by KMC (Karachi)

No.	Sector/Sub-Sector/Name of Scheme	Location	Estimated Cost (Mil. Rupees)	Approval Date
1	Providing and Laying 18" Dia Sewerage Drain from Khokrapar No.6 Naddi to Saudabad Chowk District Malir.	Malir	20.000	1-Jan-17
2	Rehabilitation/Construction of Sunkdown Strom Water Drains at intersection of Road 8000 and Road 5000 in District Korangi Karachi.	Korangi	19.999	1-Feb-17
3	Improvement/Rehabilitation of SW System and Laying of pipes for drainage of rainwater from Ayesha Manzil to Landi Kotal Chowrangi	Central	20.000	1-Jun-19
4	Improvement and Rehabilitation of drainage system and maintenance of Sohrab Goth underpass.	Central	20.000	1-Jun-19
5	Improvement/Rehabilitation of road c.c. flooring, drainage system in UC-25 & UC-26 at Gulshan-e- Iqbal, District East KMC.	East	20.000	1-Jun-19
6	Improvement/Rehabilitation of water Drainage system in different area of Scheme 36 District East	East	19.999	25-Jun-20
7	Improvement/Rehabilitation of Internal Roads and their Drainage system Various places in District Korangi, KMC.	Korangi	19.998	25-Jun-20
8	Improvement/Rehabilitation of Road &their Drainage system from Saudi College to Saudabad School No.1District Koran	Korangi	19.992	25-Jun-20
9	Improvement/Rehabilitation of Road &their Drainage system from Hurmain Chowrangi to Kala Board, District Korangi	Korangi	20.000	25-Jun-20
10	Improvement/Rehabilitation of Internal Roads and Their Drainage system in Nooruddin Goth, Shah Faisal Zone, District	Korangi	20.000	25-Jun-20
11	Improvement/Rehabilitation of Road & Drainage system from Ghanchi Hall to Saudabad Chowk, District Korangi.	Korangi	19.999	25-Jun-20
12	Providing & Laying CC Flooring/Paver Block and Drainage system in Different Streets of Naik Muhammad Goth, Jalal Mur	Malir	20.000	25-Jun-20
13	Improvement/Construction of Internal Roads and their Drainage system in the Various Places in District Korangi,	Korangi	19.999	1-Jun-20
14	Providing & Laying CC Flooring/Paver Block and Drainage System in Madina Colony, District Korangi, KMC.	Korangi	19.998	1-Jun-20
15	Improvement/Rehabilitation of Road & their Drainage system from Rana Marriage Hall to 9-C Bus Stop, District Korangi	Korangi	19.997	1-Jun-20
16	Improvement/Rehabilitation of Road and Their Drainage system from Madina Masjid to Young Football Ground, District Malir, KMC.	Malir	19.996	1-Jun-20
17	Improvement/Rehabilitation of Road and Drainage System from Muhabbat Nagar to Ali Clinic, District Malir	Malir	19.977	1-Jun-20
18	Rehabilitation/Improvement of drain from Martin Road to Guru Mandir at Jehangir Road District East KMC.	East	20.000	4-Nov-23
19	Rehabilitation/Improvement of drain form Masjid Kanzul Eman to Utility Store at Jehangir Road, District East KMC.	East	20.000	4-Nov-23
20	Construction of footpath & drainages system from Teen Hatti to Utility Storeat Jahangir road.	Central	20.000	4-Nov-23
21	Construction of footpath & drainage system from Utility Store to Guru mandir at Jahangir Road.	Central	20.000	4-Nov-23
22	Repair/Improvement of Drainage System in different Areas of Gulshan-e-Iqbal at District East, KMC	East	20.000	4-Nov-23
23	Improvement of Water & Drainage System in Scheme-33, Gulzar-e-Hijri at District East, KMC	East	20.000	4-Nov-23
24	Repair/Improvement of Water & Drainage System in Gulistan-e-Juher at District East, KMC	East	20.000	4-Nov-23
25	Repair/Improvement of Drainage System in, Block-2 & 3 at Gulistan-e-Joher District East KMC	East	20.000	4-Nov-23
26	Repair/Improvement of Roads & Drainage System in Block-16 & 17 Gulistan-e-Joher District East, KMC	East	20.000	4-Nov-23
27	Repair/Improvement of Water & Drainage System in different areas of Jamshed zone, District East, KMC	East	20.000	4-Nov-23
28	Construction of Strom Water Drain/Nallah from Building Road Department South to Wafaqi Urdu College Hera Lal Gantra Road District South Karachi	South	20.000	4-Nov-23

Source: Created by project team based on KMC Website

(2) Rainwater Drainage Project in Lahore

Rainwater Drainage Project in Lahore is managed by WASA Lahore. WASA Lahore developed the Rainwater Drainage Master Plan (MP) in 2018 through local consultants, with a target implementation period extending to 2040. Projects related to rainwater drainage have since been implemented in line with this plan. The rainwater drainage projects under the plan are shown in Table 4.1.33.

According to the interview conducted with WASA Lahore conducted on October 5, 2023, the foreign assisted rainwater drainage projects include a feasibility study on urban flood prevention and management supported by the Australian government, as well as assistance from the Danish and French governments and the Asian Infrastructure Investment Bank for the construction of sewage treatment plants and sewer pipelines.

Table 4.1.33 Rainwater Drainage Project under Current Plan (Lahore) (1/3)

MASTER PLAN ACTIVITY	SUB PROJECT	Cost (m.PKR)	Source of Funds	2018-2020	2021-2025	2026-2030	2031-2035	2036-2040	REMARKS
DEVELOPMENT EXPECTATION	Adequacy of Existing Drainage System	-	-	Adequate with some rectification.	Adequate with some rectification.	Adequate with some maintenance	Adequate with some maintenance	Adequate with some maintenance	WASA to continue service with existing system with some improvements.
	New Drainage Requirement beyond Limits of Existing System	-	-	-	Additional drainage for new developments	Additional drainage for new developments	Additional drainage for new developments	Maintenance of additional drainage	WASA to provide drainage for future private housing and industrial areas planned in Lahore district.
INSTITUTIONAL DEVELOPMENT	**Objective: Transition of WASA into Effective and Efficient Institution of Drainage System**								
	Legal Strengthening of WASA	-	-						Required to enforce levy on injection of sewage in storm water drains by private enterprises.
	Strengthening of WASA Operation and Maintenance Department		Govt of Punjab						Required to deliver Efficient and Effective Services
	Strengthening of Drainage Machinery Procurement Program.		Govt of Punjab						Required to ensure more effective Cleaning of Open and Covered Drains.
	Compliance with EPA Requirements		-						Required to Ensure WASA Contributes Responsibility to the Environment.
Existing System Components	**Objective: To Sustain and Operate the Existing Drainage Set-up Efficiently**								
	A. Administrative Matters								
	Implementation of cleaning and maintenance program of drains								
	Inspection and clearance of RoW / Encroachment of drains								
	Procurement of Special Dredging Machinery For Covered Drains								
	Liaison and coordination with Punjab Irrigation Department for maintenance and dredging of: 1. Rohi drain 2. Hudiara Drain Reach (From Crossing with BRBD to River Ravi) 3. Provision of a Waste Water Treatment (WWT) plant at end of Hudiara drain.								
	B. Structural Rectification								
	Remodeling of Low Level Bridges on Main Drains	13	Govt of Punjab						Medium Priority
	Rehabilitation of Narrow Sections of Main Drains	31	Govt.of Punjab						High Priority

Source: Preparation of Master Plan for Water Supply, Sewerage and Drainage System for Lahore, Final Master Plan Report-Drainage, Main Report, February 2019

Table 4.1.33 Rainwater Drainage Project under Current Plan (Lahore) (2/3)

MASTER PLAN ACTIVITY	SUB PROJECT	Cost (m.PKR)	Source of Funds	2018-2020	2021-2025	2026-2030	2031-2035	2036-2040	REMARKS
	Mitigation of Urban Inundation Across City Areas	617	Govt.of Punjab						High Priority
	Remodeling of Central Drain due to construction of new disposal station at Lakshmi Chowk to discharge in Central Drain and acquisition of ROW in commercial belt along drain	3,735	Govt of Punjab						High Priority
	Acquisition of ROW in Residential belt along Lower Chotta Ravi Drain	1,311	Govt.of Punjab						Medium Priority
	Construction of New Drain from Barket Town to River Ravi along GT Road in Shahdara Drain Region	95	Govt of Punjab						High Priority
	Construction of New Drain from Begum Kot Chowk to Shahdara Drain at Bara Dari Road	55	Govt.of Punjab						Medium Priority
	Rehabilitation of Shalimar Escape Channel	54	Govt.of Punjab						Medium Priority
	Rehabilitation of College Road drain in Township	38	Govt.of Punjab						Medium Priority
	Drainage Proposals from Previous Studies	10,305	Govt.of Punjab						Medium Priority
	Additional pumps for i. At Babu Sabu Pumping station (for 250 cusecs) ii. At Mahmood Booti Pumping Station (for 150 cusecs)	232	Govt.of Punjab						Medium Priority
	Procurement of Modern Cleaning Machinery for open and covered drains (Provisional)	500	Govt.of Punjab						High Priority
	Sub Total	**16,986**							
New Development Components	**Objective: To Cater for Drainage Facilities for New Development Areas in Lahore Distrcit**								
	A. Planning Options								
	Outfall Drain - Use of existing Hudiara Drain for New Areas - New Outfall Drain to meet Hudiara at its end. - new outfall drain to open in River Ravi						**Outfall Drain** - Hudiara Drain has sufficient discharge capacity to accommodate additional runoff from new areas - New Outfall Drain to meet Hudiara at its end Requires additional Funds. - new outfall drain to open River Ravi requires Additional Funds		

Source: Preparation of Master Plan for Water Supply, Sewerage and Drainage System for Lahore, Final Master Plan Report-Drainage, Main Report, February 2019

Table 4.1.33 Rainwater Drainage Project under Current Plan (Lahore) (3/3)

MASTER PLAN ACTIVITY	SUB PROJECT	Cost (m.PKR)	Source of Funds	2018-2020	2021-2025	2026-2030	2031-2035	2036-2040	REMARKS
	Planning Options of Drainage Network - Main drain along main road in new developments. - Secondary drains within roads of new development areas.						**Planning Options of Drainage Network** - Drain along roads in new developments will not require acquisition of ROW.		
	Delineation of Drainage Zones in Area beyond Existing set-up - North of Barki Road - North East of BRBD canal - East of Hudiara Drain - South East of Hudiara drain						**Delineation of Drainage Zones** - Drainage Zones to be delineated within prominent limits and availability of some outfall drain - New drains to cover areas with increasing population trend		
	Type of Drain ***Options*** *- Open drain (earthen)* *- Concrete drain (open top)* *- Concrete drain (covered in commercial area)* *- Brick masnary drain*						Open unlined drain requires wider strip of land - Open Concrete lined drain requires narrow strip of land - Covered Concrete lined drain requires additional cost of covering. -Brick masonry drain requires wider strip of land and cumbersome construction. -Cost of RCC drain and Cost of Brick Masonry Drain are quite match able. As such RCC Drain is adopted for New Developing Areas and Brick Masonry Drain is adopted for Existing Drainage area.		
	B. Structural Development								
	Drainage Zone I - Proposed Drainage System with outfall at Hudiara Drain	4,252	Govt.of Punjab						High priority
	Drainage Zone II - Proposed Drainage System with outfall at Hudiara Drain	2,972	Govt.of Punjab						Medium Priority
	Drainage Zone III -Proposed Drainage System with outfall at Hudiara Drain and Exising Nullahs	1,130.74	Govt.of Punjab				In view of meager population increase trend and the existing natural drains no additional drainage network is proposed in drainage Zone III		
	Drainage Zone IV - Proposed Drainage System with outfall at Hudiara Drain	3,218.79	Govt.of Punjab						Medium Priority
	Drainage Zone V - Proposed Drainage System with outfall at Hudiara Drain	3,323.09	Govt.of Punjab						High priority
	Drainage Zone VI - Proposed Drainage System with outfall at existing Rohi By-pass Drain	827.38	Govt.of Punjab						Medium Priority
	Drainage Zone VII - Proposed Drainage System with outfall in Existing Nullahs	1,729.26	Govt.of Punjab						Medium Priority
	Sub Total	**17453.3**							

Source: Preparation of Master Plan for Water Supply, Sewerage and Drainage System for Lahore, Final Master Plan Report-Drainage, Main Report, February 2019

4.1.6 EWS

4.1.6.1 Planned Disaster Prevention Investments in PMD

(1) List of Projects to Strengthen EWS

The list of proposed projects by PMD for strengthening flood early warning systems was obtained during meeting with PMD Islamabad on November 24 and 27, 2023, is shown in Table 4.1.34.

Table 4.1.34 List of Projects to Strengthen EWS of PMD

No	Project Name	Cost (mil. PKR)	Remarks
1	Establishment of Urban Flood and Flash Flood Forecast Warning Centers in Major Cities of Pakistan.	2,087	PC-I under process
2	Establishment of Weather Radars in Kalat, Mangla, Zhob, Dalbandin, Gilgit, Swat, Mansehra and Muzaffarabad.	7,500	PC-I under process
3	Establishment of Extreme Weather Early Warning System	1,000	PC-I under process
4	Strengthening and expanding earthquake monitoring networks with advanced analysis capabilities	1,000	
5	Establishment of tsunami early warning systems, including installation of tsunami detection buoys and sensors in coastal areas	1,200	
6	Advances in aviation weather services with specialized weather products and forecasts tailored to aviation needs	600	
7	Ensuring timely warning and evacuation through sediment disaster monitoring and early warning systems in high-risk areas	800	
8	Comprehensive meteorological and hydrological capacity development training and programs for meteorological and hydrological staff, including in collaboration with international organizations and experts	500	
9	Development of a weather forecast guidance system to enhance the accuracy and reliability of weather forecasts	1,500	
10	Development of a national early warning system framework, including a comprehensive national policy, Standard Operating Procedure (SOP) and early warning system guidelines	1,300	
11	Establishment of an upper-air observation network including radiosondes and LIDAR to enhance the quality and coverage of observations	2,000	
12	Establishment of a coastal tide monitoring system to monitor sea level changes and issue storm surge warnings	3,000	
13	Community awareness and evacuation training programs for effective response to meteorological disasters	800	
14	Extension of the GLOF early warning system in the GB, AJK and KP regions, including improved evacuation and response strategies for vulnerable areas in the event of a GLOF	4,000	
—	**Total**	**27,287**	

Source: Prepared by the project team based on PMD materials

(2) MHSP

Table 4.1.35 shows the outline of the Modernization of Hydro-met Services Project (MHSP) planned by PMD, based on the information obtained durings the meeting with PMD Islamabad on November 24 and 27, 2023, and Table 4.1.36 provides details of the project cost.

Figure 4.1.3 shows the observation station and radar network of PMD, and Figure 4.1.4 shows the AWS and radar expansion sites proposed by MHSP.

Table 4.1.35 Overview of MHSP

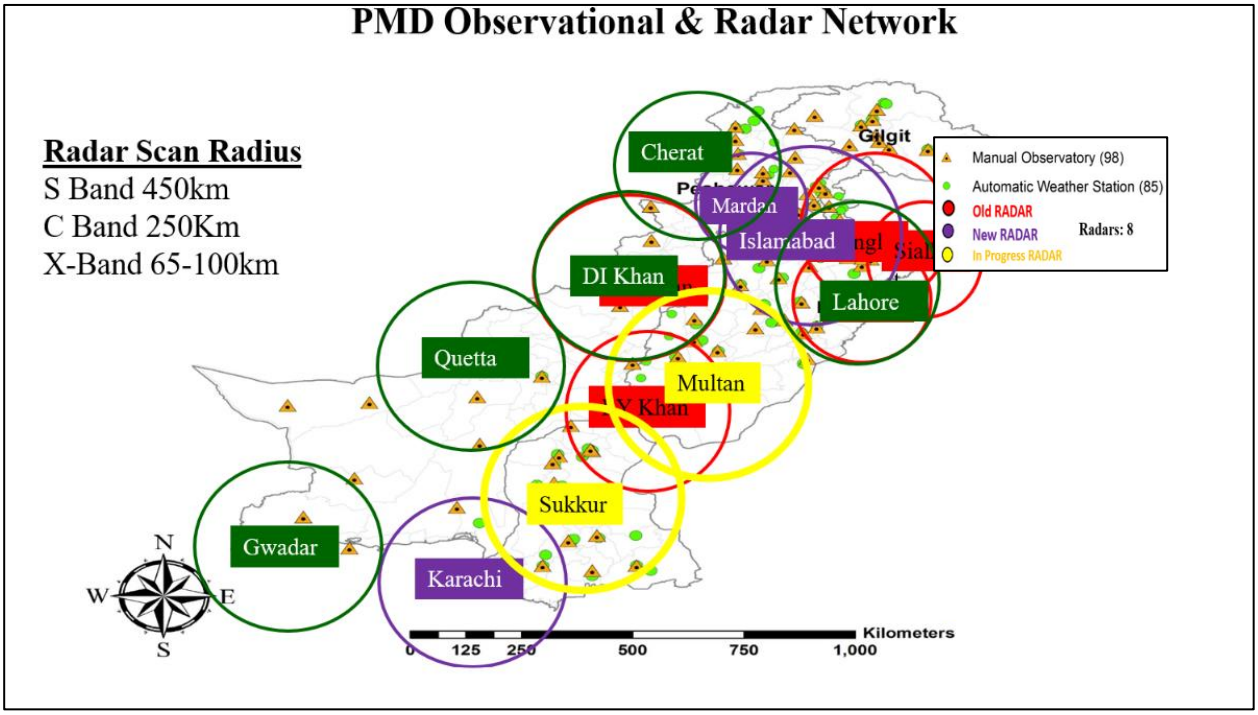
Item	Contents
Date of Approval	ECNEC: July 19, 2023
	Ministry of Aviation: September 6, 2023
Loan Agreement Signed	July 3, 2023
Total cost	Approx. 14,499 mil PKR
Implementation schedule	Start: July 2023 Completion: June 2027 (48 months)
Key projects	① Install or replace 5 weather radars (Gwadar, Quetta, D.I. Khan, Lahore, Cherat) ② Add 300 AWS observatories ③ Introduction of 220 Tera Flops high-performance computers ④ Introduction of 3 mobile weather radars ⑤ Addition of 50 agricultural weather observation centers ⑥ Strengthening the organization of IMG, CDPC and RCDPC ⑦ Improvement of PMD service delivery system

Source: Prepared by project team based on PMD data

Table 4.1.36 MHSP Project Costs

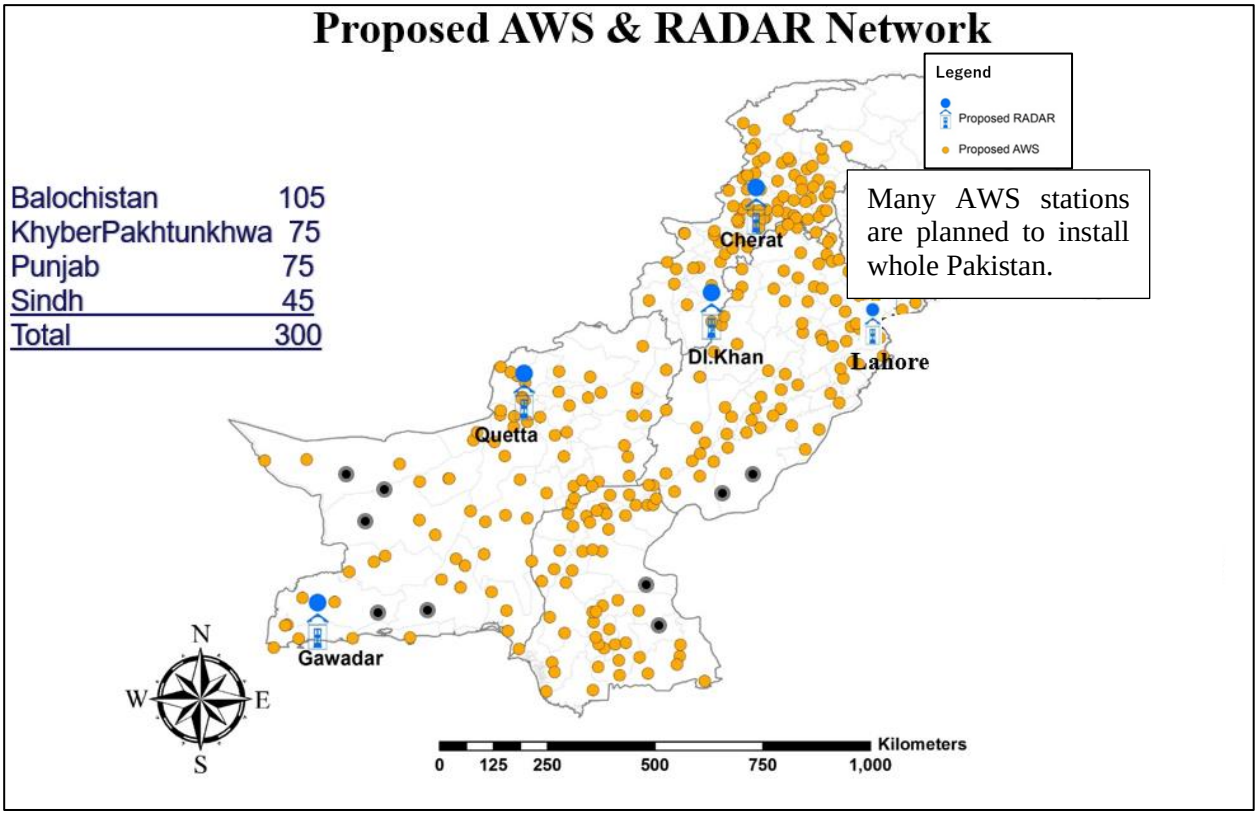
No	Project name	Cost (mil. PKR)
A	Organizational Strengthening and Capacity Building	1,402
A-1	Organizational Strengthening of IMG and Karachi	623
A-2	Organizational Strengthening of CDPC and RCDPC	164
A-3	Strengthening Meteorological Workshop, Karachi	128
A-4	Strengthening Meteorological Observatory Equipment	400
A-5	Training and Capacity Building	87
B	Improvement of Hydrological and Agricultural Systems	12,600
B-1	Improvement of weather forecasting system	3,910
B-2	Improvement of flood forecasting system	8,447
B-3	Improvement of agricultural weather forecasting system	142
B-4	Improvement of PMD service delivery system	100
C	Project management costs	497
C-1	PMC staff hire	274
C-2	Operation and Maintenance (O&M) costs	165
C-3	Consultancy/Meeting/Seminar/Workshop, etc.	30
C-4	Emergency costs	28
—	Total	14,499

Source: Prepared by the project team based on PMD data



Source: PMD

Figure 4.1.3 PMD Observatory and Radar Network



Source: PMD

Figure 4.1.4 Additional AWS and Radar Locations Proposed by MHSP

(3) Interviews

During the interview regarding the current NDMP utilization and achievement status described in Chapter 3, the interview about the Projects for the investment for disaster risk reduction is also held. Table 4.1.37 shows the overview of the interviews and Table 4.1.38 shows the results of the interviews regarding the Projects for the investment for disaster risk reduction.

Table 4.1.37 Overview of the EWS Interviews

Date of interview	Organization to be interviewed	Details of interview
November 14, 2023	WAPDA	・Current NDMP utilization and achievement ・Research activities related to glaciers, glacial lakes, and GLOF
November 15, 2023	PMD FFD Lahore	・Current NDMP utilization and achievement ・Activities mainly related to forecasting (Weather forecast, river water level forecast, etc.)
November 24, 2023 November 27, 2023	PMD Islamabad	・Current NDMP utilization and achievement status ・Contents related to EWS in general and Future Project

Source: Project Team

Table 4.1.38 (1) Interview Results on Projects for the investment for disaster risk reduction (EWS)

EWS	Current status	Project for the investment for Disaster risk reduction under consideration
Numerical Weather Forecast System	Weather and flood forecasting are carried out by PMD-FFD Lahore. Weather forecasting is carried out using numerical weather forecasting model with weather observation network of PMD as input condition. This system also includes the following two systems. ・Heavy Rain Warning System ・Heatwave Early Warning System	The model and system will be improved. In particular, higher resolution weather forecasting is realized using supercomputers. (The resolution is planned to be improved from the current 5 km to 1 km).
Flood Early Warning System	Weather forecasting and flood forecasting are operated by PMD-FFD Lahore. A runoff model, a river channel model, and an empirical formula are used to predict river channel discharge with rainfall in weather forecasting as input conditions.	Models and systems will be improved.
Earthquake & Tsunami Early Warning System	Earthquakes are continuously monitored and Tsunami prediction (simulation) is also conducted following seismic events.	Models and systems will be improved. In particular, there are only 5~6 tsunami observation points along the coastline, so it is necessary to increase the number of observation points.
Cyclone Early Warning System	Tropical Cyclone Warning Center (TCWC), Karachi has issued an alert on cyclone through PMD's website.	The model and system will be improved.
GLOF Early Warning System	Based on the information of glacial lakes with a possibility of breaching and the information of temperature, the alerts are issued for areas where GLOF may occur.	The model and system will be improved. PMD considers strengthening the observation network in particular. In addition, WAPDA is carrying out more advanced research, and it is thought that a better EWS can be realized by cooperating with WAPDA.
Drought Monitoring & Early Warning System	It was developed in 2005; however, it has been noted that an update is necessary.	The model and system will be improved.

Source: Project Team

Table 4.1.38 (2) Interview Results on Projects for the investment for disaster risk reduction (Systems that Influence EWS Forecasts)

Systems that Influence EWS Forecasts	Current status	Project for the investment for Disaster risk reduction under consideration
Upper Air Observation	Although aerial observations are carried out, they are limited to monitoring wind direction, with a low number of observation points and infrequent monitoring.	Weather elements other than wind direction are observed by flying radiosonde. It is necessary to increase observation points and observation frequency.
Satellite Information	Satellite images are used but not digitized.	Since they are not digitized, they cannot be incorporated into the Numerical Weather System.
River Flood Forecast and Warning System	Weather and flood prediction are performed by PMD-FFD Lahore. Global models, WRF Model, IFAS Model, FUSE Model, PHAST Model, regression model, etc. are used for these predictions.	The model and system will be improved.
Flash Flood Forecast and Warning System	Weather and flood prediction are performed by PMD-FFD Lahore. Global models, WRF Model, IFAS Model, FUSE Model, PHAST Model, regression model, etc. are used for these predictions.	The model and system will be improved.
Landslide Forecast and Warning System	Currently, this system is not built.	PMD Islamabad thinks it is necessary to build this system.
Tide Level Monitoring Network	This system is not currently operational. However, according to the results of a findings of 2022 hearing, PNHD appears to be in the process of developing it.	—
Snowmelt Flash Flood Forecast and Warning System	Based on the information of glacial lakes with the possibility of breaching and the information of temperature, they call for caution in areas where GLOF may occur.	The model and system will be improved. PMD considers strengthening the observation network. WAPDA is carrying out more advanced research, and it seems to realize a better EWS through collaboration with WAPDA.
Avalanche Advisory Information System	Currently, this system is not built.	PMD Islamabad considers is the establishment of this system necessary
Seismic Intensity Reporting System	This system is in operation, but it needs improvement.	The model and system will be improved.
Automatic Weather Observation System (AWS)	There are currently 85 AWS Stations. However, there is a problem that each AWS Station is too far from the WMO standard.	The MHSP will add 300 AWS Stations.
Real-Time Rainfall & Water Level Observation Network	The response indicated that PMD is responsible for monitoring rainfall, while WAPDA monitors water levels.	In order to construct and operate the Urban Flood Model, a system that collects both rainfall and water level in real time is necessary.
Weather Forecast Guidance System	The Weather Forecast Guidance System is in operation, but it needs to be improved because the available information is limited at present.	The model and system will be improved.

Source: Project Team

Table 4.1.38 (3) Interview Results on Projects for the investment for disaster risk reduction (Communication System)

Communication System	Current status	Project for the investment for Disaster risk reduction under consideration
Communication System between PMD and NDMA	Communication by mobile phone, internet, fax, WhatsApp, etc.	Expansion of content and destination of warnings and announcements.
Weather Information Broadcasting System	In PMD, weather information etc. is transmitted through social media and Website in ordinary time. On the other hand, NDMA transmits information only in case of emergency.	Expansion of content and destination of broadcasting weather information etc.
Global Telecommunication System (GTS)	Data is collected by CDPC Karachi every three hours. It is a global system and meets the standards of WMO.	The system will be improved.

Source: Project Team

Table 4.1.38 (4) Interview Results on Projects for the investment for disaster risk reduction (EWS Related Facilities)

EWS Related Facilities	Current status	Project for the investment for Disaster risk reduction under consideration
Specialized Medium Range Forecasting Center (SMRFC)	It has already been established.	Improvement is needed.
Regional Flood Forecasting and Warning Center (RFFWC)	Not yet established.	Aiming to establish in major cities.
Urban/Flash Flood Early Warning Center *Current NDMP calls it Local Flash Flood Forecasting Center (LFFFC).	Not yet established outside of Lai Nullah.	Aiming to establish in major cities.

Source: Project Team

Table 4.1.38 (5) Interview Results on Projects for the investment for disaster risk reduction (Capacity Development)

Capacity Development	Current status	Project for the investment for Disaster risk reduction under consideration
Community Enlightenment for EWS with Execution of Training and Drills	PMD has 31 staff members who holds PhDs obtained specialized education programmes.	Need to continue similar activities to further develop the skills of PMD engineers.
Education Program for Advanced Meteorology and Hydrology for PMD Staff	Regarding the education of local residents, PMD recognizes that it is the responsibilities of NDMA, PDMAs and DDMAs, rather than PMD.	—
Capacity Development against Local Flash Floods in Vulnerable Areas	Regarding the education of local residents, PMD recognizes that it is the responsibilities of NDMA, PDMAs and DDMAs, rather than PMD.	—

Source: Project Team

4.1.6.2 Recognition of NDMA Issues and Challenges and Priority Actions

Among the content described in “Challenges & Priority Areas” in the presentation of "Early Warning & DRM System of Pakistan, Learning from Good Practices" by Mr. Idrees from NDMA at South Asia Hydro-met Forum, recognition of issues and challenges related to EWS is shown in Table 4.1.39 and priority actions are shown in Table 4.1.40.

Table 4.1.39 Recognition of NDMA Issues and Challenges

No	Recognition of Issues and Challenges
1	Lack of aerial meteorological data for effective storm tracking
2	No Urban Flood Forecasting System other than Lai Nullah
3	Lack of weather information on the Indian side
4	Strengthening the Disaster Management System at the District Levels

Source: Compiled by the project team based on NDMA's Idrees presentation

Table 4.1.40 NDMA’s Priority Actions

No	Priority Actions
1	Community Based Disaster Risk Management (CBDRM), Enhancing Community Disaster Management Capacity through Awareness Raising Activities
2	Development of Disaster Management Information Systems
3	Capacity Development of Government Officials
4	Enhancing Response and Logistics Capacity
5	Establishment of Smart National Disaster Response Force (SNDRF)
6	Establishment of Pakistan School Safety Framework

Source: Prepared by the project team based on the presentation by Mr. Idrees of NDMA

4.2 Activity 3-2

The results of the risk assessment of Activity 1-4 and the results of the information collection of Activity 3-1 will be analyzed to identify additional projects that should be considered, and a draft project including budget size will be considered.

4.2.1 Flood

Based on the data collected so far and the results of interview surveys, etc., the current status, problem, challenge, and countermeasures of flood countermeasures in Pakistan were studied on the major areas and summarized as Table 4.2.1

Table 4.2.1 Current Status, Problem, Challenge, and Countermeasures of Flood Countermeasures in Pakistan (Draft)

Current Status	Problem	Challenge	Countermeasures (Draft)
・ The PIDs are leading the O&M of the structures.	・ As it is carried out on a provincial basis with a limited budget, only symptomatic repairs have been carried out, and the effect as an pre-investment for disaster risk reduction is insufficient. ・ Even for rivers that cross the provinces, the implementation from a bird's eye view of the entire river has not been carried out.	・ Introduce a "prevention" perspective instead of symptomatic maintenance ・ Realization of implementation from a bird's eye view of the entire river basin, not only on a provincial basis ・ Consideration of climate change measures	・ Formulation of a implementation plan based on the characteristics of the river channel and the concentration of assets in the entire basin ・ Improvement of barrages ・ Improvement of various technical standards including climate change measures
・ PMD is in charge of flood forecast warning and PMD's forecast warning is referenced nationwide.	・ In regions affected by Hill Torrent and downstream areas of Indus River, ensuring sufficient accuracy in forecast warnings remains challenging.	・ Improvement of accuracy of forecast warning	・ Enhancement of weather observation network ・ Enhancement of situation sharing and cooperation among concerned organizations
・ Flood risk map and zoning of inundation area are assigned to FFC. ・ PMD is in charge of rainfall observation and WAPDA is in charge of river discharge observation.	・ Although it was prepared once in the study for the formulation of NFPP-IV, its accuracy remains insufficient. ・ Various data sources are not integrated.	・ Enhancement of flood risk management	・ Enhacenment of data management ・ Flood hazard and flood risk assessment based on probabilistic analysis
・ Urban drainage is assigned to the government agency responsible for the development of each city.	・ The speed of urban development is rapid, and the drainage capacity is insufficient. ・ In some cases, such as Karachi, the government agency in charge is unclear.	・ Improvement of urban drainage ・ Consideration of climate change response measures	・ Extraction and analysis of urban drainage problems in each city ・ Formulation of comprehensive measures including non-structural approaches

Source: Project team

Based on the above, the draft project groups for the investment for disaster risk reduction in the flood field was studied. The results are shown in Table 4.2.2.

Table 4.2.2 Draft of Disaster Prevention Investment Project Groups in the Flood Field

Major Items (Strategy)	Middle Items (Interventions)	Minor Items (Action)
Flood Risk Reduction on Inter-Provincial Basins and Populated area toward NFPP-V	Promotion of projects stipulated in NFPP-IV	・ Management of listed Non-Structural measures in FPSPIII
	Update Runoff and inundation analysis model	・ Aggregation and management of data ・ Update of run-off-model ・ Establishment of probable maximum inundation analysis by bund breach
	Update the National hazard & risk assessment and its database	・ National database development of the historical flood disasters and its impact ・ Development of risk atlases including flood hazard, risk maps with catalogues at the National Level ・ Development of probabilistic flood hazard and risk maps along the Indus River with Climate change Impact ・ Research studies on selected cities prone to urban flooding ・ Mapping the Nullahs/hill torrents responsible for flash flooding
	Plannings for better risk management	・ National Drainage Master Plan (Undertake hydrological modeling and food-plain mapping and zoning of the Indus River system using climate change scenarios to estimate various projected flood levels) ・ O&M Plan of the Bunds of Indus River ・ Update of flood early warning (including flash flood early warning in Hill Torrent and Nullar) ・ Study of interventions for GLOF and coastal flooding ・ Installation of flood early warning (including GLOF and coastal flooding) ・ Plan for Improvement of Barrages along Indus River (seeing Balance of UP- and Downstream Discharge) ・ Review of Plan on Large Dams Construction
	Update and formulation of Laws and (Technical-) Guidelines and Act	・ Development of the Bund Training Manuals ・ Development of planning & design criteria ・ Guidelines of CC Impact involvement to Flood Protection Plan and Design ・ Revision of National River Act
	Establishment of NFPP-V	・ Preparation of NFPP-V
Basin-wise Flood Protection Measures	Implementation of (non-) Structural measures based on NFPP-V	・ ※Based on FPSP-III
	Revising the Monsoon Contingency Plans	・ Provincial monsoon contingency plans(if needed after disaster)
	Review of Annual Flood Protection Plan	・ Survey and Prioritization
	Implementation of contingency/annual plan	・ Construction & Improvement of regional drainage system ・ Installation of rainwater- harvesting and storage systems ・ Construction and Improvement of the hill torrent & River structures ・ Modernization of Barrages structure/operation system
Urban Drainage planning	Comprehensive urban drainage plan and implementation	・ Basic survey and studies on urban drainage issues ・ Planning on comprehensive urban flood manage plan
Adaptation to CC Impact	Preparation of Technical Guidelines	・ Determination of specific scenario and target year to set the conditions for planning and design adaptation measures in Pakistan ・ Update of Dam Operation Guideline for Flood ・ Integration of Guideline for Breach Effects ・ Guideline for Medium and Small Dams/ponds ・ Design Guideline for River Structures ・ Consolidation of Planning & Design Guideline
	Implementation of climate change adaptation plans	・ Promotion of climate change adaptation by engaging local communities ・ Implementation of climate change adaptation measures by (non-) structural programs

Major Items (Strategy)	Middle Items (Interventions)	Minor Items (Action)
	Monitoring the Effects on Adaptations	・ Formulation of climate risk assessment guidelines
Institutional Improvement	Empowerment of Integrated Flood Management	・ Planning on Asset Management of River Facility ・ Preparation of Tools and System for Projects' Management ・ Establishment of Project Management System
	Development of Flood Disaster Risk Management Information System and sharing	・ Update of hydro-metrological Information sharing mechanism and system
	Provincial-level institutional strengthening for robust flood disaster risk management	・ Provision of technical experts/staff for flood risk analysis on Hill torrent and regional rivers ・ Capacity enhancement to understand flood disaster mechanism and hydrological analysis ・ Provincial Level Flood Hazard & Risk Map ・ Strengthen the capacities to address on the Flood DRM system

Source: Project team

4.2.2 Earthquake

Based on the interviews conducted in the previous section, the outline of projects that should be considered as earthquake countermeasures is shown in Table 4.2.3.

Table 4.2.3 Projects that should be Considered as Earthquake Countermeasures

Plan	Specific methods	Priority
Further advancement of earthquake monitoring, development of a real-time earthquake risk assessment system, and implementation of scenario-based earthquake risk assessment to contribute to local disaster prevention and mitigation measures	・ Update probabilistic earthquake hazard maps ・ Assess earthquake faults and conduct scenario-based earthquake hazard and risk assessment ・ Develop structural inventory data and investigate damage functions for structural vulnerability assessment	1
Formulation of local urban resilience plans based on risk assessment	・ Formulation of urban resilience plans based on risk assessments, focusing on areas with severe buildings damage ・ Proposal plans tailored to regional characteristics	2
Seismic strengthening of important facilities and introduction of seismic design and building regulations in urban development	・ Implement seismic strengthening of critical facilities such as schools and hospitals, in a severely affected areas as a model projects. ・ Conduct training to support the design and construction of buildings with sufficient seismic resistance and seismic reinforcement.	2
Human resource development that contributes to earthquake disaster prevention	・ Support the education of earthquake and earthquake engineering specialists at universities and other institutions ・ Provide training for engineers such as architects and carpenters. ・ Conduct training to enhance disaster management education for school students.	1

4.2.3 Tsunami

Based on the interviews conducted in the previous section, Table 4.2.4 outlines additional projects to be considered as tsunami countermeasures.

Table 4.2.4 Additional Projects to be Considered as Tsunami Countermeasures

Plan	Specific methods	Priority
Development of tsunami hazard and risk assessment system	・ Assess earthquake faults in the Makran Sub-duction zone that cause tsunamis ・ Conduct a scenario-based earthquake hazard risk assessment using assessment results of active earthquake faults ・ Conduct a probabilistic tsunami inundation simulation	1
Formulate a regional urban resilience plans based on tsunami risk assessment	・ Consider measures to mitigate flood damage to buildings based on tsunami risk assessment ・ When formulating plans, ensure they are tailored to regional characteristics	2
Implement disaster prevention and mitigation measures against tsunamis based on tsunami evacuation plans, evacuation site layout plans, and hard measures based on tsunami hazard assessment	・ Formulate evacuation plans for major cities, such as Karachi, using both tsunami evacuation tower layout plans and evacuation plans ・ The project team propose hard countermeasures by installing the minimum necessary number of seawalls, which are expensive, together with green infrastructure that fully incorporates the effects of shallow water. ・ The project team propose tsunami countermeasures that combine both hardware and software, and implement pilot projects.	2
Human resource development to contribute to tsunami disaster management	・ Support for education of tsunami experts at universities and other institutions ・ Conduct training programmes to enhance disaster management education for school students ・ Carryout public awareness-raising activities, such as distributing hazard maps to citizens	1

4.2.4 Human Resource Development

Based on the review of the current NDMP;s implementation status and the future plans of relevant organizations, the following three ongoing activities have been identified as requiring continued support in the future, and will be added as priority projects moving forward.

- DRM workshops for parliamentarian and policy makers
- DRM workshops for media
- Implementation of mock exercises for various disasters at local governments

4.2.5 Urban Disasters and Land Use

4.2.5.1 Urban Disasters in General

Based on the results of 4.1.5, the present status, problems, issues and countermeasures of urban disaster mitigation and land use planning considering disasters in Pakistan were examined and summarized as follows.

Table 4.2.5 Present Status, Problems, Issues and Countermeasures of Urban Disaster Prevention and Land Use Planning Considering Disasters in Pakistan

Present status	Problems	Issues	Countermeasures
・ There is an examination system for building and development, and it also functions. ・ There are also examination standards (Technical standards, location standards, etc.) ・ There are examples of damage caused by natural disasters	・ The basic principle is not prevention, but response after the occurrence of a disaster.	・ Introduce the perspective of "prevention"	・ Countermeasures based on existing legal systems ・ Introducing new legal systems (Both regulatory and inductive aspects) ・ Market guidance

Source: Project Team

Based on the above, the Projects for the investment for disaster risk reduction in the field of "urban disaster and land use planning" was examined. The results are shown in Table 4.2.6.

Table 4.2.6 Projects for the investment for disaster risk reduction in the Field of "Urban Disaster and Land Use Planning"

Major items	Middle items	Minor items
[I] Safe and secure evacuation	[I-1] Development of hazard maps and evacuation plans	・ Development of hazard maps ・ Development of evacuation plans
	[I-2] Securing evacuation spaces	・ Securing and improving evacuation areas ・ Eliminating areas of difficulty in evacuation (including wide-area evacuation) ・ Securing evacuation routes
	[I-3] Securing reliable evacuation actions	・ Establishment of equipment and methods for information collection and transmission ・ Cooperation with voluntary disaster prevention organizations ・ Evacuation support for people requiring assistance in the event of disaster ・ Earthquake-resistant housing, buildings, and ancillary structures
	[I-4] Facilities needed for tsunami evacuation measures	・ Evacuation guidance signs, guides, etc. ・ Securing the shortest evacuation route
[II] Constructing urban structures resistant to earthquakes, tsunamis and flood damage	[II-1] Measures for facilities that require measures against earthquakes, tsunamis, and flood damage	・ Promoting earthquake-resistant and water-resistant buildings ・ Promoting earthquake-resistant and water-resistant lifelines ・ Strengthening regional disaster management centers (evacuation places, government facilities, etc.) ・ Relocation of important critical facilities for disaster preparedness and response ・ Relocation of facilities for use by people requiring support in the event of disaster ・ Raising residential land for tsunami and flood preparedness ・ Reinforcement and enhancement of functions of river management facilities for bund and other areas
	[II-2] Strengthening urban areas against earthquake, tsunami, and floods	・ Reducing congestion in densely populated urban areasSecuring fire prevention zones ・ Securing evacuation areas and routes ・ Making houses and buildings combustible and flame-retardant ・ Securing water supply for fire fighting ・ Tsunami inundation measures ・ Reinforcement and enhancement of functions such as bund ・ Floating debris countermeasures ・ Coastal forest countermeasures
	[II-3] Land use planning considering disaster risk	・ Linking disaster management plans with city planning ・ Regulation and guidance of land use in disaster hazard areas

Major items	Middle items	Minor items
	[II-4] Measures for isolation and insufficient relief and relief in the event of a disaster	· Heliport development · Stockpiling of food and daily necessities (stockpiling warehouses) · Establishment of reliable means of collecting and providing information · Securing emergency transportation roads
[III] Constructing disaster-resistant organizations and human resources	[III-1] Strengthening regional disaster management capabilities	· Formation of voluntary disaster management committees and enhancement of preparedness activities · Formulation and operation of district disaster management plans · Promotion of awareness raising and disaster management education
	[III-2] Cooperation between local governments and companies	· Disaster management agreement · Support for the formulation of BCP plans by private companies · Collaboration with donors, NGOs, etc.
	[III-3] Strengthening disaster management capabilities of local governments	· Promotion of the formulation of BCP · Awareness raising, disaster management education, and capacity building of staff · Formulation of temporary housing plans in anticipation of emergencies · Formulation of transportation plans for emergency relief supplies in the event of a disaster · Formulation of response plans for waste generated in the event of a disaster
[IV] Prior reconstruction	[IV-1] Pre-reconstruction community planning	· Pre-reconstruction community planning

Source: Project Team

4.2.5.2 Rainwater Drainage (Urban Flood)

Based on the data collected in Activity 3-1, the rainwater drainage measures currently implemented in the country can be summarized as follows: renovation and enhancement of existing facilities utilizing existing sewerage systems and rainwater drainage channel; countermeasures against illegal dumping in drainage channel; construction of new rainwater drainage channels and pumping stations; construction of rainwater storage facilities to prevent and mitigate flooding at an early stage; and seepage promotion measures utilizing vacant and green areas to reduce total runoff.

In Pakistan, the construction of rainwater drainage facilities has been delayed, even in major cities. According to the current rainwater drainage M/Ps collected in Lahore, Rawalpindi, and Multan, the existing rainwater drainage facilities is insufficient in capacity for 2-year probability rainfall. The current plan has a target period of about 20 years, and the target of the plan is 5-year probability rainfall for Lahore trunk line drainage channel, 2-year probability rainfall for drainage channel branch line, and 3-year probability rainfall for Rawalpindi (the current M/P of Multan does not describe the specific target).

In Pakistan, heavy rainfall is expected to increase in both frequency and intensity due to climate change. The National Climate Change Policy (2012) identifies the revision and upgrading of rainwater drainage planning standards for major cities, such as Karachi and Lahore. However, the present rate of improvement of rainwater drainage facility is low, and according to the interview survey of the rainwater drainage department in Lahore and Karachi, although the necessity of climate change countermeasures is recognized, the plan standards have not been reviewed. On the other hand, as a part of climate change adaptation measures, the construction of underground rainwater storage facilities is proceeding, and it is expected that the increment of peak runoff due to climate change will be reduced by the rainwater storage facilities.

Based on the above situation, it is important to promote the renovation of the existing rainwater drainage facility and the construction of a new rainwater drainage facility with the aim of early reduction of inundation damage. On the other hand, rainwater drainage systems require extensive infrastructure and long-term development. Their planning, design, construction, and maintenance must be coordinated with sewerage and other related facilities. Therefore, it takes a great deal of effort, time, and cost to raise the planning target after the construction. Therefore, if the planning target is likely to be raised in the future based on the effects of urban planning and climate change, it is recommended to consider additional rainwater drainage measures that are necessary when the planning target is raised. As an example, the

increased discharge if the planned target rainfall were raised should be estimated, and it will be compared with the flow capacity of the original planned target rainfall. It is necessary to estimate the cost and feasibility of dealing the raised target discharge with new rainwater drainage facilities, as well as the damage that would occur if there were no measurements.

Regarding the hard structural measures, since the present rate of improvement of rainwater drainage facility is very low, it is necessary to promote the improvement of a new rainwater drainage facilities along with the renovation of the existing rainwater drainage facilities at an early stage. In addition, according to the interviews conducted in Karachi and Lahore, both cities are moving forward with the transition from the combined sewerage system to the split sewerage system. However, the transition to the split sewerage system requires a huge budget, and the majority of this is currently dependent on overseas assistance. In Japan, there are cases where water quality improvement measures have been implemented with the combined sewerage system in place, and cases where the plan to the split sewerage system was reviewed and the combined sewerage system was adopted in the past. Therefore, the case of improvement measures of the combined sewerage system in Japan is considered to be useful in Pakistan. It is recommended to promote investment in the hard structural measures, including the improvement to the combined sewerage system, to enhance the efficiency and expedite the implementation of rainwater drainage facilities.

As mentioned in Activities 2-2 and 2-3, Pakistan currently lacks hazard maps, pre-warning systems, and city-level emergency response plans for inland flooding. Therefore, measures should be promoted the goals of eliminating casualties, minimizing damages, and promoting self-help and mutual assistance among the residents. In addition, measures should be taken against illegal dumping in drainage channels (including the construction of waste treatment plants if necessary), and land use regulations such as development regulations and making it mandatory to convert inundation damage areas to houses on stilts should be considered.

Table 4.2.7 shows the proposals for additional Projects for the investment for disaster risk reduction in the "rainwater drainage" field.

Table 4.2.7 Proposals for Additional Projects for the investment for disaster risk reduction in the "Rainwater Drainage" Field

Classification	Projects for the investment for disaster risk reduction
Research and planning	Case study of rainwater drainage planning based on the impact of climate change
	Case study of combined sewer system improvement measures
Hard measures	Construction of rainwater seepage facilities (seepage trough, seepage trench, permeable pavement, etc.)
	Water resistance of pump facilities
	Installation of water shutoff plates in underground facilities
Soft measures	Preparation of inland water hazard maps
	Construction of a pre-warning system for inland water flooding
	Development of urban flood emergency response plans
	Measures against illegal dumping in drainage channel
	Land use regulations (building regulations in areas where flooding is common, etc.)

4.2.6 EWS

Based on the issues and improvement measures of the current NDMP in Section 3.3.6 and the information gathering results in Section 4.1.6, a detailed long list (draft) of EWS up to project list level is shown in Table 4.2.9. At the operational level, annual plans are expected to be developed based on the detailed long list (draft) of AWS presented here. However, this long list (draft) is simplified in Section 4.3.6 due to its complexity as a national disaster management operation plan.

This detailed long list (draft) was edited mainly on the following four items:

- Included in PMD's list of projects to strengthen the EWS
- Included in Modernization of Hydro-met Service Project (MHSP)
- Included in interviews
- Included in the Presentation by Mr. Idrees from NDMA

However, there are items that are not included in these four but have been added by the project team at the discretion of Table 4.2.8.

Table 4.2.8 Items Added by the Project Team

No	Project List	Reason for addition
1	2.2.3. Develop Guidelines and SOPs for Health Emergency Preparedness and Response (HEPR).	According to the former PID Chief Engineer, safety management during floods is very important in Pakistan. In addition to direct flood damage, there is the possibility of a large outbreak of dangerous organisms such as snakes and the spread of infectious diseases. Therefore, this item was added to the detailed long list (draft) related to EWS.
2	3.1.3. Make Collaboration and Information Sharing between Organizations in Pakistan (PMD, FFC, and WAPDA)	In section 3.3.6, "strengthening of inter-organizational cooperation" is considered as the organizational and institutional issue. Every year during monsoon season, PMD, WAPDA, PID, and PDMAs collaborate together at PMD-FFD Lahore to share information, however, cooperation beyond remains inadequate. Strengthening inter-governmental cooperation over the next decade is essential.

Source: Project Team

Table 4.2.9 (1) Detailed Long List of AWS (Draft)

Strategy	Action	Program	Project List		Responsible Organization	
					Main	Support
Multi-Hazard Early Warning System.	1. Strengthen Forecasting and Early Warning System.	1.1. Strengthen Weather Observation Network.	1.1.1.	Expand the AWS (Automatic Weather Observation System) Network	PMD	NDMA, PDMAs
			1.1.2.	Establish the Integration System of Satellite Information into Numerical Weather Forecasting System	PMD	NDMA, PDMAs
			1.1.3.	Establish of Upper Air Observation Network Including Radiosondes and LIDAR	PMD	NDMA, PDMAs
			1.1.4.	Establishment of Radar Stations	PMD	NDMA, PDMAs
			1.1.5.	Replacement of Radar Stations	PMD	NDMA, PDMAs
		1.2. Establish Forecasting Centers.	1.2.1.	Strengthen the System of SMRFC (Specialized Medium Range Forecasting Center)	PMD	NDMA, PDMAs, FFC, IRSA
			1.2.2.	Establish RFFWC (Regional Flood Forecasting and Warning Center)	PMD	NDMA, PDMAs, FFC, IRSA
			1.2.3.	Establish Urban/Flash Flood Early Warning Center in Major Cities of Pakistan	PMD	NDMA, PDMAs, FFC, IRSA
		1.3. Strengthen Flood Forecasting and Early Warning System.	1.3.1.	Improve Numerical Weather Forecast System by Installing HPC (High Performance Computer)	PMD	NDMA, PDMAs, FFC, IRSA
			1.3.2.	Improve and Expand Flood Early Warning System	PMD	NDMA, PDMAs, FFC, IRSA
			1.3.3.	Establish Trans-Boundary Early Warning Mechanisms	PMD, NDMA	PDMAs, FFC, IRSA
			1.3.4.	Develop Weather Forecast Guidance System	PMD	NDMA, PDMAs, FFC, IRSA
			1.3.5.	Expand GLOF Early Warning System in the GB, AJK and KP Regions	WAPDA, PMD	NDMA, PDMAs, FFC, IRSA
			1.3.6.	Establish Avalanche Advisory Information System	WAPDA, PMD	NDMA, PDMAs, FFC, IRSA
			1.3.7.	Research Activity for Snow/Glacier/Glacial Lakes	WAPDA, PMD	NDMA, PDMAs, FFC, IRSA

: Included in PMD's list of projects to strengthen EWS
: Included in Modernization of Hydro-met Service Project (MHSP)
: Included in interviews
: Included in the Presentation by Mr. Idrees from NDMA

Source: Project Team

Table 4.2.9 (2) Detailed Long List of AWS (Draft)

Strategy	Action	Program	Project List	Responsible Organization	
				Main	Support
Multi-Hazard Early Warning System.	1. Strengthen Forecasting and Early Warning System.	1.4. Strengthen Earthquake & Tsunami Forecasting and Early Warning System.	1.4.1. Improve Tsunami Early Warning System with Installation of Tsunami Detection Buoys and Sensors along Coastal Regions	PMD	GSP, NDMA, PDMAs
			1.4.2. Strength and Expand the Seismic Monitoring Network with Advanced Analysis Capabilities	PMD	GSP, NDMA, PDMAs
			1.4.3. Establish Coastal Tide Level Monitoring System	PNHD, PMD	GSP, NDMA, PDMAs
		1.5. Strengthen Forecasting and Early Warning System for Other Disasters.	1.5.1. Establish Landslide Hazard Monitoring and Early Warning System in High-Risk Area	PMD	NDMA, PDMAs
			1.5.2. Improve and Expand Cyclone Early Warning System	PMD	NDMA, PDMAs
			1.5.3. Improve and Expand Drought Monitoring Early Warning System	PMD	NDMA, PDMAs
			1.5.4. Establish Extreme Weather Early Warning System	PMD	NDMA, PDMAs

: Included in PMD's list of projects to strengthen EWS
: Included in Modernization of Hydro-met Service Project (MHSP)
: Included in interviews
: Included in the Presentation by Mr. Idrees from NDMA

Source: Project Team

Table 4.2.9 (3) Detailed Long List of AWS (Draft)

Strategy	Action	Program	Project List	Responsible Organization	
				Main	Support
Multi-Hazard Early Warning System.	2. Prepare Emergency Response.	2.1. Strengthen Early Warning Dissemination System.	2.1.1. Strengthen and Expand the Warning Transmission System by PMD.	PMD	NDMA, PDMAs
			2.1.2. Strengthen Communication System between DMAs (NDMA-PDMAs-DDMAs)	NDMA, PDMA, DDMA	PMD
			2.1.3. Communication System between NDMA and Federal Agencies	NDMA	PMD, PDMAs
			2.1.4. Communication System between PDMAs and Provincial Line Departments	PDMA	PMD, NDMA
			2.1.5. Communication System between DDMA and District Line Departments, Tehsil & UC Level Local Administration, Communities	DDMA	PMD, NDMA, PDMAs
			2.1.6. Expand Weather Information Broadcasting System	PMD, NDMA	PDMAs, DDMA
			2.1.7. Improve Global Telecommunication System (GTS).	PMD	NDMA, PDMAs
		2.2. Develop National Plan, Guidelines and SOPs for Early Warning System and Health Emergency Preparedness and Response (HEPR).	2.2.1. Develop National Early Warning System Framework and Comprehensive National Policies	PMD, NDMA	PDMAs
			2.2.2. Develop Guidelines and SOPs for Early Warning Systems	PMD, NDMA	PDMAs
			2.2.3. Develop Guidelines and SOPs for Health Emergency Preparedness and Response (HEPR).	NHSRC	NDMA, PDMAs, Provincial Health Department

: Included in PMD's list of projects to strengthen EWS
: Included in Modernization of Hydro-met Service Project (MHSP)
: Included in interviews
: Included in the Presentation by Mr. Idrees from NDMA

Source: Project team

Table 4.2.9 (4) Detailed Long List of AWS (Draft)

Strategy	Action	Program	Project List	Responsible Organization	
				Main	Support
Multi-Hazard Early Warning System.	3. Develop Capacity of Early Warning and Evacuation System.	3.1. Develop Capacity for Staff and Officials.	3.1.1. Provide Meteo-Hydrology Capacity Building Program for Professional Staff	PMD, FFC, WAPDA	NDMA, PDMAs
			3.1.2. Make Collaboration with International Agencies and Experts.	PMD, FFC, WAPDA	NDMA, PDMAs
			3.1.3. Make Collaboration and Information Sharing between Organizations in Pakistan (PMD, FFC, WAPDA)	PMD, FFC, WAPDA	NDMA, PDMAs
			3.1.4. Develop Capacity for Government Officials	NDMA, PDMAs, DDMAs	PMD, FFC
		3.2. Develop Capacity for Community.	3.2.1. Conduct Community Awareness Training and Drills Program for Effective Response to Weather Related Hazards.	NDMA, PDMAs, DDMAs	PMD, FFC
			3.2.2. Enhance Resilience Capacity of Community Through CBDRM (Community Based Disaster Risk Management)	NDMA, PDMAs, DDMAs	PMD, FFC

: Included in PMD's list of projects to strengthen EWS
: Included in Modernization of Hydro-met Service Project (MHSP)
: Included in interviews
: Included in the Presentation by Mr. Idrees from NDMA

Source: Project Team

4.3 Activity 3-3

Based on Activity 3-1 and 3-2, A long list (draft) is prepared for each disaster type and the priority is considered.

4.3.1 Flood

A long list (draft) for floods based on the project groups arranged in Table 4.2.2 was prepared. As of December 2023, a short-term priority action plan was discussed with the FFC at the request of the NDMA. The priority action plan discussed is shown in the attached document "National Disaster Mitigation Plan Pakistan 2024."

4.3.2 Earthquake

Based on the discussion of the outline of projects to be added as earthquake countermeasures in the previous section, the long list (draft) was discussed at the program level. At the program level, what kind of projects will be carried out is vague, so the red parts are added to give more specifics. The long list (draft) of disaster prevention investment projects compiled in this way is shown in Table 4.3.1.

Table 4.3.1 Long List (Draft) Related to Earthquakes

Strategy	Interventions	Actions	Execution Agency		Short term for 2024				Long-Term planning								
			Main	Support organizations	1st Qtr	2nd Qtr	3rd Qtr	4th Qtr	25	26	27	28	29	30	31	32	33
Enhancing Earthquake Monitoring and Assessment	Establishment of a committee for the evaluation of earthquake faults and the practice of fault evaluation.	Announcement and selection of the committee members	NDMA	Academic, PEC		X	X										
		Hold regular meetings for institutionalization	NDMA	Academic, PEC				X	X	X	X	X	X	X	X	X	X
		Assessing the active and non-active faults and their mapping	NDMA	Academic, PEC				X	X	X	X	X	X	X	X	X	X
	Maintaining record and publication of epicenter rapture (real-time) and estimated intensity data	Collect historical data of earthquake disasters, and update it with epicenter rupture / estimated intensity data	PMD	GSP, NDMA, PDMA, DDMA			X	X	X	X							
		Present and approval from the committee	PMD	GSP, NDMA, PDMA, DDMA			X	X	X								
		Information sharing about the seismic data to relevant government authorities (e.g. to EOCs)	PMD	GSP, NDMA, PDMA, DDMA			X	X	X	X							
		Publication and dissemination of data and reports	PMD	GSP, NDMA, PDMA, DDMA				X	X	X							
	Scenario earthquake based seismic risk assessment of selected earthquake-prone cities deriving the geotechnical information including urban-level details, liquefaction, etc.	Studying local amplification factors	PMD	PGS, NDMA, Academia							X	X	X	X	X	X	X
		Data collections for analysis	PDMA	NDMA, PMD, , DDMA, Academia							X	X	X				
		Analyzing damage estimation	PMD	PDMA, DDMA, Academia								X	X				
		Identify at-risk areas to earthquakes and prioritize promoting earthquake disaster resilience	PDMA	Province, DDMA, Academia								X	X				
		Develop an earthquake hazard / risk atlas and associated database	PDMA	Province, DDMA, Academia								X	X				
		Developing measures and investment planning (e.g. landuse plan, priority road designations, roadside risks, bridge assessments, evacuation plans, prioritizing retrofitting of the public buildings)	Provincial concerned dep.	PDMA, DDMA								X	X				

	Publicizing Atlases for wide dissemination	PDMA	NDMA, DDMA, PMD, Academia									X					
Upgrading the Damage and Loss Assessments Procedures	Updating hazard assessment by probabilistic seismic risk assessment using soil amplification factors	Studying national level amplification factors	PEC	PMD, NDMA, PDMAs, Academia, NESPAK										X	X	X	X
		Updating a hazard assessment of selected cities	PEC	PMD, NDMA, PDMAs, Academia, NESPAK										X	X	X	X
	Develop guidelines for an earthquake damage assessment (scenario-based earthquake microzoning assessment)	Develop standard guidelines for conducting an earthquake damage assessment for rural areas	NDMA	PDMAs, PMD, NESPAK, SUPARCO, Academia									X	X			
		Develop separate formats and guidelines for conducting an earthquake damage assessment in the urban areas	NDMA	PDMAs, PMD, NESPAK, SUPARCO, Academia									X	X			
		Organize training and awareness raising sessions on the use of template for damage assessment	NDMA	PDMAs, PMD, NESPAK, SUPARCO, Academia									X	X			
Capacity Enhancement of Human Resources for Earthquake Safer Construction	Training of the concerned technical officers of the government sector	Develop training toolkits	NDMA	PDMAs, C&Ws, PMD, GSP, NESPAK, PEC, Academia											X	X	X
		Conduct training for the technical officers on safer building construction and building bylaws	NDMA	PDMAs, C&Ws, PMD, GSP, NESPAK, PEC, Academia											X	X	X
		Conduct foreign training for learning and experience sharing	NDMA	PDMAs, C&Ws, PMD, GSP, NESPAK, PEC, Academia											X	X	X
		Developing a database of trained officers	NDMA	PDMAs, C&Ws, PMD, GSP, NESPAK, PEC, Academia											X	X	X
	Training of building engineers and contractors from the private sector to promote earthquake-resistant construction.	Develop training toolkits	District Administration	PDMAs, MCs, District Development Authorities, Housing sicieties, DDMAs											X	X	X
		Organize training for the contractors/building engineers on safer building construction and following building bylaws	District Administration	PDMAs, MCs, District Development Authorities, Housing sicieties, DDMAs											X	X	X

| | | |

earthquake at risk areas	schools that are more vulnerable to earthquake hazard	Select the most vulnerable schools and do the retrofitting and repairing providing further strengthening of the building	NESPAK	Academic, PEC, PDMAs, NDMA									X	X	X	X
		Detail reporting of the retrofitting for model replications in other areas / districts	NESPAK	Academic, PEC, PDMAs, NDMA									X	X	X	X
		Seismically safer upgradation of public buildings as a long-term initiative through retrofitting and repairing	NESPAK	Academic, PEC, PDMAs, NDMA									X	X	X	X

4.3.3 Tsunami

Based on the outline of tsunami countermeasures projects described in the previous section, the project team reviewed the long list (draft) at the program level. At the program level, what kind of projects will be carried out is vague, so the project team added the red parts to make it more specific. The long list (draft) of disaster prevention investment projects compiled in Table 4.3.2.

Table 4.3.2 Long List (Draft) Related to Tsunami

Strategy	Interventions	Actions	Execution Agency		Short term for 2024				Long-Term planning								
			Main	Support organizations	1st Qtr	2nd Qtr	3rd Qtr	4th Qtr	25	26	27	28	29	30	31	32	33
Developing/Revising the Tsunami Hazard / Risk Assessment Database	Establishment of a committee for the evaluation of earthquake faults and the practice of fault evaluation.	Announcement and selection of the committee members	NDMA	Academic, PEC		X	X										
		Hold regular meetings for institutionalization	NDMA	Academic, PEC				X	X	X	X	X	X	X	X	X	X
		Assessing the active and non-active faults and their mapping	NDMA	Academic, PEC				X	X	X	X	X	X	X	X	X	X
	Updating tsunami and inundation simulation (by scenario earthquakes)	Compilation of geological and roughness data	PMD	SUPARCO, GSP, Academia,NDMA, PDMA Sindh, PDMA Baluchistan						X	X						
		Improving program codes	PMD	SUPARCO, GSP, Academia,NDMA, PDMA Sindh, PDMA Baluchistan						X	X						
	Developing simulation models for damage and loss estimation on the basis of tsunami risk assessment (by scenario earthquakes)	Updating the tsunami hazard data	PMD	SUPARCO, Academia, NDMA, PDMA Sindh, PDMA Baluchistan							X						
		Implementation of the tsunami damage assessment simulation (by scenario earthquakes)	PMD	SUPARCO, Academia, NDMA, PDMA Sindh, PDMA Baluchistan							X	X					
		Developing guidelines for tsunami damage assessment simulation	PMD	SUPARCO, Academia, NDMA, PDMA Sindh, PDMA Baluchistan								X	X				
		Training of concerned technical staff	PMD	SUPARCO, Academia, NDMA, PDMA Sindh, PDMA Baluchistan									X				
	Develop tsunami risk assessment for Gwadar city / Karachi city at micro scale	Conducting a risk assessment of Gwadar / Karachi city	DDMA Gwadar/Karachi	PDMAs, GDA, KDA, TMAs								X	X				
		Identify critical infrastructure needs retrofitting / further strengthening	DDMA Gwadar/Karachi	PDMAs, GDA, KDA, TMAs								X	X				
		Develop a tsunami hazard atlas and database of the Gwadar / Karachi city	DDMA Gwadar/Karachi	PDMAs, GDA, KDA, TMAs								X	X				

| | | |

Enhancing Tsunami Early Warning System	Engaging electronic and print media personnel for tsunami EWS and its reach out to the tsunami hazard-prone areas	Identifying relevant media platforms for early warning dissemination, study bottlenecks, and suggest recommendations for improvement of EWS	PEMRA	PDMA Baluchistan, DDMA and TMA	x									x	x			
		MOUs signing with the TVs, radios, and important newspapers	PEMRA	PDMAs, Press clubs, electronic and print media agencies		x	x											
		Enhancing communication networks through seminars, drills, and training	PEMRA	PDMAs, Press clubs, electronic and print media agencies		x	x	x	x	x								
		Defining SOPs disseminating the tsunami EWS and developing thresholds for evacuation	PEMRA	PDMAs, Press clubs, electronic and print media agencies		x	x											
Reducing the Impact of Tsunami Hazard	Reforestation through mangrove plantation	Survey of the target areas and their prioritization for project implementation	Forest department Sindh, Coastal Development authority in Baluchistan	PDMAs, District administration, INGOs, UNO				x	x	x								
		Plantation of Mangrove trees together with the target communities in the prioritized villages of Sindh and Baluchistan Province	Forest department Sindh, Coastal Development authority in Baluchistan	PDMAs, District administration, INGOs, UNO					x	x								
		Conduct training of communities for taking care of mangrove plantations	Forest department Sindh, Coastal Development authority in Baluchistan	PDMAs, District administration, INGOs, UNO					x	x								
		Organize exposure visits among both provinces to learn from the best practices	Forest department Sindh, Coastal Development authority in Baluchistan	PDMAs, District administration, INGOs, UNO					x	x								
		Documentation of the best practices and lesson learn	Forest department Sindh, Coastal Development authority in Baluchistan	PDMAs, District administration, INGOs, UNO						x								
	Reducing the Impact of Tsunami Hazard	Setting residual risks and protection concept	Coastal Development authority	PDMAs, District administration, INGOs,										x				

													x
													x
												x	
												x	

4.3.4 Human Resource Development

Based on the implementation status of the current NDMP, although some progress have been made, the efforts remain insufficient, and many of the same challenges identified in the existing NDMP still persist. Therefore, it is considered that the long list in this field should basically follow the priority projects of the current NDMP. The following table shows the proposed priority projects under Intervention 5: Human Resource Development Plan.

Although the priority projects are basically from the current NDMP, the outputs are redefined based on the priorities of the Sendai Framework for Disaster Risk Reduction which was adopted after the formulation of the current NDMP, and the priority projects of the current NDMP are reorganized. The concept is to set the outputs which promote priority 1 (understanding disaster risk) as the basis for implementing priorities 2 and 3 of the Sendai Framework for Disaster Risk Reduction, and promote priority 4 (Enhancing disaster preparedness for effective response, and to "Build Back Better" in recovery, rehabilitation and reconstruction) in order to prepare for disasters that may occur soon. The wording of some priority projects has been revised to reflect the actual situation and future plans. In addition, the priority projects of the current NDMP, "DRM Workshop for TMA staff" will be removed from the priority projects this time because the implementation has not been confirmed, the number of TMAs is large, and the training of PDMAs and DDMAs has not yet been sufficiently implemented. The implementing agencies have been reviewed in light of the current NDMP's implementation status. Since NIDM is not yet fully functioning, the name of the NDMA has been designated for projects originally assigned to NIDM. Furthermore, DDMA was changed to PDMA because the implementation of the activities by DDMA was not confirmed.

Table 4.3.3 Draft Long List for Intervention 5 (Human Resources Development Plan)

Output	Sendai Framework	Activity	Responsible Organization
Build the knowledge of government officials at all levels	Priority 1, 24, (g)	Setting up of organization of NIDM	NDMA
		Construction of NIDM	NDMA
		Enhancement of coordination and partnership for human resource development (*1)	NDMA/ NIDM
		DRM courses for NDMA, PDMAs and DDMAs staff	NDMA/ NIDM, PDMA
		DRM workshops for parliamentarian and policy makers (*2)	NDMA/ NIDM, PDMA
		DRM workshops for relevant ministries of Federal government	NDMA/ NIDM
		DRM workshops for relevant departments of provincial governments	PDMA
		DRM workshops for relevant departments at district level	PDMA
		Improvement of DRM training quality	NDMA/ NIDM
Promote investments in innovation and technology development in long term, multi hazard and solution driven research in disaster risk management	Priority 1, 24 (k)	Establishment of library on DRM	NDMA/ NIDM
		Promotion of research	NDMA/ NIDM
Promote the incorporation of disaster risk knowledge in education	Priority 1, 24, (l)	DRM subjects into curriculum of training institutes for government officials (*1)	NDMA/ NIDM, Establishment Division/ National School of Public Policy
		Promotion of disaster education at schools	NDMA/ NIDM, PDMA, Provincial Education Department
		Promotion of disaster education in higher education	NDMA/ NIDM, HEC, PDMA, Provincial Education Department

Output	Sendai Framework	Activity	Responsible Organization
Promote national strategies to strengthen public education and awareness in disaster risk reduction	Priority 1, 24, (m)	Awareness raising (*1)	NDMA, PDMA, Rescue 1122, Civil Defence, DDMA
		DRM workshops for media (*2)	NDMA, PDMA
Train the existing workforce and voluntary workers in disaster response	Priority 4, 33, (f)	Capacity enhancement of urban search and rescue teams	NDMA, City District Governments
		Implementation of regular training of rescue teams (*1)	Rescue 1122, Civil Defence
		Rescue training for members of community emergency response teams (*1)	Rescue 1122, Civil Defence, NGO
Promote regular disaster preparedness, response and recovery exercises	Priority 4, 33, (h)	Implementation of mock exercises for various disasters at local governments (*2)	NDMA, PDMA, DDMA, Rescue 1122, Provincial line departments
		Implementation of evacuation drills/ mock exercises at schools (*1)	NDMA, PDMA, Rescue 1122, Provincial Education Department

*1: Activities existing in the current NDMP but reworded, restructured, or integrated.
*2: New activities

4.3.5 Urban Disasters and Land Use

4.3.5.1 Urban Disasters in General

Based on the findings described in Section 4.2.5, a draft long list of investment projects for DRRM (Disaster Risk Reduction and Mitigation) projects in the area of "Urban disasters and land use planning" is currently under review. The interim results of this assessment are shown in the table below.

Table 4.3.4 Long List (Draft) and Priority Projects in the Field of "Urban Disasters and Land Use Planning"

Group of Projects for the investment for disaster risk reduction			Long list (draft) and priority projects
Major item	**Middle item**	**Major item**	**Middle item**
[I] Safe evacuation	[I-2] Securing evacuation space	Safe evacuation	・ Formulation of "Disaster Prevention Urban Planning" ・ Agreement on mutual use of evacuation shelters in the event of disasters
		Elimination of areas where evacuation is difficult (including wide-area evacuation)	・ Development of evacuation facilities (tower type) ・ Designation of evacuation buildings and support for the development of evacuation buildings ・ Evacuation facilities (high ground) improvement ・ Community disaster prevention centers for tsunamis
		Securing evacuation sites	・ Evacuation routes improvement ・ Disaster prevention roads (wide roads) improvement
[II] Constructing urban structures resistant to earthquakes, tsunamis and floods	[II-1] Measures for "facilities" required for earthquake, tsunami and flood prevention measures	Constructing urban structures resistant to earthquakes, tsunamis and floods	・ Designation of disaster risk areas under local ordinances
	[II-2] Resilience of regions against earthquakes, tsunamis, and floods	Elimination of crowded sites	・ Disaster prevention measures based on "risk assessment" for each space below the fundamental local government
		Securing evacuation sites and routes	・ Development of regional disaster prevention centers
	[II-3] Land use planning considering disaster risk	Linkage between disaster prevention plans and city planning	・ Introduction of disaster risk management guidelines in municipal urban planning Master Plan
		Regulation and guidance of land use in disaster areas	・ Designation of disaster risk areas under the Building Standards Act and municipal ordinances ・ Introduction of disaster risk management guidelines in municipal urban planning Master Plan ・ Promotion of housing relocation in high-risk areas

4.3.5.2 Rainwater Drainage (Urban Flood)

Based on the results of the study described in Section 4.2.5, the following table shows the proposed draft long list of Projects for the investment for disaster risk reduction in the field of "Rainwater Drainage (Urban Flood)."

Table 4.3.5 Long List (Draft) and Priority Projects in the Field of "Rainwater Drainage (Urban Flood)"

Classification	Projects for the investment for disaster risk reduction
Research and Planning	A case study of rainwater drainage planning based on the impact of climate change
	A case study of improvement measures for combined sewerage systems
Hard Measures	Rehabilitation and enhancement of existing rainwater drainage facility (drainage channel pump station)
	Construction of new rainwater drainage facilities (drainage channel pump stations)
	Construction of rainwater storage facilities (adjustment ponds, storage pipes, etc.)
	Construction of rainwater seepage facilities (seepage trough, seepage trench, permeable pavement, etc.)
	Water-resistant pumping facilities
	Installation of water shutoff board in underground facilities
Soft Measures	Preparation of inland water hazard maps
	Construction of prewarning system for inland water flooding
	Development of urban flood emergency response plans
	Measures against illegal dumping in drainage channels
	Land use regulations (building regulations in areas where flooding is common)

4.3.6 EWS

Based on the finding described in Section 4.2.6, the long list (draft) of Projects for the investment for disaster risk reduction in the EWS is based on the detailed long list (draft) up to the program level. However, since the types of projects to be implemented at the program level are not clearly defined, the red-highlighted sections have been added to provide greater specificity . Table 4.3.6 shows the long list (draft) of Projects for the investment for disaster risk reduction compiled.

Table 4.3.6(1) Long List (Draft) Related to AWS

Strategy	Action	Program	Responsible Organization		2024				Long Term		
			Main	Support	1Q	2Q	3Q	4Q	~2026	~2028	~2033
Multi-Hazard Early Warning System.	1. Strengthen Forecasting and Early Warning System.	1.1. Strengthen Weather Observation Network Including AWS, Radars, Upper Air Observation and Satellite.	PMD	NDMA, PDMAs	✓	✓	✓	✓	✓	✓	✓
		1.2. Establish Forecasting Centers Including RFFWC and Urban/Flash Flood Early Warning Center in Major Cities.	PMD	NDMA, PDMAs, FFC, IRSA		✓			✓		
		1.3. Strengthen Flood Forecasting and Early Warning System Including GLOF.	PMD, WAPDA, NDMA	PDMAs, FFC, IRSA		✓	✓		✓	✓	✓
		1.4. Strengthen Earthquake & Tsunami Forecasting and Early Warning System.	PMD, PNHD	GSP, NDMA, PDMAs	✓	✓	✓	✓	✓	✓	✓
		1.5. Strengthen Forecasting and Early Warning System for Other Disasters: Landslide, Cyclone, Drought and Extreme Weather.	PMD	NDMA, PDMAs	✓	✓	✓	✓	✓	✓	

Source: Project Team

Table 4.3.6(2) Long List (Draft) Related to AWS

Strategy	Action	Program	Responsible Organization		2024				Long Term		
			Main	Support	1Q	2Q	3Q	4Q	~2026	~2028	~2033
Multi-Hazard Early Warning System.	2. Prepare Emergency Response.	2.1. Strengthen Early Warning Dissemination System between PMD, DMAs and Other Agencies or Departments.	PMD, NDMA, PDMAs, DDMA	—	✓	✓	✓	✓	✓	✓	
		2.2. Develop National Plan, Guidelines and SOPs for Early Warning System and Health Emergency Preparedness and Response (HEPR).	PMD, NDMAs, NHSRC	PDMA, Provincial Health Department	✓	✓	✓	✓	✓		
	3. Develop Capacity of Early Warning and Evacuation System.	3.1. Develop Capacity for Staff and Officials by Education Courses and Collaboration with Other Organizations.	PMD, FFC, WAPDA, NDMAs, PDMA, DDMAs	—	✓	✓	✓	✓	✓	✓	✓
		3.2. Develop Capacity for Community by Training and Drill Programs and Community Based Disaster Risk Management (CBDRM).	NDMAs, PDMA, DDMAs	PMD, FFC		✓	✓		✓	✓	✓

Source: Project Team

4.4 Activity 3-4

The implementation promotion method of priority projects is examined, and the technical examination material of NDMP-II is prepared.

4.4.1 Activities by Disaster Type

4.4.1.1 Earthquake

The implementation of priority projects was examined through field activities. Regarding the implementation promotion method of hazard and risk assessment, a real-time earthquake damage estimation system is proposed considering the introduction to "NEOC" which is currently a pressing priority for the NDMA. In Japan, the National Research Institute for Disaster Prevention is currently developing of a system incorporating AI based analysis. An overview of this system is presented in Figure 4.4.1.

（https://www.bosai.go.jp/activity_special/disasterresponse/detail003.html

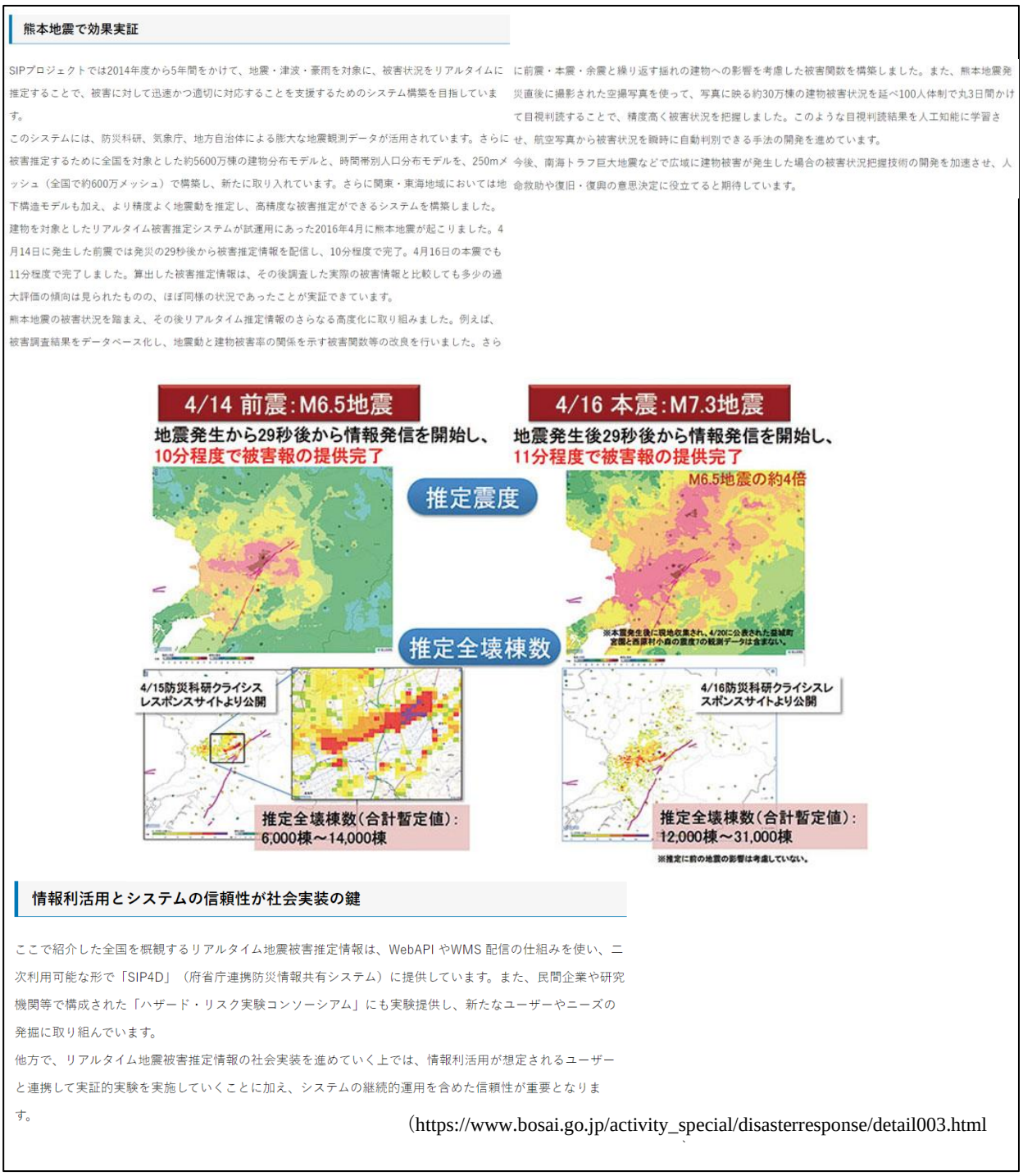


Figure 4.4.1 Real-Time Earthquake Damage Estimation System Developed by the National Institute of Disaster Prevention and Science

《 Future Directions 》

The NEOC demonstrates the capability to estimate real-time earthquake damage. During normal times, it can also conduct damage predictions in advance based on scenario earthquakes . Based on the damage prediction results, the NEOC can serve as a foundational tool for disaster prevention and mitigation planning by identifying high-risk areas, to implement disaster prevention investments in advance, such as the enhancement of important disaster prevention facilities, and to provide logistical support to the military and rescue teams that rush to rescue people in the event of an emergency, and to facilitate the evacuation of residents.

4.4.1.2 Tsunami

During the field activities for the implementation of the priority project, the technical transfer of key technologies and discussions were carried out with the staff at PMD Islamabad Headquarters. These focused on probabilistic tsunami inundation simulation, which is considered as essential foundation data for designing tsunami seawall and related infrastructure, etc., as outlined below.

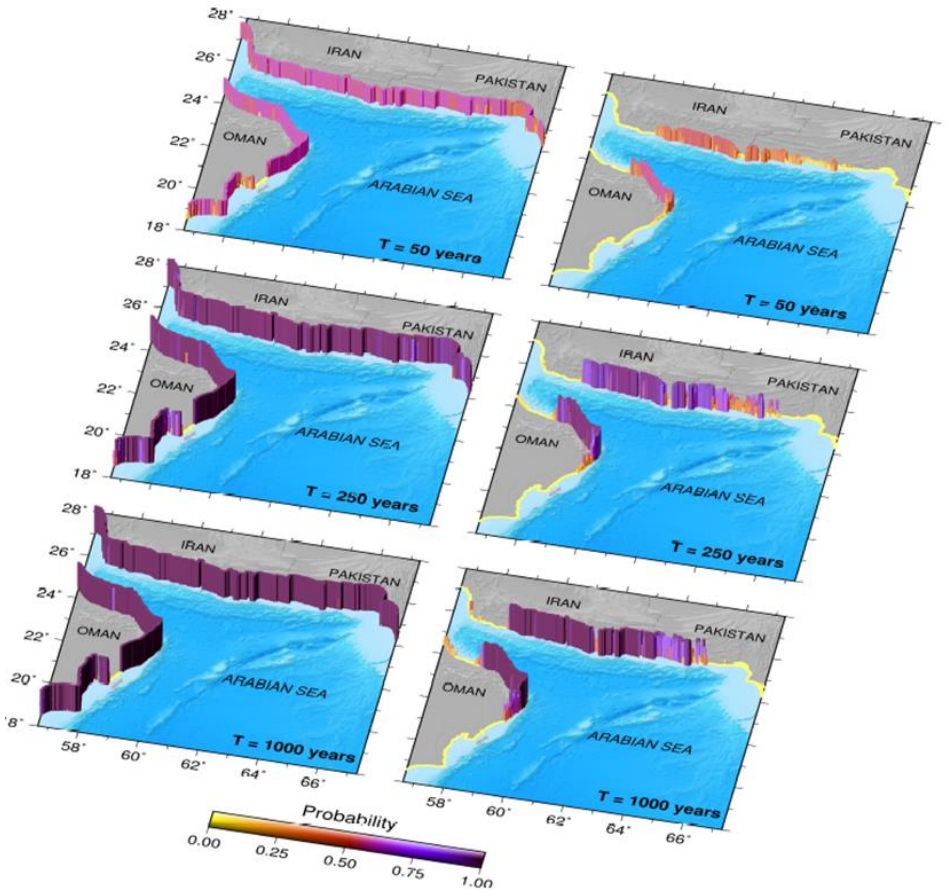
(1) Future Consideration of Tsunami Inundation Simulation

It has been pointed out that high-resolution topographic data are necessary for future tsunami simulations. An interview survey conducted by PMD on probabilistic tsunami hazards revealed that, while PMD has previously performed tsunami hazard assessments using deterministic methods, it has not yet conducted probabilistic tsunami hazard assessments. Since probabilistic tsunami hazard information is not included in the current NDMP, the project team introduced the probabilistic tsunami hazard assessment conducted in Japan to PMD. As the evaluation target area, it is necessary to evaluate the hazard of the entire coast of Pakistan, not only 1 coastal city described in the current NDMP-1. Furthermore, since no domestic research institutes or ministries have conducted probabilistic tsunami hazard assessments, there is need to develop a framework and programming for this type of assessment. In order to implement probabilistic tsunami hazard, it is necessary to set the probability of occurrence of fault models corresponding to trench-type earthquakes. Regarding this point, when the project team checked with the PMD staff, it was confirmed that there are no supporting documents such as the "long-term assessment" published by the Japan Earthquake Research Commission in Pakistan. Based on these domestic circumstances, the project team presents a technical policy for implementing probabilistic tsunami hazard assessment in Pakistan.

(2) Program Overview for the Development of Probabilistic Tsunami Hazard Assessment

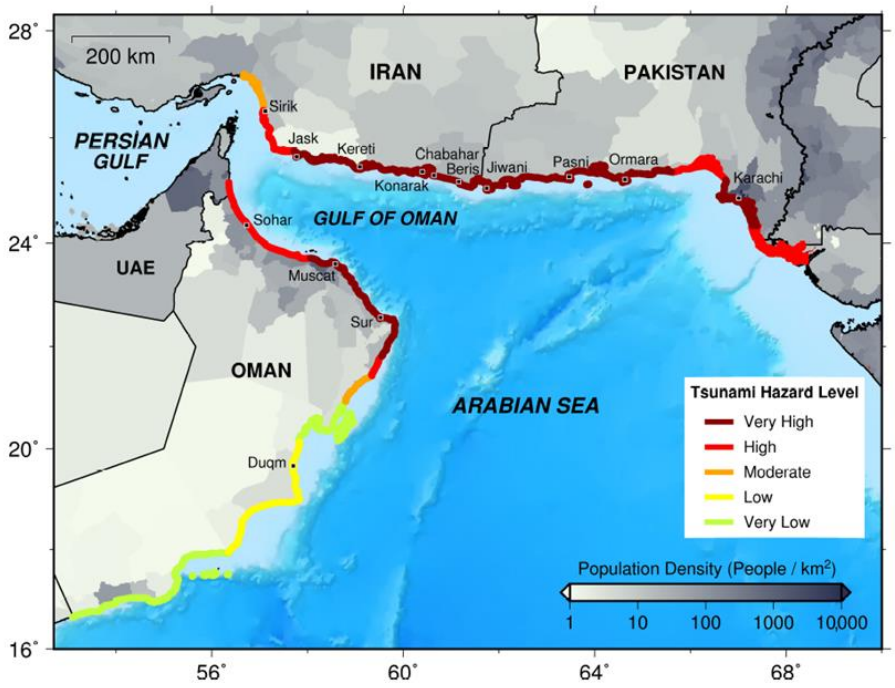
1) Background and Overview

In recent years, the limitations of the scenario-based tsunami hazard assessment have become apparent, leading to the promotion of probabilistic approach across various fields and countries. However, in Pakistan, the scenario-based approach is still dominant, and the adoption of the probabilistic tsunami hazard assessment (Probabilistic Tsunami Hazard Assessment: PTHA) has been slow to advance. Researchers from Iran and other countries have conducted studies presenting tsunami hazards along the coast of Pakistan. (see Figure 4.4.2, 4.4.3)



Source: A Review of Tsunami Hazards in the Makran Subduction Zone (2020)
Amin Rashidi , Denys Dutykh, Zaher Hossein Shomali, Nasser Keshavarz Farajkhah and Mohammadsadegh Nouri

Figure 4.4.2 Probability Distribution of Tsunami Heights Exceeding 0.5 M (Left Panel) and 3 M (Right Panel) Corresponding to Simulated Periods of 50, 250, and 1000 Years Along the Coasts of Iran, Pakistan, and Oman



Source: A Review of Tsunami Hazards in the Makran Subduction Zone (2020)
Amin Rashidi , Denys Dutykh, Zaher Hossein Shomali, Nasser Keshavarz Farajkhah and Mohammadsadegh Nouri

Figure 4.4.3 Population Density and Tsunami Hazard Distribution Along the Makran Area (In Pakistan, the Tsunami Hazard Level is High in Karachi and the Western Coast.)

Probabilistic Tsunami Hazard Assessment (PTHA) mainly consists of three components: developing fault models, performing tsunami simulations, and generating hazard curves. The quantity of fault models required often reaches several thousand cases when they are set along a subduction zone of several hundred kilometers. In the case of PTHA for Pakistan, an element fault (a small fault of 5 km square) is set on the upper surface of the Arabian Plate subducting into the Eurasian Plate, and a simplified fault model called "Characterized Source Fault Model" is constructed to cover various types of earthquakes that may occur in the Makran Subduction Zone. For the generation of tsunami hazard curves, it is considered that the method of giving earthquake probabilities and the method of constructing hazard curves in the preceding earthquake hazard map can be referred to. However, earthquake hazard maps mainly focus on seismic events occurring in land, while earthquakes occurring in sea areas, specifically in the Makran Subduction Zone, have not been studied sufficiently. Since the evaluation of earthquakes occurring between plates in sea areas is the most important in PTHA, it is necessary to study the evaluation of these earthquakes.

After the fault models are constructed, tsunami simulations are performed for each fault model to obtain tsunami heights along the coast. Since the number of fault models reaches several thousand, tsunami simulations are also performed for several thousand cases, and it is necessary to secure computational resources. Using the tsunami height distribution along the coast, hazard curves are generated for key locations. These curves are then analyzed to identify earthquake groups that may represent potential future risks. The process is illustrated in the figure on the right. The flow is shown in the right figure.

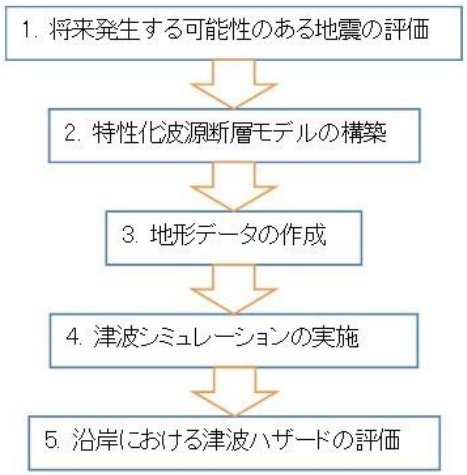
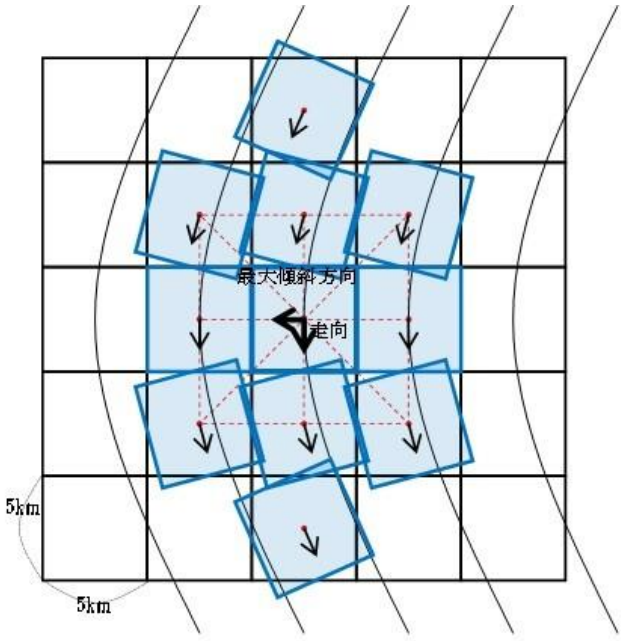


Figure 4.4.4 Flow Chart Showing the Procedure for Conducting Probabilistic Tsunami Hazard Assessment

2) Elemental Faults

As the first step in constructing a fault model, an elemental fault of about 5 km by 5 km in size (Figure 4.4.5) is established. The elemental fault is set on the upper surface of the subducting plate. The slip angle of each elemental fault is determined based on the direction of relative motion between the subducting plate (Arabian plate) and the continental plate (Eurasian plate). Element faults are extracted within the range of the fault model to be set and used as fault parameters for tsunami simulation.



Source: Research Paper No. 400, Disaster Prevention Science and Technology Research Institute, Study of the method of probabilistic tsunami hazard assessment due to earthquakes occurring in the Japan Trench

Figure 4.4.5 Image of Elemental Faults

3) Constructing a Characterized Source Fault Model

It is well established that earthquake slip distribution is inherently inhomogeneous. However, it is not practical to construct fault models considering all patterns of the slip inhomogeneity of earthquakes and perform tsunami simulation on them. Therefore, the slip distribution is simplified through a process called "characterization" to construct a Characterized Source Fault Model. This reduces the number of fault models to construct and the computational cost of tsunami simulation. The Characterized Source Fault Model consists of three different areas: "background area," "large slip area," and "super large slip area." The large slip area accounts for 30% of the source (fault) area and has twice the average slip amount, while the super large slip area covers 10% of the source area with four times the average slip amount. The large slip area is arranged while shifting at about half-pitch intervals

with respect to one source area. When the source area extends to the trench, a scenario incorporating a "super large slip area" is incorporated to represent the the possibility of very large slip occurring along the trench. An image of the characterized source fault model set in the Makran subduction zone is shown in Figure 4.4.6.

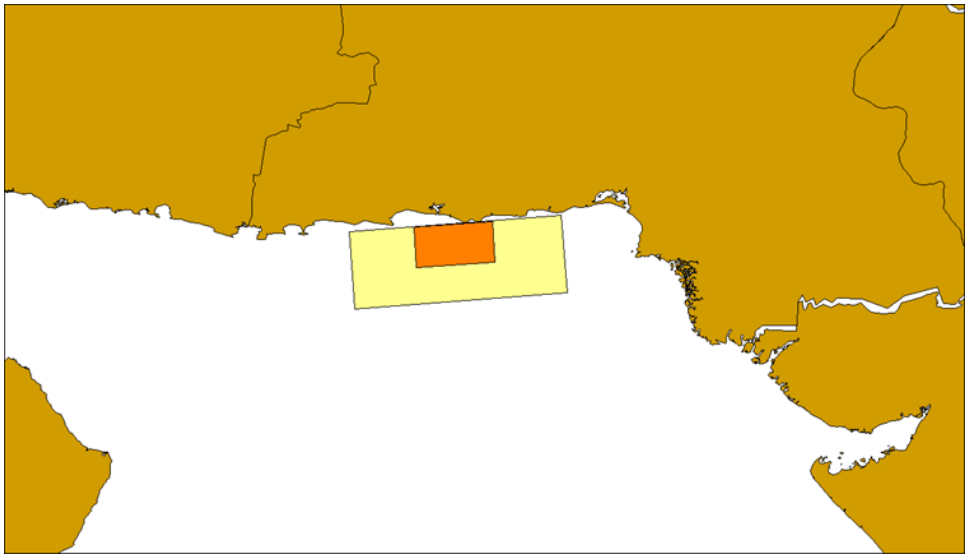


Figure 4.4.6 Image of the Characterized Source Fault Model in Makran Subduction Zone for the Probabilistic Tsunami Hazard Assessment in Pakistan

4) Tsunami Simulation

After the fault model is constructed, the topographic data are prepared for the tsunami simulation. A nesting system is used to generate high-resolution topographic data (preferably 50 m or less) for the coastal area. In the tsunami simulation, the crustal deformation of each fault model is calculated using Okada's equation, and the initial water level distribution is obtained by applying the Kajiura filter (the green function representing the sea level change to the delta function input for the time and space of seafloor displacement) to the amount of crustal deformation. Based on the initial water level distribution, the tsunami propagation calculation is performed by solving the nonlinear long wave equation including the advection term and the bottom friction term.

5) Tsunami Hazard Analysis

There are two types of probabilistic hazard assessment: current hazard assessment and long-term average hazard assessment. By evaluating the most recent occurrence time and the average recurrence interval of the target earthquake, and applying the BPT distribution as the probability model, a time-dependent hazard assessment can be conducted based on the current point in time. As the details of earthquake occurrence probability in the Makran subduction zone are unknown, it is appropriate to calculate earthquake occurrence probability using the stationary Poisson process and perform long-term average hazard assessment.

Figure 4.4.7 shows a sample Tsunami hazard curve used for analysis of Tsunami hazard. These hazard curves incorporate uncertainties, which are mainly classified into the following two types.

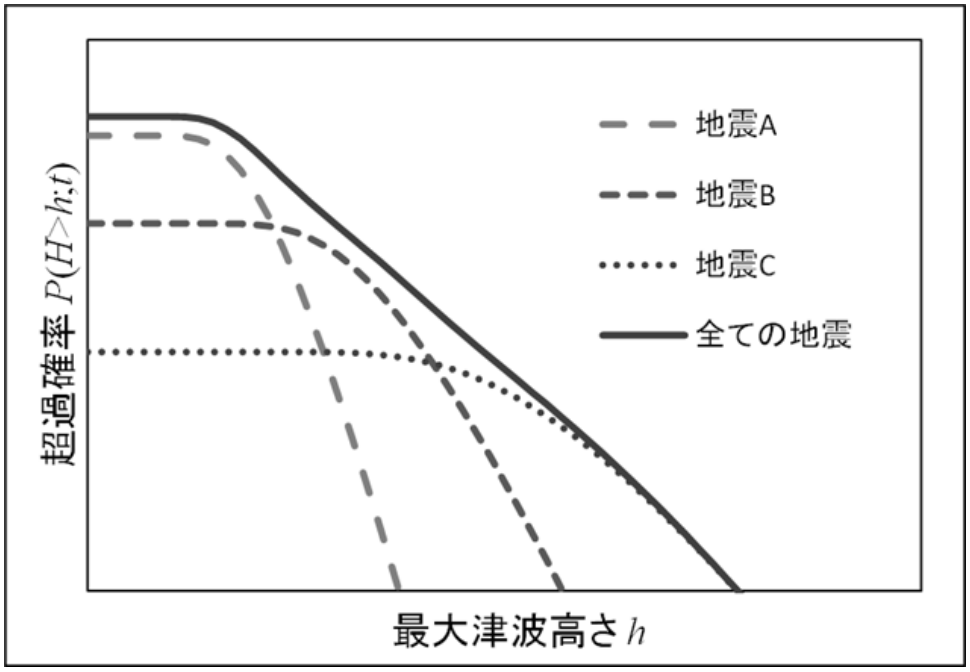
[Contingent uncertainty]

These include natural uncertainties such as earthquake magnitude, rupture processes, and seismic intensity. As they are inherent to natural phenomena, they are both inevitable and unpredictable.

[Epistemological uncertainty]

This type of uncertainty is due to lack of data and knowledge. It is not predictable at present, but it will become clear as research progresses. Examples include the presence of earthquake faults and the magnitude range. Logic trees are one of the popular tools for quantifying epistemic uncertainty

in PTHA, and this uncertainty is taken into account when creating hazard curves.



Source: Research Paper No. 400, Disaster Prevention Science and Technology Research Institute, Study of Probabilistic Tsunami Hazard Assessment Methods for Earthquakes in the Japan Trench

Figure 4.4.7 Image of Tsunami Hazard Curve

6) Tsunami Hazard Map

There are three parameters related to tsunami hazard maps:

- Coastal tsunami height
- Exceeding probability
- Period

A tsunami hazard map is developed by fixing either the coastal tsunami height or the exceedance probability over a period of time (for example, the next 30 years) and plotting the remaining one parameter. For example, the distribution of the coastal tsunami height with a 30-year exceedance probability of 10%, or the distribution of the exceedance probability over the next 30 years with a tsunami height of 3 m.

7) PMD Questionnaire-PTHA Implementation

A survey on Probabilistic Tsunami Hazard Assessment (PTHA) in Pakistan revealed that it has not been implemented in Japan. Therefore, in order to clarify the issues in implementing Pakistan's original PTHA in the future, a questionnaire survey was conducted by PMD, which is considered to be one of the most important organizations in the PTHA study. The results of the questions and answers are described below.

1) Construction of many fault models along the Makran Subduction Zone
   - Q: What are the technical problems related to this work?
   - A: The PMD needs an ocean floor survey utilizing software (tsunami propagation calculation program) or tools, detailed seafloor topographic data (NOAA, IOC), and OBS (At present, PMD has seismic stations only on land.).
   - A: The PMD wants the survey to be carried out by a Japanese research institution.
   - A: The PMD has four GPS stations but has technical problems dealing with their data.
   - Q: Building a large number of fault models is time-consuming and costly. Which institution or organization in Pakistan can support this?
   - A: Various institutions in Pakistan, including PMD, Survey of Pakistan, SUPARCO, Pakistan Navy, National Institute of Oceanography Pakistan, NESPAK, NED and NUST, think this is feasible.

2) How to assign probability of occurrence to each fault model
   - Q: One way to assign probability of occurrence to a fault model is to refer to the probability of occurrence used in PSHA (Probabilistic Seismic Hazard Assessment). The GR rule may also be used. Any other ideas?
   - A: The probability of occurrence of an earthquake along the Makran subduction zone is an important point of PTHA. Deterministic seismic hazard assessment does not exist in Pakistan.
   - Q: Do you think this is feasible?
   - A: Probably. However, the project team would like to receive training courses from NIED (National Research Institute for Disaster Reduction, Japan) and others who are familiar with probabilistic hazard assessment methods.

3) Implementation of a number of tsunami simulations
   - Q: Are there enough computational resources in Pakistan?
   - A: No. The project team thinks it is necessary to introduce supercomputers.
   - A: PMD would like the TSUNAMI code (tsunami propagation calculation program) created by Professor Imamura (Tohoku University) and its technical support.

4) Development of tsunami hazard curves
   - Q: Is it necessary to hold seminars, etc. to create tsunami hazard curves? What kind of program is needed to develop human resources who can create tsunami hazard curves?
   - A: In this regard, PMD wishes to provide training courses, etc. from NIED (National Research Institute for Disaster Prevention, Japan).

5) Other Comments
   - A: Regarding the early warning system, PMD staff has developed an early warning system for tsunamis by participating in Training-Workshop Standard Operating Procedures for Tsunami Early Warning, but does not have an early warning system for earthquakes.
   - For future efforts, the document provided by PMD lists the following issues:
   - Upgrading PMD's Earthquake and Tsunami Warning Center
   - Capacity building based on new GPS data
   - The previous SOP is based on the old Erice and GUITAR Tsunami Tool THEREFORE. The new SOP will be revised with a new upgraded system.
   - The seismic project with CENC (China) will enter a new phase and will establish new stations in the north and south of the country.

(3) Specific Examples of Countermeasures to Prevent Tsunami Inundation, Gain Time to Evacuate, and Reduce the Area and Depth of Inundation

In the Great East Japan Earthquake, the seawall in the Taro district of Miyako City, Iwate Prefecture, was severely damaged. (For example, https://www.city.miyako.iwate.jp/kanko/tarobochotei.html)

Based on such lessons, Japan's Ministry of Land, Infrastructure, Transport and Tourism is leading initiatives to prevent tsunami inundation, extend evacuation time, and reduce the extent and depth of inundation, recognizing that hard countermeasures such as seawalls alone are not suffecient. Figure 4.4.8 shows an example of a "green seawall."

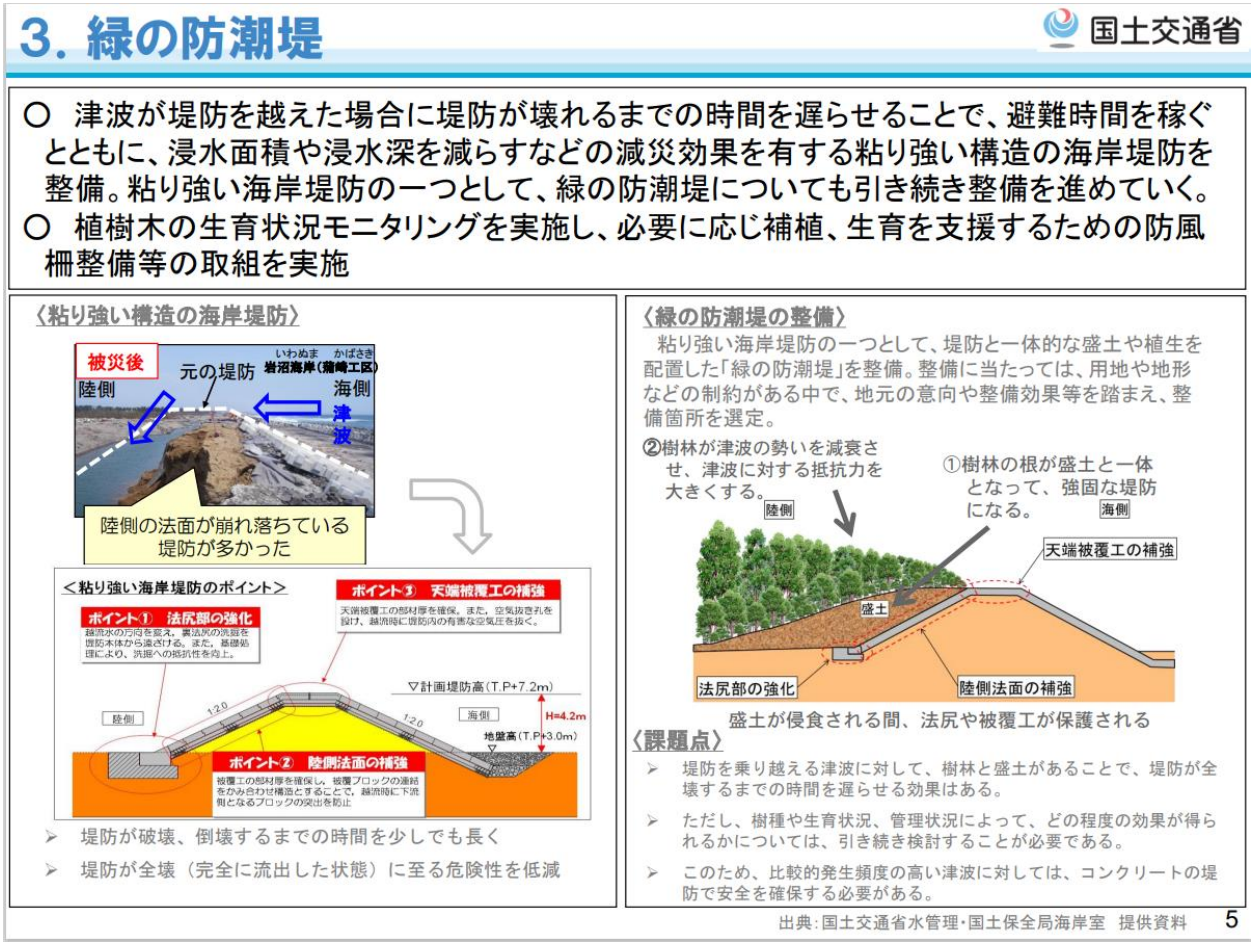


Figure 4.4.8 An Example of the Development of a Tenacious Coastal Bund and a Green Seawall

4.4.1.3 Urban Disasters and Land Use

This section discusses methods for promoting implementation of priority projects in the field of "urban disasters and land use planning" based on "Activity 3-3: Preparation of a long list (draft) and consideration of priorities for each type of disaster" and prepares technical study materials for NDMP II.

Specifically, the following pages present technical study materials for the following projects proposed in Section 4.2.5, including a long list (draft) and priority projects in the field of "urban disasters and land use planning").

① Formulation of "disaster prevention community development plan"

② Agreement on mutual use of evacuation sites in the event of disasters

③ Development of evacuation facilities

④ Development of evacuation routes

⑤ Designation of disaster risk areas by local government ordinance

⑥ Detailed risk assessment on a spatial basis (e.g., by block)

⑦ Development of regional disaster prevention centers

⑧ Formulation of guidelines for "disaster prevention community development" by the federal government and reflection in the development Master Plan by local governments

⑨ Promotion of relocation of housing in high-risk areas

Table 4.4.1 Technical Study Material 1: Formulation of "Community Based Disaster Management Plan"

Project Name	**Development of Community Based Disaster Management Plan**
Overview	In order to ensure the safety of residents, a Community Based Disaster Management Plan is formulated based on the assumption that a large-scale natural disaster occurs. The main contents of the plan are as follows. • Basic policy: Public assistance, mutual assistance, and self-help are included. • Development of facilities such as evacuation sites and evacuation routes in each district • Improvement of daily disaster prevention awareness • Disaster prevention measures
Main Points of the Project	• Formulation of a Community Based Disaster Management Plan through workshops with participation of community members. Residents, together with relevant departments and organizations of Division/District/Tehsil, exchange opinions through Community Based Disaster Management Plan consultation workshops, form a common understanding of the current situation and issues in the region, and compile countermeasures. • Prioritization of disaster management projects For each community, residents, in cooperation with the government, examine both hard and soft disaster prevention projects (Examples: Securing evacuation sites, preparing evacuation routes, preparing evacuation signs and information, raising awareness of disaster prevention, conducting disaster prevention drills, etc.). The considered hard and soft projects are evaluated in terms of urgency and effectiveness, priority projects are selected, and it is compiled into a disaster prevention community development plan. • Establishment of a system for implementing disaster prevention projects After the Community Based Disaster Management Plan is formulated, an organization for implementing the project is established in cooperation with stakeholders such as the government, local governments, residents, and private companies in order to promote its implementation. At this time, attention should be paid to cooperation with related organizations according to the content of the project. (Example: In the case of river development, river managers, people involved in water utilization and flood control, local governments in the basin, residents in the vicinity, private companies, etc.)
Reference Image	 Example of disaster prevention community development plan formulation workshop *Source: Guidelines for Building Resilient Cities against Earthquakes and Tsunamis, the Ministry of Land, Infrastructure, Transport and Tourism Chubu Regional Development Bureau*

Table 4.4.2 Technical Study Material 2: Agreement on Mutual Use of Evacuation Sites in the Event of Disasters

Project Name	**Agreement on Mutual Use of Evacuation Centers in the Event of Disasters**
Overview	• Conclude an agreement so that residents of neighboring municipalities/union councils can mutually use evacuation centers. • Utilize private facilities as temporary shelters. For this purpose, conclude an agreement on the use of private facilities by municipalities, facility owners, and local communities. • In the event of a wide-area disaster (floods, landslide, earthquake, etc.) affecting multiple municipalities across municipal boundaries, neighboring municipalities conclude an agreement so that residents can evacuate across municipal boundaries.
Main Points of the Project	• This agreement on the mutual use of evacuation centers makes it possible to evacuate from each residence to the nearest evacuation site and improves safety. • To accommodate evacuees until there is no risk of damage from a disaster, agreements must be signed with private organizations, local communities, and local governments. • A list of facilities that have signed agreements on the mutual use of evacuation sites, location maps, photographs, and easy-to-understand evacuation routes should be made available on local governments' websites.

Table 4.4.3 Technical Study Material 3: Development of Evacuation Facilities

Project Name	**Development of Evacuation Facilities**
Overview	• Identify areas with high risk of tsunami and flooding where evacuation is difficult, and develop tower-type evacuation facilities to serve as safe shelters for individuals with limited evacuation options. • Promote the development of elevated evacuation sites and designate buildings suitable for vertical evacuation to higher ground in the event of tsunami and flooding. • Subsidies are provided to private organizations that are engaged in the development of evacuation buildings within the budget of local governments. • For the purpose of promoting disaster prevention activities in the region, disaster prevention community development base will be developed. This facility will function as a base for disaster prevention related volunteer activities in ordinary time, as a place for disaster prevention education, awareness raising, and disaster prevention training, and as a base for evacuation and emergency response activities in times of disaster.
Main Points of the Project	• Development of tsunami evacuation facilities utilizing road space: In order to reduce the time and cost required for land acquisition and construction, evacuation facilities will be developed as a shared structure integrated with structures in road space (e.g., pedestrian bridges). • Formulation of design criteria for evacuation facilities by committee method: It is difficult for local governments to plan and design evacuation facilities that are structurally strong against natural hazards such as earthquakes, tsunamis, heavy rain, flooding, and landslides. Therefore, a committee on design criteria for evacuation facilities will be established, consisting of local government officials, experts, central government agencies, and officials from relevant departments of Division/District/Tehsil. • Designation of evacuation facilities based on community level evacuation plans: Evacuation plans is formulated at the community level , and facilities are designated accordingly to ensure effective and accessible evacuation during emergencies.. • Support for the development of evacuation facilities in the community: Financial support will be provided for the development of buildings designated as evacuation facilities in the target areas of evacuation drills. Examples of financial support include subsidies for the costs of exterior stairs, handrail, evacuation signs, emergency lights that can be turned on in case of power outages, etc., at designated facilities. • The artificial mountain will be developed as a space with a park function. This will make the facilities known to residents during normal times, so that residents can evacuate smoothly in the event of disasters. It is also expected to be used for disaster prevention education for children. • Evacuation facilities and structures need to be updated due to aging. Although artificial mountain structures require regular maintenance, they do not require maintenance like large-scale repair of high-rise buildings, and thus it contributes to long-term safe and secure community building.
Reference Image	 Example of tsunami evacuation tower Effective area/capacity: 426 m^2/800 persons Deck height (assumed inundation depth): 8.1 m (3.0 m) *Source: Shizuoka Prefecture Yoshida Town website* Example of artificial mountain construction Site area/capacity: 6,433m^2/1,300 persons Total floor area: 1,300m^2, 7.2 m above ground level (10 m above sea level) *Source: Website of Fukuroi City, Shizuoka Prefecture*

Table 4.4.4 Technical Study Material 4: Development of Evacuation Routes

Project Name	**Development of Evacuation Routes**
Overview	• Development of evacuation routes, including the installation of evacuation stairs leading to higher ground and other necessary infrastructure improvements, etc.) • Development of disaster-resistant roads, including road widening, e.g. (upgrading existing roads to a width of 6 m wide road) to enhance accessibility and emergency response during disasters. • The local government formulates an action plan for the "Community Based Disaster Management Plan." Based on this action plan, evacuation routes are improved and the local government also provides subsidies to support the community led construction of evacuation stairs. • In dense areas with narrow road width, road blockage due to collapse of buildings in the event of disasters is problematic. The road is widened to facilitate evacuation, rescue, relief, and recovery and reconstruction activities.
Main Points of the Project	• Development of tsunami evacuation facilities utilizing road space: In order to reduce the time and cost required for land acquisition and construction, evacuation facilities will be developed as a shared structure integrated with structures in road space (e.g., pedestrian bridges). • Formulation of design criteria for evacuation facilities by committee method: It is difficult for local governments to plan and design evacuation facilities that are structurally strong against external forces such as earthquakes, tsunamis, heavy rain, flooding, and landslides. Therefore, a committee on design criteria for evacuation facilities will be established, consisting of local government officials, experts, central government departments, and officials from relevant departments of Division/District/Tehsil. • Designation of evacuation facilities based on community evacuation plans: An evacuation plan is formulated at the community level. Based on this plan, evacuation facilities will be designated.
Reference Image	 Image of development of evacuation routes (Left: Evacuation stairs to higher ground, Right: Road widening) *Source: Guidelines for Building Resilient Cities against Earthquakes and Tsunamis, the Ministry of Land, Infrastructure, Transport and Tourism Chubu Regional Development Bureau*

Table 4.4.5 Technical Study Material 5: Designation of Disaster Risk Areas by Local Government Ordinance

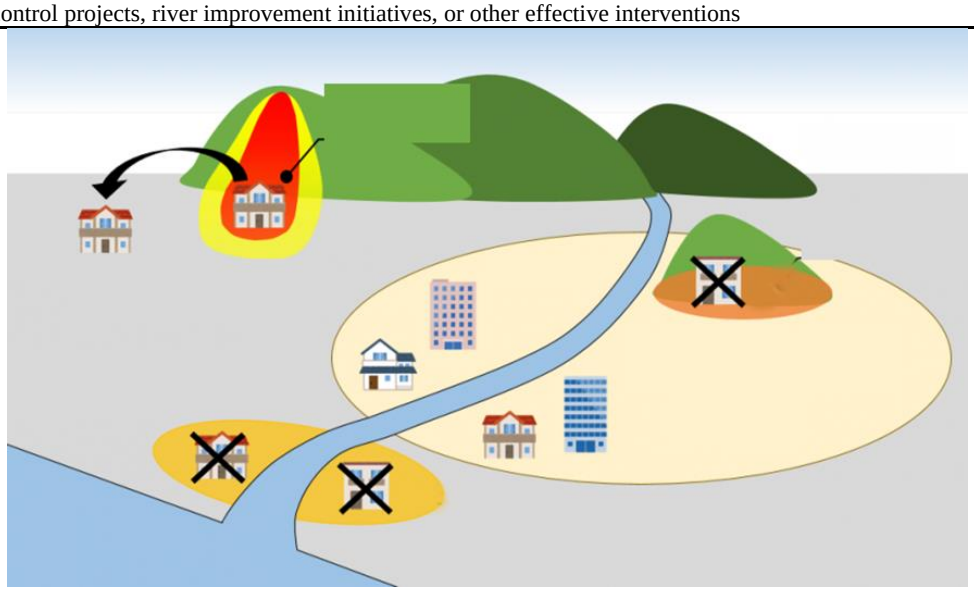
Project Name	**Designation of Disaster Risk Areas by Local Government Ordinance**
Overview	• Local government ordinances designates areas with a particularly high risk of tsunamis, floods, and landslides as "Disaster Risk Areas." In these hazard zones, the construction of residential buildings is restricted. Other land use and building and development activities are regulated. Disaster risk areas include: o Areas identified by the governor as vulnerable to landslide disasters o In addition to the above, areas designated by the governor where there is a risk of danger due to tsunamis, storm surges, floods, etc. • The construction of residential buildings is prohibited in disaster risk areas. However, this does not apply if the buildings are reinforced or disaster mitigation measures are implemented and the governor judges that there is no risk of damage due to disasters, etc.
Main Points of the Project	• Handling of existing buildings in disaster risk areas: The use and residence of existing buildings existing before the enactment of the ordinance is not prohibited. • Disaster Risk Area designations can be lifted if the hazards are mitigated through measures such as erosion control projects, river improvement initiatives, or other effective interventions
Reference Image	 Image of designation of disaster risk areas by local government ordinance *Source: Strengthening regulations in areas with high risk of disasters, Sendai City, Miyagi Prefecture website*

Table 4.4.6 Technical Study Material 6: Detailed Risk Assessment on a Spatial Basis (e.g., by Block)

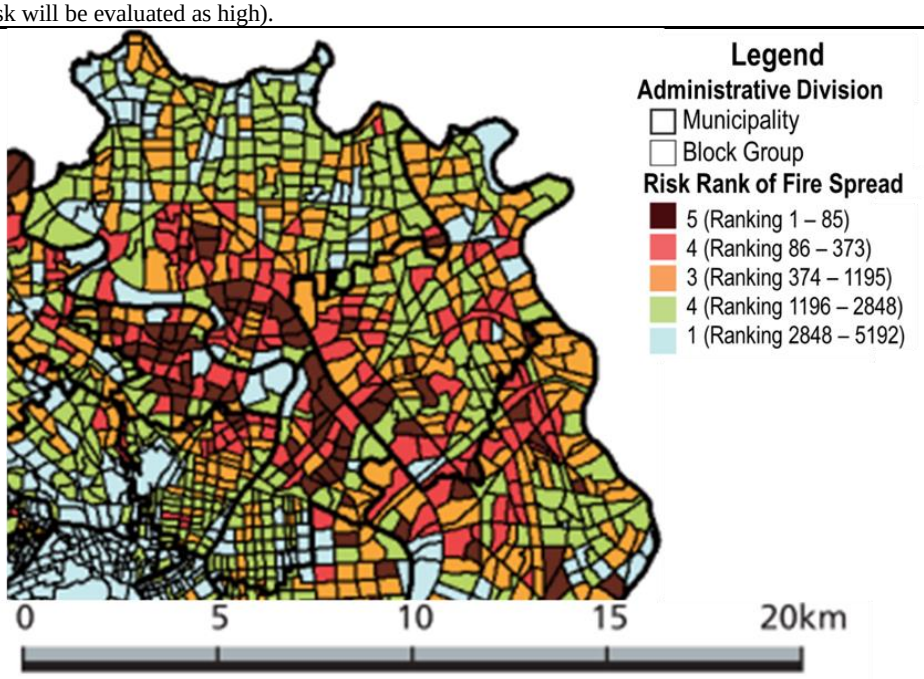
Project Name	**Detailed Risk Assessment on a Spatial Basis (e.g., by Block)**
Overview	• Disaster risk at the block level is assessed, and disaster-related issues in residents' living environments are identified. • Example of evaluation criteria: o Building collapse risk: Assessed based on building structure types (wooden, concrete, steel, block, etc.), age of building, characteristics of ground shaking, liquefaction, and risks associated with large-scale embankment developments. o Fire risk: Assessed considering building structure types (wooden, concrete, steel, block, etc.), road width, open space, etc. o Difficulty of activities during disasters: Evaluated by the shortage of usable activity spaces such as roads, parks, and open spaces during disasters, the effectiveness of evacuation road networks, the development status of fire and water supply, the probability of road blockage calculated from road width and earthquake resistance of adjacent buildings, and the development status of evacuation facilities.
Main Points of the Project	• Difference between "risk assessment" and "damage assumption": "damage assumption" assumes a specific disaster (Example: epicenter, scale, time of occurrence, etc.). Therefore, the impact of shaking and the scale of damage are affected by the distance from the assumed epicenter. On the other hand, "risk assessment" evaluates the relative vulnerability of each city section to a disaster at the city block level. This differs from damage assumption in that it does not assume a specific earthquake disaster but instead evaluates the risk of the same earthquake occurring in all city blocks. • This risk assessment is a relative assessment. Therefore, if the safety of one city block is improved due to earthquake-proofing of buildings or widening of roads, the evaluation results of other city blocks will decrease (the risk will be evaluated as high).
Reference Image	 Example of risk assessment results for each city block *Source: Regional risk measurement survey on earthquakes - Tokyo Metropolitan Government Bureau of Urban Development, 2022*

Table 4.4.7 Technical Study Material 7: Development of Regional Disaster Management Centers

Project Name	**Development of Regional Disaster Management Centers**
Overview	Regional disaster centers: include designated open spaces, green areas, regional disaster management centers, and parks that serve as evacuation centers and emergency shelters, etc. • In order to improve urban structures that are vulnerable to disasters, parks are developed as evacuation sites based on disaster management plans prepared by the local governments, such as the Regional Disaster Management Plan and Urban Development Plan. • Parks are designated as official evacuation sites in the municipal city Master Plan. Projects that contribute to disaster prevention and mitigation, such as the development of access roads to these evacuation sites and constructions of bund to enhance the safety of evacuation sites, are strategically coordinated.
Main Points of the Project	• Roads and uplands will be integrated. In the event of a flood, a multifunctional disaster prevention base will be established, utilizing the road network and major roads located on the opposite side of the river. This will ensure disaster prevention functions that can be accessed from uplands even in the event of flooding. • Small-scale parks at the community level will be established to function as evacuation sites, food distribution centers, recovery centers for necessary services, and information provision centers. Large-scale parks will also be established with parking lots and open spaces that serve as bases for gathering and distributing necessary goods and supporting activities. • Parks will be equipped with storage warehouses, earthquake-resistant water tanks, broadcasting facilities, information and communication facilities, heliports, power generation facilities, and water discharge facilities to ensure disaster prevention functions.
Reference Image	 Image of the development of regional disaster prevention centers *Source: Ministry of Land, Infrastructure, Transport and Tourism Chubu Regional Development Bureau website*

Table 4.4.8 Technical Study Material 8: Formulation of Guidelines for "Disaster Mitigation through Community Development"

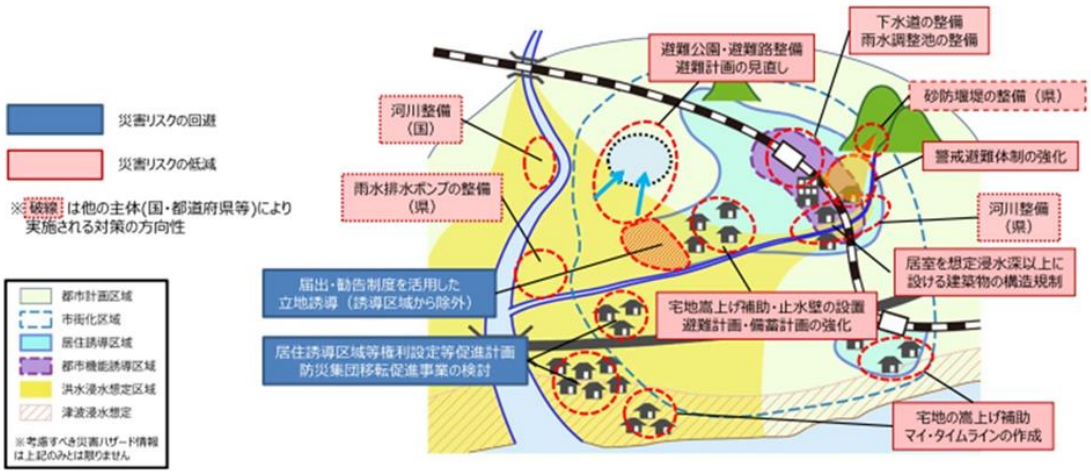
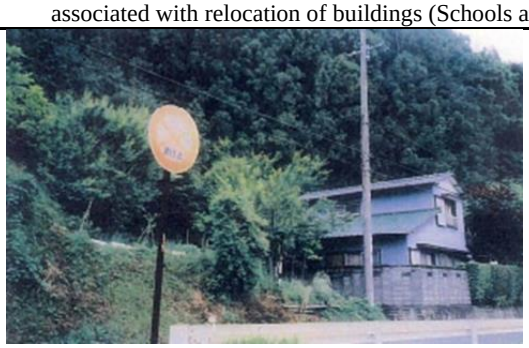
Project Name	**Formulation of Guidelines for Disaster Mitigation through Community Development**
Overview	The federal and provincial governments prepare guidelines for disaster mitigation through community development. Based on these guidelines, local governments incorporate the content of disaster mitigation through community development into their Development Master Plan (Examples: Karachi Strategic Development Plan 20XX, Lahore Master Plan 20XX, etc.). By doing so, they aim to realize and promote disaster prevention projects.
Main Points of the Project	Example of guidelines for disaster mitigation through community development 1. Disaster risk analysis and extraction of challenges for disaster mitigation through community development a. Collect and organize information on topography, geology, and frequency and scale of past flood disasters, including data on inundation depth, duration of inundation, inundation, tsunami, storm surge, landslide, liquefaction, and flooding, etc. b. Extracting areas with high disaster risk - Setting the perspective of risk assessment: For example, the relationship between inundation depth and human damage, duration of inundation, fluid force and damage to houses, etc. - Extracting issues through community-level analysis (micro analysis): Overlaying multiple pieces of information such as inundation depth × building distribution × building height, inundation depth × inundation arrival time × urban function, inundation arrival time × road network, etc. to extract issues. - Quantitative assessment of disaster risk: Population in the expected inundation area, estimated value of property damage due to flood damage, extension of major roads that will be interrupted due to disaster, etc. c. Organizing disaster management issues for each district 2. Examining future vision and policy for disaster mitigation through community development 3. Examining concrete initiatives, schedules, and target values a. Examination of specific hardware and software initiatives based on disaster management guidelines b. Examination of initiative schedules and target values c. Utilization of systems related to disaster mitigation through community development
Reference Image	 Description of specific initiatives for disaster prevention community development in Development Master Plan *Source: Review of Disaster Prevention Guidelines, Guide for Preparation of Site Optimization Plans (revised March 2023), the Ministry of Land, Infrastructure, Transport and Tourism*

Table 4.4.9 Technical Study Material 9: Promotion of Relocation of Housing in High-Risk Areas

Project Name	**Promotion of Relocation of Housing in High-Risk Areas**
Overview	Promote the relocation of existing buildings identified as posing a high risk to the lives and safety of residents, particularly in areas classified as high disaster risk zones.
Main Points of the Project	Promotion of relocation from disaster risk areas through the following support system. • Subsidy through projects to promote safe housing construction: The national and provincial governments subsidize the cost of demolishing houses in disaster risk areas. In addition, nterest rate subsidies are provided by the national government for loans taken from financial institutions to rebuild homes using disaster-resilient designs in safer locations.. • Loans from the Housing Finance Organization: Loans are provided to support housing relocation based on official relocation recommendations, including the construction of alternative housing, and acquisition of land. • Tax incentives to encourage relocation from disaster risk areas: A certain percentage of real estate acquisition tax is deducted when new houses or land are acquired outside disaster risk areas. • Financing by public policy financial institutions: Financing for land acquisition and construction costs for disaster prevention measures related to development, Financing for land acquisition and construction costs associated with relocation of buildings (Schools and medical facilities)
Reference Image	 (Before relocation) Houses at risk of landslide disasters (After relocation) Buildings in the red frame have been removed *Source: Outline of the Sediment Disaster Prevention Act, the Ministry of Land, Infrastructure, Transport and Tourism*

4.4.1.4 EWS

(1) EWS Operated by the PMD

Table 4.4.10 is reproduced in types of EWS and their overview.

Table 4.4.10 Types of EWS and Their Overview.

Types of EWS	Overview
Numerical Weather Forecast System	**Numerical Weather Prediction System** Weather prediction is carried out using numerical weather prediction model with the weather observation network owned by PMD as input condition. This system also includes the following two systems. ・Heavy Rainfall Warning System ・Heat Waves Early Warning System
Flood Early Warning System	**Flood Early Warning System** The river discharge forecast are generated using the runoff model, river channel model, empirical formula, etc. as input condition of the rainfall of the weather forecast.
Earthquake & Tsunami Early Warning System	**Earthquake and Tsunami Early Warning System** Earthquake monitoring is carried out. Tsunami prediction (simulation) is also carried out in event of earthquake.
Cyclone Early Warning System	**Cyclone Early Warning System** Tropical Cyclone Warning Center (TCWC), Karachi at PMD issues cyclone warnings through PMD's website.
GLOF Early Warning System	**Glacial Lake Outburst Flood (GLOF) Early Warning System** Alerts for potential glacial lakes outburst are issued based on data regarding glacial lakes and temperature information, targetting areas at risk.
Drought Monitoring & Early Warning System	**Drought Monitoring and Early Warning System** National Drought Monitoring Center, Islamabad in PMD monitors drought situations and issues bulletins and recommendations.

Source: Prepared by the project team based on the PMD introduction presentation

(2) Proposals for EWS improvement

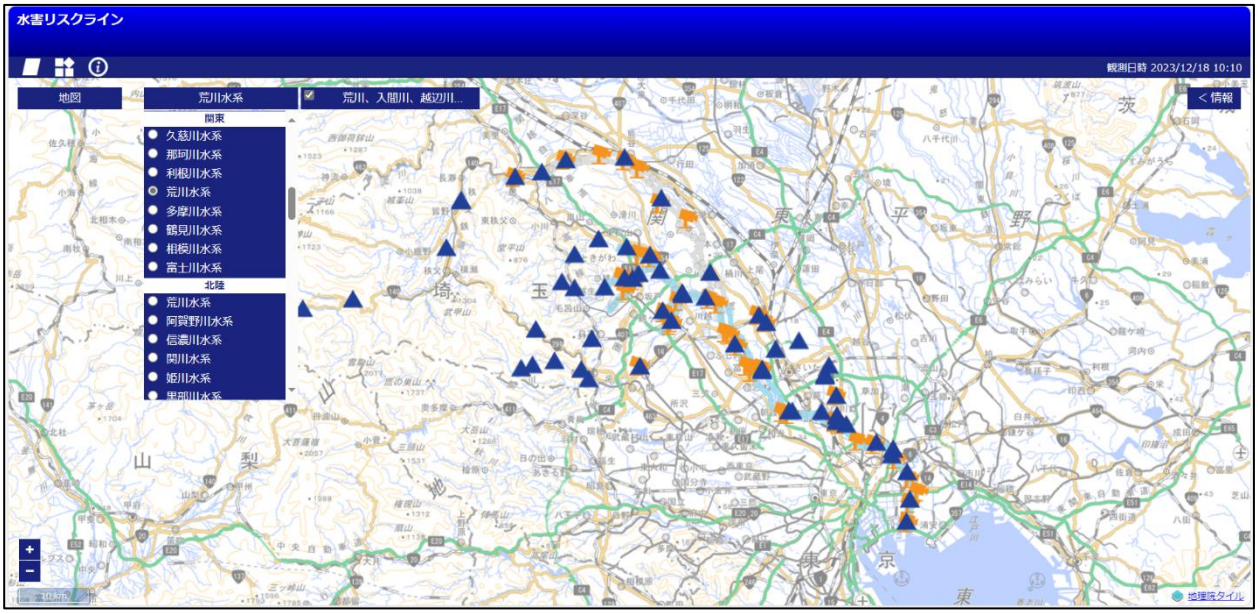
Japanese technologies for improving EWS are introduced. It is recommended to share these technologies with PMD technical staff in a dedicated meeting and explore the feasibility of their adoption in Pakistan.

1) Flood Risk Line

Since 2020, the Ministry of Land, Infrastructure, Transport and Tourism has been providing flood risk distributions (named as “Flood Risk Line”) for each 200 m on the left and right banks by calculating the longitudinal water level (water surface shape), in addition to flood forecasts covering areas ranging from several km to several tens of kilometers for national rivers. The estimated water level at approximately every 200 m exceeds the height of bund, etc., and areas where there is a possibility of flooding are displayed in black (information corresponding to alert level 5). This means that the degree of danger in each area can be grasped in detail and can be used as a reference for issuing evacuation information.

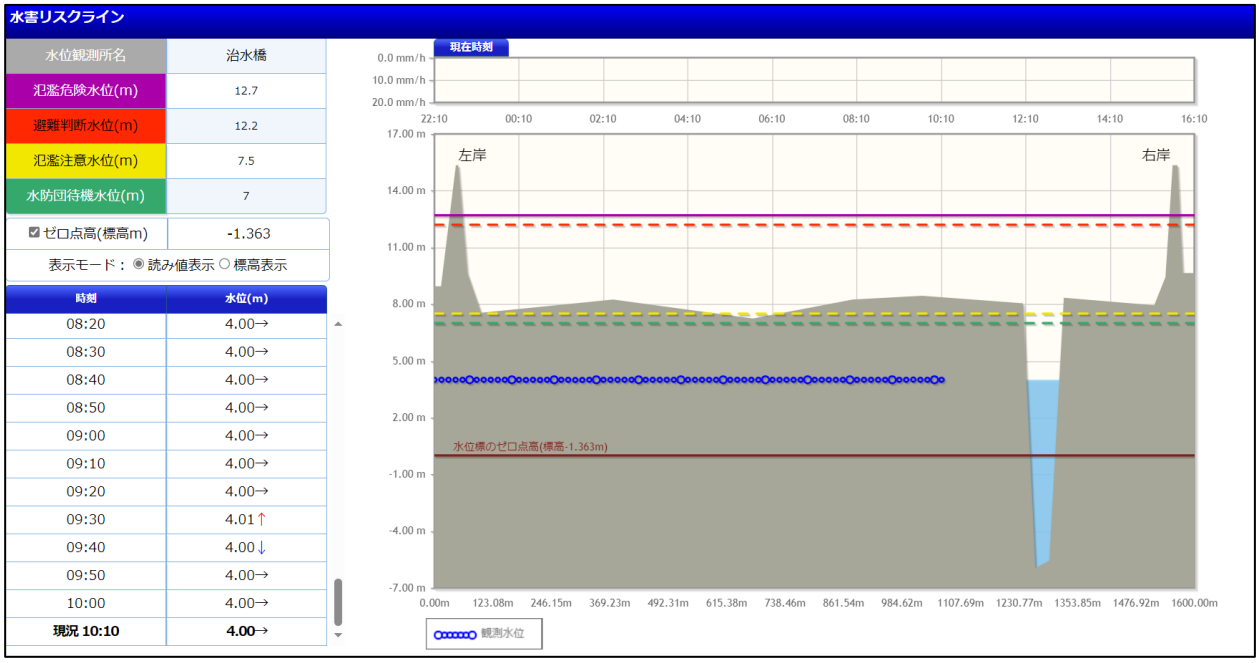
Figure 4.4.9 shows the front screen of the Flood Risk Line. In this screen, the flood risk level can be checked and the observation station and camera to be displayed can be selected. Figure 4.4.10 shows the water level observation station screen, and Figure 4.4.11 shows the river camera screen. General users can check the history of rainfall and water level up to the current time. River managers can check the forecast for up to 6 hours ahead. Figure 4.4.12 shows the forecast and actual results of the risk distribution for the flood that occurred on July 10, 2023. The forecast and actual results were generally consistent, and it can be said that the risk of rivers several hours ahead could be predicted.

The flood information can be shared more effectively by reflecting the parts that can be referenced in the Flood Risk Line into the Flood Early Warning System in Pakistan.



Source: Flood Risk Line (https://frl.river.go.jp/)

Figure 4.4.9 Flood Risk Line (Top Screen)

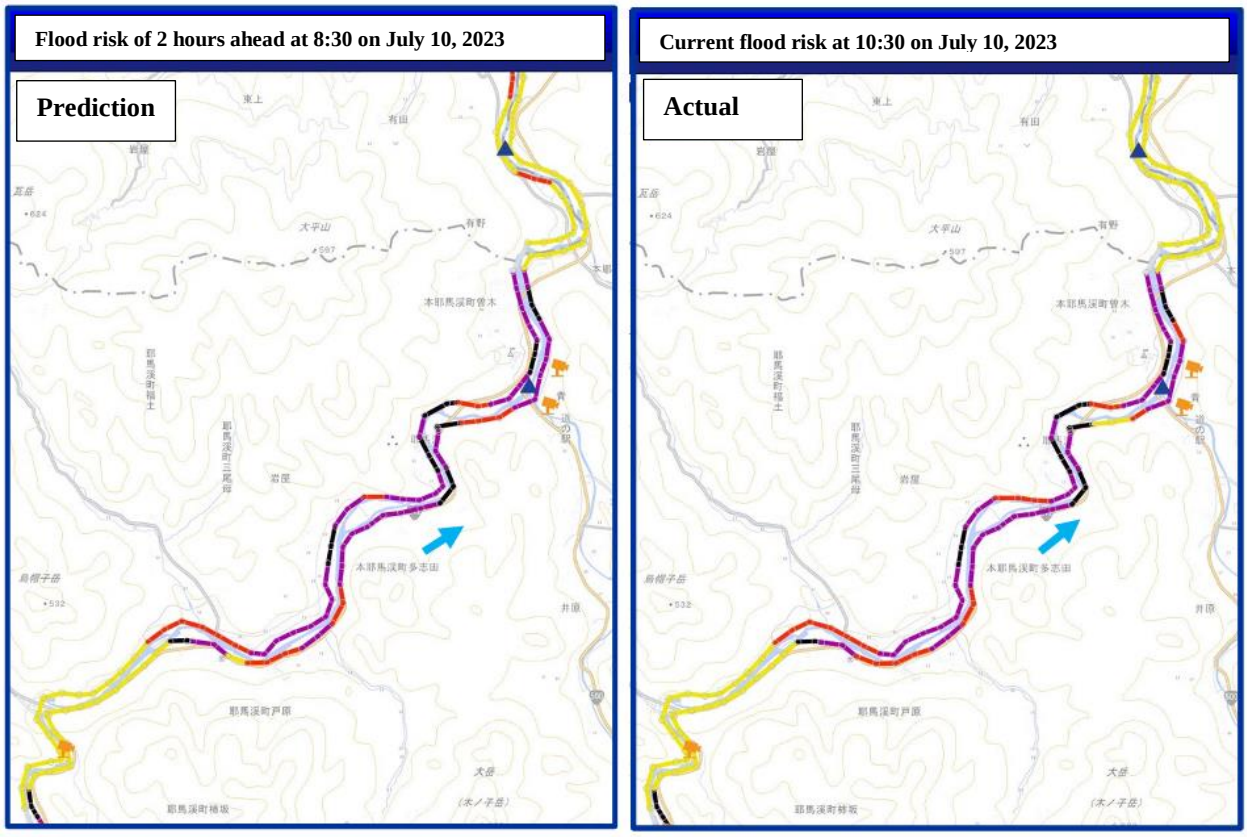


Source: Flood Risk Line (https://frl.river.go.jp/)

Figure 4.4.10 Flood Risk Line (Water Level Observation Station Screen)

Source: Flood Risk Line (https://frl.river.go.jp/)

Figure 4.4.11 Flood Risk Line (River Camera Screen)



Source: Training Materials by Foundation of River & basin Integrated Communications

Figure 4.4.12 Flood Risk Line (Comparison between Forecast and Actual)

2) Utilization of River Cameras and AI (River Water Level Determination)

By installing river cameras in high-risk flood areas near homes and critical facilities, realistic flood conditions can be shared with residents as images alongside traditional water level data, helping them make informed evacuation decisions. Posting the images on the Flood Early Warning System can also contribute to reducing human damage in the event of flooding in Pakistan.

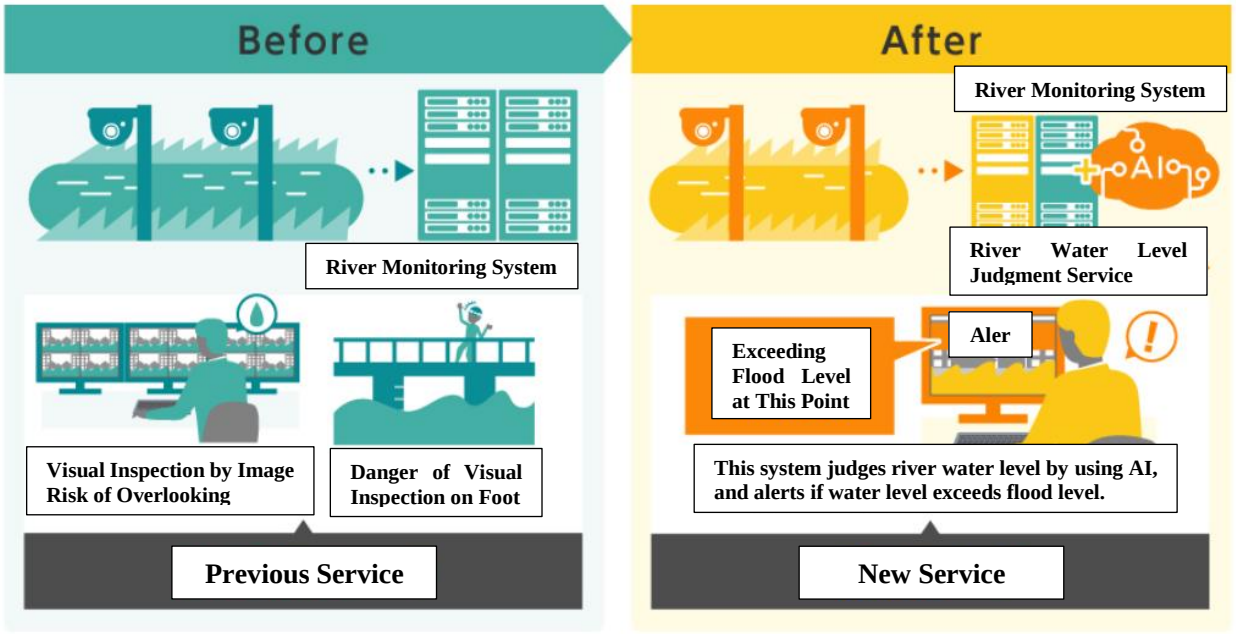
As shown in Table 4.4.11, there are simplified river monitoring cameras as well as CCTV cameras. Although simplified river monitoring cameras are still images and do not rotate, they are cheaper than CCTV cameras, so they are suitable for introduction in Pakistan.

Furthermore, technology exists that can monitor river water levels from river camera images. Figure 4.4.13 shows an image of a river water level determination service developed by Hitachi Solutions Create Co., Ltd. Developing an AI specialized in determining river water level can reduce the workload of river water level monitoring staff, enhance their safety, and facilitate the early evacuation of residents in the surrounding areas.

Table 4.4.11 Comparison of Simplified River Monitoring Cameras and CCTV Cameras

Item	Simplified River Monitoring Camera	CCTV Camera
Image		
Quality	Full HD Quality (1920×1080)	Full HD Quality (1920×1080)
Shooting range	Fixed camera (No turning)	Turning camera (With turning)
Shooting range	Solar panels, etc.	Commercial power
Field	Horizontal 180° Vertical 117°	Horizontal 62.9° to 2.2° Vertical 37.0° to 1.3°
Transmission method	Operator lines, etc.	Optical, etc.
Minimum subject degree	0.002 lux	0.0002 lux
Construction cost	About 1 million JPY	About 8 million JPY

Source: Training Materials by Foundation of River & basin Integrated Communications



Source: Hitachi Solutions Create, Inc.
(https://www.hitachi-solutions-create.co.jp/company/newsrelease/2023/0112_01/)

Figure 4.4.13 Utilization Image of River Water Level Determination Service

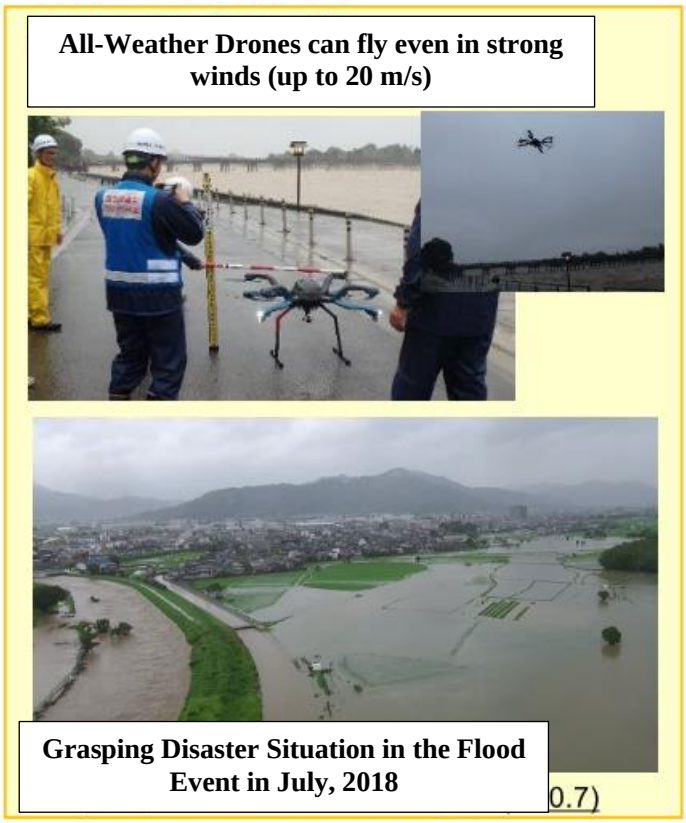
3) Grasping Disaster Situation Using All-Weather Drones and Satellite Images with AI Technology

When a disaster occurs, it is important to quickly grasp the disaster situation. Two methods are introduced for grasping disaster situation: all-weather drones and satellite imagery with AI technology.

Figure 4.4.14 shows an image of grasping disaster situation using all-weather drones. Since all-weather drones can fly under strong winds of 20 m/s, they can quickly grasp the disaster situation without waiting for the weather to recover even in the event of a flood. (Helicopters are difficult to fly at wind speeds of several m/s).

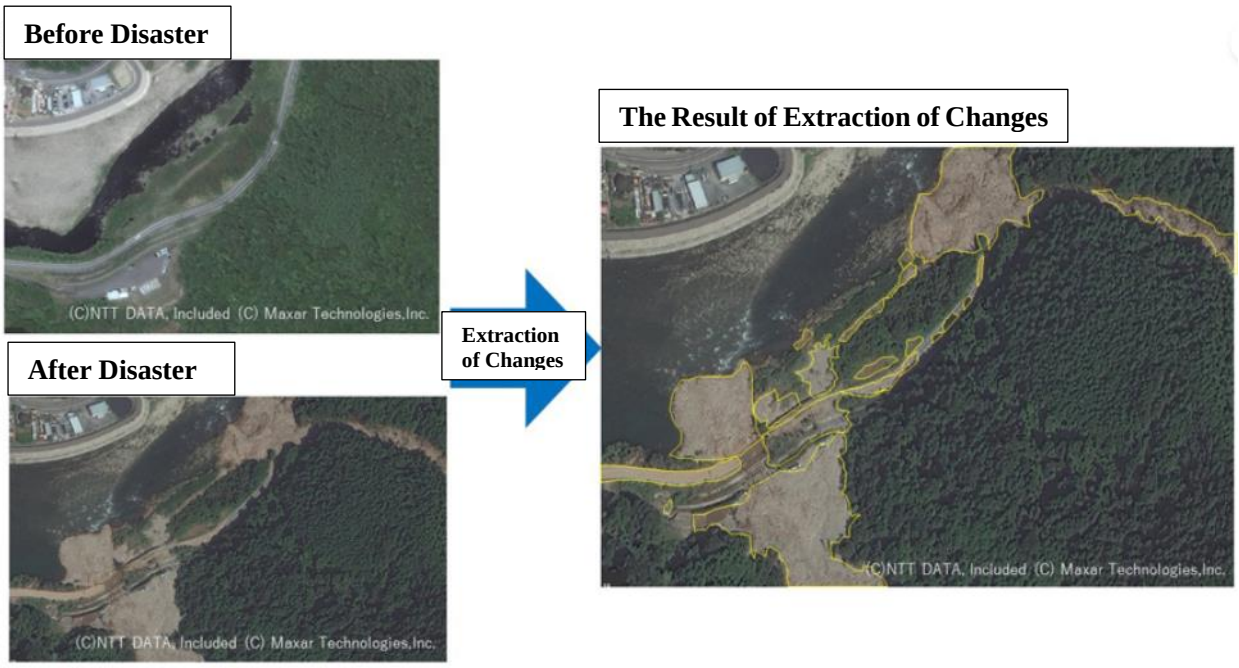
Figure 4.4.15 shows an image of grasping disaster situation using satellite images with AI technology. NTT DATA has developed technology to identify disaster areas in as little as half a day by using AI technology to compare satellite images before and after the disaster.

By combining the above two technologies, it is possible to quickly collect both the overall picture and details of the disaster situation and to plan and implement disaster countermeasures at an early stage.



Source: Training Materials by Foundation of River & basin Integrated Communications

Figure 4.4.14 Grasping Disaster Situation by All-Weather Drones



Source: NTT DATA (https://www.nttdata.com/global/ja/news/services_info/2020/102701/)

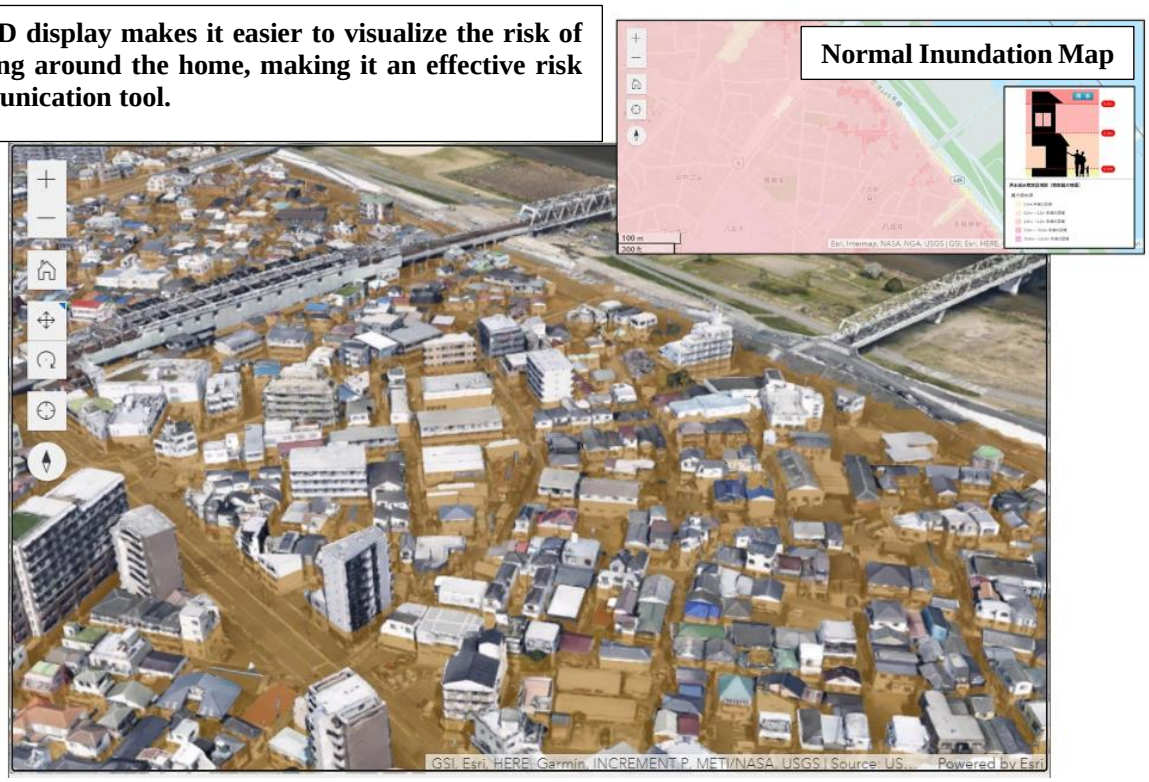
Figure 4.4.15 Grasping Damage Situation Using Satellite Images and AI Technology

4) 3D Display of Inundation Risk

It is very important to inform residents about the risk of flood damage in order to minimize human causalities. However, a key challenge is that information such as river water levels and discharge is difficult for the general public to understand, except for those with technical expertise.

As an example of a solution to this problem, Figure 4.4.16 shows a 3D display of flood risk (3D flood inundation area map of Arakawa Ward). Displaying flood risk in 3Dallows for a more intuitive and accessible way to convey the risk of flood damage more easily. For example, by making this information available online or by incorporating it into for evacuation drills, it is expected to enhance residents' understanding and preparedness for flood risks.

The 3D display makes it easier to visualize the risk of flooding around the home, making it an effective risk communication tool.



Source: 3D Flood Inundation Area Map of Arakawa Ward
(https://experience.arcgis.com/experience/a14b9a7cee8943889babc2096f5a5fe7/page/page_0/?views=view_1)

Figure 4.4.16 3D Display of Flood Risk (3D Flood Inundation Area Map of Arakawa Ward)

5) Utilization of Sediment Disaster and Inundation Sensors

Because of the increasing frequency of heavy rainfall caused by climate change and the effect of urbanization, the risk of flood disasters and landslides is rising. By installing landslides and inundation sensors in high risk areas and monitoring the situation in real time, EWS can be established to response to landslides and urban floods, including inland flooding.

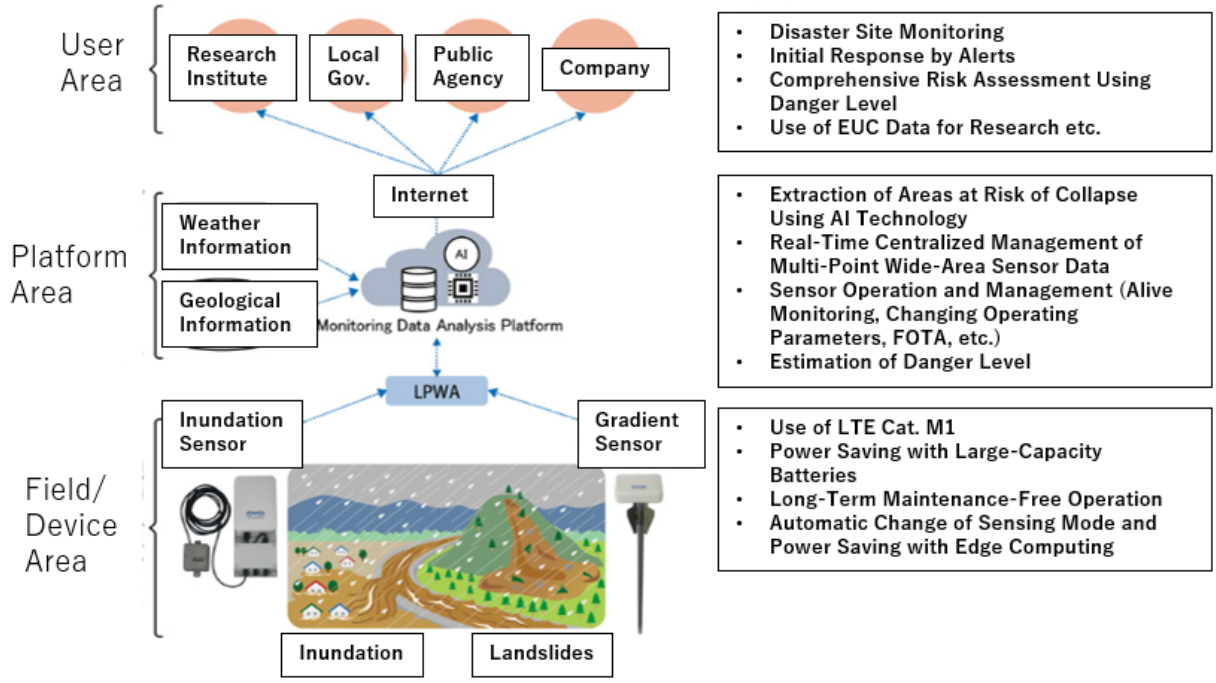
As an example, the system of hazard mapping sensor solution by OYO Corporation is introduced. It consists of a multi-point disaster prevention sensor using edge computing, LPWA, IoT/cloud (see Figure 4.4.17 and Figure 4.4.18), and a sensor location extraction function using AI technology that incorporates knowledge from geotechnical investigation and disaster prevention and mitigation projects. It enables multi-point, planar, wide-area, real-time monitoring with reduced operating costs and human burdens. An overview of this system is shown in Figure 4.4.19.

Source: Climate Change Adaptation Information Platform A-PLAT, OYO Corporation (https://adaptation-platform.nies.go.jp/private_sector/database/opportunities/report_079.html)

Figure 4.4.17 Sediment Disaster Sensor

Source: Climate Change Adaptation Information Platform A-PLAT, OYO Corporation (https://adaptation-platform.nies.go.jp/private_sector/database/opportunities/report_079.html)

Figure 4.4.18 Inundation Sensor and Ballard



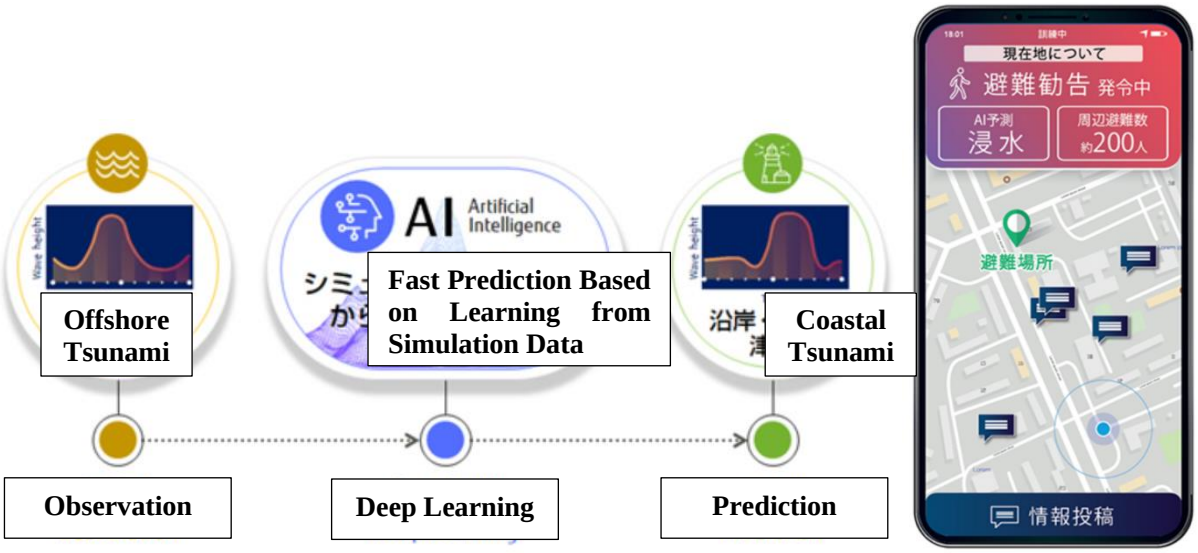
Source: Climate Change Adaptation Information Platform A-PLAT, OYO Corporation
(https://adaptation-platform.nies.go.jp/private_sector/database/opportunities/report_079.html)

Figure 4.4.19 System Overview of Hazard Mapping Sensor Solution

6) Application for Determining Tsunami Inundation Potential

If the potential of tsunami inundation potential can be assessed in advance and evacuation measures can be implemented promptly following an earthquake, it can significantly contribute to reduce human causalities.

As an example, the inundation possibility judgment application developed by Fujitsu is introduced in Figure 4.4.20. In addition to tsunami warnings and evacuation advisories based on hazard maps, this application displays the possibility of inundation with the current location based on AI predictions. By providing a map from the current location to the evacuation site, along with AI based assessment of inundation risk and real time data on the number of people who completed evacuation at nearby shelters, the system thereby supports informed evacuation decision-making. It also provides a function for evacuees to post and share information about difficult points along the evacuation route, which was confirmed to be effective in a demonstration test conducted in December 2018.



Source: Fujitsu Corporation (https://pr.fujitsu.com/jp/news/2019/10/24.html)

Figure 4.4.20 Image of the Inundation Possibility Judgment Application and Its Screen

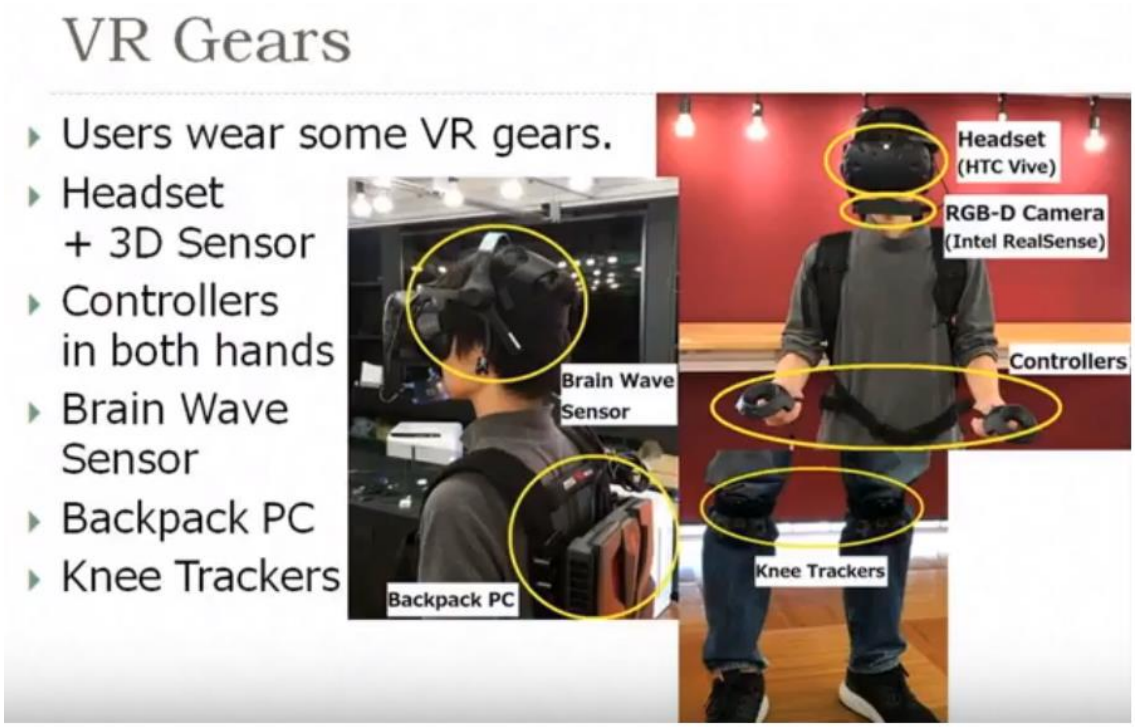
7) Virtual Evacuation Training

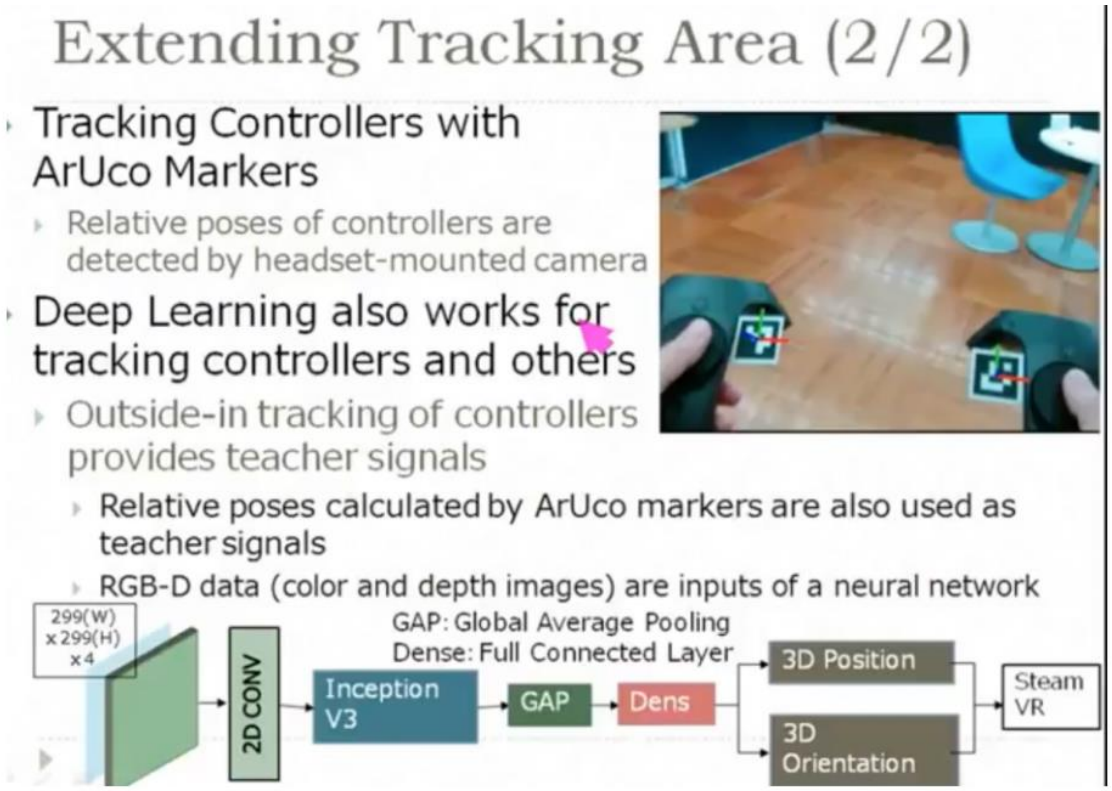
Virtual evacuation training using VR systems is also becoming increasingly practical. Microsoft Research has developed “Building Scale VR”, a disaster simulation system that enables users to experience disasters scenarios with real buildings using VR systems. Figure 4.4.21 shows an image of this disaster simulation system.

By implementing disaster simulation using such VR systems, it is possible to learn in advance what to do in case of any disaster, which can lead to strengthening of disaster response by both government officials and residents.

Source: Seamless article https://shiropen.com/2018/01/16/31544/

Figure 4.4.21(1) Image of Disaster Simulation System





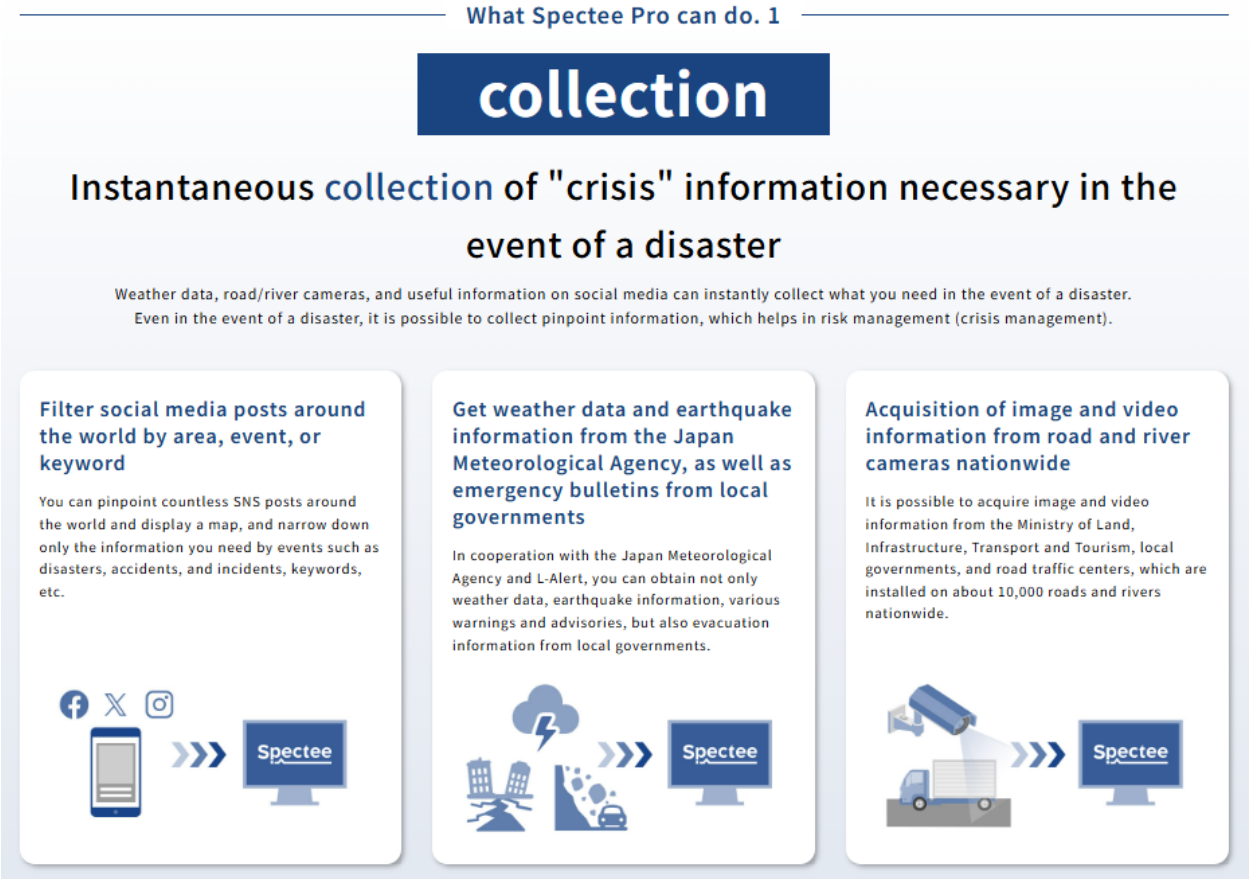
Source: Seamless article https://shiropen.com/2018/01/16/31544/

Figure 4.4.21(2) Image of Disaster Simulation System

8) AI-Powered Disaster Information Collection Platform

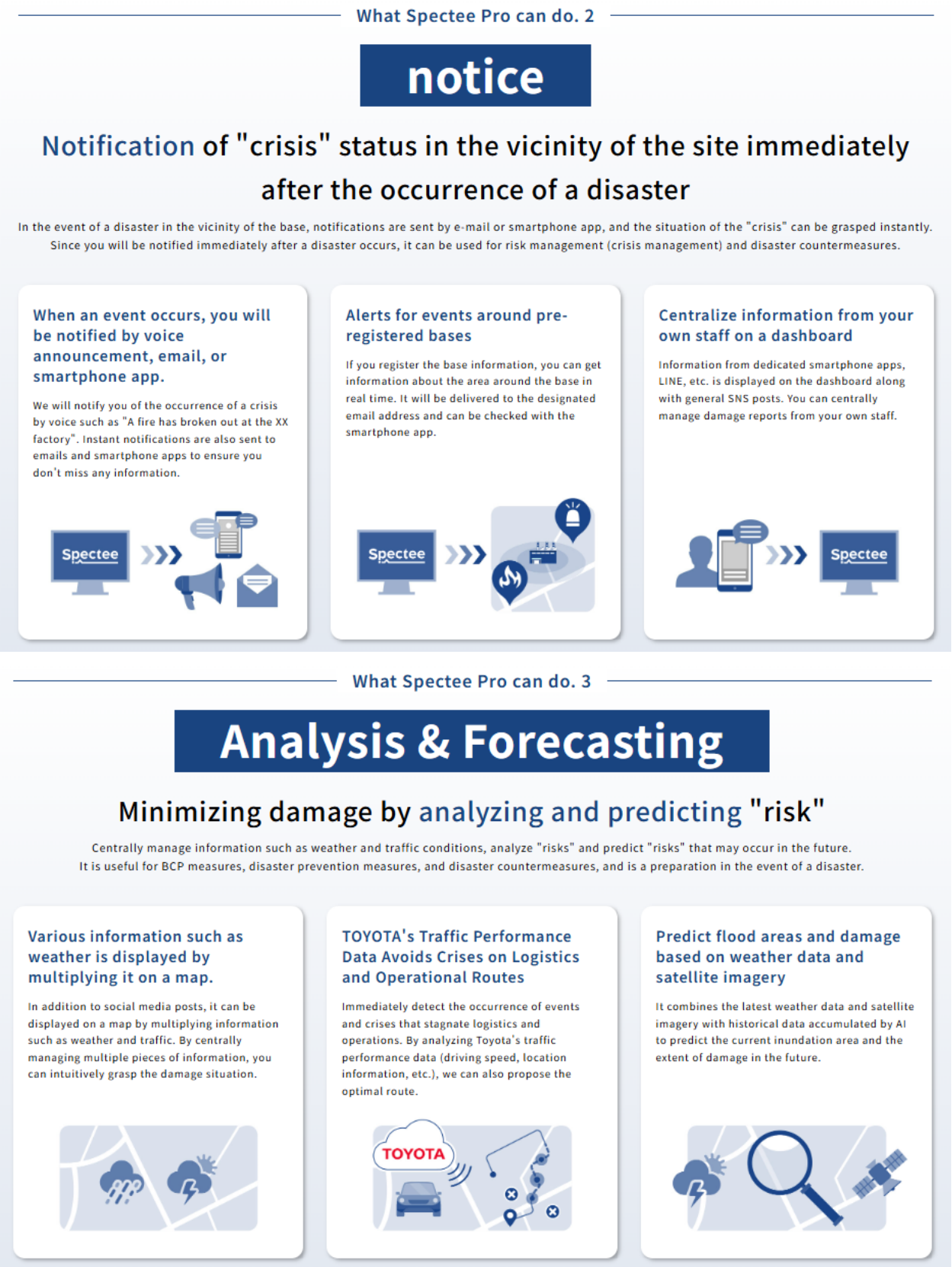
As an example of disaster information collection using AI, a platform operated by Spectee Corporation which is named “Spectee Pro” is introduced shown in Figure 4.4.22.

This platform uses AI to collect, disseminate, analyze, and predict disaster related information. It gathers data not only from government sources but also from nationwide road and river cameras, as well as social media posts. Furthermore, the obtained information is used for disaster response by using alerts, emails, mobile applications, and other communication tools. The collected information is also used for disaster preparedness by analyzing and predicting possible disaster risks.



Source: Spectee Corporation (https://spectee.co.jp/service/spectee/)

Figure 4.4.22(1) AI-Powered Disaster Information Collection Platform



Source: Spectee Corporation (https://spectee.co.jp/service/spectee/)

Figure 4.4.22(2) AI-Powered Disaster Information Collection Platform

CHAPTER 5 ACTIVITIES RELATED TO OUTCOME 4

Outcome 4	Capacity as a central disaster prevention organization through disaster prevention centers is improved.

5.1 Activities 4-1

Knowledge sharing on flood and earthquake related to capacity strengthening of the National Emergency Operation Center (NEOC) is conducted.

5.1.1 Knowledge Sharing in Flood Field

5.1.1.1 Understanding of Challenges in Risk Analysis of NEOC

Prior to the first and second workshops in Pakistan, (described later), the project team visited the office of the NEOC and conducted an interview survey to assess the current status and challenges related to risk analysis of NEOC. Figure 5.1.1 shows protos taken during the visit to the NEOC office.

Table 5.1.1 summarizes the current status and challenges in risk analysis as identified during the interview with NEOC. The following two major issues were highlighted.

- NEOC's in-house analysis capabilities are limited, and they mainly relies on forecasts based on global data.
- There is inadequate coordination with disaster management organizations within Pakistan, leading to challenges in information sharing among these organizations.

Table 5.1.1 Current Status and Challenges in Risk Analysis Based on the Interview with NEOC

Current Status
● NEOC utilizes the National Common Operation Picture (NCOP) Futuristic Portal, a GIS-based system that overlays essential data (geographic information, administrative boundaries, rivers, dams, infrastructure, etc.), hazard maps, and prediction results, to visualize risk analysis results. ● NEOC also manages a mobile application called “Pak NDMA Disaster Alert”, which includes features such as a warning interface (Pak NDMA Alerts), hazard projections, and safety guidelines, enabling effective information sharing during disasters.
Challenges at NEOC
● Although NEOC has systems in place for visualizing risk analysis results and sharing information, the forecast data are displayed using global data such as GLOFAS (Rainfall Probability Prediction) and ECMWF (River Discharge Prediction). These global datasets are difficult to handle as reliable data for local analysis, and they are not necessarily accurate because they are not prepared with detailed topographic conditions around Pakistan. ● During the second workshop in Pakistan, the NEOC staff emphasized the need to strengthen the response capabilities for Flash Floods and Coastal Inundation. In the case of Flash Floods, special attention is needed for Hill Torrents, which now occur almost annually due to climate change. For the Coastal Inundation, NEOC needs to be able to predict cyclones in coastal areas and tsunamis in the event of earthquakes.
Challenges Considered by the Project Team
● In addition to the issues mentioned above, NEOC has limited collaboration with other key organizations in Pakistan. In the event of a disaster, NEOC es expected to coordinate effectively with NDMA (National Disaster Management Authority), FFC (Federal Flood Commission), WAPDA (Water and Power Development Authority), PID (Provincial Irrigation Department), and PMD (Pakistan Meteorological Department). However, at present, information sharing is minimal with organizations other than NDMA.

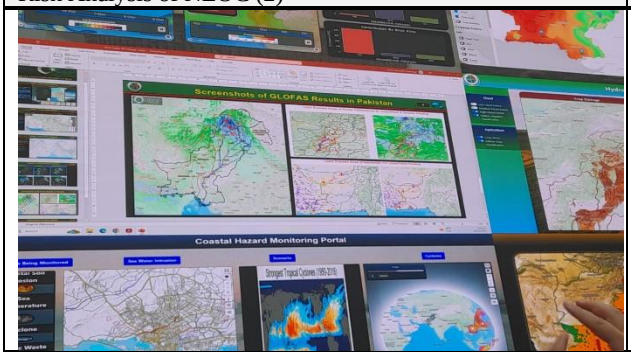
NEOC's Office	Risk Analysis by NEOC (1)
Risk Analysis of NEOC (2)	Meeting Prior to the Workshop
	

Figure 5.1.1 Photos Taken during Visit to the NEOC Office

5.1.1.2 Effects of Workshops and Training and Future Directions

By a series of four-day workshops held in Pakistan, the project team provided NEOC staff with hands-on training in flood risk analysis focusing particularly on the use GIS tools and hydraulic modelling techniques. As a result, NEOC is better equip now to develop its own hydraulic analysis models while prioritizing areas with high flood risk. In addition, NEOC will be able to handle risk assessment under various scenarios, such as hydraulic analysis under the assumption of bund breach and hydraulic analysis under inland flooding.

In addition, through the training conducted in Japan, NEOC staff gained valuable knowledge of the latest Japanese disaster risk reduction technologies, including the use of satellite imagery for disaster information sharing, as well as hands-on experience with hydraulic analysis modeling. Furthermore, from the global trends in disaster risk reduction and the role of Japan's disaster management agencies, it will be possible to learn the importance of disaster management agencies in Pakistan and how to collaborate among organizations. It is expected to improve the collaboration between NEOC and other disaster management organizations in Pakistan, more efficient information sharing and fostering the development of joint initiatives to strengthen disaster preparedness and response across the country.

5.1.2 Knowledge Sharing in Earthquakes Field

5.1.2.1 Understanding of Challenges in Risk Analysis of NEOC

Prior to this Workshop, the Early Warning Unit within NEOC had requested support for technology transfer related to real-time earthquake damage estimation.. Considering this as one of the important issues in earthquakes risk management, in the first Workshop, the project team practiced seismic hazard assessment and assessment of exposed population using GIS. Following the Workshop, the project team installed a complete set of data and calculation tools at NEOC, enabling participants to practice during the WS to review the data and process it immediately after the earthquake event if necessary. The calculation method is a distance attenuation equation based on the setting of a foundation point epicenter and seismic hazard assessment based on the ground amplification factor based on Vs30. The expression of exposed population is based on the census data constructed in the first half of this project. The construction of population data is a task of the National Statistical Office, but it is desirable to disclose (or collaborate with) GIS data with NEOC.

Another issue in the earthquake domain at NEOC is the immediate estimation of building damages. During the second workshop, the project team explained the concepts and structure of damage functions developed in Japan, followed by a practical exercise using damage functions provided by the NED University of Engineering and Technology in Pakistan as a case study. However, this raises the issue of building database. Since damage functions are classified based on structural type of buildings, it was necessary to construct a database of building inventory. Initially, the project team considered showing the necessity of steady work, However, following the suggestion to explore AI applications during the second workshop, the team opted to conduct a practical session on extracting building footprints from satellite imagery using ArcGIS Deep Learning tools. Although this exercise was not a case of Pakistan, it was possible to implement learning with a parameter of Class (applied to classification of building construction, etc.) using ArcGIS tools in the target area, and showed the possibility of using AI for building database construction in Pakistan in the future.

An issue in terms of earthquake observation is the limited strong motion observation network of the Japan Meteorological Agency (JMA). In Japan, since the Hyogo-ken Nanbu Earthquake, the Disaster Management Research Institute and the Japan Meteorological Agency have developed a high-density strong motion observation network, enabling the estimation of areal intensity distributions at the engineering bedrock level. In Pakistan, it is important to gradually strengthen the strong motion observation network. This progressive enhancement will significantly improve the accuracy of real-time earthquake damage estimation at NEOC, particularly through close collaboration with the PMD.

The presentation by NEOC indicates that the NEOC aims to apply not only the immediate damage estimation system but also advance disaster management investment. (See Figure 5.1.2) However, to what extent the DRR & Planning Section in NEOC can strengthen the cooperation with the EW Unit and whether it is possible to make advance disaster management investment in cooperation with other ministries and agencies remains an important issue for the future.

Figure 5.1.3 and Figure 5.1.4 shows some outcome and activities of the first WS.

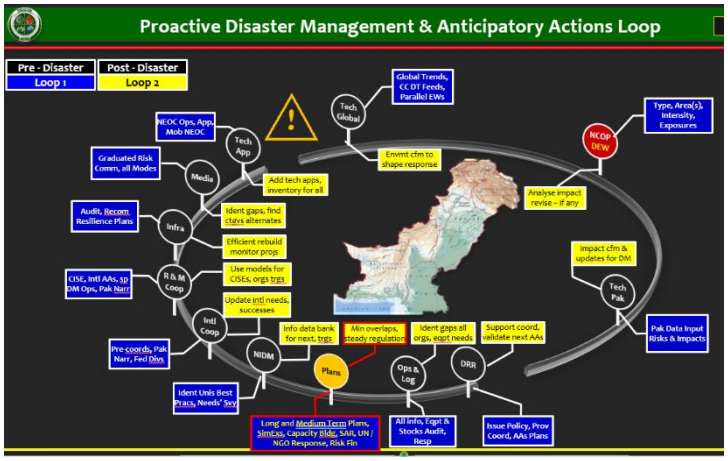


Figure 5.1.2 Materials Presented by the Participants At this Interactive Workshop (Explanation of the Importance of Being Linked by NEOC Organizations from Early Warning to Advance Disaster Management)

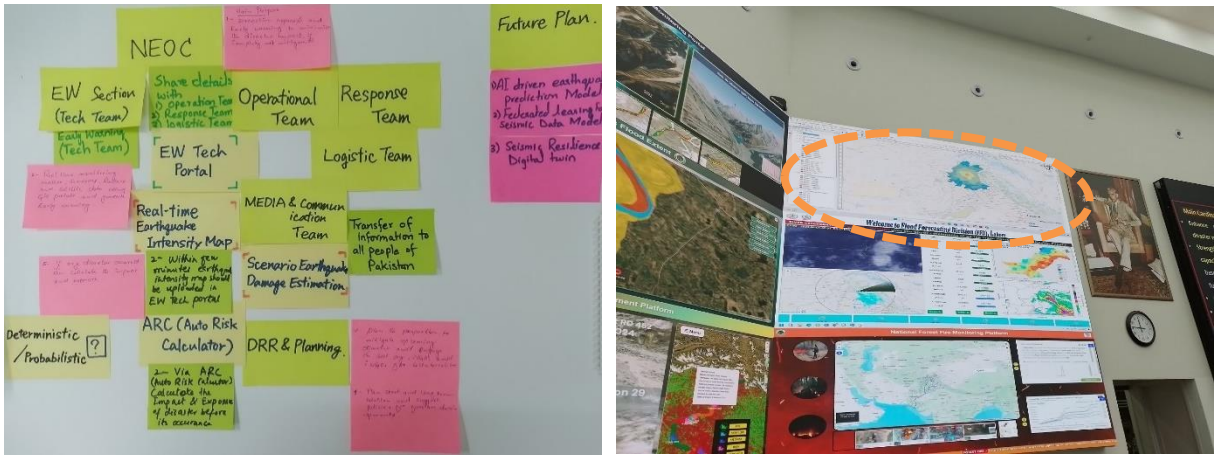


Figure 5.1.3 First WS Outcome (Left: Summary of the Roles and Interrelationships of NEOC Organizations; Right: Example of Earthquake Hazard Assessment and Exposed Population GIS, Which Was Practiced in Workshop Displayed on the NEOC Screen)

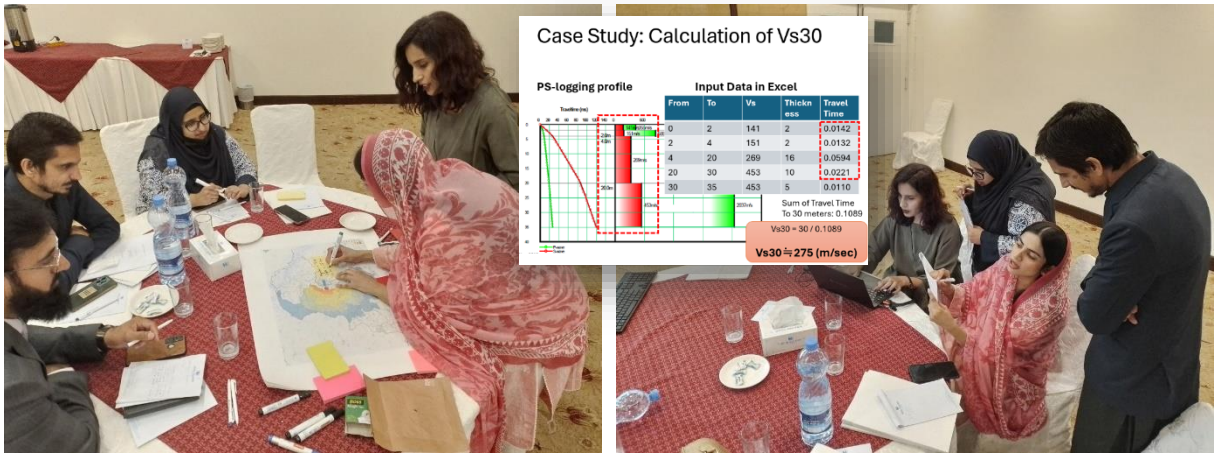


Figure 5.1.4 Results of the First Workshop (Left: Examination of NEOC Response to a Hypothetical Earthquake; Right: Calculation Exercise for Vs30)

5.1.2.2 Effects of the Workshop and Training and Future Directions

In the field of earthquake management, the project team emphasized that the interactive exchange of opinions, rager than a lecture-based formal, allowed for the extraction of valuable feedback and proposals from the participants, infomed by inout from the Japanese side.

Figure 5.1.5 shows a slide from the proposal presentation delivered from NEOC during the 2nd Workshop. The main points are as following, (1) development of an immediate damage estimation system, (2) utilization of AI technology for decision support such as prioritization of disaster relief and material support, and (3) utilization of the system for advance investment in disaster management (Example: Promotion of investment in disaster management in advance to prevent emergency assistance due to road blockages in areas where damage to buildings is expected to be severe due to an earthquake).

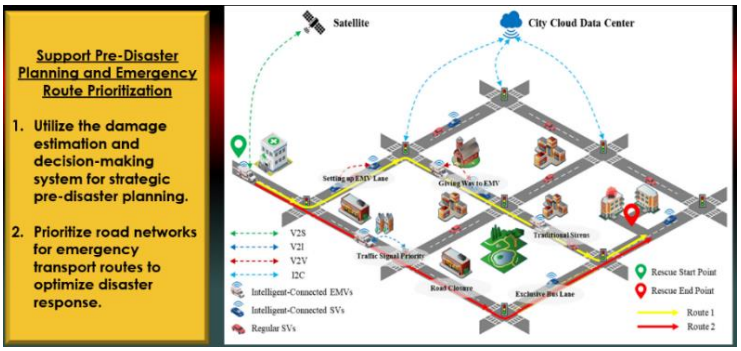


Figure 5.1.5 Results of the 2nd WS (Proposal in the Final Presentation from NEOC)

5.2 Activities 4-2

Workshops and training will be conducted in Pakistan and Japan.

5.2.1 Overview of Workshops and Training

Workshops and training were conducted on the schedule shown in Table 5.2.1 to improve the risk analysis skills of NEOC staff in the field of floods and earthquakes.

Table 5.2.1 Overview of Workshops and Training

Training Name	Date and Time	Location	Main Themes
1st Workshop	October 25, 2024 November 1, 2024	Hotel in Islamabad	● Floods: Workshop on how to use GIS software and inundation analysis models ● Earthquakes: Workshop on seismic hazard and risk assessment using basic concepts and ArcGIS
2nd Workshop	February 19, 2025 February 25, 2025	Hotel in Islamabad	● Flood: Workshop on runoff analysis models and calibration ● Earthquakes: Application of AI Technology Using basic concepts and ArcGIS for seismic risk assessment
Training in Japan (Cancelled)	May 9 to 15, 2025	Multiple Training Organizations in Tokyo and Ibaraki Prefecture	● Lectures and workshops on the use of satellite images in the field of disaster management ● Lectures and workshops on the use of hydrological models ● Lectures on the importance of disaster management authorities in Pakistan, global trends in disaster management and the role of disaster management agencies in Japan ● Opinion exchange meeting based on this project and training

5.2.2 First Workshop in Pakistan

The first workshop took place on October 25 and November 1, 2024, at a hotel in Islamabad. As shown in In the field of earthquake management, the participants used population data based on the tehsil-wise census data and Vs30 (average S-wave velocity of the ground) constructed in the first half of the project, and practiced foundation type content (Attenuation equation and ground amplification degree, using Vs30 data of USGS) related to earthquake hazard and risk assessment. Following this, the participants practiced hazard assessment and analysis of the exposed population using ArcGIS, and the data and calculation tool were introduced to the participants for homework. The session also highlighted the importance of enhancing the accuracy of Vs30 data through physical exploration was also discussed.

Table 5.2.2, the workshop was divided into two groups: the flood group and the earthquake group. The list of participants is provided in Table 5.2.3. Photos of the first workshop are shown in Figure 5.2.1.

In the flood group, the workshop focused on the use of GIS software and inundation analysis models, essential for river risk analysis. This session provided NEOC staff with basic knowledge of these tools, significantly enhancing their practical skills and improving their ability to apply them in real-world scenarios.. In addition, the workshop included topics such as introduction of the latest Japanese river technologies and discussion on these technologies, presentation from NEOC staff (introduction of systems and applications operated by NEOC), and information sharing with NEOC staff.

In the field of earthquake management, the participants used population data based on the tehsil-wise census data and Vs30 (average S-wave velocity of the ground) constructed in the first half of the project, and practiced foundation type content (Attenuation equation and ground amplification degree, using Vs30 data of USGS) related to earthquake hazard and risk assessment. Following this, the participants practiced hazard assessment and analysis of the exposed population using ArcGIS, and the data and calculation tool were introduced to the participants for homework. The session also highlighted the importance of enhancing the accuracy of Vs30 data through physical exploration was also discussed.

Table 5.2.2 Contents of the First Workshop

Day-1 (2024/10/25 Friday)					
Time	Activity	Duration	Time	Activity	Duration
10:00	Briefing				20 min
Flood Session			Earthquake Session		
10:20	Lecture: Introduction of Flood Control Measures in Japan	40 min	10:20	Lecture: Overview of earthquake hazard and risk assessment and mapping for scenario earthquakes	50 min
11:00	Break time	20 min	11:10	Break time	20 min
11:20	Discussion: Necessary Technologies for Flood Control for NEOC	60 min	11:30	Group Discussion	50 min
12:20	Lunch Break	90 min	12:20	Lunch Break	90 min
13:50	Workshop: Instruction of Analysis Using GIS Software ~QGIS~ (1)	50 min	13:50	Lecture: GIS Database for simplified earthquake hazard and risk assessment	50 min
14:40	Break time	20 min	14:40	Break time	20 min
15:00	Workshop: Instruction of Analysis Using GIS Software ~QGIS~ (2)	50 min	15:00	Workshop: How to calculate earthquake hazard using ArcGIS	50 min
15:50	Break time	20 min	15:50	Break time	20 min
16:10	Workshop: Instruction of Analysis Using GIS Software ~QGIS~ (3)	50 min	16:10	Workshop: Summary of Today's topics	50 min
17:00	Day-1 Finish	–	17:00	Day-1 Finish	–

Day-2 (2024/11/1 Friday)					
Time	Activity	Duration	Time	Activity	Duration
10:00	Briefing				5 min
Flood Session			Earthquake Session		
10:10	Workshop: Instruction of Building Flood Inundation Model ~HEC-RAS Simulation~ (1)	60 min	10:10	Group Discussion: Probabilistic earthquake hazard vs Scenario earthquake	60 min
11:10	Break time	20 min	11:10	Break time	20 min
11:30	Workshop: Instruction of Building Flood Inundation Model ~HEC-RAS Simulation~ (2)	50 min	11:30	Group Discussion: Topics on the Noto earthquake in Japan	50 min
12:20	Lunch Break	90 min	12:20	Lunch Break	90 min
13:50	Workshop: Instruction of Analyzing Flood Inundation Model ~HEC-RAS Simulation~	70 min	13:50	Workshop: Real-time earthquake damage estimation and DRR plan	50 min
15:00	Break time	20 min	14:40	Break time	20 min
15:20	Presentation of Example of HEC-RAS Simulation by Mr. Umair	30 min	15:00	Group Discussion: disaster prevention investments and ideas for smooth implementation of NEOC rescue activities	50 min
15:50	Break time	20 min	15:50	Break time	20 min
16:10	Presentation from NEOC and Discussion by Group 1	25 min	16:10	Presentation from NEOC	50 min
16:35	Presentation from NEOC and Discussion by Group 2	25 min			
Time	All Participants				Duration
17:00	Crosing Celemony, Group Photo				20 min
17:20	Day-2 Finish				–

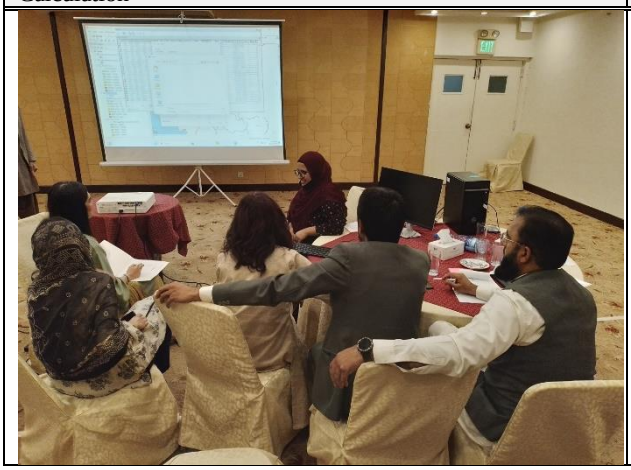
Table 5.2.3 Participants of the First Workshop

No	Name Participants of Flood Session	Designation	No	Name Participants of Earthquake Session	Designation
1	Engr. M. Umair Afzal	DM Hydrology	1	Anum Bashir	DM Seismology
2	Baber Faiz	DM Meteorology	2	Tanveer Ahmed	DM Risk
3	Sajawal Rehman	DM Climatology	3	Kashif Iqbal	DM Backend
4	Dr. Shazia Akhtar	DM Research	4	Faiza Ghous	AM Meteorology
5	Syed Waqar Hussain Shah	AM GIS Analysis	5	Dur-e-Maknoon	AM Backend
6	Saba Shahzadi	AM Hydrology	6	Talyaa Najam (Day 2 only)	Earthquake
7	Syeda Saleha Fatim Ali	AM Climatology			
8	Ayman Fatima	AM Data Support			
9	Nawaz Shareef	AM Infra			

Flood Group: Lectures on Flood Analysis Models

Flood Group: Discussions on Using the Latest Technologies

Earthquake Group: Exercise on Earthquake Hazard Calculation



Earthquake Group: Interactive Exchange of Opinions by NEOC

Figure 5.2.1 Photos of the First Workshop

5.2.3 Second Workshop in Pakistan

The second workshop was conducted on February 19 and 25, 2025, at a hotel in Islamabad. The workshop agenda and contents are shown in During the earthquake related workshop, in response to the suggestion from NEOC that AI should be considered, part of the original plan of the earthquake risk exercise ((1) damage calculations using building damage functions, (2) Building inventory data construction by GIS, (3) Thematic map visualization) was changed, and (2) building footprint data construction exercise using ArcGIS using Deep Learning (a kind of AI) (using a sample from the United States) was carried out. In addition, The workshop also included a discussion on how AI could bridge early warning systems with proactive disaster management investments, such as measures to prevent road blockages from building collapses.

Table 5.2.4. As shown in Table 5.2.5, the participants remained as those who attended the first workshop. Photos from the second workshop are shown in Figure 5.2.2.

In the flood group, the participants deepened their understanding of the hydraulic analysis model by experiencing calibration of the runoff analysis model focusing on parameter adjustment. In particular, NEOC staff gained understanding of the key parameters of the runoff analysis model and the reason why the calculation results changed when the parameters were changed. In addition, NEOC staff delivered presentations outlining the issues they would like to challenge in the future and the parts requiring technical cooperation.

During the earthquake related workshop, in response to the suggestion from NEOC that AI should be considered, part of the original plan of the earthquake risk exercise ((1) damage calculations using building damage functions, (2) Building inventory data construction by GIS, (3) Thematic map visualization) was changed, and (2) building footprint data construction exercise using ArcGIS using Deep Learning (a kind of AI) (using a sample from the United States) was carried out. In addition, The workshop also included a discussion on how AI could bridge early warning systems with proactive disaster management investments, such as measures to prevent road blockages from building collapses.

Table 5.2.4 Contents of the Second Workshop

Day-1 (2025/02/19 Wednesday)							
Time	Activity	Moderator	Duration	Time	Activity	Moderator	Duration
All Participants							
10:00	Briefing					Goto	5 min
Flood Session				Earthquake Session			
10:10	Presentation & Discussion: HEC-RAS Simulation Results (The homework of Previous Workshop) from NEOC Side	Sueishi	40 min	10:10	Review of Seismic Hazard Assessment	Hasegawa	40 min
10:50	Break time	–	20 min	10:50	Break time	–	20 min
11:10	Lecture: Breach Analysis for Indus River downstream of Tarbela	Sueishi	50 min	11:10	Earthquake Damage Assessment using a Fragility Curve	Hasegawa	50 min
12:00	Lunch Break	–	90 min	12:00	Lunch Break	–	90 min
13:30	Lecture: Brief Explanation about RRI Model and the Sample Model in Tonegawa River	Sueishi	30 min	13:30	AI Technologies in Earthquake and Tsunami DRR	Hasegawa	40 min
14:00	Break time	–	15 min	14:10	Break time	–	15 min
14:15	Workshop: Test Run of the Sample Model in Tonegawa River (1) Conduct the Test Run and Evaluation of the Results	Sueishi	30 min	14:25	Let's use AI!	Hasegawa	40 min
14:45	Workshop: Test Run of the Sample Model in Tonegawa River (2) Understand How to Change the Parameters	Sueishi	30 min	15:05	Break time	–	15 min
15:15	Break time	–	15 min	15:20	Discussion and Explanation on Homework	Hasegawa	40 min
15:30	Workshop: Test Run of the Sample Model in Tonegawa River (3) Discuss the Strategy of Setting Parameters	Sueishi	30 min				
16:00	Day-1 Finish					–	–

Day-2 (2025/02/25 Tuesday)							
Time	Activity	Moderator	Duration	Time	Activity	Moderator	Duration
Flood Session				Earthquake Session			
10:00	Workshop: Parameter Adjustment of the Sample Model in Tonegawa River (1)	Sueishi	50 min	10:00	AI Applications in Disaster Phases	Hasegawa	50 min
10:50	Break time	–	20 min	10:50	Break time	–	20 min
11:10	Workshop: Parameter Adjustment of the Sample Model in Tonegawa River (2)	Sueishi	50 min	11:10	ArcGIS Application for Earthquake Damage Assessment	Hasegawa	50 min
12:00	Lunch Break	–	90 min	12:00	Lunch Break	–	90 min
13:30	Presentation from Each Group Parameter Adjustment Competition	Sueishi	30 min	13:30	ArcGIS Application for Earthquake Damage Assessment	Hasegawa	40 min
14:00	Lecture: Notes on Calibration	Sueishi	20 min	14:10	Discussion / Additional Exercise	Hasegawa	30 min
14:20	Discussion of Parameter Adjustment	Sueishi	20 min				
All Participants							
14:40	Presentation & Discussion for Suggestions for Cooperation from JICA					Goto	50 min
15:30	Crosing Celemony, Group Photo					Goto	20 min
15:50	Day-2 Finish					–	–

Table 5.2.5 Participants of the Second Workshop

No	Name Participants of Flood Session	Designation	No	Name Participants of Earthquake Session	Designation
1	Engr. M. Umair Afzal	DM Hydrology	1	Anum Bashir	DM Seismology
2	Baber Faiz	DM Meteorology	2	Tanveer Ahmed	DM Risk
3	Sajawal Rehman	DM Climatology	3	Kashif Iqbal	DM Backend
4	Dr. Shazia Akhtar	DM Research	4	Faiza Ghous	AM Meteorology
5	Syed Waqar Hussain Shah	AM GIS Analysis	5	Dur-e-Maknoon	AM Backend
6	Saba Shahzadi	AM Hydrology	6	Talyaa Najam	Earthquake
7	Syeda Saleha Fatim Ali	AM Climatology			
8	Ayman Fatima	AM Data Support			
9	Nawaz Shareef	AM Infra			

Flood Group: Hands on Practice in Workshop

Flood Group: Presentation on Parameter Adjustment

Earthquake Group: Lecture before Exercise

Earthquake Group: Deep Learning Exercise Using GIS

Figure 5.2.2 Photos of the Second Workshop

5.2.4 Training in Japan

5.2.4.1 Reasons for the Cancellation of the Training in Japan

The training in Japan, originally scheduled for May 9 to May 15, 2025, was canceled due to the escalating conflict between Pakistan and India. The reasons for the cancellation of the training in Japan are summarized in Table 5.2.6.

Table 5.2.6 Reasons for the Cancellation of the Training in Japan

Date	Incident
April 22 (Tue)	A terrorist incident took place in Pahalgam, located in the Union Territory of Jammu and Kashmir in northern India (an area under Indian control in the disputed Kashmir region), resulting in the deaths of 26 civilians, including foreign tourists. Reported indicated that The Resistance Front (TRF), believed to be an off short of the Pakistan-based Islamic extremist group Lashkar-e-Taiba (LeT), was involved.
May 7 (Wed) Early Morning	The government of Pakistan announced that the Indian military had launched missile attacks, airstrikes, and drone attacks on several locations within Pakistan's sovereign territory. The targeted areas included Sialkot, Shakargarh, Muridke, Bahawalpur in Punjab province, and Kotli and Muzaffarabad in the Azad Jammu and Kashmir region, causing damage to civilians and civilian infrastructure.
May 7 (Wed) Early Morning	According to Pakistani aviation authorities, the country airspace was closed for at least 48 hours following the recent airstrikes by India. Additionally, Pakistan Airlines announced that its flight operations had been suspended since the early morning of May 7th, while other airlines were reported to have individually contacted their booked passengers regarding the flight disruptions.
May 7 (Wed) Night	The trainees scheduled to attend training in Japan were set to depart on the night of May 7th; however, their flight was canceled due to the closure of Pakistani airspace, as mentioned above.
May 10 (Sat) Early Morning	In the early hours of May 10th, the Pakistani military announced that the Nur Khan Airbase (Rawalpindi), Muridke Airbase (Chakwal District, Punjab Province), and Shorkot Airbase (Jhang District, Punjab Province) of the Pakistan Air Force were targeted by the missiles and drones attacks launched by the Indian military. Additionally, according to the government of Pakistan, a total of 77 Indian drones had been shot down as of May 10th. Reports also indicated that some civilian casualties occurred as a result of downed drones.
May 10 (Saat) Evening	On the evening of May 10th, both the Pakistani and Indian governments announced that a ceasefire had been agreed upon between the two countries. Following this development, Pakistani airport authorities confirmed the country airspace has been reopened for all types of flights.
May 16 (Fri)	Since both countries agreed to an immediate ceasefire on May 10th, no significant deterioration in the security situation has been observed, and a certain level of stabilization is expected moving forward. However, the situation remains fluid, and the risk of sudden escalation cannot be ruled out. Therefore, after thorough consideration from various perspectives within JICA, it has been decided to fully cancel he planned training program in Japan.

Source: Cautionary email from the Embassy, Ministry of Foreign Affairs (Overseas Safety Website) – as of May 17, 2025

5.2.4.2 Contents of the Planned Training in Japan

The contents of the planned training in Japan are shown in Table 5.2.7. The purpose of the training in Japan was to learn about Japan's latest disaster management technology through lectures and hands-on workshops, with the goal of adapting and applying these innovations to our work in Pakistan. In addition, lectures and exchanges of opinions were scheduled to hold on the role of disaster management authorities in Pakistan based on trends and methods of disaster management in the world and Japan, with the aim of contributing to raising awareness of disaster management officials in Pakistan. The participants are shown in Table 5.2.8.

Table 5.2.7 Schedule of the Japanese training

Date	Time	Training Destination	Instructor	Training Content
Thursday, May 8	—	—	—	Arrival
Friday, May 9	10:00~12:00	ICA Tokyo	JICA and Consultant	Briefing and Orientation
	14:00~17:00	RESTEC	RESTEC	Workshop on the Use of Satellite Images in Disaster Management
Saturday, May 10	—	—	—	Review
Sunday, May 11	—	—	—	Review
Monday, May 12	9:00~11:00	—	—	Transportation (Tokyo ⇒ Ibaraki)
	14:00~15:30	JAXA	JAXA	Sharing Disaster Information Using Satellite Images
Tuesday, May 13	10:00~12:00	JICA Tsukuba	Former NDMA Expert Mr. Hasegawa	The Importance of Disaster Management Authorities in Pakistan
	13:30~15:00	JICA Tsukuba	JICA Disaster Management Group Mr. Nishikawa	Global Trends in Disaster Management Authorities and the Role of Disaster Management Organizations in Japan
	15:00~17:00	JICA Tsukuba	JICA and Consultant	Exchange of Opinions Based on This Project and Training
Wednesday, May 14	10:00~16:00	ICHARM	ICHARM	Lectures and Workshops on the Overview and Utilization of Hydrological and Hydraulic Models
Thursday, May 15	9:30~11:30	NIED	NIED	Information Sharing System Using Satellite Data, Collaboration between Organizations in Disaster Management Research
	13:30~16:00	JICA Tsukuba	JICA and Consultant	Evaluation Meeting and Closing Ceremony
Friday, May 16	—	—	—	Departure

RESTEC: Remote Sensing Technology Center of Japan
JAXA: Japan Aerospace Exploration Agency
ICHARM: International Center for Water Hazard and Risk Management
NIED: National research Institute for Earth science and Disaster prevention

Table 5.2.8 Participants of Training in Japan

No	Name	Organization	Designation
1	Brig. Badar Zaman	NEOC	Director General NEOC
2	Ms. Zahra Hassan	NEOC	General Manager GIS
3	Mr. Syed Muhammad Tayyab Shah	NEOC	General Manager Risk Assessment
4	Mr. Sajawal Rehman	NEOC	Deputy Manager Climatology
5	Mr. Umair Afzal	NEOC	Deputy Manager Hydrology
6	Ms. Anum Bashir	NEOC	Deputy Manager Seismology
7	Mr. Hafiz Muhammad Haris Mir	NEOC	Deputy Manager GIS Analysis
8	Mr. Abdul Hanan	NEOC	Deputy Manager GIS Analysis
9	Dr. Tanveer Ahmed	NEOC	Deputy Manager Seismology
10	Mr. Sajid Inam	NEOC	Deputy Manager Risk Assessment

5.2.4.3 Impact of Training in Japan on the Project

In light of Outcome 4 “Capacity as a central disaster management organization through disaster management centers is improved”, although the training in Japan was canceled due to the international conflict, the two workshops held in Pakistan contributed to strengthening the capacity of NEOC staff. Furthermore, through the implementation of these workshops and the preparations for the training in Japan, the project team was able to assess the current capabilities of NEOC staff and clarify the technical skills and knowledge they need to acquire going forward.

Going forward, it is desirable to implement training programs aimed at enhancing the capacity of NEOC staff through other projects or country-focused training programs, utilizing the content originally planned for the training in Japan.

In addition, considering the current state of disaster management capacity in Pakistan, it is believed that NEOC and its parent organization, NDMA, are expected to fulfill the following roles.

- To enhance the national disaster management capacity, NEOC and NDMA are expected to take a long-term and mid-term perspective, clearly define the roles, goals, and necessary activities of each disaster-related agency, and manage the activities required to achieve those goals. (Management of disaster planning and implementation of countermeasures)
- As the central disaster management agency, NEOC and NDMA are expected to identify disaster risk areas based on information and outputs organized by other ministries and agencies, in order to enable a swift response following a disaster in Pakistan. It should also identify challenges related to risk assessment by disaster type and collaborate with relevant organizations to address those challenges. (Promotion of efficient disaster risk management and disaster response)
- To strengthen the disaster management capacity of staff in both central and local agencies, NEOC and NDMA are expected to consolidate, update, and implement disaster education programs, thereby enhancing national disaster management capabilities that contribute to risk management and mitigation. (Capacity building of disaster management agencies)

CHAPTER 6 OTHERS

6.1 Issues, Ideas, and Lessons in Project Implementation

6.1.1 Challenges in Providing Assistance Due to the Organizational Characteristics of NDMA

The content of this project was finalized following a detailed planning survey conducted in June 2022. This agreement was reached through the exchange of the Record of Discussions (R/D) and an explanation of the Project outline to the counterpart (C/P), based on the Workshop Record (W/R). However, during the kickoff meeting held in December 2022, the current Chairman of NDMA, who assumed office around September of the same year, requested substantial changes to the scope of work previously agreed upon in the R/D. The nature of these requests deviated from the project's original core approach, which focused on updating the existing medium- to long-term plan, known as the NDMP. Despite a series of intermittent discussions over the following year, no consensus could be reached on revising the Chairman's proposed policy direction.

Within NDMA, a significant number of senior and mid-level officials come from military backgrounds. Similarly, many stakeholders involved in this project, particularly those above the level of direct counterparts, also had military experience, reinforcing a strong top-down decision-making structure. Furthermore, in the aftermath of the 2022 flood disaster, NDMA increasingly prioritized investments in initiatives deemed more urgent than, for instance, the establishment of the National Emergency Operations Center (NEOC). Under these circumstances, while the project team was able to gain the understanding of C/P personnel working in close coordination, it was difficult to collaboratively propose compromise solutions that did not fully align with the Chairman's direction.

Ultimately, with the involvement of the JICA mission team, the Japanese side proceeded with the provision of assistance in a manner consistent with the original objectives of the project. Going forward, when engaging with organizations that have similar decision-making structures in countries in countries like Pakistan, it is important to anticipate the possibility of arising such situations and incorporate appropriate risk mitigation measures during the planning and implementation phases of technical cooperation projects.

6.1.2 Engagement in the Update of the Medium- to Long-Term Plan

Among the counterpart (C/P) personnel involved in the Project were individuals who had previously contributed to the formulation of the current National Disaster Management Plan (NDMP). Leveraging their experience, these personnel actively organized stakeholder meetings to discuss the progress and potential update of the NDMP, and proactively shared relevant information. In addition, they continued discussions with the project team to support the ongoing implementation of the project in line with the Record of Discussions (R/D). However, a consensus with the Chairman could ultimately not be reached regarding the core issue of updating the medium- to long-term NDMP.

On the other hand, through repeated consultations between the Japanese side, the Chairman, and C/P personnel, a growing recognition emerged among C/P staff and associated personnel of the importance of long-term planning, informed by disaster management practices and results observed in Japan and other countries. As a result, even after the kickoff meeting, the C/P personnel proposed options such as extending of the project or temporary suspension to mitigate the risk of project disruption and ensure implementation remained consistent in line with the R/D.

Furthermore, the C/P personnel were aware that other entities, including those involved in Post-Disaster Needs Assessment (PDNA) and flood-related sectors, were progressing with the development of medium- to long-term plans in their respective domains. This awareness contributed to a broader understanding of the need to update the current NDMP in order to align with and coordinate among related institutions.

This shift in perception represents a significant outcome, as it is expected to meaningfully support sustained investment in the disaster risk reduction sector in Pakistan.

6.1.3 Support for NEOC

Although direct support to the National Emergency Operations Center (NEOC) was not originally included within the scope of this project, the activities being undertaken by NEOC at the time represented an early stage in the development of a disaster response decision-making system. In light of this, training sessions were organized as an extension of the project's activities, focusing on relevant technical elements such as strengthening capabilities for disaster information management and risk assessment.

Through these training sessions, the project team gained a certain level of understanding regarding the NEOC's operational status and progress in system development. In addition, by presenting case studies from Japan, the training emphasized the importance of inter-agency coordination in the collection, analysis, and dissemination of disaster-related information and risk assessments. This contributed to broadening perspectives among participants, particularly regarding the conventional approach of NEOC as solely responsible for all such functions.

While the extent to which the insights of trainees were reflected in higher-level decision-making remains to be seen, the training effectively contributed to fostering a more practical and realistic understanding at the operational level. This progress is expected to enhance the overall efficiency of both preparedness efforts (including the management of pre-disaster investments) and disaster response operations in the future.

6.1.4 Frequent Early Morning Meetings With the C/P In Charge (Issues and Ideas)

Before formulating the new NDMP, the C/P designated a lead person at the Deputy Director General level and held brief meetings (typically about an hour in the morning) as needed. At the initial stages of the project, these meetings mainly addressed basic discussions and requests, such as questions from members, appointments for meetings, requests for data submission, and office maintenance . These meetings were held about three days a week. In addition, discussions were held with the second-tier officer (Member) once a month. Although the meetings were mainly logistical, these were found useful for promptly considering and implementing the project team's responses inline with the NDMA's situation.

6.1.5 Long-Term Employment of Local Mercenaries (Issues and Ideas)

Unlike the current NDMP-I, this plan was supposed to address a broader range of disasters, such as natural and man-made disasters other than floods and earthquakes. On the other hand, the CP requested from the beginning of the new M/M handover that "It is difficult for NDMA staff to work together, and we hope that the project team will develop the plan as it did in the previous project." Therefore, the project team has no choice but to take care of the cooperation and collaboration in technology transfer activities. This work intends to supplement technology transfer with opportunities through seminars and workshops, but the challenge remains how to overcome the work burden. To address this, long-term local employees were hired particularly to support efforts in the flood related areas, where the ratio of pre-investment projects is high, and in the multi-disaster field other than floods and earthquakes.

The local employees were responsible for reporting the progress of NDMP-I to the relevant ministries and department, and organizing information, etc., even when the survey team members were not present. Furthermore, since the 2 local employees were retired from the disaster management related organizations (One is from a flood preparedness organization, and one is from National Disaster Management Authority_NDMA with expertise in disaster preparedness, response and earthquake management.), they were able to handle information gaps from the hearing. Through their government counterpart connections, they efficiently processed and shaped this information into the NDMP2024 proposal activities and projects.. In addition, as the NDMA planned to circulate NDMP2024 to other ministries and departments without holding prior consultation meetings, the project team took proactive steps to engage the relevant department in advance, measures were confirmed with the relevant department in advance for disasters in the flood and earthquake fields with significant cumulative damages or with high frequency. Since the NDMA chairman did not convene an official meetings with the relevant department, the project team held series of meeting with the relevant department after confirmation from the JICA Pakistan office. During these meetings, the local employees, due to their familiarity and trusted relationships, represented the

project team in sensitive discussions and facilitated the smooth exchange of feedback from the relevant departments.

6.2 Achievement of Project Goals

The objective of the project is to strengthen the disaster management planning process and thereby contribute to the promotion of advance disaster prevention investment by preparing Substantial Contents for analyzing disaster risk, updating national disaster management plans, reviewing the draft pre-disaster prevention investment plan in the flood field, and coordinating with the related organizations.

However, the achievement of the project objectives was brought into question due to the management approach of the current NDMA Chairman. While the Chairman is not opposed to pre-disaster investment, the policy adopted is based on the idea of entrusting NDMA and related agencies without exercising effective control, rather than strengthening the national-level disaster management system. In other words, there is prevailing belief that disaster management efforts by local governments and other department fall under the jurisdiction of NDMA.

Amid these circumstances, the project underwent significant scope reductions in consultation with the counterpart government, particularly revising the tasks under Outcome 4. Despite these adjustments, the incorporated technical support programs aimed to raise awareness of the importance of pre-disaster investment in coordination with relevant organizations. As a result, all planned activities were successfully completed, and the Project objectives were achieved.

6.3 Proposals for Achievement of Top Goals

The Overall Goal of the Project is "Based on the updated National Disaster Management Plan (NDMP-II), priority projects emphasizing proactive disaster prevention investments will be identified, and their implementation will commence within the country."

Reflecting on the situation to date, ensuring the achievement of the Overall Goal requires that, during project formulation, careful attention is given to external factors identified in the PDM, such as: (1) the organizations from which senior officials of NDMA are appointed or dispatched; (2) the decision-making process within NDMA; and (3) the feasibility of meaningful consultations between NDMA staff and senior officials. The project should be designed and implemented with activities that effectively address and adapt to these external conditions.

Appendix

1. Project Design Matrix (PDM)

Annex 1

Project Design Matrix

Project ; Technical support project to review and update National Disaster Management Plan (NDMP)

Implementing Agency: National Disaster Management Authority (NDMA)

Version 1.0

Period of Project: March 2022 to May 2025 (tentative)

Date 2022 June 17

Narrative Summary	Objectively Verifiable Indicators	Means of Verification	Important Assumption
Overall Goal			
Based on the revised National Disaster Management Plan (NDMP-II), the priority projects will be defined and initiated for implementation in Pakistan with emphasis on pre-disaster investments in disaster risk reduction	The budget amount for pre-disaster investment projects will increase.	List of pre-disaster investment plans/programme which are allocated by federal and/or local government budget.	
Project Purpose			
Planning process of disaster management at national level is strengthened through revising NDMP to facilitate, monitor and coordinate implementation led by NDMA	The substantial contents for NDMP II, including measures to improve the current NDMP, are developed.	Approval of NDMP II including actions/programme by NDMA	
Outputs			
1. NDMA's capabilities of information collection and analysis of natural hazards, situation of exposures and vulnerabilities are strengthened.	Disaster risk is assessed on a national scale.	Revision of "NDMA policy guidelines" for conduct of hazard, exposures, vulnerability and risk assessment by the risk assessment unit of NDMA.	
2. NDMA's capabilities of formulation for more effective NDMP-Ⅱ is strengthened while grasping the progress and issues of the current NDMP.	The overall sturcture for NDMP II is formulated.	The validity evaluation report regarding the overall structure of NDMP II by the steering committeee of the Project	
3. A draft long list of priority projects for each disaster type is prepared, and a substantial contents for NDMP II is formulated.	Methods to promote the implementation of priority projects are considered and the substantial contents for	The priotity project list selected on the basis of the substantial contents including the procedure of	
4. The capacity to promote pre-disaster investment is enhanced through the study of a draft pre-disaster investment plan for flood disasters and coordination with relevant organizations.	Draft pre-disaster investment plan in flood sector is prepared.	The formulated investment plan in the substantial document	
Activities	**Inputs**		**Important Assumption**
	The Japanese Side	**The Pakistani Side**	
1-1. Confirm hazard intensity and probability of occurrence as the basis for disaster risk information in the current NDMP.	(1) Dispatching the Consultant Team (Tentative) -Chief Consultant/Comprehensive Disaster Risk Reduction plan/ Donor Coordination -Pre-Disaster Investment Plan/Budget Planning -Disaster Risk Reduction/ Local DRR Plan -Hydrological analysis / flood inundation analysis / flood runoff analysis -River Structural Measures -Meteorology/ Hydrology/ Climate Change impact assessment -Urban Disaster/ Drainage System/ Land Use Plan -Earthquake and Tsunami -Institutional Development/ Human Resources Development Plan -Environmental and Social Considerations -Database/ GIS/ Equipment Planning (2) Knowledge Co-creation Program in Japan	(1) The Government of Pakistan shall accord privileges and other benefits to the Team in accordance with the Agreement on Technical Cooperation between the Government of Japan and the Goverrnment of Pakistan signed on June 8, 2021. (2) The Government of Pakistan shall bear claims, if any arise against the member(s) of the Team resulting from, occurring in the course of, or otherwise connected with the discharge of their duties in the implementation of the Project, except when such claims arise from gross negligence or willful misconduct on the part of the Team. (3) The Government of Pakistan shall assist the Team, if necessary, in customs clearance, exempt from any duties with respect to equipment,	
1-2. Confirm of the vulnerability and exposure data as the basis for disaster risk information in the current NDMP.			
1-3. Identify the need to update and review disaster risk information in the current NDMP.			
1-4. Assess the risk of disaster risk information identified as need to be updated and reviewed in Activity 1-3.			
2-1. Review the basic strategy of DRR and its effectiveness of the current NDMP based on the results of the risk assessment of Activities 1-4.			
2-2. Review and analyze the progress and challenges of priority projects and the Human Resources Development Plan of the current NDMP for the each line ministry/agency			
2-3. Collect information on issues of the current NDMP and analyze it, and consider how to improve the current NDMP based on information that were colected and analyzed.			**Pre-Conditions**
2-4. Examine the overall substantial contents for NDMP II based on the Activities above.			
3-1. Collect information on pre-DRR projects currently planned by relevant ministries and agencies.			
3-2. Identify needs for additional project(s) to be considered, based on analysis on the risk assessment in Activity 1-4 and information collection in Activity 3-1, and consider a draft of the projects, including the rough cost estimation.			
3-3. Compile a draft long list for each hazard type and discuss priority projects with order for each ministry and agency, based on the Activity 3-1 and 3-2.			
3-4. Develop the substantial contents for NDMP II while considering a method to promote the implementation of the priority projects			
3-5. Based on the experience for preparing the draft pre-disaster investment plan for flood disasters formulated in Activity 4, the following two points will be clarified: ① Basic principles and concepts in the pre-DRR investment plan for other hazard types than flood after reviewing the pre-DRR investment plan for flood, ② Roles of NDMA in coordination with relevant line ministries and agencies toward the implementation of projects.			
4-1. Collect existing flood information, damage status and evaluate existing flood countermeasures, etc. to accurately grasp the status of flood hazards.			
4-2. Identify priority project(s) with high urgency and relevance of flood structural countermeasures, taking into account the effects of climate change.			
4-3.Conduct pre-F/S on the priority project(s) identified by Activity 4-2, including natural conditions survey, outline design, cost estimation, and procurement / construction planning			
4-4. Select small-scale projects that are not undertaken by the relevant ministries and agencies, and formulate prototype of local DRR plans for flood in the target Province and District.			
4-5.Integrate all the above-mentioned activities and prepare a draft pre-disaster Investment plan for flood.			

Annex 1

Project Design Matrix

Project : Technical support project to review and update National Disaster Management Plan (NDMP)

Implementing Agency: National Disaster Management Authority (NDMA) — **Version 2**

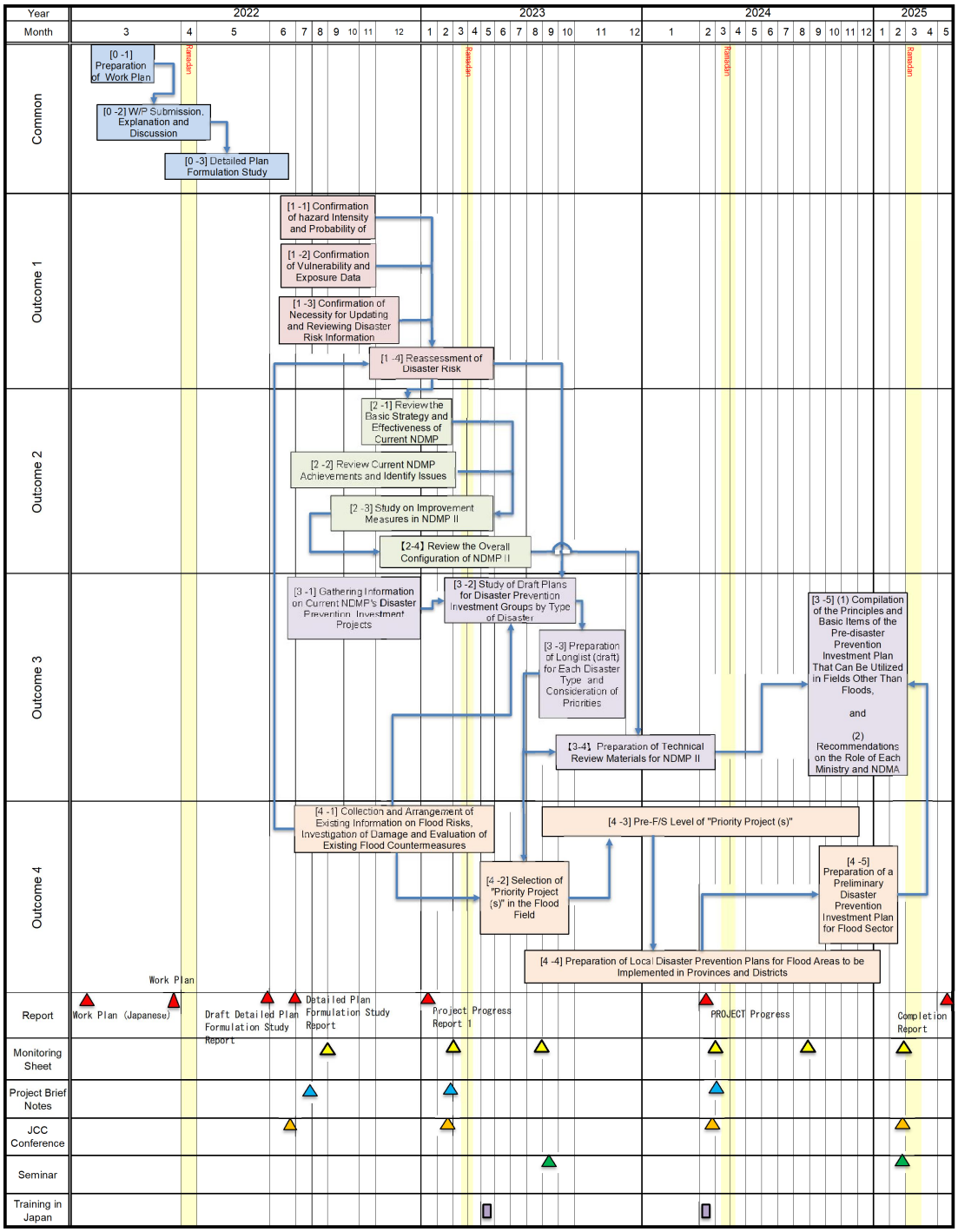
Period of Project: March 2022 to May 2025 (tentative) — **Date 2024 June 11**

今後

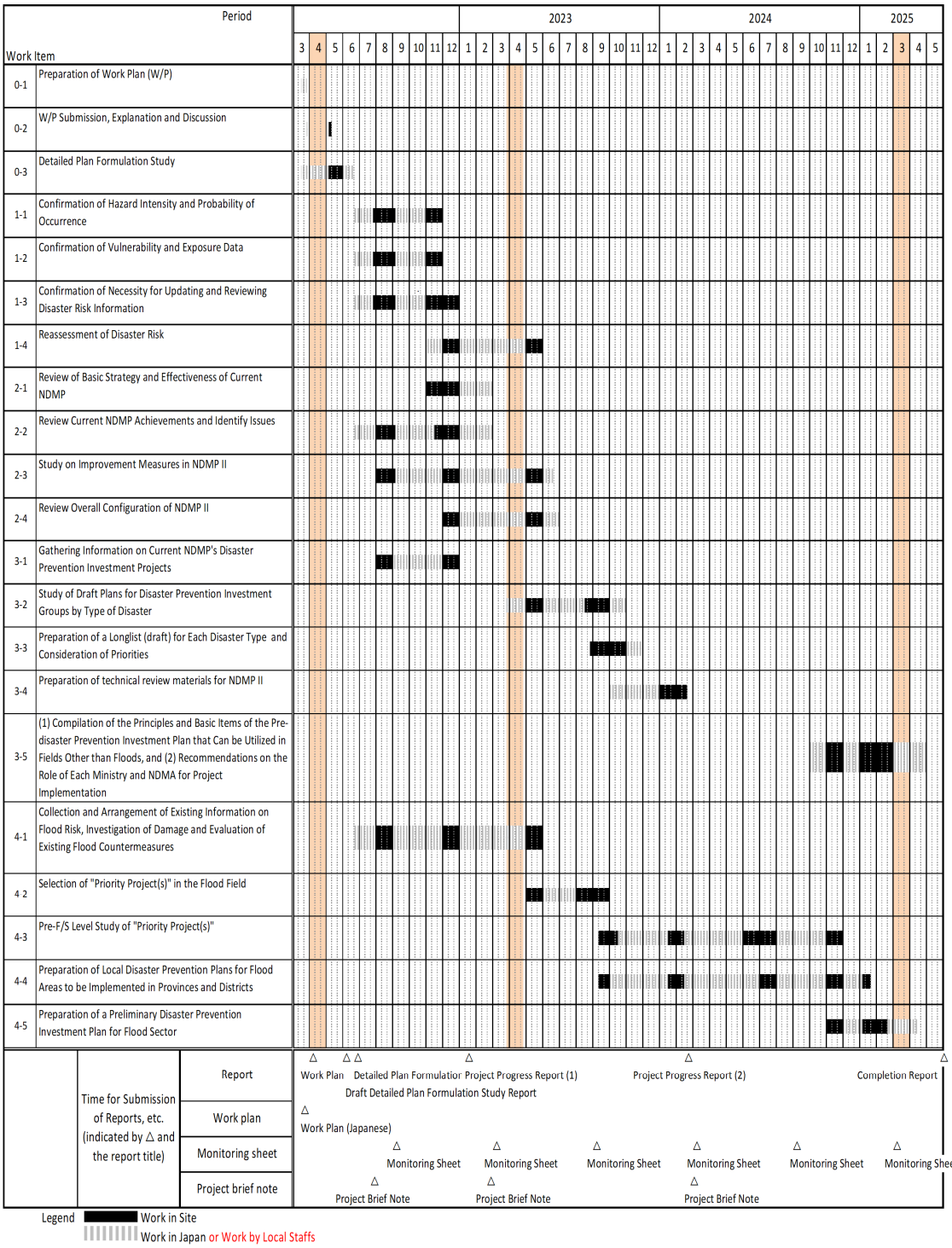
Narrative Summary	Objectively Verifiable Indicators	Means of Verification	Important Assumption
Overall Goal			
Based on the National Disaster Management Plan(NDMP), the priority projects will be defined and initiated for implementation in Pakistan with emphasis on pre-disaster investments in disaster risk reduction	The budget amount for pre-disaster investment projects and activitie swill increase.	List of pre-disaster investment plans/programme which are allocated by federal and/or local government budget.	
Project Purpose			
Planning process of disaster management at national level is strengthened through formulating annual NDMP to facilitate, monitor and coordinate implementation led by NDMA	The draft Annual NDMP including measures to improve the current NDMP are developed and finalized by NDMA.	Approval of NDMP II including actions/programme by NDMA	
Outputs			
1. NDMA's capabilities of information collection and analysis of natural hazards, situation of exposures and vulnerabilities are strengthened.	Disaster risk is assessed on a national scale.	Revision of "NDMA policy guidelines" for conduct of hazard, exposures, vulnerability and risk assessment by the risk assessment unit of NDMA.	
2. NDMA's capabilities of formulation for more effective NDMP-Ⅱ is strengthened while grasping the progress and issues of the current NDMP.	The overall sturcture for NDMP II is formulated.	The validity evaluation report regarding the overall structure of NDMP II by the steering committeee of the Project	
3. A draft long list of priority projects for each disaster type is prepared, and a draft annual NDMP is formulated.	Methods to promote the implementation of priority projects are considered and the substantial contents for NDMP II are developed.	The priotity project list selected on the basis of the substantial contents including the procedure of selection of priority projects	
4. The capacity of NDMA will be improved through the Development of National Emergency Operation Center (NEOC)	Workshops are completed	Programme of workshops and participants list	

Activities	Inputs		Important Assumption
	The Japanese Side	**The Pakistani Side**	
1-1. Confirm hazard intensity and probability of occurrence as the basis for disaster risk information in the current NDMP.	(1) Dispatching the Consultant Team (Tentative) -Chief Consultant/Comprehensive Disaster Risk Reduction plan/ Donor Coordination -Pre-Disaster Investment Plan/Budget Planning -Disaster Risk Reduction/ Local DRR Plan -Hydrological analysis / flood inundation analysis / flood runoff analysis -River Structural Measures -Meteorology/ Hydrology/ Climate Change impact assessment -Urban Disaster/ Drainage System/ Land Use Plan -Earthquake and Tsunami -Institutional Development/ Human Resources Development Plan -Environmental and Social Considerations -Database/ GIS/ Equipment Planning (2) Knowledge Co-creation Program in Japan (If needed)	(1) The Government of Pakistan shall accord privileges and other benefits to the Team in accordance with the Agreement on Technical Cooperation between the Government of Japan and the Government of Pakistan signed on June 8, 2021. (2) The Government of Pakistan shall bear claims, if any arise against the member(s) of the Team resulting from, occurring in the course of, or otherwise connected with the discharge of their duties in the implementation of the Project, except when such claims arise from gross negligence or willful misconduct on the part of the Team. (3) The Government of Pakistan shall assist the Team, if necessary, in customs clearance, exempt from any duties with respect to equipment,	
1-2. Confirm of the vulnerability and exposure data as the basis for disaster risk information in the current NDMP.			
1-3. Identify the need to update and review disaster risk information in the current NDMP.			
1-4. Assess the risk of disaster risk information identified as need to be updated and reviewed in Activity 1-3.			
2-1. Review the basic strategy of DRR and its effectiveness of the current NDMP based on the results of the risk assessment of Activities 1-4.			
2-2. Review and analyze the progress and challenges of priority projects and the Human Resources Development Plan of the current NDMP for the each line ministry/agency			
2-3. Collect information on issues of the current NDMP and analyze it, and consider how to improve the current NDMP based on information that were colected and analyzed.			**Pre-Conditions**
3-1. Collect information on pre-DRR projects currently planned by relevant ministries and agencies.			
3-2. Identify needs for additional project(s) to be considered, based on analysis on the risk assessment in Activity 1-4 and information collection in Activity 3-1, and consider a draft of the projects, including the direction of cost estimation.			
3-3. Compile a draft long list for each hazard type and discuss priority projects with order for each ministry and agency, based on the Activity 3-1 and 3-2.			
3-4. Develop the draft annual NDMP while considering a method to promote the implementation of the priority projects			
4-1. Sharing of knowledge and information in the field of floods and earthquakes disaster risk management for the capacity building of NEOC.			
4-2.Workshops and training will be held in Pakistan and in Japan.			

2. Flowchart of the Project


Year
Month
2022
2023
2024
2025
3 4 5 6 7 8 9 10 11 12 1 2 3 4 5 6 7 8 9 10 11 12 1 2 3 4 5 6 7 8 9 10 11 12 1 2 3 4 5
Ramadan
Common
Outcome 1
Outcome 2
Outcome 3
Outcome 4
[0 -1] Preparation of Work Plan
[0 -2] W/P Submission, Explanation and Discussion
[0 -3] Detailed Plan Formulation Study
[1 -1] Confirmation of hazard Intensity and Probability of
[1 -2] Confirmation of Vulnerability and Exposure Data
[1 -3] Confirmation of Necessity for Updating and Reviewing Disaster Risk Information
[1 -4] Reassessment of Disaster Risk
[2 -1] Review the Basic Strategy and Effectiveness of Current NDMP
[2 -2] Review Current NDMP Achievements and Identify Issues
[2 -3] Study on Improvement Measures in NDMP II
【2-4】 Review the Overall Configuration of NDMP II
[3 -1] Gathering Information on Current NDMP's Disaster Prevention Investment Projects
[3 -2] Study of Draft Plans for Disaster Prevention Investment Groups by Type of Disaster
[3 -3] Preparation of Longlist (draft) for Each Disaster Type and Consideration of Priorities
【3-4】 Preparation of Technical Review Materials for NDMP II
[3 -5] (1) Compilation of the Principles and Basic Items of the Pre-disaster Prevention Investment Plan That Can Be Utilized in Fields Other Than Floods,
and
(2) Recommendations on the Role of Each Ministry and NDMA
[4 -1] Collection and Arrangement of Existing Information on Flood Risks, Investigation of Damage and Evaluation of Existing Flood Countermeasures
[4 -2] Selection of "Priority Project (s)" in the Flood Field
[4 -3] Pre-F/S Level of "Priority Project (s)"
[4 -4] Preparation of Local Disaster Prevention Plans for Flood Areas to be Implemented in Provinces and Districts
[4 -5] Preparation of a Preliminary Disaster Prevention Investment Plan for Flood Sector
Work Plan
Report
Work Plan (Japanese)
Draft Detailed Plan Formulation Study Report
Detailed Plan Formulation Study Report
Project Progress Report 1
PROJECT Progress
Completion Report
Monitoring Sheet
Project Brief Notes
JCC Conference
Seminar
Training in Japan

3. Detailed Activity Plan, Plan of Operation


Period
Work Item
2023
2024
2025
3 4 5 6 7 8 9 10 11 12 1 2 3 4 5 6 7 8 9 10 11 12 1 2 3 4 5 6 7 8 9 10 11 12 1 2 3 4 5
0-1 Preparation of Work Plan (W/P)
0-2 W/P Submission, Explanation and Discussion
0-3 Detailed Plan Formulation Study
1-1 Confirmation of Hazard Intensity and Probability of Occurrence
1-2 Confirmation of Vulnerability and Exposure Data
1-3 Confirmation of Necessity for Updating and Reviewing Disaster Risk Information
1-4 Reassessment of Disaster Risk
2-1 Review of Basic Strategy and Effectiveness of Current NDMP
2-2 Review Current NDMP Achievements and Identify Issues
2-3 Study on Improvement Measures in NDMP II
2-4 Review Overall Configuration of NDMP II
3-1 Gathering Information on Current NDMP's Disaster Prevention Investment Projects
3-2 Study of Draft Plans for Disaster Prevention Investment Groups by Type of Disaster
3-3 Preparation of a Longlist (draft) for Each Disaster Type and Consideration of Priorities
3-4 Preparation of technical review materials for NDMP II
3-5 (1) Compilation of the Principles and Basic Items of the Pre-disaster Prevention Investment Plan that Can be Utilized in Fields Other than Floods, and (2) Recommendations on the Role of Each Ministry and NDMA for Project Implementation
4-1 Collection and Arrangement of Existing Information on Flood Risk, Investigation of Damage and Evaluation of Existing Flood Countermeasures
4 2 Selection of "Priority Project(s)" in the Flood Field
4-3 Pre-F/S Level Study of "Priority Project(s)"
4-4 Preparation of Local Disaster Prevention Plans for Flood Areas to be Implemented in Provinces and Districts
4-5 Preparation of a Preliminary Disaster Prevention Investment Plan for Flood Sector
Time for Submission of Reports, etc. (indicated by △ and the report title)
Report
Work plan
Monitoring sheet
Project brief note
△ Work Plan
△ △ Detailed Plan Formulatior
△ Project Progress Report (1)
△ Project Progress Report (2)
△ Completion Report
Draft Detailed Plan Formulation Study Report
△ Work Plan (Japanese)
△ Monitoring Sheet
△ Monitoring Sheet
△ Monitoring Sheet
△ Monitoring Sheet
△ Monitoring Sheet
△ Monitoring Sheet
△ Project Brief Note
△ Project Brief Note
△ Project Brief Note
Legend
Work in Site
Work in Japan or Work by Local Staffs

Tentative Plan of Operation

Version 2

Project Title: Technical support project to review and updae National Disaster Management Plan (NDMP)

Date 2024 June 11

Activities	Year	1st Year(2022)				2nd Year(2023)				3rd Year(2024)				4th Year(2025)			
		I	II	III	IV	I	II	III	IV	I	II	III	IV	I	II	III	IV
Output 1: NDMA's capabilities of information collection and analysis of natural hazards, situation of exposures and vulnerabilities is strengthened.																	
1-1. Confirmation of hazard intensity and probability of occurrence as the basis for disaster risk information in the current NDMP.	Plan																
	Actual																
1-2. Confirmation of the vulnerability and exposure data as the basis for disaster risk information in the current NDMP.	Plan																
	Actual																
1-3. Identify the need to update and review disaster risk information in the current NDMP.	Plan																
	Actual																
1-4. Assess the risk of disaster risk information identified as need to be updated and reviewed in Activity 1-3.	Plan																
	Actual																
Output 2:The progress and issues of the current NDMP will be grasped.																	
2-1. Review the basic strategy of DRR and its effectiveness of the current NDMP based on the results of the risk assessment of Activities 1-4.	Plan																
	Actual																
2-2. Review and analyze the progress and challenges of priority projects and Human Resources Development Plan of the current NDMP for disaster concerned line ministries and agencies, and identify issues and challenges.	Plan																
	Actual																
2-3. Collect information on issues of the current NDMP and analyze it, and consider measures to improve them for development of NDMP II.	Plan																
	Actual																
Output 3: A draft long list of priority projects for each disaster type is prepared, and a substantial contents for NDMP II is formulated.																	
3-1. Collect information on DRR projects currently planned by relevant ministries and agencies.	Plan																
	Actual																
3-2. Identify needs for additional project(s) to be considered, based on analysis on the risk assessment in Activity 1-4 and information collection in Activity 3-1, and consider a draft of the projects, including the rough cost estimation.	Plan																
	Actual																
3-3. Compile a draft long list for each disaster type and discuss priority projects with order by each ministry and agency, based on the Activity 3-1 and 3-2.	Plan																
	Actual																
3-4. Consider methods for promoting the implementation of priority projects and develop the substantial contents for NDMP II.	Plan																
	Actual																
Output 4: The capacity of NDMA will be improved through the Development of National Emergency Operation Center (NEOC)																	
4-1. Sharing of knowledge and information in the field of floods and earthquakes disaster risk management for the capacity building of NEOC.	Plan																
	Actual																
4-2.Workshops and training will be held in Pakistan and in Japan.	Plan																
	Actual																

4. Expert Dispatch Record

Form 4-3

≪Whether there is a business management group system≫
( )We propose the operation management system as the operation management group (operation chief + deputy operation chief).
(○)An operation management system will be proposed by the operation chief alone.

Engagement Plan/Actual Table

	Area of Expertise	Name	Rating	Plan/ Actual	No of Trips	Total In days	Total In Month
Work in Pakistan	Chief Consultant/Comprehensive Disaster Risk Reduction Plan/ Donor Coordination	Toshihiro Goto	2	Plan	8	262.00	8.73
				Actual	10	262.00	8.73
	Pre-Disaster Investment Plan/Budget Planning	Makoto Yajima	3	Plan	2	74.00	2.47
				Actual	2	74.00	2.47
	Flood Control / River Planning	Kohei Takayama	3	Plan	7	152.00	5.07
				Actual	5	152.00	5.07
	Deputy Chief Consultant/Disaster Risk Reduction/ Local DRR Plan	Tsuyoshi Matsushita	2	Plan	7	30.00	1.00
				Actual	1	30.00	1.00
	Hydrological Analysis / Flood Inundation Analysis / Flood Runoff Analysis	Takafumi Sueishi	5	Plan	3	104.00	3.47
				Actual	3	104.00	3.47
	River Structural Measures	Naoto Nakamura	4	Plan	5	0.00	0.00
				Actual	0	0.00	0.00
	Construction Planning/Procurement Planning/Cost Estimate	Rahmat Ullah	4	Plan	4	116.00	3.87
				Actual	2	116.00	3.87
	Meteorology/ Hydrology	Cristelle San Antonio	5	Plan	2	53.00	1.77
				Actual	2	53.00	1.77
	Climate Change Impact Assessment	Toshihide Endo	4	Plan	1	31.00	1.03
				Actual	1	31.00	1.03
	Urban Disaster/Land Use Plan	Yoshizumi Gonai	3	Plan	2	56.00	1.87
				Actual	2	56.00	1.87
	Drainage System	Hirokazu Sakai	4	Plan	1	30.00	1.00
				Actual	1	30.00	1.00
	Institutional Development/ Human Resources Development Plan	Harumi Tsukahara	3	Plan	2	28.00	0.93
				Actual	1	28.00	0.93
	Environmental and Social Considerations	Shumpei Ichikawa	4	Plan	5	14.00	0.47
				Actual	1	14.00	0.47
	Earthquake and Tsunami Hazard Assessment	Tadashi Kitou	4	Plan	2	60.00	2.00
				Actual	2	60.00	2.00
	Earthquake and Tsunami Risk Assessment/ GIS/ Database	Koichi Hasegawa	4	Plan	3	143.00	4.77
				Actual	6	143.00	4.77
	Earthquake and Tsunami Integrated Assessment	Shaw Tomoko	4	Plan	2	71.00	2.37
				Actual	3	71.00	2.37
				Subtotal	Plan	1224	40.82
					Actual	1224	40.82

Engagement Plan/Actual Table

	Area of Expertise	Name	Rating	Plan/ Actual	No of Trips	2022 3 4 5 6 7 8 9 10 11 12	2023 1 2 3 4 5 6 7 8 9 10 11 12	2024 1 2 3 4 5 6 7 8 9 # # #	2025 1 2 3 4 5	Total In days	Total In Month
Work in Japan	Chief Consultant/Comprehensive Disaster Risk Reduction Plan/ Donor Coordination	Toshihiro Goto	2	Plan						5.00	0.25
				Actual		4/28 (1day)	3/13 (1day) 5/9 (1day)	4/22-23 (2days)		5.00	0.25
	Pre-Disaster Investment Plan/Budget Planning	Makoto Yajima	3	Plan						10.00	0.50
				Actual			12/11-15 (5days)	3/25-29 (5days)		10.00	0.50
	Flood Control / River Planning	Kohei Takayama	3	Plan						1.00	0.05
				Actual		4/28 (1day)				1.00	0.05
	Deputy Chief Consultant/Disaster Risk Reduction/ Local DRR Plan	Tsuyoshi Matsushita	2	Plan						0.00	0.00
				Actual						0.00	0.00
	Hydrological Analysis / Flood Inundation Analysis / Flood Runoff Analysis	Takafumi Sueishi	5	Plan						20.00	1.00
				Actual			12/4-8,11-15 (10days)	9/6,9-13,17-20 (10days)		20.00	1.00
	River Structural Measures	Naoto Nakamura	4	Plan						0.00	0.00
				Actual						0.00	0.00
	Construction Planning/Procurement Planning/Cost Estimate	Rahmat Ullah	4	Plan						0.00	0.00
				Actual						0.00	0.00
	Meteorology/ Hydrology	Cristelle San Antonio	5	Plan						4.67	0.23
				Actual		1/9-1/17 (4.67days)				4.67	0.23
	Climate Change Impact Assessment	Toshihide Endo	4	Plan						0.00	0.00
				Actual						0.00	0.00
	Urban Disaster/Land Use Plan	Yoshizumi Gonai	3	Plan						0.00	0.00
				Actual						0.00	0.00
	Drainage System	Hirokazu Sakai	4	Plan						0.00	0.00
				Actual						0.00	0.00
	Institutional Development/ Human Resources Development Plan	Harumi Tsukahara	3	Plan						0.00	0.00
				Actual						0.00	0.00
	Environmental and Social Considerations	Shumpei Ichikawa	4	Plan						0.00	0.00
				Actual						0.00	0.00
	Earthquake and Tsunami Hazard Assessment	Tadashi Kitou	4	Plan						0.00	0.00
				Actual						0.00	0.00
	Earthquake and Tsunami Risk Assessment/ GIS/ Database	Koichi Hasegawa	4	Plan						10.00	0.50
				Actual				9/13, 10/9-10, 16-17, 1/20, 27, 2/3-5 (5days) (5days)		10.00	0.50
	Earthquake and Tsunami Integrated Assessment	Shaw Tomoko	4	Plan						3.40	0.17
				Actual			2/22-23 (2days) 3/14-15 (1.4days)			3.40	0.17
									Subtotal Plan	54.07	2.70
									Actual	54.07	2.70

Plan	43.52
Actual	43.52

5. Equipment Provided Record

List of Property Lending

Name of Project : Technical Support Project to Review and Update Nagtional Disaster Management Plan
Country : Pakistan
Division in Charge : Global Environment Department

(As of May, 2025)

Name of Property	Standard, Part Number	Quantity	Purchase Price			Date of Inspection Passed	After Completion of Project: Handover/Return
			Purchase Price	Currency	In Japanese Yen		
Desktop PC	Dell Opti Plex 7090(Core I7) , 24inch LED	1	444,600	PKR	252,942	2022/8/16	Handover to NEOC
Inkjet Printer	HPInk Tank 415	1	58,500	PKR	36,823	2022/10/12	Same as the above
ArcGIS (Installed in Desktop PC)	DesktopBasic, 3DAnalyst, Spatial Analyst	1	3,568,500	PKR	2,385,828	2022/11/24	Same as the above
Laptop PC	Lenovo IP Ci5-1135G4	1	149,262	PKR	92,115	2022/12/7	Same as the above
Multi-function Printer	Canon Copier iR-ADV DX C3822i 230V S.No. 2YW03061	1	1,006,200	PKR	498,653	2023/2/9	Same as the above

CERTIFICATE OF DONATION OF OFFICE EQUIPMENT

To: JICA Pakistan Office

Re: Technical Support Project to Review and Update National Disaster Management Plan (NDMP)

This certificate of donation of office equipment is to certify that the equipment in the attached list, which shall be utilized for the "Technical Support Project to Review and Update National Disaster Management Plan (NDMP)" (hereinafter referred to as "Project"), have been handed over properly to NEOC/NDMA (National Emergency Operation Center/National Disaster Management Authority), as of 26th February, 2025.

Attached: List of Equipment

No.	Name of Item		Qty	Place of installation	Date of handover
1	Desktop PC	Dell Opti Plex 7090(Core I7), 24inch LED	1	Office of NEOC/NDMA	26th February, 2025
2	Inkjet Printer	HPInk Tank 415	1	Same as the above	26th February, 2025
3	ArcGIS (Installed in Desktop PC)	DesktopBasic, 3DAnalyst, Spatial Analyst	1	Same as the above	26th February, 2025
4	Laptop PC	Lenovo IP Ci5-1135G4	1	Same as the above	26th February, 2025
5	Multi-function Printer	Canon Copier iR-ADV DX C3822i 230V S.No. 2YW03061	1	Same as the above	26th February, 2025

26th February, 2025

Mr. Toshihiro Goto
Project Manager
CTI Engineering International Co., Ltd. Japan

Brig. Badar Zaman
Director General,
NEOC/NDMA

6. Others
(Memorandum of Meeting)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	30 May 2022, 11:00 ~ 12:00(PST)	
Interviewee		
Place	NDMA	
Participants	NDMA	Abdul Latif (Deputy Director)
	Project Team (PT)	Goto, Matsushita, Mukhtar Ahmed Khan
Memo	・ Project Team (PT) asked about the current situation about the communication system, and NDMA answered as follows. ➢ NDMA： Mr. Abdul Latif explained that NDMA received information from PMD. NDMA send information to the PDMA and DDMA. Then DDMA send officer to that area and through mosque loudspeaker also used. ➢ NDMA： Early Warning from NDMA composes the SMS and request to PTA (Pakistan Telecommunication Authority) send Mobile SMS to common people of specific area. ➢ NDMA： He explained that JICA implemented very good project for flash flood in Rawalpindi on Lai Nullah. He also explained the very good system proposed NEWS Plan (Page No.120). However, this system has not implemented due to the lack of resources and also because NDMA don't have their own building. ・ NDMA： Mr. Abdul Latif also shared the concept note, construction of NEOC including NDMA Offices, NIDM and NDRF facilities. This concept note has been already shared with JICA Pakistan office. NDMA has own land in Red zone area. ・ NDMA： Mr. Abdul Latif also explained that the land which now NDMA is working is property of ERRA office and this ERRA land is not proper site also, because this land is originally fixed for green belt area. However, in 2005 Earthquake, the government allocated this land in emergency use for ERRA office.	
Meeting Materials		

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	1 June 2022, 11:00 ~ 12:00(PST)	
Interviewee		
Place	Pakistan Meteorological Department (PMD)	
Participants	PMD	Zahid Rafi (Director, Seismic Division)
	Project Team (PT)	Matsushita, Mukhtar Ahmed Khan
Memo	・ Project Team (PT) explained about the Project. And request support of PMD and participation to Technical Committee and Steering Committee. ➢ PMD: Mr. Zahid Rafi said when the PT needs the support, PMD will be available any time, and they will join the meeting of committees above. ・ PT asked about the total numbers of monitoring system in Pakistan for Earthquake. ➢ PMD: Total number of earthquake morning station is 31. However, some stations are not working due to some technical issue. This seismic system is installed by German. Some monitoring stations are directly connected with PMD Islamabad. Some stations send data through sim based internet and the others send data through satellite. ➢ There are 31 seismometers (observation stations) throughout Pakistan. Measurement data is sent and received via the Internet, mobile phones (mobile WiFi), and satellite communications (contract with a satellite communications company. PKR10,000/month/station. Installed at 7-8 observatories that are outside the coverage area of the Internet and are very difficult to access). ➢ PMD: Mr. Zahid Rafi promised to share coordinates of all earthquake monitoring station to the PT. ・ PT asked present status of updating work of seismic monitoring system. ➢ PMD: Four (4: Gwadar, Ormara Navy Base, Karachi, Pasni) GPSs installed along with Makran Coastal (Makran Subduction Zone), Balochistan, for monitoring the Tsunami, because in history 1945 there was big Tsunami. This GPS system was installed with aid of Turkish Expert. It is monitoring the subduction zone and also it gives PMD the value of stress. In Gwadar port, 2 seismic station and 1 GPS system was installed. ➢ PMD: Mr. Zahid Rafi informed that he got JICA's scholarship for Master Degree program at Tsukuba University in Japan about 20 years ago. ➢ PMD: In 2011 big Tsunami hit Japan and, after that calamity, Japan invited many experts from whole of the world. Mr. Zahid Rafi also invited to Japan to see the influence of Tsunami in Japan.	

	➢ PMD: He also informed the PT that there is one event in Northern Area of GLOF (Glacial Lake Outburst Flood) washed out the bridge. · Both parties moved to National Seismic Monitoring Centre ➢ PMD: Mr. Zahid Rafi informed that PMD is monitoring present earthquake situation of the world. Present monitoring system was set up by German company (Seiscom) in 2021 under UNDP fund. (Of the seismological observation stations in Japan, only data from the Inuyama observation station is shown.) ➢ Earthquake monitoring system had been established in 2008 by Government of Pakistan budget. The system was updated in 2017 with support of Islamic Development Bank (IDB) and in 2021 with help of UNDP. ➢ PMD could update the system when they have budget with support of Government of Pakistan or international organization. · PMD: Mr. Zahid Rafi insisted that they need one more seismic system then they can run both and compare which is efficient. · PMD: Mr. Zahid Rafi shows the PT the morning system and also details briefs about system.
Meeting Materials	

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	03 June 2022, 11:00 ~ 12:00(PST)	
Interviewee		
Place	NDMA	
Participants	NDMA	Abdul Latif (Deputy Director)
	Project Team (PT)	Goto, Matsushita, Takayama, Mukhtar Ahmed Khan
Memo	· Project Team (PT) asked the NDMA's opinion about the current draft of R/D, M/M and NDMP-1 as follows. ➢ PT : PT asked that what portions of current NDMA-I NDMP thought insufficient. ➢ NDMA : Mr. Abdul Latif explained that DRM is very old and its related to response. The current NDMP consists of only natural disasters, it includes the industrial hazards only as the reference. In addition, comparing to floods, concrete actions for the other kinds of disasters are not described in detail. Mr. Abdul Latif would like the PT to include entire all kinds of disasters at all levels in the investment plan, ➢ NDMA : Mr. Abdul Latif recommended the PT to hire local consultants. He also suggested that PT prepare the TOR for the procurement of local consultants and NDMA advertise it on their website and in daily newspapers also (Advertisement in Newspaper shall be provided by PT). Then the NDMA and PT team will take the interview jointly. ➢ PT : PT asked that what NDMA thought "All Levels" stood for. ➢ NDMA : NDMA thinks that "all levels" are including provinces and districts. NDMA would like to show the concrete action plan in such level. · PT : PT asked about data of damages from disasters. ➢ NDMA : Mr. Latif relied that they don't have the data management system, they used manually recorded data. Mr. Latif added that priority shall be given to the building of NDMA and NEOC.	
Meeting Materials		

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	03 June 2022, 11:00 ~ 12:00(PST)	
Interviewee		
Place	ONLINE	
Participants	UNDP	Mr. Aman Ullah, Mr. Usman
	Project Team (PT)	Goto, Matsushita, Takayama, Mukhtar Ahmed Khan
Memo	· Project Team (PT): PT explained about the project. and ask questions about ongoing projects of UNDP in Pakistan in the field of Disaster and second that PT request to UNDP to introduce ongoing or past good projects on disaster management sector. · UNDP: UNDP replied on the question as follows ➢ UNDP works on DRR on two points, one is on National Climate Change Policy and second is the DRR Strategy Program. They are working on the DRR Strategy Program closely working with NDMA. Previously the project was funded by STO & EU and now the current project on DRR is funded by Japan. This current project focus on Tsunami Warning, Capacity Building ➢ UNDP is also implementing the GOLF flood project. This project is implemented in the 10 valleys of KPK and the 14 valleys of GB. The project focuses on Early warning systems. These EWS are to be placed on high reaches of glaciers lake and downstream. This system will be connected with NDMA and GBDMA and PDMA. PMD will be the custodian of this EWS and responsible for connectivity after the project. ➢ Pakistan has 14 climate change international agreements. One of them is sustainable work on SFDRR. UNDP sends a report to NDMA every year. ➢ UNDP did a study not directly with flood protection but the effects of climate change on Pakistan water scarcity and water provision. this study conducted by the World bank. ➢ UNDP is working on risk finance, no project for this but discussion with ministry of finance is ongoing. ➢ Green line project by WWF in Punjab and Sindh is on going. · PT: National Climate Change Policy was updated last year. Was the previous version of the policy studied by UNDP? · UNDP: Yes, UNDP drafted the previous version in 2012. The update last year was to meet the current requirement such as change in the federal government. · PT: During this project, especially in the Pre-F/S on flood control project, the influence of the climate change will be considered. UNDP also has the data	

	and analysis results which are the basis of the updated climate change policy, doesn't it? · UNDP: UNDP has some of them. However, GCISC office of MoCC has good data. · UNDP: UNDP advised that the activity in each province also is important to resist the climate change. · PT: If UNDP has good practice on community-based DRM activity, PT would like to ask UNDP to share the information. · UNDP: UNDP can give support to the team. · PT: PMD has an early warning system for GLOF. Is UNDP in charge of establishing this early warning system? · UNDP: Yes, however the main department is PMD and they have early warning system for other kinds of disasters. · PT: PT would like UNDP to join the steering committee in this project. · UNDP: Noted it.
Meeting Materials	

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	15 September 2022, 15:30 ~ 16:30(PST)	
Interviewee	Mr. Muhammad Tauqeer Nawaz (Director Planning, Urban Planning Directorate)[1]	
Place	CDA (Capital Development Authority) office	
Participants	CDA	Mr. Muhammad Tauqeer Nawaz (Director Planning, Urban Planning Directorate)
	Project Team (PT)	Mukhtar san, Gonai
Memo	1. Major disaster types recognized by CDA are floods, earthquakes (and recently heatwaves also). 2. In 1990s, the CDA experienced large-scale floods, and since then there have been no large-scale floods, but in recent years it has experienced large-scale floods. 3. Flooding occurs in the settlement development area near rivers in the urban area. 4. Based on recent flood damage, the Federal Government (cabinet office) has instructed all Development Authorities to conduct a Hydrogel Study. 2 Hydrogel Study were conducted by 2 experts (Mukhtar san took a memo of their names). 5. Zoning map and Islamabad laws (including Zoning Regulation) can be provided by CDA through official letter from NDMA or JICA. 6. Mr. Nawaz introduced a DG of Architecture Directorate who is familiar with land use, building code and disaster risk management in the planning. (Mukhtar san is going to make an appointment of the DG) 7. Other than the above, the CDA-Urban Planning Directorate has not implemented any land use regulation/control for the disaster risk reduction and management	
Meeting Materials		

[1] Ph: 051-9252035, Cell: 0334-5223499, urfee1977@gmail.com

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	15 September 2022, 9:45 ~ 10:45 (PST)	
Interviewee		
Place	Pakistan Meteorological Department (PMD)	
Participants	PMD	Zahid Rafi (Director, Seismic Division) Mr. Nasir (Seismic Monitoring Centre)
	Project team (PT)	Hasegawa、Kito、Muktar
Memo	・ We explained the revision of seismic hazard data in this project to the Earthquake division of the PMD. ・ We asked whether they have a PGA hazard map like the earthquake hazard map (Spectral acceleration) in the appendix of Building Code 2021. It has been revealed that there is an updated version (2021) of Seismic Hazard and Zonation (2007), and we will be able to receive the data soon. ・ The latest earthquake catalog data (-2022) will be provided. ・ Observational data of the 1945 Makran tsunami (including old documents, etc.) can also be provided. ・ Regarding the data transfer, we will make a list of necessary data to request and send it to the staffs in the seismic monitoring center by e-mail. ・ Regarding the data format, ASCII (GMT format) files will be provided in addition to the image format. ・ We have received an offer to hold a half-day technical discussion including tsunami modeling from the next meeting onwards. The date of the meeting is not fixed yet. ・ PMD explained that in1945 there is big earthquake in Pakistan, but that time no system installed. And more explained that they interviewed the people who live at that area and get the information. PMD doesn't have documents of this event. ・ For Tsunami modeling, PMD does a lot of work on Gawader and Karachi with help of NGI Norway. PMD is also a member of IOC, France, NOVA - USA and MOST with the data of Arabian Sea. JICA Expert use their own way of work and compare it with PMD system.	
Meeting Materials		

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	16 September 2022, 09:00 ~ 12:00(PST)	
Interviewee	Dr. Tahir	
Place	PMD Planning Division	
Participants	PMD	Dr. Tahir (Director for Planning), Mr. Ismail
	IMC	Uchida
	Project Team	Takayama, Endo, San Antonio, Mukhtar
Memo	Notes: The main purpose of the meeting was to inquire about the updates in PMD in relation to the NDMP in 2012 (refer to progress table prepared by Endo san). Some information was added to better set the context of the flooding situation in Pakistan as well as other problems faced by the PMD. a. Progress on proposed projects/programs in the NDMP 1. Strengthening forecasting and early warning systems 1.1. *SMRFC*: Completed 1.2. *Satellite image + hazard maps*: Completed; IFAS System, UNESCO GFS Data, ICON 1.3. *RFFWC*: No update 1.4. *LFFFC*: No update 1.5. *Landslide EWS*: No update 1.6. *Tide level monitoring system*: No update 1.7. *GLOF forecast and EWS*: Scaling Up of GLOF in Northern Pakistan; Phase 1 is completed (one district each from Gilgit-Baltistan and Khyber Pakhtunkhwa); Phase 2 is ongoing (expanded to 18 districts in GB and KP) 1.8. *Seismic intensity reporting system*: Project: MRC Project (15 station, 2 gps stations) 1.9. *Radar system along coastal*: Gawadar (?) radar system 1.10. *Radar system at designated locations:* Multan (ongoing, under negotiation) 1.11. *Replacement of radars*: Karachi: done; RY Khan (under negotiation); DI Khan (approved but no funds yet) 1.12. *Expansion of AWS*: 300 AWS (PHCSP, World Bank), 45 stations, 65 station requested from Japan to replace existing stations 1.13. *Expansion of realtime rainfall and water level monitoring*: WAPDA, FFC for confirmation; based on stations in 1.12 2. Hazard Maps in Target Locations 2.1. *Local hazard maps*: To be confirmed (FFD) 2.2. *Landslide hazard maps*: No update 2.3. *Training on tsunami simulation and hazard maps*: UNDP Project: Strengthening of Tsunami EWS for Coastal Areas of Pakistan 12.5M (2021), 5 stations seismic monitoring stations installed, 5 warning posts (2 in Karachi); UNDP: training in Karachi (3-4 times); MRC: training	

	of 12 personnel in March 2022, another group in 2023 3. Strengthen EWS Dissemination 3.1. *Innovation in communication*: Whatsapp chat group for dissemination, established by PMD 3.2. *Development of EWS for health*: N/A 3.3. *Weather info broadcasting system:* From Japan Project: completed; 2 TV studio (1 in Karachi, 1 in Islamabad), FM radio stations, website, YouTube channel; refer to PHCSP project 3.4. *Finalization of SOP of cyclone EWS:* No update 4. Develop Capacity of EWS and Evacuation System 4.1. *Research on glacial lakes:* GLOF project 4.2. *Weather forecast guidance system*: Japan project (Endo san) 4.3. *Community EWS drills*: N/A 4.4. *Education program for PMD staff:* Partially b. <u>Other information</u> · Raining in 2022 is due to too much rain, southern areas are inundated. There is urgent need for flash flood warning center. · For the "No update" items above, the reason is mostly due to funding issues; proposal was submitted in 2013 (proposed by PMD, submitted to FFC, ~36 B project for FFWC). · PMD requested WMO to help PMD for assessment and was referred to JICA. · 2022 monsoons: more than 300 million affected, several vulnerable sectors (poor), 40billion dollars of loss · Last NDMP is outdated, there have been several changes such as land use, urban sprawl (merging cities), growing population and other problems. Both monsoon and heat waves are major disasters experienced in Pakistan. · From Uchida san: Lahore station can only observe the northern Pakistan area, there is need to add other radar monitoring systems. 80% of storms from India go to north, 20% go to west or south of Pakistan. · There had already been more than 30 floods in 2022, including GLOF. Flood control works are not sufficient; large volumes of water were not anticipated.
Prepared by	Cristelle San Antonio (Meteorology/Hydrology)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	19 September 2022, 14:00 ~ 15:00(PST)	
Interviewee	Mr. Azam Khan Lodhi (Director General of Design, CDA)[1]	
Place	CDA (Capital Development Authority) office	
Participants	CDA	Mr. Azam Khan Lodhi (Director General of Design, CDA)
	Project Team (PT)	Mukhtar san, Gonai
Memo	1) There were 2 causalities by flood disaster, 2021 July in in E11 of Islamabad. E11 is a private development block. The ROW[2] of river in E11 was narrowed by private development, which was not complied with the regulation/standard of CDA's development permit. According to the JICA's study[3], a structural countermeasure for flood was recommended on E11. After the disaster, "E11 Hydrological Study" has been started by the army's Directorate of Design and Consultancy (DD&C).Flood risk areas were identified by the first draft interim report of the "E11 Hydrological Study". CDA has advised to residents in E11 to avoid living in the risk areas on ground level through Deputy Commissioner Office. The report of "E11 Hydrological Study" can be shared with JICA team once it was passed the official protocol. (Mukhtar san keeps to following to obtain the report) 2) By-Laws & Regulations managed by CDA can be confirmed CDA's website[4]. Submission of TIA (Traffic Impact Assessment) and HIA (Hydrological Impact Assessment) is obligatory in order to obtain a building/development permit. For more detail of the regulation/protocol of the building/development permit, Arshad Chohan (Director of Housing Societies) would be helpful for JICA team. 3) There is no land use control considering disasters other than the above. There are concerns about the impact on land price/value if land use restriction based on hazard maps is introduced. 4) The precipitation assumptions are not appropriate for existing development/infrastructure standards 5) Relationship between CDA and Ministries: Although CDA complies with the federal legal system, but the federal legal system is General. The guidelines for development/building permit, official protocols and others can be determined by each Development Authority. CDA can implement CDA's project by own budget. CDA does not need to ask Ministries to obtain budget. Partly due to the progress of decentralization, each development authority implements a master plan/land use control by its own budget and regulation.	
Meeting Materials		

[1] 9252987, 0332-5143311, azlodhi@hotmail.com
[2] Right of Way
[3] The Study on Comprehensive Flood Mitigation and Environmental Improvement Plan for the Lai Nullah Basin, 2003
[4] https://www.cda.gov.pk/resource_center/bylaws.asp

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	20 September 2022, 13:00 ~ 14:00(PST)	
Interviewee	Mr. Tahir Meo, Director Land Use, Building, RDA)	
Place	WASA/RDA (Rawalpindi Development Authority)	
Participants	RDA	Mr. Tahir Meo (TEL: 03005158795, lookat2030@gmail.com) Mr. Ali Raza (GIS specialist)
	Project Team (PT)	Mukhtar san, Kitou, Hasegawa
Memo	1) Confirmation of earthquake hazard map usage in local level, especially in cases of inspection of building construction based on Building Code ・ He said all buildings are judged on the basis of 9.2 on the Richter scale[1]. ・ A copy of the document stating this can be provided. ・ RDA understand Building Code but did not clearly recognize seismic hazard map referred in the Building Code. ・ RDA has a GIS system to confirm site of construction. But it seems not to have the seismic hazard map in it. 2) Preparation of seismic hazard map, building data ・ RDA doesn't have seismic hazard map. PEC (Pakistan Engineering Council) might have it. ・ RDA has a kind of building database, which seems to be confidential. It is better to discuss again on how to share it. 3) Request for sharing the materials or data ・ Latest technical document on construction procedure including seismic ground motion to be considered will be shared through Mukhtar. He said it is PDF file. ・ Regarding the GIS system in RDA, very short discussion with GIS specialist using their laptop PC was done. But it was not enough to communicate. So additional survey on GIS in RDA was proposed through Mukhtar. 4) Request of introduction of responsible person ・ For nationwide building database, it is advised to ask Planning Commission[2]. ・ Regarding PEC, it is better to continue to search the responsible person for further information.	
Meeting Materials	Computer Screen	

[1] 9.2 is the moment magnitude specific to one earthquake. So it is inappropriate as a standard seismic ground motion for buildings.
[2] RDA did not mention details. But it is an important point who check seismic hazard parameter for new construction of buildings.

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	20 September 2022, 13:00 ~ 14:00(PST)	
Interviewee	Mr. Tahir Meo, Director Land Use, Building, RDA)[1]	
Place	WASA/RDA (Rawalpindi Development Authority)	
Participants	RDA	Mr. Tahir Meo
	Project Team (PT)	Mukhtar san, Gonai
Memo	1) Promotion of use of hazard risk information in land-use planning and zoning programs ・ The concerned hazards are flood, water pollution ・ RDA does not have the flood hazard map. 2) Preparation of the guidelines for land use planning in vulnerable areas ・ RDA does not make the guidelines for DRRM, because RDA territory does not have the vulnerable areas[2]. 3) Request for sharing the materials and data: , land use (existing, proposed), documents of urban planning and legal system such as land use regulations, guidelines ・ Land use map and following documents will be shared through Mukhtar san ・ Panjab Development Authority Land Use (Classification, Reclassification and Redevelopment) Rules, 2021 ・ RDA Building and Zoning Regulations 2019 ・ Panjab Development Authorities Private Housing Schemes Rules 2021 4) Request of introduction of responsible person who works on the land use planning/control for disaster risk management ・ There are technical support and budget support from Panjab government (not Federal) ・ Secretary of Housing and Urban Development of Panjab would be a suitable person to collect the information. ・ Mukhtar san makes meeting appointment of the secretary through the Chairman of RDA.	
Meeting Materials		

[1] TEL: 03005158795

[2] RDA territory is adjacent area of Municipal Corporation, Cantonment. The area of Municipal Corporation, Cantonment contains the flood prone area. The territory of RDA does not have the flood prone area

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	21 September 2022, 10:00 ~ 10:30(PST)	
Interviewee	Dr. Muhammad Wazir	
Place	PMD GLOF Unit (under Research and Development)	
Participants	PMD	Wazir (Director)
	Project Team	Takayama, Endo, Cristelle, Mukhtar
Memo	Agenda: - Inquire about PMD's progress in relation to the 2012 NDMP, specifically for GLOF - Obtain other information related to GLOF Minutes: a. Information on Phase 1 of the "Scaling Up of GLOF Risk Reduction in Northern Pakistan" Project - Project by the UNDP which started in 2009 and ended in 2014 - Involved three (3) valleys in Gilgit-Baltistan (GB) and Khyber Pakhtunkhwa (KP) - Project Components: o 51 AWS specifically for GLOF o Rainfall, discharge, and lake water level gauges were also installed in the three valleys b. Reasons for implementing Phase 2 of the GLOF Project - The three valleys are not sufficient to properly assess risks from GLOF in the northern areas - Just this year, there had already been more than forty (40) reported flash floods that are attributed to GLOF; it should be noted that apart from the glacial lake outburst, other factors such as land degradation, climate change, etc. also contributed to the flash floods - There are apparent data gaps that could be filled in with the installation of more rainfall stations and other instruments c. Information on Phase 2 of the GLOF Project - Continued by UNDP, with funding from the Global Climate Fund (GCF) - Started in 2016 and aimed to be completed in January 2024 (this was supposed to end earlier, but was already extended due to delays from unforeseen circumstances such as the pandemic) - The project is expanded from three (3) valleys in Phase 1 to a total of	

twenty-four (24) valleys in Phase 2, all within GB and KP

- Project Components:
  - 50 additional meteorological stations (not just AWS)
  - 400 more instruments/gauges
  - The EWS would be more integrated and holistic, with three (3) proposed Control Centers in Peshawar, GB, and Islamabad
  - One of the considerations in the study is the determination of threshold values for temperature in relation to GLOF forecasting/EWS, for all 24 valleys
  - PMD is only in charge of the technical assistance for installation and management of the stations. The main body involved in this project is the Ministry of Climate Change (MoCC), in coordination with other government agencies related to irrigation, forestry, land use, etc.

d. Challenges faced by PMD in terms of GLOF

- The data gap is a major issue. There are times when the flow data recorded do not match the rainfall observed in the stations, which implies that some information is not captured by the existing stations/gauges.
- The sustainability of equipment and stations (i.e., operations and maintenance or O&M), is also an issue. There are problems with downloading the data due to accessibility issues, and some instruments are not maintained/monitored often.
- In relation to the O&M issue, there is a need for a specialized unit for GLOF, particularly in data collection. The current GLOF unit is only a small portion of the PMD, and there is not enough trained manpower to assist in GLOF

e. Progress on NDMP Intervention 3 (Establish national hazard and vulnerability assessment

- Item 1.1. Preparation of vulnerability atlas and Item 2.1. Preparation of vulnerability analysis and hazard maps: the studies related to GLOF risks and vulnerability assessment are done by Karakoram International University (KIU) located in GB
- Item 3.1. Research on impact of climate change on glaciers and ice caps: there are long-term climate data from the existing stations, but glacial information is limited (data only started in 2010). There are data gaps and thus it is difficult to determine the relationship between temperature rise and GLOF. When looking at the history of GLOF events, however, there is evidence that the lake water level, as well as the intensity and frequency of GLOF increased over the years.
- Item 3.2. Establishment of GLOF EWS: Data are collected from the existing stations in the three valleys (from GLOF Project Phase 1). They then issue alerts to the communities, through their website, and through agencies like the GB DMA, NDMA, etc.

f. Recommendations for NDMP

- Strengthen the PMD, particularly EWS and data collection. It should be

	emphasized that disaster preparedness (through the PMD and other related agencies) is important in disaster management. - For GLOF, in particular, there is a need for more trained human resources. There are not that many graduates of glacial studies/glaciology that could assist in GLOF.
Prepared by	Cristelle San Antonio (Meteorology/Hydrology)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	22 September 2022, 10:00 ~ 10:30(PST)	
Interviewee	Dr. Tahir	
Place	PMD Planning	
Participants	PMD	Dr. Tahir, Director
	Project Team	Takayama, Endo, Cristelle, Mukhtar
Memo	Agenda: - Clarifications on information mentioned during the last meeting (16 Sept) and other information Minutes: *Intervention 4:* Item 1.3. Regional FFWC - Establishment of Regional Flood Early Warning Centers in Quetta, Karachi, Peshawar, Gilgit, Multan, and Muzaffarabad - six offices, one for each district - equipment, IFAS system, data and satellite communication system - previous planning for only four districts completed, but planning for the six districts is ongoing Item 1.4. Local FFWC - Establishment of Urban Flash Flood Guidance System in Major Cities of Pakistan - planning has just started - includes observation system, early warning system Item 1.5. Landslide EWS: no planning started Item 1.6. Tide level monitoring - tide level gauge; looking for funding (China proposed before, but was eventually scrapped by the government) - existing two HFRs (high frequency radar) not enough - need buoys to monitor; sometimes there is no visibility due to swell Item 1.7. GLOF - the GLOF Project involves other agencies for disaster rehabilitation and resilience. For PMD, they are more focused on the early warning systems Item 1.8. Seismic monitoring - Reverse Linkage between Marmara Research Centre (MRC), Turkey and Pakistan Meteorological Department (PMD), Pakistan - Seiscomp3 seismic analysis software, 5 GPS stations at Karachi, Gwadar,	

	Pasni & Ormara, 5 seismic sensors in Balochistan, and training of PMD personnel Item 1.9. Coastal radars - planning completed; project is partially completed - Planning already completed for Gwadar, Sialkot, Chera, DI Khan, Quetta, - 2 radars still ongoing ("hanging" for past four years) Item 1.10 Radars at designated locations: hanging Item 1.11 Replacement of radars - Planning is completed, but rejected - New project proposal: Replacement of Existing C-Band Analog Ordinary Radar with C-Band Dual Polarized Doppler Weather Radar at Dera Ismail Khan, Khyben Pakhtunkhwa (KPK) Item 1.12 Expansion of AWS - World Bank project - budget for only several stations at a time - total of 635 stations in subdistricts - JICS total 110 stations; 200+ remaining - 32 stations; 5 installed, 27 remaining - 45 hybrid stations from JICS; 65 additional stations (not yet sure, will depend on decision of JICA) - some stations are not calibrated Item 2.3. Training on tsunami simulation - UNDP Project: Strengthening of Tsunami EWS for Coastal Areas of Pakistan 12.5M (2021), 5 stations seismic monitoring stations installed, 5 warning posts (2 in Karachi) Item 3.3 Broadcasting system - CMA donation from China, not a project (completed in 2020) - 2 stations: 1 in Karachi, 1 in Islamabad *Intervention 3:* Item 1.1. Vulnerability atlas: seismic hazard maps available Item 3.2. Establishment of GLOF EWS: GLOF Project We also met with the chief of FFD for a possible visit to Lahore. The tentative date of the meeting is on Monday (September 26), any time of the day.
Prepared by	Cristelle San Antonio (Meteorology/Hydrology)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	26 September 2022, 10:30 ~ 11:30(PST)	
Interviewee		
Place	Bureau of Statistics, Statistics Section	
Participants	BoS	Mr. Sarfaraz, Mr. Saeed
	Project Team (PT)	Mukhtar san, Hasegawa
Memo	1) Discussion on purchasing Census Data ・ PT requested to purchase Population and Housing Census Tables in Excel format in District and Tehsile level. ・ PT also explained using ArcMap that shapefile of administrative boundary is Tehsile level. ・ BoS agreed the request that the data in Excel format in Tehsile level. 2) Preparation of purchase Census ・ PT will prepare a Request Letter and NDMA submit it to BoS. ・ 3) Request for materials or data ・ TABLE OF CONTENTS. (Hard copy is received) ・ Price of the data is presented only verbally, 2500 PKR per District. 4) Request of introduction of responsible person ・	
Meeting Materials		

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	26 September 2022, 15:00 ~ 16:00(PST)	
Interviewee	Mahr Sahibzad Khan	
Place	FFD Lahore	
Participants	FFD	Mahr Sahibzad Khan (Director General), Dr. Khalid Malik
	Project Team	Goto, Matsushita, Takayama, San Antonio, Saeed
Memo	Agenda: - Inquire on progress of NDMP-I (Interventions 3 and 4) - Acquire other information and relevant data Minutes: ***a. Intervention 3:*** 1.1 Preparation of vulnerability atlas (for flooding, GLOF) - FFD has no atlas 2.1 Preparation of vulnerability analysis and hazard maps (for flooding, GLOF) - FFD has no vulnerability analysis - Flood routing model was shown to us instead; this contains a schematic diagram of the river network and the estimated travel time from one location (barrage) to another - Estimation of travel time was done through manual recording of water level; factors that may affect travel time are river morphology and change in river bed - According to FFD, there is a need to revise the flood limits set by the irrigation department for low, medium, high, and very high flood level 3.1 Research on impacts of climate change, and 3.2 Establishment of GLOF EWS - Refer to Scaling-up of GLOF Risk Reduction in Northern Pakistan Project - There are existing studies on the relationship of temperature and GLOF - According to FFD, temperature rises during February; this year, there had been six heat warnings, and GLOF was already expected due to the observed snow melt - This year, the Tarbela reservoir reached dead level during the heat waves, and the reservoir was filled again due to snow melt (not precipitation) ***b. Intervention 4:*** 1.5 Establishment of landslide forecast and warning system - Integrated in the GLOF early warning system (including flash flood) 1.6 Establishment of a new tide level monitoring network including a data	

	communication system - There are two HFRs (high frequency radar) along the coast for tide monitoring - There is an existing cyclone center in Karachi (Tropical Cyclone Warning Center or TCWC) 1.7 Establishment of a GLOF and snow melt flash flood forecast and warning system with an avalanche advisory information system - Refer to Scaling-up of GLOF Risk Reduction in Northern Pakistan Project 1.13 Expansion of the real-time rainfall & water level observation network - Refer to JICS project (mentioned in previous interviews) 2.1 Preparation of hazard maps and capacity development against local flash floods in vulnerable areas and 2.2 Preparation of landslide hazard maps based on the topographical and geological analysis at vulnerable locations - There are no hazard maps; they only have a list of areas that are vulnerable to landslide and flash flood based on location (hilly areas and steep slopes) 3.4 Finalization of SOP of cyclone EWS - Refer to TCWC in Karachi 4.1 Enhancement of research activities for snow/glacier/glacial lakes - No research done by FFD 4.5 Education Program for advanced meteorology and hydrology for PMD staff - No action ***c. Other Information:*** a. Flood Simulation Model Procedure: No simulations done b. Flood Forecasting and Sharing Procedure: FFD will share the procedure document c. Data Communication/Transfer System: Precipitation and flow data are transmitted to more than 300 stakeholders through website, social network, etc. d. Activities of on-time Flood Monitoring Team during Flood: FFD showed us the existing flood monitoring center where there are monitors showing the real-time satellite and radar images, as well as flows within some rivers e. Activities and views regarding Climate Change and f. Challenges faced by the agency and recommendations for improvement: - Climate change makes flooding disasters worse - Pakistan has very limited resources; there is a need to improve the technology, infrastructure, and capacity building of institutions - Simulation models and increased radars are also necessary to better prepare for the flood disasters

	d. Data Collection a. GIS data of stations' locations, list of stations which are used for flood forecasting b. Updated list of historical floods (with corresponding info on rainfall, flow, and extent) c. Climate classification (include basis/analysis used) d. Climate data: normals and extremes e. Daily rainfall data f. Annual maximum rainfall g. Rainfall analysis h. River basins and network i. Flood hazard maps j. GLOF data k. Water level/flow data - All these data could be provided by the FFD. Dr. Khalid works in Islamabad and we could meet him to get the data.
Prepared by	Cristelle San Antonio (Meteorology/Hydrology)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	27 September 2022, 10:00 ~ 11:00 (PST)	
Interviewee		
Place	Pakistan Meteorological Department (PMD)	
Participants	PMD	Mr. Nasir (Seismic Monitoring Centre)
	Project team (PT)	Hasegawa、Kito、Mukhtar
Memo	・ We had discussions with Mr. Nasir to clarify the PGA data format. They gave us plot tools to create PGA maps. ・ It turned out that they do not have PGA data (2021), and they explained that one of students (NED University of Engineering and Technology, Karachi, Pakistan) might have the PGA data. They said that the work of the student has been published as a research paper entitled "Probabilistic Seismic Hazard Analysis Based Zoning Map of Pakistan". ・ We asked them whether they have a map of active faults in Pakistan, and we received only shape files of the major faults without information of any occurrence probability and return period. They have suggested that we could get more detailed data if we ask Geological Survey of Pakistan (GSP). ・ The PMD can provide us the tsunami inundation data (calculated one) for 4 major cities along the coast. ・ Regarding the tsunami hazard-related work, the PMD has not done any tsunami hazard with probabilistic approach. ・ We asked about the seismic network of the PMD and it turned out that their network consists of 22-25 broad-band seismic stations and no strong seismometers. They have, however, a plan to increase the number of seismic stations in the near future. ・ We will send a list of data request as a formal letter to the PMD, and they will prepare the data that we have not received yet, if they have.	
Meeting Materials		

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	27 September 2022, 10:00 ~ 12:30(PST)	
Interviewee	Syed Khursheed Ahmed Shah, Qazi Mubasher Maqsood	
Place	WAPDA, Lahore	
Participants	WAPDA	Syed Ahmed and other personnel in hydrology and meteorology
	Project Team	Goto, Matsushita, Takayama, San Antonio, Saeed
Memo	Agenda: - Inquire on progress of NDMP-I (Interventions 3 and 4) - Acquire other information and relevant data Minutes: ***a. Presentation on WAPDA Organization and Functions*** *2022 Flooding* - Rainfall is definitely larger than average. Flooding is expected in the northern areas since historical floods were observed there. However, this year, flooding was observed in the southern areas as well. - Punjab and Sindh, which typically experience 4-5 in/year, experienced up to 500% increase in rainfall this year - Inland drainage is also a problem in Sindh; due to its low gradient/slope, the floodwaters are retained - Climate change worsened the floods; Pakistan is proposing to have penalty on biggest CO_2 producers to aid Pakistan - Local district management aided in minimizing disaster risk, but inundation was still large. Damage was experienced along the rivers due to encroachment. *Presentation by WAPDA* - Pakistan experiences both extremes: drought and inundation - Main projects/programs to address flooding: FPSP I, II, III, and NFPP-IV - Apart from riverine flooding, “Hill Torrents” from the Suleman Mountain Range also caused flooding (particularly in DG Khan, and Rajanpur) - Tarbela, Mangla, and Chashma reservoirs are regulated by WAPDA; flood monitoring are based on the routing model and rule curves - Stream flows, sediment, water quality, and climate data are collected by WAPDA, then transmitted to other stakeholders and analyzed for planning, projection, forecasting, and research (particularly for water and power) - WAPDA issues daily water situation (river inflows, outflow, water requirements, discharge at barrages), as well as flood situation during flood days (Water Resource Management Department) - 2010 flooding is observed to be due to riverine flooding (i.e., due to	

	overflowing of the major rivers), while the 2022 flooding has more widespread inundation. The 2022 inundation map was based on satellite image analysis. - Some of the WAPDA Hydromet Telemetric Stations were damaged during the 2022 flood - NFPP-IV in 2017 (PC-I) consists of structural and non-structural flood schemes, including upgradation of gauging stations and EWS - FPSP-III funds were reduced; funding is revisited in light of the 2022 floods *Meeting in 2022* - National Water Policy was completed in 2018, which includes real time monitoring of river flows used for transparent water accounting system - Master Plan for Flood Telemetry System was prepared by FFC and WAPDA to improve data collection and monitoring - One data center is proposed at WAPDA, to be connected to regional centers - 679 hydro-met telemetry stations (237 upgraded, 442 new) are proposed - Phase 1 has more emphasis on Balochistan, still up for funding *JICA Meeting in 2021* - JICA showed willingness to fund for 45 stations - Hilly torrents are emphasized, but JICA showed hesitation because of security stations (110 stations, 84 from Punjab, others in KPK) *WAPDA Org Structure* - Departments: Surface Water, Hydro & Research (telemetry), Glacier Monitoring, Water Resource - The stations are projects by several organizations such as AFD, NJHPC, WCAP, TIKA, JICA, China, ADB - 20 Glacier Monitoring AWS are available, for 10-day water forecast for water distribution; the Data Collection Platforms record snow water, temperature, etc. - WRMD: hourly river level and rainfall, 21 high frequency radio stations, 21 stations in different rivers ***b. Interventions 3 and 4*** Intervention 3: 3.1 Research on impacts of climate change, and 3.2 Establishment of GLOF EWS - Refer to the presentation Intervention 4: 1.3 Establishment of a river flood forecast and warning system, including establishment of regional flood forecasting and warning centers (RFFWC) and 1.7 Establishment of a GLOF and snow melt flash flood forecast and warning system with an avalanche advisory information system - Not for WAPDA

	c. Other Information: 1. Operation of Flood Facility (DAMs) - Data from stations in the upstream and perimeter of catchments are used for forecasting; amount of rainfall is then estimated and will be the basis for dam release if needed - Dam Operation Manual and other protocols are available 2. Progress of Telemetry M/P and its implementation (March2022) - Refer to the presentation - Assistance to IRSA (Indus River System Authority) for telemetric stations and monitoring 3. Relationship with PMD during Monsoon/Flood Stage - PMD has radars for monitoring, but ground level monitoring is based on WAPDA stations (PMD has no manpower to monitor on field). These data are then used as input to IFAS model used by PMD for flood forecasting - During flood days (June 15 to October 15), WAPDA closely monitors flood situations with FFD Lahore. The flood monitoring is 24/7. - WAPDA generally collects data from their own stations and from other agencies like PIDS (irrigation). WAPDA then processes the data and sends the refined information to PMD for their use in forecasting 4. Activities and views regarding Climate Change - Refer to the presentation 5. Challenges faced by the agency and recommendations for improvement - WAPDA needs more real-time data for close monitoring, and one data center for data collection and integration - Trainings and capacity building is an issue faced by WAPDA and other agencies. There should also be focus on training field personnel, not just those in the academe. - Availability of equipment/technology is also a problem. The equipment are hard to maintain because they are not produced locally. - WAPDA is one of the oldest agencies (established in 1958) whose mandate is mostly for water and power. WAPDA is now transitioning to focus more on hydropower only for the power sector, but still most of the work for the water sector is shouldered by WAPDA - Strengthen glacial monitoring through additional stations (note: WAPDA is not involved in the GLOF project by UNDP) ***d. Data Collection*** 1. GIS data of stations' locations 2. Updated list of historical floods (with corresponding info on rainfall, flow, and extent) 3. Water level/flow data 4. River basins and network

	(If WAPDA has the items below.) 5. Daily rainfall data 6. Annual maximum rainfall 7. Rainfall analysis: Available, but only on a project basis (on selected dams/barrages only, not for entire Pakistan) 8. Flood hazard maps/Flood risk map 9. GLOF data - The data are available, but a formal request letter was preferred by WAPDA so that they will know our needs. The letter should include list of data, specifications, and purpose.
Prepared by	Cristelle San Antonio (Meteorology/Hydrology)

Memorandum of Meeting

<table>
<tr><td>Project</td><td colspan="2">Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)</td></tr>
<tr><td>Date and Time</td><td colspan="2">29 September 2022, 11:00 ~ 12:30 (PST)</td></tr>
<tr><td>Interviewee</td><td colspan="2">Tanveer, Jappa</td></tr>
<tr><td>Place</td><td colspan="2">WASA Rawalpindi</td></tr>
<tr><td rowspan="3">Participants</td><td>WASA</td><td>Muhammad Tanveer (Managing Director)</td></tr>
<tr><td>RDA</td><td>Muhammad Jappa (Director General)</td></tr>
<tr><td>Project Team</td><td>Takayama, San Antonio, Mukhtar</td></tr>
<tr><td>Memo</td><td colspan="2">
<u>Agenda:</u>
- Inquire on flood control activity, existing conditions, and other matters
- Data collection
<u>Minutes:</u>
a. Inquire about flood control activity
Background of Rawalpindi:
- Two types of flooding: urban/inland and riverine flooding
- Upper catchment comes from Islamabad; water enters Rawalpindi via Nullah Lai
- 2001 flood event is the major flood in Rawalpindi
- In Pakistan in general, the floodwater is released into the rivers, and only three locations so far (including Lahore) have storage tanks for rainfall harvesting
- When rainfall is more than 50 mm/hr, low-lying areas in Rawalpindi are ponded; drainage canals experience backflow when the water level is high at Nullah Lai
- Babu Sabu drainage station in Lahore is a sample solution for the drainage problem in Rawalpindi
Current issues:
- Drainage outlets at Nullah Lai experience backflow
- Encroachment of canals, as well as Nullah Lai, is also a problem
- Canals are hard to maintain and desilt due to the encroachment and due to lack of funds for equipment
- Illegal dumping of construction/industrial waste at the banks leads to the decrease in the river’s capacity
- There is no bank erosion; the issue is more on deposition
Expected/Proposed countermeasures:
- Flood mitigation project in 2004 is still being used
- There is a proposal to increase the design return period from 25 years to 100 years; studies have already been done to increase the capacity of the
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	drainage system, but the implementation is still on hold due to funding issues - Rainfall harvesting can help augment water supply in Rawalpindi, since the groundwater level in the area is low; proposed storage tanks could hold 6 million gallons of floodwater for reuse - Floodwaters coming from upstream (Islamabad) may also be stored *Non-structural measures:* - Contingency plans and comprehensive awareness campaign are prepared during monsoon season - Improvement of the design and modelling of the drainage system through a more comprehensive drainage plan (the current drainage designs are just rough calculations/estimations) - Flood maps are based on a past JICA study (But it needs update, according to WASA). These maps are said to be unavailable to the public. - Regular maintenance of the Nullah Lai is being done, but there are not enough funds for better equipment b. Inquire about natural conditions *Inundation time duration and area:* - No study is carried out for inundation time specifically; from observation, inundation usually only lasts up to 30 minutes, then drained easily to the Nullah *Flood forecasting and sharing procedure:* - This is mostly for PMD, and not WASA. PMD issues alerts every 10 minutes during a flood, through SMS and website - WASA monitors the water levels and flows from the data being transmitted by the PMD. WASA does not have their own gauging stations. *Data communication:* - There is communication with PMD for flood, and with WAPDA for power situation - Communication with community in terms of flooding is mostly through PMD, with coordination of the DDMA (District Disaster Management Authority) c. Other information - Source of funds is a major issue for WASA. Annual budget for flood control is only up to 50 million Rupees - Budget/Projects are proposed through the provincial governments for funding. If PPP, proposals are escalated to the federal government. - Acquisition of equipment is difficult because the investors (donor agencies) are reluctant about the projects due to the encroachment / presence of informal settlers

	d. Data collection - River and drainage channel network: Available - 2015 Master Plan: Available - List of projects implemented for flood: No more project aside from the 2004 flood mitigation project *Meeting with RDA Director General* - It was reiterated that rainwater could be harvested for reuse since both water scarcity and flooding are issues in Rawalpindi - Nulla@h Lai is the major problem of Rawalpindi in terms of flooding - Data are available but would need approval of the agency
Prepared by	Cristelle San Antonio (Meteorology/Hydrology)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	03 October 2022, 10:00 ~ 11:00 (PST)	
Interviewee	Muhammad Goheer	
Place	GCISC, Islamabad	
Participants	GCISC	Muhammad Goheer (Head, Agriculture & Coordination)
	Project Team	Endo, Takayama, San Antonio, Mukhtar
Memo	Agenda: - Inquire on climate models, methods, and coordination with other agencies regarding disaster management Minutes: a. Background of GCISC - The GCISC is the 'think tank' of the MoCC, which focuses more on research, and uses the IPCC AR6 as reference - They assisted in the preparation of the National Communication on Climate Change, as well as the Nationally Determined Contributions (GHG contributions) of Pakistan; they are also assisting in the preparation of the National Adaptation Plan, which is still ongoing - GCISC has been conducting research since 2010, and the list of their publications are available on their website. The Project Team could select the publications needed for the study, and GCISC will provide the copies. b. Climate Models - They have completed research based on climate projections in relation to Representative Concentration Pathways (RCP), and now they are also conducting research based on Shared Socioeconomic Pathways (SSP) - They have numerical models of up to 5m x 5m resolution, as well as national and district-level scenarios - There is a dedicated section in the GCISC for global and regional climate models, as well as statistical downscaling of models; PMD uses some of these information for seasonal forecasting - GCISC has digital servers for the following: climate modelling, national forest monitoring system, greenhouse gas inventories, crop management, and others - When asked about the authorized climate model in Pakistan, Mr. Goheer explained that several models are used for different parameters (i.e., some models are more accurate for temperature, while others are better for precipitation). A separate contact person was requested to discuss this further. c. Research Studies - GCISC has strong ties with academic institutes regarding research on	

	climate change (20 university research centers) - Research conferences are held annually, and GCISC is also involved in international workshops and preparation of reports on GHG contributions and urban resilience - According to Mr. Goheer, the most important research done by GCISC are the following: o Preparation of the NDC o Submission of the BUR (Biennial Update Report) to the UNFCCC o Collaboration for Asian Climate Change Impacts - GCISC has also conducted studies regarding effects of climate change to GLOF - Every ministry is bound to publish a yearbook; MoCC, which includes GCISC, has a published yearbook on their website d. Regarding NDMP and Institutional Linkages - NDMP was not properly implemented due to poor coordination among agencies, as well as overlapping and duplicating mandates of the agencies - Pakistan is the first country in the region (South Asia) to have a dedicated agency for climate change; despite this, weak institutional linkages hinder the proper implementation of DRRM throughout its four dimensions (Preparedness, Prevention, Response, and Rehabilitation) - MoCC requests the GCISC for technical assistance, preparation of project briefs, representation of MoCC in conventions, and others. The approach is mostly top-down (GCISC does research based on needs of MoCC), and not the other way around (GCISC has little influence on MoCC policies and decisions). - One of the issues to resolve is how agencies involved in climate change should meet more regularly and incorporate the research findings of GCISC; Mr. Goheer also mentioned that GCISC should be involved in the decision-making at the national level - When asked about interaction with the community, Mr. Goheer mentioned that it is part of their mandate to spread awareness regarding the threats of climate change to the country. GCISC, however, has not fully met that mandate yet, which is said to be mostly due to the lack of dedicated personnel for communications (most of them focus on research)
Prepared by	Cristelle San Antonio (Meteorology/Hydrology)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	04 October 2022, 11:00 ~ 12:00(PST)	
Interviewee	Dr. Jehangir Awan (Deputy Director)	
Place	PMD Office of Dr. Jehangir	
Participants	PMD	Dr. Jehangir Awan
	Project Team	Endo
Memo	Agenda: - Organizations working on climate change analysis in Pakistan - Possibility of sharing climate change analysis data Minutes: a. Organizations related to climate change analysis and countermeasures in Pakistan - The highest organization dealing with climate change in Pakistan is MoCC (Ministry of Climate Change). - Other organizations related to Climate Chgange are GCISC (Global Change Impact Studies Centre), a research institute under MoCC, - and CIIC (Climate Change Impact and Integration Cell) in R&D of PMD (Pakistan Meteorological Department). - In addition, there would be research centers at the university level and local government level, but PMD doesn't know much about them. b. About cooperation between related organizations - No formal mechanism has been established in Pakistan to integrate the research output of different research institutes and researchers. - Since GCISC is an affiliated organization of MoCC, it naturally conducts research mainly in line with MoCC policies. They are also preparing analysis data to be published in the Year Book published by MoCC. - On the other hand, PMD has never received a direct research request from MoCC or GCISC. They have the experience to participate in Climate Change related projects with the cooperation of other countries by sharing roles with GCISC, but the proportion of our activities is small. - PMD has a high-performance computer (Cluster PC) installed at the Specialized Medium Range Weather Forecasting Cente, which was established with Japanese grant aid, so it is possible to perform more complex and sophisticated simulations and analyzes than GCISC. - PMD is able to freely determine its own theme and carry out research without relying on requests or instructions from related organizations. c. Possibility of sharing climate change analysis data - PMD has published quantitative prediction results in the results of its research so far, so please refer to them. If more detailed data is required, PMD will confirm whether or not it can be disclosed externally, so please let us know what kind of data you specifically need.	
Prepared by	Toshihide ENDO	

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	05 October 2022, 10:00 ~ 11:00 (PST)	
Interviewee		
Place	Pakistan Meteorological Department (PMD)	
Participants	PMD	Mr. Nasir (Seismic Monitoring Centre)
	Project team (PT)	Hasegawa、Kito
Memo	・ We have introduced and explained Probabilistic Tsunami Hazard Assessment (PTHA) carried out in Japan to Mr. Nasir, and it seemed to be that Mr. Nasir understood the importance of probabilistic approach for tsunami hazard assessment. ・ In order to carry out PTHA in Pakistan, it seems to be that long term evaluation for possible earthquakes, especially for inter-plate earthquakes, is lacking. ・ Mr. Nasir carried out tsunami simulation for us and explained his result. He gave us his inundation data for 4 major cities along the coast (Gwadar, Karachi, Ormara and Pasni). In addition, we got PPT material that explains his tsunami simulation. ・ The tsunami inundation data was not included in the present NDMP, and therefore the acquired data should be integrated into the NDMP-Ⅱ. ・ We asked him how to determine the earthquake probability for their seismic hazard map. He suggested that we could get more information from National Engineering Services Pakistan Limited (NESPAK) about it.	
Meeting Materials		

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	6 October 2022, 10:15 ~ 11:00(PST)	
Interviewee	Mr. Aziz Asiam (General Manager)	
Place	National Engineering Services Pakistan Limited (NESPAK)	
Participants	NESPAK	Mr. Aziz Asiam
	Project Team (PT)	Mukhtar san, Hasegawa, Kito
Memo	1) Discussion on Data request ・ We discussed on seismic probability of occurrence in Pakistan which is updated in Building Code 2021. The general manager said that NESPAK has the team for the purpose. (The person in the team joined later.) ・ He promised to share the data requested below in several days. ・ 2) Request for materials or data ・ Seismic Probabilistic Acceleration (PGA) map data for 2021. ・ Tsunami Hazard Data (Gwadar city, scenario base). ・ Geological map in Pakistan for site effect. ・ Reports for the above data. ・ 3) Request of introduction of responsible person ・ He will introduce a manager of Pakistan Engineering Council (PEC) for having a meeting on the original PGA map data, if necessary. ・	
Meeting Materials		

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	30 November 2022, 09:50 ~ 11:10 (PST)	
Interviewee	Mr. Aziz Asiam (General Manager), Mr. Hafeez Abdul Rehman (DRR Expert / Project Manager), Dr. Nuhammad Rizwam	
Place	National Engineering Services Pakistan Limited (NESPAK)	
Participants	NESPAK	Mr. Aziz Asiam, Mr. Hafeez Abdul Rehman, Dr. Nuhammad Rizwam
	Project Team (PT)	Mukhtar, Kito
Memo	1) Discussion on Seismic and tsunami hazard Data ・ We discussed on seismic hazard data with the technical staffs at NESPAK. They have given us a shapefile of Seismic zones of Pakistan, as per Seismic Zoning Map of Pakistan, Building Code of Pakistan, Seismic Provisions – 2007. They have explained that recent seismic hazard assessment is based on spectral response acceleration data in Pakistan, and they do not use PGA data for recent assessment. To collect spectral response acceleration data, we need to ask PEC people. ・ We have asked them if it is possible to share the tsunami hazard data. They answered that they have tsunami hazard data in one city (GWADAR), but it is not possible to share the data with us because the data is the result of another project. Instead, they have suggested us that we should send an official letter through NDMA to PDMA Balochistan (Provincial Disaster Management Authority, Balochistan) in order to request the tsunami hazard data for many cities along the coast. ・ NESPAK requests us that we must share the tsunami hazard from PDMA with NESPAK in return for the shapefile of Seismic zones of Pakistan that NESPAK has provided us. ・ We will share this data exchange condition with NDMA officers (Mr. Abdul Latif, Deputy Director).	
Meeting Materials		

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	13 December 2022, 10:00 ~ 13:00 (PST)	
Interviewee		
Place	Pakistan Meteorological Department (PMD)	
Participants	PMD	Mr. Nasir (Seismic Monitoring Centre) Mr. Zahid Rafi (Director, Seismic Monitoring Centre)
	Project team (PT)	Kito
Memo	・ I have discussed Probabilistic Tsunami Hazard Assessment (PTHA) with Mr. Nasir Mahmood and Mr. Zahid Rafi using Questionnaire. They answered to many of the questions. ・ Regarding Probabilistic Seismic Hazard Assessment (PSHA), Mr. Nasir has called one of the authors (Muhammad Ahmed) to ask for the new PGA data, because the author is his former student. The new PGA data may be obtained through PMD. ・ PMD seems to be willing to perform PTHA, and, at the same time, PMD expects a lot of technical and material support such as providing programs and computers and conducting survey from Japanese institutions. ・ The answers from PMD are documented in the file entitled “Questionnaire about PTHA.docx”. Please see the file for more details.	
Meeting Materials	Questionnaire about PTHA.docx	

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	16 December 2022, 15:30 ~ 16:30 (PST)	
Interviewee		
Place	Federal Flood Committee (FFC)	
Participants	FFC	Mr. Shigri (Senior Engineer (Floods))
	Project team (PT)	Takayama, Qasim Saeed
Memo	・ Project Team had an interview to grasp the progress of NDMP-I in accordance with the activities listed in the interventions which relating to FFC. The progress of each activity concerned to FFC is summarized in the attached tables.	
Material	Table of Intervention 3, 4 , 7 and 10	

Intervention-3: Establish a National Hazard and Vulnerability Assessment

NDMP 2012			Progress					
Strategies	**Priority Actions /Programs**	**Responsible Organizations**	**Status as of October 2022**	**Project Name**	**Responsible Organization**	**Start/End**	**Components of the Project**	**Name of Donor**
1. Conduct detailed multi–hazard vulnerability and risk analysis/assessments at national level	1.1. Preparation of vulnerability atlas at the national level	NDMA in association with F/G/S/PDMAs, DDMA, FFC, PMD, GSP and ERRA	Partially Completed	NFPP-IV	FFC	2017-2018 2012-2022	• Flood Risk Map • No activity for GLOF • Only Lai Nullah project for Urban Flooding.	WB
WB	1.2. Digitization of vulnerability atlas and the preparation of database	NDMA and F/G/S/PDMAs						
2. Conduct detailed multi–hazard vulnerability and risk analysis/assessments at local levels	2.1. Preparation of vulnerability analysis and creation of hazard maps for selected districts and cities*	NDMA in association with F/G/S/PDMAs, DDMA, FFC, PMD, GSP and ERRA	Partially Completed	NFPP-IV	FFC		• Flood Risk Map • No activity for GLOF Only Lai Nullah project for Urban Flooding	-WB
	2.2. Digitization of vulnerability/hazard maps and the preparation of database	NDMA and F/G/S/PDMAs						
3. Conduct research and studies on impact of climate change on glaciers and ice caps	3.1. Conduct research on impact of climate change on glaciers and ice caps	WAPDA, PMD, FFC and GCISC in coordination with NDMA	On going	Taking part in the project implemented by WAPDA under T4 Project. .	WAPDA	2017- 2022	GLOF Activities	WB
	3.2. Establishment of GLOF early warning system for selected vulnerable areas	PMD in association with NDMA, FFC and WAPDA	On going	Scaling-up of Glacial Lake Outburst Flood (GLOF) Risk Reduction in Northern Pakistan (GLOF II)	MOCC, PMD (only for the operation of 50 monitoring stations)	2017 – 2024	• FFC reviews and gives advice to the project	UNDP

Intervention-4: Establish a Multi-hazard Early Warning System

NDMP 2012			Progress					
Strategies	**Priority Actions /Programs**	**Responsible Organizations**	**Status as of October 2022**	**Project Name**	**Responsible Organization**	**Period**	**Components of the Project**	**Name of Donor**
1. Strengthen forecasting and early warning systems	1.1. Establishment of a specialized medium range forecasting centre (SMRFC) with the meteorological radar station at Islamabad, upper air observations at designated locations	PMD					•	
	1.2. Strengthening of flood forecasting by satellite information and hazard maps of Indus River	PMD/FFC/SUPARCO						
			Completed; Operational	Space Application Centre for Response in Emergency and Disasters (SACRED-SUPARCO)	SUPARCO	2013	• FFC coordinated with SPARCO	
	1.3. Establishment of a river flood forecast and warning system, including establishment of regional flood forecasting and warning centres (RFFWC)	PMD/WAPDA in consultation with NDMA/PDMAs						
	1.4. Establishment of a flash flood forecasting and warning system including local flash flood forecasting centres (LFFFC)	PMD in consultation with NDMA/F/G/S/PDMAs						
	1.5. Establishment of landslide forecast and warning system	PMD in consultation with NDMA/F/G/S/ PDMAs						
	1.6. Establishment of a new tide level monitoring network including a data communication system	PMD						
	1.7. Establishment of a GLOF and snow melt flash flood forecast and warning system with an avalanche advisory information system	PMD/WAPDA						
	1.8. Establishment of a seismic intensity reporting system including a data communication system	PMD						

NDMP 2012			Progress					
Strategies	**Priority Actions /Programs**	**Responsible Organizations**	**Status as of October 2022**	**Project Name**	**Responsible Organization**	**Period**	**Components of the Project**	**Name of Donor**
	1.9. Establishment of a meteorological radar system along coastal areas	PMD						
	1.10. Establishment of a meteorological radar system at designated locations	PMD						
	1.11. Replacement of the meteorological radar systems at D.I. Khan, R.Y. Khan and Karachi	PMD						
	1.12. Expansion and additional installation of an automatic weather observation system (AWS)	PMD in consultation with NDMA/F/G/S/ PDMAs						
	1.13. Expansion of the real-time rainfall & water level observation network	PMD						
2. Prepare hazard maps at local scale in targeted locations	2.1. Preparation of hazard maps and capacity development against local flash floods in vulnerable areas	PMD/FFC	Not Yet Prepared		FFC		• The program is stated in NFPP-IV	
	2.2. Preparation of landslide hazard maps based on the topographical and geological analysis at vulnerable locations	PMD/FFC GSP/NDMA	Not Yet Prepared IN Progress	FPSP-III			• No maps • F/S for Water Shed Management for Indus Basin includes it in FPSP-III	
	2.3. Training on tsunami simulation and preparation of hazard maps	PMD, NDMA/PDMAs/ DDMA/Related Agencies						
3. Strengthen early warning dissemination system	3.1. Innovation of the communication system between PMD and NDMA, and among DMAs (NDMA-PDMA-DDMA) and a communication system utilizing cellular phone and radio broadcast networks	NDMA/F/G/S/PDMAs / DDMAs/PMD/PID/ NGOs/PTA/PEMRA/ Local Governments						
	3.2. Development of the EWS National Plan, Guidelines and SOPs for Health Emergency	NDMA/PDMAs Ministry of Health						

NDMP 2012			Progress					
Strategies	**Priority Actions /Programs**	**Responsible Organizations**	**Status as of October 2022**	**Project Name**	**Responsible Organization**	**Period**	**Components of the Project**	**Name of Donor**
	Preparedness and Response (HEPR)							
	3.3. Weather information broadcasting system and weather information broadcasting Program production system and installation of GTS in SMRFC Project	PMD						
							•	
	3.4. Finalization of SOP of cyclone EWS	PMD						
4. Develop capacity of early warning and evacuation systems	4.1. Enhancement of research activities for snow/glacier/glacial lakes	PMD/GCISC /WAPDA						
	4.2. Establishment of weather forecast guidance system	PMD						
	4.3. Enhancement of community enlightenment for EWS with execution of training and drills	NDMA/PDMAs/ DDMAs/PMD/FFC/ NGOs	No Activity by FFC					
	4.4. Education Program for advanced meteorology and hydrology for PMD staff	PMD						

Intervention-7: Infrastructure Development for Disaster Risk Reduction

NDMP 2012			Progress					
Strategies	**Priority Actions /Programs**	**Responsible Organizations**	**Status as of October 2022**	**Project Name**	**Responsible Organization**	**Start/End**	**Components of the Project**	**Name of Donor**
1. Develop schools, hospitals and other important public facilities to be safe against disasters	1.1. Structural vulnerability evaluation for schools and hospitals against earthquakes, tsunamis and floods in Pakistan	NDMA/ F/G/S/PDMAs/DDMAs					•	
	1.2. Preparation of guidelines for new public building construction in the areas vulnerable to disasters	NDMA/ F/G/S/PDMAs/DDMAs						
	1.3. Retrofitting works of important public facilities (schools and hospitals)	NDMA/ F/G/S/PDMAs /DDMAs Ministry of Education Ministry of Health						
2. Develop important coastal facilities to be safe against disasters taking into account climate change	2.1. Construction of coastal dikes along major public facilities against tsunamis and storm surges (cyclones)	Ministry of Ports and Shipping						
	2.2. Construction of DRM centres in vulnerable areas to disasters	NDMA/ F/G/S/PDMAs/DDMAs						
3. Enforce the building code in construction of buildings	3.1. Preparation of guidelines for housing construction in the areas vulnerable to disasters	NDMA/ F/G/S/PDMAs /DDMAs						
4. Implement appropriate structural measures in flood prone areas taking into account comprehensive and integrated flood management plans	4.1. Establishment of comprehensive and integrated flood management/protection plan in Pakistan • A review of breaching mechanism of the river embankments (flood bunds) and a review of designated breaching points	FFC/PIDs/Pakistan Army	On-going	NFPP-IV	PIDs	2017-2027	• Provincial Irrigation Departments are conducting study.	

NDMP 2012			Progress					
Strategies	**Priority Actions /Programs**	**Responsible Organizations**	**Status as of October 2022**	**Project Name**	**Responsible Organization**	**Start/End**	**Components of the Project**	**Name of Donor**
	• Revision and updating of NFPP-IV	FFC/PIDs/WAPDA/PMD /Planning Commission	Concept Already Approved by CCI with cost of 332 Billion PKR	NFPP-IV	FFC	2017-2027		
	• Revision of SOPs of Tarbela Dam for flood mitigation	FFC/WAPDA	Completed by WAPDA	Flood Management Manual 2020, Tarbera Dam Project	FFC/WAPDA		• Stated in NFPP-IV • Supported by FFC •	
	• Flood plain zoning and management	FFC/PIDs/PDMAs	Partially Completed	NFPP-IV	FFC		• Flood Risk Map was prepared in NFPP-IV • Management is not yet done, mentioned in NFPP-IV	
	• Capacity building of the stakeholders in flood mitigation	FFC/PIDs/PMD/WAPDA	On-Going	Technical Advisor on Flood Management	FFC	2021.10 – 2023.9	• Seminar to FFC Officials	JICA
	4.2. Construction and rehabilitation of flood control/mitigation structures • Implementation of NFPP-IV	FFC/PID/WAPDA/ PMDs /Planning Commission	Not yet Implemented					
	• Construction of DRM centres in flood prone areas	NDMA/ F/G/S/PDMAs / DDMAs /						
	• Retrofitting of existing dams, barrages and flood protection works to increase floodwater retarding capacity and ability to accommodate design discharges	FFC/PIDs/WAPDA Planning Commission	On-Going	Projects proposed in NFPP-IV	FFC			
			Completed	4 barrages were rehabilitated in Panjab	PID		• 4 number of barrages (Jinnah, Taunsa, Trimmu, Punjnad,) have already been rehabilitated and upgraded, • Islam barrage is also near to completion	ADB
	• Constructing of new flood protection works,	FFC/PID/WAPDA/	Not yet Implemented	Projects proposed in NFPP-IV	FFC, WAPDA		•	

NDMP 2012			Progress					
Strategies	**Priority Actions /Programs**	**Responsible Organizations**	**Status as of October 2022**	**Project Name**	**Responsible Organization**	**Start/End**	**Components of the Project**	**Name of Donor**
	dams and barrages to increase flood mitigation/protection capacity	/Planning Commission	Multi-purpose dam Projects On-Going	Mohmad Dam Project[1] by WAPDA Diamer Basha Dam [2]by WAPDA	WAPDA	2019-2025 2020-2029	• FFC reviews and gives advices to WAPDA's dam projects	
5. Enhance disaster risk management capacity in urban areas	5.1. Formulation of urban disaster management plan to propose corresponding countermeasures against natural hazard risk in the urban areas	NDMA/ F/G/S/PDMAs/DDMAs						
	5.2. Enforcement of effective land use control and regulations based on urban disaster management plan; introduction of the space needed for evacuation and disaster relief into land utilization program	NDMA/ F/G/S/PDMAs /DDMAs/Planning Commission						
	5.3. Construction of DRM centres in areas vulnerable to urban disasters	NDMA/ F/G/S/PDMAs/DDMAs						

[1] http://www.wapda.gov.pk/index.php/projects/water-sector/under-construction/mohmand-dam-hydropower-project
[2] http://www.wapda.gov.pk/index.php/projects/water-sector/under-construction/diamer-basha-dam

Intervention-10: Capacity Development for Post-Disaster Recovery

NDMP 2012			Progress					
Strategies	**Priority Actions /Programs**	**Responsible Organizations**	**Status as of October 2022**	**Project Name**	**Responsible Organization**	**Start/End**	**Components of the Project**	**Name of Donor**
1. Prepare guidelines for post-disaster recovery programs and activities	1.1. Preparation of guidelines for the formulation of recovery and rehabilitation plans	NDMA, SPU in consultation with F/G/S/PDMAs					•	
	1.2. Documents of lessons learnt regarding recovery from the Pakistan 2010 Flood by related agencies	NDMA, F/G/S/PDMAs/ DDMAs and PMD/FFC	Completed	NFPP-IV	FFC	2017-2027	• Major document is only the study for NFPP-IV,	GOP ADB
	1.3. Establishment of a funding system for post-disaster recovery and rehabilitation	NDMA, SPU in consultation with F/G/S/PDMAs						
2. Develop capacity of stakeholders in post-disaster recovery	2.1. Holding of orientation workshops for line ministries and other stakeholders on post-disaster recovery program design and implementation	NDMA in association with F/G/S/PDMAs						
	2.2. Database on technical capacity of relevant stakeholders in designing and implementing recovery programs	NDMA in association with F/G/S/PDMAs						
	2.3. Set-up of a system to coordinate and monitor flood early recovery activities	NDMA, F/G/S/PDMAs, DDMAs						
	2.4. Capacity development of research activities for new techniques for recovery and rehabilitation	Research Institutes						
3. Develop system and methodology for recovery needs assessment	3.1. Preparation of guidelines for recovery needs assessment and recovery program design and management for multiple sectors	NDMA, F/G/S/PDMAs, SPUs						

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	26 January 2023, 12:20 ~ 13:30(PST)	
Interviewee	・Mr. Syed M Tayyab Shah (Team leader, Risk Assessment Unit, NDMA) ・Ms. Zahra Hassan (MHVRA Expert, Risk Assessment Unit, NDMA) ・Mr. Haris Mir (GIS Officer, Risk Assessment Unit, NDMA) ・Mr. Sheikh Rafay Ehsan (Geological Expert, Risk Assessment Unit, NDMA	
Place	National Disaster Management Agency (NDMA)	
Participants	NDMA	Above four members
	Project Team (PT)	Hasegawa, Mukhtar san
Memo	1) Discussion on Data request ・ Overview of earthquake risk assessment was explained to the counterpart members. ・ RA Unit members explained overview of their activity, especially focusing on MHVRA project. ・ They understand the issues of building inventory and advised to use satellite image analysis. This time, Census 2017 data is used. But we can compare both. ・ Regarding to site amplification factor, they do not have such information as Vs30 to estimate it in Pakistan. So we agreed to use USGS or other worldwide open data. 2) Request for materials or data ・ Seismic Probabilistic Acceleration (PGA) map data for Building Code 2021 in GIS format. ・ Vulnerability functions that is used in Pakistan officially. ・ Building inventory data estimated by satellite image. It may not include whole nation. ・ Technical Report for MHVRA 2018 in PDF. 3) Request of introduction of responsible person ・ Team leader knows the contact persons of NED University of Engineering and Technology in Karachi. He can contact the person for requesting PGA data and vulnerability function. 4) Next schedule ・ Next meeting will be held on 2 February (Thursday) at the GIS room of RA Unit. We will compare Census based building inventory and Satellite based one in Tehsil level nationwide. ・	
Contact Info.	・Mr. Syed M Tayyab Shah (Team leader, Risk Assessment Unit, NDMA) ・Ms. Zahra Hassan (MHVRA Expert, Risk Assessment Unit, NDMA) ・Mr. Haris Mir (GIS Officer, Risk Assessment Unit, NDMA) ・Mr. Sheikh Rafay Ehsan (Geological Expert, Risk Assessment Unit, NDMA	

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	1 February 2023, 10:50 ~ 11:20(PST)	
Interviewee	・Mr. Yousaf Zia (Additional Register) ・Ms. Rida Nasir (Assistant Register)	
Place	Pakistan Engineering Council (PEC)	
Participants	PEC	Above members
	Project Team (PT)	Hasegawa, Mukhtar san
Memo	1) Discussion on Data request ・ Explanation about Request Letter for seismic hazard map in Building Code-2021. Peak Ground Acceleration (PGA) data is just we request for seismic risk assessment, not spectral data. ・ Requesting data to NED University might be the same as that to PEC. So, apologizing the matter and ask PEC staff to facilitate data collection for JICA project. ・ 2) Request for materials or data ・ Seismic Probabilistic Acceleration (PGA) map data in GIS format. ・ This data is not shown in Building Code but source of the data is the same, NED University. ・ 3) Request of introduction of responsible person ・ Asked PEC contact professors of NED University of Engineering and Technology in Karachi. They are focal members of Building Code 2021. 4) Next schedule ・	
Contact Info.	・Mr. Yousaf Zia (Additional Register), _____ ・Ms. Rida Nasir (Assistant Register) _____	

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	20 February 2023, 11:30 ~ 12:00(PST)	
Interviewee	・Ms. Zahra Hassan (MHVRA Expert, Risk Assessment Unit, NDMA) ・Mr. Haris Mir (GIS Officer, Risk Assessment Unit, NDMA) ・	
Place	National Disaster Management Agency (NDMA)	
Participants	NDMA	Above members
	Project Team (PT)	Hasegawa
Memo	1) Discussion on Data request ・ Some achievements are reported as following, ・ JICA Expert Team completed Tehsil level Census GIS data. ・ NED University kindly provided materials for fragility curves of buildings, and PGA GIS map data. ・ We need to calculate earthquake ・ 2) Request for technical discussion, next time ・ As Census Housing type data is based on household, we need to compare it with actual building numbers. ・ We would like to discuss how to use fragility curves to estimate earthquake building damage. ・ 3) Request of introduction of responsible person ・ 4) Next schedule ・ When Hasegawa comes again, we will discuss technically on actual earthquake damage estimation.	
Contact Info.	・Mr. Syed M Tayyab Shah (Team leader, Risk Assessment Unit, NDMA) ・Ms. Zahra Hassan (MHVRA Expert, Risk Assessment Unit, NDMA) ・Mr. Haris Mir (GIS Officer, Risk Assessment Unit, NDMA) ・Mr. Sheikh Rafay Ehsan (Geological Expert, Risk Assessment Unit, NDMA	

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	26 September 2023, 1400 ~ 1800 (PST), 27 September 2023, 1000 ~ 1730 (PST),	
Interviewee	Mr. Niaz Hassan Abbasi, Secretary Sindh Irrigation Karachi Mr. Jamal ud din, special Secretary Technical Sindh Irrigation Mr. Sajid Ali Bhutto, Chief Engineer Barrage Management Project, Sindh Irrig.	
Place	Karachi	
Participants	PID Sindh	Sajid Ali CE, Jamal ud din, Sp. Secretary Technical, Ayub Somroo, Add. Secretary Technical, Rahim Qureshi SE Kotri
	Project Team	Kohei Takayama, Takafumi Sueishi, Qasim Saeed
Memo	Agenda: - Inquire on progress of NDMP-I significant milestone and Interventions 2, 4 and 7 - Acquire other information and relevant data Minutes: JICA Study Team members explained that JICA expert team is currently working on the project entitled "Technical Support Project to Review and update The National Disaster Management Plan (NDMP)" under the Japan International Cooperation Agency (JICA) with the Coordination of National Disaster Management Authority (NDMA), which aims to study the current conditions of disaster management in Pakistan and to update the NDMP as desired. ***a. Presentation on PID Sindh Organization and Functions*** - Secretary Sindh Irrigation, desired that if possible JICA Project Team may also study the Maleer Development Project PC-1 (Karachi) and may support to the Sindh Government for the implementation of above said project. - On discussion of NFPP IV, Secretary, Irrigation desired that as the PC-1 for NFPP IV were prepared before 2017-18 for next 10 years. Six years have already been passed away and site conditions of proposed projects have already been changed partially or completely. Some of the flood protection structures have already been constructed or rehabilitated at site under the Sindh Government own finance or from International Donors financial assistance. He also emphasized that now this is the proper time to review, upgrade and reformulate the NFPP IV, PC-1s as per present site conditions with new integrated approaches keeping in view the flood damages during flood 2022. - Secretary, also informed and suggested to the JICA study team to understand the proper flood management in Sindh by visiting the all possible sites effected by floods and see the available facilities. - Secretary Irrigation mentioned the successful urban flood management methods in Japan and requested that JICA may also provide some technical support for urban flood management in Karachi City. JICA Project Team replied and informed the prevailing procedures adopted in Japan currently.	

	- Secretary, also seeks help from Government of Japan under JICA Grant in Aid Program, for rehabilitation and modernization of Kotri Barrage. For the purpose JICA Project Team may visit Kotri Barrage at Hyderabad at any convenient date. - Pakistan experiences both extremes, inundation and as result drought conditions in Sindh during and after 2022 flood. - Main projects/programs to address flooding: FPSP III, and NFPP-IV, Sindh Emergency Rehabilitation Projects (SERP). - Apart from riverine flooding, "Hill Torrents" from the Kirthar range and Suleman Mountain Range also caused flooding. Particularly Balochistan and Punjab Hill Torrents ultimately drained into river Indus and finally enters to Sindh Province. - The 2010 Flood is observed to be due to riverine flooding (overflowing of the Indus river), while the 2022 flooding has more widespread inundation mainly due to hill torrents and partly from major rivers. - Chief Engineer Mr. Sajid Bhutto Chief Engineer explained that flood 2022 created a huge inundation in the flood plain areas along the river Indus in Sindh Province which needs immediate attention and support from JICA. - The major damages and inundation took place to the Public and Private Infra structures along the right bank of river Indus was mainly due to hill torrent floods. - After the flood 2022, it was really emphasized that there is dire need of drainage plan for the Kairpur, Nawabshah, Naushero Feroze and Sukker districts so that the huge damages due to hill torrents can be avoided in future. - Chief Engineer explained that Sindh province condition is very typical in case of floods as a lower tear. - According to Mr. Sajid there are 2300 kilometer long flood embankments in Sindh province maintained by PID but in case of any kind of emergency no bund have been managed to breach however, due to the encroachment of free board sometimes some bunds have been leaked or over topped which ultimately causes a breach of flood bunds and inundation of surrounding areas. - Mr. Jamal ud Din Special Secretary Technical appreciated the interest shown by JICA for the flood management and rehabilitation of flood embankments in Sindh. He also emphasized that, in addition to the Dike management and flood management along River Indus, there is necessity of Hill torrents management also because of the all hill torrent flood comes out from Balochistan are out of control and no management plan is available with PID Sindh, which is very much important and keeping in view 2022 flood the major damages took place in Sindh Irrigation due to the Hill torrent floods, Indus river flood was not so high in 2022. - Mr. Ayub Somroo additional Secretary Technical Sindh Irrigation Dept. and Sajid Ali Chief Engineer also second the opinion of Mr. Jamal ud Din Special Secretary Technical. During the discussion following important points discussed in details. - According to the Special Secretary Sindh, Kotri barrage the last barrage of Indus River and located close to the Arabian Sea, was initially getting 84,000 Million acr feet (MAF) of water supplies, while after the Indus water Treaty with India this reduced to 13 MAF which is also a one of the big disaster for Sindh Province having 1400 number of Irrigation Channels.

	- Out of 13.1 million Acre of cultureable command areas, only 4-5 Million acres agriculture command area in Sindh Province having drainage facilities. After 2022 lot of damages occurred to farm lands and standing crops, which ultimately created drought conditions specifically along right bank of Indus River in Sindh Province. JICA may take the necessary steps to help in such challenging conditions. - While discussion upon the Kotri Barrage all piezometers showing underground water pressure are totally blocked and not working since several years, which can be a big threat to the safety of barrage structure. - Upstream of the Barrage River approach is not centralized towards main body of the Barrage structure, due to oblique type approach, river main stream hit to the barrage left bank side river training works. Such oblique type approach needs to be addressed and straightened for smooth movement of river flows upstream side of barrage. - Design discharge of Barrage is about one Million Cu. Feet / sec while barrage downstream discharge from October to June every year remains as low as Nil, however upstream discharge is hardly sufficient to run the three main off taking canals from Kotri Barrage left side. - During the flood 2010, 1.1 million Cubic feet /sec discharge passed safely from Kotri Barrage. Already installed 500 Mega Watts generation unit at Kotri Barrage, of Solar system is not working and needs immediate rehabilitation. Barrage infra structures need major rehabilitation because of the aged structures. Existing Upstream apron of the Barrage structure needs rehabilitation and extension because of the oblique approach of river floor condition is not good at site. Downstream side apron of Barrage is also in poor condition due to less discharges of water and remains dry most of the time throughout the year. Some places it has been observed that Downstream side apron is also drifted at many places which needs immediate attention and rehabilitation. ***b. Interventions 2, 4, and 7*** **Intervention 2:** **2.1.** Formulate /update disaster management plans at National, Provincial, District, and community or TMA levels. Preparation of Annual Flood fighting Plans March-April every year **2.2.** Develop Hazard Specific Contingency Plans Preparation Flood Contingency plans, after every super flood event **Intervention 4:** **4.3.** Strengthen Early warning dissemination system After getting flood discharge at first control point, in Pakistan warnings issued to all flood stake holders by PID through FFD Lahore. However, Sindh PID is planning their own flood warning system for only Sindh province flood stake holders. Sindh PID is actually interested in timely and correct information about hill torrents flood generation from Balochistan

	and Punjab Province. **Intervention 7:** 7.4. Implement appropriate structure measures in Flood prone areas taking into account comprehensive and integrated flood management plans. Although no new considerable project is undertaken during last ten years however, rehabilitation and modernization is took place for barrages and other hydraulic and flood protection structures. 7.4.1. A review of breaching Mechanism of the river embankments and a review of designed breaching points. According to the Chief of Barrage Management unit there is no planned breaching section under the control of Sindh PID. - Revision and up dation of NFPP IV The process of up dation of NFPP IV is on the way and PID has submitted the revised version of NFPP IV projects to FFC for further processing. - Flood Plain zoning and Management This is responsibility of FFC and FFC is already working on it. - Capacity Building of stake holders in flood mitigation Mainly several training courses have been conducted before the start of flood season every year inside of PID. **7.4.2.** Construction and Rehabilitation of flood control mitigation structures Several flood works undertaken under the umbrella of NFPP IV and FPSP-III Projects. - Implementation of NFPP IV NFPP IV umbrella project was formulated 2017-18, but due to flood 2022 lot site conditions have already been changed, so revision of NFPP IV is in progress and after approval from competent authority projects will be executed at site accordingly. However, several projects are in progress under SBIP. - Retrofitting of existing dams, Barrages, and flood protection works to increase flood water retarding capacity and ability to accommodate design discharge. Several projects are necessary for this purpose but due to porosity of funds no considerable project has been taken in hand. - Construction of new flood protection works, Dams and Barrages to increase flood mitigation/protection capacity. Several projects are necessary for the rehabilitation of Barrages but due to porosity of funds no considerable project has been taken in hand. However, several flood protection works have already been started under Sindh Emergency and Resillience Project (SERP).

	c. Other Information: - Balochistan Province have decided to prepare drainage plan for about 20,000 small and big hill torrents draining into Punjab and Sindh Provinces and finally all discharges flowed down to Sindh. Punjab and Balochistan Provinces never care about the drainage effluents, and hill torrents flood flows released every year to Sindh province being a lower tear. - According to Mr. Sajid Ali in the past, there were about 13-15 active river creeks in Indus River tail end near Arabian Sea, but after the flood 2022 this number is decreased to 2 which shows serious reduction in river supplies. - As the three major southern rivers have controlled by India under the Indus water Treaty 1960. This also causes considerable reduction in downstream River discharges to Sindh province and Sindh is facing sometimes drought conditions after flood every year which also needs immediate attention. - During the flood days (June 15 to October 15), PID closely monitors flood situations with FFD Lahore and flood monitoring is 24/7. ***d. Data Collection*** - Some of the available data have already been provided by PID Sindh, however for the remaining data a formal request letter from JICA Project Team was preferred by PID so that they will know our needs. The letter should include list of data, specifications and purpose.
Prepared by	M. Qasim Saeed

Memorandum of Meeting

<table>
<tr><td>Project</td><td colspan="2">Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)</td></tr>
<tr><td>Date and Time</td><td colspan="2">27 September 2023, 10:00 ~ 12:00(PST)</td></tr>
<tr><td>Interviewee</td><td colspan="2">Master Plan Department, Sindh Master Plan Authority (SMPA)</td></tr>
<tr><td>Place</td><td colspan="2">9th Floor, Karachi Civic Center</td></tr>
<tr><td rowspan="3">Participants</td><td>Master Plan Department, Sindh Master Plan Authority (SMPA)</td><td>1. Mr. Akhtar Ali, Deputy Director, 0321-2774312
2. Muhammad Onangbeb, Assistant Director / GIS expert, 0333-03471154
3. Mr. Abdul Bayoom Bacut, Director, Administration & Accounts</td></tr>
<tr><td>Karachi Development Authority</td><td>4. Rafiq Ahmed Khan, Director (P&UD), 0333-2308066, 031-2596168</td></tr>
<tr><td>Project Team (PT)</td><td>Sakai, Gonai</td></tr>
<tr><td>Memo</td><td colspan="2">▪ Regarding the organization of the Sindh Master Plan Authority (SMPA) (For detailed information, refer to the attached document 'About SMPA – 2023'): The Sindh Master Plan Authority (SMPA) is responsible for developing the Strategic Development Master Plan (SDMP) at the macro level. The spatial coverage of the master plan extends over the entire City District of Karachi, comprising 18 administrative towns, 6 cantonments, and the Federal and Provincial government land-holding agencies. The Karachi Development Authority (KDA) is responsible for formulating micro-level planning based on the SDMP and overseeing the implementation of the plan.
▪ The Project Team (PT) had sent the questionnaire to SMPA in advance. SMPA provided the completed questionnaire to PT during the actual interview survey. Please refer to the attached document for the completed questionnaire. Below are the survey results and supplementary information based on the completed questionnaire.
1) Utilization of hazard/risk information in land use planning and zoning programs
・ SMPA does not have natural disaster hazard/risk information such as hazard/risk maps. There is no coordination mechanism for National/Provincial Disaster Management Plans and related plans. Consequently, the results of natural disaster risk assessments conducted by the Provincial Government of Sindh are not shared between the province and SMPA. Due to the absence of hazard/risk maps, it is not possible to identify high-risk areas, and there are no regulations in place for construction and development in high-risk areas.
2) Guidelines for land use planning in vulnerable areas
・ The Building/Development Permit System within the SMPA planning area is based on the 'Karachi Building and Town Planning Regulations-2002.' This regulation does not include criteria for land use planning in vulnerable areas. In other words, construction and development are permitted even in areas with a high risk of flooding.
3) Request for sharing the materials and data
・ The documents indicated in the 'Material' below have been provided.
4) Specific projects for disaster risk management
・ A list of specific projects has been provided. Please refer to '7_Project List' in the 'Material' below.
・ While there are survey results and plans available for the above-mentioned projects, SMPA/Karachi City/Local Authority does not have</td></tr>
</table>

	a budget for 'Implementation.' There is no financial support from the provincial or national budgets for the execution of these projects. They rely solely on donors such as the UN and JICA. Some projects have received UN support in the past and have been implemented. 5) Enforcement of effective land use control and regulations for disaster risk management ・ The building permit system has been operated by the Building Authority of the Sindh province. This inspection is based on the SDMP and the 'Karachi Building and Town Planning Regulations-2002 (KBTPR2002).' Neither the SDMP nor the KBTPR2002 specifically regulate construction and development in high-risk disaster areas. Therefore, the enforcement of effective land use control and regulations for disaster risk management is also not being conducted. ■ In addition to the topics along the questionnaire, PT had a discussion on urban flooding and its countermeasures as follows. 6) Progress on urban flooding countermeasures in NDMP ・ Contingency plan for urban flooding has not yet been developed. (Intervention-2) ・ SMPA doesn't have information on hazard and vulnerability analysis for urban flooding. As described above, there is a problem in communication between SMPA and PDMA. (Intervention-3) ・ Early warning system for urban flooding has not yet been developed. (Intervention-4) ・ Establishment of disaster management cell to plan for and response to emergencies effectively in the City District Government Karachi (CDGK) was proposed in Karachi Strategic Development Plan 2020, but it has not yet been realized. (Intervention-7) 7) Storm water drainage planning and projects ・ Storm water drainage planning is jurisdiction of Karachi Water and Sewerage Board/Corporation (KWSB). ・ Currently, no storm water drainage projects related to SMPA is in progress, but SMPA gave us their ideas to mitigate urban flooding such as (a) channelization of drains/nullahs, (b) development of rain storage tanks, (c) establishment of monitoring system for 3-main and 38-small drains/nullahs to detect illegal solid waste dumping. 8) Urban flooding damage data ・ SMPA does not collect damage data by urban flooding due to lack of manpower and budget.
Material	https://cti-group.box.com/s/31r0ukw3s7j7ih1xhfe83pbgmluy7zu7 0_ About SMPA – 2023（Word） 1_Questionnair Sheet_Filled_SMPA（Word） 2_Karachi Strategic Master Plan 2020（PDF） 3_Karachi Building and Town Planning Regulations-2002（PDF） 4_Greater Karachi Plan – 1952（PDF） 5_Karachi Development Plan (1974-85)（PDF） 6_Karachi Development Plan 2000（PDF） 7_Project List （hard copy / Scanned PDF） 8_Land Use Zoning Maps（5 maps, hard copy）

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	06 October 2023, 8:30 - 10:00(PST)	
Interviewee	Housing, Urban Development & Public Health Department, Government of The Punjab	
Place	2-Lake Road, Mozang Chungi, Lahore	
Participants	Housing Urban Development & Public Health Engineering Department, Government of The Punjab	1. Mr. Soman Khalid Sindhu, Deputy Secretary (Technical), 0332-4537842, souman.khalid@gmail.com
	Project Team (PT)	Sakai, Gonai
Memo	Prior to the meeting, an explanation of the organizational structure under the Housing Urban Development & Public Health Engineering Department, Government of The Punjab (HUD & PHED) was provided. • Development Authority: There are nine of them, responsible for development in different areas. • Water and Sanitation Agencies: Responsible for water supply and sanitation. • Public Health Engineering Department: Handles water supply and sanitation projects in the province. • Housing and Town Planning Agency: Involved in housing projects for the poor. • Punjab Land Management and Land Development Company (BLDC): Its functions have been transferred to the Punjab housing and town planning agency. • Clean drinking water companies. 1) Utilization of hazard/risk information in land use planning and zoning programs ・ The responsible agencies for hazard mapping related to general disasters are the PDMA (Provincial Disaster Management Authority) and DDMA (District Disaster Management Authority). The responsible agency for "flood" hazard mapping is PID. HUD & PHED believes that these agencies should each integrate hazard maps into their respective planning documents. 2) Guidelines for land use planning in vulnerable areas ・ The guideline will be provided from Deputy Secretary Khan Sattar Abdus, HUD & PHED. 3) Request for sharing the materials and data. ・ The materials will be provided from Deputy Secretary Khan Sattar Abdus, HUD & PHED. 4) Specific projects for disaster risk management ・ In response to recent water-related disasters, river management facilities have been installed at elevations exceeding the anticipated flood levels. For example, electrical machinery and equipment in drainage pump facilities are positioned at higher elevations to prepare for power outages, and solar power generation is utilized for electricity supply. 5) Enforcement of effective land use control and regulations for disaster risk management ・ Efforts are being made to promote regulations on illegal construction throughout the state. However, there is still a challenge in regulating housing development in disaster-prone areas.	

	6) Urban flooding issues ・ Early warning system for fluvial flood is operated by the Punjab Irrigation Department (PID). PID gives alerts based on the observation of water level at control points along rivers. There is no early warning system for pluvial flood in Lahore. ・ Separated sewerage and drainage system is proposed for land development project in the future, but the current system almost consists of combined sewerage system.
Meeting Materials	

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	06 October 2023, 11:00 - 12:00(PST)	
Interviewee	Housing, Urban Development & Public Health Department, Government of The Punjab	
Place	2-Lake Road, Mozang Chungi, Lahore	
Participants	Housing Urban Development & Public Health Engineering Department, Government of The Punjab	1. Mr. Khan Sattar Abdus, Deputy Secretary, 0317 4491700
	Project Team (PT)	Sakai, Gonai
Memo	1) Utilization of hazard/risk information in land use planning and zoning programs ・ 2) Guidelines for land use planning in vulnerable areas ・ In Chapter II (Application for Approval) of the "Notification in exercise of the powers conferred under section 43 of the Punjab Development of Cities Act," there is Section 7 (Evaluation Criteria) which includes the criterion "It is not prone to flooding." ・ The determination of "It is not prone to flooding" is based on the assessment conducted by the Irrigation Department of the Punjab state, considering areas at risk of water-related disasters. 3) Request for sharing the materials and data. ・ “Notification in exercise of the powers conferred under section 43 of the Punjab Development of Cities Act” was shared with Project Team from DS Khan. 4) Specific projects for disaster risk management ・ The Housing Urban Development & Public Health Engineering Department, Government of The Punjab (HUD&PHED), supported by the World Bank (WB), is currently implementing a housing supply project through the Affordable Housing Scheme to address a demand-supply gap of 5 million housing units. This housing supply project takes into consideration disaster preparedness and mitigation. 5) Enforcement of effective land use control and regulations for disaster risk management ・	
Meeting Materials	1. Notification in exercise of the powers conferred under section 43 of the Punjab Development of Cities Act	

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	06 October 2023, 15:00 - 16:00(PST)	
Interviewee	Lahore Development Authority	
Place	467-DII Johar Town Lahore	
Participants	Lahore Development Authority	1. Mr. Aghar, Chief Town Planner, 042-99262321 2. Muhammad Nadeem, Assistant Director, Town Planner, 0304 7863910 3. Usman Tahir, Deputy Director Master Planning
	Project Team (PT)	Sakai, Gonai
Memo	1) Utilization of hazard/risk information in land use planning and zoning programs ・ Refer to the “Master Plan of Lahore 2050” 2) Guidelines for land use planning in vulnerable areas ・ The Lahore Development Authority (LDA) examines construction and development activities based on the "LAND USE REGULATIONS 2020." However, there are limited criteria related to Disaster Risk Management outlined in these regulations. Specifically, development examination is conducted only for the "Health and Educational Project in Agriculture Area" and the "Establishment of Industrial Project and Industrial Estate in Agricultural Area" when the area is determined to be "not prone to flooding." 3) Request for sharing the materials and data. ・ If the Project Team (PT) submits an Official Letter to the Lahore Development Authority (LDA) requesting documents, the LDA can provide the "Master Plan of Lahore 2050" "Planning Document" to the PT. On the LDA's website, only the "Land Use Map" and the "Public Notification" of this master plan are available for viewing. 4) Specific projects for disaster risk management ・ There are instances where Buffer Zones for development regulations have been designated within a certain range from rivers. ・ There are projects for Open Space/Green Space development. These projects are believed to contribute to Disaster Risk Management. ・ In the riverbank area along the Ravi River in Lahore, the jurisdiction falls under the "Ravi Urban Development Authority (RUDA)." For disaster preparedness projects in this area, contact Umer Zulfiqar, Deputy Director of Urban Planning at RUDA, at 03234140094. 5) Enforcement of effective land use control and regulations for disaster risk management ・ LDA basically has a policy of not allowing development in disaster-prone areas. Specifically, if a building or development application submitted to the LDA is subject to review at a location that is considered to be at risk of flood damage, such as along a river, the LDA will contact the Punjab Irrigation Department (PID). If the PID assesses the area as a flood hazard area, the LDA will reject the application. ・ The above is an official procedure. There are construction and development activities in areas that are at risk of flooding (Illegal building construction/development). Illegal building shall be demolished based on the “LAND USE REGULATIONS”	

	6) Countermeasures against urban flooding in Lahore ・ Contingency plan for urban flooding has not yet been developed. ・ LDA officers explained that the above mentioned “Buffer Zone” along rivers, canals and drainage channels and “Open/Green Space” in flood prone areas would be also effective to decrease the risk of urban flooding. ・ Under the LDA building and zoning regulations 2019, construction of storage tanks for rainwater harvesting are recommended to all buildings to be constructed in the future.
Meeting Materials	1. Land Use Map (Existing Land Use & Proposed Land Use), Master Plan of Lahore 2050 2. LAND USE REGULATIONS 2020

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	12th October 2023, 1400 ~ 1630 (PST),	
Interviewee	Mr. Sultan Mahmood, Chief Engineer Drainage and Floods (D&F) Zone, Lahore.	
Place	Islamabad	
Participants	PID Punjab	Sultan Mahmood Chief Engineer D&F, PID Punjab.
	Project Team	Kohei Takayama, Takafumi Sueishi, Qasim Saeed
Memo	Agenda: - Inquire on progress of NDMP-I significant milestone and Interventions 2, 4 and 7 - Acquire other information and relevant data Minutes: JICA Project Team members explained that JICA expert team is currently working on the project entitled "Technical Support Project to Review and update The National Disaster Management Plan (NDMP)" under the Japan International Cooperation Agency (JICA) with the Coordination of National Disaster Management Authority (NDMA), which aims to study the current conditions of disaster management in Pakistan and to update the NDMP as desired. ***a. Presentation on PID Punjab Organization and Functions*** - Chief Engineer Irrigation Punjab, desired that if possible Government of Japan may participate to resolve the management issues of D.G.Khan hill torrents through JICA Pakistan office on the same way as in 1994 in case of Mithawan hill torrent. Punjab Irrigation wants to formulate and implement the flash floods management project as early as possible for Dera Ghazi Khan (D.G.Khan). - On discussion of NFPP IV, Chief Engineer Irrigation desired that as the PC-1 for NFPP IV were prepared before 2017-18 for next 10 years. Six years have already been passed away and site conditions of proposed projects have already been changed partially or completely at site. Some of the flood protection structures have already been constructed or rehabilitated at site under the Punjab Government by its own finance or by the International Donors financial assistance. He also emphasized that now this is the proper time to review, upgrade and reformulate the NFPP IV, PC-1s as per present site conditions with new integrated approaches keeping in view the flood damages during flood 2022. - Chief Engineer also informed and desired that the JICA Project Team may visit the all possible sites effected by floods during flood 2022 and see the damages and available facilities. - Apart from riverine flooding, "Hill Torrents" from the Suleman Mountain Range from Balochistan also caused serious flooding. Particularly Balochistan Hill Torrents ultimately drained into river Indus after damaging the Public and Private Infrastructures. - 2010 flooding is observed to be due to riverine flooding (overflowing of the Indus River) in Punjab and causes wide spread inundation and serious damages,	

	while the 2022 flooding was not so swear in Punjab Province. - The damages and inundation took place during 2022 in Punjab to the Public and Private Infrastructures along the left and right bank of river Indus mainly due to hill torrent floods during 2022, instead of Riverine flooding which was not so serious. - After the flood 2022, it was really emphasized that there is dire need of drainage plan for the hill torrents in Punjab i.e. Mianwali, D.G.Khan and Rajanpur districts, so that the huge damages due to hill torrents flash floods can be avoided in future. ***b. Interventions 2, 4, and 7*** **Intervention 2:** **2.1.** Formulate /update disaster management plans at National, Provincial, District, and community or TMA levels. Preparation of Annual Flood fighting Plans during March-April every year for the period of one year, which includes the list of vulnerable sites, Flood fighting materials list, provision of reserve stock of stone at vulnerable sites, and provision of necessary day and night watching of field workers at different level of flood. **2.2.** Develop Hazard Specific Contingency Plans Preparation and implementation of Flood Contingency plans, after every super flood event, which includes necessary restoration of projects after super flood event passed away and its implementation plans is also a part of Flood fighting plans. **Intervention 4:** 4.3. Strengthen Early warning dissemination system PID Punjab already heading the Flood Forecasting Division Lahore, (FFD) from June to October every year. FFD headed by PID Punjab have the deep coordination with SPARCO, Pak Army, PMD, WAPDA, NDMA, PDMAs, PIDs and other local and international weather forecasting organizations. After getting latest updates FFD issued the flood warnings to all flood stake holder. 7.4. Implement appropriate structure measures in flood prone areas taking into account comprehensive and integrated flood management plans. In Punjab several new considerable flood control projects were undertaken during last ten years, however, rehabilitation and modernization is also took place to 13 number of barrages and other Irrigation system and river flood protection structures. 7.4.1. A review of breaching Mechanism of the river embankments and a review of designed breaching points. According to the Chief Engineer D&F Lahore a committee has already been constituted to decide the breach execution in case of any kind of flood emergency and encroachment of free board of Flood embankments. The committee comprises of experts from PID, Highways Department, Pak Army, and Pakistan Railway.

	- Revision and updation of NFPP IV The process of up dation of NFPP IV is on the way and PID has submitted the revised version of NFPP IV projects to FFC for further processing. - Flood Plain zoning and Management This is responsibility of FFC and FFC is already working on it. - Capacity Building of stake holders in flood mitigation Mainly several training courses have been conducted before the start of flood season every year inside of PID. In addition to that a Mock exercises for the handling of flood emergency also been conducted by Irrigation Engineer with the collaboration of Civil administration, Pak Army, and other flood stakeholder organizations. **7.4.2.** Construction and Rehabilitation of flood control mitigation structures Several works undertaken in Punjab under the Annual Development Program (ADP) and under the umbrella of NFPP-IV and FPSP-III Projects. - Implementation of NFPP - IV NFPP IV umbrella project was formulated 2017-18, but due to flood 2022 site conditions have already been changed partially or completely, so revision of NFPP IV is in progress and after approval from competent authority more projects can also be included and implemented at site accordingly. As discussed above the Modernization and Rehabilitation of 13 number of Barrage in Punjab is also a part of the NFPP IV. - Retrofitting of existing dams, Barrages, and flood protection works to increase flood water retarding capacity and ability to accommodate design discharge. Several projects were necessary for this purpose but due to porosity of funds some projects have already been taken in hand under Provincial, Federal and International donors financial assistance. - Construction of new flood protection works, Dams and Barrages to increase flood mitigation/protection capacity. As discussed above there are large number of projects which needs to be implemented to increase the flood mitigation/protection capacity, but due to porosity of sufficient funds PID can only rehabilitate and modernized major Hydraulic structure like Barrages only in the Punjab. ***Other Information:*** - According to the Chief Engineer D&F Lahore, Major hill torrents flash flood from Suleiman Range Balochistan enters into Punjab plane areas and created huge damage to public and private infrastructures which needs to be managed at National level on top priority otherwise not only damages occurs in Punjab but in Sindh Province too being a lower end tear of Indus river.

	- As the three major southern rivers have controlled by India under the Indus water Treaty 1960. This treaty causes considerable reduction in southern river discharges, which causes reduction in Indus River discharges and ultimately badly effect the agriculture supplies for Sindh province. - During the moon soon season (June 15 to October 15), PID closely monitors 24/7 the flood situations and river flows with the help of FFD office at Lahore. ***d. Data Collection*** - Some of the available data have already been provided by PID Punjab. however, for the remaining information JICA Project Team will request to PID Punjab for further data, time to time when ever needed.
Prepared by	M. Qasim Saeed

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	7th November 2023, 1100 ~ 1500 (PST),	
Interviewee	Mr. Syed Ali Shah Executive Engineer Sukkur barrage PID Sindh	
Place	Sukkur Barrage, Sukkur	
Participants	PID Sindh	Mr. Syed Ali shah Executive Engineer.
	Project Team	Kohei Takayama, Takafumi Sueishi, M. Qasim Saeed
Memo	Agenda: - Inquire on progress of NDMP-I significant milestone and Interventions 2, 4 and 7 - Acquire other information and relevant data Minutes: JICA Study Team members explained that JICA expert team is currently working on the project entitled "Technical Support Project to Review and update The National Disaster Management Plan (NDMP)" under the Financial Assistance of Japan International Cooperation Agency (JICA) with the Coordination of National Disaster Management Authority (NDMA), which aims to study the current conditions of disaster management in Pakistan and to update the NDMP as desired. ***a. Presentation on PID Sindh Organization and Functions*** - Executive Engineer, Sukkur Barrage Division Sindh Irrigation desired that if possible JICA team may also study and help the Sindh Irrigation Sukkur Barrage for the following major issues. - Restoration of design discharge of Sukkur Barrage by opening the closed 7 no. of gates. - Restoration and Monitoring of gauge recording system of Sukkur Barrage. - Installation of SCADA system over the Barrage. - Restoration and re-installment of Peizo meters. - Restoration and improvement of pressure pipes. - Complete automation and centralized control of Barrage gates operations. - To control the high siltation rate along right side off taking canals. - Pilot Project may be implemented for DADU Canal by concrete lining of first 10 km for the purpose to avoid excessive silt entries. - Under Sindh Barrage Improvement Project (SBIP) Desilting of Rice canal is planned but not yet implemented any project at site. - Sindh Province experiences both extremes inundation and as result drought conditions in Sindh during and after 2022 flood. - Flood fighting plans prepared by every Executive Engineer and after vetting by Secretary PID it forwarded to Federal Flood Commission Islamabad for final approval. - However, Contingency plan normally prepared by Sindh PID at Provincial level after every major flood event. ***b. Interventions 2, 4, and 7*** **Intervention 2:** **2.1.** Formulate /update disaster management plans at National, Provincial,	

	District, and community or TMA levels. Annual Flood fighting Plans or contingent flood plan during March-April every year prepared by each Executive Engineer for the period of one year. **2.2.** Develop Hazard Specific Contingency Plans Preparation of Flood Contingency plans, after every Major flood event **Intervention 4:** **4.3.** Strengthen Early warning dissemination system Its true that, in Pakistan warnings issued to all flood stake holders including PID through FFD, Lahore. However, Sindh PID is actually interested in timely and correct information about hill torrents flood generated from Balochistan and Punjab Province, for the purpose of timely adoption of flood preventive measures. Executive Engineer(XEN) mentioned that Sukkur barrage do not face any shortage of water and lack of flood warnings. He pointed out that they always have at least forecast for 3 days prediction time when water passed from Taunsa Barrage. He shows the prediction chart for flood. According to him 6 days prediction time is available from Chashma Barrage to Sukkur Barrage and similarly more time for Kotri Barrage. He also mentioned the water losses and minimum flow chart of water. XEN explained they can manage by internal regulation the available shortage of waters by closing secondary and tertiary canals one by one for 10 days period only. PID ensured at least one water turn to each farmer must be received every 10 days warabandi (water turn). Regarding weather forecast PID depends upon PMD, for future discharges depends upon upstream barrage discharges, while Rainfall /Temperature follow the forecasts from PMD/FFD. Similarly longer period prediction of flood look at the temperature at Sakardu, Kalash, and high Glaciers. **Intervention 7:** 7.4. Implement appropriate structure measures in flood prone areas taking into account comprehensive and integrated flood management plans. Although no new considerable project is undertaken during last ten years over the Barrage however, rehabilitation and improvement works took place over the barrage time to time. Vulnerable points also selected by the site engineers before flood. If water comes out of river bed than can never controlled. No breaching section available at barrage or its structure. 7.4.1. A review of emergency breaching Mechanism of the river embankments and a review of planned breaching points. According to the Executive Engineer Sukkur Barrage, there is no planned breaching section under the control of PID at Sukkur Barrage. - Revision and up dation of NFPP IV **XEN** Sukkur Barrage do not know about the NFPPP-IV umbrella PC-1. - Flood Plain zoning and Management. XEN Sukkur barrage also don not have any information about flood Zoning because its responsibility of FFC Islamabad. - Capacity Building of stake holders in flood mitigation. XEN Sukkur Barrage mentioned that, he also conduct some training courses before the start of each flood season every year with in Barrage staff and Junior engineer but no Mock exercises carried out at Barrage for flood preparation for other stake holders. May be these exercises carried out at Provincial level only. **7.4.2.** Construction and Rehabilitation of flood control mitigation structures. XEN Sukkur Barrage pointed out that, several flood protection works undertaken under Federal PSDP, Provincial PSDP and normal Operation

	& Maintenance (O&M) funds every year, over the Sukkur Barrage. - Implementation of NFPP IV As explained above XEN do not have any information about NFPP -IV, however, most of the projects implemented at Sukkur Barrge under PSDP, (Federal and Provincial), and several projects are in progress under SBIP. - Retrofitting of existing dams, barrages, and flood protection works to increase flood water retarding capacity and ability to accommodate design discharge. XEN Barrage do not have any opinion about such question, he thinks that it does not relates to him however, may be this is the portfolio of Federal Government. - Construction of new flood protection works, Dams and Barrages to increase flood mitigation/protection capacity. XEN agreed that several projects are necessary for the rehabilitation of Barrages but due to porosity of funds no considerable project has been taken in hand. However, several flood protection works have already been started under Sindh Emergency and Resillience Project (SERP) financed by world Bank which includes the replacement of barrage gates and major rehabilitation of works. ***c. Other Information:*** - ***. Data Collection*** - Most of the available data have already been provided by XEN PID Sukkur S however, but he ensured any other data if required will be provided on demand.
Prepared by	M. Qasim Saeed

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP 2)	
Date and Time	10th of November, 2023. at 1430-1600 Hours	
Interviewee	Mr. Samiullah Baloch, Provincial Coordinator, PID Balochistan Quetta.	
Place	Islamabad	
Participants	PID Balochistan	Provincial Coordinator (PC), PID Balochistan, Quetta.
	Project Team	Kohei Takayama, Takafumi Sueishi, Qasim Saeed
Memo	Agenda: - Inquire on progress of NDMP-I significant milestone and Interventions 2, 4 and 7 - Acquire other information and relevant data Minutes: JICA Study Team members explained that JICA expert team is currently working on the project entitled "Technical Support Project to Review and update The National Disaster Management Plan (NDMP)" under the Japan International Cooperation Agency (JICA) with the Coordination of National Disaster Management Authority (NDMA), which aims to study the current conditions of disaster management in Pakistan and to update the NDMP as desired. On discussion of NFPP IV, PC Balochistan Irrigation desired that as the umbrella PC-1 for NFPP-IV was prepared during 2017-18 for the next 10 years. Six years have already been passed away and site conditions of proposed projects have already been changed during 2022 flood. Some of the flood protection structures have already been constructed or rehabilitated at site under the Provincial Government by their own finance or from International Donors financial assistance. He also pointed out that this is the time to review, upgrade and reformulate the NFPP -IV umbrella PC-1s as per present site conditions with new integrated approaches keeping in view the flood damages during flood 2022. - PC Balochistan also seeks help from Government of Japan under JICA Grant in Aid Program, for construction of new small, Medium and Large Dams in Balochistan. ***a. Interventions 2, 4, and 7*** **Intervention 2:** **2.1.** Formulate/update disaster/flood management plans at National, and Provincial level on three monthly basis as explained below: A, Moon soon plan, from June – August every year. B, Winter Plan, from 15th December to 15th March, every year **2.2** Develop Hazard Specific Contingency Plans Preparation of Flood Contingency plans, after every super flood event i.e. in 1992, 2010, and 2022. **Intervention 4:** **4.3** Strengthen Early warning dissemination system - Balochistan is depending upon the PMD flood forecasting and warning system only and also utilizing cellular phone, Social media and Radio	

broadcast networks in Kharan, Turbat and Gwader Districts.

Intervention 7:

7.4. Implement appropriate structure measures in Flood prone areas taking into account comprehensive and integrated flood management plans.
Although no new considerable project is undertaken during last ten years however, rehabilitation and modernization took place for the hydraulic and flood protection structures.

7.4.1. A review of breaching Mechanism of the river embankments and a review of designed breaching points.
According to the PC Balochistan PID, there are about 500 vulnerable point in whole Balochistan and several planned/forced breaching points are under the control of Balochistan PID.

- Revision and up dation of NFPP- IV,
The process of up dation of NFPP- IV is on the way and PID Balochistan already have been submitted the revised updated version of NFPP -IV projects to FFC for further processing.
- Flood Plain zoning and Management,
This is responsibility of FFC and FFC is already working on it.
- Capacity Building of stake holders in flood mitigation
Several training courses have been conducted before the start of flood season every year inside of PID.

7.4.2. Construction and Rehabilitation of flood control mitigation structures,
Several flood works undertaken under the umbrella of NFPP- IV and FPSP-III Projects.

- Implementation of NFPP -IV
NFPP IV umbrella project was formulated 2017-18, but due to flood 2022, lot of site conditions have already been changed, so revision of NFPP -IV is
in progress and after approval from competent authority projects may be executed at site accordingly.
- Retrofitting of existing dams, Barrages, and flood protection works to increase flood water retarding capacity and ability to accommodate design discharge.
Several projects are necessary for this purpose, However, about 1040 Check/small dams are available for Ground water recharge purpose only while 8-10 Medium /Large multipurpose Dams are also available in the Province.
- Construction of new flood protection works, Dams and Barrages to Increase flood mitigation/protection capacity.Several projects are necessary for the rehabilitation of Barrages but due to porosity of funds no considerable project has been taken in hand. However, several flood protection works have already been started in the Province and about 15 Large/Medium type of dams construction has also been started in 2019 under Federal PSDP through Ministry of Water Resources Islamabad. In which 2% project cost is kept as O&M cost for the constructed structures.

Other Information:

- Balochistan Province have decided to prepare drainage plan for about 20,000 small and big hill torrents draining into Punjab and Sindh.
Balochistan Provinces never care about the drainage effluents and hill torrents flood flows released every year to Punjab and Sindh province being a lower tear because of no damages occurred in Balochistan.
- During the flood days (June 15 to October 15), PID closely monitors flood

	situations forecasts from FFD Lahore and PMD. - PID Balochistan shifted the required Machinery and equipment to all Designated Vulnerable sites before flood season starts. - There are about 18 river basins in Balochistan out of which 5 basins and 33 sub basins finally drain into river Indus. - Out of 35 districts about 25 districts are effected by floods every year in Balochistan. - River Flowing in Palochistan is North to South in 400-500 KM long basins. ***d. Data Collection*** - Some of the available data have already been provided by PID Balochistan - however, for the remaining PC Balochistan promised to share during this week.
Prepared by	M. Qasim Saeed

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	15 November 2023, 10:00 ~ 12:00(PST)	
Interviewee	Mr. Sultan Mahmood Chief Engineer (Drainage and Floods), PID (Punjab)	
Place	Chief Engineer Office, PID, Thokar Niaz Beig, Lahore	
Participants	PID (Punjab)	Sultan Mahmood, Chief Engineer D & F (PID), Umer (HSSEU), Kashif (HSSEU)
	Project Team	Takayama, Sueishi, Qasim Saeed, Rahmat
Memo	Agenda: - Vulnerable Point List - Future Project List - Contingency Plan - Annual Flood Fighting Plan Minutes: ***a. Vulnerable Point List:*** - There are two types of vulnerable points list (a) with respect to seepage (b) with respect to flood protection to save population etc. which includes National Flood Protection Plan. - The vulnerable points already have flood protection bunds, but these points get damaged due to floods and/or the damages due to breaching at these locations can be severe. - Camps are made at these vulnerable points and machinery should be procured etc. is also deployed (Labor) at these camps. - There are no structural measures at the hill torrent areas and these areas were severely damaged during 2022 hill torrent floods. Focus shall also be made in the hill torrent areas as the discharges reaches to more than 100,000 cusecs during hill torrent floods in major hill torrents. - Efforts shall be made to preserve this water by constructing water retaining structures such as dams, reservoirs etc., instead of making the bunds for flood protection letting the water to go in river/sea. Drainage only is not an appropriate solution as it take lot of land with least benefit. Instead holding the water in reservoir and narrowing the channel for less discharge about 10,000 Cusecs would be a good solution. - Model studies are in progress for construction of bunds at vulnerable points in South Punjab such as DG Khan, Layyah, Rajanpur, Muzaffargarh etc., upstream and downstream of Taunsa barrage. - Normal erosion at the country/land side areas, where there is no threat to Irrigation Infrastructure also catered and proposed by PID, while funds are provided by PDMA or Local Government and works are also carried out at site by PID. - After the Flood Plain Act (2016), PID cannot propose safety bunds to save	

	population/land of locals, it is the jurisdiction of PDMA. PID provides technical assistance mainly in such cases and constructs the bunds as a deposit work of PDMA or Local Government. However, if safety of Irrigation Infrastructures/network is the concrem, PID proposes, designs and constructs the flood protection bunds by their own finance. - Indus and Eastern rivers major flood events took place within every 10 to 15 years cycles, in case of flash/hill torrent flood events took place every year. ***b. Future Project List:*** - Flash floods are common in Mianwali and flood bund protection PC-Is are prepared for the project. - Other priority areas for the flash flood protection are Vidor, Sanghar, Kaha, Chachar and Pitok. - The future projects are mainly included in NFPP - IV, and FPSP - III umbrella PC-1s. - The list of four major hill torrent projects in D. G. Khan area (Vidour, Mithawan, Chacher and Sori Lund) are also handed over to World Bank for the arrangement of funds. - Concept paper for the new projects will be sent by PID later. - A list of hill torrent projects was provided in the meeting which is also available with FFC and included in NFPP-IV. - Construction of Chinot dam on Chenab River should be implemented at the earliest so that a huge quantity of water can be saved for agriculture supplies. Similarly, water storage of Islam Head works on Sutlej River and Sidnai Barrage over Ravi River can also be enhanced. ***c. Contingency Plan:*** - Flood fighting plan is considered as the annual contingency plan by PID. It is prepared by each Divisional Officer (XEN) during the March /April every year before the start of monsoon. A hard and soft copy of flood fighting plan for 2023 of Dera Ghazi Khan and Multan has also been provided for better understanding.
Prepared by	Rahmat Ullah (Construction & Procurement planning/Cost Estimate)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	14 November 2023, 14:00 ~ 16:00(PST)	
Interviewee	Mr. Masroor Naqvi GM (HRM)	
Place	WAPDA House, Lahore	
Participants	WAPDA	Masroor Naqvi GM (HRM), Rashid Hameed PD (GMRC), Hammad Ali Dir.
	Project Team	Takayama, Sueishi, Saeed, Rahmat
Memo	Agenda: - Inquire on progress of NDMP-I (Interventions 3, 4 and 7) - Acquire other information and relevant data Minutes: ***a. Intervention 3:*** 3.1 Conduct research on impact of climate change on glaciers and ice caps - WAPDA not yet capable to monitor the glaciers completely. - The most vulnerable glacier and glacial lakes were identified. 40 No. of glaciers and 36 No. of glacial lakes are currently under monitoring. - There are 5 mega glaciers included in this study and mass movement monitoring is studied for these glaciers. - Depletion, repletion of the glaciers and its melting is forecasted before time so that the concerned are prepared for the disaster. - WAPDA wants to extend the monitoring system. - A presentation was shown regarding the GLOF research and monitoring achievements. - The study was carried out under Tarbela 4th Extension HPP project. For further studies, funds from Tarbela 5th Extension HPP will be allocated amounting to 4 to 5 million US$ financed by World Bank and expected to be finalized in one to two months. The proposal is already with the ministry for approval. 3.2 Establishment of GLOF early warning system for selected vulnerable areas - The glacier at upstream of Golen Gol Project caused GLOF and some damages were observed in the powerhouse. To prevent any further damages, early warning system was installed and it is functioning satisfactorily. - This EWS at Golen Gol Project is a special condition and it is only for GLOF damages to WAPDA structure on this project only. - Currently, WAPDA does not install EWS as they do not have budget. The budgets available in T4 and T5 HPP is mainly for the monitoring	

	purposes only. ***b. Intervention 4:*** 4.1 Enhancement of research activities for snow/glacier/glacial lakes - Currently, there is no other project is planned except for T5 Extension HPP. If some project demands, they will carry out such studies under their funding. ***c. Intervention 7:*** 4.1 Establishment of comprehensive and integrated flood management/protection plan in Pakistan ● Revision and updating of NFPP-IV - Five proposals were updated with the increase in scope and cost and was approved in FPSP-III in June 2022. ● Revision of SOPs of Tarbela Dam for flood mitigation - This information should be collected from concerned GM in WAPDA. GM (HRM) cannot provide the latest update. ● Capacity building of the stakeholders in flood mitigation - It is part of National Master Plan (NMP). 4.2 Construction and rehabilitation of flood control/mitigation structures ● Implementation of NFPP-IV - National Master Plan for Telemetry Phase-1 is in progress. 457 sites out of 800 are selected for the Flood Early Warning System and Telemetry which is implemented by WAPDA. This project involves all the stake holders and data sharing between the organizations. O & M staff includes key staff from all departments. - Project cost is about 16 Billion PKR, 90% is ADB funding and 10% is PSDP allocation. - JICA has also taken up 45 sites for the telemetry whereas 20 sites are proposed for Turkish funding. The approval of 20 sites for the funding is still awaited.
Prepared by	Rahmat Ullah (Construction & Procurement planning/Cost Estimate)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	15 November 2023, 12:30 ~ 14:00(PST)	
Interviewee	Mr. M. Aslam (Chief Meteorologist, FFD- PMD)	
Place	Flood Forecasting Division (FFD), PMD, -Jail Road, Lahore	
Participants	FFD-PMD	M. Aslam
	Project Team	Takayama, Sueishi, Saeed, Rahmat
Memo	Agenda: - Inquire on progress of NDMP-I (Interventions 3, 4, 7 and 10) Minutes: ***a. Intervention 3:*** 2.1 Preparation of vulnerability analysis and creation of hazard maps for selected districts and cities - No such vulnerability analysis is prepared. It is not related to FFD. FFD only forecasts the floods. NDMA and/or PDMA shall do the vulnerability analysis. ***b. Intervention 4:*** 1.2 Strengthening of flood forecasting by satellite information and hazard maps of Indus River - It was a UNESCO project, and it is already finished. Hazard maps were prepared and handed over to FFD, Lahore. - The hazard maps were prepared not only satellite information but also using model. PHAST model was used which was prepared by SUPARCO. - SUPARCO did the analysis work on the model and handed over the output maps. FFD is using PHAST model since 2013. - For PHAST model, data of the India side is obtained from satellite and/or radar. This is PHAST limitation. In case of heavy rains, FUSE model is used. Sometimes regression models are also used. The outputs of the models are reviewed and using the staff experience the outputs are modified, if deemed necessary. - News related to weather on India side are also checked from news, electronic and social media. 4.4 Education Program for advanced meteorology and hydrology for PMD staff - FFD nominates the staff and training is carried out in Islamabad by PMD or in other countries. ***c. Intervention 7 and 10:*** - All the information planning, designing, infrastructure development and post-disaster recovery etc. is related to PMD- Islamabad.	
Prepared by	Rahmat Ullah (Construction & Procurement planning/Cost Estimate)	

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	20 November 2023, 14:00 ~ 15:45(PST)	
Interviewee	Mr. Sikander Ali Zardari (Chief Engineer)	
Place	KWSC Office, Karsaz, Shahra-e-Faisal, Karachi	
Participants	KWSC	Sikander Ali Zardari
	Project Team	Takayama, Sueishi, Saeed, Rahmat
Memo	<u>Agenda:</u> - Storm water drainage planning - Countermeasures for urban flooding - Urban flooding damage data <u>Minutes:</u> - Storm water drainage and urban flooding etc. are carried out by Karachi Metropolitan Corporation (KMC). - Karachi Water and Sewerage Corporation (KWSC) is mainly responsible for the water supply and sewerage disposals, sewerage from domestic and industrial. - KWSC was KWSB earlier and six months ago the Board was changed to Corporation. - KWSC has installed the network of underground conduits for carrying sewerage. KWSC looks after that and mainly responsible for this work. - The stormwater is forced to drain in the sewerage by the locals. They uplift the manhole covers and let the stormwater pass through the conduits. However, the capacity of these conduits are limited and it is choked because of high volume of water and debris. - KWSC uses pumps etc. equipment to remove this stormwater. - KMC is mainly responsible to cater the urban flooding, and big nullahs/open channels are used to drain the stormwater. However, sewerage is also drained in these nullahs, and the stormwater and sewerage are mixed. S-III Project is an ongoing project to separate the sewerage lines and stormwater drainage and 70% has already been separated. 30 % is still connected to the stormwater nullah. - S-III project presentation was shown and hardcopy was provided. - Sewerage system of Karachi was shown in the meeting. There are 3 treatment plants (TPs). - For the stormwater drainage, there are three main nullahs (i) Mahmoodabad (b) Orangi and (c) Gujar nullah. All of the nullahs have encroachments and National Logistics Corporation (NLC) and Frontier Works Organization (FWO) are working on Orangi and Gujar nullahs respectively, to remove the encroachments and construct embankments,	

	similar to Lai nullah project. - After the treatment of sewerage, the treated water is drained into sea. Currently, there is an ongoing project in which the treated water is sold to the industry. - There are three development authorities in Karachi. (a) KDA (b) LDA (c) MDA. After 1988, no development is proposed. - In Karachi, sewerage and drinking water are mixed. The only source of drinking water is Maleer Water Basin, since 1860 and water is pumped through wells. - KWSC has very old sewerage system and change in land use from residential to commercial has increased load on the current system which is not upgraded from a long time ago. In addition, increase in population has also increased the sewerage quantity. - Many sewerage pipes are sunk down due to rains and flooding of 2022. Erosion occurred and these conduits are exposed now whereas they were buried under the ground earlier. - DHA and other private housing societies, they do all the sewerage treatment by themselves through their own treatment plants. The water is supplied at one point and DHA/others supply to their housing schemes by themselves. Similarly, the Cantt Boards (11 Nos) which is looked after by Army has similar situation. - Currently KWSC holds 41 jetting machines, 41 suction machines and 3 dewatering pumps.
Prepared by	Rahmat Ullah (Construction & Procurement planning/Cost Estimate)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	22 November 2023, 14:30 ~ 15:15(PST)	
Interviewee	Mr. Amjad Saeed (Member-Punjab), IRSA	
Place	IRSA Office, Islamabad	
Participants	IRSA	Amjad Saeed
	Project Team	Goto, Takayama, Sueishi, Saeed, Rahmat
Memo	<u>Agenda:</u> - Role and Responsibility of IRSA - Water Apportionment Accord 1991 - Hydrological Models <u>Minutes:</u> - IRSA members are nominated and recommended by the provincial Chief Minister and Prime Minister of Pakistan made the appointment of the members to work under the Ministry of Water Resources. There are some water disputes between the provinces specifically between Sindh and Punjab. The water disputes were resolved in 1991 after the signing of water accord, which was agreed by all the provinces. The accord defines the water shares of all the provinces. - The water share is defined as crop-wise for Kharif and Rabi crops. At the low water time, the allocation of water is done as per the 1991 water accord. - At the start of every crop season, IRSA Advisory Committee has a meeting for the water distribution. The committee comprises of members from all provinces, agriculture department, WAPDA, PMD, and FFC representatives. - IRSA finalizes the water distribution criteria for the crop season, by forecasting the inflows. Director (Operations), IRSA make statistical analysis using historical inflows. They have databases for all the inflows, Kabul River has control at Nowshera, Indus River at Tarbela, Jhelum River at Mangla, Chenab River at Marala Barrage, Sutlej River at Ganda Singhwala and Ravi River at Jasser. - Every province (PIDs) and WAPDA etc. also prepare their own analysis using numerical hydrological models and presented during the advisory committee meeting. There are several gaps between the analysis made by each department and discussion will be done during this meeting. After discussion, the water distribution criteria between the provinces are finalized. - A separate meeting can be arranged with WAPDA and/or PIDs to discuss and understand their numerical hydrological models. - Earlier IRSA was doing all the calculations manually but recently they started to use the model Water Apportionment Accord Tool (WA Tool) developed by Australia organization (CSIRO) financed by ACIAR. The	

	tool was just a transformation of manual calculations to computer aided software. - Now the Australian government has allocated more funds and now they developed mid-season analysis tools also. - There is no authentic mechanism for the losses calculation which creates serious misunderstandings and disputes between provinces because losses cannot be properly calculated. - 142 MAF is currently available water for Pakistan and out of which 20 MAF is booked as losses but there is no calculation to show such losses. There is a dire need to rationalize the losses at the control points in the main rivers. - If telemetry system is installed at all the control points such disputes can be resolved.
Prepared by	Rahmat Ullah (Construction & Procurement planning/Cost Estimate)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	21 November 2023, 12:00 ~ 13:15(PST)	
Interviewee	Mr. Sajid Ali Bhutto (Chief Engineer), PID Sindh	
Place	PID (Sindh) Office, Sindh Secretariat, Karachi	
Participants	PID (Sindh)	Sajid Ali Bhutto, Ali Asghar Maysar
	Project Team	Takayama, Sueishi, Saeed, Rahmat
Memo	Agenda: - Information about Sindh Barrage Improvement Project (SBIP) - Document and Data Collection for Manchar Lake and Indus River - Future Projects Minutes: - Sindh PID is mainly interested in the rehabilitation / modernization of barrages in Sindh province and an institution for hydrology studies. - Embankment is not priority for Sindh PID because it is attended time to time, but barrages involve great expertise and huge finances. Sindh PID would like to request for 3 barrages improvement and 1 institution for hydrological study. - Whole barrage improvements are not required, and part of the barrage improvement / modernization would also be acceptable. For example, aprons on the upstream and downstream, piezometers installation and upgradation, installation of SCADA system etc. - SBIP is covering the rehabilitation of three barrages in the order of Guddu Barrage, Sukkar Barrage and Kotri Barrage. The works on Guddu barrage have already been started. - Guddu Barrage was constructed in 1962 and the design discharge was 1.2 million Cusecs. Since then, many components of the barrage were not attended till 2013. For example, gates were broken at the bottom but no replacement, and problems of river training work at upstream etc. Due to such issues some projects (probably three) were carried out in recent years such as construction of spurs to centralize the river flow. 2 spurs were constructed at the right and the left side to centralize the river flow. - SBIP is a World Bank project, and all the formalities were done in 2014 while the work was started in 2019. - PC-I of SBIP had many addressed items but a lot of things were also missed in PC-I which are also being carried out through Variation Orders of the Contract Agreement. For example, the glacis were inspected, and it was completely eroded. The water flows throughout the year through this barrage and such inspections can only be done in the inspection period during the annual closure, also the embedded part (Mechanical works) were not in PC-I but during inspection it was found that it was completely rusted. This was also replaced with new parts through Variation Order of the Contract. The Consultant team for this project was	

	from UK. - 10 gates are already replaced including the hoisting system, and it was imported from Turkey. Mostly the gates are imported. After covid-19 there was an issue with the supply from the same company and then it was changed to vendor in China using same specifications, including the hoisting system. - The barrage improvement / modernization projects were not included in NFPP-IV and FPSP-III because there was no provision in them. It was mainly flood protection works and costs of modernization is very high which will increase NFPP-IV drastically. But priority for the modernization/improvement of barrages is very high. - The works at Guddu barrage were carried out using bulkhead gates and for Sukkur by coffer dams. After monsoon 2024, the construction of coffer dam will be started. The major works are divided in different packages such as S-I (Major contract for gate replacement), S-II (excavation and dredging), S-III, S-IV (Desilting on L/S about 48 RDs). - Guddu Barrage has 2 packages G-II (Gate replacements), G-III (Raising of the bunds). - Guddu and Sukkar Barrages, all the gates will be replaced including the hoist system. - The Kotri Barrage is remaining and the TORs for the modernization is under preparation. After that consultants will be hired for the design and tendering. - After the rehabilitation counter weights for the gates will not be required. - For the Sukkar and Kotri Barrages, two feet height of the gates are increased but for Guddu Barrage the height of the gate is same after modernization. - For Kotri Barrage, it is expected the process is long and time taking due to the procedures by World Bank. It is requested if JICA can provide some funds for this project. - Project brief for SBIP was requested and shall be provided later. Some documents are also available on websites. - List of vulnerable points related to flood protection bunds were requested and will be sent later. - Daily discharges data of Manchar lake from 2010 to 2023 was requested and has already been received from PID Sindh. - For the Indus River daily discharges at 3 barrages from 2010 to 2023 was requested and has already been provided by PID Sindh. - SOP, cross-sectional data of 3 barrages was requested. Also, inundation maps, and material related to artificial bund breaches was also requested.
Prepared by	Rahmat Ullah (Construction & Procurement planning/Cost Estimate)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	(1) 24 November 2023, 14:30 ~ 15:30(PST) (2) 27 November 2023, 09:30 ~ 12:00(PST)	
Interviewee	Dr. M. Tahir Khan Director (Planning), PMD	
Place	PMD Office, Islamabad	
Participants	PID (Sindh)	M. Tahir Khan
	Project Team	(1) Takayama, Sueishi, Saeed, Rahmat (2) Sueishi, Saeed, Rahmat
Memo	Agenda: - Outline of each Early Warning System (EWS) and system - Present situation of each EWS and system - System Mentioned in NDMP-I - Future project of each EWS and system - Radar System - Communication System - Capacity Development of PMD Minutes: Existing EWS - At the moment, there are 85 AWS available in Pakistan. Out of which 45 were provided by JICA under JICS program while rest of the AWS were provided in several other programs in small parts, say 5 No.s at one time etc. - At present 2 projects are in pipeline for AWS. (a) Modernization of Hydro-Met Services (300 AWS by WB) (b) Expansion of Metrological Network of PMD (252 AWS by FFC). In these projects 552 AWS will be installed. - For EWS, currently several kinds of systems/framework are in use, as these systems are not wholly connected/integrated with each other. For example, 2 radars are installed with JICA assistance, and they are not yet integrated in the system. Their data are not being used for the numerical prediction models. Similarly, 2 wind profilers are installed at Islamabad and Multan and both are not integrated in the forecasting system. Also, one ex band radar installed in 2018/19 is also not integrated. - Currently, PMD is getting data from about 100 observatories and a manual forecast is prepared besides the numerical prediction facility, and both analyses are examined if they are providing similar results. - In present system, human interaction is too much for the forecasting and warning system. - It is planned to expand the AWS data points in line with the WMO criteria. WMO suggests that the distance between 2 AWS should not be	

	more than 25 kms. But currently, the distances are sometimes several 100 kms far. For example, the observation station at Quetta and Zoub are 370 km apart. Zoub to D.I. Khan is the most vulnerable area and the distance between the observatories is more than 350 km. - Currently, PMD is using models and satellite data together with the staff experience for the forecasting. - PMD wants to further increase the data points, integrate all the system at one place, and digitize the EWS by enhancing the numerical prediction facilities/simulations. The forecast can be done more reliable. - Even if all the proposed projects/radars are installed, still there will a lot of gap/distance between the radars. Furthermore, at the mountainous areas/ Northern Areas the distance between the radars shall be further reduced because of high mountains hindering the signals. - AWS are required to monitor the urban/flash flooding. Smaller scale systems are also vital during the heavy rainfall in the vulnerable areas which can provide forecasting of urban/flash flooding. - It is requested if such projects can be included in the report. Also, the future projects list shall be shared by PMD. PMD has prepared a Master Plan and shall be provided to us after some updating. - The Existing EWS has a lot of gaps and requires update. Some of the systems are already outdated and they are performing in isolation, not as an integrated system. - The details of the Existing EWS and current status will be shared by PMD. Systems Mentioned in NDMP-I - Upper Air Observation: It has only limited sites because it is very costly. Currently no flights are in operation for radio sounding observation. Once it is released it cannot be controlled and flies outside of Pakistan also. Now, the balloons are used to check the wind directions etc, using theodolites. But details such as air temp, humidity etc. are missed. - Upper air observation through radio system is urgently required to improve the forecast. - Satellite Information: PMD is getting satellite information but PMD wants to integrate the information in numerical weather prediction (NWP) - River Flood Forecast and Warning System: FFD deals with this system, and they have the models for the forecast. - Flash Flood Forecast and Warning System: There is a system called FFGS and it takes input from WRF and ICON model and makes the flash flood guidance for Pakistan and Afghanistan. It may be referred as Pak-Afghan FFGS model. - Landslide Forecast and Warning System: There is no such system, but it is required in KPK Province. - Tide Level Monitoring Network: PMD does not have such monitoring. But he will check and inform.

	- Snow Melt Flash Flood Forecast: It is being studied under GLOF-1 and GLOF-2. In this project AWS are installed, and they get satellite images. Based on these, warning is issued but they don't have model-based warning system, and their own analysis-based warning is issued, using temperature rise etc. The potential lakes that can be burst are observed. - PMD issues the warnings. - Avalanche Advisory Information System: Not available as of now. - Seismic Intensity Reporting System: PMD has the system and functioning, but it has limited stations and shall be expanded. Model shall also be improved. - Real-Time Rainfall & Water Level Observation Network: AWS is the real time. Water levels are monitored by WAPDA and rainfall by PMD. For Lei nullah, water levels are also monitored by PMD. - Weather Forecast Guidance System: PMD has this system, but it is not up to the satisfaction. It provides station-based information and has limited functionality. Existing EWS - Numerical Weather Forecast System: PMD has Numerical Weather Forecast (NWF) using WRF model. PMD wants to increase the resolution and increase the number of models. Currently, resolution is 5km but target to make the resolution to 1 km, for each super computers are required and updating the system is mandatory. - Heavy Rain Warning System: PMD don't have any separate system and it is same as NWF. - Flood Early Warning System: FFD does through the river routing models using PHAST model. It is only available for big/main rivers and not for small rivers. - Earthquake & Tsunami EWS: Earthquake is being monitored only. For Tsunami it is working but it was a pilot project, not a full scale. Only 5 to 6 warning posts were installed, therefore more stations are required. Coastal belt is more than 750 km and Makran is most vulnerable for Tsunamis. Recently, the tropical cyclones are rotating around the coastal areas and entering in Baluchistan but earlier it was ending up in the Arabian sea. - Cyclone EWS: It is available but not so strong system. It was developed in 2000 and since it was not updated. - Heat Wave & Marine EWS: There are no EWS. It is forecasted using the rise in temperature and precipitation data and it is also included in the drought EWS. The forecast is published in news etc. Currently it can be excluded and not required, as of now. - GLOF Early Warning System: Currently, 24 valleys are covered but about 76 valleys are unattended, therefore, upgradation is necessary. GLOF is predicted by observing temperatures, cloud cover, and chances of GLOF is predicted. WAPDA GLOF research has no connection with PMD and WAPDA is doing independently/separately. The warning is given through press release.

	- Drought monitoring & Early Warning System: It was established in 2005 and it is very old and outdated which needs to be improved. This can also serve as Heat Wave EWS. Radar System - Multan and Sukkar are the priority. For Multan radar system, all processes are completed, PC-I is finalized, and tendering finished. It is expected to start in Dec 2023. For Sukkar, the consultant requested additional amount of about 840 million and approval for this additional amount is awaited. - PMD is going to install 7 radars. JICA will assist 2 radars, 1 each in Multan and Sukkar whereas 5 radars are approved in WB PC-I at Gwadar, Cherat, Lahore, D.I. Khan, Quetta. In D.I. Khan the equipment will be replaced but building will be same. The equipment is 90s and it is analog. In Lahore, a new building will be constructed because the height of the building is low and there are many buildings and noise around that area, therefore, a new taller building shall be constructed at the same location with new equipment. - With regards to EWS, first priority is Lahore and then Multan. Other priorities are Cherat, D.I. Khan, Quetta, and Gwadar. - For AWS, the priority is to cover all the Tehsils and then later it will be expanded further. However, the Tehsil itself is too big which cannot cover whole area, therefore, sub-stations are necessary. For example, Rawalpindi has 3 Tehsils and these 3 are not well represented. - It is suggested that the Urban Flooding project is a big project and covers many items including flash flood warning. The provincial capitals shall be the priority. We can estimate the river floods from cities and estimates floods. - X-band radars in mountain region performs satisfactorily. Facility - SMRFC (Specialized Medium Range Forecasting Centre): It is established in Islamabad. PMD has constructed 220 teraflop Numerical Weather Prediction (NWP) facility, and it will be further improved under the ongoing MHSP program, but funds are still awaited. It does not forecast the floods but as a result the floods can be measured. This forecasting center can be regarded as EWS. - RFFWC (Regional Flood Forecasting and Warning Centre): FFD is the only available for flood forecasting at the national level. Further RFFWC are planned but not yet implemented and can be regarded as future plans. - LFFFC (Local Flash Flood Forecasting Centre): It is only available at Lei nullah, and others are planned as future plans, such as Urban Flash Flooding Centers. Communication System - Communication System between PMD and NDMA: Communication is done using mobile, fax, internet, letter, social media groups (Whatsapp). - Weather Information Broadcasting System: There is no broadcasting system available with PMD. In Islamabad, Lahore and Karachi, there is

	a video studio where they make videos for the youtube or social media etc., also on their website. Almost they make video daily informing about weather etc. The future project for the establishment of Weather Early Warning broadcast system in the provincial capitals and Islamabad is suggested by PMD to JICA team. - PDMA make videos etc related to disasters, but PMD informs on daily basis regarding the weather. - GTS (Global Telecommunication System): As per the WMO guidelines, after every 3 hours the data is collected and shared with CDPC (Computerized Data Processing Center) Karachi. CDPC is responsible to collect all the data from Pakistan. WMO has established an international regional center at New Delhi, and it is further shared with WMO. WMO further predicts and estimates the weather worldwide using all the regional offices data. - WMO provides the trainings also and the observatories are maintained using the WMO standards. - PMD suggested to have models for the data simulations for the predictions, using the country's regional data. Similarly, the regional models shall be prepared for the predictions. Capacity Development of PMD - Community Enlightenment for EWS of Training and Drills: PMD has no facility for the training and drills. PMD provided for Lei nullah only. However, this training is very important as the community is not aware of the actions to be taken. - Education Program for Advanced Met and Hydro for PMD Staff: The education program is ongoing and shall be continued - Capacity Development against Local Flash Floods in Vulnerable Areas: It shall be carried out by PDMA or DDMA.
Prepared by	Rahmat Ullah (Construction & Procurement planning/Cost Estimate)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	04 January 2024, 10:00 ~ 11:00(PST)	
Interviewee	Col. (R) Raza Iqbal, Executive Director (Planning), NDMA Mr. Abdul Latif, Deputy Director (Planning), NDMA	
Place	NDMA Office, Islamabad	
Participants	NDMA	Raza Iqbal, Abdul Latif
	Project Team	Falak, Rahmat
Memo	Agenda: 1. Opinion exchange with NDMA for amendment/revision of NDMP-2's R/D (finalization by third week of January 2024). 2. Finalizing authority and signatory of the revised amendment. 3. Selection of participants by NDMA for the second Study Tour to Japan (scheduled in May 2024). (Finalization of candidates by second week of February 2024 at the latest). 4. Discussion on E-disaster Risk Assessment and possibility to assist NDMA within this Project. (by middle of January 2024) Minutes: 1. Opinion exchange with NDMA for amendment/revision of NDMP-2's R/D. - NDMA wants to close NDMP-2 project and requested to revise the R/D accordingly. NDMA asked to propose the amendments in the track changes using the original R/D and send officially by letter to NDMA. They will seek final approval from the chairman to finalize R/D. - NDMA requested to keep the doors open for further future projects but currently, NDMA plans to update the NDMP annually and they do not require long term plan. - Whatever is completed so far in the project can be part of R/D but further no services are required for this project and that shall be excluded from R/D. - It was informed that a letter from JICA Pakistan office is received stating that JICA Japan Team would like to visit NDMA during third week of January 2024 to discuss (a) NDMP-2 and (b) Assistance for NEOC. NDMA informed that further discussion on NDMP-2 is not required but for NEOC it is welcomed. 2. Finalizing authority and signatory of the revised amendment. - The revised amendment or R/D will be approved by chairman and Member can sign the R/D after approval from chairman. 3. Selection of participants by NDMA for the second Study Tour to Japan - Since the project is discontinued so it is suggested to cancel the study	

	tour. Such study tour can be arranged in other study/project with NDMA. - It was suggested that the participants from other organizations such as PDMA etc. shall be allowed and not limited to only NDMA staff. 4. Discussion on E-disaster Risk Assessment and possibility to assist NDMA within this Project. - NDMA has requested NDRMF for the funds for the E-disaster Risk Assessment and it is expected that the funds will be released but the process is time consuming. The funds estimated for this project is about 1 billion PKR. The funds will be used for the equipment, software, and studies. - NEOC has the capacity to carry out such studies. - JICA can support NEOC by furnishing latest equipment, software, technology transfer and capacity building. However, it shall be done in another project as NDMP-2 will be discontinued, or the name of the project shall be changed.
Prepared by	Rahmat Ullah (Construction & Procurement planning/Cost Estimate)

Memorandum of Meeting

Project	Technical Support Project to Review and Update National Disaster Prevention Plan (NDMP)	
Date and Time	22 February 2024, 14:00 ~ 15:30 (PST)	
Interviewee	・Tanveer Ahmed (Assistant Manager Seismology, NEOC) ・Sajid Ali (Assistant Manager GIS, NEOC)	
Place	National Emergencies Operation Center (NEOC), National Disaster Management Agency (NDMA)	
Participants	NDMA	Above members, Ms. Zahra Hassan and Mr. Syed M Tayyab Shah
	JICA Project Team	Hasegawa, Mukhtar
Memo	1) Discussion ・ Seismology staff of NEOC explained overview of the Deterministic Seismic Hazard Mapping System on the screen of GIS. ・ Contour lines of estimated seismic intensity was drawn on the map around northern area of Islamabad. He said the target area is higher seismic potential area than others. ・ Amplification factor for site effects are seems to be set to each geological unit by observing microtremor H/V spectrum. (after KHAN Sarfraz, Seismic Hazard Map Based on Geology and Shear-wave Velocity of Rawalpindi-Islamabad, 2020) ・ Details of GIS database is not mentioned at that time, only showing layers for presentation mode. But it is highlighted that the GIS system contains building footprint polygon layer. He said that it will be used to estimate building damage due to earthquakes. ・ In the meeting, we do not have any information about structure type of building footprint, nor vulnerability function to estimate damage. ・ Hasegawa advised that grid cell system is usually used to calculate seismic hazard map, population exposure and several structural damages. ・ One of the important grid cell GIS data is geomorphologic classification map, named “Japan Engineering Geomorphologic Classification Map” (Hasegawa is one of co-authors.) It is also supplementally explained that this kind of soil amplification database is upgraded according to the latest geophysical survey result. ・ As a case study, Hasegawa demonstrated ArcGIS based calculation of deterministic seismic hazard map using USGS Vs30 data. It was prepared for another JICA project by Hasegawa. (See last page.) ・ This system can be applied to NEOC, supposing just after large earthquake occurs. PMD will provide hypocenter parameters of longitude, latitude, depth and magnitude. Just inputting the parameters to the calculation system, we obtain the PGA and MMI map in a few seconds. ・ This kind of GIS system is very important as Real-time Seismic Hazard Mapping System. ・ 2) Issues on technical discussions ・ It is very important to collaborate with PMD to complete Real-time Seismic Hazard & Risk Mapping System. Seismic parameters should be provided by PMD just after large earthquake occurs. ・ There is less discussion on database and systems to calculate deterministic seismic hazard map by NEOC side. We need further technical discussion for better upgrading. ・ There is less information of building structure type and vulnerability functions (fragility curves) by NEOC side. We need further technical	

	discussion for how to add attribute of structure type, etc. for detailed damage estimation. ・ As a prototype soul amplification data (Vs30), USGS open data would be used. It would be upgraded step by step according to future geophysical surveys. ・ H/V spectrum method is a little bit out of date. If a new project is launched, Japanese side will give some technology transfer on array microtremor analysis for reliable Vs30 values. ・ These above technical issues would be continuously discussed in the NDMP project by acceptance among JICA and NEOC chairman. ・ ・ This real-time system can be used for considering plans for dispatching troops to disaster areas. ・ The system can also be used for prioritizing vulnerable areas to earthquakes by executing several deterministic seismic hazard and risk assessment. The result of the hazard and risk estimation must be used for seismic countermeasure, like retrofitting infrastructures (road, lifelines, etc.) important facilities (government office, hospitals, evacuation shelters, etc.) ・ ・ JICA intends to conduct the pre-investment to resist future earthquake damages. Investing in disaster prevention in vulnerable areas in advance can reduce damage to affected areas in an actual earthquake and ensure that military relief operations are reduced and more efficient. ・ 3) Request of introduction of responsible person ・ It is requested to explain the above discussion result to the upper person of NEOC. ・ 4) Next schedule ・ It is welcome to further discussion to introduce the grid cell system to calculate Real-time Seismic Hazard & Risk Mapping System to NEOC. ・
Contact Info.	・Tanveer Ahmed (Assistant Manager Seismology, NEOC) ・Sajid Ali (Assistant Manager GIS, NEOC) ・Mr. Syed M Tayyab Shah (Team leader, Risk Assessment Unit, NDMA) ・Ms. Zahra Hassan (MHVRA Expert, Risk Assessment Unit, NDMA